

# Actividad - Proyecto práctico

Actividad en grupos de 4 alumnos. Nombres en orden alfabético (de apellidos):

- **Alumno 1:** Vara Rodríguez, Jorge
- **Alumno 2:** Larios Pérez, José María
- **Alumno 3:** Fernández Gordo, Javier
- **Alumno 4:** Carvajal Benítez, Pablo

**Entorno:** SpaceInvaders-v0 · **Algoritmo:** DQN (Stable-Baselines3) · **Requisito mínimo:** media de recompensa en modo test por encima del umbral indicado en el listado del grupo.

**Repositorio del proyecto:**

[https://github.com/pablocarvajal12/08\\_MIAR\\_PROYECTO\\_GRUPO22](https://github.com/pablocarvajal12/08_MIAR_PROYECTO_GRUPO22)

## Proyecto 10 - DQN con Stable-Baselines3

Entrenamiento de un agente DQN para SpaceInvaders-v0 usando Stable-Baselines3.

El notebook mantiene la idea de la práctica anterior:

- imágenes de entrada de 84x84,
- pila de 4 frames,
- red convolucional tipo DQN Atari,
- recompensa clipped,
- conteo de episodios consecutivos con recompensa clipped superior a 20,
- guardado de métricas completas en CSV/JSON, TensorBoard y guardado manual del modelo.

```
In [ ]: # =====  
# 1. Instalación de librerías  
# =====  
  
# Instalación completa para usar SpaceInvaders-v0 con Gym clásico  
  
%pip install gym==0.17.3  
%pip install git+https://github.com/Kojoley/atari-py.git  
%pip install stable-baselines3==0.10.0
```

Collecting gym==0.17.3

Downloading gym-0.17.3.tar.gz (1.6 MB)

```
0.0/1.6 MB ? eta -:--:--
1.6/1.6 MB 111.2 MB/s eta 0:00:01
1.6/1.6 MB 39.0 MB/s eta 0:00:00
```

Preparing metadata (setup.py) ... done

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from gym==0.17.3) (1.16.3)

Requirement already satisfied: numpy>=1.10.4 in /usr/local/lib/python3.12/dist-packages (from gym==0.17.3) (2.0.2)

Collecting pygamelet<=1.5.0,>=1.4.0 (from gym==0.17.3)

Downloading pygamelet-1.5.0-py2.py3-none-any.whl.metadata (7.6 kB)

Collecting cloudpickle<1.7.0,>=1.2.0 (from gym==0.17.3)

Downloading cloudpickle-1.6.0-py3-none-any.whl.metadata (4.3 kB)

Requirement already satisfied: future in /usr/local/lib/python3.12/dist-packages (from pygamelet<=1.5.0,>=1.4.0->gym==0.17.3) (1.0.0)

Downloading cloudpickle-1.6.0-py3-none-any.whl (23 kB)

Downloading pygamelet-1.5.0-py2.py3-none-any.whl (1.0 MB)

```
1.0/1.0 MB 78.0 MB/s eta 0:00:00
```

Building wheels for collected packages: gym

Building wheel for gym (setup.py) ... done

Created wheel for gym: filename=gym-0.17.3-py3-none-any.whl size=1654617 sha256=0b5e6520baaf7a136e63263f1b39b2976e202d04f4f03d4a54b66d3a3d7ecc48

Stored in directory: /root/.cache/pip/wheels/99/7e/03/69cdeb432380e05bc9a56ff68451d8063e41aa06db508f4be5

Successfully built gym

Installing collected packages: pygamelet, cloudpickle, gym

Attempting uninstall: cloudpickle

Found existing installation: cloudpickle 3.1.2

Uninstalling cloudpickle-3.1.2:

Successfully uninstalled cloudpickle-3.1.2

Attempting uninstall: gym

Found existing installation: gym 0.25.2

Uninstalling gym-0.25.2:

Successfully uninstalled gym-0.25.2

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

bigframes 2.42.0 requires cloudpickle>=2.0.0, but you have cloudpickle 1.6.0 which is incompatible.

distributed 2026.1.1 requires cloudpickle>=3.0.0, but you have cloudpickle 1.6.0 which is incompatible.

dask 2026.1.1 requires cloudpickle>=3.0.0, but you have cloudpickle 1.6.0 which is incompatible.

Successfully installed cloudpickle-1.6.0 gym-0.17.3 pygamelet-1.5.0

Collecting git+https://github.com/Kojoley/atari-py.git

Cloning https://github.com/Kojoley/atari-py.git to /tmp/pip-req-build-imub\_2zg

Running command git clone --filter=blob:none --quiet https://github.com/Kojoley/atari-py.git /tmp/pip-req-build-imub\_2zg

Resolved https://github.com/Kojoley/atari-py.git to commit 86a1e05c0a95e9e6233c3a413521fdb34ca8a089

Preparing metadata (setup.py) ... done

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from atari-py==1.2.2) (2.0.2)

Building wheels for collected packages: atari-py

Building wheel for atari-py (setup.py) ... done

Created wheel for atari-py: filename=atari\_py-1.2.2-cp312-cp312-linux\_x86\_64.whl size=4739135 sha256=48f93941e5a0515e594f3c782b419f2cb4ff36dda52493963af0baed20d5dc61

Stored in directory: /tmp/pip-ephem-wheel-cache-4sqy5dcp/wheels/cb/1f/56/35b393

898f74ae3d978063738e8eeb4b3d7357d0d7ecec6b9f

Successfully built atari-py

Installing collected packages: atari-py

Successfully installed atari-py-1.2.2

Collecting stable-baselines3==0.10.0

Downloading stable\_baselines3-0.10.0-py3-none-any.whl.metadata (3.5 kB)

Requirement already satisfied: gym>=0.17 in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (0.17.3)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.0.2)

Requirement already satisfied: torch>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.11.0+cu128)

Requirement already satisfied: cloudpickle in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (1.6.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.2.2)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (3.10.0)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from gym>=0.17->stable-baselines3==0.10.0) (1.16.3)

Requirement already satisfied: pygame<=1.5.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from gym>=0.17->stable-baselines3==0.10.0) (1.5.0)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.29.4)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (4.15.0)

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.6.1)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (2025.3.0)

Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.1)

Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (12.9.7)

Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (9.19.0.56)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (2.28.9)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.4.5)

Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.6.0)

Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.4.1)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)

Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (11.3.3.83)

Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (1.13.1.3)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)

Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (10.3.9.90)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (11.7.3.90)

Requirement already satisfied: nvidia-cusparsenvjitlink-cu12==12.5.8.93.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.5.8.93)

Requirement already satisfied: nvidia-cusolver-cu12==12.8.93.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.93)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.93)

Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.\* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform\_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (4.63.0)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (1.5.0)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (26.2)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (3.3.2)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->stable-baselines3==0.10.0) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->stable-baselines3==0.10.0) (2026.2)

Requirement already satisfied: cuda-pathfinder~1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4->torch>=1.4.0->stable-baselines3==0.10.0) (1.5.5)

Requirement already satisfied: future in /usr/local/lib/python3.12/dist-packages

(from pygame<=1.5.0,>=1.4.0->gym>=0.17->stable-baselines3==0.10.0) (1.0.0)  
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages  
(from python-dateutil>=2.7->matplotlib->stable-baselines3==0.10.0) (1.17.0)  
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages  
(from sympy>=1.13.3->torch>=1.4.0->stable-baselines3==0.10.0) (1.3.0)  
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages  
(from jinja2->torch>=1.4.0->stable-baselines3==0.10.0) (3.0.3)  
Downloading stable\_baselines3-0.10.0-py3-none-any.whl (145 kB)

146.0/146.0 kB 14.5 MB/s eta 0:00:00

Installing collected packages: stable-baselines3

Successfully installed stable-baselines3-0.10.0

```
In [ ]: import gym
        from stable_baselines3 import DQN

        ENV_ID = "SpaceInvaders-v0"
        env = gym.make(ENV_ID)

        print("Gym OK")
        print("SB3 DQN OK")
```

Gym OK

SB3 DQN OK

```
In [ ]: # =====
        # 2. Imports
        # =====

        from pathlib import Path
        import json
        import time
        from collections import deque

        import numpy as np
        import pandas as pd
        import matplotlib.pyplot as plt

        import gym
        import torch as th
        import torch.nn as nn

        from google.colab import drive

        from stable_baselines3 import DQN
        from stable_baselines3.common.env_util import make_atari_env
        from stable_baselines3.common.vec_env import VecFrameStack
        from stable_baselines3.common.callbacks import BaseCallback
        from stable_baselines3.common.torch_layers import BaseFeaturesExtractor
```

```
In [ ]: # =====
        # 3. Google Drive y rutas
        # =====

        drive.mount('/content/gdrive')

        drive_root = '/content/gdrive/My Drive/08_MIAR/sb3/conlogs'
        PROJECT_NAME = 'sb3_1_spaceinvaders_v0'

        OUTPUT_DIR = Path(drive_root) / 'datos' / PROJECT_NAME
        OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
```

```

# Modelos
MODEL_LATEST = OUTPUT_DIR / 'dqn_spaceinvaders_v0_sb3_latest.zip'
POLICY_LATEST = OUTPUT_DIR / 'dqn_spaceinvaders_v0_policy_latest.pt'

# Logs completos propios
LOG_FILENAME = OUTPUT_DIR / 'dqn_SpaceInvaders-v0_sb3_log.json'
CSV_LOG_FILENAME = OUTPUT_DIR / 'dqn_SpaceInvaders-v0_sb3_log.csv'

# Logs oficiales de SB3
TENSORBOARD_DIR = OUTPUT_DIR / 'tensorboard'
MONITOR_DIR = OUTPUT_DIR / 'monitor'
TENSORBOARD_DIR.mkdir(parents=True, exist_ok=True)
MONITOR_DIR.mkdir(parents=True, exist_ok=True)

print('OUTPUT_DIR:', OUTPUT_DIR)
print('MODEL_LATEST:', MODEL_LATEST)
print('POLICY_LATEST:', POLICY_LATEST)
print('LOG_FILENAME:', LOG_FILENAME)
print('CSV_LOG_FILENAME:', CSV_LOG_FILENAME)
print('TENSORBOARD_DIR:', TENSORBOARD_DIR)
print('MONITOR_DIR:', MONITOR_DIR)

```

Mounted at /content/gdrive

```

OUTPUT_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0
MODEL_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_sb3_latest.zip
POLICY_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_policy_latest.pt
LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json
CSV_LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv
TENSORBOARD_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/tensorboard
MONITOR_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/monitor

```

```

In [ ]: # =====
# 4. Parámetros principales
# =====

ENV_ID = 'SpaceInvaders-v0'
SEED = 42

TOTAL_TIMESTEPS = 10_000_000

LEARNING_RATE = 1e-4
BUFFER_SIZE = 100_000
LEARNING_STARTS = 50_000
BATCH_SIZE = 32
GAMMA = 0.99
TRAIN_FREQ = 4
TARGET_UPDATE_INTERVAL = 10_000

EXPLORATION_INITIAL_EPS = 1.0
EXPLORATION_FINAL_EPS = 0.001
EXPLORATION_FRACTION = 0.30

```

```

CONSECUTIVE_THRESHOLD = 20.0
TARGET_CONSECUTIVE_EPISODES = 100

SAVE_FREQ_STEPS = 100_000
PRINT_EVERY_EPISODES = 10

LOAD_MODEL = False

```

```

In [ ]: # =====
# 5. Comprobación del entorno
# =====
# Si SpaceInvaders-v0 no existe en el entorno de Colab, esta celda dará error.

test_env = gym.make(ENV_ID)
print('Entorno creado correctamente:', ENV_ID)
print('Action space:', test_env.action_space)
print('Observation space:', test_env.observation_space)
test_env.close()

```

Entorno creado correctamente: SpaceInvaders-v0  
 Action space: Discrete(6)  
 Observation space: Box(0, 255, (210, 160, 3), uint8)

```

In [ ]: # =====
# 6. Mostrar imagen original del entorno
# =====

def reset_env_compat(env):
    result = env.reset()
    if isinstance(result, tuple):
        return result[0]
    return result

raw_env = gym.make(ENV_ID)
obs = reset_env_compat(raw_env)

try:
    frame_rgb = raw_env.render(mode='rgb_array')
except TypeError:
    frame_rgb = raw_env.render()

plt.figure(figsize=(6, 6))
plt.imshow(frame_rgb)
plt.title('Imagen original del entorno')
plt.axis('off')
plt.show()

raw_env.close()

```

Imagen original del entorno



```
In [ ]: # =====
# 7. Crear entorno vectorizado para SB3
# =====
# make_atari_env aplica los wrappers habituales de Atari.
# clip_reward=True hace que la recompensa sea clipped [-1, 0, 1].
# VecFrameStack apila 4 imágenes consecutivas.
```

```
def create_env():
    env = make_atari_env(
        ENV_ID,
        n_envs=1,
        seed=SEED,
        monitor_dir=str(OUTPUT_DIR / 'monitor'),
        wrapper_kwargs=dict(clip_reward=True)
    )
    env = VecFrameStack(env, n_stack=4)
    return env

env = create_env()
print('Observation space SB3:', env.observation_space)
print('Action space SB3:', env.action_space)
```

Observation space SB3: Box(0, 255, (84, 84, 4), uint8)  
Action space SB3: Discrete(6)

```
In [ ]: # =====
# 8. Mostrar imágenes procesadas 84x84 y stack de frames
# =====

obs = env.reset()
```



```

print('Shape observación:', obs.shape)

# Normalmente con CnnPolicy/SB3 queda en formato: (n_envs, channels, height, width)
obs0 = obs[0]

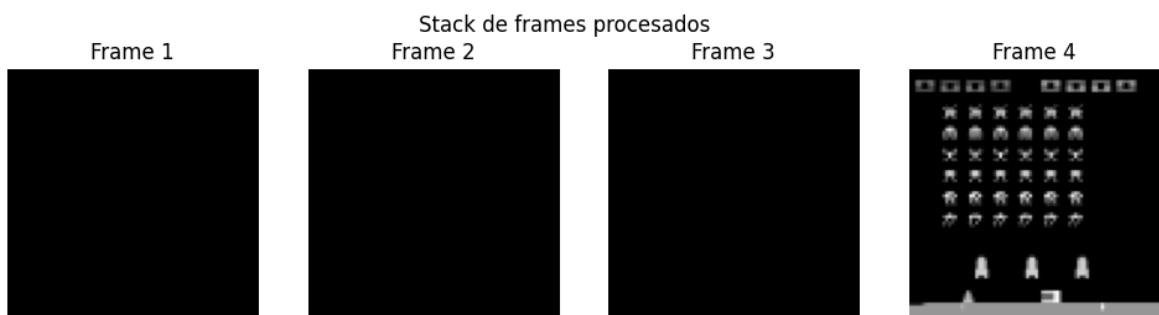
if obs0.shape[0] in [1, 3, 4, 12]:
    # channels first
    frames = obs0
else:
    # channels last
    frames = np.transpose(obs0, (2, 0, 1))

num_frames = min(4, frames.shape[0])

plt.figure(figsize=(12, 3))
for i in range(num_frames):
    plt.subplot(1, num_frames, i + 1)
    plt.imshow(frames[i], cmap='gray')
    plt.title(f'Frame {i + 1}')
    plt.axis('off')
plt.suptitle('Stack de frames procesados')
plt.show()

```

Shape observación: (1, 84, 84, 4)



```

In [ ]: # =====
# 9. Red convolucional personalizada 84x84
# =====
# Arquitectura tipo DQN Atari:
# Entrada: 4 x 84 x 84
# Conv2D 32 filtros 8x8 stride 4
# Conv2D 64 filtros 4x4 stride 2
# Conv2D 64 filtros 3x3 stride 1
# Capa densa de 512 características

class CustomCNN84(BaseFeaturesExtractor):
    def __init__(self, observation_space, features_dim=512):
        super().__init__(observation_space, features_dim)

        n_input_channels = observation_space.shape[0]

        self.cnn = nn.Sequential(
            nn.Conv2d(n_input_channels, 32, kernel_size=8, stride=4),
            nn.ReLU(),
            nn.Conv2d(32, 64, kernel_size=4, stride=2),
            nn.ReLU(),
            nn.Conv2d(64, 64, kernel_size=3, stride=1),
            nn.ReLU(),
            nn.Flatten()
        )

```

```

        with th.no_grad():
            sample = th.as_tensor(observation_space.sample()[None]).float()
            n_flatten = self.cnn(sample).shape[1]

        self.linear = nn.Sequential(
            nn.Linear(n_flatten, features_dim),
            nn.ReLU()
        )

    def forward(self, observations):
        return self.linear(self.cnn(observations))

policy_kwargs = dict(
    features_extractor_class=CustomCNN84,
    features_extractor_kwargs=dict(features_dim=512),
    net_arch=[]
)

```

```

In [ ]: # =====
# 9. Función auxiliar de guardado
# =====
# Esta función solo se usa como alternativa si model.save(...) falla por problem
# de serialización en Colab/Python. El logger oficial será el de SB3/TensorBoard

def save_policy_checkpoint(model, path):
    path = Path(path)
    path.parent.mkdir(parents=True, exist_ok=True)

    th.save(
        {
            "policy_state_dict": model.policy.state_dict(),
            "optimizer_state_dict": model.policy.optimizer.state_dict(),
            "num_timesteps": model.num_timesteps,
            "exploration_rate": getattr(model, "exploration_rate", None),
        },
        path
    )

```

```

In [ ]: # =====
# 10. Logger completo y callback personalizado
# =====
# Guarda todas las métricas propias en JSON y CSV.
# Además registra métricas custom en TensorBoard/SB3.

class SB3TrainingLogger:
    def __init__(self, json_path, csv_path):
        self.json_path = Path(json_path)
        self.csv_path = Path(csv_path)
        self.json_path.parent.mkdir(parents=True, exist_ok=True)
        self.csv_path.parent.mkdir(parents=True, exist_ok=True)

        if self.json_path.exists():
            with open(self.json_path, 'r') as f:
                self.data = json.load(f)
            print('Log JSON previo cargado:', self.json_path)
        else:
            self.data = {
                'episode': [],

```

```

        'total_steps': [],
        'episode_reward': [],
        'clipped_episode_reward': [],
        'consecutive_clipped_over_20': [],
        'max_consecutive_clipped_over_20': [],
        'avg_clip50': [],
        'duration': [],
        'exploration_rate': []
    }

def add_episode(self, episode, total_steps, reward, clipped_reward,
               consecutive, max_consecutive, avg_clip50, duration,
               exploration_rate):
    row = {
        'episode': int(episode),
        'total_steps': int(total_steps),
        'episode_reward': float(reward),
        'clipped_episode_reward': float(clipped_reward),
        'consecutive_clipped_over_20': int(consecutive),
        'max_consecutive_clipped_over_20': int(max_consecutive),
        'avg_clip50': float(avg_clip50),
        'duration': float(duration),
        'exploration_rate': float(exploration_rate) if exploration_rate is not None
    }

    for key, value in row.items():
        self.data[key].append(value)

    # CSV incremental: se guarda en cada episodio para no perder información
    pd.DataFrame([row]).to_csv(
        self.csv_path,
        mode='a',
        header=not self.csv_path.exists(),
        index=False
    )

def save_json(self):
    with open(self.json_path, 'w') as f:
        json.dump(self.data, f, indent=2)

def __len__(self):
    return len(self.data.get('episode', []))

class ConsecutiveRewardCallback(BaseCallback):
    def __init__(self, training_logger, verbose=0):
        super().__init__(verbose)
        self.training_logger = training_logger
        self.episode = len(training_logger)
        self.consecutive = 0
        self.max_consecutive = 0
        self.recent_clipped_rewards = deque(maxlen=50)
        self.start_time = time.time()
        self.last_json_save_step = 0

        previous_clipped = training_logger.data.get('clipped_episode_reward', [])
        for value in previous_clipped:
            self.recent_clipped_rewards.append(float(value))

        previous_consec = training_logger.data.get('consecutive_clipped_over_20', [])

```

```

if len(previous_consec) > 0:
    self.consecutive = int(previous_consec[-1])

previous_max = training_logger.data.get('max_consecutive_clipped_over_20')
if len(previous_max) > 0:
    self.max_consecutive = int(max(previous_max))

def _on_step(self) -> bool:
    infos = self.locals.get('infos', [])

    for info in infos:
        if 'episode' not in info:
            continue

        ep_reward = float(info['episode']['r'])

        # Métrica comparable con tu práctica anterior.
        # En esta configuración SB3 informa reward real; para Space Invaders
        clipped_reward = ep_reward / 10.0

        duration = time.time() - self.start_time
        self.start_time = time.time()

        self.recent_clipped_rewards.append(clipped_reward)

        if clipped_reward > CONSECUTIVE_THRESHOLD:
            self.consecutive += 1
        else:
            self.consecutive = 0

        if self.consecutive > self.max_consecutive:
            self.max_consecutive = self.consecutive

        avg_clip50 = (
            float(np.mean(self.recent_clipped_rewards))
            if len(self.recent_clipped_rewards) > 0
            else 0.0
        )

        exploration_rate = getattr(self.model, 'exploration_rate', np.nan)

        self.training_logger.add_episode(
            episode=self.episode,
            total_steps=self.num_timesteps,
            reward=ep_reward,
            clipped_reward=clipped_reward,
            consecutive=self.consecutive,
            max_consecutive=self.max_consecutive,
            avg_clip50=avg_clip50,
            duration=duration,
            exploration_rate=exploration_rate
        )

        # Métricas personalizadas para TensorBoard/SB3.
        self.logger.record('custom/clipped_episode_reward', clipped_reward)
        self.logger.record('custom/consecutive_clipped_over_20', self.consecutive)
        self.logger.record('custom/max_consecutive_clipped_over_20', self.max_consecutive)
        self.logger.record('custom/avg_clip50', avg_clip50)

    if self.episode % PRINT_EVERY_EPISODES == 0:

```

```

        print(
            f'EP={self.episode:,} | '
            f'STEP={self.num_timesteps:,} | '
            f'clip={clipped_reward:.0f} | '
            f'consec={self.consecutive} | '
            f'max_consec={self.max_consecutive} | '
            f'avg_clip50={avg_clip50:.2f}'
        )

        self.episode += 1

        if self.max_consecutive >= TARGET_CONSECUTIVE_EPISODES:
            print('Objetivo conseguido: 100 episodios consecutivos superando')
            self.training_logger.save_json()
            return False

        if self.num_timesteps - self.last_json_save_step >= SAVE_FREQ_STEPS:
            self.training_logger.save_json()
            self.last_json_save_step = self.num_timesteps

        return True

    def _on_training_end(self):
        self.training_logger.save_json()
        print('Entrenamiento finalizado.')
        print('max_consec alcanzado:', self.max_consecutive)
        print('Log JSON:', LOG_FILENAME)
        print('Log CSV:', CSV_LOG_FILENAME)

```

```

In [ ]: # =====
# 11. Crear o cargar modelo DQN
# =====

if LOAD_MODEL and MODEL_LATEST.exists():
    print("Cargando modelo desde:", MODEL_LATEST)

    model = DQN.load(
        str(MODEL_LATEST),
        env=env,
        device="auto",
        tensorboard_log=str(TENSORBOARD_DIR)
    )

    print("Modelo cargado correctamente.")

else:
    print("Creando modelo nuevo.")

    model = DQN(
        policy="CnnPolicy",
        env=env,
        learning_rate=LEARNING_RATE,
        buffer_size=BUFFER_SIZE,
        learning_starts=LEARNING_STARTS,
        batch_size=BATCH_SIZE,
        gamma=GAMMA,
        train_freq=TRAIN_FREQ,
        target_update_interval=TARGET_UPDATE_INTERVAL,
        exploration_initial_eps=EXPLORATION_INITIAL_EPS,
        exploration_final_eps=EXPLORATION_FINAL_EPS,

```

```

        exploration_fraction=EXPLORATION_FRACTION,
        policy_kwargs=policy_kwargs,
        tensorboard_log=str(TENSORBOARD_DIR),
        verbose=1,
        seed=SEED,
        device="auto"
    )

    print("Modelo nuevo creado correctamente.")

print("Política usada:")
print(model.policy)

```

Creando modelo nuevo.

Using cuda device

Wrapping the env in a VecTransposeImage.

Modelo nuevo creado correctamente.

Política usada:

CnnPolicy(

```

    (q_net): QNetwork(
      (features_extractor): CustomCNN84(
        (cnn): Sequential(
          (0): Conv2d(4, 32, kernel_size=(8, 8), stride=(4, 4))
          (1): ReLU()
          (2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))
          (3): ReLU()
          (4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))
          (5): ReLU()
          (6): Flatten(start_dim=1, end_dim=-1)
        )
        (linear): Sequential(
          (0): Linear(in_features=3136, out_features=512, bias=True)
          (1): ReLU()
        )
      )
    )
    (q_net): Sequential(
      (0): Linear(in_features=512, out_features=6, bias=True)
    )
  )
  (q_net_target): QNetwork(
    (features_extractor): CustomCNN84(
      (cnn): Sequential(
        (0): Conv2d(4, 32, kernel_size=(8, 8), stride=(4, 4))
        (1): ReLU()
        (2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))
        (3): ReLU()
        (4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))
        (5): ReLU()
        (6): Flatten(start_dim=1, end_dim=-1)
      )
      (linear): Sequential(
        (0): Linear(in_features=3136, out_features=512, bias=True)
        (1): ReLU()
      )
    )
  )
  (q_net): Sequential(
    (0): Linear(in_features=512, out_features=6, bias=True)
  )
)
)

```

```
In [ ]: # =====
# 12. Entrenamiento
# =====

training_logger = SB3TrainingLogger(
    json_path=LOG_FILENAME,
    csv_path=CSV_LOG_FILENAME
)

callback = ConsecutiveRewardCallback(
    training_logger=training_logger,
    verbose=1
)

model.learn(
    total_timesteps=TOTAL_TIMESTEPS,
    callback=callback,
    reset_num_timesteps=not LOAD_MODEL,
    tb_log_name='DQN_SpaceInvaders_v0',
    log_interval=100
)
```

Se han truncado las últimas 5000 líneas del flujo de salida.

EP=4,590 | STEP=802,238 | clip=22 | consec=3 | max\_consec=8 | avg\_clip50=22.83

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 22.8   |  |
| clipped_episode_r... | 16.5   |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 838    |  |
| ep_rew_mean          | 225    |  |
| exploration rate     | 0.732  |  |
| time/                |        |  |
| episodes             | 13700  |  |
| fps                  | 262    |  |
| time_elapsed         | 3057   |  |
| total timesteps      | 803356 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0988 |  |
| n_updates            | 188338 |  |

EP=4,600 | STEP=804,242 | clip=3 | consec=0 | max\_consec=8 | avg\_clip50=21.31

EP=4,610 | STEP=806,407 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=21.86

EP=4,620 | STEP=808,665 | clip=34 | consec=5 | max\_consec=8 | avg\_clip50=24.62

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 24.9   |  |
| clipped_episode_r... | 39.5   |  |
| consecutive_clipp... | 2      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 865    |  |
| ep_rew_mean          | 236    |  |
| exploration rate     | 0.73   |  |
| time/                |        |  |
| episodes             | 13800  |  |
| fps                  | 262    |  |
| time_elapsed         | 3085   |  |
| total timesteps      | 810578 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.157  |  |
| n_updates            | 190144 |  |

EP=4,630 | STEP=810,660 | clip=38 | consec=3 | max\_consec=8 | avg\_clip50=25.48

EP=4,640 | STEP=812,758 | clip=4 | consec=0 | max\_consec=8 | avg\_clip50=25.26

EP=4,650 | STEP=814,780 | clip=38 | consec=1 | max\_consec=8 | avg\_clip50=26.46

EP=4,660 | STEP=816,706 | clip=34 | consec=2 | max\_consec=8 | avg\_clip50=26.78

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 27.2  |  |
| clipped_episode_r... | 11.5  |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 8     |  |
| rollout/             |       |  |
| ep_len_mean          | 866   |  |
| ep_rew_mean          | 247   |  |
| exploration rate     | 0.728 |  |
| time/                |       |  |



|                 |        |  |
|-----------------|--------|--|
| episodes        | 13900  |  |
| fps             | 262    |  |
| time_elapsed    | 3112   |  |
| total timesteps | 817497 |  |
| train/          |        |  |
| learning_rate   | 0.0001 |  |
| loss            | 0.127  |  |
| n_updates       | 191874 |  |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,670 | STEP=818,693 | clip=30 | consec=1 | max_consec=8 | avg_clip50=24.79 |
| EP=4,680 | STEP=820,395 | clip=24 | consec=1 | max_consec=8 | avg_clip50=24.63 |
| EP=4,690 | STEP=822,592 | clip=19 | consec=0 | max_consec=8 | avg_clip50=24.28 |

|                      |        |  |
|----------------------|--------|--|
| custom/              |        |  |
| avg_clip50           | 23.9   |  |
| clipped_episode_r... | 7.5    |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 835    |  |
| ep_rew_mean          | 249    |  |
| exploration rate     | 0.726  |  |
| time/                |        |  |
| episodes             | 14000  |  |
| fps                  | 262    |  |
| time_elapsed         | 3138   |  |
| total timesteps      | 823777 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0993 |  |
| n_updates            | 193444 |  |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,700 | STEP=824,250 | clip=44 | consec=2 | max_consec=8 | avg_clip50=24.43 |
| EP=4,710 | STEP=826,413 | clip=8  | consec=0 | max_consec=8 | avg_clip50=24.32 |
| EP=4,720 | STEP=828,194 | clip=17 | consec=0 | max_consec=8 | avg_clip50=23.50 |
| EP=4,730 | STEP=830,118 | clip=30 | consec=1 | max_consec=8 | avg_clip50=24.23 |

|                      |        |  |
|----------------------|--------|--|
| custom/              |        |  |
| avg_clip50           | 23.7   |  |
| clipped_episode_r... | 11     |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 826    |  |
| ep_rew_mean          | 245    |  |
| exploration rate     | 0.723  |  |
| time/                |        |  |
| episodes             | 14100  |  |
| fps                  | 262    |  |
| time_elapsed         | 3164   |  |
| total timesteps      | 830476 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0857 |  |
| n_updates            | 195118 |  |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,740 | STEP=832,118 | clip=9  | consec=0 | max_consec=8 | avg_clip50=23.95 |
| EP=4,750 | STEP=834,449 | clip=28 | consec=1 | max_consec=8 | avg_clip50=24.89 |
| EP=4,760 | STEP=836,389 | clip=8  | consec=0 | max_consec=8 | avg_clip50=23.07 |

-----

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.4   |
| clipped_episode_r... | 20     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 846    |
| ep_rew_mean          | 247    |
| exploration rate     | 0.721  |
| time/                |        |
| episodes             | 14200  |
| fps                  | 262    |
| time_elapsed         | 3193   |
| total timesteps      | 837790 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.124  |
| n_updates            | 196947 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,770 | STEP=838,677 | clip=17 | consec=0 | max_consec=8 | avg_clip50=25.18 |
| EP=4,780 | STEP=841,062 | clip=88 | consec=1 | max_consec=8 | avg_clip50=27.49 |
| EP=4,790 | STEP=843,098 | clip=14 | consec=0 | max_consec=8 | avg_clip50=27.62 |

-----

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 27.7   |
| clipped_episode_r... | 58     |
| consecutive_clipp... | 5      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 891    |
| ep_rew_mean          | 265    |
| exploration rate     | 0.719  |
| time/                |        |
| episodes             | 14300  |
| fps                  | 262    |
| time_elapsed         | 3222   |
| total timesteps      | 845165 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.062  |
| n_updates            | 198791 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,800 | STEP=845,408 | clip=33 | consec=6 | max_consec=8 | avg_clip50=27.83 |
| EP=4,810 | STEP=847,667 | clip=38 | consec=2 | max_consec=8 | avg_clip50=29.81 |
| EP=4,820 | STEP=849,856 | clip=37 | consec=1 | max_consec=8 | avg_clip50=30.00 |
| EP=4,830 | STEP=852,072 | clip=23 | consec=4 | max_consec=8 | avg_clip50=27.77 |

-----

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 27    |
| clipped_episode_r... | 6     |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 8     |
| rollout/             |       |
| ep_len_mean          | 911   |
| ep_rew_mean          | 270   |
| exploration rate     | 0.716 |
| time/                |       |
| episodes             | 14400 |
| fps                  | 262   |
| time_elapsed         | 3251  |

|                 |        |
|-----------------|--------|
| total timesteps | 852552 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.041  |
| n_updates       | 200637 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,840 | STEP=853,640 | clip=10 | consec=0 | max_consec=8 | avg_clip50=25.91 |
| EP=4,850 | STEP=855,848 | clip=24 | consec=4 | max_consec=8 | avg_clip50=24.87 |
| EP=4,860 | STEP=857,771 | clip=22 | consec=5 | max_consec=8 | avg_clip50=24.59 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.1   |
| clipped_episode_r... | 19     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 894    |
| ep_rew_mean          | 268    |
| exploration rate     | 0.714  |
| time/                |        |
| episodes             | 14500  |
| fps                  | 262    |
| time_elapsed         | 3280   |
| total timesteps      | 859639 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0443 |
| n_updates            | 202409 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,870 | STEP=860,189 | clip=32 | consec=2 | max_consec=8 | avg_clip50=25.22 |
| EP=4,880 | STEP=862,498 | clip=34 | consec=3 | max_consec=8 | avg_clip50=25.27 |
| EP=4,890 | STEP=864,594 | clip=14 | consec=0 | max_consec=8 | avg_clip50=27.98 |
| EP=4,900 | STEP=866,662 | clip=40 | consec=3 | max_consec=8 | avg_clip50=27.68 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 27.5   |
| clipped_episode_r... | 10     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 889    |
| ep_rew_mean          | 261    |
| exploration rate     | 0.711  |
| time/                |        |
| episodes             | 14600  |
| fps                  | 262    |
| time_elapsed         | 3309   |
| total timesteps      | 867006 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.297  |
| n_updates            | 204251 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,910 | STEP=868,526 | clip=31 | consec=1 | max_consec=8 | avg_clip50=25.99 |
| EP=4,920 | STEP=870,659 | clip=28 | consec=3 | max_consec=8 | avg_clip50=24.12 |
| EP=4,930 | STEP=872,759 | clip=17 | consec=0 | max_consec=8 | avg_clip50=23.65 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 22.5 |
| clipped_episode_r... | 25.5 |

|                      |        |
|----------------------|--------|
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 879    |
| ep_rew_mean          | 248    |
| exploration rate     | 0.709  |
| time/                |        |
| episodes             | 14700  |
| fps                  | 261    |
| time_elapsed         | 3337   |
| total timesteps      | 874106 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.224  |
| n_updates            | 206026 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,940 | STEP=874,739 | clip=12 | consec=0 | max_consec=8 | avg_clip50=22.05 |
| EP=4,950 | STEP=877,285 | clip=47 | consec=3 | max_consec=8 | avg_clip50=25.04 |
| EP=4,960 | STEP=879,123 | clip=30 | consec=1 | max_consec=8 | avg_clip50=25.65 |
| EP=4,970 | STEP=881,229 | clip=32 | consec=1 | max_consec=8 | avg_clip50=26.04 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26     |
| clipped_episode_r... | 32.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 887    |
| ep_rew_mean          | 251    |
| exploration rate     | 0.707  |
| time/                |        |
| episodes             | 14800  |
| fps                  | 261    |
| time_elapsed         | 3365   |
| total timesteps      | 881229 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0561 |
| n_updates            | 207807 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,980 | STEP=883,305 | clip=18 | consec=0 | max_consec=8 | avg_clip50=25.85 |
| EP=4,990 | STEP=885,249 | clip=15 | consec=0 | max_consec=8 | avg_clip50=26.18 |
| EP=5,000 | STEP=887,343 | clip=22 | consec=2 | max_consec=8 | avg_clip50=23.16 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23     |
| clipped_episode_r... | 5.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 869    |
| ep_rew_mean          | 242    |
| exploration rate     | 0.704  |
| time/                |        |
| episodes             | 14900  |
| fps                  | 261    |
| time_elapsed         | 3392   |
| total timesteps      | 887810 |
| train/               |        |
| learning_rate        | 0.0001 |

|           |        |
|-----------|--------|
| loss      | 0.0852 |
| n_updates | 209452 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,010 | STEP=889,027 | clip=12 | consec=0 | max_consec=8 | avg_clip50=22.52 |
| EP=5,020 | STEP=890,756 | clip=55 | consec=2 | max_consec=8 | avg_clip50=21.05 |
| EP=5,030 | STEP=892,664 | clip=50 | consec=4 | max_consec=8 | avg_clip50=21.37 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.2   |
| clipped_episode_r... | 21.5   |
| consecutive_clipp... | 3      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 846    |
| ep_rew_mean          | 235    |
| exploration rate     | 0.702  |
| time/                |        |
| episodes             | 15000  |
| fps                  | 261    |
| time_elapsed         | 3417   |
| total timesteps      | 894170 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.324  |
| n_updates            | 211042 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,040 | STEP=894,690 | clip=8  | consec=0 | max_consec=8 | avg_clip50=20.84 |
| EP=5,050 | STEP=896,740 | clip=35 | consec=1 | max_consec=8 | avg_clip50=20.88 |
| EP=5,060 | STEP=898,678 | clip=14 | consec=0 | max_consec=8 | avg_clip50=21.73 |
| EP=5,070 | STEP=900,878 | clip=32 | consec=1 | max_consec=8 | avg_clip50=22.69 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.4   |
| clipped_episode_r... | 48.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 831    |
| ep_rew_mean          | 218    |
| exploration rate     | 0.7    |
| time/                |        |
| episodes             | 15100  |
| fps                  | 261    |
| time_elapsed         | 3445   |
| total timesteps      | 901192 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.449  |
| n_updates            | 212797 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,080 | STEP=903,263 | clip=22 | consec=5 | max_consec=8 | avg_clip50=23.07 |
| EP=5,090 | STEP=905,513 | clip=41 | consec=2 | max_consec=8 | avg_clip50=25.47 |
| EP=5,100 | STEP=907,410 | clip=16 | consec=0 | max_consec=8 | avg_clip50=24.39 |

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|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 24.4 |
| clipped_episode_r... | 12.5 |
| consecutive_clipp... | 0    |
| max_consecutive_c... | 8    |
| rollout/             |      |

|                  |        |
|------------------|--------|
| ep_len_mean      | 845    |
| ep_rew_mean      | 224    |
| exploration rate | 0.697  |
| time/            |        |
| episodes         | 15200  |
| fps              | 261    |
| time_elapsed     | 3474   |
| total timesteps  | 908410 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.103  |
| n_updates        | 214602 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,110 | STEP=909,150 | clip=14 | consec=0 | max_consec=8 | avg_clip50=23.56 |
| EP=5,120 | STEP=911,310 | clip=12 | consec=0 | max_consec=8 | avg_clip50=25.26 |
| EP=5,130 | STEP=913,151 | clip=14 | consec=0 | max_consec=8 | avg_clip50=24.58 |
| EP=5,140 | STEP=915,136 | clip=12 | consec=0 | max_consec=8 | avg_clip50=22.49 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.5   |
| clipped_episode_r... | 11.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 863    |
| ep_rew_mean          | 240    |
| exploration rate     | 0.695  |
| time/                |        |
| episodes             | 15300  |
| fps                  | 261    |
| time_elapsed         | 3501   |
| total timesteps      | 915346 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.111  |
| n_updates            | 216336 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,150 | STEP=917,157 | clip=18 | consec=0 | max_consec=8 | avg_clip50=22.64 |
| EP=5,160 | STEP=919,154 | clip=34 | consec=2 | max_consec=8 | avg_clip50=23.29 |
| EP=5,170 | STEP=920,662 | clip=10 | consec=0 | max_consec=8 | avg_clip50=19.90 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 18.8   |
| clipped_episode_r... | 27.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 830    |
| ep_rew_mean          | 221    |
| exploration rate     | 0.693  |
| time/                |        |
| episodes             | 15400  |
| fps                  | 261    |
| time_elapsed         | 3526   |
| total timesteps      | 921468 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0593 |
| n_updates            | 217866 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,180 | STEP=922,953 | clip=35 | consec=4 | max_consec=8 | avg_clip50=21.19 |
| EP=5,190 | STEP=925,360 | clip=62 | consec=1 | max_consec=8 | avg_clip50=23.97 |
| EP=5,200 | STEP=927,596 | clip=38 | consec=5 | max_consec=8 | avg_clip50=25.92 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.9   |
| clipped_episode_r... | 10.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 854    |
| ep_rew_mean          | 244    |
| exploration rate     | 0.691  |
| time/                |        |
| episodes             | 15500  |
| fps                  | 261    |
| time_elapsed         | 3555   |
| total timesteps      | 928835 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.108  |
| n_updates            | 219708 |

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,210 | STEP=929,255 | clip=3  | consec=0 | max_consec=8 | avg_clip50=25.45 |
| EP=5,220 | STEP=931,281 | clip=28 | consec=2 | max_consec=8 | avg_clip50=27.22 |
| EP=5,230 | STEP=933,098 | clip=8  | consec=0 | max_consec=8 | avg_clip50=24.87 |
| EP=5,240 | STEP=935,139 | clip=6  | consec=0 | max_consec=8 | avg_clip50=23.56 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.7   |
| clipped_episode_r... | 17.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 843    |
| ep_rew_mean          | 237    |
| exploration rate     | 0.688  |
| time/                |        |
| episodes             | 15600  |
| fps                  | 261    |
| time_elapsed         | 3582   |
| total timesteps      | 935436 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.353  |
| n_updates            | 221358 |

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,250 | STEP=937,146 | clip=6  | consec=0 | max_consec=8 | avg_clip50=22.18 |
| EP=5,260 | STEP=939,553 | clip=59 | consec=6 | max_consec=8 | avg_clip50=25.75 |
| EP=5,270 | STEP=941,509 | clip=18 | consec=0 | max_consec=8 | avg_clip50=26.54 |

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 27.7  |
| clipped_episode_r... | 65.5  |
| consecutive_clipp... | 2     |
| max_consecutive_c... | 8     |
| rollout/             |       |
| ep_len_mean          | 880   |
| ep_rew_mean          | 273   |
| exploration rate     | 0.686 |

|                 |        |
|-----------------|--------|
| time/           |        |
| episodes        | 15700  |
| fps             | 260    |
| time_elapsed    | 3612   |
| total timesteps | 942795 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.06   |
| n_updates       | 223198 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,280 | STEP=943,547 | clip=29 | consec=1 | max_consec=8 | avg_clip50=28.13 |
| EP=5,290 | STEP=945,503 | clip=26 | consec=1 | max_consec=8 | avg_clip50=26.00 |
| EP=5,300 | STEP=947,676 | clip=5  | consec=0 | max_consec=8 | avg_clip50=27.73 |
| EP=5,310 | STEP=949,730 | clip=36 | consec=3 | max_consec=8 | avg_clip50=25.06 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.1   |
| clipped_episode_r... | 36.5   |
| consecutive_clipp... | 3      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 864    |
| ep_rew_mean          | 254    |
| exploration rate     | 0.684  |
| time/                |        |
| episodes             | 15800  |
| fps                  | 260    |
| time_elapsed         | 3640   |
| total timesteps      | 949730 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0798 |
| n_updates            | 224932 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,320 | STEP=952,100 | clip=10 | consec=0 | max_consec=9 | avg_clip50=27.21 |
| EP=5,330 | STEP=954,287 | clip=16 | consec=0 | max_consec=9 | avg_clip50=27.54 |
| EP=5,340 | STEP=955,931 | clip=14 | consec=0 | max_consec=9 | avg_clip50=27.41 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 27.3   |
| clipped_episode_r... | 7      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 9      |
| rollout/             |        |
| ep_len_mean          | 874    |
| ep_rew_mean          | 268    |
| exploration rate     | 0.681  |
| time/                |        |
| episodes             | 15900  |
| fps                  | 260    |
| time_elapsed         | 3668   |
| total timesteps      | 956741 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.249  |
| n_updates            | 226685 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,350 | STEP=958,027 | clip=63 | consec=1 | max_consec=9 | avg_clip50=27.70 |
| EP=5,360 | STEP=959,747 | clip=34 | consec=4 | max_consec=9 | avg_clip50=26.97 |
| EP=5,370 | STEP=962,015 | clip=53 | consec=3 | max_consec=9 | avg_clip50=24.62 |



|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 25.3   |  |
| clipped_episode_r... | 42     |  |
| consecutive_clipp... | 2      |  |
| max_consecutive_c... | 9      |  |
| rollout/             |        |  |
| ep_len_mean          | 870    |  |
| ep_rew_mean          | 256    |  |
| exploration rate     | 0.679  |  |
| time/                |        |  |
| episodes             | 16000  |  |
| fps                  | 260    |  |
| time_elapsed         | 3695   |  |
| total timesteps      | 963568 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0589 |  |
| n_updates            | 228391 |  |

EP=5,380 | STEP=964,343 | clip=34 | consec=5 | max\_consec=9 | avg\_clip50=24.83  
EP=5,390 | STEP=966,296 | clip=48 | consec=3 | max\_consec=9 | avg\_clip50=26.91  
EP=5,400 | STEP=968,427 | clip=28 | consec=2 | max\_consec=9 | avg\_clip50=26.55  
EP=5,410 | STEP=970,314 | clip=22 | consec=1 | max\_consec=9 | avg\_clip50=25.96

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 26     |  |
| clipped_episode_r... | 21.5   |  |
| consecutive_clipp... | 1      |  |
| max_consecutive_c... | 9      |  |
| rollout/             |        |  |
| ep_len_mean          | 869    |  |
| ep_rew_mean          | 265    |  |
| exploration rate     | 0.677  |  |
| time/                |        |  |
| episodes             | 16100  |  |
| fps                  | 260    |  |
| time_elapsed         | 3723   |  |
| total timesteps      | 970367 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0278 |  |
| n_updates            | 230091 |  |

EP=5,420 | STEP=972,210 | clip=32 | consec=2 | max\_consec=9 | avg\_clip50=25.15  
EP=5,430 | STEP=974,608 | clip=29 | consec=6 | max\_consec=9 | avg\_clip50=24.90  
EP=5,440 | STEP=976,665 | clip=6 | consec=0 | max\_consec=9 | avg\_clip50=23.80

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 21.5  |  |
| clipped_episode_r... | 15.5  |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 9     |  |
| rollout/             |       |  |
| ep_len_mean          | 866   |  |
| ep_rew_mean          | 247   |  |
| exploration rate     | 0.675 |  |
| time/                |       |  |
| episodes             | 16200 |  |
| fps                  | 260   |  |

|                 |        |
|-----------------|--------|
| time_elapsed    | 3751   |
| total timesteps | 977346 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0391 |
| n_updates       | 231836 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,450 | STEP=978,412 | clip=23 | consec=2 | max_consec=9 | avg_clip50=21.48 |
| EP=5,460 | STEP=980,372 | clip=20 | consec=0 | max_consec=9 | avg_clip50=22.49 |
| EP=5,470 | STEP=982,249 | clip=7  | consec=0 | max_consec=9 | avg_clip50=23.42 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.9   |
| clipped_episode_r... | 39     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 9      |
| rollout/             |        |
| ep_len_mean          | 846    |
| ep_rew_mean          | 239    |
| exploration rate     | 0.672  |
| time/                |        |
| episodes             | 16300  |
| fps                  | 260    |
| time_elapsed         | 3778   |
| total timesteps      | 984106 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0411 |
| n_updates            | 233526 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,480 | STEP=984,265 | clip=14 | consec=0 | max_consec=9 | avg_clip50=22.61 |
| EP=5,490 | STEP=986,890 | clip=77 | consec=1 | max_consec=9 | avg_clip50=25.43 |
| EP=5,500 | STEP=989,275 | clip=12 | consec=0 | max_consec=9 | avg_clip50=27.17 |
| EP=5,510 | STEP=991,344 | clip=2  | consec=0 | max_consec=9 | avg_clip50=27.47 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 27.4   |
| clipped_episode_r... | 18     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 9      |
| rollout/             |        |
| ep_len_mean          | 887    |
| ep_rew_mean          | 250    |
| exploration rate     | 0.67   |
| time/                |        |
| episodes             | 16400  |
| fps                  | 260    |
| time_elapsed         | 3809   |
| total timesteps      | 991685 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.371  |
| n_updates            | 235421 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,520 | STEP=993,090 | clip=13 | consec=0 | max_consec=9 | avg_clip50=25.79 |
| EP=5,530 | STEP=995,488 | clip=12 | consec=0 | max_consec=9 | avg_clip50=26.53 |
| EP=5,540 | STEP=997,565 | clip=18 | consec=0 | max_consec=9 | avg_clip50=24.81 |

|            |      |
|------------|------|
| custom/    |      |
| avg_clip50 | 25.3 |

|  |                      |        |  |
|--|----------------------|--------|--|
|  | clipped_episode_r... | 9.5    |  |
|  | consecutive_clipp... | 0      |  |
|  | max_consecutive_c... | 9      |  |
|  | rollout/             |        |  |
|  | ep_len_mean          | 900    |  |
|  | ep_rew_mean          | 263    |  |
|  | exploration rate     | 0.667  |  |
|  | time/                |        |  |
|  | episodes             | 16500  |  |
|  | fps                  | 260    |  |
|  | time_elapsed         | 3837   |  |
|  | total timesteps      | 998867 |  |
|  | train/               |        |  |
|  | learning_rate        | 0.0001 |  |
|  | loss                 | 0.201  |  |
|  | n_updates            | 237216 |  |

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EP=5,550 | STEP=999,974 | clip=22 | consec=4 | max\_consec=9 | avg\_clip50=25.51  
EP=5,560 | STEP=1,001,919 | clip=19 | consec=0 | max\_consec=9 | avg\_clip50=24.94  
EP=5,570 | STEP=1,004,484 | clip=74 | consec=4 | max\_consec=9 | avg\_clip50=28.37  
EP=5,580 | STEP=1,006,482 | clip=8 | consec=0 | max\_consec=9 | avg\_clip50=27.24

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 27.2    |  |
|  | clipped_episode_r... | 8.5     |  |
|  | consecutive_clipp... | 0       |  |
|  | max_consecutive_c... | 9       |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 934     |  |
|  | ep_rew_mean          | 269     |  |
|  | exploration rate     | 0.665   |  |
|  | time/                |         |  |
|  | episodes             | 16600   |  |
|  | fps                  | 260     |  |
|  | time_elapsed         | 3868    |  |
|  | total timesteps      | 1006537 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.198   |  |
|  | n_updates            | 239134  |  |

-----

EP=5,590 | STEP=1,008,554 | clip=14 | consec=0 | max\_consec=9 | avg\_clip50=26.11  
EP=5,600 | STEP=1,010,530 | clip=22 | consec=1 | max\_consec=9 | avg\_clip50=22.91  
EP=5,610 | STEP=1,012,293 | clip=24 | consec=1 | max\_consec=9 | avg\_clip50=22.93

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 23.3    |  |
|  | clipped_episode_r... | 17      |  |
|  | consecutive_clipp... | 0       |  |
|  | max_consecutive_c... | 9       |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 891     |  |
|  | ep_rew_mean          | 244     |  |
|  | exploration rate     | 0.663   |  |
|  | time/                |         |  |
|  | episodes             | 16700   |  |
|  | fps                  | 260     |  |
|  | time_elapsed         | 3895    |  |
|  | total timesteps      | 1013181 |  |
|  | train/               |         |  |

|               |        |
|---------------|--------|
| learning_rate | 0.0001 |
| loss          | 0.152  |
| n_updates     | 240795 |

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=5,620 | STEP=1,014,504 | clip=12 | consec=0 | max_consec=9 | avg_clip50=20.73 |
| EP=5,630 | STEP=1,016,520 | clip=17 | consec=0 | max_consec=9 | avg_clip50=21.48 |
| EP=5,640 | STEP=1,018,627 | clip=24 | consec=2 | max_consec=9 | avg_clip50=22.27 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 24.1    |
| clipped_episode_r... | 18.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 894     |
| ep_rew_mean          | 239     |
| exploration rate     | 0.66    |
| time/                |         |
| episodes             | 16800   |
| fps                  | 260     |
| time_elapsed         | 3925    |
| total timesteps      | 1020552 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.102   |
| n_updates            | 242637  |

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=5,650 | STEP=1,021,144 | clip=30 | consec=2 | max_consec=9 | avg_clip50=24.51 |
| EP=5,660 | STEP=1,023,306 | clip=14 | consec=0 | max_consec=9 | avg_clip50=25.40 |
| EP=5,670 | STEP=1,025,286 | clip=21 | consec=1 | max_consec=9 | avg_clip50=25.61 |
| EP=5,680 | STEP=1,027,652 | clip=4  | consec=0 | max_consec=9 | avg_clip50=26.25 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26.8    |
| clipped_episode_r... | 45      |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 899     |
| ep_rew_mean          | 242     |
| exploration rate     | 0.658   |
| time/                |         |
| episodes             | 16900   |
| fps                  | 259     |
| time_elapsed         | 3955    |
| total timesteps      | 1028201 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.413   |
| n_updates            | 244550  |

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=5,690 | STEP=1,029,819 | clip=16 | consec=0 | max_consec=9 | avg_clip50=26.55 |
| EP=5,700 | STEP=1,031,771 | clip=48 | consec=1 | max_consec=9 | avg_clip50=25.86 |
| EP=5,710 | STEP=1,034,088 | clip=38 | consec=1 | max_consec=9 | avg_clip50=26.26 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 27.3 |
| clipped_episode_r... | 26   |
| consecutive_clipp... | 6    |
| max_consecutive_c... | 9    |

|                  |         |
|------------------|---------|
| rollout/         |         |
| ep_len_mean      | 933     |
| ep_rew_mean      | 263     |
| exploration rate | 0.655   |
| time/            |         |
| episodes         | 17000   |
| fps              | 259     |
| time_elapsed     | 3986    |
| total timesteps  | 1035720 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.179   |
| n_updates        | 246429  |

-----

|          |  |                |  |         |  |          |  |              |  |                  |
|----------|--|----------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,720 |  | STEP=1,036,973 |  | clip=32 |  | consec=1 |  | max_consec=9 |  | avg_clip50=28.82 |
| EP=5,730 |  | STEP=1,039,308 |  | clip=46 |  | consec=1 |  | max_consec=9 |  | avg_clip50=28.86 |
| EP=5,740 |  | STEP=1,041,363 |  | clip=19 |  | consec=0 |  | max_consec=9 |  | avg_clip50=28.83 |

-----

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29.4    |
| clipped_episode_r... | 11.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 940     |
| ep_rew_mean          | 275     |
| exploration rate     | 0.653   |
| time/                |         |
| episodes             | 17100   |
| fps                  | 259     |
| time_elapsed         | 4016    |
| total timesteps      | 1043253 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.438   |
| n_updates            | 248313  |

-----

|          |  |                |  |         |  |          |  |              |  |                  |
|----------|--|----------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,750 |  | STEP=1,043,476 |  | clip=17 |  | consec=0 |  | max_consec=9 |  | avg_clip50=28.81 |
| EP=5,760 |  | STEP=1,045,715 |  | clip=28 |  | consec=4 |  | max_consec=9 |  | avg_clip50=28.59 |
| EP=5,770 |  | STEP=1,047,846 |  | clip=35 |  | consec=4 |  | max_consec=9 |  | avg_clip50=27.96 |
| EP=5,780 |  | STEP=1,049,904 |  | clip=13 |  | consec=0 |  | max_consec=9 |  | avg_clip50=28.24 |

-----

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.9    |
| clipped_episode_r... | 13      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 919     |
| ep_rew_mean          | 276     |
| exploration rate     | 0.65    |
| time/                |         |
| episodes             | 17200   |
| fps                  | 259     |
| time_elapsed         | 4045    |
| total timesteps      | 1050458 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.296   |
| n_updates            | 250114  |

```

-----
EP=5,790 | STEP=1,051,940 | clip=17 | consec=0 | max_consec=9 | avg_clip50=27.83
EP=5,800 | STEP=1,053,775 | clip=4 | consec=0 | max_consec=9 | avg_clip50=27.90
EP=5,810 | STEP=1,056,497 | clip=50 | consec=7 | max_consec=9 | avg_clip50=30.02
-----

```

```

| custom/
|   avg_clip50           | 29.4
|   clipped_episode_r... | 34.5
|   consecutive_clipp... | 1
|   max_consecutive_c... | 9
| rollout/
|   ep_len_mean          | 906
|   ep_rew_mean          | 281
|   exploration rate     | 0.648
| time/
|   episodes             | 17300
|   fps                  | 259
|   time_elapsed         | 4075
|   total timesteps      | 1057961
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.236
|   n_updates            | 251990
-----

```

```

EP=5,820 | STEP=1,058,384 | clip=4 | consec=0 | max_consec=9 | avg_clip50=27.58
EP=5,830 | STEP=1,060,600 | clip=16 | consec=0 | max_consec=9 | avg_clip50=27.61
EP=5,840 | STEP=1,062,961 | clip=48 | consec=3 | max_consec=9 | avg_clip50=28.68
EP=5,850 | STEP=1,064,732 | clip=20 | consec=0 | max_consec=9 | avg_clip50=28.63
-----

```

```

| custom/
|   avg_clip50           | 28.4
|   clipped_episode_r... | 15.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 9
| rollout/
|   ep_len_mean          | 898
|   ep_rew_mean          | 285
|   exploration rate     | 0.645
| time/
|   episodes             | 17400
|   fps                  | 259
|   time_elapsed         | 4104
|   total timesteps      | 1065234
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.237
|   n_updates            | 253808
-----

```

```

EP=5,860 | STEP=1,066,996 | clip=6 | consec=0 | max_consec=9 | avg_clip50=28.12
EP=5,870 | STEP=1,069,138 | clip=8 | consec=0 | max_consec=9 | avg_clip50=28.46
EP=5,880 | STEP=1,071,218 | clip=16 | consec=0 | max_consec=9 | avg_clip50=27.68
-----

```

```

| custom/
|   avg_clip50           | 27.9
|   clipped_episode_r... | 15
|   consecutive_clipp... | 0
|   max_consecutive_c... | 9
| rollout/
|   ep_len_mean          | 905
|   ep_rew_mean          | 284
-----

```

|                  |         |
|------------------|---------|
| exploration rate | 0.643   |
| time/            |         |
| episodes         | 17500   |
| fps              | 259     |
| time_elapsed     | 4134    |
| total timesteps  | 1072685 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.548   |
| n_updates        | 255671  |

EP=5,890 | STEP=1,073,279 | clip=19 | consec=0 | max\_consec=9 | avg\_clip50=27.42  
EP=5,900 | STEP=1,075,571 | clip=80 | consec=1 | max\_consec=9 | avg\_clip50=29.13  
EP=5,910 | STEP=1,077,503 | clip=12 | consec=0 | max\_consec=9 | avg\_clip50=26.38  
EP=5,920 | STEP=1,079,618 | clip=5 | consec=0 | max\_consec=9 | avg\_clip50=26.19

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26.2    |
| clipped_episode_r... | 5       |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 894     |
| ep_rew_mean          | 273     |
| exploration rate     | 0.64    |
| time/                |         |
| episodes             | 17600   |
| fps                  | 259     |
| time_elapsed         | 4162    |
| total timesteps      | 1079618 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0494  |
| n_updates            | 257404  |

EP=5,930 | STEP=1,081,535 | clip=28 | consec=1 | max\_consec=9 | avg\_clip50=26.05  
EP=5,940 | STEP=1,083,903 | clip=50 | consec=2 | max\_consec=9 | avg\_clip50=26.81  
EP=5,950 | STEP=1,086,240 | clip=12 | consec=0 | max\_consec=9 | avg\_clip50=26.44

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.1    |
| clipped_episode_r... | 21      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 897     |
| ep_rew_mean          | 270     |
| exploration rate     | 0.638   |
| time/                |         |
| episodes             | 17700   |
| fps                  | 259     |
| time_elapsed         | 4193    |
| total timesteps      | 1087346 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0778  |
| n_updates            | 259336  |

EP=5,960 | STEP=1,088,747 | clip=17 | consec=0 | max\_consec=9 | avg\_clip50=29.42  
EP=5,970 | STEP=1,091,041 | clip=84 | consec=2 | max\_consec=9 | avg\_clip50=30.85

EP=5,980 | STEP=1,093,438 | clip=48 | consec=1 | max\_consec=9 | avg\_clip50=31.29

|                      |         |
|----------------------|---------|
| -----                |         |
| custom/              |         |
| avg_clip50           | 30.2    |
| clipped_episode_r... | 14.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 934     |
| ep_rew_mean          | 282     |
| exploration rate     | 0.635   |
| time/                |         |
| episodes             | 17800   |
| fps                  | 259     |
| time_elapsed         | 4226    |
| total timesteps      | 1095444 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.126   |
| n_updates            | 261360  |

EP=5,990 | STEP=1,095,512 | clip=43 | consec=1 | max\_consec=9 | avg\_clip50=30.09  
EP=6,000 | STEP=1,097,870 | clip=44 | consec=3 | max\_consec=9 | avg\_clip50=30.38  
EP=6,010 | STEP=1,099,501 | clip=14 | consec=0 | max\_consec=9 | avg\_clip50=26.69  
EP=6,020 | STEP=1,101,616 | clip=61 | consec=3 | max\_consec=9 | avg\_clip50=26.02

|                      |         |
|----------------------|---------|
| -----                |         |
| custom/              |         |
| avg_clip50           | 25.8    |
| clipped_episode_r... | 33      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 926     |
| ep_rew_mean          | 281     |
| exploration rate     | 0.633   |
| time/                |         |
| episodes             | 17900   |
| fps                  | 259     |
| time_elapsed         | 4254    |
| total timesteps      | 1102522 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0923  |
| n_updates            | 263130  |

EP=6,030 | STEP=1,103,829 | clip=15 | consec=0 | max\_consec=9 | avg\_clip50=26.59  
EP=6,040 | STEP=1,105,755 | clip=50 | consec=1 | max\_consec=9 | avg\_clip50=25.96  
EP=6,050 | STEP=1,108,014 | clip=22 | consec=1 | max\_consec=9 | avg\_clip50=24.47

|                      |       |
|----------------------|-------|
| -----                |       |
| custom/              |       |
| avg_clip50           | 26.4  |
| clipped_episode_r... | 13    |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 9     |
| rollout/             |       |
| ep_len_mean          | 904   |
| ep_rew_mean          | 266   |
| exploration rate     | 0.63  |
| time/                |       |
| episodes             | 18000 |



|                 |         |
|-----------------|---------|
| fps             | 259     |
| time_elapsed    | 4284    |
| total timesteps | 1109921 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.0688  |
| n_updates       | 264980  |

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|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=6,060 | STEP=1,110,257 | clip=36 | consec=1 | max_consec=9 | avg_clip50=26.40 |
| EP=6,070 | STEP=1,112,557 | clip=14 | consec=0 | max_consec=9 | avg_clip50=27.37 |
| EP=6,080 | STEP=1,114,470 | clip=6  | consec=0 | max_consec=9 | avg_clip50=25.22 |
| EP=6,090 | STEP=1,116,599 | clip=18 | consec=0 | max_consec=9 | avg_clip50=25.27 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26.6    |
| clipped_episode_r... | 34      |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 901     |
| ep_rew_mean          | 260     |
| exploration rate     | 0.628   |
| time/                |         |
| episodes             | 18100   |
| fps                  | 258     |
| time_elapsed         | 4314    |
| total timesteps      | 1117482 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.395   |
| n_updates            | 266870  |

-----

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=6,100 | STEP=1,118,667 | clip=16 | consec=0 | max_consec=9 | avg_clip50=25.04 |
| EP=6,110 | STEP=1,120,855 | clip=25 | consec=3 | max_consec=9 | avg_clip50=25.19 |
| EP=6,120 | STEP=1,122,549 | clip=20 | consec=0 | max_consec=9 | avg_clip50=22.86 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 24      |
| clipped_episode_r... | 4.5     |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 876     |
| ep_rew_mean          | 250     |
| exploration rate     | 0.626   |
| time/                |         |
| episodes             | 18200   |
| fps                  | 258     |
| time_elapsed         | 4340    |
| total timesteps      | 1123851 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.343   |
| n_updates            | 268462  |

-----

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=6,130 | STEP=1,124,529 | clip=18 | consec=0 | max_consec=9 | avg_clip50=24.26 |
| EP=6,140 | STEP=1,126,543 | clip=16 | consec=0 | max_consec=9 | avg_clip50=24.14 |
| EP=6,150 | STEP=1,128,624 | clip=25 | consec=1 | max_consec=9 | avg_clip50=24.08 |
| EP=6,160 | STEP=1,130,848 | clip=60 | consec=1 | max_consec=9 | avg_clip50=24.44 |

-----

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 24.4    |
| clipped_episode_r... | 60.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 870     |
| ep_rew_mean          | 248     |
| exploration rate     | 0.623   |
| time/                |         |
| episodes             | 18300   |
| fps                  | 258     |
| time_elapsed         | 4369    |
| total timesteps      | 1130923 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0686  |
| n_updates            | 270230  |

-----

EP=6,170 | STEP=1,132,933 | clip=52 | consec=1 | max\_consec=9 | avg\_clip50=24.92  
EP=6,180 | STEP=1,134,966 | clip=14 | consec=0 | max\_consec=9 | avg\_clip50=24.26  
EP=6,190 | STEP=1,136,995 | clip=6 | consec=0 | max\_consec=9 | avg\_clip50=24.34

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 24      |
| clipped_episode_r... | 2       |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 847     |
| ep_rew_mean          | 238     |
| exploration rate     | 0.621   |
| time/                |         |
| episodes             | 18400   |
| fps                  | 258     |
| time_elapsed         | 4397    |
| total timesteps      | 1137793 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0686  |
| n_updates            | 271948  |

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EP=6,200 | STEP=1,138,993 | clip=9 | consec=0 | max\_consec=9 | avg\_clip50=25.60  
EP=6,210 | STEP=1,140,903 | clip=14 | consec=0 | max\_consec=9 | avg\_clip50=24.37  
EP=6,220 | STEP=1,143,114 | clip=11 | consec=0 | max\_consec=9 | avg\_clip50=25.61

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 25.1    |
| clipped_episode_r... | 36.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 869     |
| ep_rew_mean          | 247     |
| exploration rate     | 0.619   |
| time/                |         |
| episodes             | 18500   |
| fps                  | 258     |
| time_elapsed         | 4427    |
| total timesteps      | 1144909 |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0667 |
| n_updates     | 273727 |

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=6,230 | STEP=1,145,120 | clip=16 | consec=0 | max_consec=9 | avg_clip50=25.13 |
| EP=6,240 | STEP=1,147,058 | clip=52 | consec=2 | max_consec=9 | avg_clip50=25.62 |
| EP=6,250 | STEP=1,149,154 | clip=21 | consec=3 | max_consec=9 | avg_clip50=25.70 |
| EP=6,260 | STEP=1,151,190 | clip=29 | consec=1 | max_consec=9 | avg_clip50=27.09 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26.8    |
| clipped_episode_r... | 15.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 854     |
| ep_rew_mean          | 255     |
| exploration rate     | 0.616   |
| time/                |         |
| episodes             | 18600   |
| fps                  | 258     |
| time_elapsed         | 4456    |
| total timesteps      | 1152002 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.405   |
| n_updates            | 275500  |

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=6,270 | STEP=1,153,213 | clip=50 | consec=1 | max_consec=9 | avg_clip50=26.59 |
| EP=6,280 | STEP=1,155,173 | clip=28 | consec=3 | max_consec=9 | avg_clip50=26.68 |
| EP=6,290 | STEP=1,157,038 | clip=24 | consec=3 | max_consec=9 | avg_clip50=26.96 |

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| custom/              |         |
| avg_clip50           | 26.5    |
| clipped_episode_r... | 14      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 842     |
| ep_rew_mean          | 257     |
| exploration rate     | 0.614   |
| time/                |         |
| episodes             | 18700   |
| fps                  | 258     |
| time_elapsed         | 4481    |
| total timesteps      | 1158337 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0593  |
| n_updates            | 277084  |

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=6,300 | STEP=1,158,796 | clip=27 | consec=1 | max_consec=9 | avg_clip50=25.01 |
| EP=6,310 | STEP=1,160,981 | clip=8  | consec=0 | max_consec=9 | avg_clip50=23.78 |
| EP=6,320 | STEP=1,162,839 | clip=19 | consec=0 | max_consec=9 | avg_clip50=23.44 |
| EP=6,330 | STEP=1,165,211 | clip=42 | consec=1 | max_consec=9 | avg_clip50=24.45 |

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| custom/              |      |
| avg_clip50           | 25.1 |
| clipped_episode_r... | 35   |

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| consecutive_clipp... | 2       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 852     |
| ep_rew_mean          | 259     |
| exploration rate     | 0.612   |
| time/                |         |
| episodes             | 18800   |
| fps                  | 258     |
| time_elapsed         | 4510    |
| total timesteps      | 1165419 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0867  |
| n_updates            | 278854  |

EP=6,340 | STEP=1,167,413 | clip=28 | consec=5 | max\_consec=9 | avg\_clip50=25.42  
EP=6,350 | STEP=1,169,712 | clip=12 | consec=0 | max\_consec=9 | avg\_clip50=27.39  
EP=6,360 | STEP=1,171,928 | clip=20 | consec=0 | max\_consec=9 | avg\_clip50=29.00

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|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.6    |
| clipped_episode_r... | 29      |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 868     |
| ep_rew_mean          | 260     |
| exploration rate     | 0.609   |
| time/                |         |
| episodes             | 18900   |
| fps                  | 258     |
| time_elapsed         | 4540    |
| total timesteps      | 1172925 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0958  |
| n_updates            | 280731  |

EP=6,370 | STEP=1,174,002 | clip=15 | consec=0 | max\_consec=9 | avg\_clip50=29.45  
EP=6,380 | STEP=1,176,268 | clip=51 | consec=3 | max\_consec=9 | avg\_clip50=31.98  
EP=6,390 | STEP=1,178,362 | clip=34 | consec=2 | max\_consec=9 | avg\_clip50=31.07  
EP=6,400 | STEP=1,180,591 | clip=30 | consec=3 | max\_consec=9 | avg\_clip50=31.45

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 31.4    |
| clipped_episode_r... | 29.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 917     |
| ep_rew_mean          | 294     |
| exploration rate     | 0.607   |
| time/                |         |
| episodes             | 19000   |
| fps                  | 258     |
| time_elapsed         | 4571    |
| total timesteps      | 1180591 |
| train/               |         |
| learning_rate        | 0.0001  |

|           |        |
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| loss      | 0.0547 |
| n_updates | 282647 |

EP=6,410 | STEP=1,182,726 | clip=28 | consec=4 | max\_consec=9 | avg\_clip50=30.28  
EP=6,420 | STEP=1,184,534 | clip=50 | consec=1 | max\_consec=9 | avg\_clip50=29.22  
EP=6,430 | STEP=1,186,684 | clip=29 | consec=2 | max\_consec=9 | avg\_clip50=27.15

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26.2    |
| clipped_episode_r... | 22      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 901     |
| ep_rew_mean          | 291     |
| exploration rate     | 0.605   |
| time/                |         |
| episodes             | 19100   |
| fps                  | 258     |
| time_elapsed         | 4599    |
| total timesteps      | 1187377 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.276   |
| n_updates            | 284344  |

EP=6,440 | STEP=1,188,572 | clip=8 | consec=0 | max\_consec=9 | avg\_clip50=25.65  
EP=6,450 | STEP=1,190,475 | clip=36 | consec=2 | max\_consec=9 | avg\_clip50=24.72  
EP=6,460 | STEP=1,192,696 | clip=70 | consec=5 | max\_consec=9 | avg\_clip50=25.90

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|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.6    |
| clipped_episode_r... | 71.5    |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 892     |
| ep_rew_mean          | 285     |
| exploration rate     | 0.602   |
| time/                |         |
| episodes             | 19200   |
| fps                  | 258     |
| time_elapsed         | 4628    |
| total timesteps      | 1194513 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.417   |
| n_updates            | 286128  |

EP=6,470 | STEP=1,194,942 | clip=30 | consec=1 | max\_consec=9 | avg\_clip50=27.55  
EP=6,480 | STEP=1,197,210 | clip=30 | consec=2 | max\_consec=9 | avg\_clip50=28.03  
EP=6,490 | STEP=1,199,838 | clip=16 | consec=0 | max\_consec=9 | avg\_clip50=31.44  
EP=6,500 | STEP=1,202,359 | clip=20 | consec=0 | max\_consec=9 | avg\_clip50=33.38

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 32.9 |
| clipped_episode_r... | 32.5 |
| consecutive_clipp... | 1    |
| max_consecutive_c... | 9    |
| rollout/             |      |

|                  |         |
|------------------|---------|
| ep_len_mean      | 913     |
| ep_rew_mean      | 288     |
| exploration rate | 0.599   |
| time/            |         |
| episodes         | 19300   |
| fps              | 258     |
| time_elapsed     | 4661    |
| total timesteps  | 1202706 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.461   |
| n_updates        | 288176  |

EP=6,510 | STEP=1,204,272 | clip=17 | consec=0 | max\_consec=9 | avg\_clip50=31.63  
EP=6,520 | STEP=1,206,523 | clip=44 | consec=1 | max\_consec=9 | avg\_clip50=32.16  
EP=6,530 | STEP=1,208,078 | clip=9 | consec=0 | max\_consec=9 | avg\_clip50=29.33

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.1    |
| clipped_episode_r... | 12.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 904     |
| ep_rew_mean          | 289     |
| exploration rate     | 0.597   |
| time/                |         |
| episodes             | 19400   |
| fps                  | 257     |
| time_elapsed         | 4688    |
| total timesteps      | 1209299 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0251  |
| n_updates            | 289824  |

EP=6,540 | STEP=1,210,147 | clip=10 | consec=0 | max\_consec=9 | avg\_clip50=26.81  
EP=6,550 | STEP=1,212,031 | clip=10 | consec=0 | max\_consec=9 | avg\_clip50=23.99  
EP=6,560 | STEP=1,213,916 | clip=29 | consec=3 | max\_consec=9 | avg\_clip50=23.77  
EP=6,570 | STEP=1,216,002 | clip=32 | consec=1 | max\_consec=9 | avg\_clip50=23.28

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 23.3    |
| clipped_episode_r... | 32.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 9       |
| rollout/             |         |
| ep_len_mean          | 888     |
| ep_rew_mean          | 277     |
| exploration rate     | 0.595   |
| time/                |         |
| episodes             | 19500   |
| fps                  | 257     |
| time_elapsed         | 4715    |
| total timesteps      | 1216036 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0224  |
| n_updates            | 291508  |

EP=6,580 | STEP=1,218,530 | clip=36 | consec=11 | max\_consec=11 | avg\_clip50=27.45

EP=6,590 | STEP=1,220,773 | clip=24 | consec=1 | max\_consec=12 | avg\_clip50=29.26

EP=6,600 | STEP=1,222,711 | clip=16 | consec=0 | max\_consec=12 | avg\_clip50=29.78

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|----------------------|---------|
| -----                |         |
| custom/              |         |
| avg_clip50           | 29.9    |
| clipped_episode_r... | 8       |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 856     |
| ep_rew_mean          | 269     |
| exploration rate     | 0.593   |
| time/                |         |
| episodes             | 19600   |
| fps                  | 257     |
| time_elapsed         | 4745    |
| total timesteps      | 1223327 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.172   |
| n_updates            | 293331  |

EP=6,610 | STEP=1,224,631 | clip=47 | consec=2 | max\_consec=12 | avg\_clip50=30.19

EP=6,620 | STEP=1,226,427 | clip=4 | consec=0 | max\_consec=12 | avg\_clip50=28.98

EP=6,630 | STEP=1,228,539 | clip=18 | consec=0 | max\_consec=12 | avg\_clip50=26.84

|                      |         |
|----------------------|---------|
| -----                |         |
| custom/              |         |
| avg_clip50           | 26.4    |
| clipped_episode_r... | 30      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 876     |
| ep_rew_mean          | 280     |
| exploration rate     | 0.59    |
| time/                |         |
| episodes             | 19700   |
| fps                  | 257     |
| time_elapsed         | 4774    |
| total timesteps      | 1230611 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.109   |
| n_updates            | 295152  |

EP=6,640 | STEP=1,231,052 | clip=35 | consec=4 | max\_consec=12 | avg\_clip50=26.81

EP=6,650 | STEP=1,232,985 | clip=26 | consec=2 | max\_consec=12 | avg\_clip50=26.36

EP=6,660 | STEP=1,235,327 | clip=60 | consec=2 | max\_consec=12 | avg\_clip50=28.01

EP=6,670 | STEP=1,237,921 | clip=4 | consec=0 | max\_consec=12 | avg\_clip50=30.58

|                      |      |
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| -----                |      |
| custom/              |      |
| avg_clip50           | 29.6 |
| clipped_episode_r... | 19   |
| consecutive_clipp... | 0    |
| max_consecutive_c... | 12   |
| rollout/             |      |
| ep_len_mean          | 920  |
| ep_rew_mean          | 295  |

|                  |         |
|------------------|---------|
| exploration rate | 0.587   |
| time/            |         |
| episodes         | 19800   |
| fps              | 257     |
| time_elapsed     | 4808    |
| total timesteps  | 1238773 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0616  |
| n_updates        | 297193  |

EP=6,680 | STEP=1,240,269 | clip=35 | consec=1 | max\_consec=12 | avg\_clip50=31.00  
EP=6,690 | STEP=1,242,358 | clip=25 | consec=1 | max\_consec=12 | avg\_clip50=29.87  
EP=6,700 | STEP=1,244,463 | clip=34 | consec=3 | max\_consec=12 | avg\_clip50=30.68

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.4    |
| clipped_episode_r... | 13.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 934     |
| ep_rew_mean          | 292     |
| exploration rate     | 0.585   |
| time/                |         |
| episodes             | 19900   |
| fps                  | 257     |
| time_elapsed         | 4838    |
| total timesteps      | 1246133 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.296   |
| n_updates            | 299033  |

EP=6,710 | STEP=1,246,719 | clip=11 | consec=0 | max\_consec=12 | avg\_clip50=29.73  
EP=6,720 | STEP=1,248,746 | clip=17 | consec=0 | max\_consec=12 | avg\_clip50=27.51  
EP=6,730 | STEP=1,251,278 | clip=22 | consec=1 | max\_consec=12 | avg\_clip50=28.16  
EP=6,740 | STEP=1,253,224 | clip=34 | consec=2 | max\_consec=12 | avg\_clip50=27.51

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.6    |
| clipped_episode_r... | 64.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 941     |
| ep_rew_mean          | 292     |
| exploration rate     | 0.583   |
| time/                |         |
| episodes             | 20000   |
| fps                  | 257     |
| time_elapsed         | 4868    |
| total timesteps      | 1253574 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0406  |
| n_updates            | 300893  |

EP=6,750 | STEP=1,255,475 | clip=56 | consec=1 | max\_consec=12 | avg\_clip50=28.75  
EP=6,760 | STEP=1,257,837 | clip=30 | consec=1 | max\_consec=12 | avg\_clip50=29.71



EP=6,770 | STEP=1,260,467 | clip=10 | consec=0 | max\_consec=12 | avg\_clip50=32.38

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|----------------------|---------|--|
| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 33.5    |  |
| clipped_episode_r... | 13      |  |
| consecutive_clipp... | 0       |  |
| max_consecutive_c... | 12      |  |
| rollout/             |         |  |
| ep_len_mean          | 947     |  |
| ep_rew_mean          | 303     |  |
| exploration rate     | 0.58    |  |
| time/                |         |  |
| episodes             | 20100   |  |
| fps                  | 257     |  |
| time_elapsed         | 4900    |  |
| total timesteps      | 1261644 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.112   |  |
| n_updates            | 302910  |  |

EP=6,780 | STEP=1,262,904 | clip=42 | consec=3 | max\_consec=12 | avg\_clip50=33.07

EP=6,790 | STEP=1,265,333 | clip=9 | consec=0 | max\_consec=12 | avg\_clip50=35.62

EP=6,800 | STEP=1,268,062 | clip=8 | consec=0 | max\_consec=12 | avg\_clip50=37.96

|                      |         |  |
|----------------------|---------|--|
| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 36.5    |  |
| clipped_episode_r... | 6       |  |
| consecutive_clipp... | 0       |  |
| max_consecutive_c... | 12      |  |
| rollout/             |         |  |
| ep_len_mean          | 990     |  |
| ep_rew_mean          | 331     |  |
| exploration rate     | 0.577   |  |
| time/                |         |  |
| episodes             | 20200   |  |
| fps                  | 257     |  |
| time_elapsed         | 4934    |  |
| total timesteps      | 1269835 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.164   |  |
| n_updates            | 304958  |  |

EP=6,810 | STEP=1,270,360 | clip=28 | consec=2 | max\_consec=12 | avg\_clip50=37.04

EP=6,820 | STEP=1,272,660 | clip=44 | consec=2 | max\_consec=12 | avg\_clip50=35.47

EP=6,830 | STEP=1,274,816 | clip=40 | consec=2 | max\_consec=12 | avg\_clip50=33.46

EP=6,840 | STEP=1,276,743 | clip=12 | consec=0 | max\_consec=12 | avg\_clip50=31.02

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|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 29    |  |
| clipped_episode_r... | 5.5   |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 12    |  |
| rollout/             |       |  |
| ep_len_mean          | 979   |  |
| ep_rew_mean          | 329   |  |
| exploration rate     | 0.575 |  |
| time/                |       |  |
| episodes             | 20300 |  |

|                      |                |         |          |               |                  |
|----------------------|----------------|---------|----------|---------------|------------------|
| fps                  | 257            |         |          |               |                  |
| time_elapsed         | 4963           |         |          |               |                  |
| total timesteps      | 1277040        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0395         |         |          |               |                  |
| n_updates            | 306759         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=6,850             | STEP=1,278,668 | clip=22 | consec=2 | max_consec=12 | avg_clip50=26.78 |
| EP=6,860             | STEP=1,280,611 | clip=35 | consec=1 | max_consec=12 | avg_clip50=26.28 |
| EP=6,870             | STEP=1,282,617 | clip=11 | consec=0 | max_consec=12 | avg_clip50=25.08 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 25.2           |         |          |               |                  |
| clipped_episode_r... | 9              |         |          |               |                  |
| consecutive_clipp... | 0              |         |          |               |                  |
| max_consecutive_c... | 12             |         |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          | 919            |         |          |               |                  |
| ep_rew_mean          | 294            |         |          |               |                  |
| exploration rate     | 0.572          |         |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             | 20400          |         |          |               |                  |
| fps                  | 257            |         |          |               |                  |
| time_elapsed         | 4991           |         |          |               |                  |
| total timesteps      | 1283816        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.307          |         |          |               |                  |
| n_updates            | 308453         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=6,880             | STEP=1,284,793 | clip=40 | consec=2 | max_consec=12 | avg_clip50=25.58 |
| EP=6,890             | STEP=1,287,173 | clip=64 | consec=2 | max_consec=12 | avg_clip50=27.42 |
| EP=6,900             | STEP=1,289,588 | clip=62 | consec=2 | max_consec=12 | avg_clip50=29.52 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 32.7           |         |          |               |                  |
| clipped_episode_r... | 46.5           |         |          |               |                  |
| consecutive_clipp... | 2              |         |          |               |                  |
| max_consecutive_c... | 12             |         |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          | 922            |         |          |               |                  |
| ep_rew_mean          | 294            |         |          |               |                  |
| exploration rate     | 0.57           |         |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             | 20500          |         |          |               |                  |
| fps                  | 257            |         |          |               |                  |
| time_elapsed         | 5025           |         |          |               |                  |
| total timesteps      | 1292154        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0732         |         |          |               |                  |
| n_updates            | 310538         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=6,910             | STEP=1,292,350 | clip=38 | consec=3 | max_consec=12 | avg_clip50=32.74 |
| EP=6,920             | STEP=1,294,403 | clip=16 | consec=0 | max_consec=12 | avg_clip50=32.82 |
| EP=6,930             | STEP=1,297,102 | clip=59 | consec=7 | max_consec=12 | avg_clip50=34.83 |
| EP=6,940             | STEP=1,299,508 | clip=28 | consec=8 | max_consec=12 | avg_clip50=34.68 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |

|                      |         |
|----------------------|---------|
| avg_clip50           | 34      |
| clipped_episode_r... | 12      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 958     |
| ep_rew_mean          | 311     |
| exploration rate     | 0.567   |
| time/                |         |
| episodes             | 20600   |
| fps                  | 257     |
| time_elapsed         | 5057    |
| total timesteps      | 1300022 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.554   |
| n_updates            | 312505  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,950 | STEP=1,301,273 | clip=11 | consec=0 | max_consec=12 | avg_clip50=31.81 |
| EP=6,960 | STEP=1,303,388 | clip=16 | consec=0 | max_consec=12 | avg_clip50=28.99 |
| EP=6,970 | STEP=1,305,616 | clip=22 | consec=2 | max_consec=12 | avg_clip50=30.14 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.2    |
| clipped_episode_r... | 12      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 967     |
| ep_rew_mean          | 319     |
| exploration rate     | 0.565   |
| time/                |         |
| episodes             | 20700   |
| fps                  | 256     |
| time_elapsed         | 5085    |
| total timesteps      | 1306866 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0983  |
| n_updates            | 314216  |

-----

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,980 | STEP=1,307,732 | clip=5  | consec=0 | max_consec=12 | avg_clip50=28.07 |
| EP=6,990 | STEP=1,309,653 | clip=22 | consec=2 | max_consec=12 | avg_clip50=26.01 |
| EP=7,000 | STEP=1,311,658 | clip=2  | consec=0 | max_consec=12 | avg_clip50=27.34 |
| EP=7,010 | STEP=1,313,841 | clip=32 | consec=2 | max_consec=12 | avg_clip50=27.10 |

-----

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.1    |
| clipped_episode_r... | 32      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 906     |
| ep_rew_mean          | 280     |
| exploration rate     | 0.562   |
| time/                |         |
| episodes             | 20800   |
| fps                  | 256     |
| time_elapsed         | 5114    |
| total timesteps      | 1313904 |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.419  |
| n_updates     | 315975 |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,020 | STEP=1,316,057 | clip=26 | consec=5 | max_consec=12 | avg_clip50=26.57 |
| EP=7,030 | STEP=1,318,434 | clip=14 | consec=0 | max_consec=12 | avg_clip50=26.16 |
| EP=7,040 | STEP=1,320,876 | clip=38 | consec=1 | max_consec=12 | avg_clip50=28.76 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26.8    |
| clipped_episode_r... | 15      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 901     |
| ep_rew_mean          | 273     |
| exploration rate     | 0.56    |
| time/                |         |
| episodes             | 20900   |
| fps                  | 256     |
| time_elapsed         | 5145    |
| total timesteps      | 1321661 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0516  |
| n_updates            | 317915  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,050 | STEP=1,323,275 | clip=16 | consec=0 | max_consec=12 | avg_clip50=30.20 |
| EP=7,060 | STEP=1,325,765 | clip=55 | consec=2 | max_consec=12 | avg_clip50=31.64 |
| EP=7,070 | STEP=1,328,094 | clip=21 | consec=3 | max_consec=12 | avg_clip50=32.05 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 33.1    |
| clipped_episode_r... | 16      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 953     |
| ep_rew_mean          | 294     |
| exploration rate     | 0.557   |
| time/                |         |
| episodes             | 21000   |
| fps                  | 256     |
| time_elapsed         | 5179    |
| total timesteps      | 1330055 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0728  |
| n_updates            | 320013  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,080 | STEP=1,330,421 | clip=26 | consec=2 | max_consec=12 | avg_clip50=33.02 |
| EP=7,090 | STEP=1,332,669 | clip=50 | consec=2 | max_consec=12 | avg_clip50=32.46 |
| EP=7,100 | STEP=1,335,093 | clip=41 | consec=1 | max_consec=12 | avg_clip50=32.84 |
| EP=7,110 | STEP=1,337,438 | clip=35 | consec=1 | max_consec=12 | avg_clip50=32.15 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 32   |
| clipped_episode_r... | 12.5 |
| consecutive_clipp... | 0    |

|                      |         |
|----------------------|---------|
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 983     |
| ep_rew_mean          | 316     |
| exploration rate     | 0.555   |
| time/                |         |
| episodes             | 21100   |
| fps                  | 256     |
| time_elapsed         | 5210    |
| total timesteps      | 1337634 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.488   |
| n_updates            | 321908  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,120 | STEP=1,339,849 | clip=49 | consec=1 | max_consec=12 | avg_clip50=32.52 |
| EP=7,130 | STEP=1,342,021 | clip=30 | consec=3 | max_consec=12 | avg_clip50=32.53 |
| EP=7,140 | STEP=1,344,723 | clip=34 | consec=5 | max_consec=12 | avg_clip50=33.41 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 33.5     |
| clipped_episode_r... | 42       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 12       |
| rollout/             |          |
| ep_len_mean          | 1.01e+03 |
| ep_rew_mean          | 335      |
| exploration rate     | 0.552    |
| time/                |          |
| episodes             | 21200    |
| fps                  | 256      |
| time_elapsed         | 5244     |
| total timesteps      | 1346006  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.104    |
| n_updates            | 324001   |

-----

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,150 | STEP=1,347,156 | clip=22 | consec=4 | max_consec=12 | avg_clip50=33.78 |
| EP=7,160 | STEP=1,348,873 | clip=12 | consec=0 | max_consec=12 | avg_clip50=31.60 |
| EP=7,170 | STEP=1,350,729 | clip=22 | consec=4 | max_consec=12 | avg_clip50=29.37 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.5    |
| clipped_episode_r... | 14.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 946     |
| ep_rew_mean          | 310     |
| exploration rate     | 0.55    |
| time/                |         |
| episodes             | 21300   |
| fps                  | 256     |
| time_elapsed         | 5272    |
| total timesteps      | 1352674 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.234   |
| n_updates            | 325668  |

```

-----
EP=7,180 | STEP=1,352,832 | clip=8 | consec=0 | max_consec=12 | avg_clip50=28.73
EP=7,190 | STEP=1,355,404 | clip=24 | consec=2 | max_consec=12 | avg_clip50=27.87
EP=7,200 | STEP=1,357,124 | clip=3 | consec=0 | max_consec=12 | avg_clip50=24.52
EP=7,210 | STEP=1,359,030 | clip=4 | consec=0 | max_consec=12 | avg_clip50=25.32
-----

```

```

| custom/
|   avg_clip50          | 25.3
|   clipped_episode_r... | 17.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 12
| rollout/
|   ep_len_mean        | 912
|   ep_rew_mean        | 286
|   exploration rate    | 0.547
| time/
|   episodes           | 21400
|   fps                | 256
|   time_elapsed       | 5300
|   total timesteps     | 1359541
| train/
|   learning_rate       | 0.0001
|   loss                | 0.219
|   n_updates           | 327385
-----

```

```

EP=7,220 | STEP=1,361,469 | clip=47 | consec=3 | max_consec=12 | avg_clip50=27.38
EP=7,230 | STEP=1,363,420 | clip=6 | consec=0 | max_consec=12 | avg_clip50=26.95
EP=7,240 | STEP=1,365,632 | clip=70 | consec=4 | max_consec=12 | avg_clip50=26.69
-----

```

```

| custom/
|   avg_clip50          | 27.6
|   clipped_episode_r... | 49
|   consecutive_clipp... | 1
|   max_consecutive_c... | 12
| rollout/
|   ep_len_mean        | 872
|   ep_rew_mean        | 271
|   exploration rate    | 0.545
| time/
|   episodes           | 21500
|   fps                | 256
|   time_elapsed       | 5330
|   total timesteps     | 1366777
| train/
|   learning_rate       | 0.0001
|   loss                | 0.0694
|   n_updates           | 329194
-----

```

```

EP=7,250 | STEP=1,367,861 | clip=26 | consec=2 | max_consec=12 | avg_clip50=28.35
EP=7,260 | STEP=1,369,979 | clip=13 | consec=0 | max_consec=12 | avg_clip50=29.40
EP=7,270 | STEP=1,372,295 | clip=41 | consec=1 | max_consec=12 | avg_clip50=28.98
-----

```

```

| custom/
|   avg_clip50          | 30.1
|   clipped_episode_r... | 57.5
|   consecutive_clipp... | 1
|   max_consecutive_c... | 12
| rollout/
|   ep_len_mean        | 898
|   ep_rew_mean        | 280
-----

```

|                  |         |
|------------------|---------|
| exploration rate | 0.542   |
| time/            |         |
| episodes         | 21600   |
| fps              | 256     |
| time_elapsed     | 5359    |
| total timesteps  | 1374029 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0553  |
| n_updates        | 331007  |

EP=7,280 | STEP=1,374,429 | clip=32 | consec=3 | max\_consec=12 | avg\_clip50=29.97  
EP=7,290 | STEP=1,377,459 | clip=37 | consec=2 | max\_consec=12 | avg\_clip50=31.35  
EP=7,300 | STEP=1,379,682 | clip=22 | consec=1 | max\_consec=12 | avg\_clip50=32.71  
EP=7,310 | STEP=1,382,128 | clip=70 | consec=8 | max\_consec=12 | avg\_clip50=34.57

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 35.1    |
| clipped_episode_r... | 18.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 958     |
| ep_rew_mean          | 316     |
| exploration rate     | 0.539   |
| time/                |         |
| episodes             | 21700   |
| fps                  | 256     |
| time_elapsed         | 5396    |
| total timesteps      | 1383008 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.319   |
| n_updates            | 333251  |

EP=7,320 | STEP=1,384,575 | clip=32 | consec=6 | max\_consec=12 | avg\_clip50=36.00  
EP=7,330 | STEP=1,386,849 | clip=12 | consec=0 | max\_consec=12 | avg\_clip50=36.21  
EP=7,340 | STEP=1,389,344 | clip=66 | consec=3 | max\_consec=12 | avg\_clip50=36.51

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37.6     |
| clipped_episode_r... | 26.5     |
| consecutive_clipp... | 11       |
| max_consecutive_c... | 12       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 356      |
| exploration rate     | 0.537    |
| time/                |          |
| episodes             | 21800    |
| fps                  | 256      |
| time_elapsed         | 5431     |
| total timesteps      | 1391651  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0579   |
| n_updates            | 335412   |

EP=7,350 | STEP=1,392,126 | clip=36 | consec=1 | max\_consec=12 | avg\_clip50=37.87  
EP=7,360 | STEP=1,394,384 | clip=47 | consec=3 | max\_consec=12 | avg\_clip50=36.80

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,370 | STEP=1,396,422 | clip=16 | consec=0 | max_consec=12 | avg_clip50=35.21 |
| EP=7,380 | STEP=1,398,358 | clip=22 | consec=1 | max_consec=12 | avg_clip50=34.52 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34.2    |
| clipped_episode_r... | 31      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 998     |
| ep_rew_mean          | 351     |
| exploration rate     | 0.534   |
| time/                |         |
| episodes             | 21900   |
| fps                  | 256     |
| time_elapsed         | 5460    |
| total timesteps      | 1398683 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.503   |
| n_updates            | 337170  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,390 | STEP=1,400,967 | clip=68 | consec=7 | max_consec=12 | avg_clip50=34.48 |
| EP=7,400 | STEP=1,403,107 | clip=48 | consec=1 | max_consec=12 | avg_clip50=32.00 |
| EP=7,410 | STEP=1,405,496 | clip=43 | consec=4 | max_consec=12 | avg_clip50=33.03 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 33.2    |
| clipped_episode_r... | 51.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 983     |
| ep_rew_mean          | 349     |
| exploration rate     | 0.532   |
| time/                |         |
| episodes             | 22000   |
| fps                  | 256     |
| time_elapsed         | 5492    |
| total timesteps      | 1406541 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.326   |
| n_updates            | 339135  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,420 | STEP=1,408,176 | clip=44 | consec=3 | max_consec=12 | avg_clip50=35.07 |
| EP=7,430 | STEP=1,410,197 | clip=14 | consec=0 | max_consec=12 | avg_clip50=34.14 |
| EP=7,440 | STEP=1,411,833 | clip=23 | consec=1 | max_consec=12 | avg_clip50=29.92 |

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 29.6  |
| clipped_episode_r... | 17    |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 12    |
| rollout/             |       |
| ep_len_mean          | 923   |
| ep_rew_mean          | 307   |
| exploration rate     | 0.529 |
| time/                |       |
| episodes             | 22100 |



|                      |                |         |          |               |                  |
|----------------------|----------------|---------|----------|---------------|------------------|
| fps                  | 255            |         |          |               |                  |
| time_elapsed         | 5522           |         |          |               |                  |
| total timesteps      | 1413565        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.102          |         |          |               |                  |
| n_updates            | 340891         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=7,450             | STEP=1,413,942 | clip=17 | consec=0 | max_consec=12 | avg_clip50=29.09 |
| EP=7,460             | STEP=1,415,791 | clip=25 | consec=2 | max_consec=12 | avg_clip50=26.30 |
| EP=7,470             | STEP=1,418,494 | clip=4  | consec=0 | max_consec=12 | avg_clip50=27.47 |
| EP=7,480             | STEP=1,420,448 | clip=28 | consec=3 | max_consec=12 | avg_clip50=27.90 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 28.1           |         |          |               |                  |
| clipped_episode_r... | 25.5           |         |          |               |                  |
| consecutive_clipp... | 4              |         |          |               |                  |
| max_consecutive_c... | 12             |         |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          | 929            |         |          |               |                  |
| ep_rew_mean          | 310            |         |          |               |                  |
| exploration rate     | 0.527          |         |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             | 22200          |         |          |               |                  |
| fps                  | 255            |         |          |               |                  |
| time_elapsed         | 5552           |         |          |               |                  |
| total timesteps      | 1420830        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0735         |         |          |               |                  |
| n_updates            | 342707         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=7,490             | STEP=1,422,757 | clip=36 | consec=1 | max_consec=12 | avg_clip50=30.10 |
| EP=7,500             | STEP=1,424,832 | clip=42 | consec=1 | max_consec=12 | avg_clip50=30.15 |
| EP=7,510             | STEP=1,426,411 | clip=36 | consec=4 | max_consec=12 | avg_clip50=29.09 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 28.7           |         |          |               |                  |
| clipped_episode_r... | 92.5           |         |          |               |                  |
| consecutive_clipp... | 1              |         |          |               |                  |
| max_consecutive_c... | 12             |         |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          | 881            |         |          |               |                  |
| ep_rew_mean          | 283            |         |          |               |                  |
| exploration rate     | 0.525          |         |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             | 22300          |         |          |               |                  |
| fps                  | 255            |         |          |               |                  |
| time_elapsed         | 5579           |         |          |               |                  |
| total timesteps      | 1427325        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0337         |         |          |               |                  |
| n_updates            | 344331         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=7,520             | STEP=1,428,836 | clip=49 | consec=2 | max_consec=12 | avg_clip50=28.39 |
| EP=7,530             | STEP=1,431,371 | clip=8  | consec=0 | max_consec=12 | avg_clip50=29.99 |
| EP=7,540             | STEP=1,433,378 | clip=8  | consec=0 | max_consec=12 | avg_clip50=28.53 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |

|                      |         |
|----------------------|---------|
| avg_clip50           | 29.2    |
| clipped_episode_r... | 32.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 913     |
| ep_rew_mean          | 295     |
| exploration rate     | 0.522   |
| time/                |         |
| episodes             | 22400   |
| fps                  | 255     |
| time_elapsed         | 5611    |
| total timesteps      | 1435287 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.276   |
| n_updates            | 346321  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,550 | STEP=1,435,779 | clip=70 | consec=5 | max_consec=12 | avg_clip50=30.69 |
| EP=7,560 | STEP=1,437,935 | clip=6  | consec=0 | max_consec=12 | avg_clip50=34.06 |
| EP=7,570 | STEP=1,439,969 | clip=14 | consec=0 | max_consec=12 | avg_clip50=30.62 |
| EP=7,580 | STEP=1,441,978 | clip=3  | consec=0 | max_consec=12 | avg_clip50=29.86 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29.5    |
| clipped_episode_r... | 33      |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 904     |
| ep_rew_mean          | 295     |
| exploration rate     | 0.52    |
| time/                |         |
| episodes             | 22500   |
| fps                  | 255     |
| time_elapsed         | 5641    |
| total timesteps      | 1442551 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0684  |
| n_updates            | 348137  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,590 | STEP=1,443,969 | clip=16 | consec=0 | max_consec=12 | avg_clip50=30.50 |
| EP=7,600 | STEP=1,446,564 | clip=19 | consec=0 | max_consec=12 | avg_clip50=31.47 |
| EP=7,610 | STEP=1,448,455 | clip=10 | consec=0 | max_consec=12 | avg_clip50=29.43 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.5    |
| clipped_episode_r... | 2       |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 928     |
| ep_rew_mean          | 310     |
| exploration rate     | 0.517   |
| time/                |         |
| episodes             | 22600   |
| fps                  | 255     |
| time_elapsed         | 5672    |
| total timesteps      | 1449889 |

|                      |                |         |          |               |                  |  |
|----------------------|----------------|---------|----------|---------------|------------------|--|
| train/               |                |         |          |               |                  |  |
| learning_rate        |                | 0.0001  |          |               |                  |  |
| loss                 |                | 0.73    |          |               |                  |  |
| n_updates            |                | 349972  |          |               |                  |  |
| -----                |                |         |          |               |                  |  |
| EP=7,620             | STEP=1,450,792 | clip=14 | consec=0 | max_consec=12 | avg_clip50=31.38 |  |
| EP=7,630             | STEP=1,452,909 | clip=11 | consec=0 | max_consec=12 | avg_clip50=30.83 |  |
| EP=7,640             | STEP=1,455,572 | clip=50 | consec=2 | max_consec=12 | avg_clip50=33.85 |  |
| EP=7,650             | STEP=1,457,839 | clip=20 | consec=0 | max_consec=12 | avg_clip50=31.91 |  |
| -----                |                |         |          |               |                  |  |
| custom/              |                |         |          |               |                  |  |
| avg_clip50           |                | 31.9    |          |               |                  |  |
| clipped_episode_r... |                | 19.5    |          |               |                  |  |
| consecutive_clipp... |                | 0       |          |               |                  |  |
| max_consecutive_c... |                | 12      |          |               |                  |  |
| rollout/             |                |         |          |               |                  |  |
| ep_len_mean          |                | 929     |          |               |                  |  |
| ep_rew_mean          |                | 317     |          |               |                  |  |
| exploration rate     |                | 0.515   |          |               |                  |  |
| time/                |                |         |          |               |                  |  |
| episodes             |                | 22700   |          |               |                  |  |
| fps                  |                | 255     |          |               |                  |  |
| time_elapsed         |                | 5704    |          |               |                  |  |
| total timesteps      |                | 1457839 |          |               |                  |  |
| train/               |                |         |          |               |                  |  |
| learning_rate        |                | 0.0001  |          |               |                  |  |
| loss                 |                | 0.0446  |          |               |                  |  |
| n_updates            |                | 351959  |          |               |                  |  |
| -----                |                |         |          |               |                  |  |
| EP=7,660             | STEP=1,460,284 | clip=9  | consec=0 | max_consec=12 | avg_clip50=33.58 |  |
| EP=7,670             | STEP=1,462,609 | clip=26 | consec=6 | max_consec=12 | avg_clip50=33.37 |  |
| EP=7,680             | STEP=1,465,013 | clip=14 | consec=0 | max_consec=12 | avg_clip50=34.75 |  |
| -----                |                |         |          |               |                  |  |
| custom/              |                |         |          |               |                  |  |
| avg_clip50           |                | 33.4    |          |               |                  |  |
| clipped_episode_r... |                | 33.5    |          |               |                  |  |
| consecutive_clipp... |                | 1       |          |               |                  |  |
| max_consecutive_c... |                | 12      |          |               |                  |  |
| rollout/             |                |         |          |               |                  |  |
| ep_len_mean          |                | 970     |          |               |                  |  |
| ep_rew_mean          |                | 329     |          |               |                  |  |
| exploration rate     |                | 0.512   |          |               |                  |  |
| time/                |                |         |          |               |                  |  |
| episodes             |                | 22800   |          |               |                  |  |
| fps                  |                | 255     |          |               |                  |  |
| time_elapsed         |                | 5737    |          |               |                  |  |
| total timesteps      |                | 1465880 |          |               |                  |  |
| train/               |                |         |          |               |                  |  |
| learning_rate        |                | 0.0001  |          |               |                  |  |
| loss                 |                | 0.0728  |          |               |                  |  |
| n_updates            |                | 353969  |          |               |                  |  |
| -----                |                |         |          |               |                  |  |
| EP=7,690             | STEP=1,467,732 | clip=39 | consec=3 | max_consec=12 | avg_clip50=35.04 |  |
| EP=7,700             | STEP=1,469,834 | clip=10 | consec=0 | max_consec=12 | avg_clip50=34.57 |  |
| EP=7,710             | STEP=1,472,282 | clip=19 | consec=0 | max_consec=12 | avg_clip50=35.25 |  |
| -----                |                |         |          |               |                  |  |
| custom/              |                |         |          |               |                  |  |
| avg_clip50           |                | 35.5    |          |               |                  |  |
| clipped_episode_r... |                | 39.5    |          |               |                  |  |
| consecutive_clipp... |                | 1       |          |               |                  |  |

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| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 994     |
| ep_rew_mean          | 342     |
| exploration rate     | 0.509   |
| time/                |         |
| episodes             | 22900   |
| fps                  | 255     |
| time_elapsed         | 5771    |
| total timesteps      | 1474131 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0411  |
| n_updates            | 356032  |

EP=7,720 | STEP=1,474,493 | clip=10 | consec=0 | max\_consec=12 | avg\_clip50=34.85  
EP=7,730 | STEP=1,476,863 | clip=60 | consec=1 | max\_consec=12 | avg\_clip50=35.45  
EP=7,740 | STEP=1,479,144 | clip=13 | consec=0 | max\_consec=12 | avg\_clip50=35.30  
EP=7,750 | STEP=1,481,296 | clip=44 | consec=2 | max\_consec=12 | avg\_clip50=35.07

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| custom/              |         |
| avg_clip50           | 35.5    |
| clipped_episode_r... | 39.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 975     |
| ep_rew_mean          | 347     |
| exploration rate     | 0.507   |
| time/                |         |
| episodes             | 23000   |
| fps                  | 255     |
| time_elapsed         | 5802    |
| total timesteps      | 1481678 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.278   |
| n_updates            | 357919  |

EP=7,760 | STEP=1,483,445 | clip=47 | consec=2 | max\_consec=12 | avg\_clip50=34.12  
EP=7,770 | STEP=1,485,695 | clip=45 | consec=2 | max\_consec=12 | avg\_clip50=34.91  
EP=7,780 | STEP=1,487,931 | clip=25 | consec=5 | max\_consec=12 | avg\_clip50=34.52

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| custom/              |         |
| avg_clip50           | 34.9    |
| clipped_episode_r... | 29.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 960     |
| ep_rew_mean          | 346     |
| exploration rate     | 0.504   |
| time/                |         |
| episodes             | 23100   |
| fps                  | 255     |
| time_elapsed         | 5834    |
| total timesteps      | 1489420 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0366  |

|                      |                |          |          |               |                  |
|----------------------|----------------|----------|----------|---------------|------------------|
| n_updates            |                | 359854   |          |               |                  |
| -----                |                |          |          |               |                  |
| EP=7,790             | STEP=1,490,257 | clip=26  | consec=7 | max_consec=12 | avg_clip50=33.09 |
| EP=7,800             | STEP=1,492,802 | clip=20  | consec=0 | max_consec=12 | avg_clip50=33.53 |
| EP=7,810             | STEP=1,495,576 | clip=53  | consec=4 | max_consec=12 | avg_clip50=35.68 |
| EP=7,820             | STEP=1,497,876 | clip=28  | consec=1 | max_consec=12 | avg_clip50=35.79 |
| -----                |                |          |          |               |                  |
| custom/              |                |          |          |               |                  |
| avg_clip50           |                | 35.8     |          |               |                  |
| clipped_episode_r... |                | 38.5     |          |               |                  |
| consecutive_clipp... |                | 2        |          |               |                  |
| max_consecutive_c... |                | 12       |          |               |                  |
| rollout/             |                |          |          |               |                  |
| ep_len_mean          |                | 981      |          |               |                  |
| ep_rew_mean          |                | 354      |          |               |                  |
| exploration rate     |                | 0.501    |          |               |                  |
| time/                |                |          |          |               |                  |
| episodes             |                | 23200    |          |               |                  |
| fps                  |                | 255      |          |               |                  |
| time_elapsed         |                | 5870     |          |               |                  |
| total timesteps      |                | 1498190  |          |               |                  |
| train/               |                |          |          |               |                  |
| learning_rate        |                | 0.0001   |          |               |                  |
| loss                 |                | 0.0648   |          |               |                  |
| n_updates            |                | 362047   |          |               |                  |
| -----                |                |          |          |               |                  |
| EP=7,830             | STEP=1,500,176 | clip=37  | consec=2 | max_consec=12 | avg_clip50=35.71 |
| EP=7,840             | STEP=1,502,343 | clip=11  | consec=0 | max_consec=12 | avg_clip50=35.38 |
| EP=7,850             | STEP=1,505,345 | clip=49  | consec=1 | max_consec=12 | avg_clip50=39.26 |
| -----                |                |          |          |               |                  |
| custom/              |                |          |          |               |                  |
| avg_clip50           |                | 38.6     |          |               |                  |
| clipped_episode_r... |                | 35.5     |          |               |                  |
| consecutive_clipp... |                | 2        |          |               |                  |
| max_consecutive_c... |                | 12       |          |               |                  |
| rollout/             |                |          |          |               |                  |
| ep_len_mean          |                | 1.01e+03 |          |               |                  |
| ep_rew_mean          |                | 364      |          |               |                  |
| exploration rate     |                | 0.498    |          |               |                  |
| time/                |                |          |          |               |                  |
| episodes             |                | 23300    |          |               |                  |
| fps                  |                | 255      |          |               |                  |
| time_elapsed         |                | 5905     |          |               |                  |
| total timesteps      |                | 1506561  |          |               |                  |
| train/               |                |          |          |               |                  |
| learning_rate        |                | 0.0001   |          |               |                  |
| loss                 |                | 0.434    |          |               |                  |
| n_updates            |                | 364140   |          |               |                  |
| -----                |                |          |          |               |                  |
| EP=7,860             | STEP=1,507,663 | clip=35  | consec=3 | max_consec=12 | avg_clip50=37.06 |
| EP=7,870             | STEP=1,510,122 | clip=48  | consec=3 | max_consec=12 | avg_clip50=36.96 |
| EP=7,880             | STEP=1,512,505 | clip=39  | consec=1 | max_consec=12 | avg_clip50=36.43 |
| -----                |                |          |          |               |                  |
| custom/              |                |          |          |               |                  |
| avg_clip50           |                | 37.6     |          |               |                  |
| clipped_episode_r... |                | 77       |          |               |                  |
| consecutive_clipp... |                | 2        |          |               |                  |
| max_consecutive_c... |                | 12       |          |               |                  |
| rollout/             |                |          |          |               |                  |
| ep_len_mean          |                | 1.03e+03 |          |               |                  |

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|------------------|---------|
| ep_rew_mean      | 366     |
| exploration rate | 0.496   |
| time/            |         |
| episodes         | 23400   |
| fps              | 255     |
| time_elapsed     | 5939    |
| total timesteps  | 1514830 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.358   |
| n_updates        | 366207  |

EP=7,890 | STEP=1,514,990 | clip=32 | consec=3 | max\_consec=12 | avg\_clip50=37.99  
EP=7,900 | STEP=1,517,787 | clip=44 | consec=5 | max\_consec=12 | avg\_clip50=36.68  
EP=7,910 | STEP=1,520,436 | clip=50 | consec=3 | max\_consec=12 | avg\_clip50=37.86  
EP=7,920 | STEP=1,522,831 | clip=12 | consec=0 | max\_consec=12 | avg\_clip50=38.07

|                      |          |
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| custom/              |          |
| avg_clip50           | 36.5     |
| clipped_episode_r... | 9.5      |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 12       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 371      |
| exploration rate     | 0.493    |
| time/                |          |
| episodes             | 23500    |
| fps                  | 254      |
| time_elapsed         | 5976     |
| total timesteps      | 1523985  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0515   |
| n_updates            | 368496   |

EP=7,930 | STEP=1,525,147 | clip=69 | consec=2 | max\_consec=12 | avg\_clip50=37.89  
EP=7,940 | STEP=1,527,758 | clip=34 | consec=2 | max\_consec=12 | avg\_clip50=38.08  
EP=7,950 | STEP=1,529,968 | clip=13 | consec=0 | max\_consec=12 | avg\_clip50=35.62

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| custom/              |          |
| avg_clip50           | 33.4     |
| clipped_episode_r... | 22       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 12       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 356      |
| exploration rate     | 0.49     |
| time/                |          |
| episodes             | 23600    |
| fps                  | 254      |
| time_elapsed         | 6007     |
| total timesteps      | 1531532  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0629   |
| n_updates            | 370382   |

EP=7,960 | STEP=1,532,084 | clip=50 | consec=3 | max\_consec=12 | avg\_clip50=33.46

EP=7,970 | STEP=1,534,498 | clip=66 | consec=7 | max\_consec=12 | avg\_clip50=33.53  
EP=7,980 | STEP=1,536,970 | clip=32 | consec=5 | max\_consec=12 | avg\_clip50=34.15  
EP=7,990 | STEP=1,539,482 | clip=48 | consec=15 | max\_consec=15 | avg\_clip50=34.45

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 33.8     |  |
| clipped_episode_r... | 49       |  |
| consecutive_clipp... | 1        |  |
| max_consecutive_c... | 15       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.03e+03 |  |
| ep_rew_mean          | 363      |  |
| exploration rate     | 0.487    |  |
| time/                |          |  |
| episodes             | 23700    |  |
| fps                  | 254      |  |
| time_elapsed         | 6044     |  |
| total timesteps      | 1540264  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.326    |  |
| n_updates            | 372565   |  |

EP=8,000 | STEP=1,541,786 | clip=38 | consec=1 | max\_consec=15 | avg\_clip50=34.34  
EP=8,010 | STEP=1,544,212 | clip=37 | consec=2 | max\_consec=15 | avg\_clip50=35.70  
EP=8,020 | STEP=1,546,362 | clip=68 | consec=4 | max\_consec=15 | avg\_clip50=34.74

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| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 36      |  |
| clipped_episode_r... | 32      |  |
| consecutive_clipp... | 10      |  |
| max_consecutive_c... | 15      |  |
| rollout/             |         |  |
| ep_len_mean          | 982     |  |
| ep_rew_mean          | 350     |  |
| exploration rate     | 0.485   |  |
| time/                |         |  |
| episodes             | 23800   |  |
| fps                  | 254     |  |
| time_elapsed         | 6075    |  |
| total timesteps      | 1547827 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.0594  |  |
| n_updates            | 374456  |  |

EP=8,030 | STEP=1,548,908 | clip=42 | consec=14 | max\_consec=15 | avg\_clip50=36.88  
EP=8,040 | STEP=1,551,768 | clip=56 | consec=8 | max\_consec=15 | avg\_clip50=37.58  
EP=8,050 | STEP=1,554,126 | clip=55 | consec=2 | max\_consec=15 | avg\_clip50=38.00  
EP=8,060 | STEP=1,556,626 | clip=26 | consec=1 | max\_consec=15 | avg\_clip50=39.18

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 39.2     |  |
| clipped_episode_r... | 26       |  |
| consecutive_clipp... | 1        |  |
| max_consecutive_c... | 15       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.03e+03 |  |

|                  |         |
|------------------|---------|
| ep_rew_mean      | 374     |
| exploration rate | 0.482   |
| time/            |         |
| episodes         | 23900   |
| fps              | 254     |
| time_elapsed     | 6111    |
| total timesteps  | 1556671 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.028   |
| n_updates        | 376667  |

EP=8,070 | STEP=1,559,615 | clip=100 | consec=3 | max\_consec=15 | avg\_clip50=42.78

EP=8,080 | STEP=1,562,051 | clip=8 | consec=0 | max\_consec=15 | avg\_clip50=41.09

EP=8,090 | STEP=1,564,600 | clip=13 | consec=0 | max\_consec=15 | avg\_clip50=39.16

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39.1     |
| clipped_episode_r... | 26.5     |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 15       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 377      |
| exploration rate     | 0.479    |
| time/                |          |
| episodes             | 24000    |
| fps                  | 254      |
| time_elapsed         | 6148     |
| total timesteps      | 1565599  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0602   |
| n_updates            | 378899   |

EP=8,100 | STEP=1,566,920 | clip=46 | consec=3 | max\_consec=15 | avg\_clip50=39.49

EP=8,110 | STEP=1,569,706 | clip=68 | consec=9 | max\_consec=15 | avg\_clip50=42.19

EP=8,120 | STEP=1,572,181 | clip=57 | consec=4 | max\_consec=15 | avg\_clip50=40.16

EP=8,130 | STEP=1,574,750 | clip=38 | consec=5 | max\_consec=15 | avg\_clip50=39.95

|                      |          |
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| custom/              |          |
| avg_clip50           | 40       |
| clipped_episode_r... | 38       |
| consecutive_clipp... | 5        |
| max_consecutive_c... | 15       |
| rollout/             |          |
| ep_len_mean          | 1.08e+03 |
| ep_rew_mean          | 405      |
| exploration rate     | 0.476    |
| time/                |          |
| episodes             | 24100    |
| fps                  | 254      |
| time_elapsed         | 6186     |
| total timesteps      | 1574750  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.283    |
| n_updates            | 381187   |



|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=8,140 | STEP=1,577,142 | clip=6  | consec=0 | max_consec=15 | avg_clip50=39.87 |
| EP=8,150 | STEP=1,580,048 | clip=24 | consec=3 | max_consec=15 | avg_clip50=42.33 |
| EP=8,160 | STEP=1,582,626 | clip=34 | consec=4 | max_consec=15 | avg_clip50=39.29 |

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|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 37.9     |
| clipped_episode_r... | 23       |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 15       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 399      |
| exploration rate     | 0.473    |
| time/                |          |
| episodes             | 24200    |
| fps                  | 254      |
| time_elapsed         | 6221     |
| total timesteps      | 1583429  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.207    |
| n_updates            | 383357   |

|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=8,170 | STEP=1,584,972 | clip=52 | consec=2  | max_consec=15 | avg_clip50=37.83 |
| EP=8,180 | STEP=1,587,351 | clip=43 | consec=1  | max_consec=15 | avg_clip50=37.79 |
| EP=8,190 | STEP=1,590,303 | clip=44 | consec=11 | max_consec=15 | avg_clip50=39.1  |

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|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 37.3     |
| clipped_episode_r... | 12       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 15       |
| rollout/             |          |
| ep_len_mean          | 1.08e+03 |
| ep_rew_mean          | 400      |
| exploration rate     | 0.47     |
| time/                |          |
| episodes             | 24300    |
| fps                  | 254      |
| time_elapsed         | 6256     |
| total timesteps      | 1592023  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.149    |
| n_updates            | 385505   |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=8,200 | STEP=1,592,748 | clip=52 | consec=3 | max_consec=15 | avg_clip50=37.32 |
| EP=8,210 | STEP=1,595,025 | clip=21 | consec=1 | max_consec=15 | avg_clip50=37.65 |
| EP=8,220 | STEP=1,597,290 | clip=27 | consec=1 | max_consec=15 | avg_clip50=37.16 |
| EP=8,230 | STEP=1,600,045 | clip=28 | consec=8 | max_consec=15 | avg_clip50=38.23 |

|                      |          |
|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 38.5     |
| clipped_episode_r... | 50       |
| consecutive_clipp... | 9        |
| max_consecutive_c... | 15       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 384      |

|                  |         |
|------------------|---------|
| exploration rate | 0.467   |
| time/            |         |
| episodes         | 24400   |
| fps              | 254     |
| time_elapsed     | 6291    |
| total timesteps  | 1600481 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.404   |
| n_updates        | 387620  |

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EP=8,240 | STEP=1,602,287 | clip=30 | consec=4 | max\_consec=15 | avg\_clip50=37.14  
EP=8,250 | STEP=1,605,219 | clip=24 | consec=14 | max\_consec=15 | avg\_clip50=39.0  
2  
EP=8,260 | STEP=1,607,894 | clip=28 | consec=3 | max\_consec=20 | avg\_clip50=39.02

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37.1     |
| clipped_episode_r... | 19.5     |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 377      |
| exploration rate     | 0.464    |
| time/                |          |
| episodes             | 24500    |
| fps                  | 254      |
| time_elapsed         | 6325     |
| total timesteps      | 1608706  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0379   |
| n_updates            | 389676   |

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EP=8,270 | STEP=1,610,171 | clip=50 | consec=5 | max\_consec=20 | avg\_clip50=39.48  
EP=8,280 | STEP=1,612,824 | clip=74 | consec=1 | max\_consec=20 | avg\_clip50=40.64  
EP=8,290 | STEP=1,615,562 | clip=47 | consec=11 | max\_consec=20 | avg\_clip50=42.1  
5

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39.5     |
| clipped_episode_r... | 29.5     |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 396      |
| exploration rate     | 0.461    |
| time/                |          |
| episodes             | 24600    |
| fps                  | 254      |
| time_elapsed         | 6362     |
| total timesteps      | 1617791  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.114    |
| n_updates            | 391947   |

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EP=8,300 | STEP=1,617,872 | clip=12 | consec=0 | max\_consec=20 | avg\_clip50=39.28

EP=8,310 | STEP=1,620,338 | clip=20 | consec=0 | max\_consec=20 | avg\_clip50=38.69  
EP=8,320 | STEP=1,622,476 | clip=6 | consec=0 | max\_consec=20 | avg\_clip50=38.51  
EP=8,330 | STEP=1,625,304 | clip=60 | consec=7 | max\_consec=20 | avg\_clip50=38.64

|                      |          |
|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 39       |
| clipped_episode_r... | 54.5     |
| consecutive_clipp... | 10       |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 398      |
| exploration rate     | 0.458    |
| time/                |          |
| episodes             | 24700    |
| fps                  | 254      |
| time_elapsed         | 6398     |
| total timesteps      | 1626402  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0231   |
| n_updates            | 394100   |

EP=8,340 | STEP=1,628,170 | clip=44 | consec=17 | max\_consec=20 | avg\_clip50=39.45  
EP=8,350 | STEP=1,631,058 | clip=48 | consec=5 | max\_consec=20 | avg\_clip50=41.58  
EP=8,360 | STEP=1,633,599 | clip=44 | consec=7 | max\_consec=20 | avg\_clip50=41.89

|                      |          |
|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 42.7     |
| clipped_episode_r... | 8        |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.08e+03 |
| ep_rew_mean          | 409      |
| exploration rate     | 0.455    |
| time/                |          |
| episodes             | 24800    |
| fps                  | 254      |
| time_elapsed         | 6436     |
| total timesteps      | 1635472  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0785   |
| n_updates            | 396367   |

EP=8,370 | STEP=1,636,093 | clip=35 | consec=3 | max\_consec=20 | avg\_clip50=43.78  
EP=8,380 | STEP=1,638,446 | clip=26 | consec=1 | max\_consec=20 | avg\_clip50=40.60  
EP=8,390 | STEP=1,641,197 | clip=60 | consec=8 | max\_consec=20 | avg\_clip50=39.29  
EP=8,400 | STEP=1,643,565 | clip=46 | consec=2 | max\_consec=20 | avg\_clip50=37.71

|                      |          |
|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 36.2     |
| clipped_episode_r... | 12       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 389      |

|                  |         |
|------------------|---------|
| exploration rate | 0.453   |
| time/            |         |
| episodes         | 24900   |
| fps              | 254     |
| time_elapsed     | 6470    |
| total timesteps  | 1643778 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0509  |
| n_updates        | 398444  |

EP=8,410 | STEP=1,645,856 | clip=10 | consec=0 | max\_consec=20 | avg\_clip50=36.29  
EP=8,420 | STEP=1,648,442 | clip=26 | consec=8 | max\_consec=20 | avg\_clip50=36.99  
EP=8,430 | STEP=1,651,180 | clip=58 | consec=8 | max\_consec=20 | avg\_clip50=40.34

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39.5     |
| clipped_episode_r... | 25       |
| consecutive_clipp... | 13       |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 398      |
| exploration rate     | 0.45     |
| time/                |          |
| episodes             | 25000    |
| fps                  | 253      |
| time_elapsed         | 6506     |
| total timesteps      | 1652404  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.457    |
| n_updates            | 400600   |

EP=8,440 | STEP=1,653,809 | clip=42 | consec=4 | max\_consec=20 | avg\_clip50=39.64  
EP=8,450 | STEP=1,656,119 | clip=6 | consec=0 | max\_consec=20 | avg\_clip50=38.56  
EP=8,460 | STEP=1,658,581 | clip=27 | consec=3 | max\_consec=20 | avg\_clip50=39.63

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37.3     |
| clipped_episode_r... | 32.5     |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 378      |
| exploration rate     | 0.447    |
| time/                |          |
| episodes             | 25100    |
| fps                  | 253      |
| time_elapsed         | 6540     |
| total timesteps      | 1660585  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0542   |
| n_updates            | 402646   |

EP=8,470 | STEP=1,661,015 | clip=76 | consec=4 | max\_consec=20 | avg\_clip50=38.18  
EP=8,480 | STEP=1,663,851 | clip=52 | consec=6 | max\_consec=20 | avg\_clip50=36.54  
EP=8,490 | STEP=1,666,378 | clip=44 | consec=2 | max\_consec=20 | avg\_clip50=35.93

EP=8,500 | STEP=1,669,256 | clip=54 | consec=7 | max\_consec=20 | avg\_clip50=38.82

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37       |
| clipped_episode_r... | 46.5     |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 391      |
| exploration rate     | 0.444    |
| time/                |          |
| episodes             | 25200    |
| fps                  | 253      |
| time_elapsed         | 6578     |
| total timesteps      | 1669943  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0571   |
| n_updates            | 404985   |

EP=8,510 | STEP=1,671,700 | clip=52 | consec=4 | max\_consec=20 | avg\_clip50=38.40

EP=8,520 | STEP=1,673,836 | clip=4 | consec=0 | max\_consec=20 | avg\_clip50=36.73

EP=8,530 | STEP=1,676,275 | clip=20 | consec=0 | max\_consec=20 | avg\_clip50=35.75

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 36.1     |
| clipped_episode_r... | 21.5     |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 360      |
| exploration rate     | 0.441    |
| time/                |          |
| episodes             | 25300    |
| fps                  | 253      |
| time_elapsed         | 6613     |
| total timesteps      | 1678192  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0993   |
| n_updates            | 407047   |

EP=8,540 | STEP=1,678,762 | clip=57 | consec=3 | max\_consec=20 | avg\_clip50=36.15

EP=8,550 | STEP=1,680,835 | clip=12 | consec=0 | max\_consec=20 | avg\_clip50=33.28

EP=8,560 | STEP=1,683,745 | clip=30 | consec=7 | max\_consec=20 | avg\_clip50=35.94

EP=8,570 | STEP=1,686,260 | clip=44 | consec=1 | max\_consec=20 | avg\_clip50=36.98

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37       |
| clipped_episode_r... | 33.5     |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 370      |
| exploration rate     | 0.438    |
| time/                |          |
| episodes             | 25400    |

|                 |         |
|-----------------|---------|
| fps             | 253     |
| time_elapsed    | 6648    |
| total timesteps | 1686607 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.0979  |
| n_updates       | 409151  |

EP=8,580 | STEP=1,689,132 | clip=74 | consec=11 | max\_consec=20 | avg\_clip50=40.05

EP=8,590 | STEP=1,691,581 | clip=46 | consec=1 | max\_consec=20 | avg\_clip50=39.08

EP=8,600 | STEP=1,694,207 | clip=18 | consec=0 | max\_consec=20 | avg\_clip50=41.50

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 41.2     |
| clipped_episode_r... | 60       |
| consecutive_clipp... | 6        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 386      |
| exploration rate     | 0.435    |
| time/                |          |
| episodes             | 25500    |
| fps                  | 253      |
| time_elapsed         | 6687     |
| total timesteps      | 1696141  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.185    |
| n_updates            | 411535   |

EP=8,610 | STEP=1,697,038 | clip=46 | consec=1 | max\_consec=20 | avg\_clip50=40.16

EP=8,620 | STEP=1,699,277 | clip=55 | consec=2 | max\_consec=20 | avg\_clip50=40.17

EP=8,630 | STEP=1,701,640 | clip=46 | consec=7 | max\_consec=20 | avg\_clip50=36.81

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 36.8     |
| clipped_episode_r... | 8        |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 380      |
| exploration rate     | 0.433    |
| time/                |          |
| episodes             | 25600    |
| fps                  | 253      |
| time_elapsed         | 6720     |
| total timesteps      | 1703968  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0904   |
| n_updates            | 413491   |

EP=8,640 | STEP=1,704,086 | clip=85 | consec=1 | max\_consec=20 | avg\_clip50=37.55

EP=8,650 | STEP=1,706,975 | clip=72 | consec=11 | max\_consec=20 | avg\_clip50=38.72

EP=8,660 | STEP=1,709,387 | clip=54 | consec=2 | max\_consec=20 | avg\_clip50=38.84

EP=8,670 | STEP=1,711,376 | clip=14 | consec=0 | max\_consec=20 | avg\_clip50=38.69

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39       |
| clipped_episode_r... | 12       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 387      |
| exploration rate     | 0.43     |
| time/                |          |
| episodes             | 25700    |
| fps                  | 253      |
| time_elapsed         | 6755     |
| total timesteps      | 1712437  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0711   |
| n_updates            | 415609   |

EP=8,680 | STEP=1,714,001 | clip=52 | consec=6 | max\_consec=20 | avg\_clip50=39.49  
EP=8,690 | STEP=1,716,441 | clip=37 | consec=1 | max\_consec=20 | avg\_clip50=38.73  
EP=8,700 | STEP=1,718,623 | clip=42 | consec=1 | max\_consec=20 | avg\_clip50=36.45

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 36.9     |
| clipped_episode_r... | 8.5      |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 374      |
| exploration rate     | 0.427    |
| time/                |          |
| episodes             | 25800    |
| fps                  | 253      |
| time_elapsed         | 6790     |
| total timesteps      | 1720807  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0449   |
| n_updates            | 417701   |

EP=8,710 | STEP=1,721,199 | clip=15 | consec=0 | max\_consec=20 | avg\_clip50=35.66  
EP=8,720 | STEP=1,723,739 | clip=72 | consec=6 | max\_consec=20 | avg\_clip50=36.23  
EP=8,730 | STEP=1,726,309 | clip=60 | consec=2 | max\_consec=20 | avg\_clip50=36.06  
EP=8,740 | STEP=1,728,706 | clip=38 | consec=6 | max\_consec=20 | avg\_clip50=36.62

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 36       |
| clipped_episode_r... | 11       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.03e+03 |
| ep_rew_mean          | 374      |
| exploration rate     | 0.424    |
| time/                |          |
| episodes             | 25900    |
| fps                  | 253      |

|  |                 |  |         |  |
|--|-----------------|--|---------|--|
|  | time_elapsed    |  | 6825    |  |
|  | total timesteps |  | 1729179 |  |
|  | train/          |  |         |  |
|  | learning_rate   |  | 0.0001  |  |
|  | loss            |  | 0.0307  |  |
|  | n_updates       |  | 419794  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=8,750 |  | STEP=1,730,943 |  | clip=18 |  | consec=0 |  | max_consec=20 |  | avg_clip50=35.92 |
| EP=8,760 |  | STEP=1,733,216 |  | clip=78 |  | consec=3 |  | max_consec=20 |  | avg_clip50=35.31 |
| EP=8,770 |  | STEP=1,735,608 |  | clip=32 |  | consec=2 |  | max_consec=20 |  | avg_clip50=35.38 |

|  |                      |  |          |  |
|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 34.2     |  |
|  | clipped_episode_r... |  | 21.5     |  |
|  | consecutive_clipp... |  | 8        |  |
|  | max_consecutive_c... |  | 20       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.01e+03 |  |
|  | ep_rew_mean          |  | 357      |  |
|  | exploration rate     |  | 0.422    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 26000    |  |
|  | fps                  |  | 253      |  |
|  | time_elapsed         |  | 6858     |  |
|  | total timesteps      |  | 1737061  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.0705   |  |
|  | n_updates            |  | 421765   |  |

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|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=8,780 |  | STEP=1,738,139 |  | clip=28 |  | consec=12 |  | max_consec=20 |  | avg_clip50=36.08 |
| EP=8,790 |  | STEP=1,740,676 |  | clip=36 |  | consec=7  |  | max_consec=20 |  | avg_clip50=37.43 |
| EP=8,800 |  | STEP=1,743,214 |  | clip=26 |  | consec=7  |  | max_consec=20 |  | avg_clip50=38.26 |

|  |                      |  |          |  |
|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 39.3     |  |
|  | clipped_episode_r... |  | 41.5     |  |
|  | consecutive_clipp... |  | 4        |  |
|  | max_consecutive_c... |  | 20       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.02e+03 |  |
|  | ep_rew_mean          |  | 367      |  |
|  | exploration rate     |  | 0.419    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 26100    |  |
|  | fps                  |  | 253      |  |
|  | time_elapsed         |  | 6894     |  |
|  | total timesteps      |  | 1745681  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.155    |  |
|  | n_updates            |  | 423920   |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=8,810 |  | STEP=1,745,773 |  | clip=36 |  | consec=5 |  | max_consec=20 |  | avg_clip50=38.45 |
| EP=8,820 |  | STEP=1,748,333 |  | clip=58 |  | consec=1 |  | max_consec=20 |  | avg_clip50=38.87 |
| EP=8,830 |  | STEP=1,750,847 |  | clip=52 |  | consec=2 |  | max_consec=20 |  | avg_clip50=39.19 |
| EP=8,840 |  | STEP=1,753,233 |  | clip=12 |  | consec=0 |  | max_consec=20 |  | avg_clip50=37.79 |

|  |         |  |  |  |
|--|---------|--|--|--|
|  | custom/ |  |  |  |
|--|---------|--|--|--|



|                      |          |
|----------------------|----------|
| avg_clip50           | 39       |
| clipped_episode_r... | 50       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 386      |
| exploration rate     | 0.416    |
| time/                |          |
| episodes             | 26200    |
| fps                  | 253      |
| time_elapsed         | 6930     |
| total timesteps      | 1754229  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0829   |
| n_updates            | 426057   |

EP=8,850 | STEP=1,755,833 | clip=16 | consec=0 | max\_consec=20 | avg\_clip50=38.80  
EP=8,860 | STEP=1,758,356 | clip=43 | consec=1 | max\_consec=20 | avg\_clip50=39.59  
EP=8,870 | STEP=1,760,841 | clip=26 | consec=11 | max\_consec=20 | avg\_clip50=40.00

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.7     |
| clipped_episode_r... | 38       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 392      |
| exploration rate     | 0.413    |
| time/                |          |
| episodes             | 26300    |
| fps                  | 253      |
| time_elapsed         | 6964     |
| total timesteps      | 1762362  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.1      |
| n_updates            | 428090   |

EP=8,880 | STEP=1,763,093 | clip=42 | consec=6 | max\_consec=20 | avg\_clip50=39.08  
EP=8,890 | STEP=1,765,954 | clip=50 | consec=5 | max\_consec=20 | avg\_clip50=40.93  
EP=8,900 | STEP=1,768,177 | clip=55 | consec=2 | max\_consec=20 | avg\_clip50=38.80  
EP=8,910 | STEP=1,770,407 | clip=44 | consec=3 | max\_consec=20 | avg\_clip50=37.85

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.1     |
| clipped_episode_r... | 32       |
| consecutive_clipp... | 5        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 389      |
| exploration rate     | 0.41     |
| time/                |          |
| episodes             | 26400    |
| fps                  | 252      |
| time_elapsed         | 7000     |

|                 |         |
|-----------------|---------|
| total timesteps | 1770958 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.559   |
| n_updates       | 430239  |

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EP=8,920 | STEP=1,773,203 | clip=28 | consec=13 | max\_consec=20 | avg\_clip50=39.84

EP=8,930 | STEP=1,775,854 | clip=56 | consec=2 | max\_consec=20 | avg\_clip50=40.63

EP=8,940 | STEP=1,777,995 | clip=12 | consec=0 | max\_consec=20 | avg\_clip50=37.85

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39.3     |
| clipped_episode_r... | 54       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 392      |
| exploration rate     | 0.407    |
| time/                |          |
| episodes             | 26500    |
| fps                  | 252      |
| time_elapsed         | 7037     |
| total timesteps      | 1779867  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.275    |
| n_updates            | 432466   |

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EP=8,950 | STEP=1,780,637 | clip=42 | consec=4 | max\_consec=20 | avg\_clip50=39.78

EP=8,960 | STEP=1,782,870 | clip=9 | consec=0 | max\_consec=20 | avg\_clip50=39.05

EP=8,970 | STEP=1,785,629 | clip=4 | consec=0 | max\_consec=20 | avg\_clip50=37.13

EP=8,980 | STEP=1,788,488 | clip=46 | consec=10 | max\_consec=20 | avg\_clip50=38.55

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.3     |
| clipped_episode_r... | 18       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 392      |
| exploration rate     | 0.404    |
| time/                |          |
| episodes             | 26600    |
| fps                  | 252      |
| time_elapsed         | 7075     |
| total timesteps      | 1788877  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0194   |
| n_updates            | 434719   |

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EP=8,990 | STEP=1,790,873 | clip=17 | consec=0 | max\_consec=20 | avg\_clip50=39.25

EP=9,000 | STEP=1,793,679 | clip=38 | consec=10 | max\_consec=20 | avg\_clip50=39.98

EP=9,010 | STEP=1,795,973 | clip=38 | consec=8 | max\_consec=20 | avg\_clip50=40.81

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 41.3     |
| clipped_episode_r... | 49       |
| consecutive_clipp... | 14       |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 404      |
| exploration rate     | 0.401    |
| time/                |          |
| episodes             | 26700    |
| fps                  | 252      |
| time_elapsed         | 7112     |
| total timesteps      | 1797759  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.136    |
| n_updates            | 436939   |

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EP=9,020 | STEP=1,799,006 | clip=40 | consec=1 | max\_consec=20 | avg\_clip50=41.98  
EP=9,030 | STEP=1,801,148 | clip=4 | consec=0 | max\_consec=20 | avg\_clip50=38.97  
EP=9,040 | STEP=1,802,913 | clip=52 | consec=2 | max\_consec=20 | avg\_clip50=37.87  
EP=9,050 | STEP=1,805,606 | clip=34 | consec=12 | max\_consec=20 | avg\_clip50=38.3  
1

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.9     |
| clipped_episode_r... | 41.5     |
| consecutive_clipp... | 13       |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 393      |
| exploration rate     | 0.399    |
| time/                |          |
| episodes             | 26800    |
| fps                  | 252      |
| time_elapsed         | 7146     |
| total timesteps      | 1805884  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0669   |
| n_updates            | 438970   |

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EP=9,060 | STEP=1,807,864 | clip=32 | consec=3 | max\_consec=20 | avg\_clip50=38.70  
EP=9,070 | STEP=1,810,460 | clip=13 | consec=0 | max\_consec=20 | avg\_clip50=37.34  
EP=9,080 | STEP=1,812,520 | clip=16 | consec=0 | max\_consec=20 | avg\_clip50=36.57

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 36.6  |
| clipped_episode_r... | 19    |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 20    |
| rollout/             |       |
| ep_len_mean          | 996   |
| ep_rew_mean          | 370   |
| exploration rate     | 0.396 |
| time/                |       |
| episodes             | 26900 |
| fps                  | 252   |

|  |                 |         |  |
|--|-----------------|---------|--|
|  | time_elapsed    | 7178    |  |
|  | total timesteps | 1813457 |  |
|  | train/          |         |  |
|  | learning_rate   | 0.0001  |  |
|  | loss            | 0.0958  |  |
|  | n_updates       | 440864  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=9,090 |  | STEP=1,814,651 |  | clip=20 |  | consec=2 |  | max_consec=20 |  | avg_clip50=36.32 |
| EP=9,100 |  | STEP=1,817,024 |  | clip=48 |  | consec=4 |  | max_consec=20 |  | avg_clip50=35.31 |
| EP=9,110 |  | STEP=1,819,456 |  | clip=58 |  | consec=3 |  | max_consec=20 |  | avg_clip50=35.70 |

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 35.6    |  |
|  | clipped_episode_r... | 52      |  |
|  | consecutive_clipp... | 3       |  |
|  | max_consecutive_c... | 20      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 965     |  |
|  | ep_rew_mean          | 367     |  |
|  | exploration rate     | 0.393   |  |
|  | time/                |         |  |
|  | episodes             | 27000   |  |
|  | fps                  | 252     |  |
|  | time_elapsed         | 7213    |  |
|  | total timesteps      | 1821797 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.0523  |  |
|  | n_updates            | 442949  |  |

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|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=9,120 |  | STEP=1,821,916 |  | clip=24 |  | consec=4  |  | max_consec=20 |  | avg_clip50=35.78 |
| EP=9,130 |  | STEP=1,824,364 |  | clip=54 |  | consec=4  |  | max_consec=20 |  | avg_clip50=38.11 |
| EP=9,140 |  | STEP=1,827,124 |  | clip=48 |  | consec=14 |  | max_consec=20 |  | avg_clip50=42.34 |
| EP=9,150 |  | STEP=1,829,582 |  | clip=38 |  | consec=7  |  | max_consec=20 |  | avg_clip50=41.32 |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 43.8     |  |
|  | clipped_episode_r... | 47       |  |
|  | consecutive_clipp... | 10       |  |
|  | max_consecutive_c... | 20       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.03e+03 |  |
|  | ep_rew_mean          | 390      |  |
|  | exploration rate     | 0.39     |  |
|  | time/                |          |  |
|  | episodes             | 27100    |  |
|  | fps                  | 252      |  |
|  | time_elapsed         | 7252     |  |
|  | total timesteps      | 1831000  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0993   |  |
|  | n_updates            | 445249   |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=9,160 |  | STEP=1,833,300 |  | clip=4  |  | consec=0 |  | max_consec=20 |  | avg_clip50=45.31 |
| EP=9,170 |  | STEP=1,835,916 |  | clip=38 |  | consec=2 |  | max_consec=20 |  | avg_clip50=45.52 |
| EP=9,180 |  | STEP=1,838,568 |  | clip=28 |  | consec=9 |  | max_consec=20 |  | avg_clip50=45.87 |

|  |         |  |  |
|--|---------|--|--|
|  | custom/ |  |  |
|--|---------|--|--|

|                      |         |
|----------------------|---------|
| avg_clip50           | 45.4    |
| clipped_episode_r... | 26.5    |
| consecutive_clipp... | 15      |
| max_consecutive_c... | 20      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 429     |
| exploration rate     | 0.387   |
| time/                |         |
| episodes             | 27200   |
| fps                  | 252     |
| time_elapsed         | 7291    |
| total timesteps      | 1840404 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0535  |
| n_updates            | 447600  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=9,190 | STEP=1,841,092 | clip=27 | consec=2 | max_consec=20 | avg_clip50=43.77 |
| EP=9,200 | STEP=1,843,954 | clip=38 | consec=7 | max_consec=20 | avg_clip50=45.09 |
| EP=9,210 | STEP=1,846,854 | clip=54 | consec=4 | max_consec=20 | avg_clip50=42.37 |
| EP=9,220 | STEP=1,849,386 | clip=40 | consec=2 | max_consec=20 | avg_clip50=42.20 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 42.5     |
| clipped_episode_r... | 36.5     |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.14e+03 |
| ep_rew_mean          | 436      |
| exploration rate     | 0.384    |
| time/                |          |
| episodes             | 27300    |
| fps                  | 252      |
| time_elapsed         | 7330     |
| total timesteps      | 1849810  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0731   |
| n_updates            | 449952   |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=9,230 | STEP=1,852,038 | clip=41 | consec=7  | max_consec=20 | avg_clip50=43.89 |
| EP=9,240 | STEP=1,854,756 | clip=48 | consec=6  | max_consec=20 | avg_clip50=43.90 |
| EP=9,250 | STEP=1,857,329 | clip=44 | consec=16 | max_consec=20 | avg_clip50=43.30 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 44.1     |
| clipped_episode_r... | 30.5     |
| consecutive_clipp... | 5        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 429      |
| exploration rate     | 0.381    |
| time/                |          |
| episodes             | 27400    |
| fps                  | 252      |
| time_elapsed         | 7369     |

|                 |         |
|-----------------|---------|
| total timesteps | 1859014 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.0559  |
| n_updates       | 452253  |

EP=9,260 | STEP=1,859,893 | clip=64 | consec=8 | max\_consec=20 | avg\_clip50=43.79  
EP=9,270 | STEP=1,862,967 | clip=52 | consec=9 | max\_consec=20 | avg\_clip50=46.09  
EP=9,280 | STEP=1,866,074 | clip=34 | consec=19 | max\_consec=20 | avg\_clip50=46.21

EP=9,290 | STEP=1,868,644 | clip=14 | consec=0 | max\_consec=20 | avg\_clip50=46.10

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 46.5     |
| clipped_episode_r... | 46       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.14e+03 |
| ep_rew_mean          | 452      |
| exploration rate     | 0.378    |
| time/                |          |
| episodes             | 27500    |
| fps                  | 252      |
| time_elapsed         | 7410     |
| total timesteps      | 1868993  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.068    |
| n_updates            | 454748   |

EP=9,300 | STEP=1,871,285 | clip=24 | consec=10 | max\_consec=20 | avg\_clip50=47.17

EP=9,310 | STEP=1,874,135 | clip=58 | consec=6 | max\_consec=20 | avg\_clip50=46.93

EP=9,320 | STEP=1,877,172 | clip=46 | consec=16 | max\_consec=20 | avg\_clip50=47.10

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 45.2     |
| clipped_episode_r... | 32.5     |
| consecutive_clipp... | 4        |
| max_consecutive_c... | 20       |
| rollout/             |          |
| ep_len_mean          | 1.16e+03 |
| ep_rew_mean          | 468      |
| exploration rate     | 0.374    |
| time/                |          |
| episodes             | 27600    |
| fps                  | 252      |
| time_elapsed         | 7450     |
| total timesteps      | 1878653  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0549   |
| n_updates            | 457163   |

EP=9,330 | STEP=1,879,564 | clip=6 | consec=0 | max\_consec=20 | avg\_clip50=43.89

EP=9,340 | STEP=1,882,296 | clip=49 | consec=4 | max\_consec=20 | avg\_clip50=45.16

EP=9,350 | STEP=1,885,260 | clip=30 | consec=14 | max\_consec=20 | avg\_clip50=46.56

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 47.8     |
| clipped_episode_r... | 25.5     |
| consecutive_clipp... | 23       |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.18e+03 |
| ep_rew_mean          | 474      |
| exploration rate     | 0.371    |
| time/                |          |
| episodes             | 27700    |
| fps                  | 252      |
| time_elapsed         | 7490     |
| total timesteps      | 1888189  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.556    |
| n_updates            | 459547   |

EP=9,360 | STEP=1,888,202 | clip=49 | consec=24 | max\_consec=24 | avg\_clip50=47.6  
1

EP=9,370 | STEP=1,891,050 | clip=46 | consec=4 | max\_consec=29 | avg\_clip50=46.47

EP=9,380 | STEP=1,894,027 | clip=20 | consec=0 | max\_consec=29 | avg\_clip50=48.87

EP=9,390 | STEP=1,896,330 | clip=21 | consec=1 | max\_consec=29 | avg\_clip50=47.47

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 47       |
| clipped_episode_r... | 60       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 29       |
| rollout/             |          |
| ep_len_mean          | 1.15e+03 |
| ep_rew_mean          | 462      |
| exploration rate     | 0.368    |
| time/                |          |
| episodes             | 27800    |
| fps                  | 252      |
| time_elapsed         | 7527     |
| total timesteps      | 1897104  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.455    |
| n_updates            | 461775   |

EP=9,400 | STEP=1,898,853 | clip=34 | consec=6 | max\_consec=29 | avg\_clip50=44.95

EP=9,410 | STEP=1,901,488 | clip=80 | consec=16 | max\_consec=29 | avg\_clip50=44.6  
6

EP=9,420 | STEP=1,903,800 | clip=18 | consec=0 | max\_consec=29 | avg\_clip50=41.54

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 40.2     |
| clipped_episode_r... | 2        |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 29       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 441      |
| exploration rate     | 0.365    |
| time/                |          |

|                      |                                                                 |
|----------------------|-----------------------------------------------------------------|
| episodes             | 27900                                                           |
| fps                  | 251                                                             |
| time_elapsed         | 7563                                                            |
| total timesteps      | 1905675                                                         |
| train/               |                                                                 |
| learning_rate        | 0.0001                                                          |
| loss                 | 0.0465                                                          |
| n_updates            | 463918                                                          |
| -----                |                                                                 |
| EP=9,430             | STEP=1,906,323 clip=64 consec=2 max_consec=29 avg_clip50=40.25  |
| EP=9,440             | STEP=1,909,118 clip=29 consec=3 max_consec=29 avg_clip50=41.56  |
| EP=9,450             | STEP=1,911,847 clip=94 consec=1 max_consec=29 avg_clip50=42.06  |
| EP=9,460             | STEP=1,914,423 clip=28 consec=8 max_consec=29 avg_clip50=41.53  |
| -----                |                                                                 |
| custom/              |                                                                 |
| avg_clip50           | 42.1                                                            |
| clipped_episode_r... | 55                                                              |
| consecutive_clipp... | 9                                                               |
| max_consecutive_c... | 29                                                              |
| rollout/             |                                                                 |
| ep_len_mean          | 1.1e+03                                                         |
| ep_rew_mean          | 430                                                             |
| exploration rate     | 0.362                                                           |
| time/                |                                                                 |
| episodes             | 28000                                                           |
| fps                  | 251                                                             |
| time_elapsed         | 7602                                                            |
| total timesteps      | 1914870                                                         |
| train/               |                                                                 |
| learning_rate        | 0.0001                                                          |
| loss                 | 0.0543                                                          |
| n_updates            | 466217                                                          |
| -----                |                                                                 |
| EP=9,470             | STEP=1,917,532 clip=92 consec=18 max_consec=29 avg_clip50=46.00 |
| EP=9,480             | STEP=1,919,978 clip=27 consec=3 max_consec=29 avg_clip50=45.08  |
| EP=9,490             | STEP=1,923,098 clip=74 consec=13 max_consec=29 avg_clip50=47.86 |
| -----                |                                                                 |
| custom/              |                                                                 |
| avg_clip50           | 48.8                                                            |
| clipped_episode_r... | 47.5                                                            |
| consecutive_clipp... | 20                                                              |
| max_consecutive_c... | 29                                                              |
| rollout/             |                                                                 |
| ep_len_mean          | 1.12e+03                                                        |
| ep_rew_mean          | 451                                                             |
| exploration rate     | 0.359                                                           |
| time/                |                                                                 |
| episodes             | 28100                                                           |
| fps                  | 251                                                             |
| time_elapsed         | 7644                                                            |
| total timesteps      | 1925042                                                         |
| train/               |                                                                 |
| learning_rate        | 0.0001                                                          |
| loss                 | 0.474                                                           |
| n_updates            | 468760                                                          |
| -----                |                                                                 |
| EP=9,500             | STEP=1,925,927 clip=60 consec=23 max_consec=29 avg_clip50=48.58 |



EP=9,510 | STEP=1,928,519 | clip=58 | consec=33 | max\_consec=33 | avg\_clip50=47.24

EP=9,520 | STEP=1,930,852 | clip=80 | consec=1 | max\_consec=40 | avg\_clip50=43.98

EP=9,530 | STEP=1,933,336 | clip=48 | consec=3 | max\_consec=40 | avg\_clip50=44.88

|                      |          |
|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 44.9     |
| clipped_episode_r... | 48       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 40       |
| rollout/             |          |
| ep_len_mean          | 1.12e+03 |
| ep_rew_mean          | 450      |
| exploration rate     | 0.356    |
| time/                |          |
| episodes             | 28200    |
| fps                  | 251      |
| time_elapsed         | 7680     |
| total timesteps      | 1933525  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.043    |
| n_updates            | 470881   |

EP=9,540 | STEP=1,935,787 | clip=47 | consec=2 | max\_consec=40 | avg\_clip50=41.64

EP=9,550 | STEP=1,938,535 | clip=48 | consec=12 | max\_consec=40 | avg\_clip50=41.01

EP=9,560 | STEP=1,941,153 | clip=54 | consec=4 | max\_consec=40 | avg\_clip50=41.34

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| -----                |         |
| custom/              |         |
| avg_clip50           | 41.5    |
| clipped_episode_r... | 38      |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 40      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 438     |
| exploration rate     | 0.353   |
| time/                |         |
| episodes             | 28300   |
| fps                  | 251     |
| time_elapsed         | 7719    |
| total timesteps      | 1942763 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0652  |
| n_updates            | 473190  |

EP=9,570 | STEP=1,943,917 | clip=47 | consec=1 | max\_consec=40 | avg\_clip50=42.35

EP=9,580 | STEP=1,946,526 | clip=60 | consec=11 | max\_consec=40 | avg\_clip50=42.52

EP=9,590 | STEP=1,949,202 | clip=49 | consec=9 | max\_consec=40 | avg\_clip50=42.91

EP=9,600 | STEP=1,951,472 | clip=18 | consec=0 | max\_consec=40 | avg\_clip50=41.26

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| -----                |      |
| custom/              |      |
| avg_clip50           | 41.3 |
| clipped_episode_r... | 18   |
| consecutive_clipp... | 0    |
| max_consecutive_c... | 40   |
| rollout/             |      |

|                  |          |
|------------------|----------|
| ep_len_mean      | 1.07e+03 |
| ep_rew_mean      | 411      |
| exploration rate | 0.35     |
| time/            |          |
| episodes         | 28400    |
| fps              | 251      |
| time_elapsed     | 7757     |
| total timesteps  | 1951653  |
| train/           |          |
| learning_rate    | 0.0001   |
| loss             | 0.0761   |
| n_updates        | 475413   |

EP=9,610 | STEP=1,953,995 | clip=58 | consec=10 | max\_consec=40 | avg\_clip50=40.97

EP=9,620 | STEP=1,956,182 | clip=12 | consec=0 | max\_consec=40 | avg\_clip50=39.55

EP=9,630 | STEP=1,958,840 | clip=64 | consec=10 | max\_consec=40 | avg\_clip50=40.43

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|----------------------|----------|
| custom/              |          |
| avg_clip50           | 40.3     |
| clipped_episode_r... | 24       |
| consecutive_clipp... | 14       |
| max_consecutive_c... | 40       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 414      |
| exploration rate     | 0.347    |
| time/                |          |
| episodes             | 28500    |
| fps                  | 251      |
| time_elapsed         | 7792     |
| total timesteps      | 1960127  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.043    |
| n_updates            | 477531   |

EP=9,640 | STEP=1,961,787 | clip=60 | consec=20 | max\_consec=40 | avg\_clip50=41.86

EP=9,650 | STEP=1,964,527 | clip=22 | consec=30 | max\_consec=40 | avg\_clip50=43.49

EP=9,660 | STEP=1,967,312 | clip=50 | consec=2 | max\_consec=40 | avg\_clip50=43.65

EP=9,670 | STEP=1,970,017 | clip=75 | consec=5 | max\_consec=40 | avg\_clip50=47.12

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| custom/              |          |
| avg_clip50           | 47.7     |
| clipped_episode_r... | 54.5     |
| consecutive_clipp... | 6        |
| max_consecutive_c... | 40       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 433      |
| exploration rate     | 0.344    |
| time/                |          |
| episodes             | 28600    |
| fps                  | 251      |
| time_elapsed         | 7836     |
| total timesteps      | 1970573  |
| train/               |          |

|               |        |
|---------------|--------|
| learning_rate | 0.0001 |
| loss          | 0.464  |
| n_updates     | 480143 |

EP=9,680 | STEP=1,972,810 | clip=54 | consec=3 | max\_consec=40 | avg\_clip50=47.79  
EP=9,690 | STEP=1,975,480 | clip=13 | consec=0 | max\_consec=40 | avg\_clip50=45.65  
EP=9,700 | STEP=1,978,199 | clip=58 | consec=4 | max\_consec=40 | avg\_clip50=45.09

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| custom/              |          |
| avg_clip50           | 46.5     |
| clipped_episode_r... | 30       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 40       |
| rollout/             |          |
| ep_len_mean          | 1.12e+03 |
| ep_rew_mean          | 452      |
| exploration rate     | 0.341    |
| time/                |          |
| episodes             | 28700    |
| fps                  | 251      |
| time_elapsed         | 7876     |
| total timesteps      | 1980169  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.044    |
| n_updates            | 482542   |

EP=9,710 | STEP=1,980,921 | clip=52 | consec=4 | max\_consec=40 | avg\_clip50=46.39  
EP=9,720 | STEP=1,983,081 | clip=22 | consec=3 | max\_consec=40 | avg\_clip50=43.43  
EP=9,730 | STEP=1,985,895 | clip=72 | consec=13 | max\_consec=40 | avg\_clip50=42.63  
EP=9,740 | STEP=1,988,280 | clip=48 | consec=23 | max\_consec=40 | avg\_clip50=42.44

|                      |         |
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| custom/              |         |
| avg_clip50           | 42      |
| clipped_episode_r... | 23.5    |
| consecutive_clipp... | 24      |
| max_consecutive_c... | 40      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 439     |
| exploration rate     | 0.338   |
| time/                |         |
| episodes             | 28800   |
| fps                  | 251     |
| time_elapsed         | 7912    |
| total timesteps      | 1988590 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0295  |
| n_updates            | 484647  |

EP=9,750 | STEP=1,990,867 | clip=30 | consec=6 | max\_consec=40 | avg\_clip50=42.89  
EP=9,760 | STEP=1,993,214 | clip=46 | consec=2 | max\_consec=40 | avg\_clip50=40.49  
EP=9,770 | STEP=1,995,903 | clip=66 | consec=2 | max\_consec=40 | avg\_clip50=42.99

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| custom/              |      |
| avg_clip50           | 42.3 |
| clipped_episode_r... | 24.5 |

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| consecutive_clipp... | 10       |
| max_consecutive_c... | 40       |
| rollout/             |          |
| ep_len_mean          | 1.08e+03 |
| ep_rew_mean          | 424      |
| exploration rate     | 0.335    |
| time/                |          |
| episodes             | 28900    |
| fps                  | 251      |
| time_elapsed         | 7951     |
| total timesteps      | 1997909  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0408   |
| n_updates            | 486977   |

EP=9,780 | STEP=1,998,652 | clip=56 | consec=12 | max\_consec=40 | avg\_clip50=41.92

EP=9,790 | STEP=2,001,186 | clip=21 | consec=1 | max\_consec=40 | avg\_clip50=42.24

EP=9,800 | STEP=2,004,573 | clip=46 | consec=11 | max\_consec=40 | avg\_clip50=44.07

EP=9,810 | STEP=2,007,216 | clip=54 | consec=6 | max\_consec=40 | avg\_clip50=45.15

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| custom/              |         |
| avg_clip50           | 45.1    |
| clipped_episode_r... | 51.5    |
| consecutive_clipp... | 8       |
| max_consecutive_c... | 40      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 434     |
| exploration rate     | 0.331   |
| time/                |         |
| episodes             | 29000   |
| fps                  | 251     |
| time_elapsed         | 7994    |
| total timesteps      | 2008034 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0493  |
| n_updates            | 489508  |

EP=9,820 | STEP=2,010,015 | clip=58 | consec=1 | max\_consec=40 | avg\_clip50=44.86

EP=9,830 | STEP=2,012,465 | clip=55 | consec=11 | max\_consec=40 | avg\_clip50=45.47

EP=9,840 | STEP=2,015,511 | clip=77 | consec=21 | max\_consec=40 | avg\_clip50=47.19

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| custom/              |          |
| avg_clip50           | 46       |
| clipped_episode_r... | 20       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 40       |
| rollout/             |          |
| ep_len_mean          | 1.15e+03 |
| ep_rew_mean          | 450      |
| exploration rate     | 0.328    |
| time/                |          |
| episodes             | 29100    |
| fps                  | 251      |

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|--|-----------------|--|---------|--|
|  | time_elapsed    |  | 8034    |  |
|  | total timesteps |  | 2017819 |  |
|  | train/          |  |         |  |
|  | learning_rate   |  | 0.0001  |  |
|  | loss            |  | 0.0402  |  |
|  | n_updates       |  | 491954  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=9,850 |  | STEP=2,018,378 |  | clip=46 |  | consec=2 |  | max_consec=40 |  | avg_clip50=46.36 |
| EP=9,860 |  | STEP=2,020,816 |  | clip=50 |  | consec=5 |  | max_consec=40 |  | avg_clip50=46.59 |
| EP=9,870 |  | STEP=2,023,380 |  | clip=60 |  | consec=4 |  | max_consec=40 |  | avg_clip50=44.50 |
| EP=9,880 |  | STEP=2,026,479 |  | clip=58 |  | consec=3 |  | max_consec=40 |  | avg_clip50=45.83 |

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|  | custom/              |  |          |  |
|  | avg_clip50           |  | 45       |  |
|  | clipped_episode_r... |  | 28.5     |  |
|  | consecutive_clipp... |  | 5        |  |
|  | max_consecutive_c... |  | 40       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.15e+03 |  |
|  | ep_rew_mean          |  | 453      |  |
|  | exploration rate     |  | 0.325    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 29200    |  |
|  | fps                  |  | 251      |  |
|  | time_elapsed         |  | 8073     |  |
|  | total timesteps      |  | 2027046  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.0441   |  |
|  | n_updates            |  | 494261   |  |

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|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=9,890 |  | STEP=2,029,432 |  | clip=24 |  | consec=13 |  | max_consec=40 |  | avg_clip50=47.94 |
| EP=9,900 |  | STEP=2,032,429 |  | clip=48 |  | consec=23 |  | max_consec=40 |  | avg_clip50=49.10 |
| EP=9,910 |  | STEP=2,034,598 |  | clip=52 |  | consec=3  |  | max_consec=40 |  | avg_clip50=47.57 |

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|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 51.1     |  |
|  | clipped_episode_r... |  | 73       |  |
|  | consecutive_clipp... |  | 12       |  |
|  | max_consecutive_c... |  | 40       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.15e+03 |  |
|  | ep_rew_mean          |  | 478      |  |
|  | exploration rate     |  | 0.321    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 29300    |  |
|  | fps                  |  | 251      |  |
|  | time_elapsed         |  | 8118     |  |
|  | total timesteps      |  | 2037721  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.0478   |  |
|  | n_updates            |  | 496930   |  |

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|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=9,920 |  | STEP=2,037,765 |  | clip=53 |  | consec=13 |  | max_consec=40 |  | avg_clip50=50.97 |
| EP=9,930 |  | STEP=2,040,632 |  | clip=48 |  | consec=23 |  | max_consec=40 |  | avg_clip50=49.89 |

EP=9,940 | STEP=2,043,409 | clip=81 | consec=33 | max\_consec=40 | avg\_clip50=46.6  
9

EP=9,950 | STEP=2,045,841 | clip=8 | consec=0 | max\_consec=40 | avg\_clip50=43.08

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|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 43.7     |  |
| clipped_episode_r... | 19       |  |
| consecutive_clipp... | 0        |  |
| max_consecutive_c... | 40       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.13e+03 |  |
| ep_rew_mean          | 452      |  |
| exploration rate     | 0.318    |  |
| time/                |          |  |
| episodes             | 29400    |  |
| fps                  | 250      |  |
| time_elapsed         | 8156     |  |
| total timesteps      | 2046832  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.451    |  |
| n_updates            | 499207   |  |

EP=9,960 | STEP=2,048,396 | clip=31 | consec=7 | max\_consec=40 | avg\_clip50=44.53

EP=9,970 | STEP=2,050,628 | clip=42 | consec=6 | max\_consec=40 | avg\_clip50=40.87

EP=9,980 | STEP=2,053,717 | clip=45 | consec=16 | max\_consec=40 | avg\_clip50=43.0  
3

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 43.8     |  |
| clipped_episode_r... | 26       |  |
| consecutive_clipp... | 3        |  |
| max_consecutive_c... | 40       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.13e+03 |  |
| ep_rew_mean          | 454      |  |
| exploration rate     | 0.315    |  |
| time/                |          |  |
| episodes             | 29500    |  |
| fps                  | 250      |  |
| time_elapsed         | 8194     |  |
| total timesteps      | 2055913  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0347   |  |
| n_updates            | 501478   |  |

EP=9,990 | STEP=2,056,544 | clip=54 | consec=6 | max\_consec=40 | avg\_clip50=43.61

EP=10,000 | STEP=2,059,461 | clip=58 | consec=16 | max\_consec=40 | avg\_clip50=48.  
05

EP=10,010 | STEP=2,062,012 | clip=21 | consec=7 | max\_consec=40 | avg\_clip50=47.9  
2

EP=10,020 | STEP=2,064,815 | clip=48 | consec=17 | max\_consec=40 | avg\_clip50=50.  
75

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|----------------------|------|--|
| -----                |      |  |
| custom/              |      |  |
| avg_clip50           | 50.3 |  |
| clipped_episode_r... | 10   |  |
| consecutive_clipp... | 0    |  |
| max_consecutive_c... | 40   |  |

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|------------------|----------|--|
| rollout/         |          |  |
| ep_len_mean      | 1.11e+03 |  |
| ep_rew_mean      | 454      |  |
| exploration rate | 0.312    |  |
| time/            |          |  |
| episodes         | 29600    |  |
| fps              | 250      |  |
| time_elapsed     | 8233     |  |
| total timesteps  | 2065095  |  |
| train/           |          |  |
| learning_rate    | 0.0001   |  |
| loss             | 0.0976   |  |
| n_updates        | 503773   |  |

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EP=10,030 | STEP=2,067,094 | clip=68 | consec=6 | max\_consec=40 | avg\_clip50=47.43

EP=10,040 | STEP=2,069,240 | clip=47 | consec=2 | max\_consec=40 | avg\_clip50=44.95

EP=10,050 | STEP=2,071,556 | clip=73 | consec=1 | max\_consec=40 | avg\_clip50=42.18

|                      |          |  |
|----------------------|----------|--|
| custom/              |          |  |
| avg_clip50           | 42.4     |  |
| clipped_episode_r... | 31.5     |  |
| consecutive_clipp... | 2        |  |
| max_consecutive_c... | 40       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.08e+03 |  |
| ep_rew_mean          | 452      |  |
| exploration rate     | 0.309    |  |
| time/                |          |  |
| episodes             | 29700    |  |
| fps                  | 250      |  |
| time_elapsed         | 8269     |  |
| total timesteps      | 2073588  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.516    |  |
| n_updates            | 505896   |  |

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EP=10,060 | STEP=2,074,376 | clip=34 | consec=5 | max\_consec=40 | avg\_clip50=43.75

EP=10,070 | STEP=2,077,255 | clip=14 | consec=0 | max\_consec=40 | avg\_clip50=42.94

EP=10,080 | STEP=2,080,283 | clip=70 | consec=10 | max\_consec=40 | avg\_clip50=45.71

EP=10,090 | STEP=2,083,030 | clip=54 | consec=20 | max\_consec=40 | avg\_clip50=46.49

|                      |          |  |
|----------------------|----------|--|
| custom/              |          |  |
| avg_clip50           | 46.9     |  |
| clipped_episode_r... | 46.5     |  |
| consecutive_clipp... | 21       |  |
| max_consecutive_c... | 40       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.11e+03 |  |
| ep_rew_mean          | 454      |  |
| exploration rate     | 0.306    |  |
| time/                |          |  |
| episodes             | 29800    |  |

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|--|-----------------|--|---------|--|
|  | fps             |  | 250     |  |
|  | time_elapsed    |  | 8310    |  |
|  | total timesteps |  | 2083388 |  |
|  | train/          |  |         |  |
|  | learning_rate   |  | 0.0001  |  |
|  | loss            |  | 0.519   |  |
|  | n_updates       |  | 508346  |  |

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EP=10,100 | STEP=2,085,746 | clip=12 | consec=0 | max\_consec=40 | avg\_clip50=45.96

EP=10,110 | STEP=2,089,226 | clip=64 | consec=10 | max\_consec=40 | avg\_clip50=48.37

EP=10,120 | STEP=2,092,087 | clip=26 | consec=1 | max\_consec=40 | avg\_clip50=47.81

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|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 46.7     |  |
|  | clipped_episode_r... |  | 60       |  |
|  | consecutive_clipp... |  | 1        |  |
|  | max_consecutive_c... |  | 40       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.15e+03 |  |
|  | ep_rew_mean          |  | 458      |  |
|  | exploration rate     |  | 0.303    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 29900    |  |
|  | fps                  |  | 250      |  |
|  | time_elapsed         |  | 8352     |  |
|  | total timesteps      |  | 2093347  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.0906   |  |
|  | n_updates            |  | 510836   |  |

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EP=10,130 | STEP=2,094,918 | clip=74 | consec=6 | max\_consec=40 | avg\_clip50=47.48

EP=10,140 | STEP=2,097,798 | clip=41 | consec=16 | max\_consec=40 | avg\_clip50=49.56

EP=10,150 | STEP=2,100,435 | clip=24 | consec=4 | max\_consec=40 | avg\_clip50=49.15

|  |                      |  |         |  |
|--|----------------------|--|---------|--|
|  | custom/              |  |         |  |
|  | avg_clip50           |  | 48.6    |  |
|  | clipped_episode_r... |  | 39.5    |  |
|  | consecutive_clipp... |  | 6       |  |
|  | max_consecutive_c... |  | 40      |  |
|  | rollout/             |  |         |  |
|  | ep_len_mean          |  | 1.2e+03 |  |
|  | ep_rew_mean          |  | 482     |  |
|  | exploration rate     |  | 0.3     |  |
|  | time/                |  |         |  |
|  | episodes             |  | 30000   |  |
|  | fps                  |  | 250     |  |
|  | time_elapsed         |  | 8393    |  |
|  | total timesteps      |  | 2103084 |  |
|  | train/               |  |         |  |
|  | learning_rate        |  | 0.0001  |  |
|  | loss                 |  | 0.663   |  |
|  | n_updates            |  | 513270  |  |



EP=10,160 | STEP=2,103,168 | clip=4 | consec=0 | max\_consec=40 | avg\_clip50=47.37  
 EP=10,170 | STEP=2,105,801 | clip=52 | consec=10 | max\_consec=40 | avg\_clip50=48.02  
 EP=10,180 | STEP=2,108,686 | clip=36 | consec=20 | max\_consec=40 | avg\_clip50=48.07  
 EP=10,190 | STEP=2,111,718 | clip=54 | consec=30 | max\_consec=40 | avg\_clip50=50.31

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| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 51.3    |  |
| clipped_episode_r... | 44.5    |  |
| consecutive_clipp... | 34      |  |
| max_consecutive_c... | 40      |  |
| rollout/             |         |  |
| ep_len_mean          | 1.2e+03 |  |
| ep_rew_mean          | 504     |  |
| exploration rate     | 0.296   |  |
| time/                |         |  |
| episodes             | 30100   |  |
| fps                  | 250     |  |
| time_elapsed         | 8435    |  |
| total timesteps      | 2112945 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.133   |  |
| n_updates            | 515736  |  |

EP=10,200 | STEP=2,114,596 | clip=34 | consec=40 | max\_consec=40 | avg\_clip50=52.51  
 EP=10,210 | STEP=2,117,566 | clip=36 | consec=9 | max\_consec=40 | avg\_clip50=52.04  
 EP=10,220 | STEP=2,120,344 | clip=58 | consec=19 | max\_consec=40 | avg\_clip50=52.31

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 50.1     |  |
| clipped_episode_r... | 55.5     |  |
| consecutive_clipp... | 6        |  |
| max_consecutive_c... | 40       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.17e+03 |  |
| ep_rew_mean          | 494      |  |
| exploration rate     | 0.293    |  |
| time/                |          |  |
| episodes             | 30200    |  |
| fps                  | 250      |  |
| time_elapsed         | 8475     |  |
| total timesteps      | 2122528  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0564   |  |
| n_updates            | 518131   |  |

EP=10,230 | STEP=2,123,061 | clip=54 | consec=8 | max\_consec=40 | avg\_clip50=50.32  
 EP=10,240 | STEP=2,125,678 | clip=38 | consec=4 | max\_consec=40 | avg\_clip50=49.15  
 EP=10,250 | STEP=2,128,342 | clip=58 | consec=6 | max\_consec=40 | avg\_clip50=47.32  
 EP=10,260 | STEP=2,131,483 | clip=52 | consec=16 | max\_consec=40 | avg\_clip50=47.

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 48.9     |  |
| clipped_episode_r... | 86       |  |
| consecutive_clipp... | 18       |  |
| max_consecutive_c... | 40       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.18e+03 |  |
| ep_rew_mean          | 500      |  |
| exploration rate     | 0.29     |  |
| time/                |          |  |
| episodes             | 30300    |  |
| fps                  | 250      |  |
| time_elapsed         | 8517     |  |
| total timesteps      | 2132454  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0847   |  |
| n_updates            | 520613   |  |

EP=10,270 | STEP=2,134,266 | clip=77 | consec=1 | max\_consec=40 | avg\_clip50=47.5  
4  
EP=10,280 | STEP=2,137,192 | clip=28 | consec=8 | max\_consec=40 | avg\_clip50=49.6  
3  
EP=10,290 | STEP=2,140,332 | clip=43 | consec=18 | max\_consec=40 | avg\_clip50=48.  
78

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|----------------------|---------|--|
| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 50.7    |  |
| clipped_episode_r... | 77.5    |  |
| consecutive_clipp... | 25      |  |
| max_consecutive_c... | 40      |  |
| rollout/             |         |  |
| ep_len_mean          | 1.2e+03 |  |
| ep_rew_mean          | 491     |  |
| exploration rate     | 0.286   |  |
| time/                |         |  |
| episodes             | 30400   |  |
| fps                  | 250     |  |
| time_elapsed         | 8561    |  |
| total timesteps      | 2142870 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.0642  |  |
| n_updates            | 523217  |  |

EP=10,300 | STEP=2,143,467 | clip=74 | consec=28 | max\_consec=40 | avg\_clip50=51.  
86  
EP=10,310 | STEP=2,146,267 | clip=72 | consec=1 | max\_consec=40 | avg\_clip50=52.7  
2  
EP=10,320 | STEP=2,149,472 | clip=56 | consec=11 | max\_consec=40 | avg\_clip50=53.  
26  
EP=10,330 | STEP=2,152,584 | clip=46 | consec=21 | max\_consec=40 | avg\_clip50=53.  
86

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| -----                |      |  |
| custom/              |      |  |
| avg_clip50           | 54.5 |  |
| clipped_episode_r... | 68.5 |  |
| consecutive_clipp... | 23   |  |

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|--|----------------------|----------|--|
|  | max_consecutive_c... | 40       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.22e+03 |  |
|  | ep_rew_mean          | 513      |  |
|  | exploration rate     | 0.283    |  |
|  | time/                |          |  |
|  | episodes             | 30500    |  |
|  | fps                  | 250      |  |
|  | time_elapsed         | 8605     |  |
|  | total timesteps      | 2153271  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.085    |  |
|  | n_updates            | 525817   |  |

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EP=10,340 | STEP=2,155,779 | clip=68 | consec=31 | max\_consec=40 | avg\_clip50=54.43

EP=10,350 | STEP=2,158,518 | clip=12 | consec=0 | max\_consec=40 | avg\_clip50=52.99

EP=10,360 | STEP=2,161,570 | clip=23 | consec=5 | max\_consec=40 | avg\_clip50=51.22

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|  | custom/              |          |  |
|  | avg_clip50           | 50.1     |  |
|  | clipped_episode_r... | 33.5     |  |
|  | consecutive_clipp... | 4        |  |
|  | max_consecutive_c... | 40       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.24e+03 |  |
|  | ep_rew_mean          | 512      |  |
|  | exploration rate     | 0.28     |  |
|  | time/                |          |  |
|  | episodes             | 30600    |  |
|  | fps                  | 250      |  |
|  | time_elapsed         | 8648     |  |
|  | total timesteps      | 2163244  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.266    |  |
|  | n_updates            | 528310   |  |

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EP=10,370 | STEP=2,164,164 | clip=7 | consec=0 | max\_consec=40 | avg\_clip50=48.85

EP=10,380 | STEP=2,167,101 | clip=54 | consec=6 | max\_consec=40 | avg\_clip50=47.31

EP=10,390 | STEP=2,169,897 | clip=48 | consec=16 | max\_consec=40 | avg\_clip50=46.17

EP=10,400 | STEP=2,172,718 | clip=54 | consec=26 | max\_consec=40 | avg\_clip50=46.55

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|  | custom/              |          |  |
|  | avg_clip50           | 46.5     |  |
|  | clipped_episode_r... | 53.5     |  |
|  | consecutive_clipp... | 26       |  |
|  | max_consecutive_c... | 40       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.22e+03 |  |
|  | ep_rew_mean          | 498      |  |
|  | exploration rate     | 0.276    |  |
|  | time/                |          |  |
|  | episodes             | 30700    |  |

|                 |         |
|-----------------|---------|
| fps             | 250     |
| time_elapsed    | 8689    |
| total timesteps | 2173059 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.0687  |
| n_updates       | 530764  |

EP=10,410 | STEP=2,175,889 | clip=42 | consec=36 | max\_consec=40 | avg\_clip50=47.87

EP=10,420 | STEP=2,178,346 | clip=58 | consec=3 | max\_consec=42 | avg\_clip50=49.08

EP=10,430 | STEP=2,181,726 | clip=60 | consec=13 | max\_consec=42 | avg\_clip50=50.62

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 50       |
| clipped_episode_r... | 45.5     |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 42       |
| rollout/             |          |
| ep_len_mean          | 1.21e+03 |
| ep_rew_mean          | 490      |
| exploration rate     | 0.273    |
| time/                |          |
| episodes             | 30800    |
| fps                  | 250      |
| time_elapsed         | 8732     |
| total timesteps      | 2183184  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.516    |
| n_updates            | 533295   |

EP=10,440 | STEP=2,184,897 | clip=60 | consec=7 | max\_consec=42 | avg\_clip50=51.76

EP=10,450 | STEP=2,187,596 | clip=44 | consec=2 | max\_consec=42 | avg\_clip50=51.44

EP=10,460 | STEP=2,190,289 | clip=24 | consec=12 | max\_consec=42 | avg\_clip50=49.30

EP=10,470 | STEP=2,193,241 | clip=72 | consec=22 | max\_consec=42 | avg\_clip50=50.62

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 50.6     |
| clipped_episode_r... | 46.5     |
| consecutive_clipp... | 23       |
| max_consecutive_c... | 42       |
| rollout/             |          |
| ep_len_mean          | 1.21e+03 |
| ep_rew_mean          | 501      |
| exploration rate     | 0.27     |
| time/                |          |
| episodes             | 30900    |
| fps                  | 249      |
| time_elapsed         | 8776     |
| total timesteps      | 2193510  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.358    |

|           |        |
|-----------|--------|
| n_updates | 535877 |
|-----------|--------|

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|           |                |         |           |               |                  |
|-----------|----------------|---------|-----------|---------------|------------------|
| EP=10,480 | STEP=2,196,359 | clip=54 | consec=2  | max_consec=42 | avg_clip50=49.98 |
| EP=10,490 | STEP=2,199,283 | clip=14 | consec=0  | max_consec=42 | avg_clip50=50.83 |
| EP=10,500 | STEP=2,201,977 | clip=31 | consec=10 | max_consec=42 | avg_clip50=49.31 |

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| custom/              |          |
| avg_clip50           | 49.2     |
| clipped_episode_r... | 60       |
| consecutive_clipp... | 16       |
| max_consecutive_c... | 42       |
| rollout/             |          |
| ep_len_mean          | 1.21e+03 |
| ep_rew_mean          | 500      |
| exploration rate     | 0.266    |
| time/                |          |
| episodes             | 31000    |
| fps                  | 249      |
| time_elapsed         | 8820     |
| total timesteps      | 2203969  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0568   |
| n_updates            | 538492   |

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|           |                |         |           |               |                  |
|-----------|----------------|---------|-----------|---------------|------------------|
| EP=10,510 | STEP=2,205,095 | clip=52 | consec=20 | max_consec=42 | avg_clip50=50.68 |
| EP=10,520 | STEP=2,207,860 | clip=47 | consec=30 | max_consec=42 | avg_clip50=49.46 |
| EP=10,530 | STEP=2,211,099 | clip=54 | consec=40 | max_consec=42 | avg_clip50=50.48 |
| EP=10,540 | STEP=2,213,937 | clip=44 | consec=5  | max_consec=44 | avg_clip50=48.10 |

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| custom/              |          |
| avg_clip50           | 48.3     |
| clipped_episode_r... | 60.5     |
| consecutive_clipp... | 6        |
| max_consecutive_c... | 44       |
| rollout/             |          |
| ep_len_mean          | 1.21e+03 |
| ep_rew_mean          | 498      |
| exploration rate     | 0.263    |
| time/                |          |
| episodes             | 31100    |
| fps                  | 249      |
| time_elapsed         | 8865     |
| total timesteps      | 2214513  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.231    |
| n_updates            | 541128   |

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|           |                |         |           |               |                  |
|-----------|----------------|---------|-----------|---------------|------------------|
| EP=10,550 | STEP=2,217,009 | clip=48 | consec=15 | max_consec=44 | avg_clip50=50.72 |
| EP=10,560 | STEP=2,219,562 | clip=12 | consec=0  | max_consec=44 | avg_clip50=50.05 |

EP=10,570 | STEP=2,222,699 | clip=58 | consec=10 | max\_consec=44 | avg\_clip50=51.02

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 52.1     |  |
| clipped_episode_r... | 47       |  |
| consecutive_clipp... | 17       |  |
| max_consecutive_c... | 44       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.21e+03 |  |
| ep_rew_mean          | 504      |  |
| exploration rate     | 0.259    |  |
| time/                |          |  |
| episodes             | 31200    |  |
| fps                  | 249      |  |
| time_elapsed         | 8908     |  |
| total timesteps      | 2224862  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0435   |  |
| n_updates            | 543715   |  |

EP=10,580 | STEP=2,225,591 | clip=66 | consec=20 | max\_consec=44 | avg\_clip50=50.43

EP=10,590 | STEP=2,228,616 | clip=12 | consec=0 | max\_consec=44 | avg\_clip50=51.03

EP=10,600 | STEP=2,231,647 | clip=16 | consec=0 | max\_consec=44 | avg\_clip50=50.19

EP=10,610 | STEP=2,234,701 | clip=64 | consec=9 | max\_consec=44 | avg\_clip50=51.16

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 51.7     |  |
| clipped_episode_r... | 48.5     |  |
| consecutive_clipp... | 13       |  |
| max_consecutive_c... | 44       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.23e+03 |  |
| ep_rew_mean          | 509      |  |
| exploration rate     | 0.255    |  |
| time/                |          |  |
| episodes             | 31300    |  |
| fps                  | 249      |  |
| time_elapsed         | 8955     |  |
| total timesteps      | 2235955  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0808   |  |
| n_updates            | 546488   |  |

EP=10,620 | STEP=2,237,625 | clip=65 | consec=19 | max\_consec=44 | avg\_clip50=51.24

EP=10,630 | STEP=2,240,147 | clip=60 | consec=7 | max\_consec=44 | avg\_clip50=48.94

EP=10,640 | STEP=2,242,703 | clip=60 | consec=6 | max\_consec=44 | avg\_clip50=48.32

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| -----                |      |  |
| custom/              |      |  |
| avg_clip50           | 46.2 |  |
| clipped_episode_r... | 31   |  |

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|--|----------------------|----------|--|
|  | consecutive_clipp... | 3        |  |
|  | max_consecutive_c... | 44       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.18e+03 |  |
|  | ep_rew_mean          | 485      |  |
|  | exploration rate     | 0.252    |  |
|  | time/                |          |  |
|  | episodes             | 31400    |  |
|  | fps                  | 249      |  |
|  | time_elapsed         | 8995     |  |
|  | total timesteps      | 2245185  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0398   |  |
|  | n_updates            | 548796   |  |

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EP=10,650 | STEP=2,245,521 | clip=54 | consec=4 | max\_consec=44 | avg\_clip50=46.94

EP=10,660 | STEP=2,248,669 | clip=60 | consec=3 | max\_consec=44 | avg\_clip50=47.50

EP=10,670 | STEP=2,251,693 | clip=30 | consec=13 | max\_consec=44 | avg\_clip50=48.31

EP=10,680 | STEP=2,254,616 | clip=54 | consec=2 | max\_consec=44 | avg\_clip50=49.49

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|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 50.4    |  |
|  | clipped_episode_r... | 28      |  |
|  | consecutive_clipp... | 7       |  |
|  | max_consecutive_c... | 44      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 1.2e+03 |  |
|  | ep_rew_mean          | 490     |  |
|  | exploration rate     | 0.249   |  |
|  | time/                |         |  |
|  | episodes             | 31500   |  |
|  | fps                  | 249     |  |
|  | time_elapsed         | 9041    |  |
|  | total timesteps      | 2256097 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.309   |  |
|  | n_updates            | 551524  |  |

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EP=10,690 | STEP=2,257,427 | clip=94 | consec=12 | max\_consec=44 | avg\_clip50=50.80

EP=10,700 | STEP=2,260,175 | clip=80 | consec=22 | max\_consec=44 | avg\_clip50=52.61

EP=10,710 | STEP=2,262,984 | clip=39 | consec=32 | max\_consec=44 | avg\_clip50=51.42

EP=10,720 | STEP=2,265,738 | clip=52 | consec=42 | max\_consec=44 | avg\_clip50=50.17

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 50.2     |  |
|  | clipped_episode_r... | 77.5     |  |
|  | consecutive_clipp... | 44       |  |
|  | max_consecutive_c... | 44       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.17e+03 |  |

|  |                  |         |  |
|--|------------------|---------|--|
|  | ep_rew_mean      | 497     |  |
|  | exploration rate | 0.245   |  |
|  | time/            |         |  |
|  | episodes         | 31600   |  |
|  | fps              | 249     |  |
|  | time_elapsed     | 9084    |  |
|  | total timesteps  | 2266273 |  |
|  | train/           |         |  |
|  | learning_rate    | 0.0001  |  |
|  | loss             | 0.0571  |  |
|  | n_updates        | 554068  |  |

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EP=10,730 | STEP=2,268,353 | clip=54 | consec=52 | max\_consec=52 | avg\_clip50=50.33

EP=10,740 | STEP=2,270,941 | clip=12 | consec=0 | max\_consec=53 | avg\_clip50=49.11

EP=10,750 | STEP=2,274,166 | clip=44 | consec=10 | max\_consec=53 | avg\_clip50=50.36

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 52.4     |  |
|  | clipped_episode_r... | 60       |  |
|  | consecutive_clipp... | 18       |  |
|  | max_consecutive_c... | 53       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.19e+03 |  |
|  | ep_rew_mean          | 519      |  |
|  | exploration rate     | 0.242    |  |
|  | time/                |          |  |
|  | episodes             | 31700    |  |
|  | fps                  | 249      |  |
|  | time_elapsed         | 9130     |  |
|  | total timesteps      | 2277151  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.278    |  |
|  | n_updates            | 556787   |  |

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EP=10,760 | STEP=2,277,496 | clip=46 | consec=20 | max\_consec=53 | avg\_clip50=52.28

EP=10,770 | STEP=2,280,440 | clip=43 | consec=30 | max\_consec=53 | avg\_clip50=51.39

EP=10,780 | STEP=2,283,327 | clip=54 | consec=40 | max\_consec=53 | avg\_clip50=50.32

EP=10,790 | STEP=2,286,448 | clip=98 | consec=50 | max\_consec=53 | avg\_clip50=53.00

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 52.4     |  |
|  | clipped_episode_r... | 22.5     |  |
|  | consecutive_clipp... | 53       |  |
|  | max_consecutive_c... | 53       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.21e+03 |  |
|  | ep_rew_mean          | 515      |  |
|  | exploration rate     | 0.238    |  |
|  | time/                |          |  |
|  | episodes             | 31800    |  |
|  | fps                  | 249      |  |
|  | time_elapsed         | 9174     |  |



|                 |         |
|-----------------|---------|
| total timesteps | 2287476 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.515   |
| n_updates       | 559368  |

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EP=10,800 | STEP=2,288,804 | clip=42 | consec=1 | max\_consec=53 | avg\_clip50=48.42

EP=10,810 | STEP=2,291,828 | clip=60 | consec=11 | max\_consec=53 | avg\_clip50=47.20

EP=10,820 | STEP=2,294,463 | clip=36 | consec=21 | max\_consec=53 | avg\_clip50=46.92

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 48.8     |
| clipped_episode_r... | 45.5     |
| consecutive_clipp... | 29       |
| max_consecutive_c... | 53       |
| rollout/             |          |
| ep_len_mean          | 1.21e+03 |
| ep_rew_mean          | 495      |
| exploration rate     | 0.235    |
| time/                |          |
| episodes             | 31900    |
| fps                  | 249      |
| time_elapsed         | 9215     |
| total timesteps      | 2296956  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.413    |
| n_updates            | 561738   |

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EP=10,830 | STEP=2,297,472 | clip=17 | consec=0 | max\_consec=53 | avg\_clip50=48.46

EP=10,840 | STEP=2,300,364 | clip=42 | consec=8 | max\_consec=53 | avg\_clip50=46.74

EP=10,850 | STEP=2,303,282 | clip=94 | consec=1 | max\_consec=53 | avg\_clip50=49.47

EP=10,860 | STEP=2,306,913 | clip=59 | consec=4 | max\_consec=53 | avg\_clip50=51.46

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| custom/              |          |
| avg_clip50           | 52.3     |
| clipped_episode_r... | 69       |
| consecutive_clipp... | 6        |
| max_consecutive_c... | 53       |
| rollout/             |          |
| ep_len_mean          | 1.23e+03 |
| ep_rew_mean          | 500      |
| exploration rate     | 0.231    |
| time/                |          |
| episodes             | 32000    |
| fps                  | 249      |
| time_elapsed         | 9262     |
| total timesteps      | 2307868  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0904   |
| n_updates            | 564466   |

EP=10,870 | STEP=2,310,251 | clip=50 | consec=14 | max\_consec=53 | avg\_clip50=53.84  
EP=10,880 | STEP=2,312,985 | clip=52 | consec=24 | max\_consec=53 | avg\_clip50=53.25  
EP=10,890 | STEP=2,315,938 | clip=60 | consec=2 | max\_consec=53 | avg\_clip50=52.88

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| custom/              |          |  |
| avg_clip50           | 55.1     |  |
| clipped_episode_r... | 91.5     |  |
| consecutive_clipp... | 10       |  |
| max_consecutive_c... | 53       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.24e+03 |  |
| ep_rew_mean          | 518      |  |
| exploration rate     | 0.228    |  |
| time/                |          |  |
| episodes             | 32100    |  |
| fps                  | 249      |  |
| time_elapsed         | 9309     |  |
| total timesteps      | 2318554  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0501   |  |
| n_updates            | 567138   |  |

EP=10,900 | STEP=2,319,043 | clip=60 | consec=12 | max\_consec=53 | avg\_clip50=55.19  
EP=10,910 | STEP=2,322,430 | clip=46 | consec=22 | max\_consec=53 | avg\_clip50=55.04  
EP=10,920 | STEP=2,325,154 | clip=60 | consec=1 | max\_consec=53 | avg\_clip50=52.27  
EP=10,930 | STEP=2,327,960 | clip=35 | consec=11 | max\_consec=53 | avg\_clip50=51.65

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| custom/              |          |  |
| avg_clip50           | 52.8     |  |
| clipped_episode_r... | 37.5     |  |
| consecutive_clipp... | 14       |  |
| max_consecutive_c... | 53       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.27e+03 |  |
| ep_rew_mean          | 534      |  |
| exploration rate     | 0.224    |  |
| time/                |          |  |
| episodes             | 32200    |  |
| fps                  | 248      |  |
| time_elapsed         | 9354     |  |
| total timesteps      | 2329032  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0458   |  |
| n_updates            | 569757   |  |

EP=10,940 | STEP=2,331,262 | clip=100 | consec=3 | max\_consec=53 | avg\_clip50=53.63  
EP=10,950 | STEP=2,334,213 | clip=57 | consec=13 | max\_consec=53 | avg\_clip50=51.76  
EP=10,960 | STEP=2,337,248 | clip=74 | consec=3 | max\_consec=53 | avg\_clip50=50.54

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| custom/              |          |  |
| avg_clip50           | 53.6     |  |
| clipped_episode_r... | 54.5     |  |
| consecutive_clipp... | 11       |  |
| max_consecutive_c... | 53       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.25e+03 |  |
| ep_rew_mean          | 529      |  |
| exploration rate     | 0.221    |  |
| time/                |          |  |
| episodes             | 32300    |  |
| fps                  | 248      |  |
| time_elapsed         | 9401     |  |
| total timesteps      | 2339964  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.049    |  |
| n_updates            | 572490   |  |

EP=10,970 | STEP=2,340,421 | clip=40 | consec=13 | max\_consec=53 | avg\_clip50=53.84  
EP=10,980 | STEP=2,342,902 | clip=45 | consec=4 | max\_consec=53 | avg\_clip50=52.41  
EP=10,990 | STEP=2,346,215 | clip=27 | consec=14 | max\_consec=53 | avg\_clip50=51.80  
EP=11,000 | STEP=2,349,282 | clip=74 | consec=24 | max\_consec=53 | avg\_clip50=51.96

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| custom/              |          |  |
| avg_clip50           | 52.3     |  |
| clipped_episode_r... | 80       |  |
| consecutive_clipp... | 26       |  |
| max_consecutive_c... | 53       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.25e+03 |  |
| ep_rew_mean          | 520      |  |
| exploration rate     | 0.217    |  |
| time/                |          |  |
| episodes             | 32400    |  |
| fps                  | 248      |  |
| time_elapsed         | 9444     |  |
| total timesteps      | 2350035  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0543   |  |
| n_updates            | 575008   |  |

EP=11,010 | STEP=2,351,895 | clip=38 | consec=6 | max\_consec=53 | avg\_clip50=51.67  
EP=11,020 | STEP=2,354,704 | clip=50 | consec=16 | max\_consec=53 | avg\_clip50=48.89  
EP=11,030 | STEP=2,358,141 | clip=54 | consec=26 | max\_consec=53 | avg\_clip50=52.66

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| custom/              |      |  |
| avg_clip50           | 53.5 |  |
| clipped_episode_r... | 49   |  |
| consecutive_clipp... | 33   |  |
| max_consecutive_c... | 53   |  |

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| rollout/         |          |
| ep_len_mean      | 1.28e+03 |
| ep_rew_mean      | 535      |
| exploration rate | 0.214    |
| time/            |          |
| episodes         | 32500    |
| fps              | 248      |
| time_elapsed     | 9491     |
| total timesteps  | 2360896  |
| train/           |          |
| learning_rate    | 0.0001   |
| loss             | 0.0446   |
| n_updates        | 577723   |

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EP=11,040 | STEP=2,361,654 | clip=52 | consec=2 | max\_consec=53 | avg\_clip50=53.60  
EP=11,050 | STEP=2,364,700 | clip=50 | consec=12 | max\_consec=53 | avg\_clip50=53.76  
EP=11,060 | STEP=2,367,837 | clip=54 | consec=22 | max\_consec=53 | avg\_clip50=55.65  
EP=11,070 | STEP=2,370,683 | clip=38 | consec=4 | max\_consec=53 | avg\_clip50=57.13

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| custom/              |          |
| avg_clip50           | 57.6     |
| clipped_episode_r... | 54       |
| consecutive_clipp... | 6        |
| max_consecutive_c... | 53       |
| rollout/             |          |
| ep_len_mean          | 1.26e+03 |
| ep_rew_mean          | 534      |
| exploration rate     | 0.21     |
| time/                |          |
| episodes             | 32600    |
| fps                  | 248      |
| time_elapsed         | 9536     |
| total timesteps      | 2371613  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0419   |
| n_updates            | 580403   |

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EP=11,080 | STEP=2,373,438 | clip=52 | consec=14 | max\_consec=53 | avg\_clip50=56.59  
EP=11,090 | STEP=2,376,385 | clip=96 | consec=24 | max\_consec=53 | avg\_clip50=56.20  
EP=11,100 | STEP=2,379,194 | clip=22 | consec=34 | max\_consec=53 | avg\_clip50=55.62  
EP=11,110 | STEP=2,382,481 | clip=74 | consec=44 | max\_consec=53 | avg\_clip50=56.69

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| custom/              |          |
| avg_clip50           | 56.7     |
| clipped_episode_r... | 74.5     |
| consecutive_clipp... | 44       |
| max_consecutive_c... | 53       |
| rollout/             |          |
| ep_len_mean          | 1.27e+03 |
| ep_rew_mean          | 562      |
| exploration rate     | 0.207    |

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| time/           |         |  |
| episodes        | 32700   |  |
| fps             | 248     |  |
| time_elapsed    | 9583    |  |
| total timesteps | 2382481 |  |
| train/          |         |  |
| learning_rate   | 0.0001  |  |
| loss            | 0.489   |  |
| n_updates       | 583120  |  |

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EP=11,120 | STEP=2,385,544 | clip=60 | consec=54 | max\_consec=54 | avg\_clip50=57.44

EP=11,130 | STEP=2,388,211 | clip=28 | consec=64 | max\_consec=64 | avg\_clip50=55.39

EP=11,140 | STEP=2,390,993 | clip=37 | consec=4 | max\_consec=69 | avg\_clip50=53.16

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| custom/              |          |  |
| avg_clip50           | 52.4     |  |
| clipped_episode_r... | 45       |  |
| consecutive_clipp... | 9        |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.22e+03 |  |
| ep_rew_mean          | 546      |  |
| exploration rate     | 0.203    |  |
| time/                |          |  |
| episodes             | 32800    |  |
| fps                  | 248      |  |
| time_elapsed         | 9627     |  |
| total timesteps      | 2392602  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.099    |  |
| n_updates            | 585650   |  |

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EP=11,150 | STEP=2,394,250 | clip=112 | consec=14 | max\_consec=69 | avg\_clip50=54.49

EP=11,160 | STEP=2,397,203 | clip=77 | consec=24 | max\_consec=69 | avg\_clip50=52.11

EP=11,170 | STEP=2,400,464 | clip=60 | consec=34 | max\_consec=69 | avg\_clip50=52.56

EP=11,180 | STEP=2,403,273 | clip=41 | consec=44 | max\_consec=69 | avg\_clip50=54.63

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| custom/              |          |  |
| avg_clip50           | 54.6     |  |
| clipped_episode_r... | 41       |  |
| consecutive_clipp... | 44       |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.24e+03 |  |
| ep_rew_mean          | 550      |  |
| exploration rate     | 0.2      |  |
| time/                |          |  |
| episodes             | 32900    |  |
| fps                  | 248      |  |
| time_elapsed         | 9673     |  |
| total timesteps      | 2403548  |  |
| train/               |          |  |

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| learning_rate | 0.0001 |
| loss          | 0.0606 |
| n_updates     | 588386 |

EP=11,190 | STEP=2,406,016 | clip=74 | consec=1 | max\_consec=69 | avg\_clip50=55.12

EP=11,200 | STEP=2,409,283 | clip=96 | consec=11 | max\_consec=69 | avg\_clip50=55.20

EP=11,210 | STEP=2,412,548 | clip=37 | consec=21 | max\_consec=69 | avg\_clip50=56.07

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| custom/              |          |
| avg_clip50           | 54.9     |
| clipped_episode_r... | 52       |
| consecutive_clipp... | 30       |
| max_consecutive_c... | 69       |
| rollout/             |          |
| ep_len_mean          | 1.24e+03 |
| ep_rew_mean          | 538      |
| exploration rate     | 0.196    |
| time/                |          |
| episodes             | 33000    |
| fps                  | 248      |
| time_elapsed         | 9723     |
| total timesteps      | 2415048  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0554   |
| n_updates            | 591261   |

EP=11,220 | STEP=2,415,174 | clip=18 | consec=0 | max\_consec=69 | avg\_clip50=54.11

EP=11,230 | STEP=2,417,811 | clip=52 | consec=5 | max\_consec=69 | avg\_clip50=52.22

EP=11,240 | STEP=2,420,848 | clip=38 | consec=15 | max\_consec=69 | avg\_clip50=52.47

EP=11,250 | STEP=2,423,936 | clip=57 | consec=25 | max\_consec=69 | avg\_clip50=52.55

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| custom/              |          |
| avg_clip50           | 52.2     |
| clipped_episode_r... | 45       |
| consecutive_clipp... | 30       |
| max_consecutive_c... | 69       |
| rollout/             |          |
| ep_len_mean          | 1.22e+03 |
| ep_rew_mean          | 537      |
| exploration rate     | 0.192    |
| time/                |          |
| episodes             | 33100    |
| fps                  | 248      |
| time_elapsed         | 9768     |
| total timesteps      | 2425413  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0682   |
| n_updates            | 593853   |

EP=11,260 | STEP=2,427,272 | clip=58 | consec=35 | max\_consec=69 | avg\_clip50=53.44

EP=11,270 | STEP=2,430,045 | clip=54 | consec=3 | max\_consec=69 | avg\_clip50=53.67

EP=11,280 | STEP=2,433,139 | clip=47 | consec=13 | max\_consec=69 | avg\_clip50=54.95

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| custom/              |          |  |
| avg_clip50           | 54.3     |  |
| clipped_episode_r... | 22       |  |
| consecutive_clipp... | 22       |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.24e+03 |  |
| ep_rew_mean          | 537      |  |
| exploration rate     | 0.189    |  |
| time/                |          |  |
| episodes             | 33200    |  |
| fps                  | 248      |  |
| time_elapsed         | 9812     |  |
| total timesteps      | 2435745  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0688   |  |
| n_updates            | 596436   |  |

EP=11,290 | STEP=2,435,963 | clip=60 | consec=23 | max\_consec=69 | avg\_clip50=54.73

EP=11,300 | STEP=2,438,715 | clip=55 | consec=1 | max\_consec=69 | avg\_clip50=53.27

EP=11,310 | STEP=2,442,448 | clip=54 | consec=11 | max\_consec=69 | avg\_clip50=54.30

EP=11,320 | STEP=2,445,785 | clip=55 | consec=21 | max\_consec=69 | avg\_clip50=55.55

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| custom/              |          |  |
| avg_clip50           | 56.2     |  |
| clipped_episode_r... | 57.5     |  |
| consecutive_clipp... | 25       |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.28e+03 |  |
| ep_rew_mean          | 553      |  |
| exploration rate     | 0.185    |  |
| time/                |          |  |
| episodes             | 33300    |  |
| fps                  | 248      |  |
| time_elapsed         | 9861     |  |
| total timesteps      | 2447190  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0345   |  |
| n_updates            | 599297   |  |

EP=11,330 | STEP=2,449,007 | clip=36 | consec=31 | max\_consec=69 | avg\_clip50=56.70

EP=11,340 | STEP=2,452,119 | clip=60 | consec=41 | max\_consec=69 | avg\_clip50=58.01

EP=11,350 | STEP=2,456,000 | clip=52 | consec=51 | max\_consec=69 | avg\_clip50=61.06

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| custom/ |  |  |

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|  | avg_clip50           | 58.5     |  |
|  | clipped_episode_r... | 75       |  |
|  | consecutive_clipp... | 7        |  |
|  | max_consecutive_c... | 69       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.31e+03 |  |
|  | ep_rew_mean          | 564      |  |
|  | exploration rate     | 0.181    |  |
|  | time/                |          |  |
|  | episodes             | 33400    |  |
|  | fps                  | 248      |  |
|  | time_elapsed         | 9911     |  |
|  | total timesteps      | 2458909  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0422   |  |
|  | n_updates            | 602227   |  |

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EP=11,360 | STEP=2,458,983 | clip=48 | consec=8 | max\_consec=69 | avg\_clip50=58.35

EP=11,370 | STEP=2,461,515 | clip=80 | consec=2 | max\_consec=69 | avg\_clip50=56.33

EP=11,380 | STEP=2,464,749 | clip=60 | consec=12 | max\_consec=69 | avg\_clip50=56.58

EP=11,390 | STEP=2,468,043 | clip=38 | consec=22 | max\_consec=69 | avg\_clip50=56.91

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|  | custom/              |          |  |
|  | avg_clip50           | 53.8     |  |
|  | clipped_episode_r... | 20.5     |  |
|  | consecutive_clipp... | 27       |  |
|  | max_consecutive_c... | 69       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.31e+03 |  |
|  | ep_rew_mean          | 568      |  |
|  | exploration rate     | 0.178    |  |
|  | time/                |          |  |
|  | episodes             | 33500    |  |
|  | fps                  | 248      |  |
|  | time_elapsed         | 9955     |  |
|  | total timesteps      | 2469143  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.442    |  |
|  | n_updates            | 604785   |  |

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EP=11,400 | STEP=2,470,766 | clip=38 | consec=32 | max\_consec=69 | avg\_clip50=52.63

EP=11,410 | STEP=2,473,851 | clip=39 | consec=42 | max\_consec=69 | avg\_clip50=50.97

EP=11,420 | STEP=2,477,198 | clip=50 | consec=52 | max\_consec=69 | avg\_clip50=54.55

EP=11,430 | STEP=2,479,997 | clip=57 | consec=7 | max\_consec=69 | avg\_clip50=52.37

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|  | custom/              |      |  |
|  | avg_clip50           | 52.4 |  |
|  | clipped_episode_r... | 57   |  |
|  | consecutive_clipp... | 7    |  |
|  | max_consecutive_c... | 69   |  |



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| rollout/         |          |  |
| ep_len_mean      | 1.28e+03 |  |
| ep_rew_mean      | 545      |  |
| exploration rate | 0.174    |  |
| time/            |          |  |
| episodes         | 33600    |  |
| fps              | 247      |  |
| time_elapsed     | 10002    |  |
| total timesteps  | 2479997  |  |
| train/           |          |  |
| learning_rate    | 0.0001   |  |
| loss             | 0.614    |  |
| n_updates        | 607499   |  |

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EP=11,440 | STEP=2,483,181 | clip=49 | consec=17 | max\_consec=69 | avg\_clip50=52.49

EP=11,450 | STEP=2,486,221 | clip=70 | consec=27 | max\_consec=69 | avg\_clip50=56.39

EP=11,460 | STEP=2,489,837 | clip=54 | consec=37 | max\_consec=69 | avg\_clip50=59.91

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| custom/              |          |  |
| avg_clip50           | 58.2     |  |
| clipped_episode_r... | 66.5     |  |
| consecutive_clipp... | 44       |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.28e+03 |  |
| ep_rew_mean          | 553      |  |
| exploration rate     | 0.17     |  |
| time/                |          |  |
| episodes             | 33700    |  |
| fps                  | 247      |  |
| time_elapsed         | 10053    |  |
| total timesteps      | 2491787  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.222    |  |
| n_updates            | 610446   |  |

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EP=11,470 | STEP=2,492,721 | clip=54 | consec=47 | max\_consec=69 | avg\_clip50=56.95

EP=11,480 | STEP=2,495,099 | clip=78 | consec=57 | max\_consec=69 | avg\_clip50=56.26

EP=11,490 | STEP=2,498,197 | clip=38 | consec=2 | max\_consec=69 | avg\_clip50=55.66

EP=11,500 | STEP=2,502,252 | clip=54 | consec=12 | max\_consec=69 | avg\_clip50=56.70

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| custom/              |          |  |
| avg_clip50           | 55.5     |  |
| clipped_episode_r... | 63       |  |
| consecutive_clipp... | 14       |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.31e+03 |  |
| ep_rew_mean          | 566      |  |
| exploration rate     | 0.167    |  |
| time/                |          |  |
| episodes             | 33800    |  |

|                 |         |
|-----------------|---------|
| fps             | 247     |
| time_elapsed    | 10100   |
| total timesteps | 2502853 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.325   |
| n_updates       | 613213  |

EP=11,510 | STEP=2,505,379 | clip=19 | consec=0 | max\_consec=69 | avg\_clip50=55.59

EP=11,520 | STEP=2,508,419 | clip=10 | consec=0 | max\_consec=69 | avg\_clip50=57.94

EP=11,530 | STEP=2,511,509 | clip=80 | consec=10 | max\_consec=69 | avg\_clip50=60.16

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 61       |
| clipped_episode_r... | 110      |
| consecutive_clipp... | 16       |
| max_consecutive_c... | 69       |
| rollout/             |          |
| ep_len_mean          | 1.32e+03 |
| ep_rew_mean          | 584      |
| exploration rate     | 0.163    |
| time/                |          |
| episodes             | 33900    |
| fps                  | 247      |
| time_elapsed         | 10148    |
| total timesteps      | 2513956  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0604   |
| n_updates            | 615988   |

EP=11,540 | STEP=2,514,890 | clip=49 | consec=20 | max\_consec=69 | avg\_clip50=61.18

EP=11,550 | STEP=2,517,844 | clip=42 | consec=3 | max\_consec=69 | avg\_clip50=57.70

EP=11,560 | STEP=2,520,916 | clip=58 | consec=13 | max\_consec=69 | avg\_clip50=56.00

EP=11,570 | STEP=2,524,269 | clip=60 | consec=23 | max\_consec=69 | avg\_clip50=57.65

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 57.8     |
| clipped_episode_r... | 50       |
| consecutive_clipp... | 25       |
| max_consecutive_c... | 69       |
| rollout/             |          |
| ep_len_mean          | 1.31e+03 |
| ep_rew_mean          | 579      |
| exploration rate     | 0.159    |
| time/                |          |
| episodes             | 34000    |
| fps                  | 247      |
| time_elapsed         | 10195    |
| total timesteps      | 2524902  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0842   |

|           |        |
|-----------|--------|
| n_updates | 618725 |
|-----------|--------|

---

EP=11,580 | STEP=2,527,212 | clip=34 | consec=33 | max\_consec=69 | avg\_clip50=57.87

EP=11,590 | STEP=2,530,171 | clip=44 | consec=2 | max\_consec=69 | avg\_clip50=56.26

EP=11,600 | STEP=2,533,460 | clip=54 | consec=12 | max\_consec=69 | avg\_clip50=56.85

---

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 56.8     |
| clipped_episode_r... | 38.5     |
| consecutive_clipp... | 18       |
| max_consecutive_c... | 69       |
| rollout/             |          |
| ep_len_mean          | 1.28e+03 |
| ep_rew_mean          | 567      |
| exploration rate     | 0.156    |
| time/                |          |
| episodes             | 34100    |
| fps                  | 247      |
| time_elapsed         | 10240    |
| total timesteps      | 2535267  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0522   |
| n_updates            | 621316   |

---

EP=11,610 | STEP=2,536,221 | clip=54 | consec=22 | max\_consec=69 | avg\_clip50=57.59

EP=11,620 | STEP=2,539,228 | clip=57 | consec=32 | max\_consec=69 | avg\_clip50=56.24

EP=11,630 | STEP=2,542,212 | clip=6 | consec=0 | max\_consec=69 | avg\_clip50=55.83

EP=11,640 | STEP=2,544,943 | clip=58 | consec=1 | max\_consec=69 | avg\_clip50=55.49

---

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 55.3     |
| clipped_episode_r... | 35       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 69       |
| rollout/             |          |
| ep_len_mean          | 1.24e+03 |
| ep_rew_mean          | 553      |
| exploration rate     | 0.152    |
| time/                |          |
| episodes             | 34200    |
| fps                  | 247      |
| time_elapsed         | 10285    |
| total timesteps      | 2545698  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.232    |
| n_updates            | 623924   |

---

EP=11,650 | STEP=2,547,640 | clip=74 | consec=2 | max\_consec=69 | avg\_clip50=54.11

EP=11,660 | STEP=2,550,778 | clip=44 | consec=12 | max\_consec=69 | avg\_clip50=55.90

EP=11,670 | STEP=2,553,539 | clip=74 | consec=22 | max\_consec=69 | avg\_clip50=55.

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 52.9     |  |
| clipped_episode_r... | 54.5     |  |
| consecutive_clipp... | 5        |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.19e+03 |  |
| ep_rew_mean          | 544      |  |
| exploration rate     | 0.149    |  |
| time/                |          |  |
| episodes             | 34300    |  |
| fps                  | 247      |  |
| time_elapsed         | 10328    |  |
| total timesteps      | 2555616  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.106    |  |
| n_updates            | 626403   |  |

EP=11,680 | STEP=2,555,875 | clip=24 | consec=7 | max\_consec=69 | avg\_clip50=52.9  
5  
EP=11,690 | STEP=2,559,368 | clip=52 | consec=6 | max\_consec=69 | avg\_clip50=55.0  
0  
EP=11,700 | STEP=2,562,086 | clip=60 | consec=3 | max\_consec=69 | avg\_clip50=55.5  
0  
EP=11,710 | STEP=2,565,235 | clip=103 | consec=4 | max\_consec=69 | avg\_clip50=54.  
41

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 54.3     |  |
| clipped_episode_r... | 35.5     |  |
| consecutive_clipp... | 7        |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.21e+03 |  |
| ep_rew_mean          | 552      |  |
| exploration rate     | 0.145    |  |
| time/                |          |  |
| episodes             | 34400    |  |
| fps                  | 247      |  |
| time_elapsed         | 10374    |  |
| total timesteps      | 2566170  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0418   |  |
| n_updates            | 629042   |  |

EP=11,720 | STEP=2,568,130 | clip=57 | consec=14 | max\_consec=69 | avg\_clip50=53.  
02  
EP=11,730 | STEP=2,571,459 | clip=60 | consec=24 | max\_consec=69 | avg\_clip50=59.  
03  
EP=11,740 | STEP=2,574,582 | clip=60 | consec=34 | max\_consec=69 | avg\_clip50=59.  
13  
EP=11,750 | STEP=2,577,609 | clip=60 | consec=44 | max\_consec=69 | avg\_clip50=60.  
11

|            |      |  |
|------------|------|--|
| -----      |      |  |
| custom/    |      |  |
| avg_clip50 | 60.1 |  |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | clipped_episode_r... | 60.5     |  |
|  | consecutive_clipp... | 44       |  |
|  | max_consecutive_c... | 69       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.24e+03 |  |
|  | ep_rew_mean          | 578      |  |
|  | exploration rate     | 0.142    |  |
|  | time/                |          |  |
|  | episodes             | 34500    |  |
|  | fps                  | 247      |  |
|  | time_elapsed         | 10423    |  |
|  | total timesteps      | 2577635  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0536   |  |
|  | n_updates            | 631908   |  |

-----

EP=11,760 | STEP=2,580,153 | clip=77 | consec=3 | max\_consec=69 | avg\_clip50=57.2  
2

EP=11,770 | STEP=2,583,127 | clip=60 | consec=5 | max\_consec=69 | avg\_clip50=57.5  
2

EP=11,780 | STEP=2,586,115 | clip=61 | consec=15 | max\_consec=69 | avg\_clip50=53.  
67

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 49.7     |  |
|  | clipped_episode_r... | 39.5     |  |
|  | consecutive_clipp... | 2        |  |
|  | max_consecutive_c... | 69       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.23e+03 |  |
|  | ep_rew_mean          | 549      |  |
|  | exploration rate     | 0.138    |  |
|  | time/                |          |  |
|  | episodes             | 34600    |  |
|  | fps                  | 247      |  |
|  | time_elapsed         | 10467    |  |
|  | total timesteps      | 2587582  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0394   |  |
|  | n_updates            | 634395   |  |

-----

EP=11,790 | STEP=2,588,849 | clip=54 | consec=6 | max\_consec=69 | avg\_clip50=51.1  
3

EP=11,800 | STEP=2,591,672 | clip=61 | consec=7 | max\_consec=69 | avg\_clip50=50.1  
3

EP=11,810 | STEP=2,595,334 | clip=96 | consec=17 | max\_consec=69 | avg\_clip50=54.  
97

EP=11,820 | STEP=2,598,298 | clip=36 | consec=27 | max\_consec=69 | avg\_clip50=54.  
56

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 55.5     |  |
|  | clipped_episode_r... | 54.5     |  |
|  | consecutive_clipp... | 29       |  |
|  | max_consecutive_c... | 69       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.25e+03 |  |
|  | ep_rew_mean          | 562      |  |

|                  |         |  |
|------------------|---------|--|
| exploration rate | 0.135   |  |
| time/            |         |  |
| episodes         | 34700   |  |
| fps              | 247     |  |
| time_elapsed     | 10515   |  |
| total timesteps  | 2598850 |  |
| train/           |         |  |
| learning_rate    | 0.0001  |  |
| loss             | 0.0442  |  |
| n_updates        | 637212  |  |

-----

EP=11,830 | STEP=2,601,071 | clip=18 | consec=0 | max\_consec=69 | avg\_clip50=56.17

EP=11,840 | STEP=2,604,005 | clip=31 | consec=10 | max\_consec=69 | avg\_clip50=57.27

EP=11,850 | STEP=2,607,372 | clip=95 | consec=20 | max\_consec=69 | avg\_clip50=60.09

|                      |          |  |
|----------------------|----------|--|
| custom/              |          |  |
| avg_clip50           | 60.1     |  |
| clipped_episode_r... | 80.5     |  |
| consecutive_clipp... | 28       |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.27e+03 |  |
| ep_rew_mean          | 572      |  |
| exploration rate     | 0.131    |  |
| time/                |          |  |
| episodes             | 34800    |  |
| fps                  | 247      |  |
| time_elapsed         | 10565    |  |
| total timesteps      | 2610276  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0539   |  |
| n_updates            | 640068   |  |

-----

EP=11,860 | STEP=2,610,658 | clip=66 | consec=30 | max\_consec=69 | avg\_clip50=59.45

EP=11,870 | STEP=2,613,953 | clip=52 | consec=40 | max\_consec=69 | avg\_clip50=60.41

EP=11,880 | STEP=2,617,612 | clip=54 | consec=50 | max\_consec=69 | avg\_clip50=61.55

EP=11,890 | STEP=2,621,058 | clip=60 | consec=60 | max\_consec=69 | avg\_clip50=63.25

|                      |          |  |
|----------------------|----------|--|
| custom/              |          |  |
| avg_clip50           | 63.3     |  |
| clipped_episode_r... | 75       |  |
| consecutive_clipp... | 65       |  |
| max_consecutive_c... | 69       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.33e+03 |  |
| ep_rew_mean          | 607      |  |
| exploration rate     | 0.127    |  |
| time/                |          |  |
| episodes             | 34900    |  |
| fps                  | 246      |  |
| time_elapsed         | 10618    |  |
| total timesteps      | 2622524  |  |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0272 |
| n_updates     | 643130 |

```
-----
EP=11,900 | STEP=2,624,159 | clip=52 | consec=70 | max_consec=70 | avg_clip50=61.99
EP=11,910 | STEP=2,627,060 | clip=39 | consec=80 | max_consec=80 | avg_clip50=59.96
EP=11,920 | STEP=2,630,106 | clip=72 | consec=90 | max_consec=90 | avg_clip50=60.35
EP=11,930 | STEP=2,633,424 | clip=98 | consec=100 | max_consec=100 | avg_clip50=60.29
```

Objetivo conseguido: 100 episodios consecutivos superando el umbral.

Entrenamiento finalizado.

max\_consec alcanzado: 100

Log JSON: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_SpaceInvaders-v0\_sb3\_log.json

Log CSV: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_SpaceInvaders-v0\_sb3\_log.csv

Out[ ]: <stable\_baselines3.dqn.DQN at 0x7d661b344290>

```
In [ ]: # =====
# 13. Guardado manual del modelo
# =====
# Primero se intenta guardar con SB3. Si falla por serialización ,
# se guarda únicamente la política y el optimizador con PyTorch.

try:
    model.save(str(MODEL_LATEST))
    print('Modelo completo SB3 guardado en:', MODEL_LATEST)

except Exception as e:
    print('No se pudo guardar el modelo completo con SB3.')
    print('Error:', repr(e))
    save_policy_checkpoint(model, POLICY_LATEST)
    print('Política/optimizador guardados con PyTorch en:', POLICY_LATEST)
```

No se pudo guardar el modelo completo con SB3.

Error: IndexError('tuple index out of range')

Política/optimizador guardados con PyTorch en: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_spaceinvaders\_v0\_policy\_latest.pt

```
In [ ]: # =====
# 14. Tablas y gráficas del entrenamiento
# =====

if not CSV_LOG_FILENAME.exists():
    raise FileNotFoundError(f'No existe el log CSV: {CSV_LOG_FILENAME}')

df = pd.read_csv(CSV_LOG_FILENAME)
print('Filas registradas:', len(df))
print('Columnas:', list(df.columns))

display(df.tail(10))

summary_table = pd.DataFrame({
    'metrica': [
        'episodios_registrados',
        'ultimo_step',
```

```

        'mejor_clip',
        'mejor_avg_clip50',
        'mejor_max_consec',
        'ultimo_avg_clip50',
        'ultimo_exploration_rate'
    ],
    'valor': [
        len(df),
        int(df['total_steps'].iloc[-1]) if len(df) else 0,
        float(df['clipped_episode_reward'].max()) if len(df) else 0.0,
        float(df['avg_clip50'].max()) if len(df) else 0.0,
        int(df['max_consecutive_clipped_over_20'].max()) if len(df) else 0,
        float(df['avg_clip50'].iloc[-1]) if len(df) else 0.0,
        float(df['exploration_rate'].iloc[-1]) if 'exploration_rate' in df and 1
    ]
})

display(summary_table)

steps = df['total_steps'].to_numpy()
clipped = df['clipped_episode_reward'].to_numpy()
consec = df['consecutive_clipped_over_20'].to_numpy()
max_consec = df['max_consecutive_clipped_over_20'].to_numpy()
avg50 = df['avg_clip50'].to_numpy()

plt.figure(figsize=(10, 4))
plt.plot(steps, clipped)
plt.title('Recompensa clipped por episodio')
plt.xlabel('Steps')
plt.ylabel('Clipped reward')
plt.grid(True)
plt.show()

plt.figure(figsize=(10, 4))
plt.plot(steps, avg50)
plt.title('Media de recompensa clipped en los últimos 50 episodios')
plt.xlabel('Steps')
plt.ylabel('avg_clip50')
plt.grid(True)
plt.show()

plt.figure(figsize=(10, 4))
plt.plot(steps, consec, label='consec')
plt.plot(steps, max_consec, label='max_consec')
plt.axhline(TARGET_CONSECUTIVE_EPISODES, linestyle='--')
plt.title('Episodios consecutivos con recompensa clipped > 20')
plt.xlabel('Steps')
plt.ylabel('Consecutivos')
plt.legend()
plt.grid(True)
plt.show()

if 'exploration_rate' in df.columns:
    plt.figure(figsize=(10, 4))
    plt.plot(steps, df['exploration_rate'])
    plt.title('Evolución de epsilon / exploration_rate')
    plt.xlabel('Steps')
    plt.ylabel('Exploration rate')
    plt.grid(True)
    plt.show()

```



```
print('Mejor max_consec:', int(max(max_consec)) if len(max_consec) else 0)
print('Mejor avg_clip50:', float(max(avg50)) if len(avg50) else 0.0)
```

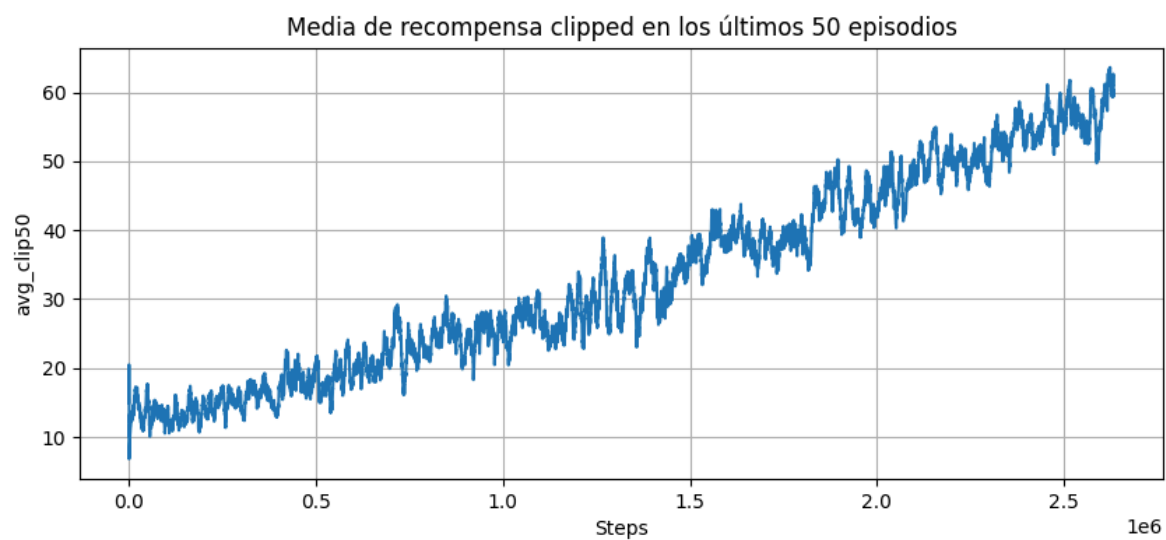
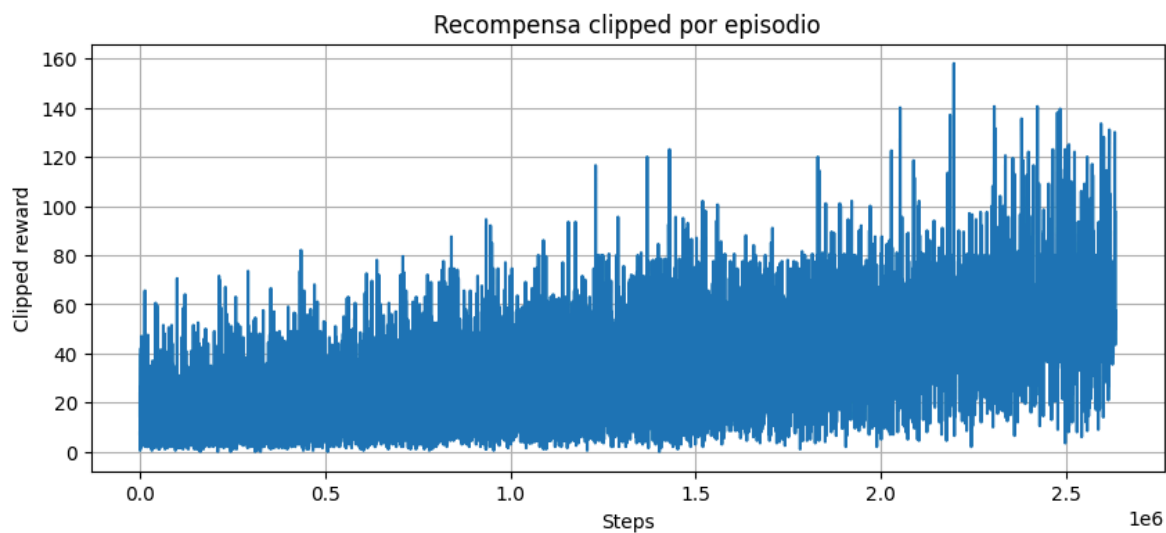
Filas registradas: 11961

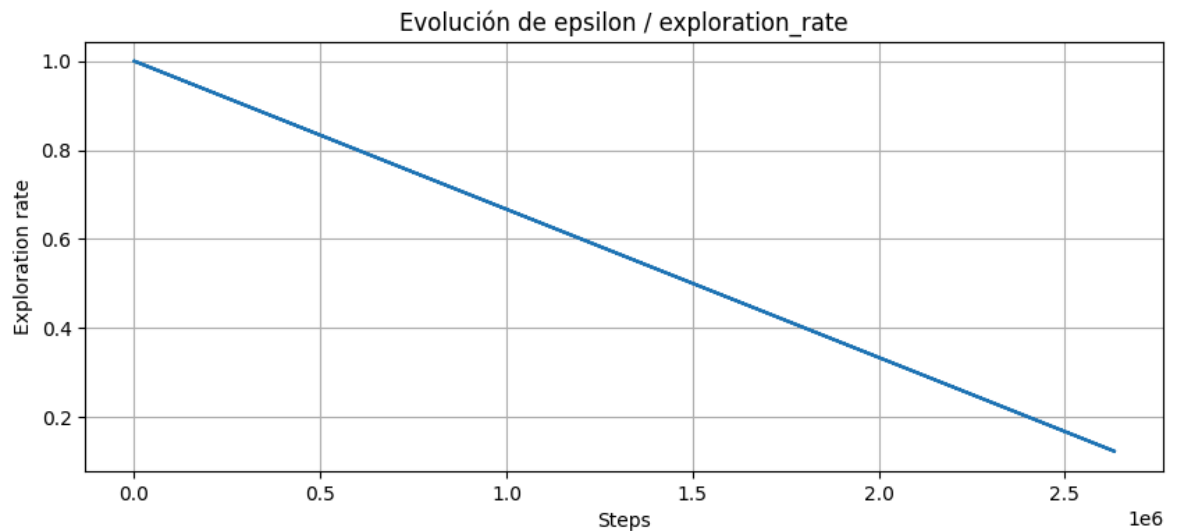
Columnas: ['episode', 'total\_steps', 'episode\_reward', 'clipped\_episode\_reward', 'consecutive\_clipped\_over\_20', 'max\_consecutive\_clipped\_over\_20', 'avg\_clip50', 'duration', 'exploration\_rate']

|              | episode | total_steps | episode_reward | clipped_episode_reward | consecutive_clipped_ |
|--------------|---------|-------------|----------------|------------------------|----------------------|
| <b>11951</b> | 11921   | 2630487     | 525.0          | 52.5                   |                      |
| <b>11952</b> | 11922   | 2630696     | 595.0          | 59.5                   |                      |
| <b>11953</b> | 11923   | 2631025     | 520.0          | 52.0                   |                      |
| <b>11954</b> | 11924   | 2631553     | 1300.0         | 130.0                  |                      |
| <b>11955</b> | 11925   | 2631852     | 530.0          | 53.0                   |                      |
| <b>11956</b> | 11926   | 2632170     | 550.0          | 55.0                   |                      |
| <b>11957</b> | 11927   | 2632426     | 500.0          | 50.0                   |                      |
| <b>11958</b> | 11928   | 2632752     | 575.0          | 57.5                   |                      |
| <b>11959</b> | 11929   | 2632938     | 435.0          | 43.5                   |                      |
| <b>11960</b> | 11930   | 2633424     | 975.0          | 97.5                   |                      |



|          | metrica                 | valor        |
|----------|-------------------------|--------------|
| <b>0</b> | episodios_registrados   | 1.196100e+04 |
| <b>1</b> | ultimo_step             | 2.633424e+06 |
| <b>2</b> | mejor_clip              | 1.580000e+02 |
| <b>3</b> | mejor_avg_clip50        | 6.361000e+01 |
| <b>4</b> | mejor_max_consec        | 1.000000e+02 |
| <b>5</b> | ultimo_avg_clip50       | 6.029000e+01 |
| <b>6</b> | ultimo_exploration_rate | 1.230701e-01 |





Mejor max\_consec: 100  
 Mejor avg\_clip50: 63.61

```
In [ ]: # =====
# 15. Evaluación visual rápida
# =====
# Si existe el modelo completo de SB3, se carga.
# Si no, se usa el modelo actual de memoria.

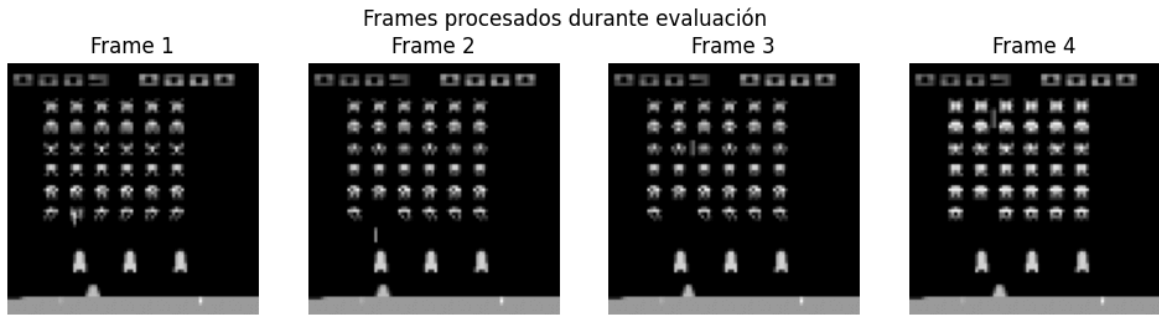
if MODEL_LATEST.exists():
    try:
        model = DQN.load(str(MODEL_LATEST), env=env, device='auto')
        print('Modelo cargado para evaluación:', MODEL_LATEST)
    except Exception as e:
        print('No se pudo cargar MODEL_LATEST. Se usa el modelo actual en memoria.')
        print('Error:', repr(e))
else:
    print('No existe MODEL_LATEST. Se usa el modelo actual en memoria.')

obs = env.reset()
for _ in range(10):
    action, _ = model.predict(obs, deterministic=True)
    obs, reward, done, info = env.step(action)

obs0 = obs[0]
if obs0.shape[0] in [1, 3, 4, 12]:
    frames = obs0
else:
    frames = np.transpose(obs0, (2, 0, 1))

num_frames = min(4, frames.shape[0])
plt.figure(figsize=(12, 3))
for i in range(num_frames):
    plt.subplot(1, num_frames, i + 1)
    plt.imshow(frames[i], cmap='gray')
    plt.title(f'Frame {i + 1}')
    plt.axis('off')
plt.suptitle('Frames procesados durante evaluación')
plt.show()
```

No se pudo cargar MODEL\_LATEST. Se usa el modelo actual en memoria.  
 Error: ValueError("Error: the file <\_io.BufferedReader name='/content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_spaceinvaders\_v0\_sb3\_la test.zip'> wasn't a zip-file")



```
In [ ]: # =====
# 16. Cerrar entorno
# =====

# env.close()
```

```
In [ ]: # =====
# 17. Comprobación final de ficheros y resultados
# =====

print('OUTPUT_DIR:', OUTPUT_DIR)
print('Existe OUTPUT_DIR:', OUTPUT_DIR.exists())
print('MODEL_LATEST:', MODEL_LATEST, '| existe:', MODEL_LATEST.exists())
print('POLICY_LATEST:', POLICY_LATEST, '| existe:', POLICY_LATEST.exists())
print('LOG_FILENAME:', LOG_FILENAME, '| existe:', LOG_FILENAME.exists())
print('CSV_LOG_FILENAME:', CSV_LOG_FILENAME, '| existe:', CSV_LOG_FILENAME.exists())
print('TENSORBOARD_DIR:', TENSORBOARD_DIR, '| existe:', TENSORBOARD_DIR.exists())
print('MONITOR_DIR:', MONITOR_DIR, '| existe:', MONITOR_DIR.exists())

if CSV_LOG_FILENAME.exists():
    df_check = pd.read_csv(CSV_LOG_FILENAME)
    print('\nResumen del log CSV')
    print('Episodios registrados:', len(df_check))

    if len(df_check) > 0:
        print('Último episodio:', int(df_check['episode'].iloc[-1]))
        print('Último step:', int(df_check['total_steps'].iloc[-1]))
        print('Mejor clipped_episode_reward:', float(df_check['clipped_episode_r']))
        print('Mejor avg_clip50:', float(df_check['avg_clip50'].max()))
        print('Mejor max_consec:', int(df_check['max_consecutive_clipped_over_20']))
        print('Último avg_clip50:', float(df_check['avg_clip50'].iloc[-1]))

        display(df_check.tail(5))
    else:
        print('\nNo se encontró el CSV de entrenamiento. Ejecuta el bloque de entren

if LOG_FILENAME.exists():
    with open(LOG_FILENAME, 'r') as f:
        data_check = json.load(f)
        print('\nResumen del log JSON')
        print('Claves:', list(data_check.keys()))
        print('Episodios en JSON:', len(data_check.get('episode', [])))
else:
    print('\nNo se encontró el JSON de entrenamiento.')

# Listado breve de ficheros creados en la carpeta del proyecto.
print('\nFicheros principales en OUTPUT_DIR:')
for p in sorted(OUTPUT_DIR.glob('*')):
    if p.is_file():

```

```

        print(f'- {p.name} ({p.stat().st_size / 1024:.1f} KB)')
    else:
        print(f'- {p.name}/')

```

OUTPUT\_DIR: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0  
 Existe OUTPUT\_DIR: True  
 MODEL\_LATEST: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_spaceinvaders\_v0\_sb3\_latest.zip | existe: True  
 POLICY\_LATEST: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_spaceinvaders\_v0\_policy\_latest.pt | existe: True  
 LOG\_FILENAME: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_SpaceInvaders-v0\_sb3\_log.json | existe: True  
 CSV\_LOG\_FILENAME: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_SpaceInvaders-v0\_sb3\_log.csv | existe: True  
 TENSORBOARD\_DIR: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/tensorboard | existe: True  
 MONITOR\_DIR: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/monitor | existe: True

Resumen del log CSV  
 Episodios registrados: 11961  
 Último episodio: 11930  
 Último step: 2633424  
 Mejor clipped\_episode\_reward: 158.0  
 Mejor avg\_clip50: 63.61  
 Mejor max\_consec: 100  
 Último avg\_clip50: 60.29

|              | episode | total_steps | episode_reward | clipped_episode_reward | consecutive_clipped |
|--------------|---------|-------------|----------------|------------------------|---------------------|
| <b>11956</b> | 11926   | 2632170     | 550.0          | 55.0                   |                     |
| <b>11957</b> | 11927   | 2632426     | 500.0          | 50.0                   |                     |
| <b>11958</b> | 11928   | 2632752     | 575.0          | 57.5                   |                     |
| <b>11959</b> | 11929   | 2632938     | 435.0          | 43.5                   |                     |
| <b>11960</b> | 11930   | 2633424     | 975.0          | 97.5                   |                     |

Resumen del log JSON

Claves: ['episode', 'total\_steps', 'episode\_reward', 'clipped\_episode\_reward', 'consecutive\_clipped\_over\_20', 'max\_consecutive\_clipped\_over\_20', 'avg\_clip50', 'duration', 'exploration\_rate']  
 Episodios en JSON: 11931

Ficheros principales en OUTPUT\_DIR:

- .DS\_Store (6.0 KB)
- dqn\_SpaceInvaders-v0\_sb3\_log.csv (819.1 KB)
- dqn\_SpaceInvaders-v0\_sb3\_log.json (1341.4 KB)
- dqn\_spaceinvaders\_v0\_policy\_latest.pt (26381.2 KB)
- dqn\_spaceinvaders\_v0\_sb3\_latest.zip (0.0 KB)
- monitor/
- tensorboard/

```

In [ ]: # =====
        # Entorno de evaluación real
        # =====

```

```

import gym
import numpy as np
from stable_baselines3.common.atari_wrappers import (
    MaxAndSkipEnv,
    FireResetEnv,
    WarpFrame
)
from stable_baselines3.common.vec_env import DummyVecEnv, VecFrameStack, VecTran

def create_eval_env():
    def make_env():
        env = gym.make(ENV_ID)

        env.seed(SEED)
        env = MaxAndSkipEnv(env, skip=4)

        if "FIRE" in env.unwrapped.get_action_meanings():
            env = FireResetEnv(env)

        env = WarpFrame(env)

        return env

    env = DummyVecEnv([make_env])
    env = VecFrameStack(env, n_stack=4)
    env = VecTransposeImage(env)

    return env

```

```

In [ ]: # =====
# Cargar policy aprendida
# =====

import torch
from stable_baselines3 import DQN

POLICY_LATEST = OUTPUT_DIR / "dqn_spaceinvaders_v0_policy_latest.pt"

eval_env = create_eval_env()

model = DQN(
    policy="CnnPolicy",
    env=eval_env,
    learning_rate=LEARNING_RATE,
    buffer_size=BUFFER_SIZE,
    learning_starts=LEARNING_STARTS,
    batch_size=BATCH_SIZE,
    gamma=GAMMA,
    train_freq=TRAIN_FREQ,
    target_update_interval=TARGET_UPDATE_INTERVAL,
    exploration_initial_eps=EXPLORATION_INITIAL_EPS,
    exploration_final_eps=EXPLORATION_FINAL_EPS,
    exploration_fraction=EXPLORATION_FRACTION,
    policy_kwargs=policy_kwargs,
    verbose=1,
    seed=SEED,
    device="auto"
)

checkpoint = torch.load(POLICY_LATEST, map_location=model.device)

```

```

model.policy.load_state_dict(checkpoint["policy_state_dict"])
model.num_timesteps = checkpoint["num_timesteps"]
model.exploration_rate = checkpoint["exploration_rate"]

print("Policy cargada correctamente")
print("Timesteps:", model.num_timesteps)
print("Exploration rate:", model.exploration_rate)

```

Using cuda device

Policy cargada correctamente

Timesteps: 2633424

Exploration rate: 0.12307014100000035

```

In [ ]: # =====
# Testing sin entrenamiento
# =====

TEST_EPISODES = 120

rewards = []
clipped_rewards = []

for ep in range(TEST_EPISODES):
    obs = eval_env.reset()
    done = False
    total_reward = 0.0

    while not done:
        action, _ = model.predict(obs, deterministic=True)
        obs, reward, done, infos = eval_env.step(action)
        total_reward += float(reward[0])
        done = bool(done[0])

    clipped_reward = total_reward / 10.0
    rewards.append(total_reward)
    clipped_rewards.append(clipped_reward)

    print(
        f"TEST EP={ep+1:03d} | "
        f"reward={total_reward:.0f} | "
        f"clip={clipped_reward:.0f}"
    )

print("Reward medio:", np.mean(rewards))
print("Clip medio:", np.mean(clipped_rewards))
print("Reward máximo:", np.max(rewards))
print("Clip máximo:", np.max(clipped_rewards))

eval_env.close()

```

|      |        |  |             |  |          |
|------|--------|--|-------------|--|----------|
| TEST | EP=001 |  | reward=900  |  | clip=90  |
| TEST | EP=002 |  | reward=600  |  | clip=60  |
| TEST | EP=003 |  | reward=575  |  | clip=58  |
| TEST | EP=004 |  | reward=570  |  | clip=57  |
| TEST | EP=005 |  | reward=800  |  | clip=80  |
| TEST | EP=006 |  | reward=600  |  | clip=60  |
| TEST | EP=007 |  | reward=545  |  | clip=54  |
| TEST | EP=008 |  | reward=555  |  | clip=56  |
| TEST | EP=009 |  | reward=360  |  | clip=36  |
| TEST | EP=010 |  | reward=575  |  | clip=58  |
| TEST | EP=011 |  | reward=1180 |  | clip=118 |
| TEST | EP=012 |  | reward=600  |  | clip=60  |
| TEST | EP=013 |  | reward=575  |  | clip=58  |
| TEST | EP=014 |  | reward=545  |  | clip=54  |
| TEST | EP=015 |  | reward=600  |  | clip=60  |
| TEST | EP=016 |  | reward=915  |  | clip=92  |
| TEST | EP=017 |  | reward=800  |  | clip=80  |
| TEST | EP=018 |  | reward=575  |  | clip=58  |
| TEST | EP=019 |  | reward=600  |  | clip=60  |
| TEST | EP=020 |  | reward=605  |  | clip=60  |
| TEST | EP=021 |  | reward=775  |  | clip=78  |
| TEST | EP=022 |  | reward=575  |  | clip=58  |
| TEST | EP=023 |  | reward=1295 |  | clip=130 |
| TEST | EP=024 |  | reward=600  |  | clip=60  |
| TEST | EP=025 |  | reward=425  |  | clip=42  |
| TEST | EP=026 |  | reward=1000 |  | clip=100 |
| TEST | EP=027 |  | reward=605  |  | clip=60  |
| TEST | EP=028 |  | reward=600  |  | clip=60  |
| TEST | EP=029 |  | reward=800  |  | clip=80  |
| TEST | EP=030 |  | reward=600  |  | clip=60  |
| TEST | EP=031 |  | reward=980  |  | clip=98  |
| TEST | EP=032 |  | reward=605  |  | clip=60  |
| TEST | EP=033 |  | reward=545  |  | clip=54  |
| TEST | EP=034 |  | reward=630  |  | clip=63  |
| TEST | EP=035 |  | reward=755  |  | clip=76  |
| TEST | EP=036 |  | reward=545  |  | clip=54  |
| TEST | EP=037 |  | reward=570  |  | clip=57  |
| TEST | EP=038 |  | reward=545  |  | clip=54  |
| TEST | EP=039 |  | reward=485  |  | clip=48  |
| TEST | EP=040 |  | reward=800  |  | clip=80  |
| TEST | EP=041 |  | reward=575  |  | clip=58  |
| TEST | EP=042 |  | reward=600  |  | clip=60  |
| TEST | EP=043 |  | reward=650  |  | clip=65  |
| TEST | EP=044 |  | reward=750  |  | clip=75  |
| TEST | EP=045 |  | reward=600  |  | clip=60  |
| TEST | EP=046 |  | reward=470  |  | clip=47  |
| TEST | EP=047 |  | reward=770  |  | clip=77  |
| TEST | EP=048 |  | reward=800  |  | clip=80  |
| TEST | EP=049 |  | reward=895  |  | clip=90  |
| TEST | EP=050 |  | reward=785  |  | clip=78  |
| TEST | EP=051 |  | reward=380  |  | clip=38  |
| TEST | EP=052 |  | reward=545  |  | clip=54  |
| TEST | EP=053 |  | reward=545  |  | clip=54  |
| TEST | EP=054 |  | reward=775  |  | clip=78  |
| TEST | EP=055 |  | reward=570  |  | clip=57  |
| TEST | EP=056 |  | reward=750  |  | clip=75  |
| TEST | EP=057 |  | reward=520  |  | clip=52  |
| TEST | EP=058 |  | reward=1155 |  | clip=116 |
| TEST | EP=059 |  | reward=570  |  | clip=57  |
| TEST | EP=060 |  | reward=745  |  | clip=74  |



|      |        |  |             |  |          |
|------|--------|--|-------------|--|----------|
| TEST | EP=061 |  | reward=580  |  | clip=58  |
| TEST | EP=062 |  | reward=545  |  | clip=54  |
| TEST | EP=063 |  | reward=570  |  | clip=57  |
| TEST | EP=064 |  | reward=415  |  | clip=42  |
| TEST | EP=065 |  | reward=550  |  | clip=55  |
| TEST | EP=066 |  | reward=815  |  | clip=82  |
| TEST | EP=067 |  | reward=445  |  | clip=44  |
| TEST | EP=068 |  | reward=600  |  | clip=60  |
| TEST | EP=069 |  | reward=1175 |  | clip=118 |
| TEST | EP=070 |  | reward=575  |  | clip=58  |
| TEST | EP=071 |  | reward=815  |  | clip=82  |
| TEST | EP=072 |  | reward=515  |  | clip=52  |
| TEST | EP=073 |  | reward=575  |  | clip=58  |
| TEST | EP=074 |  | reward=575  |  | clip=58  |
| TEST | EP=075 |  | reward=605  |  | clip=60  |
| TEST | EP=076 |  | reward=540  |  | clip=54  |
| TEST | EP=077 |  | reward=600  |  | clip=60  |
| TEST | EP=078 |  | reward=510  |  | clip=51  |
| TEST | EP=079 |  | reward=500  |  | clip=50  |
| TEST | EP=080 |  | reward=465  |  | clip=46  |
| TEST | EP=081 |  | reward=545  |  | clip=54  |
| TEST | EP=082 |  | reward=605  |  | clip=60  |
| TEST | EP=083 |  | reward=555  |  | clip=56  |
| TEST | EP=084 |  | reward=770  |  | clip=77  |
| TEST | EP=085 |  | reward=1005 |  | clip=100 |
| TEST | EP=086 |  | reward=600  |  | clip=60  |
| TEST | EP=087 |  | reward=555  |  | clip=56  |
| TEST | EP=088 |  | reward=600  |  | clip=60  |
| TEST | EP=089 |  | reward=540  |  | clip=54  |
| TEST | EP=090 |  | reward=600  |  | clip=60  |
| TEST | EP=091 |  | reward=600  |  | clip=60  |
| TEST | EP=092 |  | reward=1055 |  | clip=106 |
| TEST | EP=093 |  | reward=570  |  | clip=57  |
| TEST | EP=094 |  | reward=600  |  | clip=60  |
| TEST | EP=095 |  | reward=515  |  | clip=52  |
| TEST | EP=096 |  | reward=580  |  | clip=58  |
| TEST | EP=097 |  | reward=785  |  | clip=78  |
| TEST | EP=098 |  | reward=575  |  | clip=58  |
| TEST | EP=099 |  | reward=575  |  | clip=58  |
| TEST | EP=100 |  | reward=1030 |  | clip=103 |
| TEST | EP=101 |  | reward=550  |  | clip=55  |
| TEST | EP=102 |  | reward=1150 |  | clip=115 |
| TEST | EP=103 |  | reward=630  |  | clip=63  |
| TEST | EP=104 |  | reward=575  |  | clip=58  |
| TEST | EP=105 |  | reward=515  |  | clip=52  |
| TEST | EP=106 |  | reward=725  |  | clip=72  |
| TEST | EP=107 |  | reward=575  |  | clip=58  |
| TEST | EP=108 |  | reward=545  |  | clip=54  |
| TEST | EP=109 |  | reward=1405 |  | clip=140 |
| TEST | EP=110 |  | reward=600  |  | clip=60  |
| TEST | EP=111 |  | reward=600  |  | clip=60  |
| TEST | EP=112 |  | reward=415  |  | clip=42  |
| TEST | EP=113 |  | reward=470  |  | clip=47  |
| TEST | EP=114 |  | reward=1170 |  | clip=117 |
| TEST | EP=115 |  | reward=575  |  | clip=58  |
| TEST | EP=116 |  | reward=1205 |  | clip=120 |
| TEST | EP=117 |  | reward=585  |  | clip=58  |
| TEST | EP=118 |  | reward=570  |  | clip=57  |
| TEST | EP=119 |  | reward=600  |  | clip=60  |
| TEST | EP=120 |  | reward=520  |  | clip=52  |

Reward medio: 662.25  
Clip medio: 66.225  
Reward máximo: 1405.0  
Clip máximo: 140.5

## Vídeo del juego

Se graban dos vídeos con la política **del modelo base ya entrenado** (determinista, cargada previamente desde su checkpoint con la celda "Cargar policy aprendida"). Esto sustituye a la renderización en vivo de gym, que no es compatible con Colab:

1. **Una partida individual** (el wrapper de Atari corta el episodio al perder una vida).
2. **10 episodios seguidos** en un único mp4, para observar el comportamiento del agente de forma más representativa.

Los vídeos quedan embebidos en el notebook. Las variantes de la Parte 2 se comparan de forma **cuantitativa** mediante las gráficas y el test de 120 episodios, por lo que no aporta información adicional grabar una partida por cada variante.

```
In [ ]: # =====  
# Grabar una partida en vídeo  
# =====  
  
from stable_baselines3.common.vec_env import VecFrameStack, VecVideoRecorder  
from IPython.display import display  
import shutil  
  
VIDEO_DIR = OUTPUT_DIR / 'video_1'  
if VIDEO_DIR.exists():  
    shutil.rmtree(VIDEO_DIR)  
VIDEO_DIR.mkdir(parents=True, exist_ok=True)  
  
video_env = make_atari_env(  
    ENV_ID,  
    n_envs=1,  
    seed=SEED,  
    wrapper_kwargs=dict(clip_reward=True)  
)  
video_env = VecFrameStack(video_env, n_stack=4)  
video_env = VecVideoRecorder(  
    video_env,  
    str(VIDEO_DIR),  
    record_video_trigger=lambda step: step == 0,  
    video_length=5_000,  
    name_prefix='sb3-spaceinvaders'  
)  
  
obs = video_env.reset()  
for _ in range(5_000):  
    action, _ = model.predict(obs, deterministic=True)  
    obs, rewards, dones, infos = video_env.step(action)  
    if bool(dones[0]):  
        break  
  
video_env.close()
```

```
video_files = sorted(VIDEO_DIR.glob('*.mp4'))
print('Videos generados:', [video_file.name for video_file in video_files])
display(video_files[0] if video_files else 'No se ha generado ningún vídeo.')
```

Videos generados: ['sb3-spaceinvaders-step-0-to-step-5000.mp4']

PosixPath('/content/gdrive/My Drive/08\_MIAR/sb3/grupo/A\_Refuerzo/datos/sb3\_1\_spaceinvaders\_v0/video\_1/sb3-spaceinvaders-step-0-to-step-5000.mp4')

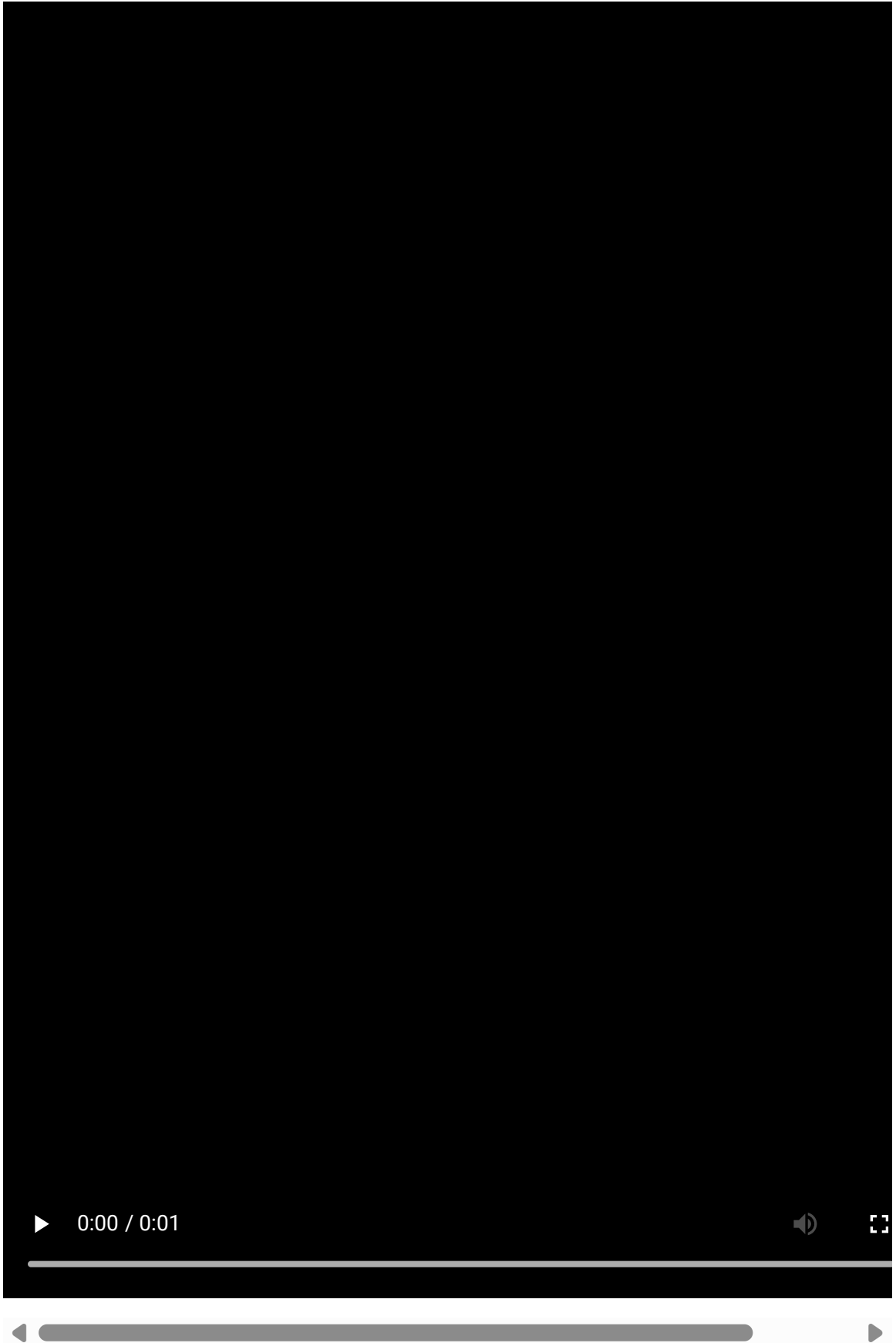
```
In [ ]: # Reproducir el MP4 generado dentro del notebook
import io
import base64
from IPython.display import HTML

video_files = sorted(VIDEO_DIR.glob('*.mp4'))
if not video_files:
    raise FileNotFoundError(f'No se encontró ningún MP4 en {VIDEO_DIR}')

video_path = video_files[0]
with io.open(video_path, 'r+b') as video_file:
    video = video_file.read()

encoded = base64.b64encode(video).decode('ascii')
HTML(
    data=f'''
    <video width="640" height="auto" alt="space-invaders" controls>
      <source src="data:video/mp4;base64,{encoded}" type="video/mp4" />
    </video>'''
)
```

Out[ ]:



```
In [ ]: # =====  
# Grabar 10 partidas en un único mp4  
# =====  
  
from stable_baselines3.common.vec_env import VecFrameStack, VecVideoRecorder  
import shutil  
  
N_EPISODES = 10
```

```

MAX_STEPS_PER_EP = 5_000 # margen por episodio, ajusta si Space Invaders dura m

VIDEO_DIR = OUTPUT_DIR / 'video_10'
if VIDEO_DIR.exists():
    shutil.rmtree(VIDEO_DIR)
VIDEO_DIR.mkdir(parents=True, exist_ok=True)

video_env = make_atari_env(
    ENV_ID,
    n_envs=1,
    seed=SEED,
    wrapper_kwargs=dict(clip_reward=False)
)
video_env = VecFrameStack(video_env, n_stack=4)
video_env = VecVideoRecorder(
    video_env,
    str(VIDEO_DIR),
    record_video_trigger=lambda step: step == 0,
    video_length=N_EPISODES * MAX_STEPS_PER_EP, # cubre las 10 partidas seguidas
    name_prefix='sb3-spaceinvaders-10eps'
)

obs = video_env.reset()
episodes_done = 0
step = 0
while episodes_done < N_EPISODES:
    action, _ = model.predict(obs, deterministic=True)
    obs, rewards, dones, infos = video_env.step(action)
    step += 1
    if bool(dones[0]):
        episodes_done += 1 # el VecEnv ya resetea internamente, obs ya es del n
video_env.close()

```

```

In [ ]: # Reproducir el MP4 generado dentro del notebook
import io
import base64
from IPython.display import HTML

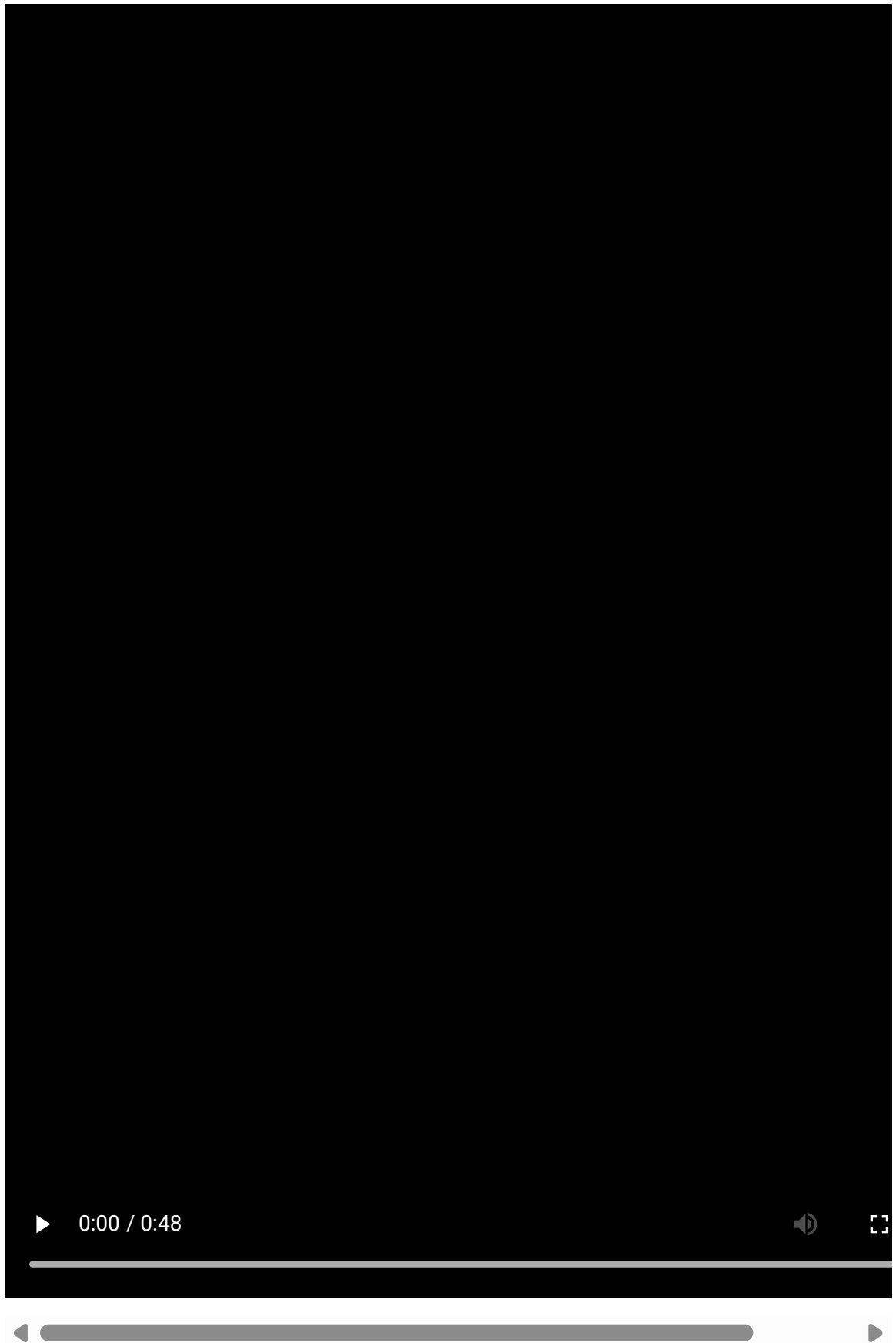
video_files = sorted(VIDEO_DIR.glob('*.mp4'))
if not video_files:
    raise FileNotFoundError(f'No se encontró ningún MP4 en {VIDEO_DIR}')

video_path = video_files[0]
with io.open(video_path, 'r+b') as video_file:
    video = video_file.read()

encoded = base64.b64encode(video).decode('ascii')
HTML(
    data=f'''
    <video width="640" height="auto" alt="space-invaders" controls>
      <source src="data:video/mp4;base64,{encoded}" type="video/mp4" />
    </video>'''
)

```

Out[ ]:



---

PARTE 2 (ampliación). Propuestas de mejora sobre el DQN base

El DQN entrenado en las secciones anteriores actúa como **referencia (baseline)**. Sobre él se implementan y comparan **tres propuestas de mejora** independientes. No se combinan entre sí a propósito: así se puede aislar y discutir el efecto de cada una.

| # | Propuesta                                       | Qué cambia                                                                                                                                                             | Motivación                                                                                                                                                                                                                        |
|---|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Double DQN</b><br>(van Hasselt et al., 2016) | El target pasa de <code>max_a Q_target(s',a)</code> a <code>Q_target(s', argmax_a Q_online(s',a))</code>                                                               | El operador <code>max</code> sobre estimaciones ruidosas sobreestima sistemáticamente los valores Q; separar la <i>selección</i> de la acción (red online) de su <i>evaluación</i> (red target) reduce ese sesgo                  |
| 2 | <b>Dueling DQN</b><br>(Wang et al., 2016)       | La cabeza de la red se divide en dos streams: <code>V(s)</code> y <code>A(s,a)</code> , con <code>Q = V + A - mean(A)</code>                                           | En muchos estados de Space Invaders la acción elegida apenas cambia el valor del estado (p. ej. cuando no hay proyectiles cerca); estimar <code>V(s)</code> por separado propaga aprendizaje a todas las acciones a la vez        |
| 3 | <b>Exploración ajustada</b>                     | <code>exploration_fraction</code> 0.30 → 0.10, <code>ε_final</code> 0.001 → 0.01, <code>target_update_interval</code> 10k → 5k, <code>learning_starts</code> 50k → 25k | El run base terminó con $\epsilon \approx 0.12$ : el 12 % de las acciones seguían siendo aleatorias incluso al final del entrenamiento. Un annealing más corto, adaptado al presupuesto real, permite explotar antes lo aprendido |

**Protocolo de comparación justa:** misma semilla (42), mismo preprocesado y misma red convolucional base, mismo presupuesto máximo de steps, mismo criterio de parada (100 episodios consecutivos con recompensa clipped > 20) y mismo test final (120 episodios con política determinista).

```
In [ ]: # =====
# M1. Infraestructura multi-run: rutas por variante y helpers
# =====
# Cada variante escribe sus logs y checkpoints en su propia carpeta de Drive,
# reutilizando SB3TrainingLogger y ConsecutiveRewardCallback sin cambios.
# La variante 'base' apunta a la carpeta del entrenamiento ya realizado.

RUN_NAMES = ['base', 'double', 'dueling', 'tuned']

RUN_LABELS = {
    'base': 'DQN base',
    'double': 'Double DQN',
    'dueling': 'Dueling DQN',
    'tuned': 'DQN exploración ajustada',
}

def setup_run_paths(run_name):
    if run_name == 'base':
        run_dir = OUTPUT_DIR
    else:
        run_dir = Path(drive_root) / 'datos' / (PROJECT_NAME + '_' + run_name)
    run_dir.mkdir(parents=True, exist_ok=True)
```

```

(run_dir / 'monitor').mkdir(parents=True, exist_ok=True)
(run_dir / 'tensorboard').mkdir(parents=True, exist_ok=True)
return {
    'dir': run_dir,
    'policy': run_dir / 'dqn_spaceinvaders_v0_policy_latest.pt',
    'model_zip': run_dir / 'dqn_spaceinvaders_v0_sb3_latest.zip',
    'json': run_dir / 'dqn_SpaceInvaders-v0_sb3_log.json',
    'csv': run_dir / 'dqn_SpaceInvaders-v0_sb3_log.csv',
    'monitor': run_dir / 'monitor',
    'tensorboard': run_dir / 'tensorboard',
    'test_json': run_dir / 'test_results.json',
}

RUN_PATHS = {name: setup_run_paths(name) for name in RUN_NAMES}

def create_train_env(monitor_dir):
    env = make_atari_env(
        ENV_ID,
        n_envs=1,
        seed=SEED,
        monitor_dir=str(monitor_dir),
        wrapper_kwargs=dict(clip_reward=True)
    )
    return VecFrameStack(env, n_stack=4)

def train_variant(run_name, model, total_timesteps=TOTAL_TIMESTEPS):
    paths = RUN_PATHS[run_name]
    training_logger = SB3TrainingLogger(json_path=paths['json'], csv_path=paths['csv'],
    callback = ConsecutiveRewardCallback(training_logger=training_logger, verbose=True)

    model.learn(
        total_timesteps=total_timesteps,
        callback=callback,
        reset_num_timesteps=True,
        tb_log_name='DQN_' + run_name,
        log_interval=100
    )

    # Siempre se guarda el checkpoint PyTorch (el formato que sabemos cargar);
    # el .zip de SB3 se intenta como extra (en el run base fallaba en Colab).
    save_policy_checkpoint(model, paths['policy'])
    print('Checkpoint PyTorch guardado en:', paths['policy'])
    try:
        model.save(str(paths['model_zip']))
        print('Modelo SB3 guardado en:', paths['model_zip'])
    except Exception as e:
        print('model.save() falló (no es crítico, se usa el .pt):', repr(e))

    return model

print('Rutas por variante creadas:')
for name in RUN_NAMES:
    print(' -', RUN_LABELS[name], '->', RUN_PATHS[name]['dir'])

```



Rutas por variante creadas:

- DQN base -> /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0
- Double DQN -> /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0\_double
- Dueling DQN -> /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0\_dueling
- DQN exploración ajustada -> /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0\_tuned

```
In [ ]: # =====
# M2. Propuesta 1: Double DQN
# =====
# DQN estándar usa como target:  $r + \gamma * \max_a Q_{\text{target}}(s', a)$ .
# El max sobre estimaciones con ruido produce sobreestimación sistemática
# de los valores Q. Double DQN (van Hasselt et al., 2016) desacopla:
#   - SELECCIÓN de la acción con la red online:  $a^* = \text{argmax}_a Q_{\text{online}}(s', a)$ 
#   - EVALUACIÓN de esa acción con la red target:  $Q_{\text{target}}(s', a^*)$ 
# Solo cambia el cálculo del target; el resto (replay buffer, epsilon-greedy,
# red target) es idéntico al DQN base, por lo que basta sobrescribir train().

import torch.nn.functional as F

try:
    from stable_baselines3.common import logger as sb3_logger
except Exception:
    sb3_logger = None

class DoubleDQN(DQN):

    def train(self, gradient_steps, batch_size=100):
        self._update_learning_rate(self.policy.optimizer)

        losses = []
        for _ in range(gradient_steps):
            replay_data = self.replay_buffer.sample(batch_size, env=self._vec_no

            with th.no_grad():
                # Selección con la red online (única diferencia con DQN)
                next_actions = self.q_net(replay_data.next_observations).argmax(
                # Evaluación con la red target
                next_q = th.gather(self.q_net_target(replay_data.next_observatio
                target_q = replay_data.rewards + (1 - replay_data.dones) * self.

            current_q = th.gather(self.q_net(replay_data.observations), dim=1, i

            loss = F.smooth_l1_loss(current_q, target_q)
            losses.append(loss.item())

            self.policy.optimizer.zero_grad()
            loss.backward()
            th.nn.utils.clip_grad_norm_(self.policy.parameters(), self.max_grad_
            self.policy.optimizer.step()

        self._n_updates += gradient_steps

        log = getattr(self, 'logger', None) or sb3_logger
        if log is not None:
            log.record('train/n_updates', self._n_updates, exclude='tensorboard')
```

```

log.record('train/loss', float(np.mean(losses)))

print('DoubleDQN definido.')

```

DoubleDQN definido.

```

In [ ]: # =====
# M3. Propuesta 2: Dueling DQN
# =====
# Q(s,a) se descompone en dos streams sobre las mismas features convolucionales:
#   V(s): valor del estado
#   A(s,a): ventaja de cada acción respecto a la media
#   Q(s,a) = V(s) + A(s,a) - mean_a' A(s,a') (Wang et al., 2016)
# Restar la media de las ventajas hace la descomposición identificable.
# Se implementa sustituyendo la cabeza lineal de QNetwork por los dos streams;
# el extractor convolucional (CustomCNN84) no cambia.

from stable_baselines3.dqn.policies import CnnPolicy as DQNCnnPolicy
from stable_baselines3.dqn.policies import QNetwork

class DuelingQNetwork(QNetwork):

    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        action_dim = self.action_space.n

        # Se anula la cabeza lineal original y se crean los dos streams
        self.q_net = nn.Identity()
        self.value_stream = nn.Sequential(
            nn.Linear(self.features_dim, 256),
            nn.ReLU(),
            nn.Linear(256, 1)
        )
        self.advantage_stream = nn.Sequential(
            nn.Linear(self.features_dim, 256),
            nn.ReLU(),
            nn.Linear(256, action_dim)
        )

    def forward(self, obs):
        features = self.extract_features(obs)
        value = self.value_stream(features)
        advantage = self.advantage_stream(features)
        return value + advantage - advantage.mean(dim=1, keepdim=True)

class DuelingCnnPolicy(DQNCnnPolicy):

    def make_q_net(self):
        net_args = self._update_features_extractor(self.net_args, features_extra)
        return DuelingQNetwork(**net_args).to(self.device)

print('DuelingCnnPolicy definida.')

```

DuelingCnnPolicy definida.

```

In [ ]: # =====
# M4. Propuesta 3 (exploración ajustada) y constructor de modelos

```

```

# =====
# El run base se detuvo (criterio de 100 consecutivos) en ~2.6M steps con
# epsilon ~ 0.12: el schedule (30% de 10M steps) era demasiado largo para el
# presupuesto real. La propuesta 3 usa el mismo algoritmo DQN base pero:
# - annealing de epsilon en el 10% del presupuesto y epsilon final 0.01
# - red target actualizada cada 5k steps (en lugar de 10k)
# - learning_starts en 25k (empieza a aprender antes)

TUNED_PARAMS = dict(
    exploration_fraction=0.10,
    exploration_final_eps=0.01,
    target_update_interval=5_000,
    learning_starts=25_000,
)

def build_model(run_name, env, tensorboard_dir=None, for_eval=False):
    # for_eval=True reduce el replay buffer a 1k transiciones para no
    # reservar ~2.8 GB de RAM cuando solo se quiere evaluar la política.
    common = dict(
        env=env,
        learning_rate=LEARNING_RATE,
        buffer_size=1_000 if for_eval else BUFFER_SIZE,
        learning_starts=LEARNING_STARTS,
        batch_size=BATCH_SIZE,
        gamma=GAMMA,
        train_freq=TRAIN_FREQ,
        target_update_interval=TARGET_UPDATE_INTERVAL,
        exploration_initial_eps=EXPLORATION_INITIAL_EPS,
        exploration_final_eps=EXPLORATION_FINAL_EPS,
        exploration_fraction=EXPLORATION_FRACTION,
        policy_kwargs=policy_kwargs,
        tensorboard_log=str(tensorboard_dir) if tensorboard_dir is not None else
        verbose=1,
        seed=SEED,
        device='auto',
    )

    if run_name == 'base':
        return DQN(policy='CnnPolicy', **common)
    if run_name == 'double':
        return DoubleDQN(policy='CnnPolicy', **common)
    if run_name == 'dueling':
        return DQN(policy=DuelingCnnPolicy, **common)
    if run_name == 'tuned':
        common.update(TUNED_PARAMS)
        return DQN(policy='CnnPolicy', **common)

    raise ValueError('Variante desconocida: ' + run_name)

print('build_model listo. Variantes disponibles:', RUN_NAMES)

```

build\_model listo. Variantes disponibles: ['base', 'double', 'dueling', 'tuned']

**Notas de ejecución.** Con la GPU estándar de Colab el run base avanzó a ~260 fps, es decir, el criterio de parada (~2.6M steps) se alcanzó en unas 3 horas. Cada variante se entrena en una celda independiente para poder repartirlas entre sesiones de Colab. Si una sesión se corta, el checkpoint

`.pt` y los logs CSV/JSON quedan guardados en Drive: para reanudar, cargar el `policy_state_dict` en un modelo nuevo y ajustar `exploration_initial_eps` al último `exploration_rate` registrado en el CSV (tal y como indica el enunciado, al retomar un entrenamiento hay que adaptar los parámetros de exploración).

```
In [ ]: # =====  
# M5. Entrenamiento Propuesta 1: Double DQN  
# =====  
  
env_double = create_train_env(RUN_PATHS['double']['monitor'])  
model_double = build_model('double', env_double, RUN_PATHS['double']['tensorboard'])  
model_double = train_variant('double', model_double)  
env_double.close()
```

Se han truncado las últimas 5000 líneas del flujo de salida.

```
-----
EP=3,490 | STEP=595,871 | clip=55 | consec=4 | max_consec=6 | avg_clip50=20.10
EP=3,500 | STEP=597,374 | clip=10 | consec=0 | max_consec=6 | avg_clip50=18.17
EP=3,510 | STEP=599,564 | clip=15 | consec=0 | max_consec=6 | avg_clip50=20.56
-----
```

```
| custom/
|   avg_clip50           | 19.9
|   clipped_episode_r... | 19.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 6
| rollout/
|   ep_len_mean         | 773
|   ep_rew_mean         | 187
|   exploration rate    | 0.8
| time/
|   episodes            | 10500
|   fps                 | 214
|   time_elapsed        | 2794
|   total timesteps     | 600641
| train/
|   learning_rate       | 0.0001
|   loss                | 0.202
|   n_updates           | 137660
-----
```

```
EP=3,520 | STEP=601,661 | clip=14 | consec=0 | max_consec=6 | avg_clip50=21.30
EP=3,530 | STEP=603,282 | clip=28 | consec=1 | max_consec=6 | avg_clip50=20.21
EP=3,540 | STEP=605,012 | clip=30 | consec=1 | max_consec=6 | avg_clip50=17.38
-----
```

```
| custom/
|   avg_clip50           | 18.4
|   clipped_episode_r... | 27.5
|   consecutive_clipp... | 2
|   max_consecutive_c... | 6
| rollout/
|   ep_len_mean         | 781
|   ep_rew_mean         | 183
|   exploration rate    | 0.798
| time/
|   episodes            | 10600
|   fps                 | 214
|   time_elapsed        | 2822
|   total timesteps     | 606646
| train/
|   learning_rate       | 0.0001
|   loss                | 0.0389
|   n_updates           | 139161
-----
```

```
EP=3,550 | STEP=606,793 | clip=14 | consec=0 | max_consec=6 | avg_clip50=18.43
EP=3,560 | STEP=608,889 | clip=6  | consec=0 | max_consec=6 | avg_clip50=17.94
EP=3,570 | STEP=610,569 | clip=8  | consec=0 | max_consec=6 | avg_clip50=16.32
EP=3,580 | STEP=612,611 | clip=32 | consec=2 | max_consec=6 | avg_clip50=18.03
-----
```

```
| custom/
|   avg_clip50           | 18.9
|   clipped_episode_r... | 48
|   consecutive_clipp... | 1
|   max_consecutive_c... | 6
| rollout/
|   ep_len_mean         | 812
-----
```

|                  |        |
|------------------|--------|
| ep_rew_mean      | 195    |
| exploration rate | 0.796  |
| time/            |        |
| episodes         | 10700  |
| fps              | 214    |
| time_elapsed     | 2852   |
| total timesteps  | 613074 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0804 |
| n_updates        | 140768 |

-----

EP=3,590 | STEP=614,818 | clip=21 | consec=2 | max\_consec=6 | avg\_clip50=19.96  
EP=3,600 | STEP=616,361 | clip=5 | consec=0 | max\_consec=6 | avg\_clip50=19.80  
EP=3,610 | STEP=618,588 | clip=50 | consec=1 | max\_consec=6 | avg\_clip50=21.13

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.1   |
| clipped_episode_r... | 9      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 815    |
| ep_rew_mean          | 198    |
| exploration rate     | 0.794  |
| time/                |        |
| episodes             | 10800  |
| fps                  | 214    |
| time_elapsed         | 2885   |
| total timesteps      | 619928 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0351 |
| n_updates            | 142481 |

-----

EP=3,620 | STEP=620,650 | clip=8 | consec=0 | max\_consec=6 | avg\_clip50=22.54  
EP=3,630 | STEP=622,779 | clip=32 | consec=2 | max\_consec=6 | avg\_clip50=22.51  
EP=3,640 | STEP=624,393 | clip=10 | consec=0 | max\_consec=6 | avg\_clip50=20.26

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.1   |
| clipped_episode_r... | 12     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 825    |
| ep_rew_mean          | 206    |
| exploration rate     | 0.791  |
| time/                |        |
| episodes             | 10900  |
| fps                  | 214    |
| time_elapsed         | 2915   |
| total timesteps      | 626195 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0409 |
| n_updates            | 144048 |

-----

EP=3,650 | STEP=626,293 | clip=14 | consec=0 | max\_consec=6 | avg\_clip50=21.31  
EP=3,660 | STEP=627,829 | clip=18 | consec=0 | max\_consec=6 | avg\_clip50=18.99

EP=3,670 | STEP=629,721 | clip=15 | consec=0 | max\_consec=6 | avg\_clip50=18.79  
EP=3,680 | STEP=631,664 | clip=20 | consec=0 | max\_consec=6 | avg\_clip50=18.70

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 18.4   |  |
| clipped_episode_r... | 5      |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 6      |  |
| rollout/             |        |  |
| ep_len_mean          | 799    |  |
| ep_rew_mean          | 201    |  |
| exploration rate     | 0.79   |  |
| time/                |        |  |
| episodes             | 11000  |  |
| fps                  | 214    |  |
| time_elapsed         | 2943   |  |
| total timesteps      | 632008 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.164  |  |
| n_updates            | 145501 |  |

EP=3,690 | STEP=633,231 | clip=20 | consec=0 | max\_consec=6 | avg\_clip50=18.55  
EP=3,700 | STEP=635,099 | clip=18 | consec=0 | max\_consec=6 | avg\_clip50=17.77  
EP=3,710 | STEP=636,858 | clip=6 | consec=0 | max\_consec=6 | avg\_clip50=18.82

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 19.2   |  |
| clipped_episode_r... | 33     |  |
| consecutive_clipp... | 1      |  |
| max_consecutive_c... | 6      |  |
| rollout/             |        |  |
| ep_len_mean          | 770    |  |
| ep_rew_mean          | 187    |  |
| exploration rate     | 0.787  |  |
| time/                |        |  |
| episodes             | 11100  |  |
| fps                  | 214    |  |
| time_elapsed         | 2973   |  |
| total timesteps      | 638268 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0665 |  |
| n_updates            | 147066 |  |

EP=3,720 | STEP=638,805 | clip=22 | consec=1 | max\_consec=6 | avg\_clip50=19.36  
EP=3,730 | STEP=640,742 | clip=17 | consec=0 | max\_consec=6 | avg\_clip50=19.84  
EP=3,740 | STEP=642,628 | clip=24 | consec=3 | max\_consec=6 | avg\_clip50=20.71  
EP=3,750 | STEP=644,771 | clip=44 | consec=1 | max\_consec=6 | avg\_clip50=22.11

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 22.1  |  |
| clipped_episode_r... | 44    |  |
| consecutive_clipp... | 1     |  |
| max_consecutive_c... | 6     |  |
| rollout/             |       |  |
| ep_len_mean          | 783   |  |
| ep_rew_mean          | 199   |  |
| exploration rate     | 0.785 |  |
| time/                |       |  |

|                 |        |  |
|-----------------|--------|--|
| episodes        | 11200  |  |
| fps             | 214    |  |
| time_elapsed    | 3004   |  |
| total timesteps | 644771 |  |
| train/          |        |  |
| learning_rate   | 0.0001 |  |
| loss            | 0.0694 |  |
| n_updates       | 148692 |  |

-----

EP=3,760 | STEP=646,505 | clip=17 | consec=0 | max\_consec=6 | avg\_clip50=21.80  
EP=3,770 | STEP=648,325 | clip=2 | consec=0 | max\_consec=6 | avg\_clip50=20.47  
EP=3,780 | STEP=650,215 | clip=16 | consec=0 | max\_consec=6 | avg\_clip50=20.15

|                      |        |  |
|----------------------|--------|--|
| custom/              |        |  |
| avg_clip50           | 19.8   |  |
| clipped_episode_r... | 10.5   |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 6      |  |
| rollout/             |        |  |
| ep_len_mean          | 789    |  |
| ep_rew_mean          | 199    |  |
| exploration rate     | 0.783  |  |
| time/                |        |  |
| episodes             | 11300  |  |
| fps                  | 214    |  |
| time_elapsed         | 3036   |  |
| total timesteps      | 651165 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0814 |  |
| n_updates            | 150291 |  |

-----

EP=3,790 | STEP=652,178 | clip=15 | consec=0 | max\_consec=6 | avg\_clip50=20.32  
EP=3,800 | STEP=654,023 | clip=19 | consec=0 | max\_consec=6 | avg\_clip50=20.16  
EP=3,810 | STEP=655,711 | clip=10 | consec=0 | max\_consec=6 | avg\_clip50=19.97

|                      |        |  |
|----------------------|--------|--|
| custom/              |        |  |
| avg_clip50           | 21.9   |  |
| clipped_episode_r... | 52     |  |
| consecutive_clipp... | 2      |  |
| max_consecutive_c... | 6      |  |
| rollout/             |        |  |
| ep_len_mean          | 807    |  |
| ep_rew_mean          | 215    |  |
| exploration rate     | 0.781  |  |
| time/                |        |  |
| episodes             | 11400  |  |
| fps                  | 214    |  |
| time_elapsed         | 3066   |  |
| total timesteps      | 657536 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0409 |  |
| n_updates            | 151883 |  |

-----

EP=3,820 | STEP=657,737 | clip=15 | consec=0 | max\_consec=6 | avg\_clip50=22.22  
EP=3,830 | STEP=660,132 | clip=12 | consec=0 | max\_consec=6 | avg\_clip50=22.35  
EP=3,840 | STEP=662,242 | clip=6 | consec=0 | max\_consec=6 | avg\_clip50=23.27  
EP=3,850 | STEP=664,398 | clip=30 | consec=2 | max\_consec=6 | avg\_clip50=23.07

-----



|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.7   |
| clipped_episode_r... | 26.5   |
| consecutive_clipp... | 4      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 836    |
| ep_rew_mean          | 220    |
| exploration rate     | 0.779  |
| time/                |        |
| episodes             | 11500  |
| fps                  | 214    |
| time_elapsed         | 3100   |
| total timesteps      | 664871 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0954 |
| n_updates            | 153717 |

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EP=3,860 | STEP=666,164 | clip=6 | consec=0 | max\_consec=6 | avg\_clip50=22.92  
EP=3,870 | STEP=668,198 | clip=32 | consec=1 | max\_consec=6 | avg\_clip50=21.74  
EP=3,880 | STEP=670,206 | clip=26 | consec=1 | max\_consec=6 | avg\_clip50=21.88

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.1   |
| clipped_episode_r... | 15     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 844    |
| ep_rew_mean          | 225    |
| exploration rate     | 0.776  |
| time/                |        |
| episodes             | 11600  |
| fps                  | 214    |
| time_elapsed         | 3131   |
| total timesteps      | 671248 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0822 |
| n_updates            | 155311 |

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EP=3,890 | STEP=672,119 | clip=19 | consec=0 | max\_consec=6 | avg\_clip50=21.04  
EP=3,900 | STEP=674,492 | clip=11 | consec=0 | max\_consec=6 | avg\_clip50=22.87  
EP=3,910 | STEP=676,715 | clip=10 | consec=0 | max\_consec=6 | avg\_clip50=23.94

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.4   |
| clipped_episode_r... | 41     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 879    |
| ep_rew_mean          | 229    |
| exploration rate     | 0.774  |
| time/                |        |
| episodes             | 11700  |
| fps                  | 214    |
| time_elapsed         | 3165   |
| total timesteps      | 678456 |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0487 |
| n_updates     | 157113 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,920 | STEP=678,578 | clip=10 | consec=0 | max_consec=6 | avg_clip50=24.01 |
| EP=3,930 | STEP=680,579 | clip=18 | consec=0 | max_consec=6 | avg_clip50=23.95 |
| EP=3,940 | STEP=682,665 | clip=26 | consec=1 | max_consec=6 | avg_clip50=24.12 |
| EP=3,950 | STEP=684,586 | clip=20 | consec=0 | max_consec=6 | avg_clip50=22.17 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.2   |
| clipped_episode_r... | 14     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 852    |
| ep_rew_mean          | 223    |
| exploration rate     | 0.772  |
| time/                |        |
| episodes             | 11800  |
| fps                  | 214    |
| time_elapsed         | 3199   |
| total timesteps      | 685210 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0219 |
| n_updates            | 158802 |

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|          |              |        |          |              |                  |
|----------|--------------|--------|----------|--------------|------------------|
| EP=3,960 | STEP=686,540 | clip=2 | consec=0 | max_consec=6 | avg_clip50=21.61 |
| EP=3,970 | STEP=688,599 | clip=8 | consec=0 | max_consec=6 | avg_clip50=21.25 |
| EP=3,980 | STEP=690,446 | clip=4 | consec=0 | max_consec=6 | avg_clip50=20.33 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.4   |
| clipped_episode_r... | 32     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 858    |
| ep_rew_mean          | 222    |
| exploration rate     | 0.77   |
| time/                |        |
| episodes             | 11900  |
| fps                  | 214    |
| time_elapsed         | 3230   |
| total timesteps      | 691698 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.122  |
| n_updates            | 160424 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,990 | STEP=692,464 | clip=44 | consec=1 | max_consec=6 | avg_clip50=20.87 |
| EP=4,000 | STEP=694,245 | clip=8  | consec=0 | max_consec=6 | avg_clip50=20.00 |
| EP=4,010 | STEP=696,334 | clip=19 | consec=0 | max_consec=6 | avg_clip50=20.16 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 19.4 |
| clipped_episode_r... | 12   |
| consecutive_clipp... | 0    |

|                      |        |
|----------------------|--------|
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 829    |
| ep_rew_mean          | 204    |
| exploration rate     | 0.768  |
| time/                |        |
| episodes             | 12000  |
| fps                  | 214    |
| time_elapsed         | 3261   |
| total timesteps      | 698085 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0799 |
| n_updates            | 162021 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,020 | STEP=698,113 | clip=12 | consec=0 | max_consec=6 | avg_clip50=19.50 |
| EP=4,030 | STEP=700,213 | clip=30 | consec=2 | max_consec=6 | avg_clip50=19.29 |
| EP=4,040 | STEP=702,096 | clip=19 | consec=0 | max_consec=6 | avg_clip50=18.29 |
| EP=4,050 | STEP=703,730 | clip=12 | consec=0 | max_consec=6 | avg_clip50=17.67 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 17.6   |
| clipped_episode_r... | 8.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 812    |
| ep_rew_mean          | 188    |
| exploration rate     | 0.765  |
| time/                |        |
| episodes             | 12100  |
| fps                  | 213    |
| time_elapsed         | 3292   |
| total timesteps      | 704370 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.104  |
| n_updates            | 163592 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,060 | STEP=705,439 | clip=11 | consec=0 | max_consec=6 | avg_clip50=17.55 |
| EP=4,070 | STEP=707,633 | clip=8  | consec=0 | max_consec=6 | avg_clip50=19.88 |
| EP=4,080 | STEP=709,821 | clip=40 | consec=2 | max_consec=6 | avg_clip50=21.80 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.9   |
| clipped_episode_r... | 31.5   |
| consecutive_clipp... | 5      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 837    |
| ep_rew_mean          | 213    |
| exploration rate     | 0.763  |
| time/                |        |
| episodes             | 12200  |
| fps                  | 213    |
| time_elapsed         | 3328   |
| total timesteps      | 711632 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.268  |

|           |        |
|-----------|--------|
| n_updates | 165407 |
|-----------|--------|

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,090 | STEP=712,084 | clip=7  | consec=0 | max_consec=6 | avg_clip50=23.45 |
| EP=4,100 | STEP=713,835 | clip=59 | consec=1 | max_consec=6 | avg_clip50=24.60 |
| EP=4,110 | STEP=716,034 | clip=14 | consec=0 | max_consec=6 | avg_clip50=25.85 |
| EP=4,120 | STEP=718,040 | clip=22 | consec=1 | max_consec=6 | avg_clip50=24.46 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.2   |
| clipped_episode_r... | 18.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 843    |
| ep_rew_mean          | 222    |
| exploration rate     | 0.761  |
| time/                |        |
| episodes             | 12300  |
| fps                  | 213    |
| time_elapsed         | 3359   |
| total timesteps      | 718231 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0669 |
| n_updates            | 167057 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,130 | STEP=719,875 | clip=36 | consec=1 | max_consec=6 | avg_clip50=22.50 |
| EP=4,140 | STEP=721,411 | clip=9  | consec=0 | max_consec=6 | avg_clip50=20.25 |
| EP=4,150 | STEP=723,186 | clip=22 | consec=1 | max_consec=6 | avg_clip50=20.14 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.1   |
| clipped_episode_r... | 21     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 6      |
| rollout/             |        |
| ep_len_mean          | 824    |
| ep_rew_mean          | 225    |
| exploration rate     | 0.759  |
| time/                |        |
| episodes             | 12400  |
| fps                  | 213    |
| time_elapsed         | 3388   |
| total timesteps      | 724026 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0235 |
| n_updates            | 168506 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,160 | STEP=724,957 | clip=10 | consec=0 | max_consec=6 | avg_clip50=18.57 |
| EP=4,170 | STEP=726,929 | clip=43 | consec=1 | max_consec=6 | avg_clip50=18.91 |
| EP=4,180 | STEP=729,069 | clip=29 | consec=3 | max_consec=6 | avg_clip50=19.36 |

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|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 21.1 |
| clipped_episode_r... | 30   |
| consecutive_clipp... | 2    |
| max_consecutive_c... | 7    |
| rollout/             |      |
| ep_len_mean          | 810  |

|                  |        |
|------------------|--------|
| ep_rew_mean      | 206    |
| exploration rate | 0.757  |
| time/            |        |
| episodes         | 12500  |
| fps              | 213    |
| time_elapsed     | 3420   |
| total timesteps  | 730879 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.114  |
| n_updates        | 170219 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,190 | STEP=731,131 | clip=4  | consec=0 | max_consec=7 | avg_clip50=21.02 |
| EP=4,200 | STEP=733,116 | clip=20 | consec=0 | max_consec=7 | avg_clip50=20.79 |
| EP=4,210 | STEP=734,970 | clip=26 | consec=2 | max_consec=7 | avg_clip50=21.44 |
| EP=4,220 | STEP=737,246 | clip=16 | consec=0 | max_consec=7 | avg_clip50=23.56 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.9   |
| clipped_episode_r... | 33     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 819    |
| ep_rew_mean          | 214    |
| exploration rate     | 0.754  |
| time/                |        |
| episodes             | 12600  |
| fps                  | 213    |
| time_elapsed         | 3452   |
| total timesteps      | 737607 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.078  |
| n_updates            | 171901 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,230 | STEP=738,993 | clip=11 | consec=0 | max_consec=7 | avg_clip50=22.94 |
| EP=4,240 | STEP=740,959 | clip=74 | consec=1 | max_consec=7 | avg_clip50=23.64 |
| EP=4,250 | STEP=742,723 | clip=35 | consec=1 | max_consec=7 | avg_clip50=23.88 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.5   |
| clipped_episode_r... | 32.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 834    |
| ep_rew_mean          | 225    |
| exploration rate     | 0.752  |
| time/                |        |
| episodes             | 12700  |
| fps                  | 213    |
| time_elapsed         | 3482   |
| total timesteps      | 744010 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.173  |
| n_updates            | 173502 |

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|          |              |        |          |              |                  |
|----------|--------------|--------|----------|--------------|------------------|
| EP=4,260 | STEP=744,864 | clip=4 | consec=0 | max_consec=7 | avg_clip50=24.05 |
|----------|--------------|--------|----------|--------------|------------------|

EP=4,270 | STEP=747,147 | clip=4 | consec=0 | max\_consec=7 | avg\_clip50=23.43  
EP=4,280 | STEP=749,092 | clip=22 | consec=1 | max\_consec=7 | avg\_clip50=23.21

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 23.1   |  |
| clipped_episode_r... | 62.5   |  |
| consecutive_clipp... | 1      |  |
| max_consecutive_c... | 7      |  |
| rollout/             |        |  |
| ep_len_mean          | 827    |  |
| ep_rew_mean          | 227    |  |
| exploration rate     | 0.75   |  |
| time/                |        |  |
| episodes             | 12800  |  |
| fps                  | 213    |  |
| time_elapsed         | 3515   |  |
| total timesteps      | 750810 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.214  |  |
| n_updates            | 175202 |  |

EP=4,290 | STEP=750,842 | clip=45 | consec=2 | max\_consec=7 | avg\_clip50=22.51  
EP=4,300 | STEP=753,144 | clip=34 | consec=5 | max\_consec=7 | avg\_clip50=24.49  
EP=4,310 | STEP=755,038 | clip=14 | consec=0 | max\_consec=7 | avg\_clip50=24.36  
EP=4,320 | STEP=756,873 | clip=50 | consec=2 | max\_consec=7 | avg\_clip50=22.85

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 22.6   |  |
| clipped_episode_r... | 12     |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 7      |  |
| rollout/             |        |  |
| ep_len_mean          | 836    |  |
| ep_rew_mean          | 233    |  |
| exploration rate     | 0.748  |  |
| time/                |        |  |
| episodes             | 12900  |  |
| fps                  | 213    |  |
| time_elapsed         | 3548   |  |
| total timesteps      | 757556 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.166  |  |
| n_updates            | 176888 |  |

EP=4,330 | STEP=758,582 | clip=15 | consec=0 | max\_consec=7 | avg\_clip50=22.23  
EP=4,340 | STEP=760,547 | clip=14 | consec=0 | max\_consec=7 | avg\_clip50=22.43  
EP=4,350 | STEP=762,687 | clip=10 | consec=0 | max\_consec=7 | avg\_clip50=22.47

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 21.3  |  |
| clipped_episode_r... | 8     |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 7     |  |
| rollout/             |       |  |
| ep_len_mean          | 835   |  |
| ep_rew_mean          | 229   |  |
| exploration rate     | 0.746 |  |
| time/                |       |  |

|                 |        |
|-----------------|--------|
| episodes        | 13000  |
| fps             | 213    |
| time_elapsed    | 3578   |
| total timesteps | 763876 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0568 |
| n_updates       | 178468 |

EP=4,360 | STEP=764,598 | clip=52 | consec=3 | max\_consec=7 | avg\_clip50=22.04  
EP=4,370 | STEP=766,523 | clip=22 | consec=1 | max\_consec=7 | avg\_clip50=22.15  
EP=4,380 | STEP=768,318 | clip=10 | consec=0 | max\_consec=7 | avg\_clip50=23.25  
EP=4,390 | STEP=770,312 | clip=12 | consec=0 | max\_consec=7 | avg\_clip50=22.66

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.7   |
| clipped_episode_r... | 12.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 825    |
| ep_rew_mean          | 225    |
| exploration rate     | 0.743  |
| time/                |        |
| episodes             | 13100  |
| fps                  | 213    |
| time_elapsed         | 3610   |
| total timesteps      | 770510 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0915 |
| n_updates            | 180127 |

EP=4,400 | STEP=772,000 | clip=16 | consec=0 | max\_consec=7 | avg\_clip50=19.56  
EP=4,410 | STEP=774,258 | clip=10 | consec=0 | max\_consec=7 | avg\_clip50=22.22  
EP=4,420 | STEP=776,422 | clip=62 | consec=3 | max\_consec=7 | avg\_clip50=23.09

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.5   |
| clipped_episode_r... | 47     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 828    |
| ep_rew_mean          | 228    |
| exploration rate     | 0.741  |
| time/                |        |
| episodes             | 13200  |
| fps                  | 213    |
| time_elapsed         | 3643   |
| total timesteps      | 777250 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.155  |
| n_updates            | 181812 |

EP=4,430 | STEP=778,252 | clip=24 | consec=1 | max\_consec=7 | avg\_clip50=22.68  
EP=4,440 | STEP=780,212 | clip=50 | consec=1 | max\_consec=7 | avg\_clip50=22.96  
EP=4,450 | STEP=781,873 | clip=6 | consec=0 | max\_consec=7 | avg\_clip50=23.76

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.6   |
| clipped_episode_r... | 28.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 836    |
| ep_rew_mean          | 230    |
| exploration rate     | 0.739  |
| time/                |        |
| episodes             | 13300  |
| fps                  | 213    |
| time_elapsed         | 3675   |
| total timesteps      | 783777 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0468 |
| n_updates            | 183444 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,460 | STEP=784,173 | clip=22 | consec=1 | max_consec=7 | avg_clip50=22.96 |
| EP=4,470 | STEP=785,931 | clip=31 | consec=3 | max_consec=7 | avg_clip50=21.81 |
| EP=4,480 | STEP=787,830 | clip=3  | consec=0 | max_consec=7 | avg_clip50=22.02 |
| EP=4,490 | STEP=789,746 | clip=28 | consec=2 | max_consec=7 | avg_clip50=21.78 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.9   |
| clipped_episode_r... | 9      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 819    |
| ep_rew_mean          | 222    |
| exploration rate     | 0.737  |
| time/                |        |
| episodes             | 13400  |
| fps                  | 213    |
| time_elapsed         | 3705   |
| total timesteps      | 790014 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0901 |
| n_updates            | 185003 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,500 | STEP=791,300 | clip=20 | consec=0 | max_consec=7 | avg_clip50=21.09 |
| EP=4,510 | STEP=793,250 | clip=9  | consec=0 | max_consec=7 | avg_clip50=19.74 |
| EP=4,520 | STEP=794,623 | clip=18 | consec=0 | max_consec=7 | avg_clip50=17.53 |

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|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 15.8  |
| clipped_episode_r... | 3.5   |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 7     |
| rollout/             |       |
| ep_len_mean          | 762   |
| ep_rew_mean          | 190   |
| exploration rate     | 0.735 |
| time/                |       |
| episodes             | 13500 |
| fps                  | 213   |
| time_elapsed         | 3729  |



|                 |        |
|-----------------|--------|
| total timesteps | 795183 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.035  |
| n_updates       | 186295 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,530 | STEP=796,365 | clip=42 | consec=2 | max_consec=7 | avg_clip50=18.35 |
| EP=4,540 | STEP=798,680 | clip=33 | consec=3 | max_consec=7 | avg_clip50=20.25 |
| EP=4,550 | STEP=800,401 | clip=20 | consec=1 | max_consec=7 | avg_clip50=20.49 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.1   |
| clipped_episode_r... | 38     |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 772    |
| ep_rew_mean          | 201    |
| exploration rate     | 0.733  |
| time/                |        |
| episodes             | 13600  |
| fps                  | 213    |
| time_elapsed         | 3763   |
| total timesteps      | 801929 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.19   |
| n_updates            | 187982 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,560 | STEP=802,179 | clip=11 | consec=0 | max_consec=7 | avg_clip50=20.07 |
| EP=4,570 | STEP=804,269 | clip=76 | consec=2 | max_consec=7 | avg_clip50=23.72 |
| EP=4,580 | STEP=806,194 | clip=9  | consec=0 | max_consec=7 | avg_clip50=23.98 |
| EP=4,590 | STEP=808,382 | clip=14 | consec=0 | max_consec=7 | avg_clip50=24.03 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.9   |
| clipped_episode_r... | 9.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 788    |
| ep_rew_mean          | 221    |
| exploration rate     | 0.731  |
| time/                |        |
| episodes             | 13700  |
| fps                  | 213    |
| time_elapsed         | 3796   |
| total timesteps      | 808667 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.328  |
| n_updates            | 189666 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,600 | STEP=810,017 | clip=4  | consec=0 | max_consec=7 | avg_clip50=24.16 |
| EP=4,610 | STEP=811,848 | clip=16 | consec=0 | max_consec=7 | avg_clip50=23.95 |
| EP=4,620 | STEP=813,835 | clip=46 | consec=1 | max_consec=7 | avg_clip50=22.92 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 22   |
| clipped_episode_r... | 13.5 |

|                      |        |
|----------------------|--------|
| consecutive_clipp... | 0      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 818    |
| ep_rew_mean          | 233    |
| exploration rate     | 0.729  |
| time/                |        |
| episodes             | 13800  |
| fps                  | 212    |
| time_elapsed         | 3826   |
| total timesteps      | 814906 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0275 |
| n_updates            | 191226 |

EP=4,630 | STEP=815,663 | clip=43 | consec=1 | max\_consec=7 | avg\_clip50=21.82  
EP=4,640 | STEP=818,180 | clip=64 | consec=1 | max\_consec=7 | avg\_clip50=22.66  
EP=4,650 | STEP=819,921 | clip=10 | consec=0 | max\_consec=7 | avg\_clip50=23.64

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.9   |
| clipped_episode_r... | 30.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 858    |
| ep_rew_mean          | 249    |
| exploration rate     | 0.726  |
| time/                |        |
| episodes             | 13900  |
| fps                  | 212    |
| time_elapsed         | 3861   |
| total timesteps      | 822358 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0499 |
| n_updates            | 193089 |

EP=4,660 | STEP=822,379 | clip=12 | consec=0 | max\_consec=7 | avg\_clip50=25.78  
EP=4,670 | STEP=824,937 | clip=64 | consec=1 | max\_consec=7 | avg\_clip50=28.05  
EP=4,680 | STEP=826,767 | clip=10 | consec=0 | max\_consec=7 | avg\_clip50=27.58  
EP=4,690 | STEP=828,857 | clip=7 | consec=0 | max\_consec=7 | avg\_clip50=24.83

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.5   |
| clipped_episode_r... | 40.5   |
| consecutive_clipp... | 3      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 884    |
| ep_rew_mean          | 243    |
| exploration rate     | 0.724  |
| time/                |        |
| episodes             | 14000  |
| fps                  | 212    |
| time_elapsed         | 3898   |
| total timesteps      | 829815 |
| train/               |        |
| learning_rate        | 0.0001 |

|           |        |
|-----------|--------|
| loss      | 0.0601 |
| n_updates | 194953 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,700 | STEP=830,994 | clip=5  | consec=0 | max_consec=7 | avg_clip50=25.95 |
| EP=4,710 | STEP=832,966 | clip=32 | consec=1 | max_consec=7 | avg_clip50=24.09 |
| EP=4,720 | STEP=835,147 | clip=14 | consec=0 | max_consec=7 | avg_clip50=21.96 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.6   |
| clipped_episode_r... | 31     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 912    |
| ep_rew_mean          | 256    |
| exploration rate     | 0.721  |
| time/                |        |
| episodes             | 14100  |
| fps                  | 212    |
| time_elapsed         | 3930   |
| total timesteps      | 836668 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.042  |
| n_updates            | 196666 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,730 | STEP=837,235 | clip=8  | consec=0 | max_consec=7 | avg_clip50=23.84 |
| EP=4,740 | STEP=839,301 | clip=70 | consec=1 | max_consec=7 | avg_clip50=25.74 |
| EP=4,750 | STEP=841,646 | clip=58 | consec=2 | max_consec=7 | avg_clip50=26.36 |
| EP=4,760 | STEP=843,754 | clip=22 | consec=3 | max_consec=7 | avg_clip50=26.39 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.9   |
| clipped_episode_r... | 3.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 894    |
| ep_rew_mean          | 250    |
| exploration rate     | 0.719  |
| time/                |        |
| episodes             | 14200  |
| fps                  | 212    |
| time_elapsed         | 3965   |
| total timesteps      | 843838 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.218  |
| n_updates            | 198459 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,770 | STEP=845,307 | clip=12 | consec=0 | max_consec=7 | avg_clip50=24.42 |
| EP=4,780 | STEP=847,271 | clip=54 | consec=2 | max_consec=7 | avg_clip50=24.30 |
| EP=4,790 | STEP=849,182 | clip=58 | consec=4 | max_consec=7 | avg_clip50=22.87 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 22.4 |
| clipped_episode_r... | 15.5 |
| consecutive_clipp... | 0    |
| max_consecutive_c... | 7    |
| rollout/             |      |

|                  |        |
|------------------|--------|
| ep_len_mean      | 850    |
| ep_rew_mean      | 239    |
| exploration rate | 0.717  |
| time/            |        |
| episodes         | 14300  |
| fps              | 212    |
| time_elapsed     | 3997   |
| total timesteps  | 850177 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.112  |
| n_updates        | 200044 |

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EP=4,800 | STEP=851,283 | clip=30 | consec=1 | max\_consec=7 | avg\_clip50=22.31  
EP=4,810 | STEP=852,953 | clip=24 | consec=1 | max\_consec=7 | avg\_clip50=21.35  
EP=4,820 | STEP=854,933 | clip=4 | consec=0 | max\_consec=7 | avg\_clip50=23.86

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.5   |
| clipped_episode_r... | 22.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 832    |
| ep_rew_mean          | 238    |
| exploration rate     | 0.715  |
| time/                |        |
| episodes             | 14400  |
| fps                  | 212    |
| time_elapsed         | 4029   |
| total timesteps      | 856825 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0284 |
| n_updates            | 201706 |

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EP=4,830 | STEP=857,207 | clip=28 | consec=4 | max\_consec=7 | avg\_clip50=24.41  
EP=4,840 | STEP=859,740 | clip=34 | consec=1 | max\_consec=7 | avg\_clip50=25.59  
EP=4,850 | STEP=861,201 | clip=2 | consec=0 | max\_consec=7 | avg\_clip50=22.99  
EP=4,860 | STEP=863,016 | clip=28 | consec=2 | max\_consec=7 | avg\_clip50=23.53

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.1   |
| clipped_episode_r... | 24.5   |
| consecutive_clipp... | 4      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 824    |
| ep_rew_mean          | 227    |
| exploration rate     | 0.712  |
| time/                |        |
| episodes             | 14500  |
| fps                  | 212    |
| time_elapsed         | 4061   |
| total timesteps      | 863547 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0606 |
| n_updates            | 203386 |

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EP=4,870 | STEP=864,973 | clip=4 | consec=0 | max\_consec=8 | avg\_clip50=23.04  
EP=4,880 | STEP=866,991 | clip=29 | consec=1 | max\_consec=8 | avg\_clip50=22.25  
EP=4,890 | STEP=868,781 | clip=6 | consec=0 | max\_consec=8 | avg\_clip50=19.96

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 19     |  |
| clipped_episode_r... | 5.5    |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 808    |  |
| ep_rew_mean          | 217    |  |
| exploration rate     | 0.71   |  |
| time/                |        |  |
| episodes             | 14600  |  |
| fps                  | 212    |  |
| time_elapsed         | 4091   |  |
| total timesteps      | 869534 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0687 |  |
| n_updates            | 204883 |  |

EP=4,900 | STEP=870,199 | clip=16 | consec=0 | max\_consec=8 | avg\_clip50=19.64  
EP=4,910 | STEP=872,467 | clip=43 | consec=1 | max\_consec=8 | avg\_clip50=21.79  
EP=4,920 | STEP=874,657 | clip=40 | consec=2 | max\_consec=8 | avg\_clip50=22.95  
EP=4,930 | STEP=876,822 | clip=46 | consec=1 | max\_consec=8 | avg\_clip50=24.48

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 24.5   |  |
| clipped_episode_r... | 46.5   |  |
| consecutive_clipp... | 1      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 829    |  |
| ep_rew_mean          | 234    |  |
| exploration rate     | 0.708  |  |
| time/                |        |  |
| episodes             | 14700  |  |
| fps                  | 212    |  |
| time_elapsed         | 4128   |  |
| total timesteps      | 877085 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0543 |  |
| n_updates            | 206771 |  |

EP=4,940 | STEP=878,827 | clip=30 | consec=1 | max\_consec=8 | avg\_clip50=25.08  
EP=4,950 | STEP=880,735 | clip=14 | consec=0 | max\_consec=8 | avg\_clip50=27.10  
EP=4,960 | STEP=882,598 | clip=29 | consec=1 | max\_consec=8 | avg\_clip50=25.94

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 25.7  |  |
| clipped_episode_r... | 44    |  |
| consecutive_clipp... | 1     |  |
| max_consecutive_c... | 8     |  |
| rollout/             |       |  |
| ep_len_mean          | 823   |  |
| ep_rew_mean          | 235   |  |
| exploration rate     | 0.706 |  |

|                 |        |
|-----------------|--------|
| time/           |        |
| episodes        | 14800  |
| fps             | 212    |
| time_elapsed    | 4159   |
| total timesteps | 883413 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.142  |
| n_updates       | 208353 |

EP=4,970 | STEP=884,766 | clip=16 | consec=0 | max\_consec=8 | avg\_clip50=25.20  
EP=4,980 | STEP=886,893 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=23.43  
EP=4,990 | STEP=888,784 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=24.13

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.8   |
| clipped_episode_r... | 19.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 876    |
| ep_rew_mean          | 262    |
| exploration rate     | 0.703  |
| time/                |        |
| episodes             | 14900  |
| fps                  | 212    |
| time_elapsed         | 4192   |
| total timesteps      | 890468 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0475 |
| n_updates            | 210116 |

EP=5,000 | STEP=890,863 | clip=30 | consec=1 | max\_consec=8 | avg\_clip50=24.86  
EP=5,010 | STEP=893,050 | clip=30 | consec=6 | max\_consec=8 | avg\_clip50=25.30  
EP=5,020 | STEP=895,353 | clip=45 | consec=4 | max\_consec=8 | avg\_clip50=24.97  
EP=5,030 | STEP=897,318 | clip=21 | consec=1 | max\_consec=8 | avg\_clip50=24.95

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.9   |
| clipped_episode_r... | 6.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 860    |
| ep_rew_mean          | 240    |
| exploration rate     | 0.701  |
| time/                |        |
| episodes             | 15000  |
| fps                  | 212    |
| time_elapsed         | 4227   |
| total timesteps      | 897508 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0679 |
| n_updates            | 211876 |

EP=5,040 | STEP=899,007 | clip=44 | consec=2 | max\_consec=8 | avg\_clip50=23.48  
EP=5,050 | STEP=901,205 | clip=36 | consec=4 | max\_consec=8 | avg\_clip50=23.17  
EP=5,060 | STEP=903,305 | clip=14 | consec=0 | max\_consec=8 | avg\_clip50=23.07

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.4   |
| clipped_episode_r... | 27.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 864    |
| ep_rew_mean          | 233    |
| exploration rate     | 0.699  |
| time/                |        |
| episodes             | 15100  |
| fps                  | 212    |
| time_elapsed         | 4261   |
| total timesteps      | 904398 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.127  |
| n_updates            | 213599 |

EP=5,070 | STEP=905,338 | clip=67 | consec=3 | max\_consec=8 | avg\_clip50=22.93  
EP=5,080 | STEP=907,260 | clip=42 | consec=2 | max\_consec=8 | avg\_clip50=22.28  
EP=5,090 | STEP=909,243 | clip=14 | consec=0 | max\_consec=8 | avg\_clip50=22.18

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.5   |
| clipped_episode_r... | 32     |
| consecutive_clipp... | 3      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 861    |
| ep_rew_mean          | 228    |
| exploration rate     | 0.697  |
| time/                |        |
| episodes             | 15200  |
| fps                  | 212    |
| time_elapsed         | 4294   |
| total timesteps      | 911209 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0839 |
| n_updates            | 215302 |

EP=5,100 | STEP=911,227 | clip=24 | consec=4 | max\_consec=8 | avg\_clip50=22.30  
EP=5,110 | STEP=913,164 | clip=6 | consec=0 | max\_consec=8 | avg\_clip50=22.28  
EP=5,120 | STEP=915,050 | clip=23 | consec=1 | max\_consec=8 | avg\_clip50=21.07  
EP=5,130 | STEP=917,303 | clip=14 | consec=0 | max\_consec=9 | avg\_clip50=23.58

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 26.4  |
| clipped_episode_r... | 34.5  |
| consecutive_clipp... | 1     |
| max_consecutive_c... | 9     |
| rollout/             |       |
| ep_len_mean          | 867   |
| ep_rew_mean          | 248   |
| exploration rate     | 0.694 |
| time/                |       |
| episodes             | 15300 |
| fps                  | 212   |

|  |                 |  |        |  |
|--|-----------------|--|--------|--|
|  | time_elapsed    |  | 4329   |  |
|  | total timesteps |  | 918558 |  |
|  | train/          |  |        |  |
|  | learning_rate   |  | 0.0001 |  |
|  | loss            |  | 0.0941 |  |
|  | n_updates       |  | 217139 |  |

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|          |  |              |  |         |  |          |  |              |  |                  |
|----------|--|--------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,140 |  | STEP=919,531 |  | clip=12 |  | consec=0 |  | max_consec=9 |  | avg_clip50=27.64 |
| EP=5,150 |  | STEP=921,799 |  | clip=16 |  | consec=0 |  | max_consec=9 |  | avg_clip50=28.91 |
| EP=5,160 |  | STEP=923,458 |  | clip=14 |  | consec=0 |  | max_consec=9 |  | avg_clip50=27.51 |

|  |                      |  |        |  |
|--|----------------------|--|--------|--|
|  | custom/              |  |        |  |
|  | avg_clip50           |  | 30.5   |  |
|  | clipped_episode_r... |  | 14.5   |  |
|  | consecutive_clipp... |  | 0      |  |
|  | max_consecutive_c... |  | 9      |  |
|  | rollout/             |  |        |  |
|  | ep_len_mean          |  | 871    |  |
|  | ep_rew_mean          |  | 262    |  |
|  | exploration rate     |  | 0.692  |  |
|  | time/                |  |        |  |
|  | episodes             |  | 15400  |  |
|  | fps                  |  | 212    |  |
|  | time_elapsed         |  | 4365   |  |
|  | total timesteps      |  | 925691 |  |
|  | train/               |  |        |  |
|  | learning_rate        |  | 0.0001 |  |
|  | loss                 |  | 0.0574 |  |
|  | n_updates            |  | 218922 |  |

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|          |  |              |  |         |  |          |  |              |  |                  |
|----------|--|--------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,170 |  | STEP=925,920 |  | clip=28 |  | consec=1 |  | max_consec=9 |  | avg_clip50=30.64 |
| EP=5,180 |  | STEP=928,175 |  | clip=7  |  | consec=0 |  | max_consec=9 |  | avg_clip50=30.79 |
| EP=5,190 |  | STEP=929,708 |  | clip=6  |  | consec=0 |  | max_consec=9 |  | avg_clip50=26.41 |
| EP=5,200 |  | STEP=931,774 |  | clip=39 |  | consec=1 |  | max_consec=9 |  | avg_clip50=25.28 |

|  |                      |  |        |  |
|--|----------------------|--|--------|--|
|  | custom/              |  |        |  |
|  | avg_clip50           |  | 26.2   |  |
|  | clipped_episode_r... |  | 24.5   |  |
|  | consecutive_clipp... |  | 4      |  |
|  | max_consecutive_c... |  | 9      |  |
|  | rollout/             |  |        |  |
|  | ep_len_mean          |  | 869    |  |
|  | ep_rew_mean          |  | 271    |  |
|  | exploration rate     |  | 0.689  |  |
|  | time/                |  |        |  |
|  | episodes             |  | 15500  |  |
|  | fps                  |  | 212    |  |
|  | time_elapsed         |  | 4398   |  |
|  | total timesteps      |  | 932494 |  |
|  | train/               |  |        |  |
|  | learning_rate        |  | 0.0001 |  |
|  | loss                 |  | 0.0419 |  |
|  | n_updates            |  | 220623 |  |

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|          |  |              |  |         |  |          |  |              |  |                  |
|----------|--|--------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,210 |  | STEP=933,521 |  | clip=15 |  | consec=0 |  | max_consec=9 |  | avg_clip50=25.50 |
| EP=5,220 |  | STEP=935,390 |  | clip=30 |  | consec=1 |  | max_consec=9 |  | avg_clip50=22.80 |
| EP=5,230 |  | STEP=937,468 |  | clip=62 |  | consec=1 |  | max_consec=9 |  | avg_clip50=21.01 |

|  |            |  |      |  |
|--|------------|--|------|--|
|  | custom/    |  |      |  |
|  | avg_clip50 |  | 22.5 |  |



|  |                      |        |  |
|--|----------------------|--------|--|
|  | clipped_episode_r... | 65     |  |
|  | consecutive_clipp... | 4      |  |
|  | max_consecutive_c... | 9      |  |
|  | rollout/             |        |  |
|  | ep_len_mean          | 851    |  |
|  | ep_rew_mean          | 251    |  |
|  | exploration rate     | 0.687  |  |
|  | time/                |        |  |
|  | episodes             | 15600  |  |
|  | fps                  | 211    |  |
|  | time_elapsed         | 4431   |  |
|  | total timesteps      | 939039 |  |
|  | train/               |        |  |
|  | learning_rate        | 0.0001 |  |
|  | loss                 | 0.0652 |  |
|  | n_updates            | 222259 |  |

-----

|          |  |              |  |         |  |          |  |              |  |                  |
|----------|--|--------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,240 |  | STEP=939,773 |  | clip=48 |  | consec=1 |  | max_consec=9 |  | avg_clip50=23.96 |
| EP=5,250 |  | STEP=941,346 |  | clip=11 |  | consec=0 |  | max_consec=9 |  | avg_clip50=22.51 |
| EP=5,260 |  | STEP=943,535 |  | clip=8  |  | consec=0 |  | max_consec=9 |  | avg_clip50=24.71 |
| EP=5,270 |  | STEP=945,749 |  | clip=54 |  | consec=1 |  | max_consec=9 |  | avg_clip50=26.40 |

-----

|  |                      |        |  |
|--|----------------------|--------|--|
|  | custom/              |        |  |
|  | avg_clip50           | 26.6   |  |
|  | clipped_episode_r... | 25.5   |  |
|  | consecutive_clipp... | 2      |  |
|  | max_consecutive_c... | 9      |  |
|  | rollout/             |        |  |
|  | ep_len_mean          | 839    |  |
|  | ep_rew_mean          | 247    |  |
|  | exploration rate     | 0.685  |  |
|  | time/                |        |  |
|  | episodes             | 15700  |  |
|  | fps                  | 211    |  |
|  | time_elapsed         | 4467   |  |
|  | total timesteps      | 946048 |  |
|  | train/               |        |  |
|  | learning_rate        | 0.0001 |  |
|  | loss                 | 0.0503 |  |
|  | n_updates            | 224011 |  |

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|          |  |              |  |         |  |          |  |              |  |                  |
|----------|--|--------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,280 |  | STEP=947,777 |  | clip=7  |  | consec=0 |  | max_consec=9 |  | avg_clip50=27.50 |
| EP=5,290 |  | STEP=949,822 |  | clip=12 |  | consec=0 |  | max_consec=9 |  | avg_clip50=26.35 |
| EP=5,300 |  | STEP=951,753 |  | clip=14 |  | consec=0 |  | max_consec=9 |  | avg_clip50=26.52 |

-----

|  |                      |        |  |
|--|----------------------|--------|--|
|  | custom/              |        |  |
|  | avg_clip50           | 25.8   |  |
|  | clipped_episode_r... | 33.5   |  |
|  | consecutive_clipp... | 1      |  |
|  | max_consecutive_c... | 9      |  |
|  | rollout/             |        |  |
|  | ep_len_mean          | 838    |  |
|  | ep_rew_mean          | 243    |  |
|  | exploration rate     | 0.683  |  |
|  | time/                |        |  |
|  | episodes             | 15800  |  |
|  | fps                  | 211    |  |
|  | time_elapsed         | 4499   |  |
|  | total timesteps      | 952638 |  |
|  | train/               |        |  |

|               |        |
|---------------|--------|
| learning_rate | 0.0001 |
| loss          | 0.0457 |
| n_updates     | 225659 |

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,310 | STEP=953,659 | clip=26 | consec=2 | max_consec=9 | avg_clip50=25.48 |
| EP=5,320 | STEP=956,046 | clip=42 | consec=4 | max_consec=9 | avg_clip50=26.34 |
| EP=5,330 | STEP=958,323 | clip=55 | consec=1 | max_consec=9 | avg_clip50=26.99 |
| EP=5,340 | STEP=960,692 | clip=34 | consec=1 | max_consec=9 | avg_clip50=27.92 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 27.9   |
| clipped_episode_r... | 33.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 9      |
| rollout/             |        |
| ep_len_mean          | 881    |
| ep_rew_mean          | 271    |
| exploration rate     | 0.68   |
| time/                |        |
| episodes             | 15900  |
| fps                  | 211    |
| time_elapsed         | 4540   |
| total timesteps      | 960999 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.115  |
| n_updates            | 227749 |

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,350 | STEP=962,766 | clip=10 | consec=0 | max_consec=9 | avg_clip50=29.52 |
| EP=5,360 | STEP=965,059 | clip=24 | consec=2 | max_consec=9 | avg_clip50=30.31 |
| EP=5,370 | STEP=967,795 | clip=82 | consec=8 | max_consec=9 | avg_clip50=30.68 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 31.8   |
| clipped_episode_r... | 23     |
| consecutive_clipp... | 12     |
| max_consecutive_c... | 12     |
| rollout/             |        |
| ep_len_mean          | 928    |
| ep_rew_mean          | 290    |
| exploration rate     | 0.677  |
| time/                |        |
| episodes             | 16000  |
| fps                  | 211    |
| time_elapsed         | 4577   |
| total timesteps      | 968635 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.127  |
| n_updates            | 229658 |

|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=5,380 | STEP=969,994 | clip=36 | consec=3 | max_consec=14 | avg_clip50=30.68 |
| EP=5,390 | STEP=972,261 | clip=53 | consec=2 | max_consec=14 | avg_clip50=31.42 |
| EP=5,400 | STEP=974,513 | clip=10 | consec=0 | max_consec=14 | avg_clip50=30.81 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 30.5 |
| clipped_episode_r... | 29   |
| consecutive_clipp... | 3    |
| max_consecutive_c... | 14   |

|                  |        |
|------------------|--------|
| rollout/         |        |
| ep_len_mean      | 973    |
| ep_rew_mean      | 304    |
| exploration rate | 0.675  |
| time/            |        |
| episodes         | 16100  |
| fps              | 211    |
| time_elapsed     | 4617   |
| total timesteps  | 976726 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.141  |
| n_updates        | 231681 |

-----

|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=5,410 | STEP=976,758 | clip=10 | consec=0 | max_consec=14 | avg_clip50=30.26 |
| EP=5,420 | STEP=978,603 | clip=12 | consec=0 | max_consec=14 | avg_clip50=28.05 |
| EP=5,430 | STEP=980,573 | clip=12 | consec=0 | max_consec=14 | avg_clip50=26.61 |
| EP=5,440 | STEP=982,833 | clip=26 | consec=2 | max_consec=14 | avg_clip50=25.49 |

-----

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.3   |
| clipped_episode_r... | 14.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 14     |
| rollout/             |        |
| ep_len_mean          | 922    |
| ep_rew_mean          | 279    |
| exploration rate     | 0.673  |
| time/                |        |
| episodes             | 16200  |
| fps                  | 211    |
| time_elapsed         | 4651   |
| total timesteps      | 983387 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.15   |
| n_updates            | 233346 |

-----

|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=5,450 | STEP=984,935 | clip=10 | consec=0 | max_consec=14 | avg_clip50=26.09 |
| EP=5,460 | STEP=987,158 | clip=36 | consec=1 | max_consec=14 | avg_clip50=27.23 |
| EP=5,470 | STEP=989,275 | clip=46 | consec=3 | max_consec=14 | avg_clip50=28.08 |

-----

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 29.5   |
| clipped_episode_r... | 16     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 14     |
| rollout/             |        |
| ep_len_mean          | 922    |
| ep_rew_mean          | 284    |
| exploration rate     | 0.67   |
| time/                |        |
| episodes             | 16300  |
| fps                  | 211    |
| time_elapsed         | 4689   |
| total timesteps      | 991226 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.123  |
| n_updates            | 235306 |

```

-----
EP=5,480 | STEP=991,877 | clip=31 | consec=1 | max_consec=14 | avg_clip50=29.46
EP=5,490 | STEP=994,258 | clip=45 | consec=5 | max_consec=14 | avg_clip50=30.65
EP=5,500 | STEP=996,143 | clip=21 | consec=2 | max_consec=14 | avg_clip50=30.54
EP=5,510 | STEP=998,373 | clip=26 | consec=1 | max_consec=14 | avg_clip50=30.76

```

```

-----
| custom/
|   avg_clip50           | 30.4
|   clipped_episode_r... | 26.5
|   consecutive_clipp... | 2
|   max_consecutive_c... | 14
| rollout/
|   ep_len_mean         | 909
|   ep_rew_mean         | 290
|   exploration rate    | 0.667
| time/
|   episodes            | 16400
|   fps                 | 211
|   time_elapsed        | 4726
|   total timesteps     | 998610
| train/
|   learning_rate       | 0.0001
|   loss                | 0.657
|   n_updates           | 237152

```

```

-----
EP=5,520 | STEP=1,001,016 | clip=56 | consec=1 | max_consec=14 | avg_clip50=32.09
EP=5,530 | STEP=1,003,131 | clip=10 | consec=0 | max_consec=14 | avg_clip50=30.70
EP=5,540 | STEP=1,005,238 | clip=58 | consec=2 | max_consec=14 | avg_clip50=30.40

```

```

-----
| custom/
|   avg_clip50           | 30.3
|   clipped_episode_r... | 10.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 14
| rollout/
|   ep_len_mean         | 938
|   ep_rew_mean         | 302
|   exploration rate    | 0.665
| time/
|   episodes            | 16500
|   fps                 | 211
|   time_elapsed        | 4761
|   total timesteps     | 1006036
| train/
|   learning_rate       | 0.0001
|   loss                | 0.042
|   n_updates           | 239008

```

```

-----
EP=5,550 | STEP=1,007,154 | clip=32 | consec=2 | max_consec=14 | avg_clip50=30.18
EP=5,560 | STEP=1,009,503 | clip=23 | consec=3 | max_consec=14 | avg_clip50=29.78
EP=5,570 | STEP=1,011,928 | clip=9  | consec=0 | max_consec=14 | avg_clip50=28.24

```

```

-----
| custom/
|   avg_clip50           | 26.5
|   clipped_episode_r... | 18.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 14
| rollout/
|   ep_len_mean         | 917
|   ep_rew_mean         | 288

```

|                  |         |
|------------------|---------|
| exploration rate | 0.663   |
| time/            |         |
| episodes         | 16600   |
| fps              | 211     |
| time_elapsed     | 4797    |
| total timesteps  | 1013284 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.135   |
| n_updates        | 240820  |

EP=5,580 | STEP=1,013,657 | clip=24 | consec=2 | max\_consec=14 | avg\_clip50=27.11  
EP=5,590 | STEP=1,015,602 | clip=25 | consec=1 | max\_consec=14 | avg\_clip50=26.36  
EP=5,600 | STEP=1,017,364 | clip=16 | consec=0 | max\_consec=14 | avg\_clip50=25.15  
EP=5,610 | STEP=1,019,595 | clip=32 | consec=2 | max\_consec=14 | avg\_clip50=24.61

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 24.6    |
| clipped_episode_r... | 33.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 895     |
| ep_rew_mean          | 273     |
| exploration rate     | 0.66    |
| time/                |         |
| episodes             | 16700   |
| fps                  | 211     |
| time_elapsed         | 4830    |
| total timesteps      | 1019890 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.287   |
| n_updates            | 242472  |

EP=5,620 | STEP=1,021,910 | clip=38 | consec=2 | max\_consec=14 | avg\_clip50=25.44  
EP=5,630 | STEP=1,023,723 | clip=11 | consec=0 | max\_consec=14 | avg\_clip50=26.27  
EP=5,640 | STEP=1,025,835 | clip=38 | consec=2 | max\_consec=14 | avg\_clip50=26.12

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26.3    |
| clipped_episode_r... | 18.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 884     |
| ep_rew_mean          | 266     |
| exploration rate     | 0.658   |
| time/                |         |
| episodes             | 16800   |
| fps                  | 211     |
| time_elapsed         | 4865    |
| total timesteps      | 1027078 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0602  |
| n_updates            | 244269  |

EP=5,650 | STEP=1,028,017 | clip=30 | consec=1 | max\_consec=14 | avg\_clip50=27.40  
EP=5,660 | STEP=1,029,595 | clip=14 | consec=0 | max\_consec=14 | avg\_clip50=24.89

EP=5,670 | STEP=1,031,295 | clip=18 | consec=0 | max\_consec=14 | avg\_clip50=22.28

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 22.3    |
| clipped_episode_r... | 27.5    |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 836     |
| ep_rew_mean          | 244     |
| exploration rate     | 0.656   |
| time/                |         |
| episodes             | 16900   |
| fps                  | 211     |
| time_elapsed         | 4896    |
| total timesteps      | 1033137 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0616  |
| n_updates            | 245784  |

EP=5,680 | STEP=1,033,525 | clip=65 | consec=1 | max\_consec=14 | avg\_clip50=23.18  
EP=5,690 | STEP=1,035,178 | clip=46 | consec=1 | max\_consec=14 | avg\_clip50=21.75  
EP=5,700 | STEP=1,037,537 | clip=10 | consec=0 | max\_consec=14 | avg\_clip50=22.30  
EP=5,710 | STEP=1,039,457 | clip=58 | consec=3 | max\_consec=14 | avg\_clip50=24.74

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 25.7    |
| clipped_episode_r... | 47.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 840     |
| ep_rew_mean          | 246     |
| exploration rate     | 0.654   |
| time/                |         |
| episodes             | 17000   |
| fps                  | 210     |
| time_elapsed         | 4931    |
| total timesteps      | 1039925 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0684  |
| n_updates            | 247481  |

EP=5,720 | STEP=1,041,366 | clip=26 | consec=3 | max\_consec=14 | avg\_clip50=26.07  
EP=5,730 | STEP=1,043,358 | clip=16 | consec=0 | max\_consec=14 | avg\_clip50=25.11  
EP=5,740 | STEP=1,045,267 | clip=28 | consec=1 | max\_consec=14 | avg\_clip50=25.99

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 25.2  |
| clipped_episode_r... | 14    |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 14    |
| rollout/             |       |
| ep_len_mean          | 820   |
| ep_rew_mean          | 242   |
| exploration rate     | 0.651 |
| time/                |       |
| episodes             | 17100 |

|                 |         |
|-----------------|---------|
| fps             | 210     |
| time_elapsed    | 4964    |
| total timesteps | 1046563 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.464   |
| n_updates       | 249140  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=5,750 | STEP=1,047,360 | clip=54 | consec=1 | max_consec=14 | avg_clip50=25.70 |
| EP=5,760 | STEP=1,050,080 | clip=63 | consec=1 | max_consec=14 | avg_clip50=28.00 |
| EP=5,770 | STEP=1,051,894 | clip=41 | consec=2 | max_consec=14 | avg_clip50=27.18 |
| EP=5,780 | STEP=1,054,057 | clip=8  | consec=0 | max_consec=14 | avg_clip50=27.92 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.9    |
| clipped_episode_r... | 8       |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 867     |
| ep_rew_mean          | 265     |
| exploration rate     | 0.649   |
| time/                |         |
| episodes             | 17200   |
| fps                  | 210     |
| time_elapsed         | 5003    |
| total timesteps      | 1054057 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.286   |
| n_updates            | 251014  |

-----

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=5,790 | STEP=1,056,233 | clip=52 | consec=1 | max_consec=14 | avg_clip50=30.28 |
| EP=5,800 | STEP=1,057,997 | clip=14 | consec=0 | max_consec=14 | avg_clip50=28.62 |
| EP=5,810 | STEP=1,059,891 | clip=4  | consec=0 | max_consec=14 | avg_clip50=25.08 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 24.9    |
| clipped_episode_r... | 19      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 853     |
| ep_rew_mean          | 261     |
| exploration rate     | 0.647   |
| time/                |         |
| episodes             | 17300   |
| fps                  | 210     |
| time_elapsed         | 5035    |
| total timesteps      | 1060500 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0435  |
| n_updates            | 252624  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=5,820 | STEP=1,061,942 | clip=4  | consec=0 | max_consec=14 | avg_clip50=25.22 |
| EP=5,830 | STEP=1,063,617 | clip=10 | consec=0 | max_consec=14 | avg_clip50=23.41 |
| EP=5,840 | STEP=1,065,639 | clip=26 | consec=4 | max_consec=14 | avg_clip50=21.05 |

|         |  |
|---------|--|
| custom/ |  |
|---------|--|

|                      |         |
|----------------------|---------|
| avg_clip50           | 20.8    |
| clipped_episode_r... | 10      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 848     |
| ep_rew_mean          | 248     |
| exploration rate     | 0.645   |
| time/                |         |
| episodes             | 17400   |
| fps                  | 210     |
| time_elapsed         | 5066    |
| total timesteps      | 1066933 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0746  |
| n_updates            | 254233  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=5,850 | STEP=1,067,629 | clip=67 | consec=1 | max_consec=14 | avg_clip50=21.76 |
| EP=5,860 | STEP=1,070,118 | clip=12 | consec=0 | max_consec=14 | avg_clip50=24.66 |
| EP=5,870 | STEP=1,072,094 | clip=34 | consec=1 | max_consec=14 | avg_clip50=24.96 |
| EP=5,880 | STEP=1,073,841 | clip=6  | consec=0 | max_consec=14 | avg_clip50=25.82 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26      |
| clipped_episode_r... | 25      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 845     |
| ep_rew_mean          | 249     |
| exploration rate     | 0.642   |
| time/                |         |
| episodes             | 17500   |
| fps                  | 210     |
| time_elapsed         | 5102    |
| total timesteps      | 1074383 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0496  |
| n_updates            | 256095  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=5,890 | STEP=1,076,054 | clip=2  | consec=0 | max_consec=14 | avg_clip50=27.20 |
| EP=5,900 | STEP=1,078,171 | clip=12 | consec=0 | max_consec=14 | avg_clip50=28.05 |
| EP=5,910 | STEP=1,080,549 | clip=24 | consec=6 | max_consec=14 | avg_clip50=26.98 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 28.1    |
| clipped_episode_r... | 42.5    |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 887     |
| ep_rew_mean          | 265     |
| exploration rate     | 0.64    |
| time/                |         |
| episodes             | 17600   |
| fps                  | 210     |
| time_elapsed         | 5140    |
| total timesteps      | 1082015 |



|                      |                |         |          |               |                  |
|----------------------|----------------|---------|----------|---------------|------------------|
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0511         |         |          |               |                  |
| n_updates            | 258003         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=5,920             | STEP=1,082,974 | clip=18 | consec=0 | max_consec=14 | avg_clip50=28.26 |
| EP=5,930             | STEP=1,085,028 | clip=16 | consec=0 | max_consec=14 | avg_clip50=28.19 |
| EP=5,940             | STEP=1,086,861 | clip=10 | consec=0 | max_consec=14 | avg_clip50=25.65 |
| EP=5,950             | STEP=1,088,948 | clip=43 | consec=1 | max_consec=14 | avg_clip50=26.48 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 26.5           |         |          |               |                  |
| clipped_episode_r... | 43             |         |          |               |                  |
| consecutive_clipp... | 1              |         |          |               |                  |
| max_consecutive_c... | 14             |         |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          | 898            |         |          |               |                  |
| ep_rew_mean          | 273            |         |          |               |                  |
| exploration rate     | 0.637          |         |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             | 17700          |         |          |               |                  |
| fps                  | 210            |         |          |               |                  |
| time_elapsed         | 5175           |         |          |               |                  |
| total timesteps      | 1088972        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.291          |         |          |               |                  |
| n_updates            | 259742         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=5,960             | STEP=1,090,756 | clip=50 | consec=1 | max_consec=14 | avg_clip50=24.65 |
| EP=5,970             | STEP=1,093,094 | clip=54 | consec=1 | max_consec=14 | avg_clip50=24.33 |
| EP=5,980             | STEP=1,095,450 | clip=43 | consec=2 | max_consec=14 | avg_clip50=26.07 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 25.8           |         |          |               |                  |
| clipped_episode_r... | 4.5            |         |          |               |                  |
| consecutive_clipp... | 0              |         |          |               |                  |
| max_consecutive_c... | 14             |         |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          | 892            |         |          |               |                  |
| ep_rew_mean          | 264            |         |          |               |                  |
| exploration rate     | 0.635          |         |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             | 17800          |         |          |               |                  |
| fps                  | 210            |         |          |               |                  |
| time_elapsed         | 5211           |         |          |               |                  |
| total timesteps      | 1096257        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0377         |         |          |               |                  |
| n_updates            | 261564         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=5,990             | STEP=1,097,362 | clip=20 | consec=0 | max_consec=14 | avg_clip50=27.68 |
| EP=6,000             | STEP=1,099,428 | clip=25 | consec=4 | max_consec=14 | avg_clip50=27.08 |
| EP=6,010             | STEP=1,101,546 | clip=60 | consec=2 | max_consec=14 | avg_clip50=28.45 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 27.8           |         |          |               |                  |
| clipped_episode_r... | 17.5           |         |          |               |                  |
| consecutive_clipp... | 0              |         |          |               |                  |

|                      |         |
|----------------------|---------|
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 864     |
| ep_rew_mean          | 257     |
| exploration rate     | 0.633   |
| time/                |         |
| episodes             | 17900   |
| fps                  | 210     |
| time_elapsed         | 5247    |
| total timesteps      | 1103301 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0912  |
| n_updates            | 263325  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,020 | STEP=1,103,569 | clip=30 | consec=1 | max_consec=14 | avg_clip50=27.34 |
| EP=6,030 | STEP=1,105,815 | clip=8  | consec=0 | max_consec=14 | avg_clip50=27.08 |
| EP=6,040 | STEP=1,107,934 | clip=28 | consec=6 | max_consec=14 | avg_clip50=28.51 |
| EP=6,050 | STEP=1,110,375 | clip=34 | consec=3 | max_consec=14 | avg_clip50=29.79 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29      |
| clipped_episode_r... | 12.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 901     |
| ep_rew_mean          | 285     |
| exploration rate     | 0.63    |
| time/                |         |
| episodes             | 18000   |
| fps                  | 210     |
| time_elapsed         | 5289    |
| total timesteps      | 1110930 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0877  |
| n_updates            | 265232  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,060 | STEP=1,112,596 | clip=44 | consec=1 | max_consec=14 | avg_clip50=29.95 |
| EP=6,070 | STEP=1,114,629 | clip=34 | consec=1 | max_consec=14 | avg_clip50=30.94 |
| EP=6,080 | STEP=1,117,065 | clip=42 | consec=1 | max_consec=14 | avg_clip50=30.42 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 31.1    |
| clipped_episode_r... | 6.5     |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 923     |
| ep_rew_mean          | 296     |
| exploration rate     | 0.628   |
| time/                |         |
| episodes             | 18100   |
| fps                  | 209     |
| time_elapsed         | 5328    |
| total timesteps      | 1118570 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.159   |

|                      |                |         |          |               |                  |
|----------------------|----------------|---------|----------|---------------|------------------|
| n_updates            |                | 267142  |          |               |                  |
|                      |                |         |          |               |                  |
| EP=6,090             | STEP=1,119,178 | clip=32 | consec=2 | max_consec=14 | avg_clip50=29.82 |
| EP=6,100             | STEP=1,121,010 | clip=20 | consec=2 | max_consec=14 | avg_clip50=27.32 |
| EP=6,110             | STEP=1,123,028 | clip=10 | consec=0 | max_consec=14 | avg_clip50=26.30 |
| EP=6,120             | STEP=1,125,072 | clip=36 | consec=1 | max_consec=14 | avg_clip50=25.32 |
|                      |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           |                | 25      |          |               |                  |
| clipped_episode_r... |                | 14      |          |               |                  |
| consecutive_clipp... |                | 0       |          |               |                  |
| max_consecutive_c... |                | 14      |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          |                | 906     |          |               |                  |
| ep_rew_mean          |                | 282     |          |               |                  |
| exploration rate     |                | 0.625   |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             |                | 18200   |          |               |                  |
| fps                  |                | 209     |          |               |                  |
| time_elapsed         |                | 5362    |          |               |                  |
| total timesteps      |                | 1125429 |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        |                | 0.0001  |          |               |                  |
| loss                 |                | 0.306   |          |               |                  |
| n_updates            |                | 268857  |          |               |                  |
|                      |                |         |          |               |                  |
| EP=6,130             | STEP=1,126,926 | clip=14 | consec=0 | max_consec=14 | avg_clip50=23.74 |
| EP=6,140             | STEP=1,129,256 | clip=20 | consec=0 | max_consec=14 | avg_clip50=22.42 |
| EP=6,150             | STEP=1,131,252 | clip=26 | consec=3 | max_consec=14 | avg_clip50=23.29 |
|                      |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           |                | 24.7    |          |               |                  |
| clipped_episode_r... |                | 34.5    |          |               |                  |
| consecutive_clipp... |                | 2       |          |               |                  |
| max_consecutive_c... |                | 14      |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          |                | 880     |          |               |                  |
| ep_rew_mean          |                | 256     |          |               |                  |
| exploration rate     |                | 0.623   |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             |                | 18300   |          |               |                  |
| fps                  |                | 209     |          |               |                  |
| time_elapsed         |                | 5396    |          |               |                  |
| total timesteps      |                | 1132115 |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        |                | 0.0001  |          |               |                  |
| loss                 |                | 0.133   |          |               |                  |
| n_updates            |                | 270528  |          |               |                  |
|                      |                |         |          |               |                  |
| EP=6,160             | STEP=1,133,395 | clip=42 | consec=4 | max_consec=14 | avg_clip50=24.32 |
| EP=6,170             | STEP=1,135,461 | clip=18 | consec=0 | max_consec=14 | avg_clip50=25.28 |
| EP=6,180             | STEP=1,137,485 | clip=18 | consec=0 | max_consec=14 | avg_clip50=26.31 |
|                      |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           |                | 27.1    |          |               |                  |
| clipped_episode_r... |                | 3.5     |          |               |                  |
| consecutive_clipp... |                | 0       |          |               |                  |
| max_consecutive_c... |                | 14      |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          |                | 871     |          |               |                  |

|                  |         |
|------------------|---------|
| ep_rew_mean      | 252     |
| exploration rate | 0.621   |
| time/            |         |
| episodes         | 18400   |
| fps              | 209     |
| time_elapsed     | 5433    |
| total timesteps  | 1139489 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.055   |
| n_updates        | 272372  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,190 | STEP=1,139,950 | clip=39 | consec=2 | max_consec=14 | avg_clip50=28.39 |
| EP=6,200 | STEP=1,141,951 | clip=23 | consec=2 | max_consec=14 | avg_clip50=28.68 |
| EP=6,210 | STEP=1,144,047 | clip=12 | consec=0 | max_consec=14 | avg_clip50=27.47 |
| EP=6,220 | STEP=1,146,296 | clip=14 | consec=0 | max_consec=14 | avg_clip50=28.49 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.9    |
| clipped_episode_r... | 1       |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 887     |
| ep_rew_mean          | 263     |
| exploration rate     | 0.618   |
| time/                |         |
| episodes             | 18500   |
| fps                  | 209     |
| time_elapsed         | 5470    |
| total timesteps      | 1146745 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0369  |
| n_updates            | 274186  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,230 | STEP=1,148,260 | clip=72 | consec=2 | max_consec=14 | avg_clip50=28.60 |
| EP=6,240 | STEP=1,150,574 | clip=92 | consec=2 | max_consec=14 | avg_clip50=28.62 |
| EP=6,250 | STEP=1,152,640 | clip=55 | consec=1 | max_consec=14 | avg_clip50=29.09 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29.7    |
| clipped_episode_r... | 31.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 911     |
| ep_rew_mean          | 290     |
| exploration rate     | 0.616   |
| time/                |         |
| episodes             | 18600   |
| fps                  | 209     |
| time_elapsed         | 5507    |
| total timesteps      | 1154050 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0497  |
| n_updates            | 276012  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,260 | STEP=1,154,871 | clip=49 | consec=1 | max_consec=14 | avg_clip50=30.58 |
|----------|----------------|---------|----------|---------------|------------------|

EP=6,270 | STEP=1,156,825 | clip=8 | consec=0 | max\_consec=14 | avg\_clip50=29.11  
EP=6,280 | STEP=1,159,125 | clip=71 | consec=5 | max\_consec=14 | avg\_clip50=30.65

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 31      |
| clipped_episode_r... | 23.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 905     |
| ep_rew_mean          | 293     |
| exploration rate     | 0.613   |
| time/                |         |
| episodes             | 18700   |
| fps                  | 209     |
| time_elapsed         | 5545    |
| total timesteps      | 1161344 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0647  |
| n_updates            | 277835  |

EP=6,290 | STEP=1,161,417 | clip=40 | consec=2 | max\_consec=14 | avg\_clip50=29.98  
EP=6,300 | STEP=1,163,390 | clip=36 | consec=1 | max\_consec=14 | avg\_clip50=28.80  
EP=6,310 | STEP=1,165,187 | clip=10 | consec=0 | max\_consec=14 | avg\_clip50=26.61  
EP=6,320 | STEP=1,167,608 | clip=16 | consec=0 | max\_consec=14 | avg\_clip50=28.78

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29      |
| clipped_episode_r... | 11.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 902     |
| ep_rew_mean          | 295     |
| exploration rate     | 0.611   |
| time/                |         |
| episodes             | 18800   |
| fps                  | 209     |
| time_elapsed         | 5579    |
| total timesteps      | 1168206 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0605  |
| n_updates            | 279551  |

EP=6,330 | STEP=1,169,571 | clip=28 | consec=3 | max\_consec=14 | avg\_clip50=26.98  
EP=6,340 | STEP=1,171,628 | clip=22 | consec=1 | max\_consec=14 | avg\_clip50=25.94  
EP=6,350 | STEP=1,174,045 | clip=54 | consec=6 | max\_consec=14 | avg\_clip50=28.48

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 29.3  |
| clipped_episode_r... | 27    |
| consecutive_clipp... | 2     |
| max_consecutive_c... | 14    |
| rollout/             |       |
| ep_len_mean          | 911   |
| ep_rew_mean          | 284   |
| exploration rate     | 0.608 |
| time/                |       |

|  |                 |         |  |
|--|-----------------|---------|--|
|  | episodes        | 18900   |  |
|  | fps             | 209     |  |
|  | time_elapsed    | 5619    |  |
|  | total timesteps | 1175920 |  |
|  | train/          |         |  |
|  | learning_rate   | 0.0001  |  |
|  | loss            | 0.0605  |  |
|  | n_updates       | 281479  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=6,360 |  | STEP=1,176,584 |  | clip=50 |  | consec=5 |  | max_consec=14 |  | avg_clip50=30.43 |
| EP=6,370 |  | STEP=1,178,584 |  | clip=21 |  | consec=1 |  | max_consec=14 |  | avg_clip50=28.80 |
| EP=6,380 |  | STEP=1,180,575 |  | clip=44 |  | consec=1 |  | max_consec=14 |  | avg_clip50=27.95 |
| EP=6,390 |  | STEP=1,183,024 |  | clip=37 |  | consec=1 |  | max_consec=14 |  | avg_clip50=29.62 |

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 29.5    |  |
|  | clipped_episode_r... | 16      |  |
|  | consecutive_clipp... | 0       |  |
|  | max_consecutive_c... | 14      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 913     |  |
|  | ep_rew_mean          | 277     |  |
|  | exploration rate     | 0.606   |  |
|  | time/                |         |  |
|  | episodes             | 19000   |  |
|  | fps                  | 209     |  |
|  | time_elapsed         | 5656    |  |
|  | total timesteps      | 1183489 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.392   |  |
|  | n_updates            | 283372  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=6,400 |  | STEP=1,185,388 |  | clip=35 |  | consec=1 |  | max_consec=14 |  | avg_clip50=27.71 |
| EP=6,410 |  | STEP=1,187,162 |  | clip=8  |  | consec=0 |  | max_consec=14 |  | avg_clip50=25.76 |
| EP=6,420 |  | STEP=1,189,100 |  | clip=25 |  | consec=2 |  | max_consec=14 |  | avg_clip50=24.30 |

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 27.4    |  |
|  | clipped_episode_r... | 65.5    |  |
|  | consecutive_clipp... | 3       |  |
|  | max_consecutive_c... | 14      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 922     |  |
|  | ep_rew_mean          | 279     |  |
|  | exploration rate     | 0.603   |  |
|  | time/                |         |  |
|  | episodes             | 19100   |  |
|  | fps                  | 209     |  |
|  | time_elapsed         | 5695    |  |
|  | total timesteps      | 1190868 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.0414  |  |
|  | n_updates            | 285216  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=6,430 |  | STEP=1,191,669 |  | clip=29 |  | consec=7 |  | max_consec=14 |  | avg_clip50=28.39 |
| EP=6,440 |  | STEP=1,193,933 |  | clip=32 |  | consec=1 |  | max_consec=14 |  | avg_clip50=28.31 |
| EP=6,450 |  | STEP=1,196,731 |  | clip=52 |  | consec=3 |  | max_consec=14 |  | avg_clip50=31.83 |
| EP=6,460 |  | STEP=1,199,121 |  | clip=43 |  | consec=2 |  | max_consec=14 |  | avg_clip50=34.43 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34.4    |
| clipped_episode_r... | 43      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 946     |
| ep_rew_mean          | 301     |
| exploration rate     | 0.601   |
| time/                |         |
| episodes             | 19200   |
| fps                  | 208     |
| time_elapsed         | 5737    |
| total timesteps      | 1199146 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0659  |
| n_updates            | 287286  |

EP=6,470 | STEP=1,201,744 | clip=62 | consec=1 | max\_consec=14 | avg\_clip50=38.46  
EP=6,480 | STEP=1,203,711 | clip=46 | consec=1 | max\_consec=14 | avg\_clip50=35.46  
EP=6,490 | STEP=1,205,702 | clip=16 | consec=0 | max\_consec=14 | avg\_clip50=34.35

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34      |
| clipped_episode_r... | 37.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 951     |
| ep_rew_mean          | 317     |
| exploration rate     | 0.598   |
| time/                |         |
| episodes             | 19300   |
| fps                  | 208     |
| time_elapsed         | 5774    |
| total timesteps      | 1206571 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0536  |
| n_updates            | 289142  |

EP=6,500 | STEP=1,207,968 | clip=22 | consec=1 | max\_consec=14 | avg\_clip50=32.89  
EP=6,510 | STEP=1,210,141 | clip=48 | consec=1 | max\_consec=14 | avg\_clip50=31.97  
EP=6,520 | STEP=1,212,456 | clip=20 | consec=0 | max\_consec=14 | avg\_clip50=30.57

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 31    |
| clipped_episode_r... | 13.5  |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 14    |
| rollout/             |       |
| ep_len_mean          | 972   |
| ep_rew_mean          | 332   |
| exploration rate     | 0.596 |
| time/                |       |
| episodes             | 19400 |
| fps                  | 208   |
| time_elapsed         | 5813  |

|                 |         |
|-----------------|---------|
| total timesteps | 1214408 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.114   |
| n_updates       | 291101  |

EP=6,530 | STEP=1,214,853 | clip=28 | consec=2 | max\_consec=14 | avg\_clip50=31.28  
EP=6,540 | STEP=1,216,948 | clip=12 | consec=0 | max\_consec=14 | avg\_clip50=29.80  
EP=6,550 | STEP=1,218,922 | clip=45 | consec=2 | max\_consec=14 | avg\_clip50=28.24  
EP=6,560 | STEP=1,221,239 | clip=56 | consec=2 | max\_consec=14 | avg\_clip50=29.25

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 28.5    |
| clipped_episode_r... | 6.5     |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 924     |
| ep_rew_mean          | 301     |
| exploration rate     | 0.593   |
| time/                |         |
| episodes             | 19500   |
| fps                  | 208     |
| time_elapsed         | 5849    |
| total timesteps      | 1221710 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0637  |
| n_updates            | 292927  |

EP=6,570 | STEP=1,223,357 | clip=22 | consec=2 | max\_consec=14 | avg\_clip50=27.57  
EP=6,580 | STEP=1,225,260 | clip=2 | consec=0 | max\_consec=14 | avg\_clip50=26.39  
EP=6,590 | STEP=1,227,625 | clip=56 | consec=2 | max\_consec=14 | avg\_clip50=28.97

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 28.4    |
| clipped_episode_r... | 15.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 922     |
| ep_rew_mean          | 288     |
| exploration rate     | 0.591   |
| time/                |         |
| episodes             | 19600   |
| fps                  | 208     |
| time_elapsed         | 5887    |
| total timesteps      | 1229089 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0393  |
| n_updates            | 294772  |

EP=6,600 | STEP=1,229,688 | clip=5 | consec=0 | max\_consec=14 | avg\_clip50=27.79  
EP=6,610 | STEP=1,232,280 | clip=12 | consec=0 | max\_consec=14 | avg\_clip50=28.42  
EP=6,620 | STEP=1,234,123 | clip=27 | consec=1 | max\_consec=14 | avg\_clip50=27.75  
EP=6,630 | STEP=1,236,025 | clip=8 | consec=0 | max\_consec=14 | avg\_clip50=28.91

|            |      |
|------------|------|
| custom/    |      |
| avg_clip50 | 28.9 |



|  |                      |         |  |
|--|----------------------|---------|--|
|  | clipped_episode_r... | 8       |  |
|  | consecutive_clipp... | 0       |  |
|  | max_consecutive_c... | 14      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 892     |  |
|  | ep_rew_mean          | 276     |  |
|  | exploration rate     | 0.588   |  |
|  | time/                |         |  |
|  | episodes             | 19700   |  |
|  | fps                  | 208     |  |
|  | time_elapsed         | 5923    |  |
|  | total timesteps      | 1236106 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.254   |  |
|  | n_updates            | 296526  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=6,640 |  | STEP=1,238,356 |  | clip=28 |  | consec=5 |  | max_consec=14 |  | avg_clip50=27.97 |
| EP=6,650 |  | STEP=1,240,197 |  | clip=30 |  | consec=2 |  | max_consec=14 |  | avg_clip50=27.44 |
| EP=6,660 |  | STEP=1,242,680 |  | clip=32 |  | consec=4 |  | max_consec=14 |  | avg_clip50=26.82 |

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 27.6    |  |
|  | clipped_episode_r... | 9.5     |  |
|  | consecutive_clipp... | 0       |  |
|  | max_consecutive_c... | 14      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 902     |  |
|  | ep_rew_mean          | 278     |  |
|  | exploration rate     | 0.586   |  |
|  | time/                |         |  |
|  | episodes             | 19800   |  |
|  | fps                  | 208     |  |
|  | time_elapsed         | 5960    |  |
|  | total timesteps      | 1243523 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.162   |  |
|  | n_updates            | 298380  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=6,670 |  | STEP=1,244,878 |  | clip=36 |  | consec=5 |  | max_consec=14 |  | avg_clip50=28.47 |
| EP=6,680 |  | STEP=1,247,054 |  | clip=23 |  | consec=2 |  | max_consec=14 |  | avg_clip50=27.93 |
| EP=6,690 |  | STEP=1,249,769 |  | clip=69 |  | consec=3 |  | max_consec=14 |  | avg_clip50=29.94 |

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 31.4    |  |
|  | clipped_episode_r... | 5       |  |
|  | consecutive_clipp... | 0       |  |
|  | max_consecutive_c... | 14      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 928     |  |
|  | ep_rew_mean          | 292     |  |
|  | exploration rate     | 0.583   |  |
|  | time/                |         |  |
|  | episodes             | 19900   |  |
|  | fps                  | 208     |  |
|  | time_elapsed         | 6001    |  |
|  | total timesteps      | 1251670 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |

|          |                      |         |          |               |                  |
|----------|----------------------|---------|----------|---------------|------------------|
|          | loss                 | 0.0669  |          |               |                  |
|          | n_updates            | 300417  |          |               |                  |
| -----    |                      |         |          |               |                  |
| EP=6,700 | STEP=1,251,889       | clip=34 | consec=1 | max_consec=14 | avg_clip50=31.50 |
| EP=6,710 | STEP=1,253,751       | clip=39 | consec=1 | max_consec=14 | avg_clip50=30.01 |
| EP=6,720 | STEP=1,256,126       | clip=49 | consec=6 | max_consec=14 | avg_clip50=30.33 |
| EP=6,730 | STEP=1,258,367       | clip=30 | consec=1 | max_consec=14 | avg_clip50=31.71 |
| -----    |                      |         |          |               |                  |
|          | custom/              |         |          |               |                  |
|          | avg_clip50           | 32.4    |          |               |                  |
|          | clipped_episode_r... | 36      |          |               |                  |
|          | consecutive_clipp... | 2       |          |               |                  |
|          | max_consecutive_c... | 14      |          |               |                  |
|          | rollout/             |         |          |               |                  |
|          | ep_len_mean          | 937     |          |               |                  |
|          | ep_rew_mean          | 296     |          |               |                  |
|          | exploration rate     | 0.581   |          |               |                  |
|          | time/                |         |          |               |                  |
|          | episodes             | 20000   |          |               |                  |
|          | fps                  | 208     |          |               |                  |
|          | time_elapsed         | 6038    |          |               |                  |
|          | total timesteps      | 1259092 |          |               |                  |
|          | train/               |         |          |               |                  |
|          | learning_rate        | 0.0001  |          |               |                  |
|          | loss                 | 0.395   |          |               |                  |
|          | n_updates            | 302272  |          |               |                  |
| -----    |                      |         |          |               |                  |
| EP=6,740 | STEP=1,260,396       | clip=16 | consec=0 | max_consec=14 | avg_clip50=29.39 |
| EP=6,750 | STEP=1,262,323       | clip=14 | consec=0 | max_consec=14 | avg_clip50=28.46 |
| EP=6,760 | STEP=1,264,727       | clip=64 | consec=2 | max_consec=14 | avg_clip50=31.74 |
| -----    |                      |         |          |               |                  |
|          | custom/              |         |          |               |                  |
|          | avg_clip50           | 31      |          |               |                  |
|          | clipped_episode_r... | 23      |          |               |                  |
|          | consecutive_clipp... | 1       |          |               |                  |
|          | max_consecutive_c... | 14      |          |               |                  |
|          | rollout/             |         |          |               |                  |
|          | ep_len_mean          | 921     |          |               |                  |
|          | ep_rew_mean          | 303     |          |               |                  |
|          | exploration rate     | 0.578   |          |               |                  |
|          | time/                |         |          |               |                  |
|          | episodes             | 20100   |          |               |                  |
|          | fps                  | 208     |          |               |                  |
|          | time_elapsed         | 6073    |          |               |                  |
|          | total timesteps      | 1266070 |          |               |                  |
|          | train/               |         |          |               |                  |
|          | learning_rate        | 0.0001  |          |               |                  |
|          | loss                 | 0.139   |          |               |                  |
|          | n_updates            | 304017  |          |               |                  |
| -----    |                      |         |          |               |                  |
| EP=6,770 | STEP=1,266,764       | clip=51 | consec=4 | max_consec=14 | avg_clip50=30.82 |
| EP=6,780 | STEP=1,268,908       | clip=6  | consec=0 | max_consec=14 | avg_clip50=29.55 |
| EP=6,790 | STEP=1,270,740       | clip=14 | consec=0 | max_consec=14 | avg_clip50=28.77 |
| EP=6,800 | STEP=1,273,524       | clip=58 | consec=5 | max_consec=14 | avg_clip50=31.83 |
| -----    |                      |         |          |               |                  |
|          | custom/              |         |          |               |                  |
|          | avg_clip50           | 31.8    |          |               |                  |
|          | clipped_episode_r... | 57.5    |          |               |                  |
|          | consecutive_clipp... | 5       |          |               |                  |
|          | max_consecutive_c... | 14      |          |               |                  |

|                  |         |
|------------------|---------|
| rollout/         |         |
| ep_len_mean      | 911     |
| ep_rew_mean      | 301     |
| exploration rate | 0.576   |
| time/            |         |
| episodes         | 20200   |
| fps              | 208     |
| time_elapsed     | 6112    |
| total timesteps  | 1273784 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0761  |
| n_updates        | 305945  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,810 | STEP=1,275,402 | clip=11 | consec=0 | max_consec=14 | avg_clip50=28.18 |
| EP=6,820 | STEP=1,277,329 | clip=22 | consec=3 | max_consec=14 | avg_clip50=27.46 |
| EP=6,830 | STEP=1,279,668 | clip=18 | consec=0 | max_consec=14 | avg_clip50=29.65 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.7    |
| clipped_episode_r... | 52.5    |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 901     |
| ep_rew_mean          | 296     |
| exploration rate     | 0.573   |
| time/                |         |
| episodes             | 20300   |
| fps                  | 208     |
| time_elapsed         | 6148    |
| total timesteps      | 1280807 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.369   |
| n_updates            | 307701  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,840 | STEP=1,281,889 | clip=18 | consec=0 | max_consec=14 | avg_clip50=30.57 |
| EP=6,850 | STEP=1,284,448 | clip=30 | consec=6 | max_consec=14 | avg_clip50=30.81 |
| EP=6,860 | STEP=1,287,187 | clip=44 | consec=2 | max_consec=14 | avg_clip50=33.99 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 37.3    |
| clipped_episode_r... | 51.5    |
| consecutive_clipp... | 11      |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 968     |
| ep_rew_mean          | 327     |
| exploration rate     | 0.571   |
| time/                |         |
| episodes             | 20400   |
| fps                  | 208     |
| time_elapsed         | 6192    |
| total timesteps      | 1289599 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.182   |
| n_updates            | 309899  |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=6,870 | STEP=1,289,794 | clip=32 | consec=12 | max_consec=14 | avg_clip50=37.49 |
| EP=6,880 | STEP=1,292,482 | clip=58 | consec=9  | max_consec=14 | avg_clip50=37.96 |
| EP=6,890 | STEP=1,295,128 | clip=14 | consec=0  | max_consec=14 | avg_clip50=40.32 |
| EP=6,900 | STEP=1,297,365 | clip=25 | consec=1  | max_consec=14 | avg_clip50=38.35 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 38.2    |
| clipped_episode_r... | 50.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 1e+03   |
| ep_rew_mean          | 345     |
| exploration rate     | 0.568   |
| time/                |         |
| episodes             | 20500   |
| fps                  | 208     |
| time_elapsed         | 6235    |
| total timesteps      | 1298027 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.309   |
| n_updates            | 312006  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,910 | STEP=1,299,659 | clip=10 | consec=0 | max_consec=14 | avg_clip50=35.87 |
| EP=6,920 | STEP=1,302,056 | clip=15 | consec=0 | max_consec=14 | avg_clip50=35.01 |
| EP=6,930 | STEP=1,304,261 | clip=26 | consec=1 | max_consec=14 | avg_clip50=33.20 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 29.8     |
| clipped_episode_r... | 3        |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 14       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 349      |
| exploration rate     | 0.565    |
| time/                |          |
| episodes             | 20600    |
| fps                  | 208      |
| time_elapsed         | 6275     |
| total timesteps      | 1305794  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0466   |
| n_updates            | 313948   |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,940 | STEP=1,306,316 | clip=50 | consec=2 | max_consec=14 | avg_clip50=30.20 |
| EP=6,950 | STEP=1,308,586 | clip=17 | consec=0 | max_consec=14 | avg_clip50=28.85 |
| EP=6,960 | STEP=1,310,854 | clip=24 | consec=4 | max_consec=14 | avg_clip50=28.82 |
| EP=6,970 | STEP=1,312,712 | clip=20 | consec=1 | max_consec=14 | avg_clip50=26.13 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 25.9 |
| clipped_episode_r... | 16.5 |
| consecutive_clipp... | 0    |
| max_consecutive_c... | 14   |
| rollout/             |      |
| ep_len_mean          | 959  |

|                  |         |
|------------------|---------|
| ep_rew_mean      | 303     |
| exploration rate | 0.563   |
| time/            |         |
| episodes         | 20700   |
| fps              | 207     |
| time_elapsed     | 6313    |
| total timesteps  | 1313095 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0694  |
| n_updates        | 315773  |

EP=6,980 | STEP=1,314,854 | clip=24 | consec=1 | max\_consec=14 | avg\_clip50=26.07  
EP=6,990 | STEP=1,317,355 | clip=14 | consec=0 | max\_consec=14 | avg\_clip50=28.43  
EP=7,000 | STEP=1,319,806 | clip=42 | consec=7 | max\_consec=14 | avg\_clip50=31.25

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 32.7    |
| clipped_episode_r... | 80      |
| consecutive_clipp... | 12      |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 951     |
| ep_rew_mean          | 309     |
| exploration rate     | 0.56    |
| time/                |         |
| episodes             | 20800   |
| fps                  | 207     |
| time_elapsed         | 6354    |
| total timesteps      | 1321242 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0641  |
| n_updates            | 317810  |

EP=7,010 | STEP=1,322,153 | clip=4 | consec=0 | max\_consec=14 | avg\_clip50=32.26  
EP=7,020 | STEP=1,324,450 | clip=23 | consec=10 | max\_consec=14 | avg\_clip50=35.0  
2  
EP=7,030 | STEP=1,327,018 | clip=46 | consec=5 | max\_consec=14 | avg\_clip50=36.11

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 33.6    |
| clipped_episode_r... | 48.5    |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 962     |
| ep_rew_mean          | 314     |
| exploration rate     | 0.557   |
| time/                |         |
| episodes             | 20900   |
| fps                  | 207     |
| time_elapsed         | 6393    |
| total timesteps      | 1329109 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0861  |
| n_updates            | 319777  |

EP=7,040 | STEP=1,329,133 | clip=40 | consec=3 | max\_consec=14 | avg\_clip50=34.19

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,050 | STEP=1,331,565 | clip=34 | consec=2 | max_consec=14 | avg_clip50=33.06 |
| EP=7,060 | STEP=1,333,956 | clip=46 | consec=1 | max_consec=14 | avg_clip50=33.49 |
| EP=7,070 | STEP=1,335,977 | clip=3  | consec=0 | max_consec=14 | avg_clip50=30.90 |

|                      |         |  |
|----------------------|---------|--|
| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 32.3    |  |
| clipped_episode_r... | 47.5    |  |
| consecutive_clipp... | 3       |  |
| max_consecutive_c... | 14      |  |
| rollout/             |         |  |
| ep_len_mean          | 988     |  |
| ep_rew_mean          | 339     |  |
| exploration rate     | 0.555   |  |
| time/                |         |  |
| episodes             | 21000   |  |
| fps                  | 207     |  |
| time_elapsed         | 6432    |  |
| total timesteps      | 1337115 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.0497  |  |
| n_updates            | 321778  |  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,080 | STEP=1,338,385 | clip=24 | consec=1 | max_consec=14 | avg_clip50=30.49 |
| EP=7,090 | STEP=1,340,814 | clip=40 | consec=1 | max_consec=14 | avg_clip50=31.81 |
| EP=7,100 | STEP=1,342,759 | clip=52 | consec=1 | max_consec=14 | avg_clip50=31.20 |

|                      |         |  |
|----------------------|---------|--|
| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 29.5    |  |
| clipped_episode_r... | 29.5    |  |
| consecutive_clipp... | 3       |  |
| max_consecutive_c... | 14      |  |
| rollout/             |         |  |
| ep_len_mean          | 944     |  |
| ep_rew_mean          | 311     |  |
| exploration rate     | 0.552   |  |
| time/                |         |  |
| episodes             | 21100   |  |
| fps                  | 207     |  |
| time_elapsed         | 6470    |  |
| total timesteps      | 1344526 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.148   |  |
| n_updates            | 323631  |  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,110 | STEP=1,344,828 | clip=44 | consec=4 | max_consec=14 | avg_clip50=29.43 |
| EP=7,120 | STEP=1,346,787 | clip=11 | consec=0 | max_consec=14 | avg_clip50=29.42 |
| EP=7,130 | STEP=1,348,816 | clip=9  | consec=0 | max_consec=14 | avg_clip50=27.40 |
| EP=7,140 | STEP=1,350,811 | clip=9  | consec=0 | max_consec=14 | avg_clip50=26.31 |

|                      |      |  |
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| -----                |      |  |
| custom/              |      |  |
| avg_clip50           | 25.2 |  |
| clipped_episode_r... | 15   |  |
| consecutive_clipp... | 0    |  |
| max_consecutive_c... | 14   |  |
| rollout/             |      |  |
| ep_len_mean          | 903  |  |
| ep_rew_mean          | 284  |  |
| exploration rate     | 0.55 |  |

|                 |         |
|-----------------|---------|
| time/           |         |
| episodes        | 21200   |
| fps             | 207     |
| time_elapsed    | 6504    |
| total timesteps | 1351258 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.0507  |
| n_updates       | 325314  |

EP=7,150 | STEP=1,352,630 | clip=20 | consec=0 | max\_consec=14 | avg\_clip50=24.12  
EP=7,160 | STEP=1,355,166 | clip=6 | consec=0 | max\_consec=14 | avg\_clip50=25.54  
EP=7,170 | STEP=1,357,078 | clip=14 | consec=0 | max\_consec=14 | avg\_clip50=25.74

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 26      |
| clipped_episode_r... | 16.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 888     |
| ep_rew_mean          | 272     |
| exploration rate     | 0.548   |
| time/                |         |
| episodes             | 21300   |
| fps                  | 207     |
| time_elapsed         | 6543    |
| total timesteps      | 1358609 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0639  |
| n_updates            | 327152  |

EP=7,180 | STEP=1,359,580 | clip=36 | consec=1 | max\_consec=14 | avg\_clip50=27.67  
EP=7,190 | STEP=1,361,898 | clip=35 | consec=3 | max\_consec=14 | avg\_clip50=28.43  
EP=7,200 | STEP=1,364,053 | clip=39 | consec=1 | max\_consec=14 | avg\_clip50=30.88

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29.3    |
| clipped_episode_r... | 23.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 901     |
| ep_rew_mean          | 278     |
| exploration rate     | 0.545   |
| time/                |         |
| episodes             | 21400   |
| fps                  | 207     |
| time_elapsed         | 6582    |
| total timesteps      | 1366082 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.376   |
| n_updates            | 329020  |

EP=7,210 | STEP=1,366,205 | clip=52 | consec=2 | max\_consec=14 | avg\_clip50=30.24  
EP=7,220 | STEP=1,368,210 | clip=8 | consec=0 | max\_consec=14 | avg\_clip50=30.28  
EP=7,230 | STEP=1,370,516 | clip=16 | consec=0 | max\_consec=14 | avg\_clip50=29.66  
EP=7,240 | STEP=1,372,581 | clip=70 | consec=1 | max\_consec=14 | avg\_clip50=28.57

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29.4    |
| clipped_episode_r... | 29      |
| consecutive_clipp... | 4       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 923     |
| ep_rew_mean          | 287     |
| exploration rate     | 0.543   |
| time/                |         |
| episodes             | 21500   |
| fps                  | 207     |
| time_elapsed         | 6619    |
| total timesteps      | 1373420 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0626  |
| n_updates            | 330854  |

EP=7,250 | STEP=1,374,681 | clip=30 | consec=3 | max\_consec=14 | avg\_clip50=27.53  
EP=7,260 | STEP=1,377,459 | clip=18 | consec=0 | max\_consec=14 | avg\_clip50=30.40  
EP=7,270 | STEP=1,379,487 | clip=56 | consec=3 | max\_consec=14 | avg\_clip50=30.12

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.4    |
| clipped_episode_r... | 16      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 14      |
| rollout/             |         |
| ep_len_mean          | 945     |
| ep_rew_mean          | 303     |
| exploration rate     | 0.54    |
| time/                |         |
| episodes             | 21600   |
| fps                  | 207     |
| time_elapsed         | 6661    |
| total timesteps      | 1381560 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.104   |
| n_updates            | 332889  |

EP=7,280 | STEP=1,382,199 | clip=68 | consec=2 | max\_consec=14 | avg\_clip50=31.75  
EP=7,290 | STEP=1,384,404 | clip=56 | consec=2 | max\_consec=14 | avg\_clip50=31.92  
EP=7,300 | STEP=1,387,020 | clip=46 | consec=7 | max\_consec=14 | avg\_clip50=34.53  
EP=7,310 | STEP=1,389,386 | clip=30 | consec=3 | max\_consec=14 | avg\_clip50=32.60

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 33.1  |
| clipped_episode_r... | 35    |
| consecutive_clipp... | 4     |
| max_consecutive_c... | 14    |
| rollout/             |       |
| ep_len_mean          | 980   |
| ep_rew_mean          | 318   |
| exploration rate     | 0.537 |
| time/                |       |
| episodes             | 21700 |
| fps                  | 207   |



|  |                 |  |         |  |
|--|-----------------|--|---------|--|
|  | time_elapsed    |  | 6703    |  |
|  | total timesteps |  | 1389674 |  |
|  | train/          |  |         |  |
|  | learning_rate   |  | 0.0001  |  |
|  | loss            |  | 0.123   |  |
|  | n_updates       |  | 334918  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=7,320 |  | STEP=1,391,640 |  | clip=24 |  | consec=1 |  | max_consec=14 |  | avg_clip50=33.65 |
| EP=7,330 |  | STEP=1,394,294 |  | clip=29 |  | consec=6 |  | max_consec=14 |  | avg_clip50=34.15 |
| EP=7,340 |  | STEP=1,396,528 |  | clip=32 |  | consec=3 |  | max_consec=14 |  | avg_clip50=34.59 |

|  |                      |  |          |  |
|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 35.5     |  |
|  | clipped_episode_r... |  | 26.5     |  |
|  | consecutive_clipp... |  | 7        |  |
|  | max_consecutive_c... |  | 14       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.01e+03 |  |
|  | ep_rew_mean          |  | 338      |  |
|  | exploration rate     |  | 0.535    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 21800    |  |
|  | fps                  |  | 207      |  |
|  | time_elapsed         |  | 6744     |  |
|  | total timesteps      |  | 1397609  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.302    |  |
|  | n_updates            |  | 336902   |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=7,350 |  | STEP=1,398,836 |  | clip=66 |  | consec=4 |  | max_consec=14 |  | avg_clip50=33.65 |
| EP=7,360 |  | STEP=1,401,040 |  | clip=24 |  | consec=5 |  | max_consec=14 |  | avg_clip50=32.90 |
| EP=7,370 |  | STEP=1,403,300 |  | clip=28 |  | consec=2 |  | max_consec=14 |  | avg_clip50=33.29 |

|  |                      |  |         |  |
|--|----------------------|--|---------|--|
|  | custom/              |  |         |  |
|  | avg_clip50           |  | 30.6    |  |
|  | clipped_episode_r... |  | 15.5    |  |
|  | consecutive_clipp... |  | 0       |  |
|  | max_consecutive_c... |  | 14      |  |
|  | rollout/             |  |         |  |
|  | ep_len_mean          |  | 986     |  |
|  | ep_rew_mean          |  | 329     |  |
|  | exploration rate     |  | 0.532   |  |
|  | time/                |  |         |  |
|  | episodes             |  | 21900   |  |
|  | fps                  |  | 207     |  |
|  | time_elapsed         |  | 6781    |  |
|  | total timesteps      |  | 1405147 |  |
|  | train/               |  |         |  |
|  | learning_rate        |  | 0.0001  |  |
|  | loss                 |  | 0.105   |  |
|  | n_updates            |  | 338786  |  |

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|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=7,380 |  | STEP=1,405,659 |  | clip=42 |  | consec=2  |  | max_consec=14 |  | avg_clip50=31.01 |
| EP=7,390 |  | STEP=1,408,651 |  | clip=24 |  | consec=12 |  | max_consec=14 |  | avg_clip50=35.1  |
| 2        |  |                |  |         |  |           |  |               |  |                  |
| EP=7,400 |  | STEP=1,410,872 |  | clip=39 |  | consec=1  |  | max_consec=18 |  | avg_clip50=34.14 |
| EP=7,410 |  | STEP=1,412,737 |  | clip=34 |  | consec=1  |  | max_consec=18 |  | avg_clip50=33.82 |

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|--|---------|--|--|--|
|  | custom/ |  |  |  |
|--|---------|--|--|--|

|                      |         |
|----------------------|---------|
| avg_clip50           | 34.8    |
| clipped_episode_r... | 47.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 984     |
| ep_rew_mean          | 336     |
| exploration rate     | 0.529   |
| time/                |         |
| episodes             | 22000   |
| fps                  | 207     |
| time_elapsed         | 6823    |
| total timesteps      | 1413419 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0758  |
| n_updates            | 340854  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,420 | STEP=1,415,172 | clip=46 | consec=3 | max_consec=18 | avg_clip50=34.36 |
| EP=7,430 | STEP=1,417,354 | clip=22 | consec=4 | max_consec=18 | avg_clip50=35.48 |
| EP=7,440 | STEP=1,419,543 | clip=38 | consec=4 | max_consec=18 | avg_clip50=31.69 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 32.1    |
| clipped_episode_r... | 15.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 978     |
| ep_rew_mean          | 341     |
| exploration rate     | 0.527   |
| time/                |         |
| episodes             | 22100   |
| fps                  | 207     |
| time_elapsed         | 6864    |
| total timesteps      | 1421392 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0523  |
| n_updates            | 342847  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,450 | STEP=1,421,966 | clip=38 | consec=1 | max_consec=18 | avg_clip50=33.30 |
| EP=7,460 | STEP=1,424,180 | clip=5  | consec=0 | max_consec=18 | avg_clip50=33.13 |
| EP=7,470 | STEP=1,426,562 | clip=18 | consec=0 | max_consec=18 | avg_clip50=34.04 |
| EP=7,480 | STEP=1,428,854 | clip=13 | consec=0 | max_consec=18 | avg_clip50=34.36 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34.6    |
| clipped_episode_r... | 20.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 968     |
| ep_rew_mean          | 347     |
| exploration rate     | 0.524   |
| time/                |         |
| episodes             | 22200   |
| fps                  | 206     |
| time_elapsed         | 6904    |
| total timesteps      | 1429030 |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.11   |
| n_updates     | 344757 |

EP=7,490 | STEP=1,430,934 | clip=5 | consec=0 | max\_consec=18 | avg\_clip50=33.18  
EP=7,500 | STEP=1,433,490 | clip=24 | consec=4 | max\_consec=18 | avg\_clip50=34.08  
EP=7,510 | STEP=1,435,745 | clip=5 | consec=0 | max\_consec=18 | avg\_clip50=35.18

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 35.5    |
| clipped_episode_r... | 40      |
| consecutive_clipp... | 4       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 952     |
| ep_rew_mean          | 338     |
| exploration rate     | 0.522   |
| time/                |         |
| episodes             | 22300   |
| fps                  | 206     |
| time_elapsed         | 6943    |
| total timesteps      | 1436904 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.146   |
| n_updates            | 346725  |

EP=7,520 | STEP=1,437,891 | clip=18 | consec=0 | max\_consec=18 | avg\_clip50=34.35  
EP=7,530 | STEP=1,440,365 | clip=40 | consec=4 | max\_consec=18 | avg\_clip50=35.26  
EP=7,540 | STEP=1,443,029 | clip=54 | consec=1 | max\_consec=18 | avg\_clip50=39.47  
EP=7,550 | STEP=1,445,378 | clip=27 | consec=5 | max\_consec=18 | avg\_clip50=37.72

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 37.7    |
| clipped_episode_r... | 27      |
| consecutive_clipp... | 5       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 982     |
| ep_rew_mean          | 359     |
| exploration rate     | 0.519   |
| time/                |         |
| episodes             | 22400   |
| fps                  | 206     |
| time_elapsed         | 6987    |
| total timesteps      | 1445567 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.148   |
| n_updates            | 348891  |

EP=7,560 | STEP=1,447,334 | clip=48 | consec=1 | max\_consec=18 | avg\_clip50=35.26  
EP=7,570 | STEP=1,449,459 | clip=40 | consec=5 | max\_consec=18 | avg\_clip50=34.09  
EP=7,580 | STEP=1,451,651 | clip=6 | consec=0 | max\_consec=18 | avg\_clip50=31.02

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 30.3 |
| clipped_episode_r... | 12.5 |
| consecutive_clipp... | 0    |

|                      |         |
|----------------------|---------|
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 953     |
| ep_rew_mean          | 332     |
| exploration rate     | 0.516   |
| time/                |         |
| episodes             | 22500   |
| fps                  | 206     |
| time_elapsed         | 7024    |
| total timesteps      | 1452775 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.162   |
| n_updates            | 350693  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,590 | STEP=1,453,964 | clip=47 | consec=2 | max_consec=18 | avg_clip50=28.23 |
| EP=7,600 | STEP=1,456,278 | clip=32 | consec=6 | max_consec=18 | avg_clip50=28.33 |
| EP=7,610 | STEP=1,458,152 | clip=12 | consec=0 | max_consec=18 | avg_clip50=28.65 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.6    |
| clipped_episode_r... | 21      |
| consecutive_clipp... | 6       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 954     |
| ep_rew_mean          | 319     |
| exploration rate     | 0.514   |
| time/                |         |
| episodes             | 22600   |
| fps                  | 206     |
| time_elapsed         | 7064    |
| total timesteps      | 1460426 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0729  |
| n_updates            | 352606  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,620 | STEP=1,460,642 | clip=51 | consec=8 | max_consec=18 | avg_clip50=30.48 |
| EP=7,630 | STEP=1,463,008 | clip=48 | consec=3 | max_consec=18 | avg_clip50=32.08 |
| EP=7,640 | STEP=1,465,606 | clip=13 | consec=0 | max_consec=18 | avg_clip50=33.04 |
| EP=7,650 | STEP=1,468,094 | clip=52 | consec=1 | max_consec=18 | avg_clip50=34.31 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34.2    |
| clipped_episode_r... | 6.5     |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 955     |
| ep_rew_mean          | 315     |
| exploration rate     | 0.511   |
| time/                |         |
| episodes             | 22700   |
| fps                  | 206     |
| time_elapsed         | 7106    |
| total timesteps      | 1468834 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0423  |

|                      |                |         |           |               |                  |  |
|----------------------|----------------|---------|-----------|---------------|------------------|--|
| n_updates            |                | 354708  |           |               |                  |  |
| -----                |                |         |           |               |                  |  |
| EP=7,660             | STEP=1,469,996 | clip=14 | consec=0  | max_consec=18 | avg_clip50=35.31 |  |
| EP=7,670             | STEP=1,472,062 | clip=30 | consec=2  | max_consec=18 | avg_clip50=33.67 |  |
| EP=7,680             | STEP=1,474,501 | clip=50 | consec=2  | max_consec=18 | avg_clip50=33.78 |  |
| -----                |                |         |           |               |                  |  |
| custom/              |                |         |           |               |                  |  |
| avg_clip50           |                | 33.3    |           |               |                  |  |
| clipped_episode_r... |                | 54.5    |           |               |                  |  |
| consecutive_clipp... |                | 6       |           |               |                  |  |
| max_consecutive_c... |                | 18      |           |               |                  |  |
| rollout/             |                |         |           |               |                  |  |
| ep_len_mean          |                | 952     |           |               |                  |  |
| ep_rew_mean          |                | 328     |           |               |                  |  |
| exploration rate     |                | 0.508   |           |               |                  |  |
| time/                |                |         |           |               |                  |  |
| episodes             |                | 22800   |           |               |                  |  |
| fps                  |                | 206     |           |               |                  |  |
| time_elapsed         |                | 7143    |           |               |                  |  |
| total timesteps      |                | 1476123 |           |               |                  |  |
| train/               |                |         |           |               |                  |  |
| learning_rate        |                | 0.0001  |           |               |                  |  |
| loss                 |                | 0.0417  |           |               |                  |  |
| n_updates            |                | 356530  |           |               |                  |  |
| -----                |                |         |           |               |                  |  |
| EP=7,690             | STEP=1,476,634 | clip=58 | consec=1  | max_consec=18 | avg_clip50=32.47 |  |
| EP=7,700             | STEP=1,479,140 | clip=18 | consec=0  | max_consec=18 | avg_clip50=32.65 |  |
| EP=7,710             | STEP=1,481,623 | clip=66 | consec=10 | max_consec=18 | avg_clip50=34.6  |  |
| 0                    |                |         |           |               |                  |  |
| EP=7,720             | STEP=1,483,937 | clip=32 | consec=7  | max_consec=18 | avg_clip50=35.73 |  |
| -----                |                |         |           |               |                  |  |
| custom/              |                |         |           |               |                  |  |
| avg_clip50           |                | 35.3    |           |               |                  |  |
| clipped_episode_r... |                | 12.5    |           |               |                  |  |
| consecutive_clipp... |                | 0       |           |               |                  |  |
| max_consecutive_c... |                | 18      |           |               |                  |  |
| rollout/             |                |         |           |               |                  |  |
| ep_len_mean          |                | 972     |           |               |                  |  |
| ep_rew_mean          |                | 343     |           |               |                  |  |
| exploration rate     |                | 0.506   |           |               |                  |  |
| time/                |                |         |           |               |                  |  |
| episodes             |                | 22900   |           |               |                  |  |
| fps                  |                | 206     |           |               |                  |  |
| time_elapsed         |                | 7184    |           |               |                  |  |
| total timesteps      |                | 1484094 |           |               |                  |  |
| train/               |                |         |           |               |                  |  |
| learning_rate        |                | 0.0001  |           |               |                  |  |
| loss                 |                | 0.233   |           |               |                  |  |
| n_updates            |                | 358523  |           |               |                  |  |
| -----                |                |         |           |               |                  |  |
| EP=7,730             | STEP=1,486,128 | clip=24 | consec=1  | max_consec=18 | avg_clip50=34.16 |  |
| EP=7,740             | STEP=1,488,143 | clip=43 | consec=1  | max_consec=18 | avg_clip50=31.91 |  |
| EP=7,750             | STEP=1,490,860 | clip=42 | consec=3  | max_consec=18 | avg_clip50=31.48 |  |
| -----                |                |         |           |               |                  |  |
| custom/              |                |         |           |               |                  |  |
| avg_clip50           |                | 31.9    |           |               |                  |  |
| clipped_episode_r... |                | 69      |           |               |                  |  |
| consecutive_clipp... |                | 1       |           |               |                  |  |
| max_consecutive_c... |                | 18      |           |               |                  |  |
| rollout/             |                |         |           |               |                  |  |

|                  |         |
|------------------|---------|
| ep_len_mean      | 964     |
| ep_rew_mean      | 328     |
| exploration rate | 0.503   |
| time/            |         |
| episodes         | 23000   |
| fps              | 206     |
| time_elapsed     | 7225    |
| total timesteps  | 1492100 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.288   |
| n_updates        | 360524  |

EP=7,760 | STEP=1,493,489 | clip=48 | consec=7 | max\_consec=18 | avg\_clip50=32.45  
EP=7,770 | STEP=1,495,642 | clip=32 | consec=3 | max\_consec=18 | avg\_clip50=32.32  
EP=7,780 | STEP=1,497,963 | clip=20 | consec=3 | max\_consec=18 | avg\_clip50=33.39

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 36.1    |
| clipped_episode_r... | 42.5    |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 1e+03   |
| ep_rew_mean          | 341     |
| exploration rate     | 0.5     |
| time/                |         |
| episodes             | 23100   |
| fps                  | 206     |
| time_elapsed         | 7265    |
| total timesteps      | 1500022 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.179   |
| n_updates            | 362505  |

EP=7,790 | STEP=1,500,446 | clip=23 | consec=4 | max\_consec=18 | avg\_clip50=35.88  
EP=7,800 | STEP=1,502,878 | clip=54 | consec=3 | max\_consec=18 | avg\_clip50=35.51  
EP=7,810 | STEP=1,504,988 | clip=45 | consec=1 | max\_consec=18 | avg\_clip50=33.07  
EP=7,820 | STEP=1,506,847 | clip=3 | consec=0 | max\_consec=18 | avg\_clip50=30.98

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29.9    |
| clipped_episode_r... | 27.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 963     |
| ep_rew_mean          | 319     |
| exploration rate     | 0.498   |
| time/                |         |
| episodes             | 23200   |
| fps                  | 206     |
| time_elapsed         | 7304    |
| total timesteps      | 1507762 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0508  |
| n_updates            | 364440  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,830 | STEP=1,509,197 | clip=34 | consec=2 | max_consec=18 | avg_clip50=32.52 |
| EP=7,840 | STEP=1,511,623 | clip=19 | consec=0 | max_consec=18 | avg_clip50=33.11 |
| EP=7,850 | STEP=1,513,943 | clip=34 | consec=1 | max_consec=18 | avg_clip50=32.88 |

|                      |         |
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| -----                |         |
| custom/              |         |
| avg_clip50           | 34.4    |
| clipped_episode_r... | 26      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 978     |
| ep_rew_mean          | 344     |
| exploration rate     | 0.495   |
| time/                |         |
| episodes             | 23300   |
| fps                  | 206     |
| time_elapsed         | 7347    |
| total timesteps      | 1516164 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.307   |
| n_updates            | 366540  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,860 | STEP=1,516,700 | clip=29 | consec=5 | max_consec=18 | avg_clip50=35.01 |
| EP=7,870 | STEP=1,519,237 | clip=54 | consec=8 | max_consec=18 | avg_clip50=37.75 |
| EP=7,880 | STEP=1,521,275 | clip=43 | consec=3 | max_consec=18 | avg_clip50=35.52 |
| EP=7,890 | STEP=1,523,623 | clip=64 | consec=3 | max_consec=18 | avg_clip50=35.71 |

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|----------------------|---------|
| -----                |         |
| custom/              |         |
| avg_clip50           | 35.6    |
| clipped_episode_r... | 18.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 18      |
| rollout/             |         |
| ep_len_mean          | 971     |
| ep_rew_mean          | 341     |
| exploration rate     | 0.492   |
| time/                |         |
| episodes             | 23400   |
| fps                  | 206     |
| time_elapsed         | 7387    |
| total timesteps      | 1524046 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0471  |
| n_updates            | 368511  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,900 | STEP=1,525,784 | clip=48 | consec=1 | max_consec=18 | avg_clip50=34.32 |
| EP=7,910 | STEP=1,528,013 | clip=12 | consec=0 | max_consec=18 | avg_clip50=33.01 |
| EP=7,920 | STEP=1,530,409 | clip=9  | consec=0 | max_consec=18 | avg_clip50=32.42 |

|                      |      |
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| -----                |      |
| custom/              |      |
| avg_clip50           | 35   |
| clipped_episode_r... | 28   |
| consecutive_clipp... | 5    |
| max_consecutive_c... | 18   |
| rollout/             |      |
| ep_len_mean          | 999  |
| ep_rew_mean          | 353  |
| exploration rate     | 0.49 |

|                 |         |
|-----------------|---------|
| time/           |         |
| episodes        | 23500   |
| fps             | 206     |
| time_elapsed    | 7429    |
| total timesteps | 1532198 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.0875  |
| n_updates       | 370549  |

EP=7,930 | STEP=1,533,275 | clip=20 | consec=10 | max\_consec=18 | avg\_clip50=34.73

EP=7,940 | STEP=1,536,257 | clip=58 | consec=5 | max\_consec=18 | avg\_clip50=36.95

EP=7,950 | STEP=1,538,649 | clip=9 | consec=0 | max\_consec=18 | avg\_clip50=39.54

EP=7,960 | STEP=1,540,960 | clip=46 | consec=2 | max\_consec=18 | avg\_clip50=39.76

|                      |          |
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| custom/              |          |
| avg_clip50           | 39.8     |
| clipped_episode_r... | 45.5     |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 18       |
| rollout/             |          |
| ep_len_mean          | 1.01e+03 |
| ep_rew_mean          | 364      |
| exploration rate     | 0.487    |
| time/                |          |
| episodes             | 23600    |
| fps                  | 206      |
| time_elapsed         | 7475     |
| total timesteps      | 1541190  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.328    |
| n_updates            | 372797   |

EP=7,970 | STEP=1,543,140 | clip=72 | consec=2 | max\_consec=18 | avg\_clip50=38.94

EP=7,980 | STEP=1,545,735 | clip=16 | consec=0 | max\_consec=18 | avg\_clip50=39.28

EP=7,990 | STEP=1,548,211 | clip=76 | consec=3 | max\_consec=18 | avg\_clip50=36.54

|                      |          |
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| custom/              |          |
| avg_clip50           | 36       |
| clipped_episode_r... | 51       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 18       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 371      |
| exploration rate     | 0.484    |
| time/                |          |
| episodes             | 23700    |
| fps                  | 206      |
| time_elapsed         | 7519     |
| total timesteps      | 1549352  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0508   |
| n_updates            | 374837   |

EP=8,000 | STEP=1,550,508 | clip=26 | consec=1 | max\_consec=18 | avg\_clip50=35.06

EP=8,010 | STEP=1,552,987 | clip=60 | consec=2 | max\_consec=18 | avg\_clip50=34.60



EP=8,020 | STEP=1,555,392 | clip=40 | consec=1 | max\_consec=18 | avg\_clip50=35.43

|                      |          |  |
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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 32.1     |  |
| clipped_episode_r... | 51       |  |
| consecutive_clipp... | 2        |  |
| max_consecutive_c... | 18       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.02e+03 |  |
| ep_rew_mean          | 360      |  |
| exploration rate     | 0.481    |  |
| time/                |          |  |
| episodes             | 23800    |  |
| fps                  | 205      |  |
| time_elapsed         | 7560     |  |
| total timesteps      | 1557221  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0317   |  |
| n_updates            | 376805   |  |

EP=8,030 | STEP=1,557,486 | clip=15 | consec=0 | max\_consec=18 | avg\_clip50=32.01

EP=8,040 | STEP=1,559,774 | clip=44 | consec=3 | max\_consec=18 | avg\_clip50=32.44

EP=8,050 | STEP=1,562,340 | clip=39 | consec=13 | max\_consec=18 | avg\_clip50=33.92

EP=8,060 | STEP=1,564,691 | clip=28 | consec=1 | max\_consec=18 | avg\_clip50=34.12

|                      |         |  |
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| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 34.5    |  |
| clipped_episode_r... | 34.5    |  |
| consecutive_clipp... | 2       |  |
| max_consecutive_c... | 18      |  |
| rollout/             |         |  |
| ep_len_mean          | 995     |  |
| ep_rew_mean          | 344     |  |
| exploration rate     | 0.479   |  |
| time/                |         |  |
| episodes             | 23900   |  |
| fps                  | 205     |  |
| time_elapsed         | 7603    |  |
| total timesteps      | 1565322 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.0425  |  |
| n_updates            | 378830  |  |

EP=8,070 | STEP=1,567,577 | clip=78 | consec=7 | max\_consec=18 | avg\_clip50=36.58

EP=8,080 | STEP=1,569,679 | clip=4 | consec=0 | max\_consec=18 | avg\_clip50=37.24

EP=8,090 | STEP=1,572,307 | clip=16 | consec=0 | max\_consec=18 | avg\_clip50=37.95

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| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 35.8  |  |
| clipped_episode_r... | 7     |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 18    |  |
| rollout/             |       |  |
| ep_len_mean          | 985   |  |
| ep_rew_mean          | 339   |  |
| exploration rate     | 0.476 |  |
| time/                |       |  |

|  |                 |  |         |  |
|--|-----------------|--|---------|--|
|  | episodes        |  | 24000   |  |
|  | fps             |  | 205     |  |
|  | time_elapsed    |  | 7646    |  |
|  | total timesteps |  | 1573396 |  |
|  | train/          |  |         |  |
|  | learning_rate   |  | 0.0001  |  |
|  | loss            |  | 0.115   |  |
|  | n_updates       |  | 380848  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=8,100 |  | STEP=1,574,473 |  | clip=65 |  | consec=4 |  | max_consec=18 |  | avg_clip50=35.42 |
| EP=8,110 |  | STEP=1,576,773 |  | clip=17 |  | consec=0 |  | max_consec=18 |  | avg_clip50=35.07 |
| EP=8,120 |  | STEP=1,579,632 |  | clip=76 |  | consec=2 |  | max_consec=18 |  | avg_clip50=34.76 |
| EP=8,130 |  | STEP=1,581,807 |  | clip=6  |  | consec=0 |  | max_consec=18 |  | avg_clip50=34.77 |

|  |                      |  |          |  |
|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 34.8     |  |
|  | clipped_episode_r... |  | 46       |  |
|  | consecutive_clipp... |  | 1        |  |
|  | max_consecutive_c... |  | 18       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.02e+03 |  |
|  | ep_rew_mean          |  | 360      |  |
|  | exploration rate     |  | 0.473    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 24100    |  |
|  | fps                  |  | 205      |  |
|  | time_elapsed         |  | 7694     |  |
|  | total timesteps      |  | 1582167  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.106    |  |
|  | n_updates            |  | 383041   |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=8,140 |  | STEP=1,584,384 |  | clip=7  |  | consec=0 |  | max_consec=18 |  | avg_clip50=34.25 |
| EP=8,150 |  | STEP=1,586,825 |  | clip=28 |  | consec=1 |  | max_consec=18 |  | avg_clip50=36.92 |
| EP=8,160 |  | STEP=1,589,152 |  | clip=28 |  | consec=3 |  | max_consec=18 |  | avg_clip50=36.57 |

|  |                      |  |          |  |
|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 36.4     |  |
|  | clipped_episode_r... |  | 38.5     |  |
|  | consecutive_clipp... |  | 4        |  |
|  | max_consecutive_c... |  | 18       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.02e+03 |  |
|  | ep_rew_mean          |  | 359      |  |
|  | exploration rate     |  | 0.47     |  |
|  | time/                |  |          |  |
|  | episodes             |  | 24200    |  |
|  | fps                  |  | 205      |  |
|  | time_elapsed         |  | 7740     |  |
|  | total timesteps      |  | 1590526  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.0388   |  |
|  | n_updates            |  | 385131   |  |

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|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=8,170 |  | STEP=1,591,658 |  | clip=24 |  | consec=4  |  | max_consec=18 |  | avg_clip50=35.38 |
| EP=8,180 |  | STEP=1,594,128 |  | clip=38 |  | consec=2  |  | max_consec=18 |  | avg_clip50=37.69 |
| EP=8,190 |  | STEP=1,596,775 |  | clip=60 |  | consec=12 |  | max_consec=18 |  | avg_clip50=38.7  |

EP=8,200 | STEP=1,599,243 | clip=44 | consec=22 | max\_consec=22 | avg\_clip50=37.6  
0

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 37.6     |  |
| clipped_episode_r... | 44.5     |  |
| consecutive_clipp... | 22       |  |
| max_consecutive_c... | 22       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.04e+03 |  |
| ep_rew_mean          | 373      |  |
| exploration rate     | 0.467    |  |
| time/                |          |  |
| episodes             | 24300    |  |
| fps                  | 205      |  |
| time_elapsed         | 7785     |  |
| total timesteps      | 1599243  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.329    |  |
| n_updates            | 387310   |  |

EP=8,210 | STEP=1,601,618 | clip=72 | consec=2 | max\_consec=23 | avg\_clip50=39.74  
EP=8,220 | STEP=1,604,153 | clip=65 | consec=6 | max\_consec=23 | avg\_clip50=39.99  
EP=8,230 | STEP=1,606,260 | clip=16 | consec=0 | max\_consec=23 | avg\_clip50=38.58

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 37.5     |  |
| clipped_episode_r... | 42.5     |  |
| consecutive_clipp... | 4        |  |
| max_consecutive_c... | 23       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.01e+03 |  |
| ep_rew_mean          | 378      |  |
| exploration rate     | 0.465    |  |
| time/                |          |  |
| episodes             | 24400    |  |
| fps                  | 205      |  |
| time_elapsed         | 7826     |  |
| total timesteps      | 1607175  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0332   |  |
| n_updates            | 389293   |  |

EP=8,240 | STEP=1,608,775 | clip=26 | consec=10 | max\_consec=23 | avg\_clip50=37.6  
5  
EP=8,250 | STEP=1,611,052 | clip=52 | consec=3 | max\_consec=23 | avg\_clip50=37.05  
EP=8,260 | STEP=1,612,849 | clip=14 | consec=0 | max\_consec=23 | avg\_clip50=33.02

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 33.4  |  |
| clipped_episode_r... | 34.5  |  |
| consecutive_clipp... | 3     |  |
| max_consecutive_c... | 23    |  |
| rollout/             |       |  |
| ep_len_mean          | 982   |  |
| ep_rew_mean          | 363   |  |
| exploration rate     | 0.462 |  |
| time/                |       |  |

|                      |                |         |          |               |                  |
|----------------------|----------------|---------|----------|---------------|------------------|
| episodes             | 24500          |         |          |               |                  |
| fps                  | 205            |         |          |               |                  |
| time_elapsed         | 7864           |         |          |               |                  |
| total timesteps      | 1614935        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0647         |         |          |               |                  |
| n_updates            | 391233         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=8,270             | STEP=1,615,156 | clip=36 | consec=4 | max_consec=23 | avg_clip50=32.85 |
| EP=8,280             | STEP=1,617,304 | clip=14 | consec=0 | max_consec=23 | avg_clip50=32.80 |
| EP=8,290             | STEP=1,620,425 | clip=69 | consec=7 | max_consec=23 | avg_clip50=35.53 |
| EP=8,300             | STEP=1,623,061 | clip=49 | consec=4 | max_consec=23 | avg_clip50=36.75 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 38.7           |         |          |               |                  |
| clipped_episode_r... | 49.5           |         |          |               |                  |
| consecutive_clipp... | 2              |         |          |               |                  |
| max_consecutive_c... | 23             |         |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          | 991            |         |          |               |                  |
| ep_rew_mean          | 367            |         |          |               |                  |
| exploration rate     | 0.459          |         |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             | 24600          |         |          |               |                  |
| fps                  | 205            |         |          |               |                  |
| time_elapsed         | 7914           |         |          |               |                  |
| total timesteps      | 1623946        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0685         |         |          |               |                  |
| n_updates            | 393486         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=8,310             | STEP=1,625,275 | clip=50 | consec=4 | max_consec=23 | avg_clip50=40.11 |
| EP=8,320             | STEP=1,627,294 | clip=34 | consec=3 | max_consec=23 | avg_clip50=37.89 |
| EP=8,330             | STEP=1,629,236 | clip=86 | consec=1 | max_consec=23 | avg_clip50=36.99 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           | 37.1           |         |          |               |                  |
| clipped_episode_r... | 65.5           |         |          |               |                  |
| consecutive_clipp... | 9              |         |          |               |                  |
| max_consecutive_c... | 23             |         |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          | 983            |         |          |               |                  |
| ep_rew_mean          | 357            |         |          |               |                  |
| exploration rate     | 0.457          |         |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             | 24700          |         |          |               |                  |
| fps                  | 205            |         |          |               |                  |
| time_elapsed         | 7955           |         |          |               |                  |
| total timesteps      | 1631746        |         |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        | 0.0001         |         |          |               |                  |
| loss                 | 0.0365         |         |          |               |                  |
| n_updates            | 395436         |         |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=8,340             | STEP=1,632,058 | clip=16 | consec=0 | max_consec=23 | avg_clip50=35.15 |
| EP=8,350             | STEP=1,634,320 | clip=12 | consec=0 | max_consec=23 | avg_clip50=33.58 |
| EP=8,360             | STEP=1,636,887 | clip=19 | consec=0 | max_consec=23 | avg_clip50=32.36 |
| EP=8,370             | STEP=1,638,903 | clip=38 | consec=2 | max_consec=23 | avg_clip50=31.88 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 32.5    |
| clipped_episode_r... | 28.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 23      |
| rollout/             |         |
| ep_len_mean          | 992     |
| ep_rew_mean          | 344     |
| exploration rate     | 0.454   |
| time/                |         |
| episodes             | 24800   |
| fps                  | 204     |
| time_elapsed         | 7996    |
| total timesteps      | 1639285 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0453  |
| n_updates            | 397321  |

EP=8,380 | STEP=1,641,130 | clip=26 | consec=3 | max\_consec=23 | avg\_clip50=31.91  
EP=8,390 | STEP=1,643,476 | clip=28 | consec=4 | max\_consec=23 | avg\_clip50=30.46  
EP=8,400 | STEP=1,645,756 | clip=32 | consec=1 | max\_consec=23 | avg\_clip50=30.36

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.7    |
| clipped_episode_r... | 38.5    |
| consecutive_clipp... | 7       |
| max_consecutive_c... | 23      |
| rollout/             |         |
| ep_len_mean          | 978     |
| ep_rew_mean          | 327     |
| exploration rate     | 0.451   |
| time/                |         |
| episodes             | 24900   |
| fps                  | 204     |
| time_elapsed         | 8042    |
| total timesteps      | 1647694 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0613  |
| n_updates            | 399423  |

EP=8,410 | STEP=1,648,381 | clip=28 | consec=11 | max\_consec=23 | avg\_clip50=31.80  
EP=8,420 | STEP=1,650,688 | clip=13 | consec=0 | max\_consec=23 | avg\_clip50=33.22  
EP=8,430 | STEP=1,653,408 | clip=12 | consec=0 | max\_consec=23 | avg\_clip50=35.24  
EP=8,440 | STEP=1,656,119 | clip=74 | consec=5 | max\_consec=23 | avg\_clip50=38.32

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.8     |
| clipped_episode_r... | 41       |
| consecutive_clipp... | 6        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.01e+03 |
| ep_rew_mean          | 342      |
| exploration rate     | 0.448    |
| time/                |          |
| episodes             | 25000    |

|                      |                |          |           |               |                  |
|----------------------|----------------|----------|-----------|---------------|------------------|
| fps                  |                | 204      |           |               |                  |
| time_elapsed         |                | 8088     |           |               |                  |
| total timesteps      |                | 1656430  |           |               |                  |
| train/               |                |          |           |               |                  |
| learning_rate        |                | 0.0001   |           |               |                  |
| loss                 |                | 0.332    |           |               |                  |
| n_updates            |                | 401607   |           |               |                  |
| -----                |                |          |           |               |                  |
| EP=8,450             | STEP=1,658,701 | clip=40  | consec=2  | max_consec=23 | avg_clip50=39.62 |
| EP=8,460             | STEP=1,661,381 | clip=18  | consec=0  | max_consec=23 | avg_clip50=39.94 |
| EP=8,470             | STEP=1,664,074 | clip=62  | consec=4  | max_consec=23 | avg_clip50=41.88 |
| -----                |                |          |           |               |                  |
| custom/              |                |          |           |               |                  |
| avg_clip50           |                | 42.9     |           |               |                  |
| clipped_episode_r... |                | 26.5     |           |               |                  |
| consecutive_clipp... |                | 10       |           |               |                  |
| max_consecutive_c... |                | 23       |           |               |                  |
| rollout/             |                |          |           |               |                  |
| ep_len_mean          |                | 1.06e+03 |           |               |                  |
| ep_rew_mean          |                | 386      |           |               |                  |
| exploration rate     |                | 0.445    |           |               |                  |
| time/                |                |          |           |               |                  |
| episodes             |                | 25100    |           |               |                  |
| fps                  |                | 204      |           |               |                  |
| time_elapsed         |                | 8138     |           |               |                  |
| total timesteps      |                | 1665904  |           |               |                  |
| train/               |                |          |           |               |                  |
| learning_rate        |                | 0.0001   |           |               |                  |
| loss                 |                | 0.411    |           |               |                  |
| n_updates            |                | 403975   |           |               |                  |
| -----                |                |          |           |               |                  |
| EP=8,480             | STEP=1,666,619 | clip=72  | consec=2  | max_consec=23 | avg_clip50=43.33 |
| EP=8,490             | STEP=1,668,950 | clip=34  | consec=1  | max_consec=23 | avg_clip50=40.46 |
| EP=8,500             | STEP=1,671,686 | clip=54  | consec=4  | max_consec=23 | avg_clip50=43.97 |
| EP=8,510             | STEP=1,674,550 | clip=31  | consec=4  | max_consec=23 | avg_clip50=45.10 |
| -----                |                |          |           |               |                  |
| custom/              |                |          |           |               |                  |
| avg_clip50           |                | 45.1     |           |               |                  |
| clipped_episode_r... |                | 54.5     |           |               |                  |
| consecutive_clipp... |                | 5        |           |               |                  |
| max_consecutive_c... |                | 23       |           |               |                  |
| rollout/             |                |          |           |               |                  |
| ep_len_mean          |                | 1.09e+03 |           |               |                  |
| ep_rew_mean          |                | 425      |           |               |                  |
| exploration rate     |                | 0.442    |           |               |                  |
| time/                |                |          |           |               |                  |
| episodes             |                | 25200    |           |               |                  |
| fps                  |                | 204      |           |               |                  |
| time_elapsed         |                | 8186     |           |               |                  |
| total timesteps      |                | 1675037  |           |               |                  |
| train/               |                |          |           |               |                  |
| learning_rate        |                | 0.0001   |           |               |                  |
| loss                 |                | 0.0442   |           |               |                  |
| n_updates            |                | 406259   |           |               |                  |
| -----                |                |          |           |               |                  |
| EP=8,520             | STEP=1,676,905 | clip=39  | consec=14 | max_consec=23 | avg_clip50=44.18 |
| EP=8,530             | STEP=1,679,324 | clip=51  | consec=6  | max_consec=23 | avg_clip50=43.37 |
| EP=8,540             | STEP=1,682,157 | clip=24  | consec=3  | max_consec=23 | avg_clip50=44.81 |

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|----------------------|----------|
| custom/              |          |
| avg_clip50           | 42.7     |
| clipped_episode_r... | 41       |
| consecutive_clipp... | 8        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 428      |
| exploration rate     | 0.439    |
| time/                |          |
| episodes             | 25300    |
| fps                  | 204      |
| time_elapsed         | 8232     |
| total timesteps      | 1683705  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.172    |
| n_updates            | 408426   |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=8,550 | STEP=1,684,834 | clip=36 | consec=1 | max_consec=23 | avg_clip50=41.74 |
| EP=8,560 | STEP=1,687,688 | clip=51 | consec=2 | max_consec=23 | avg_clip50=41.68 |
| EP=8,570 | STEP=1,689,904 | clip=32 | consec=1 | max_consec=23 | avg_clip50=40.91 |
| EP=8,580 | STEP=1,692,491 | clip=50 | consec=3 | max_consec=23 | avg_clip50=40.07 |

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| custom/              |          |
| avg_clip50           | 40.1     |
| clipped_episode_r... | 49.5     |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.08e+03 |
| ep_rew_mean          | 417      |
| exploration rate     | 0.436    |
| time/                |          |
| episodes             | 25400    |
| fps                  | 204      |
| time_elapsed         | 8278     |
| total timesteps      | 1692491  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0329   |
| n_updates            | 410622   |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=8,590 | STEP=1,695,116 | clip=45 | consec=6  | max_consec=23 | avg_clip50=39.73 |
| EP=8,600 | STEP=1,697,594 | clip=79 | consec=1  | max_consec=23 | avg_clip50=38.91 |
| EP=8,610 | STEP=1,700,228 | clip=24 | consec=11 | max_consec=23 | avg_clip50=37.8  |

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|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37.2     |
| clipped_episode_r... | 17       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 392      |
| exploration rate     | 0.434    |
| time/                |          |
| episodes             | 25500    |
| fps                  | 204      |

|  |                 |         |  |
|--|-----------------|---------|--|
|  | time_elapsed    | 8324    |  |
|  | total timesteps | 1701085 |  |
|  | train/          |         |  |
|  | learning_rate   | 0.0001  |  |
|  | loss            | 0.122   |  |
|  | n_updates       | 412771  |  |

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|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=8,620 |  | STEP=1,702,496 |  | clip=35 |  | consec=4  |  | max_consec=23 |  | avg_clip50=37.80 |
| EP=8,630 |  | STEP=1,704,657 |  | clip=14 |  | consec=0  |  | max_consec=23 |  | avg_clip50=36.94 |
| EP=8,640 |  | STEP=1,707,852 |  | clip=54 |  | consec=10 |  | max_consec=23 |  | avg_clip50=39.11 |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 38.5     |  |
|  | clipped_episode_r... | 27       |  |
|  | consecutive_clipp... | 1        |  |
|  | max_consecutive_c... | 23       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.05e+03 |  |
|  | ep_rew_mean          | 382      |  |
|  | exploration rate     | 0.431    |  |
|  | time/                |          |  |
|  | episodes             | 25600    |  |
|  | fps                  | 204      |  |
|  | time_elapsed         | 8371     |  |
|  | total timesteps      | 1709668  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0606   |  |
|  | n_updates            | 414916   |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=8,650 |  | STEP=1,709,932 |  | clip=41 |  | consec=3 |  | max_consec=23 |  | avg_clip50=38.18 |
| EP=8,660 |  | STEP=1,712,629 |  | clip=36 |  | consec=5 |  | max_consec=23 |  | avg_clip50=38.34 |
| EP=8,670 |  | STEP=1,715,126 |  | clip=72 |  | consec=2 |  | max_consec=23 |  | avg_clip50=39.67 |
| EP=8,680 |  | STEP=1,717,863 |  | clip=50 |  | consec=3 |  | max_consec=23 |  | avg_clip50=41.56 |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 41.1     |  |
|  | clipped_episode_r... | 44       |  |
|  | consecutive_clipp... | 6        |  |
|  | max_consecutive_c... | 23       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.06e+03 |  |
|  | ep_rew_mean          | 394      |  |
|  | exploration rate     | 0.428    |  |
|  | time/                |          |  |
|  | episodes             | 25700    |  |
|  | fps                  | 204      |  |
|  | time_elapsed         | 8418     |  |
|  | total timesteps      | 1718679  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0766   |  |
|  | n_updates            | 417169   |  |

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|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=8,690 |  | STEP=1,720,598 |  | clip=60 |  | consec=4  |  | max_consec=23 |  | avg_clip50=40.03 |
| EP=8,700 |  | STEP=1,723,245 |  | clip=34 |  | consec=14 |  | max_consec=23 |  | avg_clip50=41.51 |
| EP=8,710 |  | STEP=1,725,441 |  | clip=13 |  | consec=0  |  | max_consec=23 |  | avg_clip50=39.78 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39.7     |
| clipped_episode_r... | 40       |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 398      |
| exploration rate     | 0.425    |
| time/                |          |
| episodes             | 25800    |
| fps                  | 204      |
| time_elapsed         | 8463     |
| total timesteps      | 1727304  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0582   |
| n_updates            | 419325   |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=8,720 | STEP=1,728,118 | clip=21 | consec=5 | max_consec=23 | avg_clip50=39.82 |
| EP=8,730 | STEP=1,730,625 | clip=45 | consec=1 | max_consec=23 | avg_clip50=39.36 |
| EP=8,740 | STEP=1,733,102 | clip=28 | consec=8 | max_consec=23 | avg_clip50=38.41 |
| EP=8,750 | STEP=1,735,061 | clip=32 | consec=2 | max_consec=23 | avg_clip50=36.42 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 36.9     |
| clipped_episode_r... | 38.5     |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 387      |
| exploration rate     | 0.422    |
| time/                |          |
| episodes             | 25900    |
| fps                  | 204      |
| time_elapsed         | 8506     |
| total timesteps      | 1735355  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.483    |
| n_updates            | 421338   |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=8,760 | STEP=1,737,569 | clip=50 | consec=1 | max_consec=23 | avg_clip50=36.64 |
| EP=8,770 | STEP=1,740,239 | clip=56 | consec=1 | max_consec=23 | avg_clip50=36.22 |
| EP=8,780 | STEP=1,742,491 | clip=12 | consec=0 | max_consec=23 | avg_clip50=34.58 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 34.4     |
| clipped_episode_r... | 59       |
| consecutive_clipp... | 5        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.03e+03 |
| ep_rew_mean          | 370      |
| exploration rate     | 0.419    |
| time/                |          |
| episodes             | 26000    |
| fps                  | 203      |
| time_elapsed         | 8551     |

|                 |         |
|-----------------|---------|
| total timesteps | 1743928 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.396   |
| n_updates       | 423481  |

EP=8,790 | STEP=1,744,973 | clip=28 | consec=1 | max\_consec=23 | avg\_clip50=34.37  
EP=8,800 | STEP=1,747,134 | clip=30 | consec=1 | max\_consec=23 | avg\_clip50=35.54  
EP=8,810 | STEP=1,749,903 | clip=50 | consec=4 | max\_consec=23 | avg\_clip50=36.81

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37.7     |
| clipped_episode_r... | 46       |
| consecutive_clipp... | 6        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.01e+03 |
| ep_rew_mean          | 366      |
| exploration rate     | 0.416    |
| time/                |          |
| episodes             | 26100    |
| fps                  | 203      |
| time_elapsed         | 8595     |
| total timesteps      | 1752268  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0924   |
| n_updates            | 425566   |

EP=8,820 | STEP=1,752,322 | clip=67 | consec=7 | max\_consec=23 | avg\_clip50=37.90  
EP=8,830 | STEP=1,754,515 | clip=27 | consec=7 | max\_consec=23 | avg\_clip50=38.17  
EP=8,840 | STEP=1,757,306 | clip=68 | consec=17 | max\_consec=23 | avg\_clip50=41.28  
EP=8,850 | STEP=1,759,822 | clip=20 | consec=4 | max\_consec=23 | avg\_clip50=42.00

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 41.7     |
| clipped_episode_r... | 16       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 390      |
| exploration rate     | 0.414    |
| time/                |          |
| episodes             | 26200    |
| fps                  | 203      |
| time_elapsed         | 8640     |
| total timesteps      | 1760757  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0387   |
| n_updates            | 427689   |

EP=8,860 | STEP=1,762,277 | clip=34 | consec=1 | max\_consec=23 | avg\_clip50=41.72  
EP=8,870 | STEP=1,764,886 | clip=74 | consec=6 | max\_consec=23 | avg\_clip50=40.78  
EP=8,880 | STEP=1,767,270 | clip=31 | consec=2 | max\_consec=23 | avg\_clip50=42.70

|            |      |
|------------|------|
| custom/    |      |
| avg_clip50 | 41.8 |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | clipped_episode_r... | 72.5     |  |
|  | consecutive_clipp... | 2        |  |
|  | max_consecutive_c... | 23       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.04e+03 |  |
|  | ep_rew_mean          | 408      |  |
|  | exploration rate     | 0.411    |  |
|  | time/                |          |  |
|  | episodes             | 26300    |  |
|  | fps                  | 203      |  |
|  | time_elapsed         | 8686     |  |
|  | total timesteps      | 1769578  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.07     |  |
|  | n_updates            | 429894   |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=8,890 |  | STEP=1,769,956 |  | clip=27 |  | consec=4 |  | max_consec=23 |  | avg_clip50=40.70 |
| EP=8,900 |  | STEP=1,772,233 |  | clip=36 |  | consec=4 |  | max_consec=23 |  | avg_clip50=40.67 |
| EP=8,910 |  | STEP=1,774,927 |  | clip=50 |  | consec=7 |  | max_consec=23 |  | avg_clip50=42.32 |
| EP=8,920 |  | STEP=1,777,423 |  | clip=28 |  | consec=2 |  | max_consec=23 |  | avg_clip50=43.40 |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 42.8     |  |
|  | clipped_episode_r... | 48       |  |
|  | consecutive_clipp... | 4        |  |
|  | max_consecutive_c... | 23       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.05e+03 |  |
|  | ep_rew_mean          | 422      |  |
|  | exploration rate     | 0.408    |  |
|  | time/                |          |  |
|  | episodes             | 26400    |  |
|  | fps                  | 203      |  |
|  | time_elapsed         | 8730     |  |
|  | total timesteps      | 1777964  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.052    |  |
|  | n_updates            | 431990   |  |

-----

|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=8,930 |  | STEP=1,779,885 |  | clip=50 |  | consec=7  |  | max_consec=23 |  | avg_clip50=43.03 |
| EP=8,940 |  | STEP=1,782,350 |  | clip=63 |  | consec=1  |  | max_consec=23 |  | avg_clip50=43.14 |
| EP=8,950 |  | STEP=1,785,048 |  | clip=43 |  | consec=11 |  | max_consec=23 |  | avg_clip50=44.5  |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 45.4     |  |
|  | clipped_episode_r... | 69       |  |
|  | consecutive_clipp... | 17       |  |
|  | max_consecutive_c... | 23       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.06e+03 |  |
|  | ep_rew_mean          | 429      |  |
|  | exploration rate     | 0.405    |  |
|  | time/                |          |  |
|  | episodes             | 26500    |  |
|  | fps                  | 203      |  |
|  | time_elapsed         | 8777     |  |
|  | total timesteps      | 1786973  |  |

|                      |                |          |           |               |                  |
|----------------------|----------------|----------|-----------|---------------|------------------|
| -----                |                |          |           |               |                  |
| train/               |                |          |           |               |                  |
| learning_rate        |                | 0.0001   |           |               |                  |
| loss                 |                | 0.297    |           |               |                  |
| n_updates            |                | 434243   |           |               |                  |
| -----                |                |          |           |               |                  |
| EP=8,960             | STEP=1,787,513 | clip=14  | consec=0  | max_consec=23 | avg_clip50=42.47 |
| EP=8,970             | STEP=1,789,786 | clip=16  | consec=0  | max_consec=23 | avg_clip50=39.73 |
| EP=8,980             | STEP=1,792,929 | clip=35  | consec=1  | max_consec=23 | avg_clip50=41.49 |
| -----                |                |          |           |               |                  |
| custom/              |                |          |           |               |                  |
| avg_clip50           |                | 41.3     |           |               |                  |
| clipped_episode_r... |                | 23.5     |           |               |                  |
| consecutive_clipp... |                | 3        |           |               |                  |
| max_consecutive_c... |                | 23       |           |               |                  |
| rollout/             |                |          |           |               |                  |
| ep_len_mean          |                | 1.07e+03 |           |               |                  |
| ep_rew_mean          |                | 418      |           |               |                  |
| exploration rate     |                | 0.402    |           |               |                  |
| time/                |                |          |           |               |                  |
| episodes             |                | 26600    |           |               |                  |
| fps                  |                | 203      |           |               |                  |
| time_elapsed         |                | 8822     |           |               |                  |
| total timesteps      |                | 1795429  |           |               |                  |
| train/               |                |          |           |               |                  |
| learning_rate        |                | 0.0001   |           |               |                  |
| loss                 |                | 0.0566   |           |               |                  |
| n_updates            |                | 436357   |           |               |                  |
| -----                |                |          |           |               |                  |
| EP=8,990             | STEP=1,795,509 | clip=25  | consec=4  | max_consec=23 | avg_clip50=40.51 |
| EP=9,000             | STEP=1,797,972 | clip=23  | consec=3  | max_consec=23 | avg_clip50=39.72 |
| EP=9,010             | STEP=1,800,834 | clip=10  | consec=0  | max_consec=23 | avg_clip50=41.57 |
| EP=9,020             | STEP=1,803,856 | clip=26  | consec=10 | max_consec=23 | avg_clip50=45.7  |
| 6                    |                |          |           |               |                  |
| -----                |                |          |           |               |                  |
| custom/              |                |          |           |               |                  |
| avg_clip50           |                | 44.1     |           |               |                  |
| clipped_episode_r... |                | 54       |           |               |                  |
| consecutive_clipp... |                | 3        |           |               |                  |
| max_consecutive_c... |                | 23       |           |               |                  |
| rollout/             |                |          |           |               |                  |
| ep_len_mean          |                | 1.11e+03 |           |               |                  |
| ep_rew_mean          |                | 432      |           |               |                  |
| exploration rate     |                | 0.399    |           |               |                  |
| time/                |                |          |           |               |                  |
| episodes             |                | 26700    |           |               |                  |
| fps                  |                | 203      |           |               |                  |
| time_elapsed         |                | 8872     |           |               |                  |
| total timesteps      |                | 1805044  |           |               |                  |
| train/               |                |          |           |               |                  |
| learning_rate        |                | 0.0001   |           |               |                  |
| loss                 |                | 0.0839   |           |               |                  |
| n_updates            |                | 438760   |           |               |                  |
| -----                |                |          |           |               |                  |
| EP=9,030             | STEP=1,806,802 | clip=54  | consec=9  | max_consec=23 | avg_clip50=44.76 |
| EP=9,040             | STEP=1,809,249 | clip=49  | consec=3  | max_consec=23 | avg_clip50=44.88 |
| EP=9,050             | STEP=1,811,909 | clip=39  | consec=1  | max_consec=23 | avg_clip50=44.78 |
| -----                |                |          |           |               |                  |
| custom/              |                |          |           |               |                  |
| avg_clip50           |                | 41.4     |           |               |                  |
| clipped_episode_r... |                | 22       |           |               |                  |

|                      |          |
|----------------------|----------|
| consecutive_clipp... | 3        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 415      |
| exploration rate     | 0.396    |
| time/                |          |
| episodes             | 26800    |
| fps                  | 203      |
| time_elapsed         | 8919     |
| total timesteps      | 1813892  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0542   |
| n_updates            | 440972   |

```

EP=9,060 | STEP=1,814,081 | clip=40 | consec=4 | max_consec=23 | avg_clip50=42.00
EP=9,070 | STEP=1,816,392 | clip=58 | consec=2 | max_consec=23 | avg_clip50=38.67
EP=9,080 | STEP=1,818,839 | clip=54 | consec=5 | max_consec=23 | avg_clip50=38.08
EP=9,090 | STEP=1,821,049 | clip=38 | consec=6 | max_consec=23 | avg_clip50=37.64

```

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 36.6     |
| clipped_episode_r... | 38.5     |
| consecutive_clipp... | 10       |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 411      |
| exploration rate     | 0.393    |
| time/                |          |
| episodes             | 26900    |
| fps                  | 203      |
| time_elapsed         | 8963     |
| total timesteps      | 1822009  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0652   |
| n_updates            | 443002   |

```

EP=9,100 | STEP=1,823,561 | clip=26 | consec=16 | max_consec=23 | avg_clip50=36.4
3
EP=9,110 | STEP=1,826,203 | clip=44 | consec=5 | max_consec=23 | avg_clip50=38.58
EP=9,120 | STEP=1,828,720 | clip=38 | consec=1 | max_consec=23 | avg_clip50=39.62

```

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39.3     |
| clipped_episode_r... | 60.5     |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 23       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 387      |
| exploration rate     | 0.39     |
| time/                |          |
| episodes             | 27000    |
| fps                  | 203      |
| time_elapsed         | 9010     |
| total timesteps      | 1830850  |
| train/               |          |

|          |                      |          |           |               |                  |  |
|----------|----------------------|----------|-----------|---------------|------------------|--|
|          | learning_rate        | 0.0001   |           |               |                  |  |
|          | loss                 | 0.0618   |           |               |                  |  |
|          | n_updates            | 445212   |           |               |                  |  |
| EP=9,130 | STEP=1,831,110       | clip=52  | consec=2  | max_consec=23 | avg_clip50=39.26 |  |
| EP=9,140 | STEP=1,833,661       | clip=60  | consec=7  | max_consec=23 | avg_clip50=39.50 |  |
| EP=9,150 | STEP=1,836,207       | clip=15  | consec=0  | max_consec=23 | avg_clip50=39.84 |  |
| EP=9,160 | STEP=1,839,064       | clip=64  | consec=1  | max_consec=23 | avg_clip50=41.01 |  |
| custom/  |                      |          |           |               |                  |  |
|          | avg_clip50           | 40.9     |           |               |                  |  |
|          | clipped_episode_r... | 45       |           |               |                  |  |
|          | consecutive_clipp... | 4        |           |               |                  |  |
|          | max_consecutive_c... | 23       |           |               |                  |  |
| rollout/ |                      |          |           |               |                  |  |
|          | ep_len_mean          | 1.04e+03 |           |               |                  |  |
|          | ep_rew_mean          | 396      |           |               |                  |  |
|          | exploration rate     | 0.387    |           |               |                  |  |
| time/    |                      |          |           |               |                  |  |
|          | episodes             | 27100    |           |               |                  |  |
|          | fps                  | 203      |           |               |                  |  |
|          | time_elapsed         | 9060     |           |               |                  |  |
|          | total timesteps      | 1840189  |           |               |                  |  |
| train/   |                      |          |           |               |                  |  |
|          | learning_rate        | 0.0001   |           |               |                  |  |
|          | loss                 | 0.0296   |           |               |                  |  |
|          | n_updates            | 447547   |           |               |                  |  |
| EP=9,170 | STEP=1,841,793       | clip=28  | consec=11 | max_consec=23 | avg_clip50=40.4  |  |
| 7        |                      |          |           |               |                  |  |
| EP=9,180 | STEP=1,844,578       | clip=72  | consec=21 | max_consec=23 | avg_clip50=41.6  |  |
| 9        |                      |          |           |               |                  |  |
| EP=9,190 | STEP=1,846,912       | clip=34  | consec=6  | max_consec=23 | avg_clip50=41.81 |  |
| custom/  |                      |          |           |               |                  |  |
|          | avg_clip50           | 42.4     |           |               |                  |  |
|          | clipped_episode_r... | 49       |           |               |                  |  |
|          | consecutive_clipp... | 13       |           |               |                  |  |
|          | max_consecutive_c... | 23       |           |               |                  |  |
| rollout/ |                      |          |           |               |                  |  |
|          | ep_len_mean          | 1.09e+03 |           |               |                  |  |
|          | ep_rew_mean          | 413      |           |               |                  |  |
|          | exploration rate     | 0.384    |           |               |                  |  |
| time/    |                      |          |           |               |                  |  |
|          | episodes             | 27200    |           |               |                  |  |
|          | fps                  | 203      |           |               |                  |  |
|          | time_elapsed         | 9104     |           |               |                  |  |
|          | total timesteps      | 1848779  |           |               |                  |  |
| train/   |                      |          |           |               |                  |  |
|          | learning_rate        | 0.0001   |           |               |                  |  |
|          | loss                 | 0.0482   |           |               |                  |  |
|          | n_updates            | 449694   |           |               |                  |  |
| EP=9,200 | STEP=1,849,660       | clip=75  | consec=16 | max_consec=23 | avg_clip50=43.6  |  |
| 1        |                      |          |           |               |                  |  |
| EP=9,210 | STEP=1,851,855       | clip=44  | consec=2  | max_consec=23 | avg_clip50=40.50 |  |
| EP=9,220 | STEP=1,854,868       | clip=48  | consec=4  | max_consec=23 | avg_clip50=42.07 |  |
| EP=9,230 | STEP=1,857,599       | clip=46  | consec=9  | max_consec=23 | avg_clip50=41.91 |  |
| custom/  |                      |          |           |               |                  |  |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | avg_clip50           | 41.9     |  |
|  | clipped_episode_r... | 46       |  |
|  | consecutive_clipp... | 9        |  |
|  | max_consecutive_c... | 23       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.11e+03 |  |
|  | ep_rew_mean          | 418      |  |
|  | exploration rate     | 0.381    |  |
|  | time/                |          |  |
|  | episodes             | 27300    |  |
|  | fps                  | 203      |  |
|  | time_elapsed         | 9151     |  |
|  | total timesteps      | 1857833  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0662   |  |
|  | n_updates            | 451958   |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=9,240 |  | STEP=1,860,440 |  | clip=72 |  | consec=3 |  | max_consec=23 |  | avg_clip50=42.41 |
| EP=9,250 |  | STEP=1,862,950 |  | clip=41 |  | consec=9 |  | max_consec=23 |  | avg_clip50=43.15 |
| EP=9,260 |  | STEP=1,865,308 |  | clip=26 |  | consec=6 |  | max_consec=23 |  | avg_clip50=44.57 |

-----

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 45.3     |  |
|  | clipped_episode_r... | 29.5     |  |
|  | consecutive_clipp... | 11       |  |
|  | max_consecutive_c... | 23       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.09e+03 |  |
|  | ep_rew_mean          | 432      |  |
|  | exploration rate     | 0.378    |  |
|  | time/                |          |  |
|  | episodes             | 27400    |  |
|  | fps                  | 202      |  |
|  | time_elapsed         | 9196     |  |
|  | total timesteps      | 1866791  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0602   |  |
|  | n_updates            | 454197   |  |

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|          |  |                |  |         |  |           |  |               |  |                  |
|----------|--|----------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=9,270 |  | STEP=1,868,011 |  | clip=46 |  | consec=16 |  | max_consec=23 |  | avg_clip50=44.08 |
| EP=9,280 |  | STEP=1,870,681 |  | clip=46 |  | consec=26 |  | max_consec=26 |  | avg_clip50=43.57 |
| EP=9,290 |  | STEP=1,873,261 |  | clip=45 |  | consec=1  |  | max_consec=26 |  | avg_clip50=43.44 |
| EP=9,300 |  | STEP=1,875,934 |  | clip=40 |  | consec=11 |  | max_consec=26 |  | avg_clip50=43.16 |

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|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 43.2    |  |
|  | clipped_episode_r... | 66      |  |
|  | consecutive_clipp... | 13      |  |
|  | max_consecutive_c... | 26      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 1.1e+03 |  |
|  | ep_rew_mean          | 437     |  |
|  | exploration rate     | 0.375   |  |
|  | time/                |         |  |
|  | episodes             | 27500   |  |

|                 |         |
|-----------------|---------|
| fps             | 202     |
| time_elapsed    | 9248    |
| total timesteps | 1876646 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.151   |
| n_updates       | 456661  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=9,310 | STEP=1,878,377 | clip=56 | consec=1 | max_consec=26 | avg_clip50=42.92 |
| EP=9,320 | STEP=1,880,977 | clip=32 | consec=2 | max_consec=26 | avg_clip50=43.83 |
| EP=9,330 | STEP=1,883,539 | clip=86 | consec=7 | max_consec=26 | avg_clip50=44.47 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 44.6     |
| clipped_episode_r... | 15.5     |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 26       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 443      |
| exploration rate     | 0.372    |
| time/                |          |
| episodes             | 27600    |
| fps                  | 202      |
| time_elapsed         | 9293     |
| total timesteps      | 1885180  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0798   |
| n_updates            | 458794   |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=9,340 | STEP=1,885,987 | clip=41 | consec=2  | max_consec=26 | avg_clip50=43.80 |
| EP=9,350 | STEP=1,888,497 | clip=7  | consec=0  | max_consec=26 | avg_clip50=41.22 |
| EP=9,360 | STEP=1,891,481 | clip=20 | consec=0  | max_consec=26 | avg_clip50=43.46 |
| EP=9,370 | STEP=1,894,543 | clip=52 | consec=10 | max_consec=26 | avg_clip50=45.07 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 44.6     |
| clipped_episode_r... | 60       |
| consecutive_clipp... | 12       |
| max_consecutive_c... | 26       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 450      |
| exploration rate     | 0.369    |
| time/                |          |
| episodes             | 27700    |
| fps                  | 202      |
| time_elapsed         | 9344     |
| total timesteps      | 1895122  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0645   |
| n_updates            | 461280   |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=9,380 | STEP=1,897,585 | clip=92 | consec=1 | max_consec=26 | avg_clip50=46.60 |
| EP=9,390 | STEP=1,900,167 | clip=46 | consec=4 | max_consec=26 | avg_clip50=47.68 |
| EP=9,400 | STEP=1,902,637 | clip=54 | consec=4 | max_consec=26 | avg_clip50=50.60 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 47.2     |
| clipped_episode_r... | 38.5     |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 26       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 455      |
| exploration rate     | 0.366    |
| time/                |          |
| episodes             | 27800    |
| fps                  | 202      |
| time_elapsed         | 9396     |
| total timesteps      | 1904866  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0706   |
| n_updates            | 463716   |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=9,410 | STEP=1,905,289 | clip=68 | consec=5  | max_consec=26 | avg_clip50=48.58 |
| EP=9,420 | STEP=1,907,930 | clip=73 | consec=2  | max_consec=26 | avg_clip50=46.14 |
| EP=9,430 | STEP=1,910,603 | clip=60 | consec=12 | max_consec=26 | avg_clip50=44.4  |
| 4        |                |         |           |               |                  |
| EP=9,440 | STEP=1,913,090 | clip=52 | consec=9  | max_consec=26 | avg_clip50=43.68 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 42.7     |
| clipped_episode_r... | 49.5     |
| consecutive_clipp... | 12       |
| max_consecutive_c... | 26       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 458      |
| exploration rate     | 0.363    |
| time/                |          |
| episodes             | 27900    |
| fps                  | 202      |
| time_elapsed         | 9444     |
| total timesteps      | 1914009  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0536   |
| n_updates            | 466002   |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=9,450 | STEP=1,915,883 | clip=60 | consec=1 | max_consec=26 | avg_clip50=42.50 |
| EP=9,460 | STEP=1,918,743 | clip=24 | consec=8 | max_consec=26 | avg_clip50=43.66 |
| EP=9,470 | STEP=1,921,430 | clip=80 | consec=5 | max_consec=26 | avg_clip50=43.35 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 41.1    |
| clipped_episode_r... | 54      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 26      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 433     |
| exploration rate     | 0.36    |
| time/                |         |
| episodes             | 28000   |
| fps                  | 202     |

|                 |         |
|-----------------|---------|
| time_elapsed    | 9492    |
| total timesteps | 1923311 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.404   |
| n_updates       | 468327  |

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EP=9,480 | STEP=1,923,917 | clip=41 | consec=5 | max\_consec=26 | avg\_clip50=41.92  
EP=9,490 | STEP=1,926,960 | clip=50 | consec=15 | max\_consec=26 | avg\_clip50=44.1  
0  
EP=9,500 | STEP=1,929,699 | clip=60 | consec=9 | max\_consec=26 | avg\_clip50=45.83  
EP=9,510 | STEP=1,932,365 | clip=72 | consec=19 | max\_consec=26 | avg\_clip50=45.7  
7

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 46.2     |
| clipped_episode_r... | 40.5     |
| consecutive_clipp... | 22       |
| max_consecutive_c... | 26       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 450      |
| exploration rate     | 0.356    |
| time/                |          |
| episodes             | 28100    |
| fps                  | 202      |
| time_elapsed         | 9547     |
| total timesteps      | 1933423  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0565   |
| n_updates            | 470855   |

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EP=9,520 | STEP=1,934,807 | clip=14 | consec=0 | max\_consec=27 | avg\_clip50=45.64  
EP=9,530 | STEP=1,937,566 | clip=58 | consec=10 | max\_consec=27 | avg\_clip50=47.2  
5  
EP=9,540 | STEP=1,940,223 | clip=50 | consec=2 | max\_consec=27 | avg\_clip50=45.44

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 46.3     |
| clipped_episode_r... | 52       |
| consecutive_clipp... | 10       |
| max_consecutive_c... | 27       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 456      |
| exploration rate     | 0.353    |
| time/                |          |
| episodes             | 28200    |
| fps                  | 202      |
| time_elapsed         | 9596     |
| total timesteps      | 1942731  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0428   |
| n_updates            | 473182   |

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EP=9,550 | STEP=1,943,285 | clip=32 | consec=12 | max\_consec=27 | avg\_clip50=45.5  
8  
EP=9,560 | STEP=1,946,148 | clip=38 | consec=9 | max\_consec=27 | avg\_clip50=46.15

EP=9,570 | STEP=1,949,013 | clip=52 | consec=19 | max\_consec=27 | avg\_clip50=47.14

EP=9,580 | STEP=1,952,179 | clip=51 | consec=29 | max\_consec=29 | avg\_clip50=48.93

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 51       |  |
| clipped_episode_r... | 92.5     |  |
| consecutive_clipp... | 32       |  |
| max_consecutive_c... | 32       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.18e+03 |  |
| ep_rew_mean          | 485      |  |
| exploration rate     | 0.35     |  |
| time/                |          |  |
| episodes             | 28300    |  |
| fps                  | 202      |  |
| time_elapsed         | 9652     |  |
| total timesteps      | 1953396  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0557   |  |
| n_updates            | 475848   |  |

EP=9,590 | STEP=1,955,041 | clip=43 | consec=1 | max\_consec=36 | avg\_clip50=50.59

EP=9,600 | STEP=1,957,826 | clip=49 | consec=11 | max\_consec=36 | avg\_clip50=49.57

EP=9,610 | STEP=1,960,548 | clip=44 | consec=8 | max\_consec=36 | avg\_clip50=48.29

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 48.9     |  |
| clipped_episode_r... | 39.5     |  |
| consecutive_clipp... | 17       |  |
| max_consecutive_c... | 36       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.18e+03 |  |
| ep_rew_mean          | 477      |  |
| exploration rate     | 0.346    |  |
| time/                |          |  |
| episodes             | 28400    |  |
| fps                  | 202      |  |
| time_elapsed         | 9705     |  |
| total timesteps      | 1963164  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.027    |  |
| n_updates            | 478290   |  |

EP=9,620 | STEP=1,963,316 | clip=28 | consec=18 | max\_consec=36 | avg\_clip50=48.45

EP=9,630 | STEP=1,965,798 | clip=60 | consec=7 | max\_consec=36 | avg\_clip50=45.67

EP=9,640 | STEP=1,968,563 | clip=28 | consec=17 | max\_consec=36 | avg\_clip50=44.37

EP=9,650 | STEP=1,971,178 | clip=34 | consec=27 | max\_consec=36 | avg\_clip50=43.33

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| -----                |      |  |
| custom/              |      |  |
| avg_clip50           | 43.9 |  |
| clipped_episode_r... | 28   |  |
| consecutive_clipp... | 30   |  |

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| max_consecutive_c... | 36       |
| rollout/             |          |
| ep_len_mean          | 1.16e+03 |
| ep_rew_mean          | 468      |
| exploration rate     | 0.343    |
| time/                |          |
| episodes             | 28500    |
| fps                  | 202      |
| time_elapsed         | 9753     |
| total timesteps      | 1972168  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.113    |
| n_updates            | 480541   |

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EP=9,660 | STEP=1,973,780 | clip=18 | consec=0 | max\_consec=36 | avg\_clip50=42.77  
EP=9,670 | STEP=1,976,858 | clip=110 | consec=10 | max\_consec=36 | avg\_clip50=44.49  
EP=9,680 | STEP=1,979,348 | clip=60 | consec=20 | max\_consec=36 | avg\_clip50=44.45

|                      |         |
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| custom/              |         |
| avg_clip50           | 44      |
| clipped_episode_r... | 54      |
| consecutive_clipp... | 6       |
| max_consecutive_c... | 36      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 440     |
| exploration rate     | 0.34    |
| time/                |         |
| episodes             | 28600   |
| fps                  | 202     |
| time_elapsed         | 9801    |
| total timesteps      | 1981282 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0855  |
| n_updates            | 482820  |

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EP=9,690 | STEP=1,981,744 | clip=22 | consec=8 | max\_consec=36 | avg\_clip50=43.46  
EP=9,700 | STEP=1,984,683 | clip=14 | consec=0 | max\_consec=36 | avg\_clip50=44.89  
EP=9,710 | STEP=1,987,389 | clip=60 | consec=3 | max\_consec=36 | avg\_clip50=46.37  
EP=9,720 | STEP=1,990,208 | clip=64 | consec=6 | max\_consec=36 | avg\_clip50=44.46

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|----------------------|----------|
| custom/              |          |
| avg_clip50           | 45.5     |
| clipped_episode_r... | 57       |
| consecutive_clipp... | 9        |
| max_consecutive_c... | 36       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 450      |
| exploration rate     | 0.337    |
| time/                |          |
| episodes             | 28700    |
| fps                  | 202      |
| time_elapsed         | 9854     |
| total timesteps      | 1991136  |
| train/               |          |

|               |        |
|---------------|--------|
| learning_rate | 0.0001 |
| loss          | 0.381  |
| n_updates     | 485283 |

EP=9,730 | STEP=1,992,999 | clip=75 | consec=5 | max\_consec=36 | avg\_clip50=45.47  
EP=9,740 | STEP=1,995,709 | clip=37 | consec=1 | max\_consec=36 | avg\_clip50=47.82  
EP=9,750 | STEP=1,998,626 | clip=80 | consec=11 | max\_consec=36 | avg\_clip50=47.01

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 47       |
| clipped_episode_r... | 54.5     |
| consecutive_clipp... | 18       |
| max_consecutive_c... | 36       |
| rollout/             |          |
| ep_len_mean          | 1.15e+03 |
| ep_rew_mean          | 464      |
| exploration rate     | 0.334    |
| time/                |          |
| episodes             | 28800    |
| fps                  | 201      |
| time_elapsed         | 9907     |
| total timesteps      | 2000863  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0382   |
| n_updates            | 487715   |

EP=9,760 | STEP=2,001,292 | clip=41 | consec=21 | max\_consec=36 | avg\_clip50=46.42  
EP=9,770 | STEP=2,003,920 | clip=50 | consec=31 | max\_consec=36 | avg\_clip50=45.85  
EP=9,780 | STEP=2,006,711 | clip=54 | consec=41 | max\_consec=41 | avg\_clip50=46.29  
EP=9,790 | STEP=2,009,435 | clip=62 | consec=51 | max\_consec=51 | avg\_clip50=45.91

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 44.7     |
| clipped_episode_r... | 36.5     |
| consecutive_clipp... | 54       |
| max_consecutive_c... | 54       |
| rollout/             |          |
| ep_len_mean          | 1.14e+03 |
| ep_rew_mean          | 461      |
| exploration rate     | 0.331    |
| time/                |          |
| episodes             | 28900    |
| fps                  | 201      |
| time_elapsed         | 9956     |
| total timesteps      | 2009951  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.316    |
| n_updates            | 489987   |

EP=9,800 | STEP=2,011,676 | clip=60 | consec=61 | max\_consec=61 | avg\_clip50=43.96  
EP=9,810 | STEP=2,014,201 | clip=34 | consec=3 | max\_consec=64 | avg\_clip50=43.84  
EP=9,820 | STEP=2,016,877 | clip=55 | consec=7 | max\_consec=64 | avg\_clip50=44.21

|                      |         |
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| custom/              |         |
| avg_clip50           | 43.6    |
| clipped_episode_r... | 22      |
| consecutive_clipp... | 16      |
| max_consecutive_c... | 64      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 452     |
| exploration rate     | 0.328   |
| time/                |         |
| episodes             | 29000   |
| fps                  | 201     |
| time_elapsed         | 10005   |
| total timesteps      | 2019132 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0549  |
| n_updates            | 492282  |

EP=9,830 | STEP=2,019,507 | clip=58 | consec=17 | max\_consec=64 | avg\_clip50=43.71  
EP=9,840 | STEP=2,022,125 | clip=63 | consec=6 | max\_consec=64 | avg\_clip50=42.57  
EP=9,850 | STEP=2,025,022 | clip=72 | consec=16 | max\_consec=64 | avg\_clip50=45.80  
EP=9,860 | STEP=2,027,390 | clip=36 | consec=9 | max\_consec=64 | avg\_clip50=44.83

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 45.5     |
| clipped_episode_r... | 13.5     |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 64       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 444      |
| exploration rate     | 0.325    |
| time/                |          |
| episodes             | 29100    |
| fps                  | 201      |
| time_elapsed         | 10054    |
| total timesteps      | 2028194  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0772   |
| n_updates            | 494548   |

EP=9,870 | STEP=2,030,255 | clip=36 | consec=8 | max\_consec=64 | avg\_clip50=45.74  
EP=9,880 | STEP=2,033,046 | clip=55 | consec=9 | max\_consec=64 | avg\_clip50=45.99  
EP=9,890 | STEP=2,035,726 | clip=48 | consec=5 | max\_consec=64 | avg\_clip50=46.50

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 46.5     |
| clipped_episode_r... | 55       |
| consecutive_clipp... | 12       |
| max_consecutive_c... | 64       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 460      |
| exploration rate     | 0.321    |
| time/                |          |

|                 |         |
|-----------------|---------|
| episodes        | 29200   |
| fps             | 201     |
| time_elapsed    | 10105   |
| total timesteps | 2037851 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.564   |
| n_updates       | 496962  |

EP=9,900 | STEP=2,038,575 | clip=40 | consec=15 | max\_consec=64 | avg\_clip50=46.01

EP=9,910 | STEP=2,041,500 | clip=32 | consec=2 | max\_consec=64 | avg\_clip50=48.19

EP=9,920 | STEP=2,044,136 | clip=63 | consec=1 | max\_consec=64 | avg\_clip50=47.04

EP=9,930 | STEP=2,047,006 | clip=60 | consec=11 | max\_consec=64 | avg\_clip50=47.85

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 47.6     |
| clipped_episode_r... | 42       |
| consecutive_clipp... | 12       |
| max_consecutive_c... | 64       |
| rollout/             |          |
| ep_len_mean          | 1.15e+03 |
| ep_rew_mean          | 471      |
| exploration rate     | 0.318    |
| time/                |          |
| episodes             | 29300    |
| fps                  | 201      |
| time_elapsed         | 10154    |
| total timesteps      | 2047222  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0634   |
| n_updates            | 499305   |

EP=9,940 | STEP=2,049,471 | clip=54 | consec=2 | max\_consec=64 | avg\_clip50=47.42

EP=9,950 | STEP=2,052,339 | clip=58 | consec=12 | max\_consec=64 | avg\_clip50=47.60

EP=9,960 | STEP=2,055,179 | clip=57 | consec=6 | max\_consec=64 | avg\_clip50=47.07

|                      |          |
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| custom/              |          |
| avg_clip50           | 47.7     |
| clipped_episode_r... | 57       |
| consecutive_clipp... | 11       |
| max_consecutive_c... | 64       |
| rollout/             |          |
| ep_len_mean          | 1.15e+03 |
| ep_rew_mean          | 478      |
| exploration rate     | 0.315    |
| time/                |          |
| episodes             | 29400    |
| fps                  | 201      |
| time_elapsed         | 10204    |
| total timesteps      | 2056449  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.103    |
| n_updates            | 501612   |

EP=9,970 | STEP=2,058,033 | clip=54 | consec=16 | max\_consec=64 | avg\_clip50=48.3

6

EP=9,980 | STEP=2,060,553 | clip=72 | consec=3 | max\_consec=64 | avg\_clip50=46.71  
EP=9,990 | STEP=2,063,109 | clip=49 | consec=5 | max\_consec=64 | avg\_clip50=46.51  
EP=10,000 | STEP=2,065,649 | clip=56 | consec=8 | max\_consec=64 | avg\_clip50=45.3

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 45.4     |  |
| clipped_episode_r... | 55.5     |  |
| consecutive_clipp... | 8        |  |
| max_consecutive_c... | 64       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.13e+03 |  |
| ep_rew_mean          | 465      |  |
| exploration rate     | 0.312    |  |
| time/                |          |  |
| episodes             | 29500    |  |
| fps                  | 201      |  |
| time_elapsed         | 10252    |  |
| total timesteps      | 2065649  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.213    |  |
| n_updates            | 503912   |  |

EP=10,010 | STEP=2,068,298 | clip=30 | consec=6 | max\_consec=64 | avg\_clip50=45.4

3

EP=10,020 | STEP=2,071,275 | clip=25 | consec=4 | max\_consec=64 | avg\_clip50=45.2

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EP=10,030 | STEP=2,073,971 | clip=36 | consec=14 | max\_consec=64 | avg\_clip50=46.

83

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 48.5     |  |
| clipped_episode_r... | 40       |  |
| consecutive_clipp... | 18       |  |
| max_consecutive_c... | 64       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.13e+03 |  |
| ep_rew_mean          | 477      |  |
| exploration rate     | 0.309    |  |
| time/                |          |  |
| episodes             | 29600    |  |
| fps                  | 201      |  |
| time_elapsed         | 10303    |  |
| total timesteps      | 2075258  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0741   |  |
| n_updates            | 506314   |  |

EP=10,040 | STEP=2,076,560 | clip=48 | consec=24 | max\_consec=64 | avg\_clip50=48.

58

EP=10,050 | STEP=2,079,089 | clip=39 | consec=7 | max\_consec=64 | avg\_clip50=49.2

7

EP=10,060 | STEP=2,082,166 | clip=46 | consec=4 | max\_consec=64 | avg\_clip50=50.0

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| -----      |      |  |
| custom/    |      |  |
| avg_clip50 | 48.8 |  |



|  |                      |          |  |
|--|----------------------|----------|--|
|  | clipped_episode_r... | 52       |  |
|  | consecutive_clipp... | 5        |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.12e+03 |  |
|  | ep_rew_mean          | 473      |  |
|  | exploration rate     | 0.306    |  |
|  | time/                |          |  |
|  | episodes             | 29700    |  |
|  | fps                  | 201      |  |
|  | time_elapsed         | 10353    |  |
|  | total timesteps      | 2084733  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.038    |  |
|  | n_updates            | 508683   |  |

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EP=10,070 | STEP=2,085,015 | clip=50 | consec=6 | max\_consec=64 | avg\_clip50=49.34

EP=10,080 | STEP=2,087,706 | clip=46 | consec=16 | max\_consec=64 | avg\_clip50=49.00

EP=10,090 | STEP=2,090,519 | clip=55 | consec=26 | max\_consec=64 | avg\_clip50=48.12

EP=10,100 | STEP=2,093,557 | clip=47 | consec=36 | max\_consec=64 | avg\_clip50=48.67

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 47.5     |  |
|  | clipped_episode_r... | 53       |  |
|  | consecutive_clipp... | 42       |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.15e+03 |  |
|  | ep_rew_mean          | 486      |  |
|  | exploration rate     | 0.302    |  |
|  | time/                |          |  |
|  | episodes             | 29800    |  |
|  | fps                  | 201      |  |
|  | time_elapsed         | 10408    |  |
|  | total timesteps      | 2094947  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0416   |  |
|  | n_updates            | 511236   |  |

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EP=10,110 | STEP=2,096,145 | clip=33 | consec=46 | max\_consec=64 | avg\_clip50=47.85

EP=10,120 | STEP=2,098,924 | clip=48 | consec=7 | max\_consec=64 | avg\_clip50=47.58

EP=10,130 | STEP=2,101,322 | clip=66 | consec=2 | max\_consec=64 | avg\_clip50=46.24

EP=10,140 | STEP=2,104,216 | clip=52 | consec=4 | max\_consec=64 | avg\_clip50=46.92

|  |                      |      |  |
|--|----------------------|------|--|
|  | custom/              |      |  |
|  | avg_clip50           | 46.9 |  |
|  | clipped_episode_r... | 52.5 |  |
|  | consecutive_clipp... | 4    |  |
|  | max_consecutive_c... | 64   |  |
|  | rollout/             |      |  |

|  |                  |          |  |
|--|------------------|----------|--|
|  | ep_len_mean      | 1.15e+03 |  |
|  | ep_rew_mean      | 475      |  |
|  | exploration rate | 0.299    |  |
|  | time/            |          |  |
|  | episodes         | 29900    |  |
|  | fps              | 201      |  |
|  | time_elapsed     | 10459    |  |
|  | total timesteps  | 2104359  |  |
|  | train/           |          |  |
|  | learning_rate    | 0.0001   |  |
|  | loss             | 0.443    |  |
|  | n_updates        | 513589   |  |

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EP=10,150 | STEP=2,107,151 | clip=74 | consec=14 | max\_consec=64 | avg\_clip50=46.94

EP=10,160 | STEP=2,110,029 | clip=58 | consec=9 | max\_consec=64 | avg\_clip50=47.23

EP=10,170 | STEP=2,112,514 | clip=34 | consec=2 | max\_consec=64 | avg\_clip50=46.06

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 47.2     |  |
|  | clipped_episode_r... | 77.5     |  |
|  | consecutive_clipp... | 7        |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.14e+03 |  |
|  | ep_rew_mean          | 468      |  |
|  | exploration rate     | 0.296    |  |
|  | time/                |          |  |
|  | episodes             | 30000    |  |
|  | fps                  | 201      |  |
|  | time_elapsed         | 10512    |  |
|  | total timesteps      | 2113892  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.112    |  |
|  | n_updates            | 515972   |  |

-----

EP=10,180 | STEP=2,115,385 | clip=32 | consec=12 | max\_consec=64 | avg\_clip50=49.88

EP=10,190 | STEP=2,118,155 | clip=31 | consec=22 | max\_consec=64 | avg\_clip50=47.95

EP=10,200 | STEP=2,120,476 | clip=50 | consec=4 | max\_consec=64 | avg\_clip50=45.30

EP=10,210 | STEP=2,123,467 | clip=36 | consec=14 | max\_consec=64 | avg\_clip50=46.80

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 47.2     |  |
|  | clipped_episode_r... | 57.5     |  |
|  | consecutive_clipp... | 15       |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.14e+03 |  |
|  | ep_rew_mean          | 471      |  |
|  | exploration rate     | 0.293    |  |
|  | time/                |          |  |
|  | episodes             | 30100    |  |
|  | fps                  | 201      |  |

|  |                 |         |  |
|--|-----------------|---------|--|
|  | time_elapsed    | 10564   |  |
|  | total timesteps | 2123788 |  |
|  | train/          |         |  |
|  | learning_rate   | 0.0001  |  |
|  | loss            | 0.298   |  |
|  | n_updates       | 518446  |  |

-----

EP=10,220 | STEP=2,125,997 | clip=21 | consec=4 | max\_consec=64 | avg\_clip50=48.02

EP=10,230 | STEP=2,128,902 | clip=54 | consec=14 | max\_consec=64 | avg\_clip50=46.56

EP=10,240 | STEP=2,131,445 | clip=50 | consec=24 | max\_consec=64 | avg\_clip50=47.31

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 48.5     |  |
|  | clipped_episode_r... | 37       |  |
|  | consecutive_clipp... | 29       |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.13e+03 |  |
|  | ep_rew_mean          | 472      |  |
|  | exploration rate     | 0.29     |  |
|  | time/                |          |  |
|  | episodes             | 30200    |  |
|  | fps                  | 200      |  |
|  | time_elapsed         | 10613    |  |
|  | total timesteps      | 2132719  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.239    |  |
|  | n_updates            | 520679   |  |

-----

EP=10,250 | STEP=2,134,161 | clip=52 | consec=34 | max\_consec=64 | avg\_clip50=49.20

EP=10,260 | STEP=2,136,942 | clip=62 | consec=6 | max\_consec=64 | avg\_clip50=48.59

EP=10,270 | STEP=2,139,442 | clip=2 | consec=0 | max\_consec=64 | avg\_clip50=48.72

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 47.5     |  |
|  | clipped_episode_r... | 54       |  |
|  | consecutive_clipp... | 4        |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.12e+03 |  |
|  | ep_rew_mean          | 468      |  |
|  | exploration rate     | 0.287    |  |
|  | time/                |          |  |
|  | episodes             | 30300    |  |
|  | fps                  | 200      |  |
|  | time_elapsed         | 10662    |  |
|  | total timesteps      | 2142270  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.173    |  |
|  | n_updates            | 523067   |  |

-----

EP=10,280 | STEP=2,142,320 | clip=72 | consec=5 | max\_consec=64 | avg\_clip50=47.82

EP=10,290 | STEP=2,144,787 | clip=48 | consec=6 | max\_consec=64 | avg\_clip50=47.6  
9  
EP=10,300 | STEP=2,147,003 | clip=44 | consec=4 | max\_consec=64 | avg\_clip50=46.4  
7  
EP=10,310 | STEP=2,149,543 | clip=41 | consec=14 | max\_consec=64 | avg\_clip50=44.  
70

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 44.2     |  |
| clipped_episode_r... | 47.5     |  |
| consecutive_clipp... | 2        |  |
| max_consecutive_c... | 64       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.08e+03 |  |
| ep_rew_mean          | 463      |  |
| exploration rate     | 0.284    |  |
| time/                |          |  |
| episodes             | 30400    |  |
| fps                  | 200      |  |
| time_elapsed         | 10708    |  |
| total timesteps      | 2150380  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0407   |  |
| n_updates            | 525094   |  |

EP=10,320 | STEP=2,151,862 | clip=8 | consec=0 | max\_consec=64 | avg\_clip50=44.74  
EP=10,330 | STEP=2,154,866 | clip=54 | consec=10 | max\_consec=64 | avg\_clip50=45.  
15  
EP=10,340 | STEP=2,157,837 | clip=80 | consec=20 | max\_consec=64 | avg\_clip50=48.  
38

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 50.6     |  |
| clipped_episode_r... | 60       |  |
| consecutive_clipp... | 28       |  |
| max_consecutive_c... | 64       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.11e+03 |  |
| ep_rew_mean          | 488      |  |
| exploration rate     | 0.281    |  |
| time/                |          |  |
| episodes             | 30500    |  |
| fps                  | 200      |  |
| time_elapsed         | 10762    |  |
| total timesteps      | 2160528  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.26     |  |
| n_updates            | 527631   |  |

EP=10,350 | STEP=2,160,909 | clip=50 | consec=30 | max\_consec=64 | avg\_clip50=51.  
02  
EP=10,360 | STEP=2,163,813 | clip=30 | consec=40 | max\_consec=64 | avg\_clip50=52.  
34  
EP=10,370 | STEP=2,166,635 | clip=57 | consec=7 | max\_consec=64 | avg\_clip50=53.1  
7  
EP=10,380 | STEP=2,169,310 | clip=49 | consec=17 | max\_consec=64 | avg\_clip50=52.  
33  
-----

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 52.1     |
| clipped_episode_r... | 42.5     |
| consecutive_clipp... | 20       |
| max_consecutive_c... | 64       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 491      |
| exploration rate     | 0.277    |
| time/                |          |
| episodes             | 30600    |
| fps                  | 200      |
| time_elapsed         | 10812    |
| total timesteps      | 2170186  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0413   |
| n_updates            | 530046   |

EP=10,390 | STEP=2,172,097 | clip=60 | consec=6 | max\_consec=64 | avg\_clip50=49.66

EP=10,400 | STEP=2,174,521 | clip=26 | consec=16 | max\_consec=64 | avg\_clip50=48.51

EP=10,410 | STEP=2,177,174 | clip=58 | consec=5 | max\_consec=64 | avg\_clip50=48.40

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 47.6     |
| clipped_episode_r... | 28.5     |
| consecutive_clipp... | 13       |
| max_consecutive_c... | 64       |
| rollout/             |          |
| ep_len_mean          | 1.16e+03 |
| ep_rew_mean          | 496      |
| exploration rate     | 0.274    |
| time/                |          |
| episodes             | 30700    |
| fps                  | 200      |
| time_elapsed         | 10862    |
| total timesteps      | 2179487  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0423   |
| n_updates            | 532371   |

EP=10,420 | STEP=2,180,106 | clip=57 | consec=15 | max\_consec=64 | avg\_clip50=47.93

EP=10,430 | STEP=2,183,457 | clip=44 | consec=1 | max\_consec=64 | avg\_clip50=49.23

EP=10,440 | STEP=2,185,915 | clip=36 | consec=9 | max\_consec=64 | avg\_clip50=49.03

EP=10,450 | STEP=2,189,146 | clip=48 | consec=19 | max\_consec=64 | avg\_clip50=49.45

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 49.2 |
| clipped_episode_r... | 12.5 |
| consecutive_clipp... | 0    |
| max_consecutive_c... | 64   |
| rollout/             |      |

|  |                  |          |  |
|--|------------------|----------|--|
|  | ep_len_mean      | 1.17e+03 |  |
|  | ep_rew_mean      | 489      |  |
|  | exploration rate | 0.271    |  |
|  | time/            |          |  |
|  | episodes         | 30800    |  |
|  | fps              | 200      |  |
|  | time_elapsed     | 10920    |  |
|  | total timesteps  | 2190261  |  |
|  | train/           |          |  |
|  | learning_rate    | 0.0001   |  |
|  | loss             | 0.424    |  |
|  | n_updates        | 535065   |  |

-----

EP=10,460 | STEP=2,191,847 | clip=45 | consec=7 | max\_consec=64 | avg\_clip50=49.7  
1  
EP=10,470 | STEP=2,194,202 | clip=17 | consec=0 | max\_consec=64 | avg\_clip50=48.4  
5  
EP=10,480 | STEP=2,196,957 | clip=35 | consec=10 | max\_consec=64 | avg\_clip50=47.  
97

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 49.8     |  |
|  | clipped_episode_r... | 65.5     |  |
|  | consecutive_clipp... | 19       |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.16e+03 |  |
|  | ep_rew_mean          | 496      |  |
|  | exploration rate     | 0.268    |  |
|  | time/                |          |  |
|  | episodes             | 30900    |  |
|  | fps                  | 200      |  |
|  | time_elapsed         | 10970    |  |
|  | total timesteps      | 2199574  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.175    |  |
|  | n_updates            | 537393   |  |

-----

EP=10,490 | STEP=2,199,774 | clip=68 | consec=20 | max\_consec=64 | avg\_clip50=50.  
38  
EP=10,500 | STEP=2,202,587 | clip=57 | consec=1 | max\_consec=64 | avg\_clip50=50.7  
6  
EP=10,510 | STEP=2,205,520 | clip=47 | consec=11 | max\_consec=64 | avg\_clip50=51.  
49  
EP=10,520 | STEP=2,207,803 | clip=33 | consec=4 | max\_consec=64 | avg\_clip50=51.3  
7

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 49.8     |  |
|  | clipped_episode_r... | 77       |  |
|  | consecutive_clipp... | 9        |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.12e+03 |  |
|  | ep_rew_mean          | 486      |  |
|  | exploration rate     | 0.264    |  |
|  | time/                |          |  |
|  | episodes             | 31000    |  |
|  | fps                  | 200      |  |

|  |                 |         |  |
|--|-----------------|---------|--|
|  | time_elapsed    | 11021   |  |
|  | total timesteps | 2209032 |  |
|  | train/          |         |  |
|  | learning_rate   | 0.0001  |  |
|  | loss            | 0.0949  |  |
|  | n_updates       | 539757  |  |

-----

EP=10,530 | STEP=2,210,513 | clip=50 | consec=14 | max\_consec=64 | avg\_clip50=50.65

EP=10,540 | STEP=2,213,021 | clip=47 | consec=6 | max\_consec=64 | avg\_clip50=48.10

EP=10,550 | STEP=2,215,714 | clip=30 | consec=2 | max\_consec=64 | avg\_clip50=46.30

EP=10,560 | STEP=2,218,503 | clip=34 | consec=3 | max\_consec=64 | avg\_clip50=44.25

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 44.4     |  |
|  | clipped_episode_r... | 44       |  |
|  | consecutive_clipp... | 4        |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.11e+03 |  |
|  | ep_rew_mean          | 480      |  |
|  | exploration rate     | 0.261    |  |
|  | time/                |          |  |
|  | episodes             | 31100    |  |
|  | fps                  | 200      |  |
|  | time_elapsed         | 11074    |  |
|  | total timesteps      | 2218928  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0342   |  |
|  | n_updates            | 542231   |  |

-----

EP=10,570 | STEP=2,221,258 | clip=48 | consec=5 | max\_consec=64 | avg\_clip50=45.44

EP=10,580 | STEP=2,223,880 | clip=46 | consec=5 | max\_consec=64 | avg\_clip50=45.37

EP=10,590 | STEP=2,226,976 | clip=40 | consec=15 | max\_consec=64 | avg\_clip50=46.45

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 44.8     |  |
|  | clipped_episode_r... | 34       |  |
|  | consecutive_clipp... | 21       |  |
|  | max_consecutive_c... | 64       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.12e+03 |  |
|  | ep_rew_mean          | 461      |  |
|  | exploration rate     | 0.258    |  |
|  | time/                |          |  |
|  | episodes             | 31200    |  |
|  | fps                  | 200      |  |
|  | time_elapsed         | 11127    |  |
|  | total timesteps      | 2228847  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0795   |  |
|  | n_updates            | 544711   |  |

```

-----
EP=10,600 | STEP=2,229,975 | clip=56 | consec=25 | max_consec=64 | avg_clip50=46.32
EP=10,610 | STEP=2,232,524 | clip=86 | consec=35 | max_consec=64 | avg_clip50=47.69
EP=10,620 | STEP=2,235,389 | clip=33 | consec=45 | max_consec=64 | avg_clip50=47.83
EP=10,630 | STEP=2,238,404 | clip=36 | consec=55 | max_consec=64 | avg_clip50=48.25

```

```

-----
| custom/                |          |
|   avg_clip50            |    48.5  |
|   clipped_episode_r...  |    29    |
|   consecutive_clipp...  |    57    |
|   max_consecutive_c...  |    64    |
| rollout/                |          |
|   ep_len_mean           |  1.16e+03 |
|   ep_rew_mean           |    467    |
|   exploration rate      |    0.254  |
| time/                   |          |
|   episodes              |   31300   |
|   fps                   |    200    |
|   time_elapsed          |   11182   |
|   total timesteps       |  2239022  |
| train/                  |          |
|   learning_rate         |   0.0001  |
|   loss                  |   0.0549  |
|   n_updates             |   547255  |

```

```

-----
EP=10,640 | STEP=2,241,193 | clip=55 | consec=65 | max_consec=65 | avg_clip50=48.75
EP=10,650 | STEP=2,244,069 | clip=25 | consec=7  | max_consec=67 | avg_clip50=49.38
EP=10,660 | STEP=2,246,647 | clip=52 | consec=8  | max_consec=67 | avg_clip50=48.27

```

```

-----
| custom/                |          |
|   avg_clip50            |    50.1  |
|   clipped_episode_r...  |   103    |
|   consecutive_clipp...  |     2    |
|   max_consecutive_c...  |    67    |
| rollout/                |          |
|   ep_len_mean           |  1.17e+03 |
|   ep_rew_mean           |    487    |
|   exploration rate      |    0.251  |
| time/                   |          |
|   episodes              |   31400   |
|   fps                   |    200    |
|   time_elapsed          |   11231   |
|   total timesteps       |  2248511  |
| train/                  |          |
|   learning_rate         |   0.0001  |
|   loss                  |   0.0872  |
|   n_updates             |   549627  |

```

```

-----
EP=10,670 | STEP=2,249,372 | clip=58 | consec=6  | max_consec=67 | avg_clip50=49.01
EP=10,680 | STEP=2,252,395 | clip=60 | consec=6  | max_consec=67 | avg_clip50=49.82
EP=10,690 | STEP=2,255,444 | clip=30 | consec=16 | max_consec=67 | avg_clip50=50.

```



12

EP=10,700 | STEP=2,258,422 | clip=54 | consec=7 | max\_consec=67 | avg\_clip50=51.27

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 51.7     |  |
| clipped_episode_r... | 60       |  |
| consecutive_clipp... | 9        |  |
| max_consecutive_c... | 67       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.18e+03 |  |
| ep_rew_mean          | 502      |  |
| exploration rate     | 0.248    |  |
| time/                |          |  |
| episodes             | 31500    |  |
| fps                  | 200      |  |
| time_elapsed         | 11289    |  |
| total timesteps      | 2259241  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0472   |  |
| n_updates            | 552310   |  |

EP=10,710 | STEP=2,261,340 | clip=54 | consec=3 | max\_consec=67 | avg\_clip50=51.78

EP=10,720 | STEP=2,264,609 | clip=66 | consec=13 | max\_consec=67 | avg\_clip50=53.75

EP=10,730 | STEP=2,267,528 | clip=80 | consec=23 | max\_consec=67 | avg\_clip50=53.36

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 53.4     |  |
| clipped_episode_r... | 47       |  |
| consecutive_clipp... | 31       |  |
| max_consecutive_c... | 67       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.21e+03 |  |
| ep_rew_mean          | 520      |  |
| exploration rate     | 0.244    |  |
| time/                |          |  |
| episodes             | 31600    |  |
| fps                  | 200      |  |
| time_elapsed         | 11347    |  |
| total timesteps      | 2269965  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0507   |  |
| n_updates            | 554991   |  |

EP=10,740 | STEP=2,270,602 | clip=58 | consec=33 | max\_consec=67 | avg\_clip50=53.56

EP=10,750 | STEP=2,273,546 | clip=60 | consec=43 | max\_consec=67 | avg\_clip50=53.97

EP=10,760 | STEP=2,276,668 | clip=36 | consec=53 | max\_consec=67 | avg\_clip50=56.04

EP=10,770 | STEP=2,279,605 | clip=52 | consec=63 | max\_consec=67 | avg\_clip50=54.41

|            |      |  |
|------------|------|--|
| -----      |      |  |
| custom/    |      |  |
| avg_clip50 | 53.8 |  |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | clipped_episode_r... | 41.5     |  |
|  | consecutive_clipp... | 67       |  |
|  | max_consecutive_c... | 67       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.25e+03 |  |
|  | ep_rew_mean          | 539      |  |
|  | exploration rate     | 0.241    |  |
|  | time/                |          |  |
|  | episodes             | 31700    |  |
|  | fps                  | 199      |  |
|  | time_elapsed         | 11404    |  |
|  | total timesteps      | 2280681  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0551   |  |
|  | n_updates            | 557670   |  |

EP=10,780 | STEP=2,282,359 | clip=37 | consec=73 | max\_consec=73 | avg\_clip50=53.95

EP=10,790 | STEP=2,285,078 | clip=47 | consec=83 | max\_consec=83 | avg\_clip50=53.37

EP=10,800 | STEP=2,288,255 | clip=42 | consec=93 | max\_consec=93 | avg\_clip50=54.78

Objetivo conseguido: 100 episodios consecutivos superando el umbral.

Entrenamiento finalizado.

max\_consec alcanzado: 100

Log JSON: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_SpaceInvaders-v0\_sb3\_log.json

Log CSV: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_SpaceInvaders-v0\_sb3\_log.csv

Checkpoint PyTorch guardado en: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0\_double/dqn\_spaceinvaders\_v0\_policy\_latest.pt

model.save() falló (no es crítico, se usa el .pt): IndexError('tuple index out of range')

In [ ]: *# M6-bis: Dueling DQN con checkpoint periódico*

```
class PeriodicPolicyCheckpoint(BaseCallback):
    def __init__(self, save_path, every_steps=250_000, verbose=0):
        super().__init__(verbose)
        self.save_path = save_path
        self.every_steps = every_steps
        self.last_save = 0

    def _on_step(self):
        if self.num_timesteps - self.last_save >= self.every_steps:
            save_policy_checkpoint(self.model, self.save_path)
            self.last_save = self.num_timesteps
            print(f'[checkpoint periódico] guardado a {self.num_timesteps:,} steps')
        return True

paths = RUN_PATHS['dueling']
env_dueling = create_train_env(paths['monitor'])
model_dueling = build_model('dueling', env_dueling, paths['tensorboard'])

training_logger = SB3TrainingLogger(json_path=paths['json'], csv_path=paths['csv'])
callback = ConsecutiveRewardCallback(training_logger=training_logger, verbose=1)
ckpt_cb = PeriodicPolicyCheckpoint(paths['policy'], every_steps=250_000)

model_dueling.learn(
```

```
total_timesteps=TOTAL_TIMESTEPS,
callback=[callback, ckpt_cb],
reset_num_timesteps=True,
tb_log_name='DQN_dueling',
log_interval=100
)

save_policy_checkpoint(model_dueling, paths['policy'])
print('Checkpoint final guardado en:', paths['policy'])
try:
    model_dueling.save(str(paths['model_zip']))
except Exception as e:
    print('model.save() falló (no crítico):', repr(e))
env_dueling.close()
```

Se han truncado las últimas 5000 líneas del flujo de salida.

EP=3,370 | STEP=562,571 | clip=45 | consec=2 | max\_consec=8 | avg\_clip50=17.69

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 19.7   |  |
| clipped_episode_r... | 8.5    |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 756    |  |
| ep_rew_mean          | 202    |  |
| exploration rate     | 0.812  |  |
| time/                |        |  |
| episodes             | 10100  |  |
| fps                  | 218    |  |
| time_elapsed         | 2581   |  |
| total timesteps      | 564298 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.191  |  |
| n_updates            | 128574 |  |

EP=3,380 | STEP=564,367 | clip=10 | consec=0 | max\_consec=8 | avg\_clip50=19.07

EP=3,390 | STEP=565,911 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=19.03

EP=3,400 | STEP=567,605 | clip=5 | consec=0 | max\_consec=8 | avg\_clip50=17.88

EP=3,410 | STEP=569,465 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=17.98

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 18.6   |  |
| clipped_episode_r... | 31.5   |  |
| consecutive_clipp... | 1      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 743    |  |
| ep_rew_mean          | 186    |  |
| exploration rate     | 0.81   |  |
| time/                |        |  |
| episodes             | 10200  |  |
| fps                  | 218    |  |
| time_elapsed         | 2608   |  |
| total timesteps      | 570005 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0661 |  |
| n_updates            | 130001 |  |

EP=3,420 | STEP=571,034 | clip=0 | consec=0 | max\_consec=8 | avg\_clip50=17.23

EP=3,430 | STEP=573,189 | clip=14 | consec=0 | max\_consec=8 | avg\_clip50=18.07

EP=3,440 | STEP=575,025 | clip=4 | consec=0 | max\_consec=8 | avg\_clip50=18.26

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 19    |  |
| clipped_episode_r... | 23.5  |  |
| consecutive_clipp... | 1     |  |
| max_consecutive_c... | 8     |  |
| rollout/             |       |  |
| ep_len_mean          | 759   |  |
| ep_rew_mean          | 188   |  |
| exploration rate     | 0.808 |  |
| time/                |       |  |

|                 |        |  |
|-----------------|--------|--|
| episodes        | 10300  |  |
| fps             | 218    |  |
| time_elapsed    | 2638   |  |
| total timesteps | 576322 |  |
| train/          |        |  |
| learning_rate   | 0.0001 |  |
| loss            | 0.208  |  |
| n_updates       | 131580 |  |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,450 | STEP=576,894 | clip=12 | consec=0 | max_consec=8 | avg_clip50=19.44 |
| EP=3,460 | STEP=578,769 | clip=46 | consec=2 | max_consec=8 | avg_clip50=20.19 |
| EP=3,470 | STEP=580,562 | clip=10 | consec=0 | max_consec=8 | avg_clip50=21.39 |

|                      |        |  |
|----------------------|--------|--|
| custom/              |        |  |
| avg_clip50           | 21.7   |  |
| clipped_episode_r... | 12     |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 772    |  |
| ep_rew_mean          | 198    |  |
| exploration rate     | 0.806  |  |
| time/                |        |  |
| episodes             | 10400  |  |
| fps                  | 218    |  |
| time_elapsed         | 2668   |  |
| total timesteps      | 582541 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0626 |  |
| n_updates            | 133135 |  |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,480 | STEP=582,643 | clip=18 | consec=0 | max_consec=8 | avg_clip50=21.74 |
| EP=3,490 | STEP=584,728 | clip=25 | consec=1 | max_consec=8 | avg_clip50=22.80 |
| EP=3,500 | STEP=586,304 | clip=4  | consec=0 | max_consec=8 | avg_clip50=23.33 |
| EP=3,510 | STEP=588,325 | clip=29 | consec=1 | max_consec=8 | avg_clip50=23.68 |

|                      |        |  |
|----------------------|--------|--|
| custom/              |        |  |
| avg_clip50           | 24.2   |  |
| clipped_episode_r... | 17.5   |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 800    |  |
| ep_rew_mean          | 218    |  |
| exploration rate     | 0.804  |  |
| time/                |        |  |
| episodes             | 10500  |  |
| fps                  | 218    |  |
| time_elapsed         | 2699   |  |
| total timesteps      | 589098 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.209  |  |
| n_updates            | 134774 |  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,520 | STEP=590,299 | clip=17 | consec=0 | max_consec=8 | avg_clip50=23.79 |
| EP=3,530 | STEP=592,117 | clip=28 | consec=1 | max_consec=8 | avg_clip50=22.20 |
| EP=3,540 | STEP=593,673 | clip=10 | consec=0 | max_consec=8 | avg_clip50=20.28 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.8   |
| clipped_episode_r... | 30.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 789    |
| ep_rew_mean          | 217    |
| exploration rate     | 0.802  |
| time/                |        |
| episodes             | 10600  |
| fps                  | 218    |
| time_elapsed         | 2728   |
| total timesteps      | 594975 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0691 |
| n_updates            | 136243 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,550 | STEP=595,480 | clip=20 | consec=0 | max_consec=8 | avg_clip50=19.88 |
| EP=3,560 | STEP=597,565 | clip=35 | consec=5 | max_consec=8 | avg_clip50=20.31 |
| EP=3,570 | STEP=599,229 | clip=4  | consec=0 | max_consec=8 | avg_clip50=20.73 |
| EP=3,580 | STEP=601,094 | clip=32 | consec=1 | max_consec=8 | avg_clip50=20.33 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.3   |
| clipped_episode_r... | 32.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 784    |
| ep_rew_mean          | 213    |
| exploration rate     | 0.8    |
| time/                |        |
| episodes             | 10700  |
| fps                  | 218    |
| time_elapsed         | 2756   |
| total timesteps      | 601157 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.251  |
| n_updates            | 137789 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,590 | STEP=602,934 | clip=12 | consec=0 | max_consec=8 | avg_clip50=21.14 |
| EP=3,600 | STEP=604,750 | clip=42 | consec=1 | max_consec=8 | avg_clip50=20.99 |
| EP=3,610 | STEP=606,496 | clip=14 | consec=0 | max_consec=8 | avg_clip50=18.82 |

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 18.9  |
| clipped_episode_r... | 29    |
| consecutive_clipp... | 1     |
| max_consecutive_c... | 8     |
| rollout/             |       |
| ep_len_mean          | 764   |
| ep_rew_mean          | 196   |
| exploration rate     | 0.798 |
| time/                |       |
| episodes             | 10800 |
| fps                  | 217   |
| time_elapsed         | 2785  |

|                 |        |
|-----------------|--------|
| total timesteps | 607149 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0539 |
| n_updates       | 139287 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,620 | STEP=607,985 | clip=3  | consec=0 | max_consec=8 | avg_clip50=17.34 |
| EP=3,630 | STEP=610,324 | clip=14 | consec=0 | max_consec=8 | avg_clip50=19.95 |
| EP=3,640 | STEP=612,330 | clip=34 | consec=1 | max_consec=8 | avg_clip50=20.53 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.5   |
| clipped_episode_r... | 22     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 795    |
| ep_rew_mean          | 209    |
| exploration rate     | 0.796  |
| time/                |        |
| episodes             | 10900  |
| fps                  | 217    |
| time_elapsed         | 2817   |
| total timesteps      | 613882 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0282 |
| n_updates            | 140970 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,650 | STEP=614,408 | clip=18 | consec=0 | max_consec=8 | avg_clip50=20.86 |
| EP=3,660 | STEP=616,463 | clip=20 | consec=0 | max_consec=8 | avg_clip50=22.60 |
| EP=3,670 | STEP=618,534 | clip=26 | consec=3 | max_consec=8 | avg_clip50=24.11 |
| EP=3,680 | STEP=620,345 | clip=33 | consec=1 | max_consec=8 | avg_clip50=22.54 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.5   |
| clipped_episode_r... | 33     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 814    |
| ep_rew_mean          | 212    |
| exploration rate     | 0.793  |
| time/                |        |
| episodes             | 11000  |
| fps                  | 217    |
| time_elapsed         | 2848   |
| total timesteps      | 620610 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.119  |
| n_updates            | 142652 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,690 | STEP=622,443 | clip=24 | consec=1 | max_consec=8 | avg_clip50=24.23 |
| EP=3,700 | STEP=624,338 | clip=17 | consec=0 | max_consec=8 | avg_clip50=23.72 |
| EP=3,710 | STEP=626,163 | clip=82 | consec=1 | max_consec=8 | avg_clip50=23.72 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 22.6 |
| clipped_episode_r... | 9    |

|                      |        |
|----------------------|--------|
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 823    |
| ep_rew_mean          | 227    |
| exploration rate     | 0.791  |
| time/                |        |
| episodes             | 11100  |
| fps                  | 217    |
| time_elapsed         | 2876   |
| total timesteps      | 626762 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.049  |
| n_updates            | 144190 |

-----

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,720 | STEP=628,011 | clip=17 | consec=0 | max_consec=8 | avg_clip50=23.46 |
| EP=3,730 | STEP=629,973 | clip=24 | consec=2 | max_consec=8 | avg_clip50=23.02 |
| EP=3,740 | STEP=631,851 | clip=10 | consec=0 | max_consec=8 | avg_clip50=20.42 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.3   |
| clipped_episode_r... | 48.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 819    |
| ep_rew_mean          | 225    |
| exploration rate     | 0.789  |
| time/                |        |
| episodes             | 11200  |
| fps                  | 217    |
| time_elapsed         | 2907   |
| total timesteps      | 633401 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0443 |
| n_updates            | 145850 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,750 | STEP=633,744 | clip=9  | consec=0 | max_consec=8 | avg_clip50=20.89 |
| EP=3,760 | STEP=635,648 | clip=16 | consec=0 | max_consec=8 | avg_clip50=20.70 |
| EP=3,770 | STEP=637,412 | clip=12 | consec=0 | max_consec=8 | avg_clip50=21.05 |
| EP=3,780 | STEP=639,224 | clip=13 | consec=0 | max_consec=8 | avg_clip50=20.66 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.7   |
| clipped_episode_r... | 57.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 803    |
| ep_rew_mean          | 220    |
| exploration rate     | 0.787  |
| time/                |        |
| episodes             | 11300  |
| fps                  | 217    |
| time_elapsed         | 2937   |
| total timesteps      | 639789 |
| train/               |        |
| learning_rate        | 0.0001 |



|           |        |
|-----------|--------|
| loss      | 0.0664 |
| n_updates | 147447 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,790 | STEP=641,236 | clip=10 | consec=0 | max_consec=8 | avg_clip50=22.29 |
| EP=3,800 | STEP=642,889 | clip=18 | consec=0 | max_consec=8 | avg_clip50=20.99 |
| EP=3,810 | STEP=644,698 | clip=35 | consec=1 | max_consec=8 | avg_clip50=20.41 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.2   |
| clipped_episode_r... | 30     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 792    |
| ep_rew_mean          | 210    |
| exploration rate     | 0.785  |
| time/                |        |
| episodes             | 11400  |
| fps                  | 217    |
| time_elapsed         | 2963   |
| total timesteps      | 645399 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.053  |
| n_updates            | 148849 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,820 | STEP=646,451 | clip=16 | consec=0 | max_consec=8 | avg_clip50=20.31 |
| EP=3,830 | STEP=648,599 | clip=6  | consec=0 | max_consec=8 | avg_clip50=21.36 |
| EP=3,840 | STEP=650,933 | clip=18 | consec=0 | max_consec=8 | avg_clip50=21.65 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.1   |
| clipped_episode_r... | 17     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 811    |
| ep_rew_mean          | 219    |
| exploration rate     | 0.783  |
| time/                |        |
| episodes             | 11500  |
| fps                  | 217    |
| time_elapsed         | 2996   |
| total timesteps      | 652722 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0767 |
| n_updates            | 150680 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,850 | STEP=652,818 | clip=8  | consec=0 | max_consec=8 | avg_clip50=22.87 |
| EP=3,860 | STEP=654,440 | clip=62 | consec=2 | max_consec=8 | avg_clip50=22.60 |
| EP=3,870 | STEP=655,960 | clip=18 | consec=0 | max_consec=8 | avg_clip50=20.74 |
| EP=3,880 | STEP=658,152 | clip=3  | consec=0 | max_consec=8 | avg_clip50=20.88 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 20.6 |
| clipped_episode_r... | 29   |
| consecutive_clipp... | 1    |
| max_consecutive_c... | 8    |
| rollout/             |      |

|                  |        |
|------------------|--------|
| ep_len_mean      | 791    |
| ep_rew_mean      | 205    |
| exploration rate | 0.781  |
| time/            |        |
| episodes         | 11600  |
| fps              | 217    |
| time_elapsed     | 3024   |
| total timesteps  | 658694 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0712 |
| n_updates        | 152173 |

EP=3,890 | STEP=660,358 | clip=30 | consec=4 | max\_consec=8 | avg\_clip50=21.30  
EP=3,900 | STEP=662,580 | clip=14 | consec=0 | max\_consec=8 | avg\_clip50=22.37  
EP=3,910 | STEP=664,165 | clip=0 | consec=0 | max\_consec=8 | avg\_clip50=22.47

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.1   |
| clipped_episode_r... | 19     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 834    |
| ep_rew_mean          | 230    |
| exploration rate     | 0.778  |
| time/                |        |
| episodes             | 11700  |
| fps                  | 217    |
| time_elapsed         | 3054   |
| total timesteps      | 665454 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.128  |
| n_updates            | 153863 |

EP=3,920 | STEP=666,140 | clip=18 | consec=0 | max\_consec=8 | avg\_clip50=24.37  
EP=3,930 | STEP=668,053 | clip=36 | consec=1 | max\_consec=8 | avg\_clip50=23.14  
EP=3,940 | STEP=669,898 | clip=4 | consec=0 | max\_consec=8 | avg\_clip50=21.61

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 19.3   |
| clipped_episode_r... | 12     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 788    |
| ep_rew_mean          | 208    |
| exploration rate     | 0.776  |
| time/                |        |
| episodes             | 11800  |
| fps                  | 217    |
| time_elapsed         | 3081   |
| total timesteps      | 671373 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0867 |
| n_updates            | 155343 |

EP=3,950 | STEP=671,444 | clip=24 | consec=1 | max\_consec=8 | avg\_clip50=19.49

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,960 | STEP=673,410 | clip=19 | consec=0 | max_consec=8 | avg_clip50=20.05 |
| EP=3,970 | STEP=675,400 | clip=17 | consec=0 | max_consec=8 | avg_clip50=20.20 |
| EP=3,980 | STEP=677,079 | clip=29 | consec=2 | max_consec=8 | avg_clip50=19.82 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 18.9   |
| clipped_episode_r... | 8      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 795    |
| ep_rew_mean          | 212    |
| exploration rate     | 0.774  |
| time/                |        |
| episodes             | 11900  |
| fps                  | 217    |
| time_elapsed         | 3109   |
| total timesteps      | 677499 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.042  |
| n_updates            | 156874 |

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=3,990 | STEP=678,815 | clip=32 | consec=1 | max_consec=8 | avg_clip50=19.56 |
| EP=4,000 | STEP=680,523 | clip=30 | consec=1 | max_consec=8 | avg_clip50=20.33 |
| EP=4,010 | STEP=682,624 | clip=16 | consec=0 | max_consec=8 | avg_clip50=21.27 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.2   |
| clipped_episode_r... | 31     |
| consecutive_clipp... | 3      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 779    |
| ep_rew_mean          | 205    |
| exploration rate     | 0.772  |
| time/                |        |
| episodes             | 12000  |
| fps                  | 217    |
| time_elapsed         | 3137   |
| total timesteps      | 683822 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0248 |
| n_updates            | 158455 |

|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,020 | STEP=684,401 | clip=10 | consec=0 | max_consec=8 | avg_clip50=21.42 |
| EP=4,030 | STEP=686,506 | clip=57 | consec=2 | max_consec=8 | avg_clip50=23.03 |
| EP=4,040 | STEP=688,146 | clip=20 | consec=0 | max_consec=8 | avg_clip50=22.98 |
| EP=4,050 | STEP=689,756 | clip=32 | consec=2 | max_consec=8 | avg_clip50=22.13 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 22.6 |
| clipped_episode_r... | 45   |
| consecutive_clipp... | 3    |
| max_consecutive_c... | 8    |
| rollout/             |      |
| ep_len_mean          | 781  |
| ep_rew_mean          | 215  |
| exploration rate     | 0.77 |

|                 |        |
|-----------------|--------|
| time/           |        |
| episodes        | 12100  |
| fps             | 217    |
| time_elapsed    | 3166   |
| total timesteps | 690085 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0823 |
| n_updates       | 160021 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,060 | STEP=692,338 | clip=14 | consec=0 | max_consec=8 | avg_clip50=23.12 |
| EP=4,070 | STEP=694,134 | clip=26 | consec=3 | max_consec=8 | avg_clip50=22.23 |
| EP=4,080 | STEP=696,137 | clip=49 | consec=1 | max_consec=8 | avg_clip50=22.33 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.4   |
| clipped_episode_r... | 4.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 814    |
| ep_rew_mean          | 230    |
| exploration rate     | 0.768  |
| time/                |        |
| episodes             | 12200  |
| fps                  | 217    |
| time_elapsed         | 3197   |
| total timesteps      | 696980 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0537 |
| n_updates            | 161744 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,090 | STEP=697,961 | clip=29 | consec=3 | max_consec=8 | avg_clip50=23.33 |
| EP=4,100 | STEP=700,155 | clip=24 | consec=1 | max_consec=8 | avg_clip50=26.92 |
| EP=4,110 | STEP=702,059 | clip=9  | consec=0 | max_consec=8 | avg_clip50=24.92 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.7   |
| clipped_episode_r... | 25     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 823    |
| ep_rew_mean          | 234    |
| exploration rate     | 0.766  |
| time/                |        |
| episodes             | 12300  |
| fps                  | 218    |
| time_elapsed         | 3226   |
| total timesteps      | 703554 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0304 |
| n_updates            | 163388 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,120 | STEP=703,721 | clip=5  | consec=0 | max_consec=8 | avg_clip50=24.10 |
| EP=4,130 | STEP=705,668 | clip=42 | consec=1 | max_consec=8 | avg_clip50=24.28 |
| EP=4,140 | STEP=707,704 | clip=18 | consec=0 | max_consec=8 | avg_clip50=24.36 |
| EP=4,150 | STEP=709,863 | clip=22 | consec=2 | max_consec=8 | avg_clip50=23.23 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.3   |
| clipped_episode_r... | 9.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 843    |
| ep_rew_mean          | 249    |
| exploration rate     | 0.763  |
| time/                |        |
| episodes             | 12400  |
| fps                  | 218    |
| time_elapsed         | 3257   |
| total timesteps      | 710392 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0489 |
| n_updates            | 165097 |

EP=4,160 | STEP=711,693 | clip=56 | consec=2 | max\_consec=8 | avg\_clip50=23.16  
EP=4,170 | STEP=713,636 | clip=11 | consec=0 | max\_consec=8 | avg\_clip50=24.89  
EP=4,180 | STEP=715,517 | clip=14 | consec=0 | max\_consec=8 | avg\_clip50=24.47

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.1   |
| clipped_episode_r... | 9.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 815    |
| ep_rew_mean          | 243    |
| exploration rate     | 0.761  |
| time/                |        |
| episodes             | 12500  |
| fps                  | 218    |
| time_elapsed         | 3284   |
| total timesteps      | 716298 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0528 |
| n_updates            | 166574 |

EP=4,190 | STEP=716,940 | clip=23 | consec=1 | max\_consec=8 | avg\_clip50=22.60  
EP=4,200 | STEP=718,542 | clip=17 | consec=0 | max\_consec=8 | avg\_clip50=20.87  
EP=4,210 | STEP=720,442 | clip=18 | consec=0 | max\_consec=8 | avg\_clip50=21.92

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 21.1  |
| clipped_episode_r... | 20.5  |
| consecutive_clipp... | 1     |
| max_consecutive_c... | 8     |
| rollout/             |       |
| ep_len_mean          | 784   |
| ep_rew_mean          | 229   |
| exploration rate     | 0.76  |
| time/                |       |
| episodes             | 12600 |
| fps                  | 218   |
| time_elapsed         | 3310  |

|                 |        |
|-----------------|--------|
| total timesteps | 722162 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.28   |
| n_updates       | 168040 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,220 | STEP=722,290 | clip=16 | consec=0 | max_consec=8 | avg_clip50=21.16 |
| EP=4,230 | STEP=723,875 | clip=10 | consec=0 | max_consec=8 | avg_clip50=19.37 |
| EP=4,240 | STEP=725,492 | clip=14 | consec=0 | max_consec=8 | avg_clip50=20.02 |
| EP=4,250 | STEP=727,342 | clip=7  | consec=0 | max_consec=8 | avg_clip50=20.06 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 17.4   |
| clipped_episode_r... | 19     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 740    |
| ep_rew_mean          | 203    |
| exploration rate     | 0.758  |
| time/                |        |
| episodes             | 12700  |
| fps                  | 218    |
| time_elapsed         | 3336   |
| total timesteps      | 727798 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.245  |
| n_updates            | 169449 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,260 | STEP=729,506 | clip=23 | consec=2 | max_consec=8 | avg_clip50=20.01 |
| EP=4,270 | STEP=731,599 | clip=18 | consec=0 | max_consec=8 | avg_clip50=22.01 |
| EP=4,280 | STEP=732,952 | clip=18 | consec=0 | max_consec=8 | avg_clip50=21.32 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.6   |
| clipped_episode_r... | 10     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 759    |
| ep_rew_mean          | 215    |
| exploration rate     | 0.756  |
| time/                |        |
| episodes             | 12800  |
| fps                  | 218    |
| time_elapsed         | 3365   |
| total timesteps      | 734185 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.14   |
| n_updates            | 171046 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,290 | STEP=734,914 | clip=21 | consec=3 | max_consec=8 | avg_clip50=23.26 |
| EP=4,300 | STEP=736,846 | clip=12 | consec=0 | max_consec=8 | avg_clip50=24.34 |
| EP=4,310 | STEP=738,567 | clip=6  | consec=0 | max_consec=8 | avg_clip50=22.60 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 22.4 |
| clipped_episode_r... | 61   |

|                      |        |
|----------------------|--------|
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 782    |
| ep_rew_mean          | 222    |
| exploration rate     | 0.753  |
| time/                |        |
| episodes             | 12900  |
| fps                  | 218    |
| time_elapsed         | 3394   |
| total timesteps      | 740700 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0669 |
| n_updates            | 172674 |

EP=4,320 | STEP=740,721 | clip=16 | consec=0 | max\_consec=8 | avg\_clip50=22.35  
EP=4,330 | STEP=742,632 | clip=8 | consec=0 | max\_consec=8 | avg\_clip50=24.59  
EP=4,340 | STEP=744,779 | clip=28 | consec=1 | max\_consec=8 | avg\_clip50=24.13  
EP=4,350 | STEP=746,670 | clip=13 | consec=0 | max\_consec=8 | avg\_clip50=23.66

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.3   |
| clipped_episode_r... | 54.5   |
| consecutive_clipp... | 3      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 830    |
| ep_rew_mean          | 248    |
| exploration rate     | 0.751  |
| time/                |        |
| episodes             | 13000  |
| fps                  | 218    |
| time_elapsed         | 3425   |
| total timesteps      | 747468 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0419 |
| n_updates            | 174366 |

EP=4,360 | STEP=748,638 | clip=19 | consec=0 | max\_consec=8 | avg\_clip50=24.84  
[checkpoint periodico] guardado a 750,000 steps  
EP=4,370 | STEP=750,266 | clip=9 | consec=0 | max\_consec=8 | avg\_clip50=22.59  
EP=4,380 | STEP=751,971 | clip=2 | consec=0 | max\_consec=8 | avg\_clip50=21.31

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.6   |
| clipped_episode_r... | 9.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 814    |
| ep_rew_mean          | 228    |
| exploration rate     | 0.749  |
| time/                |        |
| episodes             | 13100  |
| fps                  | 218    |
| time_elapsed         | 3454   |
| total timesteps      | 753647 |
| train/               |        |

|               |        |
|---------------|--------|
| learning_rate | 0.0001 |
| loss          | 0.0951 |
| n_updates     | 175911 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,390 | STEP=754,261 | clip=33 | consec=2 | max_consec=8 | avg_clip50=22.10 |
| EP=4,400 | STEP=756,631 | clip=36 | consec=1 | max_consec=8 | avg_clip50=23.17 |
| EP=4,410 | STEP=758,510 | clip=14 | consec=0 | max_consec=8 | avg_clip50=22.49 |
| EP=4,420 | STEP=760,709 | clip=8  | consec=0 | max_consec=8 | avg_clip50=24.16 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.2   |
| clipped_episode_r... | 7.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 845    |
| ep_rew_mean          | 234    |
| exploration rate     | 0.747  |
| time/                |        |
| episodes             | 13200  |
| fps                  | 218    |
| time_elapsed         | 3487   |
| total timesteps      | 760709 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.182  |
| n_updates            | 177677 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,430 | STEP=762,286 | clip=25 | consec=1 | max_consec=8 | avg_clip50=24.30 |
| EP=4,440 | STEP=763,910 | clip=14 | consec=0 | max_consec=8 | avg_clip50=21.60 |
| EP=4,450 | STEP=766,045 | clip=16 | consec=0 | max_consec=8 | avg_clip50=22.12 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.2   |
| clipped_episode_r... | 38.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 820    |
| ep_rew_mean          | 226    |
| exploration rate     | 0.745  |
| time/                |        |
| episodes             | 13300  |
| fps                  | 218    |
| time_elapsed         | 3516   |
| total timesteps      | 767036 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.107  |
| n_updates            | 179258 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,460 | STEP=768,072 | clip=5  | consec=0 | max_consec=8 | avg_clip50=23.40 |
| EP=4,470 | STEP=769,527 | clip=6  | consec=0 | max_consec=8 | avg_clip50=21.25 |
| EP=4,480 | STEP=771,222 | clip=18 | consec=0 | max_consec=8 | avg_clip50=21.93 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 22.3 |
| clipped_episode_r... | 25.5 |
| consecutive_clipp... | 1    |
| max_consecutive_c... | 8    |



|                  |        |
|------------------|--------|
| rollout/         |        |
| ep_len_mean      | 795    |
| ep_rew_mean      | 223    |
| exploration rate | 0.743  |
| time/            |        |
| episodes         | 13400  |
| fps              | 218    |
| time_elapsed     | 3541   |
| total timesteps  | 772589 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.035  |
| n_updates        | 180647 |

EP=4,490 | STEP=773,109 | clip=61 | consec=3 | max\_consec=8 | avg\_clip50=23.62  
EP=4,500 | STEP=774,823 | clip=24 | consec=2 | max\_consec=8 | avg\_clip50=21.16  
EP=4,510 | STEP=776,589 | clip=18 | consec=0 | max\_consec=8 | avg\_clip50=19.60  
EP=4,520 | STEP=778,052 | clip=32 | consec=1 | max\_consec=8 | avg\_clip50=21.00

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.6   |
| clipped_episode_r... | 42     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 738    |
| ep_rew_mean          | 212    |
| exploration rate     | 0.741  |
| time/                |        |
| episodes             | 13500  |
| fps                  | 218    |
| time_elapsed         | 3568   |
| total timesteps      | 778416 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.191  |
| n_updates            | 182103 |

EP=4,530 | STEP=779,710 | clip=22 | consec=1 | max\_consec=8 | avg\_clip50=20.66  
EP=4,540 | STEP=781,459 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=19.39  
EP=4,550 | STEP=783,559 | clip=38 | consec=1 | max\_consec=8 | avg\_clip50=20.00

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.1   |
| clipped_episode_r... | 20     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 745    |
| ep_rew_mean          | 204    |
| exploration rate     | 0.739  |
| time/                |        |
| episodes             | 13600  |
| fps                  | 218    |
| time_elapsed         | 3597   |
| total timesteps      | 784624 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0261 |
| n_updates            | 183655 |

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EP=4,560 | STEP=785,330 | clip=6 | consec=0 | max\_consec=8 | avg\_clip50=19.88  
EP=4,570 | STEP=787,186 | clip=19 | consec=0 | max\_consec=8 | avg\_clip50=19.78  
EP=4,580 | STEP=788,745 | clip=4 | consec=0 | max\_consec=8 | avg\_clip50=18.87  
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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 19.1   |
| clipped_episode_r... | 4      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 753    |
| ep_rew_mean          | 198    |
| exploration rate     | 0.737  |
| time/                |        |
| episodes             | 13700  |
| fps                  | 218    |
| time_elapsed         | 3622   |
| total timesteps      | 790278 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.141  |
| n_updates            | 185069 |

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EP=4,590 | STEP=790,497 | clip=24 | consec=1 | max\_consec=8 | avg\_clip50=19.34  
EP=4,600 | STEP=792,381 | clip=58 | consec=1 | max\_consec=8 | avg\_clip50=19.89  
EP=4,610 | STEP=794,308 | clip=70 | consec=1 | max\_consec=8 | avg\_clip50=21.35  
EP=4,620 | STEP=796,529 | clip=34 | consec=2 | max\_consec=8 | avg\_clip50=23.14  
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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.4   |
| clipped_episode_r... | 13.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 789    |
| ep_rew_mean          | 215    |
| exploration rate     | 0.735  |
| time/                |        |
| episodes             | 13800  |
| fps                  | 218    |
| time_elapsed         | 3654   |
| total timesteps      | 797082 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0517 |
| n_updates            | 186770 |

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EP=4,630 | STEP=798,471 | clip=8 | consec=0 | max\_consec=8 | avg\_clip50=25.78  
EP=4,640 | STEP=800,514 | clip=30 | consec=1 | max\_consec=8 | avg\_clip50=28.32  
EP=4,650 | STEP=802,553 | clip=16 | consec=0 | max\_consec=8 | avg\_clip50=28.38  
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|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 28.2 |
| clipped_episode_r... | 49   |
| consecutive_clipp... | 1    |
| max_consecutive_c... | 8    |
| rollout/             |      |
| ep_len_mean          | 797  |
| ep_rew_mean          | 241  |

|                  |        |
|------------------|--------|
| exploration rate | 0.732  |
| time/            |        |
| episodes         | 13900  |
| fps              | 218    |
| time_elapsed     | 3684   |
| total timesteps  | 803703 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0488 |
| n_updates        | 188425 |

EP=4,660 | STEP=804,118 | clip=28 | consec=1 | max\_consec=8 | avg\_clip50=26.93  
EP=4,670 | STEP=806,130 | clip=16 | consec=0 | max\_consec=8 | avg\_clip50=26.18  
EP=4,680 | STEP=807,792 | clip=10 | consec=0 | max\_consec=8 | avg\_clip50=24.25  
EP=4,690 | STEP=809,413 | clip=5 | consec=0 | max\_consec=8 | avg\_clip50=21.33

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.3   |
| clipped_episode_r... | 5      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 802    |
| ep_rew_mean          | 248    |
| exploration rate     | 0.73   |
| time/                |        |
| episodes             | 14000  |
| fps                  | 218    |
| time_elapsed         | 3711   |
| total timesteps      | 809484 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0668 |
| n_updates            | 189870 |

EP=4,700 | STEP=811,527 | clip=31 | consec=3 | max\_consec=8 | avg\_clip50=21.48  
EP=4,710 | STEP=813,298 | clip=26 | consec=1 | max\_consec=8 | avg\_clip50=21.94  
EP=4,720 | STEP=814,998 | clip=30 | consec=1 | max\_consec=8 | avg\_clip50=19.96

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.4   |
| clipped_episode_r... | 23     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 781    |
| ep_rew_mean          | 229    |
| exploration rate     | 0.728  |
| time/                |        |
| episodes             | 14100  |
| fps                  | 218    |
| time_elapsed         | 3741   |
| total timesteps      | 816038 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0908 |
| n_updates            | 191509 |

EP=4,730 | STEP=816,985 | clip=18 | consec=0 | max\_consec=8 | avg\_clip50=21.29  
EP=4,740 | STEP=819,049 | clip=30 | consec=4 | max\_consec=8 | avg\_clip50=22.92

EP=4,750 | STEP=820,951 | clip=24 | consec=3 | max\_consec=8 | avg\_clip50=22.44

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| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 22.2   |  |
| clipped_episode_r... | 29     |  |
| consecutive_clipp... | 1      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 786    |  |
| ep_rew_mean          | 221    |  |
| exploration rate     | 0.726  |  |
| time/                |        |  |
| episodes             | 14200  |  |
| fps                  | 218    |  |
| time_elapsed         | 3771   |  |
| total timesteps      | 822488 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.189  |  |
| n_updates            | 193121 |  |

EP=4,760 | STEP=822,600 | clip=11 | consec=0 | max\_consec=8 | avg\_clip50=21.90  
EP=4,770 | STEP=824,374 | clip=16 | consec=0 | max\_consec=8 | avg\_clip50=22.89  
EP=4,780 | STEP=826,386 | clip=38 | consec=1 | max\_consec=8 | avg\_clip50=22.54  
EP=4,790 | STEP=828,093 | clip=26 | consec=3 | max\_consec=8 | avg\_clip50=21.18

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| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 21.2   |  |
| clipped_episode_r... | 18     |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 797    |  |
| ep_rew_mean          | 218    |  |
| exploration rate     | 0.724  |  |
| time/                |        |  |
| episodes             | 14300  |  |
| fps                  | 218    |  |
| time_elapsed         | 3801   |  |
| total timesteps      | 828909 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.255  |  |
| n_updates            | 194727 |  |

EP=4,800 | STEP=830,182 | clip=16 | consec=0 | max\_consec=8 | avg\_clip50=21.63  
EP=4,810 | STEP=831,798 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=21.39  
EP=4,820 | STEP=833,374 | clip=24 | consec=1 | max\_consec=8 | avg\_clip50=21.05

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| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 20.9  |  |
| clipped_episode_r... | 18    |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 8     |  |
| rollout/             |       |  |
| ep_len_mean          | 783   |  |
| ep_rew_mean          | 217   |  |
| exploration rate     | 0.722 |  |
| time/                |       |  |
| episodes             | 14400 |  |

|                 |        |
|-----------------|--------|
| fps             | 218    |
| time_elapsed    | 3828   |
| total timesteps | 834794 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0526 |
| n_updates       | 196198 |

EP=4,830 | STEP=835,342 | clip=29 | consec=1 | max\_consec=8 | avg\_clip50=20.29  
EP=4,840 | STEP=837,007 | clip=6 | consec=0 | max\_consec=8 | avg\_clip50=19.77  
EP=4,850 | STEP=839,339 | clip=26 | consec=2 | max\_consec=8 | avg\_clip50=20.81  
EP=4,860 | STEP=841,006 | clip=6 | consec=0 | max\_consec=8 | avg\_clip50=21.45

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.4   |
| clipped_episode_r... | 6      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 784    |
| ep_rew_mean          | 214    |
| exploration rate     | 0.72   |
| time/                |        |
| episodes             | 14500  |
| fps                  | 218    |
| time_elapsed         | 3856   |
| total timesteps      | 841006 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.048  |
| n_updates            | 197751 |

EP=4,870 | STEP=842,893 | clip=32 | consec=1 | max\_consec=8 | avg\_clip50=23.00  
EP=4,880 | STEP=845,064 | clip=61 | consec=2 | max\_consec=8 | avg\_clip50=24.80  
EP=4,890 | STEP=846,624 | clip=20 | consec=0 | max\_consec=8 | avg\_clip50=25.02

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.4   |
| clipped_episode_r... | 21     |
| consecutive_clipp... | 3      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 781    |
| ep_rew_mean          | 224    |
| exploration rate     | 0.718  |
| time/                |        |
| episodes             | 14600  |
| fps                  | 218    |
| time_elapsed         | 3886   |
| total timesteps      | 847373 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.184  |
| n_updates            | 199343 |

EP=4,900 | STEP=848,842 | clip=18 | consec=0 | max\_consec=8 | avg\_clip50=25.22  
EP=4,910 | STEP=850,833 | clip=70 | consec=2 | max\_consec=8 | avg\_clip50=26.57  
EP=4,920 | STEP=852,639 | clip=8 | consec=0 | max\_consec=8 | avg\_clip50=25.43

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| custom/ |  |
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| avg_clip50           | 24.5   |
| clipped_episode_r... | 3      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 802    |
| ep_rew_mean          | 242    |
| exploration rate     | 0.716  |
| time/                |        |
| episodes             | 14700  |
| fps                  | 218    |
| time_elapsed         | 3916   |
| total timesteps      | 853913 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0388 |
| n_updates            | 200978 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,930 | STEP=854,436 | clip=32 | consec=2 | max_consec=8 | avg_clip50=24.07 |
| EP=4,940 | STEP=856,775 | clip=50 | consec=4 | max_consec=8 | avg_clip50=26.90 |
| EP=4,950 | STEP=858,875 | clip=32 | consec=1 | max_consec=8 | avg_clip50=26.62 |
| EP=4,960 | STEP=860,547 | clip=5  | consec=0 | max_consec=8 | avg_clip50=25.30 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.5   |
| clipped_episode_r... | 20.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 822    |
| ep_rew_mean          | 256    |
| exploration rate     | 0.713  |
| time/                |        |
| episodes             | 14800  |
| fps                  | 218    |
| time_elapsed         | 3948   |
| total timesteps      | 860718 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0782 |
| n_updates            | 202679 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=4,970 | STEP=862,388 | clip=44 | consec=2 | max_consec=8 | avg_clip50=25.91 |
| EP=4,980 | STEP=864,073 | clip=12 | consec=0 | max_consec=8 | avg_clip50=24.62 |
| EP=4,990 | STEP=866,199 | clip=41 | consec=1 | max_consec=8 | avg_clip50=24.42 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.4   |
| clipped_episode_r... | 26     |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 834    |
| ep_rew_mean          | 258    |
| exploration rate     | 0.711  |
| time/                |        |
| episodes             | 14900  |
| fps                  | 217    |
| time_elapsed         | 3978   |
| total timesteps      | 867177 |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0418 |
| n_updates     | 204294 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,000 | STEP=868,081 | clip=10 | consec=0 | max_consec=8 | avg_clip50=22.55 |
| EP=5,010 | STEP=870,062 | clip=28 | consec=5 | max_consec=8 | avg_clip50=23.93 |
| EP=5,020 | STEP=872,107 | clip=31 | consec=3 | max_consec=8 | avg_clip50=24.06 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.6   |
| clipped_episode_r... | 38.5   |
| consecutive_clipp... | 6      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 846    |
| ep_rew_mean          | 259    |
| exploration rate     | 0.709  |
| time/                |        |
| episodes             | 15000  |
| fps                  | 217    |
| time_elapsed         | 4010   |
| total timesteps      | 873987 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0966 |
| n_updates            | 205996 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,030 | STEP=874,299 | clip=14 | consec=0 | max_consec=8 | avg_clip50=26.95 |
| EP=5,040 | STEP=876,156 | clip=35 | consec=2 | max_consec=8 | avg_clip50=24.90 |
| EP=5,050 | STEP=877,903 | clip=37 | consec=1 | max_consec=8 | avg_clip50=25.04 |
| EP=5,060 | STEP=879,900 | clip=38 | consec=3 | max_consec=8 | avg_clip50=24.65 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.1   |
| clipped_episode_r... | 24     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 821    |
| ep_rew_mean          | 242    |
| exploration rate     | 0.707  |
| time/                |        |
| episodes             | 15100  |
| fps                  | 217    |
| time_elapsed         | 4039   |
| total timesteps      | 880204 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0636 |
| n_updates            | 207550 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,070 | STEP=881,786 | clip=4  | consec=0 | max_consec=8 | avg_clip50=24.61 |
| EP=5,080 | STEP=883,484 | clip=10 | consec=0 | max_consec=8 | avg_clip50=23.15 |
| EP=5,090 | STEP=885,676 | clip=16 | consec=0 | max_consec=8 | avg_clip50=24.25 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 24.2 |
| clipped_episode_r... | 13.5 |
| consecutive_clipp... | 0    |

|                      |        |
|----------------------|--------|
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 817    |
| ep_rew_mean          | 245    |
| exploration rate     | 0.705  |
| time/                |        |
| episodes             | 15200  |
| fps                  | 217    |
| time_elapsed         | 4069   |
| total timesteps      | 886760 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0629 |
| n_updates            | 209189 |

EP=5,100 | STEP=887,385 | clip=40 | consec=1 | max\_consec=8 | avg\_clip50=24.32  
EP=5,110 | STEP=889,352 | clip=6 | consec=0 | max\_consec=8 | avg\_clip50=23.56  
EP=5,120 | STEP=891,367 | clip=42 | consec=2 | max\_consec=8 | avg\_clip50=22.93  
EP=5,130 | STEP=893,415 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=23.93

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 23.9   |
| clipped_episode_r... | 12     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 808    |
| ep_rew_mean          | 235    |
| exploration rate     | 0.702  |
| time/                |        |
| episodes             | 15300  |
| fps                  | 217    |
| time_elapsed         | 4100   |
| total timesteps      | 893468 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.115  |
| n_updates            | 210866 |

EP=5,140 | STEP=895,479 | clip=54 | consec=2 | max\_consec=8 | avg\_clip50=24.69  
EP=5,150 | STEP=897,961 | clip=28 | consec=1 | max\_consec=8 | avg\_clip50=26.59  
EP=5,160 | STEP=900,088 | clip=17 | consec=0 | max\_consec=8 | avg\_clip50=27.48

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 28.2   |
| clipped_episode_r... | 14.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 850    |
| ep_rew_mean          | 251    |
| exploration rate     | 0.7    |
| time/                |        |
| episodes             | 15400  |
| fps                  | 217    |
| time_elapsed         | 4135   |
| total timesteps      | 900848 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0363 |



|                      |              |         |          |              |                  |  |
|----------------------|--------------|---------|----------|--------------|------------------|--|
| n_updates            |              | 212711  |          |              |                  |  |
| EP=5,170             | STEP=901,899 | clip=12 | consec=0 | max_consec=8 | avg_clip50=26.93 |  |
| EP=5,180             | STEP=904,068 | clip=23 | consec=1 | max_consec=8 | avg_clip50=26.58 |  |
| EP=5,190             | STEP=906,320 | clip=32 | consec=1 | max_consec=8 | avg_clip50=27.29 |  |
|                      |              |         |          |              |                  |  |
| custom/              |              |         |          |              |                  |  |
| avg_clip50           |              | 26.3    |          |              |                  |  |
| clipped_episode_r... |              | 31      |          |              |                  |  |
| consecutive_clipp... |              | 4       |          |              |                  |  |
| max_consecutive_c... |              | 8       |          |              |                  |  |
| rollout/             |              |         |          |              |                  |  |
| ep_len_mean          |              | 890     |          |              |                  |  |
| ep_rew_mean          |              | 265     |          |              |                  |  |
| exploration rate     |              | 0.698   |          |              |                  |  |
| time/                |              |         |          |              |                  |  |
| episodes             |              | 15500   |          |              |                  |  |
| fps                  |              | 217     |          |              |                  |  |
| time_elapsed         |              | 4168    |          |              |                  |  |
| total timesteps      |              | 908100  |          |              |                  |  |
| train/               |              |         |          |              |                  |  |
| learning_rate        |              | 0.0001  |          |              |                  |  |
| loss                 |              | 0.0574  |          |              |                  |  |
| n_updates            |              | 214524  |          |              |                  |  |
|                      |              |         |          |              |                  |  |
| EP=5,200             | STEP=908,380 | clip=22 | consec=1 | max_consec=8 | avg_clip50=25.96 |  |
| EP=5,210             | STEP=910,428 | clip=34 | consec=4 | max_consec=8 | avg_clip50=25.75 |  |
| EP=5,220             | STEP=912,341 | clip=27 | consec=1 | max_consec=8 | avg_clip50=26.88 |  |
| EP=5,230             | STEP=914,298 | clip=8  | consec=0 | max_consec=8 | avg_clip50=27.39 |  |
|                      |              |         |          |              |                  |  |
| custom/              |              |         |          |              |                  |  |
| avg_clip50           |              | 27      |          |              |                  |  |
| clipped_episode_r... |              | 24.5    |          |              |                  |  |
| consecutive_clipp... |              | 1       |          |              |                  |  |
| max_consecutive_c... |              | 8       |          |              |                  |  |
| rollout/             |              |         |          |              |                  |  |
| ep_len_mean          |              | 888     |          |              |                  |  |
| ep_rew_mean          |              | 271     |          |              |                  |  |
| exploration rate     |              | 0.695   |          |              |                  |  |
| time/                |              |         |          |              |                  |  |
| episodes             |              | 15600   |          |              |                  |  |
| fps                  |              | 217     |          |              |                  |  |
| time_elapsed         |              | 4198    |          |              |                  |  |
| total timesteps      |              | 914607  |          |              |                  |  |
| train/               |              |         |          |              |                  |  |
| learning_rate        |              | 0.0001  |          |              |                  |  |
| loss                 |              | 0.0676  |          |              |                  |  |
| n_updates            |              | 216151  |          |              |                  |  |
|                      |              |         |          |              |                  |  |
| EP=5,240             | STEP=916,302 | clip=36 | consec=4 | max_consec=8 | avg_clip50=25.15 |  |
| EP=5,250             | STEP=918,140 | clip=26 | consec=1 | max_consec=8 | avg_clip50=25.13 |  |
| EP=5,260             | STEP=920,464 | clip=23 | consec=1 | max_consec=8 | avg_clip50=26.69 |  |
|                      |              |         |          |              |                  |  |
| custom/              |              |         |          |              |                  |  |
| avg_clip50           |              | 25      |          |              |                  |  |
| clipped_episode_r... |              | 25.5    |          |              |                  |  |
| consecutive_clipp... |              | 1       |          |              |                  |  |
| max_consecutive_c... |              | 8       |          |              |                  |  |
| rollout/             |              |         |          |              |                  |  |
| ep len mean          |              | 859     |          |              |                  |  |

|                  |        |
|------------------|--------|
| ep_rew_mean      | 260    |
| exploration rate | 0.693  |
| time/            |        |
| episodes         | 15700  |
| fps              | 217    |
| time_elapsed     | 4229   |
| total timesteps  | 921331 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0901 |
| n_updates        | 217832 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,270 | STEP=922,364 | clip=25 | consec=2 | max_consec=8 | avg_clip50=25.45 |
| EP=5,280 | STEP=924,423 | clip=26 | consec=3 | max_consec=8 | avg_clip50=26.09 |
| EP=5,290 | STEP=926,575 | clip=34 | consec=4 | max_consec=8 | avg_clip50=27.12 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.6   |
| clipped_episode_r... | 44.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 846    |
| ep_rew_mean          | 258    |
| exploration rate     | 0.691  |
| time/                |        |
| episodes             | 15800  |
| fps                  | 217    |
| time_elapsed         | 4263   |
| total timesteps      | 928252 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0357 |
| n_updates            | 219562 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,300 | STEP=928,474 | clip=28 | consec=3 | max_consec=8 | avg_clip50=26.68 |
| EP=5,310 | STEP=929,988 | clip=58 | consec=1 | max_consec=8 | avg_clip50=23.59 |
| EP=5,320 | STEP=931,780 | clip=52 | consec=2 | max_consec=8 | avg_clip50=25.06 |
| EP=5,330 | STEP=933,716 | clip=34 | consec=2 | max_consec=8 | avg_clip50=24.29 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 24.2   |
| clipped_episode_r... | 48     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 826    |
| ep_rew_mean          | 255    |
| exploration rate     | 0.689  |
| time/                |        |
| episodes             | 15900  |
| fps                  | 217    |
| time_elapsed         | 4291   |
| total timesteps      | 934234 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0714 |
| n_updates            | 221058 |

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|          |              |        |          |              |                  |
|----------|--------------|--------|----------|--------------|------------------|
| EP=5,340 | STEP=935,613 | clip=6 | consec=0 | max_consec=8 | avg_clip50=23.70 |
|----------|--------------|--------|----------|--------------|------------------|

EP=5,350 | STEP=937,886 | clip=24 | consec=3 | max\_consec=8 | avg\_clip50=24.91  
EP=5,360 | STEP=939,989 | clip=4 | consec=0 | max\_consec=8 | avg\_clip50=27.17

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 28.1   |  |
| clipped_episode_r... | 19.5   |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 834    |  |
| ep_rew_mean          | 260    |  |
| exploration rate     | 0.687  |  |
| time/                |        |  |
| episodes             | 16000  |  |
| fps                  | 217    |  |
| time_elapsed         | 4323   |  |
| total timesteps      | 941303 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0498 |  |
| n_updates            | 222825 |  |

EP=5,370 | STEP=942,549 | clip=33 | consec=5 | max\_consec=8 | avg\_clip50=29.28  
EP=5,380 | STEP=944,642 | clip=20 | consec=1 | max\_consec=8 | avg\_clip50=29.06  
EP=5,390 | STEP=946,470 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=28.31

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 27.1   |  |
| clipped_episode_r... | 16     |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 8      |  |
| rollout/             |        |  |
| ep_len_mean          | 838    |  |
| ep_rew_mean          | 260    |  |
| exploration rate     | 0.684  |  |
| time/                |        |  |
| episodes             | 16100  |  |
| fps                  | 217    |  |
| time_elapsed         | 4355   |  |
| total timesteps      | 948189 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0223 |  |
| n_updates            | 224547 |  |

EP=5,400 | STEP=948,311 | clip=36 | consec=1 | max\_consec=8 | avg\_clip50=27.36  
EP=5,410 | STEP=950,295 | clip=32 | consec=2 | max\_consec=8 | avg\_clip50=26.70  
EP=5,420 | STEP=952,175 | clip=24 | consec=4 | max\_consec=8 | avg\_clip50=24.12  
EP=5,430 | STEP=953,849 | clip=30 | consec=2 | max\_consec=8 | avg\_clip50=22.38

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 22.1  |  |
| clipped_episode_r... | 17    |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 8     |  |
| rollout/             |       |  |
| ep_len_mean          | 850   |  |
| ep_rew_mean          | 256   |  |
| exploration rate     | 0.682 |  |
| time/                |       |  |

|                 |        |
|-----------------|--------|
| episodes        | 16200  |
| fps             | 217    |
| time_elapsed    | 4384   |
| total timesteps | 954423 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.1    |
| n_updates       | 226105 |

EP=5,440 | STEP=956,169 | clip=45 | consec=2 | max\_consec=8 | avg\_clip50=24.27  
EP=5,450 | STEP=958,254 | clip=12 | consec=0 | max\_consec=8 | avg\_clip50=26.36  
EP=5,460 | STEP=960,258 | clip=32 | consec=3 | max\_consec=8 | avg\_clip50=25.97

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.3   |
| clipped_episode_r... | 43     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 844    |
| ep_rew_mean          | 256    |
| exploration rate     | 0.68   |
| time/                |        |
| episodes             | 16300  |
| fps                  | 217    |
| time_elapsed         | 4418   |
| total timesteps      | 961578 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0851 |
| n_updates            | 227894 |

EP=5,470 | STEP=962,202 | clip=66 | consec=1 | max\_consec=8 | avg\_clip50=27.05  
EP=5,480 | STEP=964,310 | clip=58 | consec=5 | max\_consec=8 | avg\_clip50=29.88  
EP=5,490 | STEP=966,341 | clip=23 | consec=1 | max\_consec=8 | avg\_clip50=28.80  
EP=5,500 | STEP=968,234 | clip=32 | consec=1 | max\_consec=8 | avg\_clip50=26.31

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.3   |
| clipped_episode_r... | 31.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 844    |
| ep_rew_mean          | 263    |
| exploration rate     | 0.678  |
| time/                |        |
| episodes             | 16400  |
| fps                  | 217    |
| time_elapsed         | 4449   |
| total timesteps      | 968304 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.272  |
| n_updates            | 229575 |

EP=5,510 | STEP=970,073 | clip=15 | consec=0 | max\_consec=8 | avg\_clip50=26.28  
EP=5,520 | STEP=972,204 | clip=31 | consec=2 | max\_consec=8 | avg\_clip50=27.57  
EP=5,530 | STEP=974,694 | clip=20 | consec=5 | max\_consec=8 | avg\_clip50=27.55

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 27.5   |
| clipped_episode_r... | 40.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 893    |
| ep_rew_mean          | 293    |
| exploration rate     | 0.675  |
| time/                |        |
| episodes             | 16500  |
| fps                  | 217    |
| time_elapsed         | 4483   |
| total timesteps      | 975719 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.127  |
| n_updates            | 231429 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,540 | STEP=976,851 | clip=52 | consec=2 | max_consec=8 | avg_clip50=28.17 |
| EP=5,550 | STEP=978,688 | clip=28 | consec=1 | max_consec=8 | avg_clip50=28.66 |
| EP=5,560 | STEP=980,804 | clip=40 | consec=4 | max_consec=8 | avg_clip50=29.55 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 29.7   |
| clipped_episode_r... | 42.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 881    |
| ep_rew_mean          | 286    |
| exploration rate     | 0.673  |
| time/                |        |
| episodes             | 16600  |
| fps                  | 217    |
| time_elapsed         | 4516   |
| total timesteps      | 982670 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0451 |
| n_updates            | 233167 |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=5,570 | STEP=983,097 | clip=35 | consec=3 | max_consec=8 | avg_clip50=29.56 |
| EP=5,580 | STEP=985,707 | clip=58 | consec=5 | max_consec=8 | avg_clip50=30.63 |
| EP=5,590 | STEP=987,739 | clip=14 | consec=0 | max_consec=8 | avg_clip50=30.21 |
| EP=5,600 | STEP=989,906 | clip=24 | consec=1 | max_consec=8 | avg_clip50=31.09 |

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|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 31.5  |
| clipped_episode_r... | 36.5  |
| consecutive_clipp... | 2     |
| max_consecutive_c... | 8     |
| rollout/             |       |
| ep_len_mean          | 921   |
| ep_rew_mean          | 302   |
| exploration rate     | 0.67  |
| time/                |       |
| episodes             | 16700 |
| fps                  | 217   |
| time_elapsed         | 4551  |

|                 |        |
|-----------------|--------|
| total timesteps | 990320 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.657  |
| n_updates       | 235079 |

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EP=5,610 | STEP=992,060 | clip=25 | consec=1 | max\_consec=8 | avg\_clip50=30.80  
EP=5,620 | STEP=994,239 | clip=6 | consec=0 | max\_consec=8 | avg\_clip50=28.64  
EP=5,630 | STEP=996,388 | clip=44 | consec=2 | max\_consec=8 | avg\_clip50=27.14

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.4   |
| clipped_episode_r... | 23     |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 8      |
| rollout/             |        |
| ep_len_mean          | 906    |
| ep_rew_mean          | 285    |
| exploration rate     | 0.668  |
| time/                |        |
| episodes             | 16800  |
| fps                  | 217    |
| time_elapsed         | 4584   |
| total timesteps      | 997348 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.037  |
| n_updates            | 236836 |

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EP=5,640 | STEP=998,599 | clip=8 | consec=0 | max\_consec=8 | avg\_clip50=28.25  
[checkpoint periodico] guardado a 1,000,000 steps  
EP=5,650 | STEP=1,000,670 | clip=7 | consec=0 | max\_consec=8 | avg\_clip50=27.72  
EP=5,660 | STEP=1,002,528 | clip=32 | consec=4 | max\_consec=8 | avg\_clip50=26.70

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.4    |
| clipped_episode_r... | 22.5    |
| consecutive_clipp... | 4       |
| max_consecutive_c... | 8       |
| rollout/             |         |
| ep_len_mean          | 912     |
| ep_rew_mean          | 283     |
| exploration rate     | 0.666   |
| time/                |         |
| episodes             | 16900   |
| fps                  | 217     |
| time_elapsed         | 4618    |
| total timesteps      | 1004462 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0336  |
| n_updates            | 238615  |

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EP=5,670 | STEP=1,004,642 | clip=25 | consec=5 | max\_consec=8 | avg\_clip50=27.75  
EP=5,680 | STEP=1,006,428 | clip=8 | consec=0 | max\_consec=8 | avg\_clip50=26.05  
EP=5,690 | STEP=1,008,264 | clip=24 | consec=2 | max\_consec=8 | avg\_clip50=23.96  
EP=5,700 | STEP=1,010,510 | clip=20 | consec=0 | max\_consec=8 | avg\_clip50=24.24

|            |      |
|------------|------|
| custom/    |      |
| avg_clip50 | 25.1 |

|  |                      |         |  |
|--|----------------------|---------|--|
|  | clipped_episode_r... | 32.5    |  |
|  | consecutive_clipp... | 1       |  |
|  | max_consecutive_c... | 8       |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 862     |  |
|  | ep_rew_mean          | 258     |  |
|  | exploration rate     | 0.663   |  |
|  | time/                |         |  |
|  | episodes             | 17000   |  |
|  | fps                  | 217     |  |
|  | time_elapsed         | 4649    |  |
|  | total timesteps      | 1011087 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.332   |  |
|  | n_updates            | 240271  |  |

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|          |  |                |  |         |  |          |  |              |  |                  |
|----------|--|----------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,710 |  | STEP=1,012,946 |  | clip=76 |  | consec=2 |  | max_consec=8 |  | avg_clip50=27.10 |
| EP=5,720 |  | STEP=1,014,806 |  | clip=6  |  | consec=0 |  | max_consec=8 |  | avg_clip50=26.37 |
| EP=5,730 |  | STEP=1,016,551 |  | clip=18 |  | consec=0 |  | max_consec=8 |  | avg_clip50=27.00 |

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 27.1    |  |
|  | clipped_episode_r... | 12.5    |  |
|  | consecutive_clipp... | 0       |  |
|  | max_consecutive_c... | 8       |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 840     |  |
|  | ep_rew_mean          | 260     |  |
|  | exploration rate     | 0.661   |  |
|  | time/                |         |  |
|  | episodes             | 17100   |  |
|  | fps                  | 217     |  |
|  | time_elapsed         | 4681    |  |
|  | total timesteps      | 1017855 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.067   |  |
|  | n_updates            | 241963  |  |

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|          |  |                |  |         |  |          |  |              |  |                  |
|----------|--|----------------|--|---------|--|----------|--|--------------|--|------------------|
| EP=5,740 |  | STEP=1,018,665 |  | clip=40 |  | consec=3 |  | max_consec=8 |  | avg_clip50=28.36 |
| EP=5,750 |  | STEP=1,020,194 |  | clip=37 |  | consec=1 |  | max_consec=8 |  | avg_clip50=26.60 |
| EP=5,760 |  | STEP=1,022,175 |  | clip=24 |  | consec=1 |  | max_consec=8 |  | avg_clip50=25.22 |
| EP=5,770 |  | STEP=1,024,040 |  | clip=11 |  | consec=0 |  | max_consec=8 |  | avg_clip50=25.08 |

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 25.1    |  |
|  | clipped_episode_r... | 11      |  |
|  | consecutive_clipp... | 0       |  |
|  | max_consecutive_c... | 8       |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 821     |  |
|  | ep_rew_mean          | 257     |  |
|  | exploration rate     | 0.659   |  |
|  | time/                |         |  |
|  | episodes             | 17200   |  |
|  | fps                  | 217     |  |
|  | time_elapsed         | 4710    |  |
|  | total timesteps      | 1024184 |  |
|  | train/               |         |  |

|          |                      |         |          |              |                  |  |
|----------|----------------------|---------|----------|--------------|------------------|--|
|          | learning_rate        | 0.0001  |          |              |                  |  |
|          | loss                 | 0.081   |          |              |                  |  |
|          | n_updates            | 243545  |          |              |                  |  |
| -----    |                      |         |          |              |                  |  |
| EP=5,780 | STEP=1,026,000       | clip=22 | consec=2 | max_consec=8 | avg_clip50=25.51 |  |
| EP=5,790 | STEP=1,028,117       | clip=32 | consec=2 | max_consec=8 | avg_clip50=26.22 |  |
| EP=5,800 | STEP=1,029,781       | clip=14 | consec=0 | max_consec=8 | avg_clip50=26.18 |  |
| -----    |                      |         |          |              |                  |  |
|          | custom/              |         |          |              |                  |  |
|          | avg_clip50           | 24.8    |          |              |                  |  |
|          | clipped_episode_r... | 33      |          |              |                  |  |
|          | consecutive_clipp... | 1       |          |              |                  |  |
|          | max_consecutive_c... | 8       |          |              |                  |  |
|          | rollout/             |         |          |              |                  |  |
|          | ep_len_mean          | 805     |          |              |                  |  |
|          | ep_rew_mean          | 257     |          |              |                  |  |
|          | exploration rate     | 0.657   |          |              |                  |  |
|          | time/                |         |          |              |                  |  |
|          | episodes             | 17300   |          |              |                  |  |
|          | fps                  | 217     |          |              |                  |  |
|          | time_elapsed         | 4741    |          |              |                  |  |
|          | total timesteps      | 1030474 |          |              |                  |  |
|          | train/               |         |          |              |                  |  |
|          | learning_rate        | 0.0001  |          |              |                  |  |
|          | loss                 | 0.105   |          |              |                  |  |
|          | n_updates            | 245118  |          |              |                  |  |
| -----    |                      |         |          |              |                  |  |
| EP=5,810 | STEP=1,031,638       | clip=20 | consec=0 | max_consec=8 | avg_clip50=24.71 |  |
| EP=5,820 | STEP=1,033,676       | clip=16 | consec=0 | max_consec=8 | avg_clip50=24.84 |  |
| EP=5,830 | STEP=1,035,779       | clip=32 | consec=3 | max_consec=8 | avg_clip50=25.67 |  |
| -----    |                      |         |          |              |                  |  |
|          | custom/              |         |          |              |                  |  |
|          | avg_clip50           | 24.7    |          |              |                  |  |
|          | clipped_episode_r... | 9.5     |          |              |                  |  |
|          | consecutive_clipp... | 0       |          |              |                  |  |
|          | max_consecutive_c... | 8       |          |              |                  |  |
|          | rollout/             |         |          |              |                  |  |
|          | ep_len_mean          | 818     |          |              |                  |  |
|          | ep_rew_mean          | 258     |          |              |                  |  |
|          | exploration rate     | 0.655   |          |              |                  |  |
|          | time/                |         |          |              |                  |  |
|          | episodes             | 17400   |          |              |                  |  |
|          | fps                  | 217     |          |              |                  |  |
|          | time_elapsed         | 4774    |          |              |                  |  |
|          | total timesteps      | 1037536 |          |              |                  |  |
|          | train/               |         |          |              |                  |  |
|          | learning_rate        | 0.0001  |          |              |                  |  |
|          | loss                 | 0.122   |          |              |                  |  |
|          | n_updates            | 246883  |          |              |                  |  |
| -----    |                      |         |          |              |                  |  |
| EP=5,840 | STEP=1,037,831       | clip=26 | consec=1 | max_consec=8 | avg_clip50=24.46 |  |
| EP=5,850 | STEP=1,040,041       | clip=14 | consec=0 | max_consec=8 | avg_clip50=26.53 |  |
| EP=5,860 | STEP=1,042,361       | clip=81 | consec=1 | max_consec=8 | avg_clip50=28.16 |  |
| EP=5,870 | STEP=1,044,580       | clip=54 | consec=2 | max_consec=8 | avg_clip50=29.38 |  |
| -----    |                      |         |          |              |                  |  |
|          | custom/              |         |          |              |                  |  |
|          | avg_clip50           | 28.5    |          |              |                  |  |
|          | clipped_episode_r... | 31.5    |          |              |                  |  |
|          | consecutive_clipp... | 3       |          |              |                  |  |
|          | max_consecutive_c... | 8       |          |              |                  |  |



|                  |         |
|------------------|---------|
| rollout/         |         |
| ep_len_mean      | 866     |
| ep_rew_mean      | 271     |
| exploration rate | 0.652   |
| time/            |         |
| episodes         | 17500   |
| fps              | 217     |
| time_elapsed     | 4808    |
| total timesteps  | 1045074 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0509  |
| n_updates        | 248768  |

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EP=5,880 | STEP=1,046,406 | clip=7 | consec=0 | max\_consec=8 | avg\_clip50=28.13  
EP=5,890 | STEP=1,048,629 | clip=10 | consec=0 | max\_consec=8 | avg\_clip50=29.14  
EP=5,900 | STEP=1,050,412 | clip=30 | consec=1 | max\_consec=8 | avg\_clip50=28.09

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 28.1    |
| clipped_episode_r... | 13.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 8       |
| rollout/             |         |
| ep_len_mean          | 870     |
| ep_rew_mean          | 276     |
| exploration rate     | 0.65    |
| time/                |         |
| episodes             | 17600   |
| fps                  | 217     |
| time_elapsed         | 4838    |
| total timesteps      | 1051186 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.045   |
| n_updates            | 250296  |

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EP=5,910 | STEP=1,052,650 | clip=50 | consec=5 | max\_consec=8 | avg\_clip50=28.21  
EP=5,920 | STEP=1,054,592 | clip=18 | consec=0 | max\_consec=8 | avg\_clip50=26.24  
EP=5,930 | STEP=1,056,875 | clip=4 | consec=0 | max\_consec=8 | avg\_clip50=28.69

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.1    |
| clipped_episode_r... | 27      |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 8       |
| rollout/             |         |
| ep_len_mean          | 885     |
| ep_rew_mean          | 284     |
| exploration rate     | 0.648   |
| time/                |         |
| episodes             | 17700   |
| fps                  | 217     |
| time_elapsed         | 4872    |
| total timesteps      | 1058540 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0516  |
| n_updates            | 252134  |

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|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=5,940 | STEP=1,058,807 | clip=38 | consec=1 | max_consec=8 | avg_clip50=27.76 |
| EP=5,950 | STEP=1,061,008 | clip=60 | consec=1 | max_consec=8 | avg_clip50=28.90 |
| EP=5,960 | STEP=1,063,614 | clip=43 | consec=4 | max_consec=8 | avg_clip50=30.66 |
| EP=5,970 | STEP=1,065,871 | clip=70 | consec=1 | max_consec=8 | avg_clip50=34.02 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 33.8    |
| clipped_episode_r... | 44.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 8       |
| rollout/             |         |
| ep_len_mean          | 896     |
| ep_rew_mean          | 300     |
| exploration rate     | 0.645   |
| time/                |         |
| episodes             | 17800   |
| fps                  | 217     |
| time_elapsed         | 4909    |
| total timesteps      | 1066537 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0716  |
| n_updates            | 254134  |

|          |                |         |          |              |                  |
|----------|----------------|---------|----------|--------------|------------------|
| EP=5,980 | STEP=1,068,226 | clip=27 | consec=2 | max_consec=8 | avg_clip50=33.18 |
| EP=5,990 | STEP=1,070,255 | clip=53 | consec=1 | max_consec=8 | avg_clip50=33.26 |
| EP=6,000 | STEP=1,072,502 | clip=25 | consec=6 | max_consec=8 | avg_clip50=33.03 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 32.3    |
| clipped_episode_r... | 24.5    |
| consecutive_clipp... | 12      |
| max_consecutive_c... | 12      |
| rollout/             |         |
| ep_len_mean          | 938     |
| ep_rew_mean          | 314     |
| exploration rate     | 0.642   |
| time/                |         |
| episodes             | 17900   |
| fps                  | 217     |
| time_elapsed         | 4943    |
| total timesteps      | 1073864 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.146   |
| n_updates            | 255965  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,010 | STEP=1,074,775 | clip=10 | consec=0 | max_consec=13 | avg_clip50=31.16 |
| EP=6,020 | STEP=1,076,830 | clip=60 | consec=5 | max_consec=13 | avg_clip50=29.87 |
| EP=6,030 | STEP=1,079,065 | clip=38 | consec=1 | max_consec=13 | avg_clip50=29.01 |
| EP=6,040 | STEP=1,081,408 | clip=36 | consec=5 | max_consec=13 | avg_clip50=30.54 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 30.5 |
| clipped_episode_r... | 36   |
| consecutive_clipp... | 5    |
| max_consecutive_c... | 13   |
| rollout/             |      |
| ep_len_mean          | 948  |
| ep_rew_mean          | 319  |

|                  |         |
|------------------|---------|
| exploration rate | 0.64    |
| time/            |         |
| episodes         | 18000   |
| fps              | 217     |
| time_elapsed     | 4979    |
| total timesteps  | 1081544 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0214  |
| n_updates        | 257885  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,050 | STEP=1,083,969 | clip=46 | consec=4 | max_consec=13 | avg_clip50=31.82 |
| EP=6,060 | STEP=1,086,274 | clip=47 | consec=4 | max_consec=13 | avg_clip50=32.54 |
| EP=6,070 | STEP=1,088,723 | clip=41 | consec=6 | max_consec=13 | avg_clip50=32.38 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 32.1    |
| clipped_episode_r... | 24.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 13      |
| rollout/             |         |
| ep_len_mean          | 953     |
| ep_rew_mean          | 308     |
| exploration rate     | 0.637   |
| time/                |         |
| episodes             | 18100   |
| fps                  | 217     |
| time_elapsed         | 5015    |
| total timesteps      | 1089478 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0333  |
| n_updates            | 259869  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,080 | STEP=1,090,880 | clip=18 | consec=0 | max_consec=13 | avg_clip50=32.67 |
| EP=6,090 | STEP=1,093,309 | clip=17 | consec=0 | max_consec=13 | avg_clip50=32.72 |
| EP=6,100 | STEP=1,095,781 | clip=24 | consec=4 | max_consec=13 | avg_clip50=34.16 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34.5    |
| clipped_episode_r... | 44.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 13      |
| rollout/             |         |
| ep_len_mean          | 965     |
| ep_rew_mean          | 330     |
| exploration rate     | 0.635   |
| time/                |         |
| episodes             | 18200   |
| fps                  | 217     |
| time_elapsed         | 5052    |
| total timesteps      | 1097295 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.269   |
| n_updates            | 261823  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,110 | STEP=1,097,910 | clip=30 | consec=4 | max_consec=13 | avg_clip50=33.82 |
| EP=6,120 | STEP=1,099,545 | clip=7  | consec=0 | max_consec=13 | avg_clip50=31.14 |
| EP=6,130 | STEP=1,101,464 | clip=16 | consec=0 | max_consec=13 | avg_clip50=30.74 |

EP=6,140 | STEP=1,103,731 | clip=40 | consec=1 | max\_consec=13 | avg\_clip50=29.84

|                      |         |  |
|----------------------|---------|--|
| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 29.8    |  |
| clipped_episode_r... | 39.5    |  |
| consecutive_clipp... | 1       |  |
| max_consecutive_c... | 13      |  |
| rollout/             |         |  |
| ep_len_mean          | 938     |  |
| ep_rew_mean          | 313     |  |
| exploration rate     | 0.632   |  |
| time/                |         |  |
| episodes             | 18300   |  |
| fps                  | 217     |  |
| time_elapsed         | 5083    |  |
| total timesteps      | 1103834 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.307   |  |
| n_updates            | 263458  |  |

EP=6,150 | STEP=1,105,430 | clip=12 | consec=0 | max\_consec=13 | avg\_clip50=25.54

EP=6,160 | STEP=1,107,550 | clip=106 | consec=2 | max\_consec=13 | avg\_clip50=24.9  
1

EP=6,170 | STEP=1,109,694 | clip=40 | consec=1 | max\_consec=13 | avg\_clip50=26.86

|                      |         |  |
|----------------------|---------|--|
| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 25.7    |  |
| clipped_episode_r... | 11.5    |  |
| consecutive_clipp... | 0       |  |
| max_consecutive_c... | 13      |  |
| rollout/             |         |  |
| ep_len_mean          | 887     |  |
| ep_rew_mean          | 292     |  |
| exploration rate     | 0.63    |  |
| time/                |         |  |
| episodes             | 18400   |  |
| fps                  | 217     |  |
| time_elapsed         | 5114    |  |
| total timesteps      | 1110599 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.344   |  |
| n_updates            | 265149  |  |

EP=6,180 | STEP=1,111,902 | clip=36 | consec=5 | max\_consec=13 | avg\_clip50=28.43

EP=6,190 | STEP=1,114,229 | clip=26 | consec=2 | max\_consec=13 | avg\_clip50=29.62

EP=6,200 | STEP=1,116,329 | clip=24 | consec=1 | max\_consec=13 | avg\_clip50=30.77

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 34    |  |
| clipped_episode_r... | 52.5  |  |
| consecutive_clipp... | 1     |  |
| max_consecutive_c... | 13    |  |
| rollout/             |       |  |
| ep_len_mean          | 884   |  |
| ep_rew_mean          | 287   |  |
| exploration rate     | 0.628 |  |
| time/                |       |  |
| episodes             | 18500 |  |

|  |                 |  |         |  |
|--|-----------------|--|---------|--|
|  | fps             |  | 217     |  |
|  | time_elapsed    |  | 5151    |  |
|  | total timesteps |  | 1118433 |  |
|  | train/          |  |         |  |
|  | learning_rate   |  | 0.0001  |  |
|  | loss            |  | 0.0678  |  |
|  | n_updates       |  | 267108  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=6,210 |  | STEP=1,118,659 |  | clip=8  |  | consec=0 |  | max_consec=13 |  | avg_clip50=31.68 |
| EP=6,220 |  | STEP=1,121,275 |  | clip=70 |  | consec=1 |  | max_consec=13 |  | avg_clip50=34.09 |
| EP=6,230 |  | STEP=1,123,047 |  | clip=8  |  | consec=0 |  | max_consec=13 |  | avg_clip50=31.63 |
| EP=6,240 |  | STEP=1,125,411 |  | clip=26 |  | consec=2 |  | max_consec=13 |  | avg_clip50=31.41 |

|  |                      |  |         |  |
|--|----------------------|--|---------|--|
|  | custom/              |  |         |  |
|  | avg_clip50           |  | 31.5    |  |
|  | clipped_episode_r... |  | 7.5     |  |
|  | consecutive_clipp... |  | 0       |  |
|  | max_consecutive_c... |  | 13      |  |
|  | rollout/             |  |         |  |
|  | ep_len_mean          |  | 911     |  |
|  | ep_rew_mean          |  | 304     |  |
|  | exploration rate     |  | 0.625   |  |
|  | time/                |  |         |  |
|  | episodes             |  | 18600   |  |
|  | fps                  |  | 217     |  |
|  | time_elapsed         |  | 5184    |  |
|  | total timesteps      |  | 1125637 |  |
|  | train/               |  |         |  |
|  | learning_rate        |  | 0.0001  |  |
|  | loss                 |  | 0.0832  |  |
|  | n_updates            |  | 268909  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=6,250 |  | STEP=1,127,650 |  | clip=41 |  | consec=2 |  | max_consec=13 |  | avg_clip50=32.68 |
| EP=6,260 |  | STEP=1,129,971 |  | clip=52 |  | consec=2 |  | max_consec=13 |  | avg_clip50=33.01 |
| EP=6,270 |  | STEP=1,131,913 |  | clip=3  |  | consec=0 |  | max_consec=13 |  | avg_clip50=29.57 |

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|--|----------------------|--|---------|--|
|  | custom/              |  |         |  |
|  | avg_clip50           |  | 31.1    |  |
|  | clipped_episode_r... |  | 6       |  |
|  | consecutive_clipp... |  | 0       |  |
|  | max_consecutive_c... |  | 13      |  |
|  | rollout/             |  |         |  |
|  | ep_len_mean          |  | 946     |  |
|  | ep_rew_mean          |  | 321     |  |
|  | exploration rate     |  | 0.623   |  |
|  | time/                |  |         |  |
|  | episodes             |  | 18700   |  |
|  | fps                  |  | 217     |  |
|  | time_elapsed         |  | 5220    |  |
|  | total timesteps      |  | 1133146 |  |
|  | train/               |  |         |  |
|  | learning_rate        |  | 0.0001  |  |
|  | loss                 |  | 0.0352  |  |
|  | n_updates            |  | 270786  |  |

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|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=6,280 |  | STEP=1,133,982 |  | clip=24 |  | consec=2 |  | max_consec=13 |  | avg_clip50=30.74 |
| EP=6,290 |  | STEP=1,136,180 |  | clip=5  |  | consec=0 |  | max_consec=13 |  | avg_clip50=30.35 |
| EP=6,300 |  | STEP=1,138,257 |  | clip=14 |  | consec=0 |  | max_consec=13 |  | avg_clip50=30.36 |

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|--|---------|--|--|--|
|  | custom/ |  |  |  |
|--|---------|--|--|--|

|                      |         |
|----------------------|---------|
| avg_clip50           | 30.8    |
| clipped_episode_r... | 23.5    |
| consecutive_clipp... | 4       |
| max_consecutive_c... | 13      |
| rollout/             |         |
| ep_len_mean          | 923     |
| ep_rew_mean          | 315     |
| exploration rate     | 0.62    |
| time/                |         |
| episodes             | 18800   |
| fps                  | 217     |
| time_elapsed         | 5255    |
| total timesteps      | 1140769 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0486  |
| n_updates            | 272692  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,310 | STEP=1,140,808 | clip=51 | consec=5 | max_consec=13 | avg_clip50=30.80 |
| EP=6,320 | STEP=1,143,033 | clip=30 | consec=1 | max_consec=13 | avg_clip50=32.89 |
| EP=6,330 | STEP=1,145,200 | clip=9  | consec=0 | max_consec=13 | avg_clip50=33.46 |
| EP=6,340 | STEP=1,147,147 | clip=56 | consec=2 | max_consec=13 | avg_clip50=32.73 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 32.5    |
| clipped_episode_r... | 18.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 13      |
| rollout/             |         |
| ep_len_mean          | 919     |
| ep_rew_mean          | 316     |
| exploration rate     | 0.618   |
| time/                |         |
| episodes             | 18900   |
| fps                  | 216     |
| time_elapsed         | 5290    |
| total timesteps      | 1147813 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.13    |
| n_updates            | 274453  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,350 | STEP=1,149,217 | clip=18 | consec=0 | max_consec=13 | avg_clip50=31.51 |
| EP=6,360 | STEP=1,151,373 | clip=16 | consec=0 | max_consec=13 | avg_clip50=30.73 |
| EP=6,370 | STEP=1,153,599 | clip=46 | consec=8 | max_consec=13 | avg_clip50=30.68 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 28.7    |
| clipped_episode_r... | 27.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 13      |
| rollout/             |         |
| ep_len_mean          | 914     |
| ep_rew_mean          | 317     |
| exploration rate     | 0.615   |
| time/                |         |
| episodes             | 19000   |
| fps                  | 216     |
| time_elapsed         | 5328    |
| total timesteps      | 1155293 |

|                      |                |         |          |               |                  |
|----------------------|----------------|---------|----------|---------------|------------------|
| train/               |                |         |          |               |                  |
| learning_rate        |                | 0.0001  |          |               |                  |
| loss                 |                | 0.224   |          |               |                  |
| n_updates            |                | 276323  |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=6,380             | STEP=1,155,767 | clip=28 | consec=4 | max_consec=13 | avg_clip50=30.12 |
| EP=6,390             | STEP=1,158,267 | clip=56 | consec=2 | max_consec=13 | avg_clip50=32.18 |
| EP=6,400             | STEP=1,160,370 | clip=12 | consec=0 | max_consec=13 | avg_clip50=32.41 |
| EP=6,410             | STEP=1,162,518 | clip=8  | consec=0 | max_consec=13 | avg_clip50=31.66 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           |                | 32.1    |          |               |                  |
| clipped_episode_r... |                | 28.5    |          |               |                  |
| consecutive_clipp... |                | 2       |          |               |                  |
| max_consecutive_c... |                | 13      |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          |                | 914     |          |               |                  |
| ep_rew_mean          |                | 311     |          |               |                  |
| exploration rate     |                | 0.613   |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             |                | 19100   |          |               |                  |
| fps                  |                | 216     |          |               |                  |
| time_elapsed         |                | 5369    |          |               |                  |
| total timesteps      |                | 1162962 |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        |                | 0.0001  |          |               |                  |
| loss                 |                | 0.0656  |          |               |                  |
| n_updates            |                | 278240  |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=6,420             | STEP=1,164,706 | clip=27 | consec=1 | max_consec=13 | avg_clip50=30.88 |
| EP=6,430             | STEP=1,166,954 | clip=38 | consec=7 | max_consec=13 | avg_clip50=31.19 |
| EP=6,440             | STEP=1,168,942 | clip=32 | consec=1 | max_consec=13 | avg_clip50=29.19 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           |                | 29      |          |               |                  |
| clipped_episode_r... |                | 27.5    |          |               |                  |
| consecutive_clipp... |                | 2       |          |               |                  |
| max_consecutive_c... |                | 13      |          |               |                  |
| rollout/             |                |         |          |               |                  |
| ep_len_mean          |                | 915     |          |               |                  |
| ep_rew_mean          |                | 309     |          |               |                  |
| exploration rate     |                | 0.61    |          |               |                  |
| time/                |                |         |          |               |                  |
| episodes             |                | 19200   |          |               |                  |
| fps                  |                | 216     |          |               |                  |
| time_elapsed         |                | 5407    |          |               |                  |
| total timesteps      |                | 1170281 |          |               |                  |
| train/               |                |         |          |               |                  |
| learning_rate        |                | 0.0001  |          |               |                  |
| loss                 |                | 0.0346  |          |               |                  |
| n_updates            |                | 280070  |          |               |                  |
| -----                |                |         |          |               |                  |
| EP=6,450             | STEP=1,170,857 | clip=4  | consec=0 | max_consec=13 | avg_clip50=28.99 |
| EP=6,460             | STEP=1,173,353 | clip=36 | consec=2 | max_consec=13 | avg_clip50=30.90 |
| EP=6,470             | STEP=1,175,315 | clip=22 | consec=1 | max_consec=13 | avg_clip50=31.03 |
| EP=6,480             | STEP=1,177,515 | clip=14 | consec=0 | max_consec=13 | avg_clip50=30.58 |
| -----                |                |         |          |               |                  |
| custom/              |                |         |          |               |                  |
| avg_clip50           |                | 30.7    |          |               |                  |
| clipped episode r... |                | 16      |          |               |                  |

|                      |         |
|----------------------|---------|
| consecutive_clipp... | 0       |
| max_consecutive_c... | 13      |
| rollout/             |         |
| ep_len_mean          | 914     |
| ep_rew_mean          | 307     |
| exploration rate     | 0.608   |
| time/                |         |
| episodes             | 19300   |
| fps                  | 216     |
| time_elapsed         | 5444    |
| total timesteps      | 1177696 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.122   |
| n_updates            | 281923  |

EP=6,490 | STEP=1,179,858 | clip=54 | consec=5 | max\_consec=13 | avg\_clip50=31.89  
EP=6,500 | STEP=1,182,209 | clip=42 | consec=8 | max\_consec=13 | avg\_clip50=34.10  
EP=6,510 | STEP=1,184,279 | clip=34 | consec=1 | max\_consec=13 | avg\_clip50=31.85

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 31.7    |
| clipped_episode_r... | 44.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 13      |
| rollout/             |         |
| ep_len_mean          | 926     |
| ep_rew_mean          | 320     |
| exploration rate     | 0.605   |
| time/                |         |
| episodes             | 19400   |
| fps                  | 216     |
| time_elapsed         | 5485    |
| total timesteps      | 1185618 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.13    |
| n_updates            | 283904  |

EP=6,520 | STEP=1,186,674 | clip=64 | consec=1 | max\_consec=13 | avg\_clip50=33.31  
EP=6,530 | STEP=1,188,809 | clip=36 | consec=11 | max\_consec=13 | avg\_clip50=34.9  
2  
EP=6,540 | STEP=1,190,844 | clip=16 | consec=0 | max\_consec=16 | avg\_clip50=34.18

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34.1    |
| clipped_episode_r... | 32.5    |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 928     |
| ep_rew_mean          | 338     |
| exploration rate     | 0.603   |
| time/                |         |
| episodes             | 19500   |
| fps                  | 215     |
| time_elapsed         | 5524    |
| total timesteps      | 1193055 |
| train/               |         |
| learning_rate        | 0.0001  |



|          |                      |         |          |               |                  |  |
|----------|----------------------|---------|----------|---------------|------------------|--|
|          | loss                 | 0.112   |          |               |                  |  |
|          | n_updates            | 285763  |          |               |                  |  |
| -----    |                      |         |          |               |                  |  |
| EP=6,550 | STEP=1,193,133       | clip=54 | consec=3 | max_consec=16 | avg_clip50=34.38 |  |
| EP=6,560 | STEP=1,195,186       | clip=12 | consec=0 | max_consec=16 | avg_clip50=34.35 |  |
| EP=6,570 | STEP=1,197,541       | clip=30 | consec=4 | max_consec=16 | avg_clip50=33.95 |  |
| EP=6,580 | STEP=1,200,068       | clip=32 | consec=9 | max_consec=16 | avg_clip50=35.35 |  |
| -----    |                      |         |          |               |                  |  |
|          | custom/              |         |          |               |                  |  |
|          | avg_clip50           | 34.5    |          |               |                  |  |
|          | clipped_episode_r... | 21.5    |          |               |                  |  |
|          | consecutive_clipp... | 2       |          |               |                  |  |
|          | max_consecutive_c... | 16      |          |               |                  |  |
|          | rollout/             |         |          |               |                  |  |
|          | ep_len_mean          | 942     |          |               |                  |  |
|          | ep_rew_mean          | 349     |          |               |                  |  |
|          | exploration rate     | 0.6     |          |               |                  |  |
|          | time/                |         |          |               |                  |  |
|          | episodes             | 19600   |          |               |                  |  |
|          | fps                  | 215     |          |               |                  |  |
|          | time_elapsed         | 5562    |          |               |                  |  |
|          | total timesteps      | 1200689 |          |               |                  |  |
|          | train/               |         |          |               |                  |  |
|          | learning_rate        | 0.0001  |          |               |                  |  |
|          | loss                 | 0.0911  |          |               |                  |  |
|          | n_updates            | 287672  |          |               |                  |  |
| -----    |                      |         |          |               |                  |  |
| EP=6,590 | STEP=1,202,110       | clip=24 | consec=9 | max_consec=16 | avg_clip50=34.56 |  |
| EP=6,600 | STEP=1,204,477       | clip=33 | consec=1 | max_consec=16 | avg_clip50=33.49 |  |
| EP=6,610 | STEP=1,206,367       | clip=56 | consec=1 | max_consec=16 | avg_clip50=34.04 |  |
| -----    |                      |         |          |               |                  |  |
|          | custom/              |         |          |               |                  |  |
|          | avg_clip50           | 35.2    |          |               |                  |  |
|          | clipped_episode_r... | 50      |          |               |                  |  |
|          | consecutive_clipp... | 3       |          |               |                  |  |
|          | max_consecutive_c... | 16      |          |               |                  |  |
|          | rollout/             |         |          |               |                  |  |
|          | ep_len_mean          | 930     |          |               |                  |  |
|          | ep_rew_mean          | 340     |          |               |                  |  |
|          | exploration rate     | 0.598   |          |               |                  |  |
|          | time/                |         |          |               |                  |  |
|          | episodes             | 19700   |          |               |                  |  |
|          | fps                  | 215     |          |               |                  |  |
|          | time_elapsed         | 5600    |          |               |                  |  |
|          | total timesteps      | 1208119 |          |               |                  |  |
|          | train/               |         |          |               |                  |  |
|          | learning_rate        | 0.0001  |          |               |                  |  |
|          | loss                 | 0.0824  |          |               |                  |  |
|          | n_updates            | 289529  |          |               |                  |  |
| -----    |                      |         |          |               |                  |  |
| EP=6,620 | STEP=1,208,905       | clip=34 | consec=2 | max_consec=16 | avg_clip50=33.95 |  |
| EP=6,630 | STEP=1,211,234       | clip=55 | consec=4 | max_consec=16 | avg_clip50=32.47 |  |
| EP=6,640 | STEP=1,213,621       | clip=44 | consec=8 | max_consec=16 | avg_clip50=35.13 |  |
| EP=6,650 | STEP=1,215,652       | clip=12 | consec=0 | max_consec=16 | avg_clip50=34.22 |  |
| -----    |                      |         |          |               |                  |  |
|          | custom/              |         |          |               |                  |  |
|          | avg_clip50           | 34.2    |          |               |                  |  |
|          | clipped_episode_r... | 12.5    |          |               |                  |  |
|          | consecutive_clipp... | 0       |          |               |                  |  |
|          | max_consecutive_c... | 16      |          |               |                  |  |

|                  |         |
|------------------|---------|
| rollout/         |         |
| ep_len_mean      | 946     |
| ep_rew_mean      | 339     |
| exploration rate | 0.595   |
| time/            |         |
| episodes         | 19800   |
| fps              | 215     |
| time_elapsed     | 5639    |
| total timesteps  | 1215829 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0539  |
| n_updates        | 291457  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,660 | STEP=1,218,264 | clip=45 | consec=2 | max_consec=16 | avg_clip50=37.09 |
| EP=6,670 | STEP=1,220,446 | clip=49 | consec=6 | max_consec=16 | avg_clip50=36.39 |
| EP=6,680 | STEP=1,222,452 | clip=18 | consec=0 | max_consec=16 | avg_clip50=34.58 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34.6    |
| clipped_episode_r... | 15.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 943     |
| ep_rew_mean          | 338     |
| exploration rate     | 0.593   |
| time/                |         |
| episodes             | 19900   |
| fps                  | 215     |
| time_elapsed         | 5676    |
| total timesteps      | 1223277 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0259  |
| n_updates            | 293319  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,690 | STEP=1,224,641 | clip=54 | consec=1 | max_consec=16 | avg_clip50=33.12 |
| EP=6,700 | STEP=1,226,677 | clip=46 | consec=3 | max_consec=16 | avg_clip50=34.01 |
| EP=6,710 | STEP=1,228,714 | clip=28 | consec=1 | max_consec=16 | avg_clip50=30.40 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 29.8    |
| clipped_episode_r... | 19.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 921     |
| ep_rew_mean          | 327     |
| exploration rate     | 0.59    |
| time/                |         |
| episodes             | 20000   |
| fps                  | 215     |
| time_elapsed         | 5712    |
| total timesteps      | 1230250 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0297  |
| n_updates            | 295062  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,720 | STEP=1,230,951 | clip=16 | consec=0 | max_consec=16 | avg_clip50=30.32 |
| EP=6,730 | STEP=1,233,365 | clip=26 | consec=3 | max_consec=16 | avg_clip50=31.98 |
| EP=6,740 | STEP=1,235,414 | clip=44 | consec=1 | max_consec=16 | avg_clip50=31.29 |
| EP=6,750 | STEP=1,237,477 | clip=32 | consec=1 | max_consec=16 | avg_clip50=30.35 |

|                      |         |
|----------------------|---------|
| -----                |         |
| custom/              |         |
| avg_clip50           | 30.2    |
| clipped_episode_r... | 11      |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 914     |
| ep_rew_mean          | 317     |
| exploration rate     | 0.588   |
| time/                |         |
| episodes             | 20100   |
| fps                  | 215     |
| time_elapsed         | 5750    |
| total timesteps      | 1237801 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0628  |
| n_updates            | 296950  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,760 | STEP=1,239,821 | clip=48 | consec=2 | max_consec=16 | avg_clip50=31.72 |
| EP=6,770 | STEP=1,242,231 | clip=17 | consec=0 | max_consec=16 | avg_clip50=32.56 |
| EP=6,780 | STEP=1,244,464 | clip=30 | consec=2 | max_consec=16 | avg_clip50=31.63 |

|                      |         |
|----------------------|---------|
| -----                |         |
| custom/              |         |
| avg_clip50           | 31      |
| clipped_episode_r... | 60.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 927     |
| ep_rew_mean          | 319     |
| exploration rate     | 0.585   |
| time/                |         |
| episodes             | 20200   |
| fps                  | 215     |
| time_elapsed         | 5791    |
| total timesteps      | 1245539 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0297  |
| n_updates            | 298884  |

|                                                   |                |         |          |               |                  |
|---------------------------------------------------|----------------|---------|----------|---------------|------------------|
| EP=6,790                                          | STEP=1,246,601 | clip=49 | consec=2 | max_consec=16 | avg_clip50=31.34 |
| EP=6,800                                          | STEP=1,249,430 | clip=32 | consec=3 | max_consec=16 | avg_clip50=34.54 |
| [checkpoint periodico] guardado a 1,250,000 steps |                |         |          |               |                  |
| EP=6,810                                          | STEP=1,251,506 | clip=16 | consec=0 | max_consec=16 | avg_clip50=33.87 |

|                      |      |
|----------------------|------|
| -----                |      |
| custom/              |      |
| avg_clip50           | 33.5 |
| clipped_episode_r... | 21.5 |
| consecutive_clipp... | 2    |
| max_consecutive_c... | 16   |
| rollout/             |      |
| ep_len_mean          | 964  |
| ep_rew_mean          | 331  |

|                  |         |
|------------------|---------|
| exploration rate | 0.583   |
| time/            |         |
| episodes         | 20300   |
| fps              | 214     |
| time_elapsed     | 5834    |
| total timesteps  | 1253512 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0737  |
| n_updates        | 300877  |

EP=6,820 | STEP=1,253,690 | clip=18 | consec=0 | max\_consec=16 | avg\_clip50=32.75  
EP=6,830 | STEP=1,256,467 | clip=84 | consec=1 | max\_consec=16 | avg\_clip50=36.33  
EP=6,840 | STEP=1,258,501 | clip=68 | consec=2 | max\_consec=16 | avg\_clip50=36.65  
EP=6,850 | STEP=1,261,076 | clip=44 | consec=7 | max\_consec=16 | avg\_clip50=36.09

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 35.9    |
| clipped_episode_r... | 39.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 987     |
| ep_rew_mean          | 350     |
| exploration rate     | 0.58    |
| time/                |         |
| episodes             | 20400   |
| fps                  | 214     |
| time_elapsed         | 5875    |
| total timesteps      | 1261666 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0467  |
| n_updates            | 302916  |

EP=6,860 | STEP=1,263,159 | clip=34 | consec=1 | max\_consec=16 | avg\_clip50=35.85  
EP=6,870 | STEP=1,265,036 | clip=22 | consec=2 | max\_consec=16 | avg\_clip50=35.09  
EP=6,880 | STEP=1,267,185 | clip=20 | consec=0 | max\_consec=16 | avg\_clip50=31.93

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 31.3    |
| clipped_episode_r... | 12.5    |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 943     |
| ep_rew_mean          | 336     |
| exploration rate     | 0.578   |
| time/                |         |
| episodes             | 20500   |
| fps                  | 214     |
| time_elapsed         | 5908    |
| total timesteps      | 1268340 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0336  |
| n_updates            | 304584  |

EP=6,890 | STEP=1,269,027 | clip=32 | consec=3 | max\_consec=16 | avg\_clip50=30.99  
EP=6,900 | STEP=1,271,171 | clip=10 | consec=0 | max\_consec=16 | avg\_clip50=30.06

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,910 | STEP=1,273,311 | clip=27 | consec=3 | max_consec=16 | avg_clip50=29.23 |
| EP=6,920 | STEP=1,275,578 | clip=54 | consec=3 | max_consec=16 | avg_clip50=31.74 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 32.4    |
| clipped_episode_r... | 53      |
| consecutive_clipp... | 4       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 923     |
| ep_rew_mean          | 336     |
| exploration rate     | 0.575   |
| time/                |         |
| episodes             | 20600   |
| fps                  | 214     |
| time_elapsed         | 5946    |
| total timesteps      | 1275965 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0604  |
| n_updates            | 306491  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,930 | STEP=1,277,526 | clip=32 | consec=1 | max_consec=16 | avg_clip50=30.92 |
| EP=6,940 | STEP=1,279,656 | clip=14 | consec=0 | max_consec=16 | avg_clip50=31.50 |
| EP=6,950 | STEP=1,281,639 | clip=11 | consec=0 | max_consec=16 | avg_clip50=29.86 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 30.7    |
| clipped_episode_r... | 47      |
| consecutive_clipp... | 5       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 869     |
| ep_rew_mean          | 302     |
| exploration rate     | 0.573   |
| time/                |         |
| episodes             | 20700   |
| fps                  | 214     |
| time_elapsed         | 5980    |
| total timesteps      | 1282720 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0711  |
| n_updates            | 308179  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=6,960 | STEP=1,284,004 | clip=54 | consec=3 | max_consec=16 | avg_clip50=32.15 |
| EP=6,970 | STEP=1,286,001 | clip=44 | consec=3 | max_consec=16 | avg_clip50=31.62 |
| EP=6,980 | STEP=1,288,193 | clip=12 | consec=0 | max_consec=16 | avg_clip50=33.25 |

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 34.8  |
| clipped_episode_r... | 43.5  |
| consecutive_clipp... | 1     |
| max_consecutive_c... | 16    |
| rollout/             |       |
| ep_len_mean          | 912   |
| ep_rew_mean          | 333   |
| exploration rate     | 0.57  |
| time/                |       |
| episodes             | 20800 |

|          |                      |                |         |                  |
|----------|----------------------|----------------|---------|------------------|
|          | fps                  |                | 214     |                  |
|          | time_elapsed         |                | 6019    |                  |
|          | total timesteps      |                | 1290475 |                  |
|          | train/               |                |         |                  |
|          | learning_rate        |                | 0.0001  |                  |
|          | loss                 |                | 0.102   |                  |
|          | n_updates            |                | 310118  |                  |
| -----    |                      |                |         |                  |
| EP=6,990 |                      | STEP=1,290,661 |         | clip=25          |
|          |                      |                |         | consec=2         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=35.03 |
| EP=7,000 |                      | STEP=1,292,471 |         | clip=14          |
|          |                      |                |         | consec=0         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=34.42 |
| EP=7,010 |                      | STEP=1,294,805 |         | clip=11          |
|          |                      |                |         | consec=0         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=33.48 |
| EP=7,020 |                      | STEP=1,297,246 |         | clip=36          |
|          |                      |                |         | consec=6         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=33.33 |
| -----    |                      |                |         |                  |
|          | custom/              |                |         |                  |
|          | avg_clip50           |                | 32.6    |                  |
|          | clipped_episode_r... |                | 16.5    |                  |
|          | consecutive_clipp... |                | 0       |                  |
|          | max_consecutive_c... |                | 16      |                  |
|          | rollout/             |                |         |                  |
|          | ep_len_mean          |                | 909     |                  |
|          | ep_rew_mean          |                | 321     |                  |
|          | exploration rate     |                | 0.568   |                  |
|          | time/                |                |         |                  |
|          | episodes             |                | 20900   |                  |
|          | fps                  |                | 214     |                  |
|          | time_elapsed         |                | 6056    |                  |
|          | total timesteps      |                | 1297707 |                  |
|          | train/               |                |         |                  |
|          | learning_rate        |                | 0.0001  |                  |
|          | loss                 |                | 0.112   |                  |
|          | n_updates            |                | 311926  |                  |
| -----    |                      |                |         |                  |
| EP=7,030 |                      | STEP=1,299,220 |         | clip=10          |
|          |                      |                |         | consec=0         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=31.63 |
| EP=7,040 |                      | STEP=1,301,556 |         | clip=14          |
|          |                      |                |         | consec=0         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=31.23 |
| EP=7,050 |                      | STEP=1,303,744 |         | clip=41          |
|          |                      |                |         | consec=1         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=33.33 |
| -----    |                      |                |         |                  |
|          | custom/              |                |         |                  |
|          | avg_clip50           |                | 32.8    |                  |
|          | clipped_episode_r... |                | 30.5    |                  |
|          | consecutive_clipp... |                | 6       |                  |
|          | max_consecutive_c... |                | 16      |                  |
|          | rollout/             |                |         |                  |
|          | ep_len_mean          |                | 927     |                  |
|          | ep_rew_mean          |                | 338     |                  |
|          | exploration rate     |                | 0.565   |                  |
|          | time/                |                |         |                  |
|          | episodes             |                | 21000   |                  |
|          | fps                  |                | 214     |                  |
|          | time_elapsed         |                | 6094    |                  |
|          | total timesteps      |                | 1305217 |                  |
|          | train/               |                |         |                  |
|          | learning_rate        |                | 0.0001  |                  |
|          | loss                 |                | 0.0463  |                  |
|          | n_updates            |                | 313804  |                  |
| -----    |                      |                |         |                  |
| EP=7,060 |                      | STEP=1,305,870 |         | clip=32          |
|          |                      |                |         | consec=9         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=32.96 |
| EP=7,070 |                      | STEP=1,307,519 |         | clip=8           |
|          |                      |                |         | consec=0         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=30.30 |
| EP=7,080 |                      | STEP=1,309,492 |         | clip=8           |
|          |                      |                |         | consec=0         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=29.71 |
| EP=7,090 |                      | STEP=1,311,613 |         | clip=30          |
|          |                      |                |         | consec=1         |
|          |                      |                |         | max_consec=16    |
|          |                      |                |         | avg_clip50=28.71 |
| -----    |                      |                |         |                  |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 27.5    |
| clipped_episode_r... | 33.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 884     |
| ep_rew_mean          | 301     |
| exploration rate     | 0.563   |
| time/                |         |
| episodes             | 21100   |
| fps                  | 214     |
| time_elapsed         | 6130    |
| total timesteps      | 1312075 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0559  |
| n_updates            | 315518  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,100 | STEP=1,313,804 | clip=32 | consec=1 | max_consec=16 | avg_clip50=26.98 |
| EP=7,110 | STEP=1,315,673 | clip=35 | consec=1 | max_consec=16 | avg_clip50=26.69 |
| EP=7,120 | STEP=1,318,185 | clip=74 | consec=8 | max_consec=16 | avg_clip50=30.08 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 32.2    |
| clipped_episode_r... | 51      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 890     |
| ep_rew_mean          | 309     |
| exploration rate     | 0.561   |
| time/                |         |
| episodes             | 21200   |
| fps                  | 213     |
| time_elapsed         | 6169    |
| total timesteps      | 1319734 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0504  |
| n_updates            | 317433  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,130 | STEP=1,320,681 | clip=22 | consec=6 | max_consec=16 | avg_clip50=33.22 |
| EP=7,140 | STEP=1,322,936 | clip=24 | consec=7 | max_consec=16 | avg_clip50=34.78 |
| EP=7,150 | STEP=1,325,666 | clip=41 | consec=1 | max_consec=16 | avg_clip50=38.11 |
| EP=7,160 | STEP=1,327,370 | clip=21 | consec=1 | max_consec=16 | avg_clip50=36.64 |

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|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 36.6  |
| clipped_episode_r... | 21    |
| consecutive_clipp... | 1     |
| max_consecutive_c... | 16    |
| rollout/             |       |
| ep_len_mean          | 902   |
| ep_rew_mean          | 317   |
| exploration rate     | 0.558 |
| time/                |       |
| episodes             | 21300 |
| fps                  | 213   |
| time_elapsed         | 6207  |

|                 |         |
|-----------------|---------|
| total timesteps | 1327370 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.397   |
| n_updates       | 319342  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,170 | STEP=1,329,640 | clip=7  | consec=0 | max_consec=16 | avg_clip50=35.17 |
| EP=7,180 | STEP=1,331,886 | clip=27 | consec=1 | max_consec=16 | avg_clip50=33.26 |
| EP=7,190 | STEP=1,334,387 | clip=11 | consec=0 | max_consec=16 | avg_clip50=34.02 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 34.5    |
| clipped_episode_r... | 100     |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 965     |
| ep_rew_mean          | 356     |
| exploration rate     | 0.555   |
| time/                |         |
| episodes             | 21400   |
| fps                  | 213     |
| time_elapsed         | 6252    |
| total timesteps      | 1335830 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.304   |
| n_updates            | 321457  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,200 | STEP=1,337,010 | clip=42 | consec=3 | max_consec=16 | avg_clip50=34.06 |
| EP=7,210 | STEP=1,339,295 | clip=50 | consec=3 | max_consec=16 | avg_clip50=36.66 |
| EP=7,220 | STEP=1,341,719 | clip=40 | consec=6 | max_consec=16 | avg_clip50=38.13 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 37.3    |
| clipped_episode_r... | 33.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 973     |
| ep_rew_mean          | 357     |
| exploration rate     | 0.553   |
| time/                |         |
| episodes             | 21500   |
| fps                  | 213     |
| time_elapsed         | 6292    |
| total timesteps      | 1343436 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.169   |
| n_updates            | 323358  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,230 | STEP=1,343,817 | clip=50 | consec=1 | max_consec=16 | avg_clip50=37.85 |
| EP=7,240 | STEP=1,346,488 | clip=50 | consec=4 | max_consec=16 | avg_clip50=38.51 |
| EP=7,250 | STEP=1,348,718 | clip=30 | consec=2 | max_consec=16 | avg_clip50=36.63 |
| EP=7,260 | STEP=1,351,036 | clip=29 | consec=5 | max_consec=16 | avg_clip50=37.86 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 38.4 |
| clipped_episode_r... | 52   |



|                      |         |
|----------------------|---------|
| consecutive_clipp... | 2       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 988     |
| ep_rew_mean          | 373     |
| exploration rate     | 0.55    |
| time/                |         |
| episodes             | 21600   |
| fps                  | 213     |
| time_elapsed         | 6336    |
| total timesteps      | 1351883 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0461  |
| n_updates            | 325470  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,270 | STEP=1,353,221 | clip=50 | consec=1 | max_consec=16 | avg_clip50=37.30 |
| EP=7,280 | STEP=1,355,655 | clip=52 | consec=4 | max_consec=16 | avg_clip50=39.37 |
| EP=7,290 | STEP=1,357,725 | clip=33 | consec=1 | max_consec=16 | avg_clip50=36.96 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 37.2    |
| clipped_episode_r... | 26      |
| consecutive_clipp... | 8       |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 971     |
| ep_rew_mean          | 372     |
| exploration rate     | 0.547   |
| time/                |         |
| episodes             | 21700   |
| fps                  | 213     |
| time_elapsed         | 6376    |
| total timesteps      | 1359578 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0998  |
| n_updates            | 327394  |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=7,300 | STEP=1,360,159 | clip=63 | consec=11 | max_consec=16 | avg_clip50=38.13 |
| EP=7,310 | STEP=1,362,378 | clip=41 | consec=8  | max_consec=16 | avg_clip50=37.41 |
| EP=7,320 | STEP=1,364,701 | clip=57 | consec=3  | max_consec=16 | avg_clip50=37.68 |
| EP=7,330 | STEP=1,367,384 | clip=31 | consec=13 | max_consec=16 | avg_clip50=38.38 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 38.4    |
| clipped_episode_r... | 22      |
| consecutive_clipp... | 14      |
| max_consecutive_c... | 16      |
| rollout/             |         |
| ep_len_mean          | 985     |
| ep_rew_mean          | 386     |
| exploration rate     | 0.545   |
| time/                |         |
| episodes             | 21800   |
| fps                  | 213     |
| time_elapsed         | 6417    |
| total timesteps      | 1367582 |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0508 |
| n_updates     | 329395 |

EP=7,340 | STEP=1,370,016 | clip=52 | consec=23 | max\_consec=23 | avg\_clip50=40.32

EP=7,350 | STEP=1,372,473 | clip=17 | consec=0 | max\_consec=28 | avg\_clip50=39.83

EP=7,360 | STEP=1,374,390 | clip=10 | consec=0 | max\_consec=28 | avg\_clip50=38.34

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 39.5    |
| clipped_episode_r... | 68.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 985     |
| ep_rew_mean          | 381     |
| exploration rate     | 0.542   |
| time/                |         |
| episodes             | 21900   |
| fps                  | 213     |
| time_elapsed         | 6459    |
| total timesteps      | 1375874 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.319   |
| n_updates            | 331468  |

EP=7,370 | STEP=1,377,161 | clip=80 | consec=3 | max\_consec=28 | avg\_clip50=39.86

EP=7,380 | STEP=1,379,674 | clip=36 | consec=5 | max\_consec=28 | avg\_clip50=39.45

EP=7,390 | STEP=1,381,593 | clip=22 | consec=3 | max\_consec=28 | avg\_clip50=37.15

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 36      |
| clipped_episode_r... | 45.5    |
| consecutive_clipp... | 6       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 991     |
| ep_rew_mean          | 383     |
| exploration rate     | 0.539   |
| time/                |         |
| episodes             | 22000   |
| fps                  | 212     |
| time_elapsed         | 6498    |
| total timesteps      | 1383656 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0636  |
| n_updates            | 333413  |

EP=7,400 | STEP=1,383,964 | clip=42 | consec=8 | max\_consec=28 | avg\_clip50=37.40

EP=7,410 | STEP=1,386,078 | clip=42 | consec=1 | max\_consec=28 | avg\_clip50=38.26

EP=7,420 | STEP=1,388,085 | clip=46 | consec=1 | max\_consec=28 | avg\_clip50=35.90

EP=7,430 | STEP=1,390,335 | clip=12 | consec=0 | max\_consec=28 | avg\_clip50=34.52

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 35.5 |
| clipped_episode_r... | 77.5 |

|                      |         |
|----------------------|---------|
| consecutive_clipp... | 2       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 971     |
| ep_rew_mean          | 376     |
| exploration rate     | 0.537   |
| time/                |         |
| episodes             | 22100   |
| fps                  | 212     |
| time_elapsed         | 6537    |
| total timesteps      | 1391201 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0431  |
| n_updates            | 335300  |

EP=7,440 | STEP=1,392,754 | clip=26 | consec=6 | max\_consec=28 | avg\_clip50=36.53  
EP=7,450 | STEP=1,395,018 | clip=40 | consec=5 | max\_consec=28 | avg\_clip50=36.39  
EP=7,460 | STEP=1,397,445 | clip=24 | consec=1 | max\_consec=28 | avg\_clip50=36.64

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 38.2    |
| clipped_episode_r... | 28      |
| consecutive_clipp... | 7       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 957     |
| ep_rew_mean          | 374     |
| exploration rate     | 0.534   |
| time/                |         |
| episodes             | 22200   |
| fps                  | 212     |
| time_elapsed         | 6577    |
| total timesteps      | 1398969 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0354  |
| n_updates            | 337242  |

EP=7,470 | STEP=1,399,966 | clip=24 | consec=11 | max\_consec=28 | avg\_clip50=38.84  
EP=7,480 | STEP=1,402,393 | clip=48 | consec=3 | max\_consec=28 | avg\_clip50=39.92  
EP=7,490 | STEP=1,404,729 | clip=60 | consec=2 | max\_consec=28 | avg\_clip50=38.36  
EP=7,500 | STEP=1,406,758 | clip=5 | consec=0 | max\_consec=28 | avg\_clip50=37.79

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 37.8    |
| clipped_episode_r... | 5       |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 956     |
| ep_rew_mean          | 371     |
| exploration rate     | 0.532   |
| time/                |         |
| episodes             | 22300   |
| fps                  | 212     |
| time_elapsed         | 6617    |
| total timesteps      | 1406758 |
| train/               |         |

|               |        |
|---------------|--------|
| learning_rate | 0.0001 |
| loss          | 0.0278 |
| n_updates     | 339189 |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,510 | STEP=1,409,273 | clip=74 | consec=3 | max_consec=28 | avg_clip50=39.63 |
| EP=7,520 | STEP=1,411,393 | clip=36 | consec=6 | max_consec=28 | avg_clip50=36.58 |
| EP=7,530 | STEP=1,413,700 | clip=53 | consec=5 | max_consec=28 | avg_clip50=35.84 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 35.3    |
| clipped_episode_r... | 27      |
| consecutive_clipp... | 8       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 968     |
| ep_rew_mean          | 367     |
| exploration rate     | 0.529   |
| time/                |         |
| episodes             | 22400   |
| fps                  | 212     |
| time_elapsed         | 6656    |
| total timesteps      | 1414659 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.033   |
| n_updates            | 341164  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,540 | STEP=1,416,038 | clip=16 | consec=0 | max_consec=28 | avg_clip50=35.78 |
| EP=7,550 | STEP=1,418,444 | clip=27 | consec=1 | max_consec=28 | avg_clip50=36.80 |
| EP=7,560 | STEP=1,421,061 | clip=28 | consec=7 | max_consec=28 | avg_clip50=35.47 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 36      |
| clipped_episode_r... | 22.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 979     |
| ep_rew_mean          | 364     |
| exploration rate     | 0.526   |
| time/                |         |
| episodes             | 22500   |
| fps                  | 212     |
| time_elapsed         | 6697    |
| total timesteps      | 1422700 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0583  |
| n_updates            | 343174  |

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,570 | STEP=1,423,196 | clip=22 | consec=2 | max_consec=28 | avg_clip50=35.51 |
| EP=7,580 | STEP=1,425,519 | clip=41 | consec=3 | max_consec=28 | avg_clip50=35.46 |
| EP=7,590 | STEP=1,427,952 | clip=44 | consec=2 | max_consec=28 | avg_clip50=36.20 |
| EP=7,600 | STEP=1,430,193 | clip=16 | consec=0 | max_consec=28 | avg_clip50=35.93 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 35.6 |
| clipped_episode_r... | 46   |
| consecutive_clipp... | 2    |
| max_consecutive_c... | 28   |

|                  |         |
|------------------|---------|
| rollout/         |         |
| ep_len_mean      | 982     |
| ep_rew_mean      | 364     |
| exploration rate | 0.524   |
| time/            |         |
| episodes         | 22600   |
| fps              | 212     |
| time_elapsed     | 6739    |
| total timesteps  | 1430746 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0629  |
| n_updates        | 345186  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,610 | STEP=1,432,471 | clip=32 | consec=1 | max_consec=28 | avg_clip50=35.37 |
| EP=7,620 | STEP=1,435,117 | clip=52 | consec=6 | max_consec=28 | avg_clip50=37.91 |
| EP=7,630 | STEP=1,437,383 | clip=58 | consec=2 | max_consec=28 | avg_clip50=38.80 |

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|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 38.7    |
| clipped_episode_r... | 49      |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 992     |
| ep_rew_mean          | 375     |
| exploration rate     | 0.521   |
| time/                |         |
| episodes             | 22700   |
| fps                  | 212     |
| time_elapsed         | 6781    |
| total timesteps      | 1438895 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.211   |
| n_updates            | 347223  |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=7,640 | STEP=1,439,660 | clip=34 | consec=5  | max_consec=28 | avg_clip50=38.73 |
| EP=7,650 | STEP=1,442,400 | clip=54 | consec=15 | max_consec=28 | avg_clip50=40.00 |
| EP=7,660 | STEP=1,445,182 | clip=18 | consec=0  | max_consec=28 | avg_clip50=41.97 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 40.8     |
| clipped_episode_r... | 57.5     |
| consecutive_clipp... | 4        |
| max_consecutive_c... | 28       |
| rollout/             |          |
| ep_len_mean          | 1.01e+03 |
| ep_rew_mean          | 391      |
| exploration rate     | 0.518    |
| time/                |          |
| episodes             | 22800    |
| fps                  | 212      |
| time_elapsed         | 6823     |
| total timesteps      | 1447202  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0375   |
| n_updates            | 349300   |

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EP=7,670 | STEP=1,447,312 | clip=18 | consec=0 | max\_consec=28 | avg\_clip50=40.13  
EP=7,680 | STEP=1,449,728 | clip=52 | consec=3 | max\_consec=28 | avg\_clip50=39.35  
EP=7,690 | STEP=1,452,532 | clip=50 | consec=13 | max\_consec=28 | avg\_clip50=41.0  
4  
EP=7,700 | STEP=1,454,876 | clip=34 | consec=5 | max\_consec=28 | avg\_clip50=38.77  
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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.3     |
| clipped_episode_r... | 57       |
| consecutive_clipp... | 9        |
| max_consecutive_c... | 28       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 398      |
| exploration rate     | 0.515    |
| time/                |          |
| episodes             | 22900    |
| fps                  | 212      |
| time_elapsed         | 6867     |
| total timesteps      | 1456020  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.474    |
| n_updates            | 351504   |

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EP=7,710 | STEP=1,457,136 | clip=10 | consec=0 | max\_consec=28 | avg\_clip50=37.42  
EP=7,720 | STEP=1,459,549 | clip=36 | consec=1 | max\_consec=28 | avg\_clip50=38.25  
EP=7,730 | STEP=1,461,971 | clip=22 | consec=2 | max\_consec=28 | avg\_clip50=38.74  
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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.1     |
| clipped_episode_r... | 47       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 28       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 392      |
| exploration rate     | 0.513    |
| time/                |          |
| episodes             | 23000    |
| fps                  | 211      |
| time_elapsed         | 6907     |
| total timesteps      | 1463886  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0521   |
| n_updates            | 353471   |

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EP=7,740 | STEP=1,464,151 | clip=31 | consec=5 | max\_consec=28 | avg\_clip50=37.63  
EP=7,750 | STEP=1,466,539 | clip=54 | consec=3 | max\_consec=28 | avg\_clip50=38.39  
EP=7,760 | STEP=1,468,802 | clip=22 | consec=13 | max\_consec=28 | avg\_clip50=38.4  
3  
EP=7,770 | STEP=1,470,776 | clip=46 | consec=2 | max\_consec=28 | avg\_clip50=36.61  
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|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 36   |
| clipped_episode_r... | 38.5 |
| consecutive_clipp... | 5    |
| max_consecutive_c... | 28   |

|                  |         |
|------------------|---------|
| rollout/         |         |
| ep_len_mean      | 984     |
| ep_rew_mean      | 376     |
| exploration rate | 0.51    |
| time/            |         |
| episodes         | 23100   |
| fps              | 211     |
| time_elapsed     | 6947    |
| total timesteps  | 1471650 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0809  |
| n_updates        | 355412  |

EP=7,780 | STEP=1,473,072 | clip=12 | consec=0 | max\_consec=28 | avg\_clip50=35.62  
EP=7,790 | STEP=1,475,729 | clip=64 | consec=10 | max\_consec=28 | avg\_clip50=38.0  
5  
EP=7,800 | STEP=1,478,309 | clip=39 | consec=6 | max\_consec=28 | avg\_clip50=37.90

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 37.6    |
| clipped_episode_r... | 26.5    |
| consecutive_clipp... | 4       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 985     |
| ep_rew_mean          | 380     |
| exploration rate     | 0.507   |
| time/                |         |
| episodes             | 23200   |
| fps                  | 211     |
| time_elapsed         | 6990    |
| total timesteps      | 1480224 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.126   |
| n_updates            | 357555  |

EP=7,810 | STEP=1,480,646 | clip=24 | consec=1 | max\_consec=28 | avg\_clip50=37.48  
EP=7,820 | STEP=1,483,268 | clip=38 | consec=11 | max\_consec=28 | avg\_clip50=40.6  
6  
EP=7,830 | STEP=1,485,479 | clip=16 | consec=0 | max\_consec=28 | avg\_clip50=40.40  
EP=7,840 | STEP=1,487,702 | clip=12 | consec=0 | max\_consec=28 | avg\_clip50=38.19

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 37.7    |
| clipped_episode_r... | 31      |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 982     |
| ep_rew_mean          | 379     |
| exploration rate     | 0.504   |
| time/                |         |
| episodes             | 23300   |
| fps                  | 211     |
| time_elapsed         | 7031    |
| total timesteps      | 1488187 |
| train/               |         |
| learning_rate        | 0.0001  |

|           |        |
|-----------|--------|
| loss      | 0.306  |
| n_updates | 359546 |

---

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,850 | STEP=1,490,050 | clip=26 | consec=5 | max_consec=28 | avg_clip50=38.39 |
| EP=7,860 | STEP=1,492,283 | clip=78 | consec=5 | max_consec=28 | avg_clip50=39.48 |
| EP=7,870 | STEP=1,494,286 | clip=6  | consec=0 | max_consec=28 | avg_clip50=37.34 |

---

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 36.6    |
| clipped_episode_r... | 22      |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 971     |
| ep_rew_mean          | 387     |
| exploration rate     | 0.502   |
| time/                |         |
| episodes             | 23400   |
| fps                  | 211     |
| time_elapsed         | 7068    |
| total timesteps      | 1495369 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0567  |
| n_updates            | 361342  |

---

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,880 | STEP=1,496,321 | clip=31 | consec=4 | max_consec=28 | avg_clip50=36.60 |
| EP=7,890 | STEP=1,498,028 | clip=8  | consec=0 | max_consec=28 | avg_clip50=32.97 |

[checkpoint periodico] guardado a 1,500,000 steps

|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=7,900 | STEP=1,500,563 | clip=92 | consec=2 | max_consec=28 | avg_clip50=35.64 |
|----------|----------------|---------|----------|---------------|------------------|

---

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 35.2    |
| clipped_episode_r... | 29.5    |
| consecutive_clipp... | 11      |
| max_consecutive_c... | 28      |
| rollout/             |         |
| ep_len_mean          | 927     |
| ep_rew_mean          | 368     |
| exploration rate     | 0.5     |
| time/                |         |
| episodes             | 23500   |
| fps                  | 211     |
| time_elapsed         | 7106    |
| total timesteps      | 1502559 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0565  |
| n_updates            | 363139  |

---

|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=7,910 | STEP=1,502,799 | clip=66 | consec=12 | max_consec=28 | avg_clip50=34.99 |
| EP=7,920 | STEP=1,505,037 | clip=9  | consec=0  | max_consec=28 | avg_clip50=37.07 |
| EP=7,930 | STEP=1,507,361 | clip=92 | consec=4  | max_consec=28 | avg_clip50=39.42 |
| EP=7,940 | STEP=1,509,180 | clip=12 | consec=0  | max_consec=28 | avg_clip50=39.37 |

---

|                      |    |
|----------------------|----|
| custom/              |    |
| avg_clip50           | 38 |
| clipped_episode_r... | 54 |
| consecutive_clipp... | 2  |



|  |                      |         |  |
|--|----------------------|---------|--|
|  | max_consecutive_c... | 28      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 914     |  |
|  | ep_rew_mean          | 365     |  |
|  | exploration rate     | 0.497   |  |
|  | time/                |         |  |
|  | episodes             | 23600   |  |
|  | fps                  | 211     |  |
|  | time_elapsed         | 7144    |  |
|  | total timesteps      | 1510343 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.0569  |  |
|  | n_updates            | 365085  |  |

-----

|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=7,950 |  | STEP=1,511,515 |  | clip=50 |  | consec=4 |  | max_consec=28 |  | avg_clip50=36.40 |
| EP=7,960 |  | STEP=1,513,324 |  | clip=12 |  | consec=0 |  | max_consec=28 |  | avg_clip50=34.10 |
| EP=7,970 |  | STEP=1,515,116 |  | clip=44 |  | consec=2 |  | max_consec=28 |  | avg_clip50=31.61 |

-----

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 32.4    |  |
|  | clipped_episode_r... | 48.5    |  |
|  | consecutive_clipp... | 6       |  |
|  | max_consecutive_c... | 28      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 893     |  |
|  | ep_rew_mean          | 353     |  |
|  | exploration rate     | 0.495   |  |
|  | time/                |         |  |
|  | episodes             | 23700   |  |
|  | fps                  | 211     |  |
|  | time_elapsed         | 7180    |  |
|  | total timesteps      | 1517252 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.301   |  |
|  | n_updates            | 366812  |  |

-----

|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=7,980 |  | STEP=1,517,637 |  | clip=70 |  | consec=8 |  | max_consec=28 |  | avg_clip50=31.93 |
| EP=7,990 |  | STEP=1,520,265 |  | clip=41 |  | consec=1 |  | max_consec=28 |  | avg_clip50=36.28 |
| EP=8,000 |  | STEP=1,522,719 |  | clip=20 |  | consec=1 |  | max_consec=28 |  | avg_clip50=36.90 |
| EP=8,010 |  | STEP=1,525,005 |  | clip=45 |  | consec=4 |  | max_consec=28 |  | avg_clip50=39.11 |

-----

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 39      |  |
|  | clipped_episode_r... | 47.5    |  |
|  | consecutive_clipp... | 6       |  |
|  | max_consecutive_c... | 28      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 932     |  |
|  | ep_rew_mean          | 364     |  |
|  | exploration rate     | 0.492   |  |
|  | time/                |         |  |
|  | episodes             | 23800   |  |
|  | fps                  | 211     |  |
|  | time_elapsed         | 7220    |  |
|  | total timesteps      | 1525459 |  |
|  | train/               |         |  |
|  | learning_rate        | 0.0001  |  |
|  | loss                 | 0.0639  |  |

|           |                |         |          |               |                  |
|-----------|----------------|---------|----------|---------------|------------------|
| n_updates |                | 368864  |          |               |                  |
| EP=8,020  | STEP=1,527,685 | clip=39 | consec=5 | max_consec=28 | avg_clip50=40.71 |
| EP=8,030  | STEP=1,530,079 | clip=52 | consec=6 | max_consec=28 | avg_clip50=41.25 |
| EP=8,040  | STEP=1,532,127 | clip=42 | consec=4 | max_consec=28 | avg_clip50=37.98 |

|                      |         |  |
|----------------------|---------|--|
| custom/              |         |  |
| avg_clip50           | 36.3    |  |
| clipped_episode_r... | 34.5    |  |
| consecutive_clipp... | 10      |  |
| max_consecutive_c... | 28      |  |
| rollout/             |         |  |
| ep_len_mean          | 970     |  |
| ep_rew_mean          | 374     |  |
| exploration rate     | 0.489   |  |
| time/                |         |  |
| episodes             | 23900   |  |
| fps                  | 211     |  |
| time_elapsed         | 7262    |  |
| total timesteps      | 1533760 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.119   |  |
| n_updates            | 370939  |  |

|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=8,050 | STEP=1,534,543 | clip=32 | consec=1  | max_consec=28 | avg_clip50=36.95 |
| EP=8,060 | STEP=1,537,391 | clip=40 | consec=2  | max_consec=28 | avg_clip50=39.62 |
| EP=8,070 | STEP=1,539,795 | clip=24 | consec=2  | max_consec=28 | avg_clip50=40.09 |
| EP=8,080 | STEP=1,542,628 | clip=64 | consec=12 | max_consec=28 | avg_clip50=41.25 |

|                      |          |  |
|----------------------|----------|--|
| custom/              |          |  |
| avg_clip50           | 41.2     |  |
| clipped_episode_r... | 63.5     |  |
| consecutive_clipp... | 12       |  |
| max_consecutive_c... | 28       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.05e+03 |  |
| ep_rew_mean          | 412      |  |
| exploration rate     | 0.486    |  |
| time/                |          |  |
| episodes             | 24000    |  |
| fps                  | 211      |  |
| time_elapsed         | 7306     |  |
| total timesteps      | 1542746  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0924   |  |
| n_updates            | 373186   |  |

|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=8,090 | STEP=1,545,189 | clip=51 | consec=5  | max_consec=28 | avg_clip50=44.87 |
| EP=8,100 | STEP=1,547,993 | clip=60 | consec=15 | max_consec=28 | avg_clip50=47.33 |
| EP=8,110 | STEP=1,550,514 | clip=60 | consec=3  | max_consec=28 | avg_clip50=45.63 |

|                      |      |  |
|----------------------|------|--|
| custom/              |      |  |
| avg_clip50           | 46.1 |  |
| clipped_episode_r... | 55.5 |  |
| consecutive_clipp... | 1    |  |
| max_consecutive_c... | 28   |  |

|                  |          |
|------------------|----------|
| rollout/         |          |
| ep_len_mean      | 1.07e+03 |
| ep_rew_mean      | 431      |
| exploration rate | 0.483    |
| time/            |          |
| episodes         | 24100    |
| fps              | 211      |
| time_elapsed     | 7351     |
| total timesteps  | 1551741  |
| train/           |          |
| learning_rate    | 0.0001   |
| loss             | 0.0306   |
| n_updates        | 375435   |

-----

|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=8,120 |  | STEP=1,553,046 |  | clip=46 |  | consec=6 |  | max_consec=28 |  | avg_clip50=46.73 |
| EP=8,130 |  | STEP=1,555,418 |  | clip=22 |  | consec=5 |  | max_consec=28 |  | avg_clip50=43.90 |
| EP=8,140 |  | STEP=1,557,665 |  | clip=15 |  | consec=0 |  | max_consec=28 |  | avg_clip50=41.72 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39.6     |
| clipped_episode_r... | 26       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 28       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 432      |
| exploration rate     | 0.481    |
| time/                |          |
| episodes             | 24200    |
| fps                  | 210      |
| time_elapsed         | 7392     |
| total timesteps      | 1559681  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0575   |
| n_updates            | 377420   |

-----

|          |  |                |  |         |  |          |  |               |  |                  |
|----------|--|----------------|--|---------|--|----------|--|---------------|--|------------------|
| EP=8,150 |  | STEP=1,559,970 |  | clip=66 |  | consec=4 |  | max_consec=28 |  | avg_clip50=39.67 |
| EP=8,160 |  | STEP=1,562,496 |  | clip=58 |  | consec=5 |  | max_consec=28 |  | avg_clip50=39.53 |
| EP=8,170 |  | STEP=1,564,819 |  | clip=24 |  | consec=2 |  | max_consec=28 |  | avg_clip50=36.39 |
| EP=8,180 |  | STEP=1,567,390 |  | clip=54 |  | consec=8 |  | max_consec=28 |  | avg_clip50=38.33 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.4     |
| clipped_episode_r... | 69.5     |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 28       |
| rollout/             |          |
| ep_len_mean          | 1.03e+03 |
| ep_rew_mean          | 412      |
| exploration rate     | 0.478    |
| time/                |          |
| episodes             | 24300    |
| fps                  | 210      |
| time_elapsed         | 7435     |
| total timesteps      | 1568029  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0369   |
| n_updates            | 379507   |

```

-----
EP=8,190 | STEP=1,569,776 | clip=44 | consec=3 | max_consec=28 | avg_clip50=38.79
EP=8,200 | STEP=1,572,156 | clip=39 | consec=7 | max_consec=28 | avg_clip50=38.68
EP=8,210 | STEP=1,574,576 | clip=34 | consec=2 | max_consec=28 | avg_clip50=39.60
-----

```

```

| custom/
|   avg_clip50           | 42.1
|   clipped_episode_r... | 35
|   consecutive_clipp... | 8
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.01e+03
|   ep_rew_mean          | 397
|   exploration rate     | 0.475
| time/
|   episodes             | 24400
|   fps                  | 210
|   time_elapsed         | 7477
|   total timesteps      | 1576361
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.0397
|   n_updates            | 381590
-----

```

```

EP=8,220 | STEP=1,577,081 | clip=24 | consec=12 | max_consec=28 | avg_clip50=41.7
2
EP=8,230 | STEP=1,579,326 | clip=18 | consec=0 | max_consec=28 | avg_clip50=39.74
EP=8,240 | STEP=1,581,817 | clip=26 | consec=2 | max_consec=28 | avg_clip50=39.83
EP=8,250 | STEP=1,584,591 | clip=77 | consec=2 | max_consec=28 | avg_clip50=42.23
-----

```

```

| custom/
|   avg_clip50           | 42.1
|   clipped_episode_r... | 54.5
|   consecutive_clipp... | 3
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.03e+03
|   ep_rew_mean          | 405
|   exploration rate     | 0.472
| time/
|   episodes             | 24500
|   fps                  | 210
|   time_elapsed         | 7521
|   total timesteps      | 1585035
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.0596
|   n_updates            | 383758
-----

```

```

EP=8,260 | STEP=1,587,641 | clip=34 | consec=1 | max_consec=28 | avg_clip50=43.00
EP=8,270 | STEP=1,590,366 | clip=36 | consec=4 | max_consec=28 | avg_clip50=43.98
EP=8,280 | STEP=1,592,242 | clip=6 | consec=0 | max_consec=28 | avg_clip50=41.22
-----

```

```

| custom/
|   avg_clip50           | 41.2
|   clipped_episode_r... | 51
|   consecutive_clipp... | 5
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.05e+03
-----

```

|                  |         |
|------------------|---------|
| ep_rew_mean      | 408     |
| exploration rate | 0.469   |
| time/            |         |
| episodes         | 24600   |
| fps              | 210     |
| time_elapsed     | 7565    |
| total timesteps  | 1593804 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0582  |
| n_updates        | 385950  |

-----

EP=8,290 | STEP=1,594,871 | clip=43 | consec=3 | max\_consec=28 | avg\_clip50=42.19  
EP=8,300 | STEP=1,597,867 | clip=51 | consec=13 | max\_consec=28 | avg\_clip50=43.0  
1  
EP=8,310 | STEP=1,600,712 | clip=50 | consec=1 | max\_consec=28 | avg\_clip50=41.70  
EP=8,320 | STEP=1,603,263 | clip=50 | consec=9 | max\_consec=28 | avg\_clip50=40.50

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 40.5     |
| clipped_episode_r... | 49.5     |
| consecutive_clipp... | 9        |
| max_consecutive_c... | 28       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 422      |
| exploration rate     | 0.466    |
| time/                |          |
| episodes             | 24700    |
| fps                  | 210      |
| time_elapsed         | 7613     |
| total timesteps      | 1603478  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0228   |
| n_updates            | 388369   |

-----

EP=8,330 | STEP=1,606,106 | clip=54 | consec=19 | max\_consec=28 | avg\_clip50=45.2  
1  
EP=8,340 | STEP=1,608,668 | clip=43 | consec=1 | max\_consec=28 | avg\_clip50=43.93  
EP=8,350 | STEP=1,611,210 | clip=58 | consec=4 | max\_consec=28 | avg\_clip50=41.84

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 42       |
| clipped_episode_r... | 38.5     |
| consecutive_clipp... | 4        |
| max_consecutive_c... | 28       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 425      |
| exploration rate     | 0.463    |
| time/                |          |
| episodes             | 24800    |
| fps                  | 210      |
| time_elapsed         | 7661     |
| total timesteps      | 1612610  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0828   |
| n_updates            | 390652   |

```

-----
EP=8,360 | STEP=1,613,704 | clip=26 | consec=3 | max_consec=28 | avg_clip50=41.48
EP=8,370 | STEP=1,616,082 | clip=34 | consec=5 | max_consec=28 | avg_clip50=40.67
EP=8,380 | STEP=1,618,639 | clip=64 | consec=1 | max_consec=28 | avg_clip50=41.19
-----

```

```

| custom/
|   avg_clip50           | 42.2
|   clipped_episode_r... | 44.5
|   consecutive_clipp... | 3
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean         | 1.09e+03
|   ep_rew_mean         | 431
|   exploration rate     | 0.46
| time/
|   episodes            | 24900
|   fps                 | 210
|   time_elapsed        | 7706
|   total timesteps     | 1620990
| train/
|   learning_rate       | 0.0001
|   loss                | 0.0399
|   n_updates           | 392747
-----

```

```

EP=8,390 | STEP=1,621,120 | clip=46 | consec=4 | max_consec=28 | avg_clip50=42.28
EP=8,400 | STEP=1,624,092 | clip=34 | consec=9 | max_consec=28 | avg_clip50=43.21
EP=8,410 | STEP=1,626,875 | clip=34 | consec=8 | max_consec=28 | avg_clip50=43.21
EP=8,420 | STEP=1,628,895 | clip=6  | consec=0 | max_consec=28 | avg_clip50=42.50
-----

```

```

| custom/
|   avg_clip50           | 42.6
|   clipped_episode_r... | 60
|   consecutive_clipp... | 3
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean         | 1.08e+03
|   ep_rew_mean         | 416
|   exploration rate     | 0.457
| time/
|   episodes            | 25000
|   fps                 | 210
|   time_elapsed        | 7750
|   total timesteps     | 1629867
| train/
|   learning_rate       | 0.0001
|   loss                | 0.0538
|   n_updates           | 394966
-----

```

```

EP=8,430 | STEP=1,631,703 | clip=110 | consec=3 | max_consec=28 | avg_clip50=41.8
4
EP=8,440 | STEP=1,634,150 | clip=34 | consec=8 | max_consec=28 | avg_clip50=40.98
EP=8,450 | STEP=1,636,777 | clip=50 | consec=7 | max_consec=28 | avg_clip50=39.83
-----

```

```

| custom/
|   avg_clip50           | 41
|   clipped_episode_r... | 72
|   consecutive_clipp... | 13
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean         | 1.07e+03
-----

```

|  |                  |  |         |  |
|--|------------------|--|---------|--|
|  | ep_rew_mean      |  | 421     |  |
|  | exploration rate |  | 0.454   |  |
|  | time/            |  |         |  |
|  | episodes         |  | 25100   |  |
|  | fps              |  | 210     |  |
|  | time_elapsed     |  | 7794    |  |
|  | total timesteps  |  | 1638595 |  |
|  | train/           |  |         |  |
|  | learning_rate    |  | 0.0001  |  |
|  | loss             |  | 0.0342  |  |
|  | n_updates        |  | 397148  |  |

-----

EP=8,460 | STEP=1,639,794 | clip=110 | consec=17 | max\_consec=28 | avg\_clip50=43.77

EP=8,470 | STEP=1,642,283 | clip=54 | consec=27 | max\_consec=28 | avg\_clip50=44.74

EP=8,480 | STEP=1,644,948 | clip=41 | consec=37 | max\_consec=37 | avg\_clip50=44.80

EP=8,490 | STEP=1,647,779 | clip=27 | consec=1 | max\_consec=45 | avg\_clip50=45.34

|  |                      |  |          |  |
|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 45.3     |  |
|  | clipped_episode_r... |  | 27       |  |
|  | consecutive_clipp... |  | 1        |  |
|  | max_consecutive_c... |  | 45       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.11e+03 |  |
|  | ep_rew_mean          |  | 432      |  |
|  | exploration rate     |  | 0.451    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 25200    |  |
|  | fps                  |  | 210      |  |
|  | time_elapsed         |  | 7841     |  |
|  | total timesteps      |  | 1647779  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.124    |  |
|  | n_updates            |  | 399444   |  |

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EP=8,500 | STEP=1,649,861 | clip=36 | consec=2 | max\_consec=45 | avg\_clip50=43.80

EP=8,510 | STEP=1,652,301 | clip=52 | consec=3 | max\_consec=45 | avg\_clip50=38.87

EP=8,520 | STEP=1,654,795 | clip=46 | consec=13 | max\_consec=45 | avg\_clip50=39.52

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|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 38.9     |  |
|  | clipped_episode_r... |  | 20       |  |
|  | consecutive_clipp... |  | 0        |  |
|  | max_consecutive_c... |  | 45       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.08e+03 |  |
|  | ep_rew_mean          |  | 422      |  |
|  | exploration rate     |  | 0.449    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 25300    |  |
|  | fps                  |  | 209      |  |
|  | time_elapsed         |  | 7884     |  |
|  | total timesteps      |  | 1655796  |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |

|           |        |
|-----------|--------|
| loss      | 0.0739 |
| n_updates | 401448 |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=8,530 | STEP=1,657,333 | clip=58 | consec=6 | max_consec=45 | avg_clip50=38.78 |
| EP=8,540 | STEP=1,659,501 | clip=41 | consec=1 | max_consec=45 | avg_clip50=37.00 |
| EP=8,550 | STEP=1,661,760 | clip=50 | consec=2 | max_consec=45 | avg_clip50=38.03 |

|                      |          |
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| custom/              |          |
| avg_clip50           | 38.3     |
| clipped_episode_r... | 48       |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 392      |
| exploration rate     | 0.446    |
| time/                |          |
| episodes             | 25400    |
| fps                  | 209      |
| time_elapsed         | 7925     |
| total timesteps      | 1663444  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.127    |
| n_updates            | 403360   |

|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=8,560 | STEP=1,664,010 | clip=48 | consec=4  | max_consec=45 | avg_clip50=38.94 |
| EP=8,570 | STEP=1,666,406 | clip=38 | consec=4  | max_consec=45 | avg_clip50=39.38 |
| EP=8,580 | STEP=1,668,897 | clip=48 | consec=5  | max_consec=45 | avg_clip50=39.55 |
| EP=8,590 | STEP=1,671,320 | clip=59 | consec=15 | max_consec=45 | avg_clip50=41.7  |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 41.9    |
| clipped_episode_r... | 23.5    |
| consecutive_clipp... | 16      |
| max_consecutive_c... | 45      |
| rollout/             |         |
| ep_len_mean          | 984     |
| ep_rew_mean          | 392     |
| exploration rate     | 0.443   |
| time/                |         |
| episodes             | 25500   |
| fps                  | 209     |
| time_elapsed         | 7968    |
| total timesteps      | 1671869 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0489  |
| n_updates            | 405467  |

|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=8,600 | STEP=1,674,073 | clip=52 | consec=25 | max_consec=45 | avg_clip50=43.9  |
| EP=8,610 | STEP=1,676,774 | clip=47 | consec=6  | max_consec=45 | avg_clip50=44.77 |
| EP=8,620 | STEP=1,679,417 | clip=53 | consec=16 | max_consec=45 | avg_clip50=45.2  |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 44.3 |
| clipped_episode_r... | 54.5 |



|                      |          |
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| consecutive_clipp... | 1        |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 425      |
| exploration rate     | 0.44     |
| time/                |          |
| episodes             | 25600    |
| fps                  | 209      |
| time_elapsed         | 8013     |
| total timesteps      | 1680712  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.197    |
| n_updates            | 407677   |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=8,630 | STEP=1,681,516 | clip=37 | consec=2  | max_consec=45 | avg_clip50=44.22 |
| EP=8,640 | STEP=1,683,758 | clip=21 | consec=12 | max_consec=45 | avg_clip50=43.37 |
| EP=8,650 | STEP=1,686,064 | clip=46 | consec=2  | max_consec=45 | avg_clip50=40.65 |
| EP=8,660 | STEP=1,688,778 | clip=31 | consec=1  | max_consec=45 | avg_clip50=40.64 |

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|                      |          |
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| custom/              |          |
| avg_clip50           | 40.6     |
| clipped_episode_r... | 31       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.03e+03 |
| ep_rew_mean          | 427      |
| exploration rate     | 0.438    |
| time/                |          |
| episodes             | 25700    |
| fps                  | 209      |
| time_elapsed         | 8055     |
| total timesteps      | 1688851  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0509   |
| n_updates            | 409712   |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=8,670 | STEP=1,691,243 | clip=50 | consec=3 | max_consec=45 | avg_clip50=39.54 |
| EP=8,680 | STEP=1,693,838 | clip=26 | consec=1 | max_consec=45 | avg_clip50=40.56 |
| EP=8,690 | STEP=1,696,640 | clip=35 | consec=3 | max_consec=45 | avg_clip50=42.89 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 44.2     |
| clipped_episode_r... | 53       |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 433      |
| exploration rate     | 0.435    |
| time/                |          |
| episodes             | 25800    |
| fps                  | 209      |
| time_elapsed         | 8100     |
| total timesteps      | 1697757  |
| train/               |          |

|          |                      |                |          |         |  |                                              |
|----------|----------------------|----------------|----------|---------|--|----------------------------------------------|
|          | learning_rate        |                | 0.0001   |         |  |                                              |
|          | loss                 |                | 0.093    |         |  |                                              |
|          | n_updates            |                | 411939   |         |  |                                              |
| -----    |                      |                |          |         |  |                                              |
| EP=8,700 |                      | STEP=1,699,062 |          | clip=72 |  | consec=8   max_consec=45   avg_clip50=43.76  |
| EP=8,710 |                      | STEP=1,701,616 |          | clip=54 |  | consec=1   max_consec=45   avg_clip50=43.43  |
| EP=8,720 |                      | STEP=1,704,123 |          | clip=55 |  | consec=3   max_consec=45   avg_clip50=45.25  |
| -----    |                      |                |          |         |  |                                              |
|          | custom/              |                |          |         |  |                                              |
|          | avg_clip50           |                | 44.2     |         |  |                                              |
|          | clipped_episode_r... |                | 12       |         |  |                                              |
|          | consecutive_clipp... |                | 0        |         |  |                                              |
|          | max_consecutive_c... |                | 45       |         |  |                                              |
|          | rollout/             |                |          |         |  |                                              |
|          | ep_len_mean          |                | 1.05e+03 |         |  |                                              |
|          | ep_rew_mean          |                | 425      |         |  |                                              |
|          | exploration rate     |                | 0.432    |         |  |                                              |
|          | time/                |                |          |         |  |                                              |
|          | episodes             |                | 25900    |         |  |                                              |
|          | fps                  |                | 209      |         |  |                                              |
|          | time_elapsed         |                | 8145     |         |  |                                              |
|          | total timesteps      |                | 1706546  |         |  |                                              |
|          | train/               |                |          |         |  |                                              |
|          | learning_rate        |                | 0.0001   |         |  |                                              |
|          | loss                 |                | 0.059    |         |  |                                              |
|          | n_updates            |                | 414136   |         |  |                                              |
| -----    |                      |                |          |         |  |                                              |
| EP=8,730 |                      | STEP=1,706,796 |          | clip=44 |  | consec=1   max_consec=45   avg_clip50=44.60  |
| EP=8,740 |                      | STEP=1,709,529 |          | clip=50 |  | consec=11   max_consec=45   avg_clip50=43.86 |
| EP=8,750 |                      | STEP=1,711,663 |          | clip=60 |  | consec=1   max_consec=45   avg_clip50=42.73  |
| EP=8,760 |                      | STEP=1,714,333 |          | clip=60 |  | consec=5   max_consec=45   avg_clip50=42.64  |
| -----    |                      |                |          |         |  |                                              |
|          | custom/              |                |          |         |  |                                              |
|          | avg_clip50           |                | 42.9     |         |  |                                              |
|          | clipped_episode_r... |                | 27       |         |  |                                              |
|          | consecutive_clipp... |                | 11       |         |  |                                              |
|          | max_consecutive_c... |                | 45       |         |  |                                              |
|          | rollout/             |                |          |         |  |                                              |
|          | ep_len_mean          |                | 1.08e+03 |         |  |                                              |
|          | ep_rew_mean          |                | 434      |         |  |                                              |
|          | exploration rate     |                | 0.429    |         |  |                                              |
|          | time/                |                |          |         |  |                                              |
|          | episodes             |                | 26000    |         |  |                                              |
|          | fps                  |                | 209      |         |  |                                              |
|          | time_elapsed         |                | 8195     |         |  |                                              |
|          | total timesteps      |                | 1716042  |         |  |                                              |
|          | train/               |                |          |         |  |                                              |
|          | learning_rate        |                | 0.0001   |         |  |                                              |
|          | loss                 |                | 0.0295   |         |  |                                              |
|          | n_updates            |                | 416510   |         |  |                                              |
| -----    |                      |                |          |         |  |                                              |
| EP=8,770 |                      | STEP=1,717,169 |          | clip=47 |  | consec=15   max_consec=45   avg_clip50=42.94 |
| EP=8,780 |                      | STEP=1,719,509 |          | clip=34 |  | consec=5   max_consec=45   avg_clip50=42.69  |
| EP=8,790 |                      | STEP=1,721,818 |          | clip=52 |  | consec=1   max_consec=45   avg_clip50=40.25  |
| EP=8,800 |                      | STEP=1,724,235 |          | clip=66 |  | consec=11   max_consec=45   avg_clip50=42.35 |
| -----    |                      |                |          |         |  |                                              |
|          | custom/              |                |          |         |  |                                              |

|                      |          |
|----------------------|----------|
| avg_clip50           | 42.8     |
| clipped_episode_r... | 61       |
| consecutive_clipp... | 12       |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 427      |
| exploration rate     | 0.426    |
| time/                |          |
| episodes             | 26100    |
| fps                  | 209      |
| time_elapsed         | 8239     |
| total timesteps      | 1724537  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0445   |
| n_updates            | 418634   |

EP=8,810 | STEP=1,726,902 | clip=44 | consec=21 | max\_consec=45 | avg\_clip50=42.63

EP=8,820 | STEP=1,729,464 | clip=17 | consec=0 | max\_consec=45 | avg\_clip50=41.15

EP=8,830 | STEP=1,731,947 | clip=52 | consec=10 | max\_consec=45 | avg\_clip50=42.45

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 42.4     |
| clipped_episode_r... | 22.5     |
| consecutive_clipp... | 4        |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 418      |
| exploration rate     | 0.423    |
| time/                |          |
| episodes             | 26200    |
| fps                  | 209      |
| time_elapsed         | 8284     |
| total timesteps      | 1733419  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0339   |
| n_updates            | 420854   |

EP=8,840 | STEP=1,734,442 | clip=47 | consec=8 | max\_consec=45 | avg\_clip50=43.63

EP=8,850 | STEP=1,736,634 | clip=6 | consec=0 | max\_consec=45 | avg\_clip50=40.85

EP=8,860 | STEP=1,738,934 | clip=50 | consec=8 | max\_consec=45 | avg\_clip50=40.06

EP=8,870 | STEP=1,741,266 | clip=15 | consec=0 | max\_consec=45 | avg\_clip50=39.00

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 39       |
| clipped_episode_r... | 15       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.01e+03 |
| ep_rew_mean          | 401      |
| exploration rate     | 0.42     |
| time/                |          |
| episodes             | 26300    |
| fps                  | 209      |

|  |                 |         |  |
|--|-----------------|---------|--|
|  | time_elapsed    | 8326    |  |
|  | total timesteps | 1741403 |  |
|  | train/          |         |  |
|  | learning_rate   | 0.0001  |  |
|  | loss            | 0.0699  |  |
|  | n_updates       | 422850  |  |

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EP=8,880 | STEP=1,743,714 | clip=42 | consec=10 | max\_consec=45 | avg\_clip50=38.51

EP=8,890 | STEP=1,746,421 | clip=38 | consec=20 | max\_consec=45 | avg\_clip50=39.84

EP=8,900 | STEP=1,749,002 | clip=61 | consec=4 | max\_consec=45 | avg\_clip50=43.56  
[checkpoint periodico] guardado a 1,750,000 steps

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 44.2     |  |
|  | clipped_episode_r... | 57       |  |
|  | consecutive_clipp... | 11       |  |
|  | max_consecutive_c... | 45       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.02e+03 |  |
|  | ep_rew_mean          | 417      |  |
|  | exploration rate     | 0.417    |  |
|  | time/                |          |  |
|  | episodes             | 26400    |  |
|  | fps                  | 209      |  |
|  | time_elapsed         | 8374     |  |
|  | total timesteps      | 1750713  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.078    |  |
|  | n_updates            | 425178   |  |

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EP=8,910 | STEP=1,751,350 | clip=58 | consec=14 | max\_consec=45 | avg\_clip50=43.68

EP=8,920 | STEP=1,753,576 | clip=42 | consec=1 | max\_consec=45 | avg\_clip50=43.20

EP=8,930 | STEP=1,756,016 | clip=20 | consec=0 | max\_consec=45 | avg\_clip50=41.98

EP=8,940 | STEP=1,758,874 | clip=62 | consec=7 | max\_consec=45 | avg\_clip50=43.28

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 43.5     |  |
|  | clipped_episode_r... | 51       |  |
|  | consecutive_clipp... | 8        |  |
|  | max_consecutive_c... | 45       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.03e+03 |  |
|  | ep_rew_mean          | 418      |  |
|  | exploration rate     | 0.414    |  |
|  | time/                |          |  |
|  | episodes             | 26500    |  |
|  | fps                  | 208      |  |
|  | time_elapsed         | 8419     |  |
|  | total timesteps      | 1759394  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0384   |  |
|  | n_updates            | 427348   |  |

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EP=8,950 | STEP=1,761,554 | clip=32 | consec=17 | max\_consec=45 | avg\_clip50=42.43

EP=8,960 | STEP=1,764,038 | clip=58 | consec=27 | max\_consec=45 | avg\_clip50=42.8  
1  
EP=8,970 | STEP=1,766,875 | clip=49 | consec=37 | max\_consec=45 | avg\_clip50=46.9  
1

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|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 48.2     |  |
| clipped_episode_r... | 37       |  |
| consecutive_clipp... | 43       |  |
| max_consecutive_c... | 45       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.06e+03 |  |
| ep_rew_mean          | 452      |  |
| exploration rate     | 0.411    |  |
| time/                |          |  |
| episodes             | 26600    |  |
| fps                  | 208      |  |
| time_elapsed         | 8466     |  |
| total timesteps      | 1768480  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0448   |  |
| n_updates            | 429619   |  |

EP=8,980 | STEP=1,769,175 | clip=8 | consec=0 | max\_consec=45 | avg\_clip50=47.79  
EP=8,990 | STEP=1,771,731 | clip=56 | consec=1 | max\_consec=45 | avg\_clip50=47.01  
EP=9,000 | STEP=1,774,319 | clip=80 | consec=5 | max\_consec=45 | avg\_clip50=47.93  
EP=9,010 | STEP=1,777,119 | clip=65 | consec=15 | max\_consec=45 | avg\_clip50=49.3  
7

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 48.6     |  |
| clipped_episode_r... | 49       |  |
| consecutive_clipp... | 17       |  |
| max_consecutive_c... | 45       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.08e+03 |  |
| ep_rew_mean          | 463      |  |
| exploration rate     | 0.408    |  |
| time/                |          |  |
| episodes             | 26700    |  |
| fps                  | 208      |  |
| time_elapsed         | 8514     |  |
| total timesteps      | 1777684  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0435   |  |
| n_updates            | 431920   |  |

EP=9,020 | STEP=1,779,632 | clip=46 | consec=6 | max\_consec=45 | avg\_clip50=46.73  
EP=9,030 | STEP=1,781,796 | clip=14 | consec=0 | max\_consec=45 | avg\_clip50=46.11  
EP=9,040 | STEP=1,784,231 | clip=13 | consec=0 | max\_consec=45 | avg\_clip50=44.59

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 44.8     |  |
| clipped_episode_r... | 26.5     |  |
| consecutive_clipp... | 6        |  |
| max_consecutive_c... | 45       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.05e+03 |  |

|                  |         |
|------------------|---------|
| ep_rew_mean      | 457     |
| exploration rate | 0.405   |
| time/            |         |
| episodes         | 26800   |
| fps              | 208     |
| time_elapsed     | 8557    |
| total timesteps  | 1785860 |
| train/           |         |
| learning_rate    | 0.0001  |
| loss             | 0.0586  |
| n_updates        | 433964  |

EP=9,050 | STEP=1,786,705 | clip=50 | consec=3 | max\_consec=45 | avg\_clip50=43.41  
EP=9,060 | STEP=1,789,665 | clip=47 | consec=4 | max\_consec=45 | avg\_clip50=43.91  
EP=9,070 | STEP=1,792,315 | clip=16 | consec=0 | max\_consec=45 | avg\_clip50=44.57  
EP=9,080 | STEP=1,795,036 | clip=50 | consec=4 | max\_consec=45 | avg\_clip50=46.68

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 46.7     |
| clipped_episode_r... | 50.5     |
| consecutive_clipp... | 4        |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.08e+03 |
| ep_rew_mean          | 464      |
| exploration rate     | 0.402    |
| time/                |          |
| episodes             | 26900    |
| fps                  | 208      |
| time_elapsed         | 8604     |
| total timesteps      | 1795036  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0725   |
| n_updates            | 436258   |

EP=9,090 | STEP=1,797,442 | clip=52 | consec=2 | max\_consec=45 | avg\_clip50=47.29  
EP=9,100 | STEP=1,800,128 | clip=44 | consec=12 | max\_consec=45 | avg\_clip50=48.4  
5  
EP=9,110 | STEP=1,802,057 | clip=32 | consec=2 | max\_consec=45 | avg\_clip50=44.99

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| custom/              |          |
| avg_clip50           | 46.2     |
| clipped_episode_r... | 38.5     |
| consecutive_clipp... | 7        |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 453      |
| exploration rate     | 0.399    |
| time/                |          |
| episodes             | 27000    |
| fps                  | 208      |
| time_elapsed         | 8648     |
| total timesteps      | 1803531  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.228    |
| n_updates            | 438382   |

EP=9,120 | STEP=1,804,734 | clip=25 | consec=12 | max\_consec=45 | avg\_clip50=46.14  
EP=9,130 | STEP=1,807,242 | clip=40 | consec=6 | max\_consec=45 | avg\_clip50=45.78  
EP=9,140 | STEP=1,809,787 | clip=60 | consec=2 | max\_consec=45 | avg\_clip50=46.39  
EP=9,150 | STEP=1,812,323 | clip=34 | consec=6 | max\_consec=45 | avg\_clip50=45.19

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| custom/              |          |  |
| avg_clip50           | 45.2     |  |
| clipped_episode_r... | 34.5     |  |
| consecutive_clipp... | 6        |  |
| max_consecutive_c... | 45       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.07e+03 |  |
| ep_rew_mean          | 468      |  |
| exploration rate     | 0.396    |  |
| time/                |          |  |
| episodes             | 27100    |  |
| fps                  | 208      |  |
| time_elapsed         | 8695     |  |
| total timesteps      | 1812424  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0985   |  |
| n_updates            | 440605   |  |

EP=9,160 | STEP=1,814,766 | clip=74 | consec=2 | max\_consec=45 | avg\_clip50=47.39  
EP=9,170 | STEP=1,817,065 | clip=47 | consec=5 | max\_consec=45 | avg\_clip50=44.48  
EP=9,180 | STEP=1,819,884 | clip=32 | consec=15 | max\_consec=45 | avg\_clip50=44.33

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| custom/              |          |  |
| avg_clip50           | 45       |  |
| clipped_episode_r... | 39       |  |
| consecutive_clipp... | 1        |  |
| max_consecutive_c... | 45       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.04e+03 |  |
| ep_rew_mean          | 454      |  |
| exploration rate     | 0.394    |  |
| time/                |          |  |
| episodes             | 27200    |  |
| fps                  | 208      |  |
| time_elapsed         | 8739     |  |
| total timesteps      | 1820977  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0467   |  |
| n_updates            | 442744   |  |

EP=9,190 | STEP=1,822,584 | clip=53 | consec=7 | max\_consec=45 | avg\_clip50=44.64  
EP=9,200 | STEP=1,825,233 | clip=20 | consec=0 | max\_consec=45 | avg\_clip50=45.54  
EP=9,210 | STEP=1,827,818 | clip=54 | consec=10 | max\_consec=45 | avg\_clip50=45.37

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| custom/              |      |  |
| avg_clip50           | 48.8 |  |
| clipped_episode_r... | 67.5 |  |
| consecutive_clipp... | 18   |  |
| max_consecutive_c... | 45   |  |
| rollout/             |      |  |

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| ep_len_mean      | 1.07e+03 |
| ep_rew_mean      | 465      |
| exploration rate | 0.391    |
| time/            |          |
| episodes         | 27300    |
| fps              | 208      |
| time_elapsed     | 8786     |
| total timesteps  | 1830032  |
| train/           |          |
| learning_rate    | 0.0001   |
| loss             | 0.125    |
| n_updates        | 445007   |

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EP=9,220 | STEP=1,830,652 | clip=73 | consec=20 | max\_consec=45 | avg\_clip50=49.35

EP=9,230 | STEP=1,833,167 | clip=67 | consec=7 | max\_consec=45 | avg\_clip50=48.20

EP=9,240 | STEP=1,836,060 | clip=41 | consec=17 | max\_consec=45 | avg\_clip50=48.09

EP=9,250 | STEP=1,838,583 | clip=30 | consec=27 | max\_consec=45 | avg\_clip50=46.59

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| custom/              |          |
| avg_clip50           | 46.3     |
| clipped_episode_r... | 31.5     |
| consecutive_clipp... | 28       |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 457      |
| exploration rate     | 0.388    |
| time/                |          |
| episodes             | 27400    |
| fps                  | 208      |
| time_elapsed         | 8832     |
| total timesteps      | 1839012  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0484   |
| n_updates            | 447252   |

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EP=9,260 | STEP=1,841,131 | clip=36 | consec=7 | max\_consec=45 | avg\_clip50=45.36

EP=9,270 | STEP=1,843,509 | clip=60 | consec=17 | max\_consec=45 | avg\_clip50=42.86

EP=9,280 | STEP=1,845,901 | clip=34 | consec=27 | max\_consec=45 | avg\_clip50=42.76

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| custom/              |          |
| avg_clip50           | 43.6     |
| clipped_episode_r... | 51.5     |
| consecutive_clipp... | 33       |
| max_consecutive_c... | 45       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 459      |
| exploration rate     | 0.385    |
| time/                |          |
| episodes             | 27500    |
| fps                  | 208      |
| time_elapsed         | 8879     |
| total timesteps      | 1847744  |



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| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.026  |
| n_updates     | 449435 |

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EP=9,290 | STEP=1,848,748 | clip=34 | consec=37 | max\_consec=45 | avg\_clip50=43.31

EP=9,300 | STEP=1,851,596 | clip=46 | consec=47 | max\_consec=47 | avg\_clip50=44.42

EP=9,310 | STEP=1,854,430 | clip=54 | consec=57 | max\_consec=57 | avg\_clip50=46.13

EP=9,320 | STEP=1,856,807 | clip=57 | consec=8 | max\_consec=58 | avg\_clip50=45.17

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| custom/              |          |
| avg_clip50           | 45.2     |
| clipped_episode_r... | 57       |
| consecutive_clipp... | 8        |
| max_consecutive_c... | 58       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 440      |
| exploration rate     | 0.382    |
| time/                |          |
| episodes             | 27600    |
| fps                  | 207      |
| time_elapsed         | 8928     |
| total timesteps      | 1857086  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0513   |
| n_updates            | 451771   |

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EP=9,330 | STEP=1,860,431 | clip=124 | consec=18 | max\_consec=58 | avg\_clip50=50.65

EP=9,340 | STEP=1,862,966 | clip=26 | consec=7 | max\_consec=58 | avg\_clip50=49.37

EP=9,350 | STEP=1,865,494 | clip=48 | consec=4 | max\_consec=58 | avg\_clip50=48.07

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| custom/              |          |
| avg_clip50           | 47.8     |
| clipped_episode_r... | 38       |
| consecutive_clipp... | 9        |
| max_consecutive_c... | 58       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 471      |
| exploration rate     | 0.378    |
| time/                |          |
| episodes             | 27700    |
| fps                  | 207      |
| time_elapsed         | 8980     |
| total timesteps      | 1867054  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0344   |
| n_updates            | 454263   |

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EP=9,360 | STEP=1,868,194 | clip=60 | consec=14 | max\_consec=58 | avg\_clip50=47.97

EP=9,370 | STEP=1,871,114 | clip=48 | consec=24 | max\_consec=58 | avg\_clip50=51.17

EP=9,380 | STEP=1,873,860 | clip=53 | consec=8 | max\_consec=58 | avg\_clip50=46.40  
EP=9,390 | STEP=1,876,486 | clip=38 | consec=18 | max\_consec=58 | avg\_clip50=46.5  
2

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 46.4     |  |
| clipped_episode_r... | 44       |  |
| consecutive_clipp... | 20       |  |
| max_consecutive_c... | 58       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.15e+03 |  |
| ep_rew_mean          | 479      |  |
| exploration rate     | 0.375    |  |
| time/                |          |  |
| episodes             | 27800    |  |
| fps                  | 207      |  |
| time_elapsed         | 9030     |  |
| total timesteps      | 1876996  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.401    |  |
| n_updates            | 456748   |  |

EP=9,400 | STEP=1,879,197 | clip=68 | consec=6 | max\_consec=58 | avg\_clip50=46.95  
EP=9,410 | STEP=1,881,535 | clip=6 | consec=0 | max\_consec=58 | avg\_clip50=45.98  
EP=9,420 | STEP=1,884,530 | clip=72 | consec=4 | max\_consec=58 | avg\_clip50=45.08

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 47.2     |  |
| clipped_episode_r... | 77       |  |
| consecutive_clipp... | 10       |  |
| max_consecutive_c... | 58       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.14e+03 |  |
| ep_rew_mean          | 479      |  |
| exploration rate     | 0.372    |  |
| time/                |          |  |
| episodes             | 27900    |  |
| fps                  | 207      |  |
| time_elapsed         | 9079     |  |
| total timesteps      | 1886376  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0528   |  |
| n_updates            | 459093   |  |

EP=9,430 | STEP=1,887,231 | clip=26 | consec=14 | max\_consec=58 | avg\_clip50=46.8  
6  
EP=9,440 | STEP=1,889,913 | clip=60 | consec=24 | max\_consec=58 | avg\_clip50=47.2  
6  
EP=9,450 | STEP=1,892,567 | clip=36 | consec=34 | max\_consec=58 | avg\_clip50=49.4  
8  
EP=9,460 | STEP=1,895,268 | clip=46 | consec=44 | max\_consec=58 | avg\_clip50=49.8  
9

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| -----                |      |  |
| custom/              |      |  |
| avg_clip50           | 49.5 |  |
| clipped_episode_r... | 23   |  |
| consecutive_clipp... | 45   |  |
| max_consecutive_c... | 58   |  |

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| rollout/         |          |
| ep_len_mean      | 1.12e+03 |
| ep_rew_mean      | 475      |
| exploration rate | 0.369    |
| time/            |          |
| episodes         | 28000    |
| fps              | 207      |
| time_elapsed     | 9127     |
| total timesteps  | 1895487  |
| train/           |          |
| learning_rate    | 0.0001   |
| loss             | 0.0809   |
| n_updates        | 461371   |

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EP=9,470 | STEP=1,897,745 | clip=58 | consec=4 | max\_consec=58 | avg\_clip50=48.77  
EP=9,480 | STEP=1,900,672 | clip=48 | consec=7 | max\_consec=58 | avg\_clip50=48.35  
EP=9,490 | STEP=1,903,949 | clip=29 | consec=17 | max\_consec=58 | avg\_clip50=50.6  
5

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|----------------------|----------|
| custom/              |          |
| avg_clip50           | 50       |
| clipped_episode_r... | 52       |
| consecutive_clipp... | 22       |
| max_consecutive_c... | 58       |
| rollout/             |          |
| ep_len_mean          | 1.16e+03 |
| ep_rew_mean          | 498      |
| exploration rate     | 0.365    |
| time/                |          |
| episodes             | 28100    |
| fps                  | 207      |
| time_elapsed         | 9181     |
| total timesteps      | 1905490  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0595   |
| n_updates            | 463872   |

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EP=9,500 | STEP=1,906,674 | clip=54 | consec=2 | max\_consec=58 | avg\_clip50=50.45  
EP=9,510 | STEP=1,909,139 | clip=45 | consec=12 | max\_consec=58 | avg\_clip50=49.0  
3  
EP=9,520 | STEP=1,912,476 | clip=46 | consec=22 | max\_consec=58 | avg\_clip50=52.5  
6  
EP=9,530 | STEP=1,914,953 | clip=50 | consec=4 | max\_consec=58 | avg\_clip50=52.22

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 52.2     |
| clipped_episode_r... | 49.5     |
| consecutive_clipp... | 4        |
| max_consecutive_c... | 58       |
| rollout/             |          |
| ep_len_mean          | 1.15e+03 |
| ep_rew_mean          | 503      |
| exploration rate     | 0.362    |
| time/                |          |
| episodes             | 28200    |
| fps                  | 207      |
| time_elapsed         | 9231     |
| total timesteps      | 1914953  |
| train/               |          |

|               |        |
|---------------|--------|
| learning_rate | 0.0001 |
| loss          | 0.0339 |
| n_updates     | 466238 |

EP=9,540 | STEP=1,917,313 | clip=55 | consec=14 | max\_consec=58 | avg\_clip50=49.35

EP=9,550 | STEP=1,920,285 | clip=58 | consec=24 | max\_consec=58 | avg\_clip50=48.68

EP=9,560 | STEP=1,922,812 | clip=37 | consec=9 | max\_consec=58 | avg\_clip50=49.95

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| custom/              |          |
| avg_clip50           | 47.5     |
| clipped_episode_r... | 52       |
| consecutive_clipp... | 13       |
| max_consecutive_c... | 58       |
| rollout/             |          |
| ep_len_mean          | 1.14e+03 |
| ep_rew_mean          | 493      |
| exploration rate     | 0.359    |
| time/                |          |
| episodes             | 28300    |
| fps                  | 207      |
| time_elapsed         | 9277     |
| total timesteps      | 1923772  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0937   |
| n_updates            | 468442   |

EP=9,570 | STEP=1,925,313 | clip=40 | consec=19 | max\_consec=58 | avg\_clip50=45.80

EP=9,580 | STEP=1,927,817 | clip=20 | consec=0 | max\_consec=58 | avg\_clip50=45.50

EP=9,590 | STEP=1,930,628 | clip=74 | consec=10 | max\_consec=58 | avg\_clip50=46.77

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| custom/              |          |
| avg_clip50           | 47.6     |
| clipped_episode_r... | 36       |
| consecutive_clipp... | 18       |
| max_consecutive_c... | 58       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 486      |
| exploration rate     | 0.356    |
| time/                |          |
| episodes             | 28400    |
| fps                  | 207      |
| time_elapsed         | 9328     |
| total timesteps      | 1933173  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0697   |
| n_updates            | 470793   |

EP=9,600 | STEP=1,933,619 | clip=44 | consec=20 | max\_consec=58 | avg\_clip50=47.59

EP=9,610 | STEP=1,936,051 | clip=16 | consec=0 | max\_consec=58 | avg\_clip50=47.58

EP=9,620 | STEP=1,938,677 | clip=55 | consec=10 | max\_consec=58 | avg\_clip50=48.88

EP=9,630 | STEP=1,941,366 | clip=22 | consec=20 | max\_consec=58 | avg\_clip50=48.9

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| -----                |         |  |
| custom/              |         |  |
| avg_clip50           | 49.9    |  |
| clipped_episode_r... | 52.5    |  |
| consecutive_clipp... | 22      |  |
| max_consecutive_c... | 58      |  |
| rollout/             |         |  |
| ep_len_mean          | 1.1e+03 |  |
| ep_rew_mean          | 471     |  |
| exploration rate     | 0.353   |  |
| time/                |         |  |
| episodes             | 28500   |  |
| fps                  | 207     |  |
| time_elapsed         | 9375    |  |
| total timesteps      | 1942151 |  |
| train/               |         |  |
| learning_rate        | 0.0001  |  |
| loss                 | 0.0631  |  |
| n_updates            | 473037  |  |

EP=9,640 | STEP=1,944,227 | clip=55 | consec=4 | max\_consec=58 | avg\_clip50=49.39  
 EP=9,650 | STEP=1,947,008 | clip=25 | consec=5 | max\_consec=58 | avg\_clip50=47.58  
 EP=9,660 | STEP=1,949,666 | clip=25 | consec=15 | max\_consec=58 | avg\_clip50=47.4  
 1

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 47.2     |  |
| clipped_episode_r... | 50       |  |
| consecutive_clipp... | 21       |  |
| max_consecutive_c... | 58       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.13e+03 |  |
| ep_rew_mean          | 477      |  |
| exploration rate     | 0.35     |  |
| time/                |          |  |
| episodes             | 28600    |  |
| fps                  | 207      |  |
| time_elapsed         | 9425     |  |
| total timesteps      | 1951370  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0768   |  |
| n_updates            | 475342   |  |

EP=9,670 | STEP=1,952,392 | clip=28 | consec=25 | max\_consec=58 | avg\_clip50=47.2  
 1  
 EP=9,680 | STEP=1,955,231 | clip=21 | consec=35 | max\_consec=58 | avg\_clip50=47.6  
 9  
 EP=9,690 | STEP=1,958,165 | clip=37 | consec=45 | max\_consec=58 | avg\_clip50=47.6  
 5  
 EP=9,700 | STEP=1,960,465 | clip=52 | consec=55 | max\_consec=58 | avg\_clip50=47.9  
 7

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| -----                |      |  |
| custom/              |      |  |
| avg_clip50           | 48   |  |
| clipped_episode_r... | 51.5 |  |
| consecutive_clipp... | 55   |  |
| max_consecutive_c... | 58   |  |
| rollout/             |      |  |

|  |                  |          |  |
|--|------------------|----------|--|
|  | ep_len_mean      | 1.12e+03 |  |
|  | ep_rew_mean      | 478      |  |
|  | exploration rate | 0.347    |  |
|  | time/            |          |  |
|  | episodes         | 28700    |  |
|  | fps              | 206      |  |
|  | time_elapsed     | 9473     |  |
|  | total timesteps  | 1960665  |  |
|  | train/           |          |  |
|  | learning_rate    | 0.0001   |  |
|  | loss             | 0.0664   |  |
|  | n_updates        | 477666   |  |

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EP=9,710 | STEP=1,963,207 | clip=46 | consec=65 | max\_consec=65 | avg\_clip50=49.70

EP=9,720 | STEP=1,965,961 | clip=30 | consec=75 | max\_consec=75 | avg\_clip50=50.20

EP=9,730 | STEP=1,968,919 | clip=80 | consec=85 | max\_consec=85 | avg\_clip50=51.43

|  |                      |          |  |
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|  | custom/              |          |  |
|  | avg_clip50           | 50       |  |
|  | clipped_episode_r... | 54.5     |  |
|  | consecutive_clipp... | 1        |  |
|  | max_consecutive_c... | 87       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.14e+03 |  |
|  | ep_rew_mean          | 486      |  |
|  | exploration rate     | 0.344    |  |
|  | time/                |          |  |
|  | episodes             | 28800    |  |
|  | fps                  | 206      |  |
|  | time_elapsed         | 9524     |  |
|  | total timesteps      | 1970193  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.052    |  |
|  | n_updates            | 480048   |  |

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EP=9,740 | STEP=1,971,549 | clip=38 | consec=7 | max\_consec=87 | avg\_clip50=49.48

EP=9,750 | STEP=1,974,162 | clip=9 | consec=0 | max\_consec=87 | avg\_clip50=48.72

EP=9,760 | STEP=1,976,893 | clip=67 | consec=10 | max\_consec=87 | avg\_clip50=47.39

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 44.5     |  |
|  | clipped_episode_r... | 34.5     |  |
|  | consecutive_clipp... | 4        |  |
|  | max_consecutive_c... | 87       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.12e+03 |  |
|  | ep_rew_mean          | 473      |  |
|  | exploration rate     | 0.341    |  |
|  | time/                |          |  |
|  | episodes             | 28900    |  |
|  | fps                  | 206      |  |
|  | time_elapsed         | 9569     |  |
|  | total timesteps      | 1978755  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |

|           |        |
|-----------|--------|
| loss      | 0.0383 |
| n_updates | 482188 |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=9,770 | STEP=1,979,074 | clip=48 | consec=1  | max_consec=87 | avg_clip50=44.10 |
| EP=9,780 | STEP=1,981,654 | clip=55 | consec=2  | max_consec=87 | avg_clip50=41.96 |
| EP=9,790 | STEP=1,984,192 | clip=48 | consec=4  | max_consec=87 | avg_clip50=42.30 |
| EP=9,800 | STEP=1,986,950 | clip=78 | consec=14 | max_consec=87 | avg_clip50=43.71 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 43.8     |
| clipped_episode_r... | 44       |
| consecutive_clipp... | 18       |
| max_consecutive_c... | 87       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 459      |
| exploration rate     | 0.338    |
| time/                |          |
| episodes             | 29000    |
| fps                  | 206      |
| time_elapsed         | 9617     |
| total timesteps      | 1988151  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.289    |
| n_updates            | 484537   |

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|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=9,810 | STEP=1,989,887 | clip=52 | consec=24 | max_consec=87 | avg_clip50=44.35 |
| EP=9,820 | STEP=1,992,772 | clip=78 | consec=34 | max_consec=87 | avg_clip50=48.98 |
| EP=9,830 | STEP=1,995,544 | clip=72 | consec=44 | max_consec=87 | avg_clip50=50.17 |

|                      |         |
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| custom/              |         |
| avg_clip50           | 50      |
| clipped_episode_r... | 45.5    |
| consecutive_clipp... | 3       |
| max_consecutive_c... | 87      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 460     |
| exploration rate     | 0.335   |
| time/                |         |
| episodes             | 29100   |
| fps                  | 206     |
| time_elapsed         | 9668    |
| total timesteps      | 1997835 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0754  |
| n_updates            | 486958  |

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|          |                |         |          |               |                  |
|----------|----------------|---------|----------|---------------|------------------|
| EP=9,840 | STEP=1,998,041 | clip=50 | consec=4 | max_consec=87 | avg_clip50=50.03 |
|----------|----------------|---------|----------|---------------|------------------|

[checkpoint periodico] guardado a 2,000,000 steps

|          |                |         |           |               |                  |
|----------|----------------|---------|-----------|---------------|------------------|
| EP=9,850 | STEP=2,000,795 | clip=44 | consec=14 | max_consec=87 | avg_clip50=51.00 |
| EP=9,860 | STEP=2,003,886 | clip=58 | consec=24 | max_consec=87 | avg_clip50=53.10 |

EP=9,870 | STEP=2,006,541 | clip=52 | consec=5 | max\_consec=87 | avg\_clip50=50.13

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 49.9     |  |
| clipped_episode_r... | 18       |  |
| consecutive_clipp... | 0        |  |
| max_consecutive_c... | 87       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.14e+03 |  |
| ep_rew_mean          | 493      |  |
| exploration rate     | 0.331    |  |
| time/                |          |  |
| episodes             | 29200    |  |
| fps                  | 206      |  |
| time_elapsed         | 9719     |  |
| total timesteps      | 2007589  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.284    |  |
| n_updates            | 489397   |  |

EP=9,880 | STEP=2,009,263 | clip=32 | consec=6 | max\_consec=87 | avg\_clip50=48.68

EP=9,890 | STEP=2,012,015 | clip=36 | consec=1 | max\_consec=87 | avg\_clip50=49.02

EP=9,900 | STEP=2,014,707 | clip=57 | consec=11 | max\_consec=87 | avg\_clip50=47.8

4

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 45.9     |  |
| clipped_episode_r... | 37.5     |  |
| consecutive_clipp... | 19       |  |
| max_consecutive_c... | 87       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.15e+03 |  |
| ep_rew_mean          | 496      |  |
| exploration rate     | 0.328    |  |
| time/                |          |  |
| episodes             | 29300    |  |
| fps                  | 206      |  |
| time_elapsed         | 9769     |  |
| total timesteps      | 2016924  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.108    |  |
| n_updates            | 491730   |  |

EP=9,910 | STEP=2,017,156 | clip=16 | consec=0 | max\_consec=87 | avg\_clip50=44.77

EP=9,920 | STEP=2,019,603 | clip=51 | consec=10 | max\_consec=87 | avg\_clip50=45.5

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EP=9,930 | STEP=2,022,148 | clip=39 | consec=4 | max\_consec=87 | avg\_clip50=45.09

EP=9,940 | STEP=2,024,835 | clip=52 | consec=9 | max\_consec=87 | avg\_clip50=45.49

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 46.3     |  |
| clipped_episode_r... | 68       |  |
| consecutive_clipp... | 13       |  |
| max_consecutive_c... | 87       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.13e+03 |  |
| ep_rew_mean          | 480      |  |
| exploration rate     | 0.325    |  |



|                 |         |
|-----------------|---------|
| time/           |         |
| episodes        | 29400   |
| fps             | 206     |
| time_elapsed    | 9818    |
| total timesteps | 2026271 |
| train/          |         |
| learning_rate   | 0.0001  |
| loss            | 0.0367  |
| n_updates       | 494067  |

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EP=9,950 | STEP=2,027,872 | clip=56 | consec=19 | max\_consec=87 | avg\_clip50=47.12

EP=9,960 | STEP=2,030,160 | clip=23 | consec=6 | max\_consec=87 | avg\_clip50=46.52

EP=9,970 | STEP=2,032,464 | clip=77 | consec=7 | max\_consec=87 | avg\_clip50=45.88

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| custom/              |          |
| avg_clip50           | 47.6     |
| clipped_episode_r... | 40       |
| consecutive_clipp... | 5        |
| max_consecutive_c... | 87       |
| rollout/             |          |
| ep_len_mean          | 1.08e+03 |
| ep_rew_mean          | 463      |
| exploration rate     | 0.322    |
| time/                |          |
| episodes             | 29500    |
| fps                  | 206      |
| time_elapsed         | 9863     |
| total timesteps      | 2034886  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0493   |
| n_updates            | 496221   |

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EP=9,980 | STEP=2,035,047 | clip=40 | consec=6 | max\_consec=87 | avg\_clip50=47.63

EP=9,990 | STEP=2,037,638 | clip=52 | consec=16 | max\_consec=87 | avg\_clip50=46.02

EP=10,000 | STEP=2,040,294 | clip=42 | consec=26 | max\_consec=87 | avg\_clip50=43.87

EP=10,010 | STEP=2,042,880 | clip=37 | consec=36 | max\_consec=87 | avg\_clip50=45.66

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| custom/              |          |
| avg_clip50           | 46.4     |
| clipped_episode_r... | 28       |
| consecutive_clipp... | 40       |
| max_consecutive_c... | 87       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 456      |
| exploration rate     | 0.319    |
| time/                |          |
| episodes             | 29600    |
| fps                  | 206      |
| time_elapsed         | 9912     |
| total timesteps      | 2043945  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0604   |
| n_updates            | 498486   |

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EP=10,020 | STEP=2,045,820 | clip=70 | consec=46 | max\_consec=87 | avg\_clip50=47.88  
EP=10,030 | STEP=2,048,792 | clip=58 | consec=6 | max\_consec=87 | avg\_clip50=47.59  
EP=10,040 | STEP=2,051,396 | clip=31 | consec=16 | max\_consec=87 | avg\_clip50=48.56  
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|----------------------|---------|
| custom/              |         |
| avg_clip50           | 50.5    |
| clipped_episode_r... | 31      |
| consecutive_clipp... | 24      |
| max_consecutive_c... | 87      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 474     |
| exploration rate     | 0.316   |
| time/                |         |
| episodes             | 29700   |
| fps                  | 206     |
| time_elapsed         | 9964    |
| total timesteps      | 2053755 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0495  |
| n_updates            | 500938  |

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EP=10,050 | STEP=2,054,149 | clip=52 | consec=26 | max\_consec=87 | avg\_clip50=50.48  
EP=10,060 | STEP=2,056,260 | clip=25 | consec=2 | max\_consec=87 | avg\_clip50=48.33  
EP=10,070 | STEP=2,059,004 | clip=40 | consec=12 | max\_consec=87 | avg\_clip50=47.53  
EP=10,080 | STEP=2,061,657 | clip=21 | consec=22 | max\_consec=87 | avg\_clip50=47.97  
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|----------------------|----------|
| custom/              |          |
| avg_clip50           | 48.1     |
| clipped_episode_r... | 77       |
| consecutive_clipp... | 26       |
| max_consecutive_c... | 87       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 484      |
| exploration rate     | 0.313    |
| time/                |          |
| episodes             | 29800    |
| fps                  | 206      |
| time_elapsed         | 10013    |
| total timesteps      | 2062900  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0768   |
| n_updates            | 503224   |

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EP=10,090 | STEP=2,064,268 | clip=18 | consec=0 | max\_consec=87 | avg\_clip50=48.49  
EP=10,100 | STEP=2,067,461 | clip=54 | consec=10 | max\_consec=87 | avg\_clip50=49.00  
EP=10,110 | STEP=2,069,910 | clip=47 | consec=3 | max\_consec=87 | avg\_clip50=51.1

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|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 48.9     |  |
| clipped_episode_r... | 13.5     |  |
| consecutive_clipp... | 0        |  |
| max_consecutive_c... | 87       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.11e+03 |  |
| ep_rew_mean          | 485      |  |
| exploration rate     | 0.31     |  |
| time/                |          |  |
| episodes             | 29900    |  |
| fps                  | 205      |  |
| time_elapsed         | 10063    |  |
| total timesteps      | 2072315  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0536   |  |
| n_updates            | 505578   |  |

EP=10,120 | STEP=2,072,475 | clip=60 | consec=1 | max\_consec=87 | avg\_clip50=49.35  
 EP=10,130 | STEP=2,074,829 | clip=47 | consec=11 | max\_consec=87 | avg\_clip50=47.43  
 EP=10,140 | STEP=2,077,348 | clip=58 | consec=21 | max\_consec=87 | avg\_clip50=47.67  
 EP=10,150 | STEP=2,080,458 | clip=33 | consec=31 | max\_consec=87 | avg\_clip50=46.79

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 46.9     |  |
| clipped_episode_r... | 55       |  |
| consecutive_clipp... | 36       |  |
| max_consecutive_c... | 87       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.12e+03 |  |
| ep_rew_mean          | 490      |  |
| exploration rate     | 0.307    |  |
| time/                |          |  |
| episodes             | 30000    |  |
| fps                  | 205      |  |
| time_elapsed         | 10116    |  |
| total timesteps      | 2082103  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.294    |  |
| n_updates            | 508025   |  |

EP=10,160 | STEP=2,083,512 | clip=70 | consec=41 | max\_consec=87 | avg\_clip50=47.48  
 EP=10,170 | STEP=2,086,070 | clip=16 | consec=0 | max\_consec=87 | avg\_clip50=49.02  
 EP=10,180 | STEP=2,088,524 | clip=27 | consec=6 | max\_consec=87 | avg\_clip50=48.99  
 EP=10,190 | STEP=2,091,507 | clip=32 | consec=16 | max\_consec=87 | avg\_clip50=49.40

|            |      |  |
|------------|------|--|
| -----      |      |  |
| custom/    |      |  |
| avg_clip50 | 49.4 |  |

|  |                      |          |  |
|--|----------------------|----------|--|
|  | clipped_episode_r... | 32.5     |  |
|  | consecutive_clipp... | 16       |  |
|  | max_consecutive_c... | 87       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.13e+03 |  |
|  | ep_rew_mean          | 485      |  |
|  | exploration rate     | 0.303    |  |
|  | time/                |          |  |
|  | episodes             | 30100    |  |
|  | fps                  | 205      |  |
|  | time_elapsed         | 10167    |  |
|  | total timesteps      | 2091677  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0451   |  |
|  | n_updates            | 510419   |  |

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EP=10,200 | STEP=2,094,305 | clip=46 | consec=26 | max\_consec=87 | avg\_clip50=48.85

EP=10,210 | STEP=2,097,076 | clip=57 | consec=36 | max\_consec=87 | avg\_clip50=47.57

EP=10,220 | STEP=2,100,067 | clip=58 | consec=46 | max\_consec=87 | avg\_clip50=49.59

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 50.3     |  |
|  | clipped_episode_r... | 12       |  |
|  | consecutive_clipp... | 0        |  |
|  | max_consecutive_c... | 87       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.14e+03 |  |
|  | ep_rew_mean          | 493      |  |
|  | exploration rate     | 0.3      |  |
|  | time/                |          |  |
|  | episodes             | 30200    |  |
|  | fps                  | 205      |  |
|  | time_elapsed         | 10220    |  |
|  | total timesteps      | 2101340  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.0607   |  |
|  | n_updates            | 512834   |  |

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EP=10,230 | STEP=2,102,646 | clip=50 | consec=5 | max\_consec=87 | avg\_clip50=50.77

EP=10,240 | STEP=2,105,440 | clip=60 | consec=15 | max\_consec=87 | avg\_clip50=50.40

EP=10,250 | STEP=2,107,896 | clip=57 | consec=5 | max\_consec=87 | avg\_clip50=48.97

EP=10,260 | STEP=2,110,639 | clip=58 | consec=2 | max\_consec=87 | avg\_clip50=49.41

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 49.4     |  |
|  | clipped_episode_r... | 57.5     |  |
|  | consecutive_clipp... | 2        |  |
|  | max_consecutive_c... | 87       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.13e+03 |  |
|  | ep_rew_mean          | 485      |  |

|  |                  |         |  |
|--|------------------|---------|--|
|  | exploration rate | 0.297   |  |
|  | time/            |         |  |
|  | episodes         | 30300   |  |
|  | fps              | 205     |  |
|  | time_elapsed     | 10273   |  |
|  | total timesteps  | 2110930 |  |
|  | train/           |         |  |
|  | learning_rate    | 0.0001  |  |
|  | loss             | 0.0416  |  |
|  | n_updates        | 515232  |  |

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EP=10,270 | STEP=2,113,376 | clip=46 | consec=12 | max\_consec=87 | avg\_clip50=47.58

EP=10,280 | STEP=2,116,049 | clip=30 | consec=22 | max\_consec=87 | avg\_clip50=48.78

EP=10,290 | STEP=2,118,571 | clip=46 | consec=6 | max\_consec=87 | avg\_clip50=47.01

|  |                      |          |  |
|--|----------------------|----------|--|
|  | custom/              |          |  |
|  | avg_clip50           | 47.2     |  |
|  | clipped_episode_r... | 27       |  |
|  | consecutive_clipp... | 13       |  |
|  | max_consecutive_c... | 87       |  |
|  | rollout/             |          |  |
|  | ep_len_mean          | 1.11e+03 |  |
|  | ep_rew_mean          | 478      |  |
|  | exploration rate     | 0.294    |  |
|  | time/                |          |  |
|  | episodes             | 30400    |  |
|  | fps                  | 205      |  |
|  | time_elapsed         | 10323    |  |
|  | total timesteps      | 2120365  |  |
|  | train/               |          |  |
|  | learning_rate        | 0.0001   |  |
|  | loss                 | 0.049    |  |
|  | n_updates            | 517591   |  |

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EP=10,300 | STEP=2,121,126 | clip=10 | consec=0 | max\_consec=87 | avg\_clip50=47.54

EP=10,310 | STEP=2,123,915 | clip=100 | consec=10 | max\_consec=87 | avg\_clip50=47.91

EP=10,320 | STEP=2,126,551 | clip=19 | consec=0 | max\_consec=87 | avg\_clip50=48.29

EP=10,330 | STEP=2,128,896 | clip=40 | consec=8 | max\_consec=87 | avg\_clip50=45.57

|  |                      |         |  |
|--|----------------------|---------|--|
|  | custom/              |         |  |
|  | avg_clip50           | 46.3    |  |
|  | clipped_episode_r... | 74.5    |  |
|  | consecutive_clipp... | 11      |  |
|  | max_consecutive_c... | 87      |  |
|  | rollout/             |         |  |
|  | ep_len_mean          | 1.1e+03 |  |
|  | ep_rew_mean          | 476     |  |
|  | exploration rate     | 0.291   |  |
|  | time/                |         |  |
|  | episodes             | 30500   |  |
|  | fps                  | 205     |  |
|  | time_elapsed         | 10375   |  |
|  | total timesteps      | 2129911 |  |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0437 |
| n_updates     | 519977 |

EP=10,340 | STEP=2,131,714 | clip=42 | consec=18 | max\_consec=87 | avg\_clip50=47.12

EP=10,350 | STEP=2,134,175 | clip=60 | consec=5 | max\_consec=87 | avg\_clip50=47.71

EP=10,360 | STEP=2,136,997 | clip=74 | consec=15 | max\_consec=87 | avg\_clip50=47.73

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 48.6    |
| clipped_episode_r... | 60      |
| consecutive_clipp... | 23      |
| max_consecutive_c... | 87      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 477     |
| exploration rate     | 0.288   |
| time/                |         |
| episodes             | 30600   |
| fps                  | 205     |
| time_elapsed         | 10424   |
| total timesteps      | 2139225 |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0419  |
| n_updates            | 522306  |

EP=10,370 | STEP=2,139,888 | clip=56 | consec=25 | max\_consec=87 | avg\_clip50=47.90

EP=10,380 | STEP=2,142,634 | clip=54 | consec=4 | max\_consec=87 | avg\_clip50=50.18

EP=10,390 | STEP=2,145,722 | clip=50 | consec=14 | max\_consec=87 | avg\_clip50=52.02

EP=10,400 | STEP=2,148,512 | clip=60 | consec=9 | max\_consec=87 | avg\_clip50=53.01

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 54.2     |
| clipped_episode_r... | 77.5     |
| consecutive_clipp... | 12       |
| max_consecutive_c... | 87       |
| rollout/             |          |
| ep_len_mean          | 1.14e+03 |
| ep_rew_mean          | 507      |
| exploration rate     | 0.284    |
| time/                |          |
| episodes             | 30700    |
| fps                  | 205      |
| time_elapsed         | 10479    |
| total timesteps      | 2149535  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0608   |
| n_updates            | 524883   |

EP=10,410 | STEP=2,151,580 | clip=57 | consec=19 | max\_consec=87 | avg\_clip50=54.

80  
EP=10,420 | STEP=2,154,621 | clip=47 | consec=29 | max\_consec=87 | avg\_clip50=55.06  
EP=10,430 | STEP=2,157,342 | clip=52 | consec=4 | max\_consec=87 | avg\_clip50=54.69

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 52.5     |  |
| clipped_episode_r... | 45.5     |  |
| consecutive_clipp... | 4        |  |
| max_consecutive_c... | 87       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.17e+03 |  |
| ep_rew_mean          | 522      |  |
| exploration rate     | 0.281    |  |
| time/                |          |  |
| episodes             | 30800    |  |
| fps                  | 205      |  |
| time_elapsed         | 10533    |  |
| total timesteps      | 2159601  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0386   |  |
| n_updates            | 527400   |  |

EP=10,440 | STEP=2,159,950 | clip=57 | consec=5 | max\_consec=87 | avg\_clip50=52.65  
EP=10,450 | STEP=2,162,584 | clip=19 | consec=0 | max\_consec=87 | avg\_clip50=51.48  
EP=10,460 | STEP=2,165,657 | clip=61 | consec=10 | max\_consec=87 | avg\_clip50=50.98  
EP=10,470 | STEP=2,168,217 | clip=32 | consec=1 | max\_consec=87 | avg\_clip50=49.21

|                      |          |  |
|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 48.7     |  |
| clipped_episode_r... | 50       |  |
| consecutive_clipp... | 5        |  |
| max_consecutive_c... | 87       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.18e+03 |  |
| ep_rew_mean          | 519      |  |
| exploration rate     | 0.278    |  |
| time/                |          |  |
| episodes             | 30900    |  |
| fps                  | 204      |  |
| time_elapsed         | 10585    |  |
| total timesteps      | 2169347  |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0499   |  |
| n_updates            | 529836   |  |

EP=10,480 | STEP=2,170,912 | clip=74 | consec=11 | max\_consec=87 | avg\_clip50=49.52  
EP=10,490 | STEP=2,173,932 | clip=55 | consec=21 | max\_consec=87 | avg\_clip50=50.73  
EP=10,500 | STEP=2,176,535 | clip=78 | consec=31 | max\_consec=87 | avg\_clip50=50.83  
-----

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 49       |
| clipped_episode_r... | 50       |
| consecutive_clipp... | 39       |
| max_consecutive_c... | 87       |
| rollout/             |          |
| ep_len_mean          | 1.15e+03 |
| ep_rew_mean          | 498      |
| exploration rate     | 0.275    |
| time/                |          |
| episodes             | 31000    |
| fps                  | 204      |
| time_elapsed         | 10633    |
| total timesteps      | 2178565  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0567   |
| n_updates            | 532141   |

-----

EP=10,510 | STEP=2,179,124 | clip=60 | consec=41 | max\_consec=87 | avg\_clip50=48.77

EP=10,520 | STEP=2,181,804 | clip=34 | consec=8 | max\_consec=87 | avg\_clip50=49.81

EP=10,530 | STEP=2,184,701 | clip=68 | consec=18 | max\_consec=87 | avg\_clip50=50.95

EP=10,540 | STEP=2,187,450 | clip=26 | consec=7 | max\_consec=87 | avg\_clip50=50.68

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 51.1     |
| clipped_episode_r... | 47.5     |
| consecutive_clipp... | 11       |
| max_consecutive_c... | 87       |
| rollout/             |          |
| ep_len_mean          | 1.14e+03 |
| ep_rew_mean          | 505      |
| exploration rate     | 0.271    |
| time/                |          |
| episodes             | 31100    |
| fps                  | 204      |
| time_elapsed         | 10686    |
| total timesteps      | 2188522  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0472   |
| n_updates            | 534630   |

-----

EP=10,550 | STEP=2,190,301 | clip=63 | consec=17 | max\_consec=87 | avg\_clip50=51.10

EP=10,560 | STEP=2,193,216 | clip=52 | consec=27 | max\_consec=87 | avg\_clip50=52.57

EP=10,570 | STEP=2,196,196 | clip=56 | consec=37 | max\_consec=87 | avg\_clip50=53.89

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 53.6 |
| clipped_episode_r... | 27.5 |
| consecutive_clipp... | 45   |
| max_consecutive_c... | 87   |
| rollout/             |      |



|                  |          |
|------------------|----------|
| ep_len_mean      | 1.17e+03 |
| ep_rew_mean      | 524      |
| exploration rate | 0.268    |
| time/            |          |
| episodes         | 31200    |
| fps              | 204      |
| time_elapsed     | 10739    |
| total timesteps  | 2198768  |
| train/           |          |
| learning_rate    | 0.0001   |
| loss             | 0.0709   |
| n_updates        | 537191   |

```
-----
EP=10,580 | STEP=2,199,140 | clip=56 | consec=47 | max_consec=87 | avg_clip50=53.78
EP=10,590 | STEP=2,201,809 | clip=42 | consec=57 | max_consec=87 | avg_clip50=53.14
EP=10,600 | STEP=2,204,634 | clip=44 | consec=67 | max_consec=87 | avg_clip50=53.26
EP=10,610 | STEP=2,207,348 | clip=54 | consec=77 | max_consec=87 | avg_clip50=52.90
-----
```

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 53.6     |
| clipped_episode_r... | 71       |
| consecutive_clipp... | 79       |
| max_consecutive_c... | 87       |
| rollout/             |          |
| ep_len_mean          | 1.19e+03 |
| ep_rew_mean          | 534      |
| exploration rate     | 0.265    |
| time/                |          |
| episodes             | 31300    |
| fps                  | 204      |
| time_elapsed         | 10790    |
| total timesteps      | 2207991  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.399    |
| n_updates            | 539497   |

```
-----
EP=10,620 | STEP=2,210,192 | clip=80 | consec=87 | max_consec=87 | avg_clip50=53.35
EP=10,630 | STEP=2,212,713 | clip=36 | consec=97 | max_consec=97 | avg_clip50=52.27
-----
```

```
Objetivo conseguido: 100 episodios consecutivos superando el umbral.
Entrenamiento finalizado.
max_consec alcanzado: 100
Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json
Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv
Checkpoint final guardado en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_dueling/dqn_spaceinvaders_v0_policy_latest.pt
model.save() falló (no crítico): IndexError('tuple index out of range')
```

```
In [ ]: # =====
# M7. Entrenamiento Propuesta 3: exploración ajustada
# =====
```

```
env_tuned = create_train_env(RUN_PATHS['tuned']['monitor'])
model_tuned = build_model('tuned', env_tuned, RUN_PATHS['tuned']['tensorboard'])
model_tuned = train_variant('tuned', model_tuned)
env_tuned.close()
```

Using cuda device

Wrapping the env in a VecTransposeImage.

Logging to /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0\_tuned/tensorboard/DQN\_tuned\_1

EP=0 | STEP=267 | clip=18 | consec=0 | max\_consec=0 | avg\_clip50=18.00

EP=10 | STEP=1,894 | clip=66 | consec=1 | max\_consec=1 | avg\_clip50=18.45

EP=20 | STEP=3,135 | clip=8 | consec=0 | max\_consec=1 | avg\_clip50=13.05

EP=30 | STEP=4,765 | clip=16 | consec=0 | max\_consec=1 | avg\_clip50=14.29

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 14.1  |  |
| clipped_episode_r... | 15.5  |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 1     |  |
| rollout/             |       |  |
| ep_len_mean          | 659   |  |
| ep_rew_mean          | 141   |  |
| exploration rate     | 0.995 |  |
| time/                |       |  |
| episodes             | 100   |  |
| fps                  | 341   |  |
| time_elapsed         | 15    |  |
| total timesteps      | 5126  |  |

EP=40 | STEP=6,186 | clip=16 | consec=0 | max\_consec=1 | avg\_clip50=13.74

EP=50 | STEP=7,696 | clip=8 | consec=0 | max\_consec=1 | avg\_clip50=13.15

EP=60 | STEP=9,261 | clip=4 | consec=0 | max\_consec=2 | avg\_clip50=11.81

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 13.1  |  |
| clipped_episode_r... | 14    |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 2     |  |
| rollout/             |       |  |
| ep_len_mean          | 660   |  |
| ep_rew_mean          | 136   |  |
| exploration rate     | 0.99  |  |
| time/                |       |  |
| episodes             | 200   |  |
| fps                  | 333   |  |
| time_elapsed         | 30    |  |
| total timesteps      | 10233 |  |

EP=70 | STEP=10,844 | clip=32 | consec=1 | max\_consec=2 | avg\_clip50=13.80

EP=80 | STEP=12,464 | clip=34 | consec=2 | max\_consec=2 | avg\_clip50=14.04

EP=90 | STEP=14,027 | clip=4 | consec=0 | max\_consec=2 | avg\_clip50=13.89

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 14.8  |  |
| clipped_episode_r... | 21    |  |
| consecutive_clipp... | 1     |  |
| max_consecutive_c... | 3     |  |
| rollout/             |       |  |
| ep_len_mean          | 665   |  |
| ep_rew_mean          | 141   |  |
| exploration rate     | 0.985 |  |
| time/                |       |  |
| episodes             | 300   |  |
| fps                  | 326   |  |
| time_elapsed         | 47    |  |

|  |                 |       |  |  |  |  |
|--|-----------------|-------|--|--|--|--|
|  | total timesteps | 15452 |  |  |  |  |
|--|-----------------|-------|--|--|--|--|

---

|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=100 | STEP=15,660 | clip=24 | consec=2 | max_consec=3 | avg_clip50=15.15 |
| EP=110 | STEP=17,107 | clip=6  | consec=0 | max_consec=3 | avg_clip50=15.14 |
| EP=120 | STEP=18,403 | clip=13 | consec=0 | max_consec=3 | avg_clip50=13.64 |
| EP=130 | STEP=20,065 | clip=20 | consec=1 | max_consec=3 | avg_clip50=12.83 |

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|                      |       |  |
|----------------------|-------|--|
| custom/              |       |  |
| avg_clip50           | 12.9  |  |
| clipped_episode_r... | 17.5  |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 3     |  |
| rollout/             |       |  |
| ep_len_mean          | 657   |  |
| ep_rew_mean          | 135   |  |
| exploration rate     | 0.98  |  |
| time/                |       |  |
| episodes             | 400   |  |
| fps                  | 329   |  |
| time_elapsed         | 61    |  |
| total timesteps      | 20359 |  |

---

|        |             |        |          |              |                  |
|--------|-------------|--------|----------|--------------|------------------|
| EP=140 | STEP=21,723 | clip=2 | consec=0 | max_consec=3 | avg_clip50=13.47 |
| EP=150 | STEP=23,061 | clip=9 | consec=0 | max_consec=3 | avg_clip50=11.68 |
| EP=160 | STEP=24,659 | clip=6 | consec=0 | max_consec=3 | avg_clip50=12.61 |

---

|                      |        |  |
|----------------------|--------|--|
| custom/              |        |  |
| avg_clip50           | 13.6   |  |
| clipped_episode_r... | 15     |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 3      |  |
| rollout/             |        |  |
| ep_len_mean          | 664    |  |
| ep_rew_mean          | 137    |  |
| exploration rate     | 0.975  |  |
| time/                |        |  |
| episodes             | 500    |  |
| fps                  | 318    |  |
| time_elapsed         | 80     |  |
| total timesteps      | 25714  |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0302 |  |
| n_updates            | 178    |  |

---

|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=170 | STEP=26,416 | clip=8  | consec=0 | max_consec=3 | avg_clip50=14.13 |
| EP=180 | STEP=28,258 | clip=11 | consec=0 | max_consec=3 | avg_clip50=15.15 |
| EP=190 | STEP=29,915 | clip=2  | consec=0 | max_consec=3 | avg_clip50=14.92 |

---

|                      |       |  |
|----------------------|-------|--|
| custom/              |       |  |
| avg_clip50           | 15.8  |  |
| clipped_episode_r... | 3     |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 3     |  |
| rollout/             |       |  |
| ep_len_mean          | 685   |  |
| ep_rew_mean          | 139   |  |
| exploration rate     | 0.969 |  |
| time/                |       |  |
| episodes             | 600   |  |

|                 |        |
|-----------------|--------|
| fps             | 298    |
| time_elapsed    | 105    |
| total timesteps | 31423  |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.015  |
| n_updates       | 1605   |

EP=200 | STEP=31,578 | clip=11 | consec=0 | max\_consec=3 | avg\_clip50=15.83  
EP=210 | STEP=33,317 | clip=42 | consec=1 | max\_consec=3 | avg\_clip50=16.29  
EP=220 | STEP=35,104 | clip=14 | consec=0 | max\_consec=3 | avg\_clip50=16.11  
EP=230 | STEP=36,620 | clip=18 | consec=0 | max\_consec=3 | avg\_clip50=14.94

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 14.7   |
| clipped_episode_r... | 7      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 710    |
| ep_rew_mean          | 149    |
| exploration rate     | 0.963  |
| time/                |        |
| episodes             | 700    |
| fps                  | 289    |
| time_elapsed         | 127    |
| total timesteps      | 36986  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.047  |
| n_updates            | 2996   |

EP=240 | STEP=38,324 | clip=10 | consec=0 | max\_consec=3 | avg\_clip50=15.30  
EP=250 | STEP=40,087 | clip=22 | consec=1 | max\_consec=3 | avg\_clip50=15.10  
EP=260 | STEP=41,556 | clip=2 | consec=0 | max\_consec=3 | avg\_clip50=14.03

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 13.2   |
| clipped_episode_r... | 8      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 717    |
| ep_rew_mean          | 151    |
| exploration rate     | 0.958  |
| time/                |        |
| episodes             | 800    |
| fps                  | 280    |
| time_elapsed         | 151    |
| total timesteps      | 42468  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0321 |
| n_updates            | 4366   |

EP=270 | STEP=42,957 | clip=12 | consec=0 | max\_consec=3 | avg\_clip50=13.15  
EP=280 | STEP=44,848 | clip=24 | consec=2 | max\_consec=3 | avg\_clip50=13.51  
EP=290 | STEP=46,375 | clip=10 | consec=0 | max\_consec=3 | avg\_clip50=12.98

|         |  |
|---------|--|
| custom/ |  |
|---------|--|

|                      |         |
|----------------------|---------|
| avg_clip50           | 12.9    |
| clipped_episode_r... | 28.5    |
| consecutive_clipp... | 1       |
| max_consecutive_c... | 3       |
| rollout/             |         |
| ep_len_mean          | 697     |
| ep_rew_mean          | 139     |
| exploration rate     | 0.953   |
| time/                |         |
| episodes             | 900     |
| fps                  | 275     |
| time_elapsed         | 174     |
| total timesteps      | 47973   |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.00471 |
| n_updates            | 5743    |

-----

|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=300 | STEP=48,053 | clip=30 | consec=2 | max_consec=3 | avg_clip50=13.10 |
| EP=310 | STEP=49,489 | clip=18 | consec=0 | max_consec=3 | avg_clip50=13.36 |
| EP=320 | STEP=50,932 | clip=11 | consec=0 | max_consec=3 | avg_clip50=13.68 |
| EP=330 | STEP=52,675 | clip=30 | consec=1 | max_consec=3 | avg_clip50=13.44 |

-----

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 13.4   |
| clipped_episode_r... | 20     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 694    |
| ep_rew_mean          | 138    |
| exploration rate     | 0.947  |
| time/                |        |
| episodes             | 1000   |
| fps                  | 270    |
| time_elapsed         | 196    |
| total timesteps      | 53221  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0188 |
| n_updates            | 7055   |

-----

|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=340 | STEP=54,260 | clip=12 | consec=0 | max_consec=3 | avg_clip50=13.34 |
| EP=350 | STEP=55,733 | clip=12 | consec=0 | max_consec=3 | avg_clip50=13.71 |
| EP=360 | STEP=57,284 | clip=18 | consec=0 | max_consec=3 | avg_clip50=13.80 |

-----

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 13.7  |
| clipped_episode_r... | 8     |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 3     |
| rollout/             |       |
| ep_len_mean          | 673   |
| ep_rew_mean          | 133   |
| exploration rate     | 0.942 |
| time/                |       |
| episodes             | 1100  |
| fps                  | 267   |
| time_elapsed         | 218   |
| total timesteps      | 58230 |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0372 |
| n_updates     | 8307   |

-----

|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=370 | STEP=58,710 | clip=20 | consec=1 | max_consec=3 | avg_clip50=13.03 |
| EP=380 | STEP=60,286 | clip=11 | consec=0 | max_consec=3 | avg_clip50=13.69 |
| EP=390 | STEP=61,898 | clip=28 | consec=1 | max_consec=3 | avg_clip50=14.69 |
| EP=400 | STEP=63,860 | clip=44 | consec=2 | max_consec=3 | avg_clip50=15.80 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 15.8   |
| clipped_episode_r... | 44     |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 677    |
| ep_rew_mean          | 148    |
| exploration rate     | 0.937  |
| time/                |        |
| episodes             | 1200   |
| fps                  | 264    |
| time_elapsed         | 241    |
| total timesteps      | 63910  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0208 |
| n_updates            | 9727   |

-----

|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=410 | STEP=65,407 | clip=14 | consec=0 | max_consec=3 | avg_clip50=15.06 |
| EP=420 | STEP=66,901 | clip=19 | consec=0 | max_consec=3 | avg_clip50=15.20 |
| EP=430 | STEP=68,666 | clip=26 | consec=1 | max_consec=3 | avg_clip50=15.04 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 14.9   |
| clipped_episode_r... | 0      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 678    |
| ep_rew_mean          | 140    |
| exploration rate     | 0.931  |
| time/                |        |
| episodes             | 1300   |
| fps                  | 261    |
| time_elapsed         | 264    |
| total timesteps      | 69193  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0444 |
| n_updates            | 11048  |

-----

|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=440 | STEP=70,104 | clip=7  | consec=0 | max_consec=3 | avg_clip50=13.49 |
| EP=450 | STEP=71,969 | clip=14 | consec=0 | max_consec=3 | avg_clip50=12.75 |
| EP=460 | STEP=73,480 | clip=4  | consec=0 | max_consec=3 | avg_clip50=13.45 |

|                      |    |
|----------------------|----|
| custom/              |    |
| avg_clip50           | 15 |
| clipped_episode_r... | 21 |
| consecutive_clipp... | 1  |

|                      |        |
|----------------------|--------|
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 699    |
| ep_rew_mean          | 150    |
| exploration rate     | 0.926  |
| time/                |        |
| episodes             | 1400   |
| fps                  | 260    |
| time_elapsed         | 286    |
| total timesteps      | 74665  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0264 |
| n_updates            | 12416  |

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EP=470 | STEP=75,037 | clip=24 | consec=1 | max\_consec=3 | avg\_clip50=15.08  
EP=480 | STEP=76,431 | clip=6 | consec=0 | max\_consec=3 | avg\_clip50=13.45  
EP=490 | STEP=78,061 | clip=5 | consec=0 | max\_consec=3 | avg\_clip50=14.29  
EP=500 | STEP=79,683 | clip=14 | consec=0 | max\_consec=3 | avg\_clip50=13.68

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 13.7   |
| clipped_episode_r... | 13.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 680    |
| ep_rew_mean          | 132    |
| exploration rate     | 0.921  |
| time/                |        |
| episodes             | 1500   |
| fps                  | 258    |
| time_elapsed         | 308    |
| total timesteps      | 79781  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0365 |
| n_updates            | 13695  |

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EP=510 | STEP=81,247 | clip=4 | consec=0 | max\_consec=3 | avg\_clip50=12.86  
EP=520 | STEP=82,788 | clip=10 | consec=0 | max\_consec=3 | avg\_clip50=11.81  
EP=530 | STEP=84,802 | clip=44 | consec=1 | max\_consec=3 | avg\_clip50=14.58

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 14.8   |
| clipped_episode_r... | 1.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 700    |
| ep_rew_mean          | 143    |
| exploration rate     | 0.915  |
| time/                |        |
| episodes             | 1600   |
| fps                  | 256    |
| time_elapsed         | 333    |
| total timesteps      | 85527  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0533 |



|           |       |
|-----------|-------|
| n_updates | 15131 |
|-----------|-------|

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|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=540 | STEP=86,482 | clip=20 | consec=0 | max_consec=3 | avg_clip50=14.47 |
| EP=550 | STEP=88,115 | clip=26 | consec=1 | max_consec=3 | avg_clip50=14.75 |
| EP=560 | STEP=89,878 | clip=44 | consec=2 | max_consec=3 | avg_clip50=15.30 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 14.5   |
| clipped_episode_r... | 20     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 699    |
| ep_rew_mean          | 135    |
| exploration rate     | 0.91   |
| time/                |        |
| episodes             | 1700   |
| fps                  | 256    |
| time_elapsed         | 355    |
| total timesteps      | 90977  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0367 |
| n_updates            | 16494  |

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|        |             |         |          |              |                  |
|--------|-------------|---------|----------|--------------|------------------|
| EP=570 | STEP=91,296 | clip=12 | consec=0 | max_consec=3 | avg_clip50=14.65 |
| EP=580 | STEP=92,914 | clip=5  | consec=0 | max_consec=3 | avg_clip50=12.60 |
| EP=590 | STEP=94,395 | clip=26 | consec=1 | max_consec=3 | avg_clip50=12.18 |
| EP=600 | STEP=95,856 | clip=4  | consec=0 | max_consec=3 | avg_clip50=12.42 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 12.5   |
| clipped_episode_r... | 11     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 693    |
| ep_rew_mean          | 136    |
| exploration rate     | 0.905  |
| time/                |        |
| episodes             | 1800   |
| fps                  | 254    |
| time_elapsed         | 376    |
| total timesteps      | 95971  |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0151 |
| n_updates            | 17742  |

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|        |              |         |          |              |                  |
|--------|--------------|---------|----------|--------------|------------------|
| EP=610 | STEP=97,868  | clip=26 | consec=3 | max_consec=3 | avg_clip50=14.14 |
| EP=620 | STEP=99,520  | clip=40 | consec=1 | max_consec=3 | avg_clip50=14.76 |
| EP=630 | STEP=100,982 | clip=21 | consec=1 | max_consec=3 | avg_clip50=14.54 |

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|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 14.7 |
| clipped_episode_r... | 14.5 |
| consecutive_clipp... | 0    |
| max_consecutive_c... | 3    |
| rollout/             |      |
| ep_len_mean          | 687  |

|                  |        |
|------------------|--------|
| ep_rew_mean      | 134    |
| exploration rate | 0.899  |
| time/            |        |
| episodes         | 1900   |
| fps              | 254    |
| time_elapsed     | 398    |
| total timesteps  | 101534 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0384 |
| n_updates        | 19133  |

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|        |              |         |          |              |                  |
|--------|--------------|---------|----------|--------------|------------------|
| EP=640 | STEP=102,485 | clip=9  | consec=0 | max_consec=3 | avg_clip50=15.20 |
| EP=650 | STEP=104,078 | clip=15 | consec=0 | max_consec=3 | avg_clip50=15.18 |
| EP=660 | STEP=105,853 | clip=11 | consec=0 | max_consec=3 | avg_clip50=15.01 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 15.3   |
| clipped_episode_r... | 28.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 693    |
| ep_rew_mean          | 148    |
| exploration rate     | 0.894  |
| time/                |        |
| episodes             | 2000   |
| fps                  | 253    |
| time_elapsed         | 423    |
| total timesteps      | 107278 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0303 |
| n_updates            | 20569  |

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|        |              |         |          |              |                  |
|--------|--------------|---------|----------|--------------|------------------|
| EP=670 | STEP=107,745 | clip=17 | consec=0 | max_consec=3 | avg_clip50=16.02 |
| EP=680 | STEP=109,479 | clip=18 | consec=0 | max_consec=3 | avg_clip50=17.76 |
| EP=690 | STEP=111,139 | clip=10 | consec=0 | max_consec=3 | avg_clip50=18.09 |
| EP=700 | STEP=112,773 | clip=13 | consec=0 | max_consec=3 | avg_clip50=17.88 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 17.9   |
| clipped_episode_r... | 7      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 725    |
| ep_rew_mean          | 165    |
| exploration rate     | 0.888  |
| time/                |        |
| episodes             | 2100   |
| fps                  | 252    |
| time_elapsed         | 447    |
| total timesteps      | 112947 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0261 |
| n_updates            | 21986  |

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|        |              |         |          |              |                  |
|--------|--------------|---------|----------|--------------|------------------|
| EP=710 | STEP=114,374 | clip=10 | consec=0 | max_consec=3 | avg_clip50=16.51 |
|--------|--------------|---------|----------|--------------|------------------|

EP=720 | STEP=115,992 | clip=8 | consec=0 | max\_consec=3 | avg\_clip50=15.40  
EP=730 | STEP=117,645 | clip=4 | consec=0 | max\_consec=3 | avg\_clip50=14.80

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 15.1   |
| clipped_episode_r... | 28     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 718    |
| ep_rew_mean          | 165    |
| exploration rate     | 0.883  |
| time/                |        |
| episodes             | 2200   |
| fps                  | 252    |
| time_elapsed         | 469    |
| total timesteps      | 118412 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.024  |
| n_updates            | 23352  |

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EP=740 | STEP=119,236 | clip=5 | consec=0 | max\_consec=3 | avg\_clip50=14.24  
EP=750 | STEP=120,691 | clip=3 | consec=0 | max\_consec=3 | avg\_clip50=14.33  
EP=760 | STEP=122,453 | clip=4 | consec=0 | max\_consec=3 | avg\_clip50=15.01

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 15.3   |
| clipped_episode_r... | 19     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 708    |
| ep_rew_mean          | 160    |
| exploration rate     | 0.878  |
| time/                |        |
| episodes             | 2300   |
| fps                  | 251    |
| time_elapsed         | 492    |
| total timesteps      | 123687 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0377 |
| n_updates            | 24671  |

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EP=770 | STEP=123,986 | clip=10 | consec=0 | max\_consec=3 | avg\_clip50=15.46  
EP=780 | STEP=125,411 | clip=8 | consec=0 | max\_consec=3 | avg\_clip50=14.41  
EP=790 | STEP=126,908 | clip=8 | consec=0 | max\_consec=3 | avg\_clip50=14.03  
EP=800 | STEP=128,299 | clip=12 | consec=0 | max\_consec=3 | avg\_clip50=13.73

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|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 13.7  |
| clipped_episode_r... | 9.5   |
| consecutive_clipp... | 0     |
| max_consecutive_c... | 3     |
| rollout/             |       |
| ep_len_mean          | 668   |
| ep_rew_mean          | 141   |
| exploration rate     | 0.873 |
| time/                |       |

|                 |        |
|-----------------|--------|
| episodes        | 2400   |
| fps             | 251    |
| time_elapsed    | 511    |
| total timesteps | 128486 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0869 |
| n_updates       | 25871  |

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|        |              |         |          |              |                  |
|--------|--------------|---------|----------|--------------|------------------|
| EP=810 | STEP=129,812 | clip=19 | consec=0 | max_consec=3 | avg_clip50=13.10 |
| EP=820 | STEP=131,258 | clip=20 | consec=0 | max_consec=3 | avg_clip50=12.53 |
| EP=830 | STEP=132,729 | clip=14 | consec=0 | max_consec=3 | avg_clip50=12.02 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 11.6   |
| clipped_episode_r... | 22.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 640    |
| ep_rew_mean          | 129    |
| exploration rate     | 0.868  |
| time/                |        |
| episodes             | 2500   |
| fps                  | 249    |
| time_elapsed         | 532    |
| total timesteps      | 133225 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.123  |
| n_updates            | 27056  |

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|        |              |         |          |              |                  |
|--------|--------------|---------|----------|--------------|------------------|
| EP=840 | STEP=134,044 | clip=3  | consec=0 | max_consec=3 | avg_clip50=11.62 |
| EP=850 | STEP=135,627 | clip=26 | consec=1 | max_consec=3 | avg_clip50=11.53 |
| EP=860 | STEP=137,007 | clip=2  | consec=0 | max_consec=3 | avg_clip50=11.70 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 12.6   |
| clipped_episode_r... | 31.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 629    |
| ep_rew_mean          | 125    |
| exploration rate     | 0.863  |
| time/                |        |
| episodes             | 2600   |
| fps                  | 249    |
| time_elapsed         | 554    |
| total timesteps      | 138289 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.128  |
| n_updates            | 28322  |

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|        |              |         |          |              |                  |
|--------|--------------|---------|----------|--------------|------------------|
| EP=870 | STEP=138,530 | clip=8  | consec=0 | max_consec=3 | avg_clip50=12.39 |
| EP=880 | STEP=140,026 | clip=24 | consec=1 | max_consec=3 | avg_clip50=12.95 |
| EP=890 | STEP=141,653 | clip=10 | consec=0 | max_consec=3 | avg_clip50=14.40 |
| EP=900 | STEP=143,184 | clip=12 | consec=0 | max_consec=3 | avg_clip50=14.73 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 14.7   |
| clipped_episode_r... | 14     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 640    |
| ep_rew_mean          | 132    |
| exploration rate     | 0.858  |
| time/                |        |
| episodes             | 2700   |
| fps                  | 248    |
| time_elapsed         | 576    |
| total timesteps      | 143456 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0933 |
| n_updates            | 29613  |

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EP=910 | STEP=144,803 | clip=5 | consec=0 | max\_consec=3 | avg\_clip50=14.64  
EP=920 | STEP=146,331 | clip=6 | consec=0 | max\_consec=3 | avg\_clip50=13.10  
EP=930 | STEP=147,748 | clip=12 | consec=0 | max\_consec=3 | avg\_clip50=13.25

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 12.3   |
| clipped_episode_r... | 1.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 646    |
| ep_rew_mean          | 131    |
| exploration rate     | 0.853  |
| time/                |        |
| episodes             | 2800   |
| fps                  | 248    |
| time_elapsed         | 596    |
| total timesteps      | 148277 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0835 |
| n_updates            | 30819  |

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EP=940 | STEP=149,147 | clip=57 | consec=1 | max\_consec=3 | avg\_clip50=12.82  
EP=950 | STEP=150,818 | clip=37 | consec=2 | max\_consec=3 | avg\_clip50=12.66  
EP=960 | STEP=152,217 | clip=11 | consec=0 | max\_consec=3 | avg\_clip50=11.58

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 12.5   |
| clipped_episode_r... | 18.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 656    |
| ep_rew_mean          | 129    |
| exploration rate     | 0.848  |
| time/                |        |
| episodes             | 2900   |
| fps                  | 247    |
| time_elapsed         | 620    |
| total timesteps      | 153656 |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.132  |
| n_updates     | 32163  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=970   | STEP=154,099 | clip=48 | consec=1 | max_consec=3 | avg_clip50=13.56 |
| EP=980   | STEP=155,409 | clip=10 | consec=0 | max_consec=3 | avg_clip50=12.70 |
| EP=990   | STEP=156,785 | clip=19 | consec=0 | max_consec=3 | avg_clip50=11.98 |
| EP=1,000 | STEP=158,172 | clip=2  | consec=0 | max_consec=3 | avg_clip50=11.24 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 11     |
| clipped_episode_r... | 7.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 647    |
| ep_rew_mean          | 119    |
| exploration rate     | 0.843  |
| time/                |        |
| episodes             | 3000   |
| fps                  | 247    |
| time_elapsed         | 640    |
| total timesteps      | 158487 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0879 |
| n_updates            | 33371  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,010 | STEP=159,853 | clip=9  | consec=0 | max_consec=3 | avg_clip50=12.40 |
| EP=1,020 | STEP=161,233 | clip=4  | consec=0 | max_consec=3 | avg_clip50=11.15 |
| EP=1,030 | STEP=162,560 | clip=10 | consec=0 | max_consec=3 | avg_clip50=11.31 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 10.7   |
| clipped_episode_r... | 11     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 642    |
| ep_rew_mean          | 120    |
| exploration rate     | 0.838  |
| time/                |        |
| episodes             | 3100   |
| fps                  | 246    |
| time_elapsed         | 661    |
| total timesteps      | 163187 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0462 |
| n_updates            | 34546  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,040 | STEP=164,032 | clip=30 | consec=1 | max_consec=3 | avg_clip50=11.24 |
| EP=1,050 | STEP=165,422 | clip=2  | consec=0 | max_consec=3 | avg_clip50=11.23 |
| EP=1,060 | STEP=167,327 | clip=9  | consec=0 | max_consec=3 | avg_clip50=12.48 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 12.4 |
| clipped_episode_r... | 9    |
| consecutive_clipp... | 0    |

|                      |        |
|----------------------|--------|
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 645    |
| ep_rew_mean          | 123    |
| exploration rate     | 0.833  |
| time/                |        |
| episodes             | 3200   |
| fps                  | 246    |
| time_elapsed         | 684    |
| total timesteps      | 168614 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0985 |
| n_updates            | 35903  |

EP=1,070 | STEP=168,846 | clip=10 | consec=0 | max\_consec=3 | avg\_clip50=12.68  
EP=1,080 | STEP=170,581 | clip=33 | consec=1 | max\_consec=3 | avg\_clip50=14.84  
EP=1,090 | STEP=172,267 | clip=60 | consec=1 | max\_consec=3 | avg\_clip50=16.33  
EP=1,100 | STEP=173,479 | clip=6 | consec=0 | max\_consec=3 | avg\_clip50=15.92

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 15.7   |
| clipped_episode_r... | 9      |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 654    |
| ep_rew_mean          | 136    |
| exploration rate     | 0.828  |
| time/                |        |
| episodes             | 3300   |
| fps                  | 245    |
| time_elapsed         | 706    |
| total timesteps      | 173806 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0365 |
| n_updates            | 37201  |

EP=1,110 | STEP=174,822 | clip=4 | consec=0 | max\_consec=3 | avg\_clip50=13.56  
EP=1,120 | STEP=176,217 | clip=4 | consec=0 | max\_consec=3 | avg\_clip50=13.16  
EP=1,130 | STEP=178,067 | clip=14 | consec=0 | max\_consec=3 | avg\_clip50=12.53

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 12.3   |
| clipped_episode_r... | 16     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 663    |
| ep_rew_mean          | 139    |
| exploration rate     | 0.823  |
| time/                |        |
| episodes             | 3400   |
| fps                  | 245    |
| time_elapsed         | 728    |
| total timesteps      | 178748 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0194 |

|           |       |
|-----------|-------|
| n_updates | 38436 |
|-----------|-------|

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,140 | STEP=179,428 | clip=20 | consec=1 | max_consec=3 | avg_clip50=11.37 |
| EP=1,150 | STEP=181,228 | clip=7  | consec=0 | max_consec=3 | avg_clip50=12.27 |
| EP=1,160 | STEP=182,748 | clip=10 | consec=0 | max_consec=3 | avg_clip50=12.94 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 14     |
| clipped_episode_r... | 24.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 670    |
| ep_rew_mean          | 137    |
| exploration rate     | 0.818  |
| time/                |        |
| episodes             | 3500   |
| fps                  | 245    |
| time_elapsed         | 752    |
| total timesteps      | 184305 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0791 |
| n_updates            | 39826  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,170 | STEP=184,555 | clip=32 | consec=3 | max_consec=3 | avg_clip50=14.58 |
| EP=1,180 | STEP=185,994 | clip=6  | consec=0 | max_consec=3 | avg_clip50=13.62 |
| EP=1,190 | STEP=187,600 | clip=8  | consec=0 | max_consec=3 | avg_clip50=13.78 |
| EP=1,200 | STEP=189,436 | clip=11 | consec=0 | max_consec=3 | avg_clip50=14.76 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 14.4   |
| clipped_episode_r... | 7.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 684    |
| ep_rew_mean          | 135    |
| exploration rate     | 0.812  |
| time/                |        |
| episodes             | 3600   |
| fps                  | 244    |
| time_elapsed         | 776    |
| total timesteps      | 189798 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.076  |
| n_updates            | 41199  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,210 | STEP=191,105 | clip=14 | consec=0 | max_consec=3 | avg_clip50=14.88 |
| EP=1,220 | STEP=193,010 | clip=34 | consec=2 | max_consec=3 | avg_clip50=15.10 |
| EP=1,230 | STEP=194,681 | clip=8  | consec=0 | max_consec=3 | avg_clip50=16.81 |

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|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 17.1 |
| clipped_episode_r... | 14   |
| consecutive_clipp... | 0    |
| max_consecutive_c... | 3    |
| rollout/             |      |
| ep_len_mean          | 723  |



|                  |        |
|------------------|--------|
| ep_rew_mean      | 156    |
| exploration rate | 0.806  |
| time/            |        |
| episodes         | 3700   |
| fps              | 243    |
| time_elapsed     | 802    |
| total timesteps  | 195699 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0499 |
| n_updates        | 42674  |

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EP=1,240 | STEP=196,337 | clip=6 | consec=0 | max\_consec=3 | avg\_clip50=17.07  
EP=1,250 | STEP=197,936 | clip=14 | consec=0 | max\_consec=3 | avg\_clip50=16.19  
EP=1,260 | STEP=199,666 | clip=57 | consec=1 | max\_consec=3 | avg\_clip50=17.94

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 18.5   |
| clipped_episode_r... | 51.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 732    |
| ep_rew_mean          | 168    |
| exploration rate     | 0.801  |
| time/                |        |
| episodes             | 3800   |
| fps                  | 243    |
| time_elapsed         | 827    |
| total timesteps      | 201501 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.177  |
| n_updates            | 44125  |

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EP=1,270 | STEP=201,616 | clip=15 | consec=0 | max\_consec=3 | avg\_clip50=18.11  
EP=1,280 | STEP=203,436 | clip=16 | consec=0 | max\_consec=3 | avg\_clip50=18.48  
EP=1,290 | STEP=204,953 | clip=6 | consec=0 | max\_consec=3 | avg\_clip50=17.93  
EP=1,300 | STEP=206,592 | clip=12 | consec=0 | max\_consec=3 | avg\_clip50=18.04

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 17.8   |
| clipped_episode_r... | 7.5    |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 3      |
| rollout/             |        |
| ep_len_mean          | 732    |
| ep_rew_mean          | 171    |
| exploration rate     | 0.795  |
| time/                |        |
| episodes             | 3900   |
| fps                  | 243    |
| time_elapsed         | 850    |
| total timesteps      | 206990 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0896 |
| n_updates            | 45497  |

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EP=1,310 | STEP=208,272 | clip=16 | consec=0 | max\_consec=3 | avg\_clip50=16.97

EP=1,320 | STEP=209,632 | clip=26 | consec=1 | max\_consec=3 | avg\_clip50=14.89  
EP=1,330 | STEP=211,130 | clip=15 | consec=0 | max\_consec=3 | avg\_clip50=13.17

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 13.4   |  |
| clipped_episode_r... | 10.5   |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 3      |  |
| rollout/             |        |  |
| ep_len_mean          | 704    |  |
| ep_rew_mean          | 156    |  |
| exploration rate     | 0.79   |  |
| time/                |        |  |
| episodes             | 4000   |  |
| fps                  | 242    |  |
| time_elapsed         | 873    |  |
| total timesteps      | 212194 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0752 |  |
| n_updates            | 46798  |  |

EP=1,340 | STEP=212,761 | clip=4 | consec=0 | max\_consec=3 | avg\_clip50=14.07  
EP=1,350 | STEP=214,780 | clip=16 | consec=0 | max\_consec=3 | avg\_clip50=15.36  
EP=1,360 | STEP=216,639 | clip=4 | consec=0 | max\_consec=3 | avg\_clip50=15.19

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 16.3   |  |
| clipped_episode_r... | 3      |  |
| consecutive_clipp... | 0      |  |
| max_consecutive_c... | 3      |  |
| rollout/             |        |  |
| ep_len_mean          | 715    |  |
| ep_rew_mean          | 155    |  |
| exploration rate     | 0.784  |  |
| time/                |        |  |
| episodes             | 4100   |  |
| fps                  | 242    |  |
| time_elapsed         | 899    |  |
| total timesteps      | 218225 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0892 |  |
| n_updates            | 48306  |  |

EP=1,370 | STEP=218,317 | clip=20 | consec=0 | max\_consec=3 | avg\_clip50=16.20  
EP=1,380 | STEP=220,180 | clip=16 | consec=0 | max\_consec=4 | avg\_clip50=17.59  
EP=1,390 | STEP=221,947 | clip=22 | consec=2 | max\_consec=4 | avg\_clip50=17.78  
EP=1,400 | STEP=223,825 | clip=6 | consec=0 | max\_consec=4 | avg\_clip50=17.78

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 17.4  |  |
| clipped_episode_r... | 8     |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 4     |  |
| rollout/             |       |  |
| ep_len_mean          | 734   |  |
| ep_rew_mean          | 166   |  |
| exploration rate     | 0.778 |  |
| time/                |       |  |

|                 |        |
|-----------------|--------|
| episodes        | 4200   |
| fps             | 242    |
| time_elapsed    | 926    |
| total timesteps | 224245 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0416 |
| n_updates       | 49811  |

EP=1,410 | STEP=225,489 | clip=12 | consec=0 | max\_consec=4 | avg\_clip50=17.54  
EP=1,420 | STEP=227,076 | clip=24 | consec=2 | max\_consec=4 | avg\_clip50=17.76  
EP=1,430 | STEP=228,874 | clip=22 | consec=1 | max\_consec=4 | avg\_clip50=17.31

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 17.2   |
| clipped_episode_r... | 19.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 4      |
| rollout/             |        |
| ep_len_mean          | 759    |
| ep_rew_mean          | 177    |
| exploration rate     | 0.772  |
| time/                |        |
| episodes             | 4300   |
| fps                  | 241    |
| time_elapsed         | 952    |
| total timesteps      | 230017 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.112  |
| n_updates            | 51254  |

EP=1,440 | STEP=230,752 | clip=54 | consec=1 | max\_consec=4 | avg\_clip50=17.75  
EP=1,450 | STEP=232,602 | clip=20 | consec=0 | max\_consec=4 | avg\_clip50=16.85  
EP=1,460 | STEP=234,440 | clip=26 | consec=1 | max\_consec=4 | avg\_clip50=17.54  
EP=1,470 | STEP=236,293 | clip=32 | consec=1 | max\_consec=4 | avg\_clip50=17.68

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 17.7   |
| clipped_episode_r... | 32.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 4      |
| rollout/             |        |
| ep_len_mean          | 765    |
| ep_rew_mean          | 177    |
| exploration rate     | 0.766  |
| time/                |        |
| episodes             | 4400   |
| fps                  | 241    |
| time_elapsed         | 980    |
| total timesteps      | 236293 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.183  |
| n_updates            | 52823  |

EP=1,480 | STEP=238,344 | clip=4 | consec=0 | max\_consec=4 | avg\_clip50=17.68  
EP=1,490 | STEP=240,366 | clip=9 | consec=0 | max\_consec=4 | avg\_clip50=17.22  
EP=1,500 | STEP=242,201 | clip=52 | consec=2 | max\_consec=4 | avg\_clip50=18.51

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 19.7   |
| clipped_episode_r... | 21     |
| consecutive_clipp... | 5      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 794    |
| ep_rew_mean          | 184    |
| exploration rate     | 0.759  |
| time/                |        |
| episodes             | 4500   |
| fps                  | 240    |
| time_elapsed         | 1009   |
| total timesteps      | 243067 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0774 |
| n_updates            | 54516  |

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EP=1,510 | STEP=244,173 | clip=15 | consec=0 | max\_consec=5 | avg\_clip50=19.01  
EP=1,520 | STEP=246,113 | clip=7 | consec=0 | max\_consec=5 | avg\_clip50=19.79  
EP=1,530 | STEP=247,971 | clip=13 | consec=0 | max\_consec=5 | avg\_clip50=20.28

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 20.7   |
| clipped_episode_r... | 25.5   |
| consecutive_clipp... | 4      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 820    |
| ep_rew_mean          | 192    |
| exploration rate     | 0.753  |
| time/                |        |
| episodes             | 4600   |
| fps                  | 240    |
| time_elapsed         | 1037   |
| total timesteps      | 249534 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0331 |
| n_updates            | 56133  |

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EP=1,540 | STEP=249,956 | clip=24 | consec=1 | max\_consec=5 | avg\_clip50=20.66  
EP=1,550 | STEP=251,863 | clip=35 | consec=1 | max\_consec=5 | avg\_clip50=20.26  
EP=1,560 | STEP=253,552 | clip=24 | consec=1 | max\_consec=5 | avg\_clip50=19.04  
EP=1,570 | STEP=255,137 | clip=38 | consec=1 | max\_consec=5 | avg\_clip50=17.76

|                      |       |
|----------------------|-------|
| custom/              |       |
| avg_clip50           | 17.8  |
| clipped_episode_r... | 38.5  |
| consecutive_clipp... | 1     |
| max_consecutive_c... | 5     |
| rollout/             |       |
| ep_len_mean          | 799   |
| ep_rew_mean          | 188   |
| exploration rate     | 0.747 |
| time/                |       |
| episodes             | 4700  |
| fps                  | 240   |
| time_elapsed         | 1063  |

|                 |        |
|-----------------|--------|
| total timesteps | 255285 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.119  |
| n_updates       | 57571  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,580 | STEP=257,411 | clip=16 | consec=0 | max_consec=5 | avg_clip50=18.62 |
| EP=1,590 | STEP=259,462 | clip=6  | consec=0 | max_consec=5 | avg_clip50=19.44 |
| EP=1,600 | STEP=261,394 | clip=10 | consec=0 | max_consec=5 | avg_clip50=19.40 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 19.8   |
| clipped_episode_r... | 14.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 800    |
| ep_rew_mean          | 193    |
| exploration rate     | 0.741  |
| time/                |        |
| episodes             | 4800   |
| fps                  | 239    |
| time_elapsed         | 1092   |
| total timesteps      | 262112 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.103  |
| n_updates            | 59277  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,610 | STEP=263,311 | clip=7  | consec=0 | max_consec=5 | avg_clip50=20.18 |
| EP=1,620 | STEP=265,042 | clip=12 | consec=0 | max_consec=5 | avg_clip50=20.37 |
| EP=1,630 | STEP=266,580 | clip=6  | consec=0 | max_consec=5 | avg_clip50=18.51 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 16.6   |
| clipped_episode_r... | 11     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 766    |
| ep_rew_mean          | 179    |
| exploration rate     | 0.735  |
| time/                |        |
| episodes             | 4900   |
| fps                  | 239    |
| time_elapsed         | 1117   |
| total timesteps      | 267617 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.078  |
| n_updates            | 60654  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,640 | STEP=267,885 | clip=8  | consec=0 | max_consec=5 | avg_clip50=16.16 |
| EP=1,650 | STEP=269,804 | clip=34 | consec=1 | max_consec=5 | avg_clip50=15.74 |
| EP=1,660 | STEP=271,964 | clip=4  | consec=0 | max_consec=5 | avg_clip50=16.20 |
| EP=1,670 | STEP=273,733 | clip=11 | consec=0 | max_consec=5 | avg_clip50=16.37 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 16.4 |
| clipped_episode_r... | 16.5 |

|                      |        |
|----------------------|--------|
| consecutive_clipp... | 0      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 788    |
| ep_rew_mean          | 184    |
| exploration rate     | 0.729  |
| time/                |        |
| episodes             | 5000   |
| fps                  | 239    |
| time_elapsed         | 1144   |
| total timesteps      | 273849 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0422 |
| n_updates            | 62212  |

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EP=1,680 | STEP=275,687 | clip=34 | consec=4 | max\_consec=5 | avg\_clip50=18.25  
EP=1,690 | STEP=277,533 | clip=28 | consec=3 | max\_consec=5 | avg\_clip50=19.98  
EP=1,700 | STEP=279,342 | clip=8 | consec=0 | max\_consec=5 | avg\_clip50=19.40

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 19.2   |
| clipped_episode_r... | 11.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 767    |
| ep_rew_mean          | 176    |
| exploration rate     | 0.723  |
| time/                |        |
| episodes             | 5100   |
| fps                  | 239    |
| time_elapsed         | 1172   |
| total timesteps      | 280119 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.259  |
| n_updates            | 63779  |

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EP=1,710 | STEP=281,156 | clip=20 | consec=0 | max\_consec=5 | avg\_clip50=18.72  
EP=1,720 | STEP=282,694 | clip=10 | consec=0 | max\_consec=5 | avg\_clip50=17.78  
EP=1,730 | STEP=284,432 | clip=8 | consec=0 | max\_consec=5 | avg\_clip50=15.27

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 15.3   |
| clipped_episode_r... | 21     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 771    |
| ep_rew_mean          | 170    |
| exploration rate     | 0.717  |
| time/                |        |
| episodes             | 5200   |
| fps                  | 238    |
| time_elapsed         | 1195   |
| total timesteps      | 285708 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0755 |

|                      |              |         |          |              |                  |  |
|----------------------|--------------|---------|----------|--------------|------------------|--|
| n_updates            |              | 65176   |          |              |                  |  |
|                      |              |         |          |              |                  |  |
| EP=1,740             | STEP=286,304 | clip=18 | consec=0 | max_consec=5 | avg_clip50=15.26 |  |
| EP=1,750             | STEP=288,110 | clip=14 | consec=0 | max_consec=5 | avg_clip50=16.42 |  |
| EP=1,760             | STEP=290,415 | clip=14 | consec=0 | max_consec=5 | avg_clip50=17.82 |  |
| EP=1,770             | STEP=292,416 | clip=40 | consec=1 | max_consec=5 | avg_clip50=19.98 |  |
|                      |              |         |          |              |                  |  |
| custom/              |              |         |          |              |                  |  |
| avg_clip50           |              | 19.9    |          |              |                  |  |
| clipped_episode_r... |              | 8.5     |          |              |                  |  |
| consecutive_clipp... |              | 0       |          |              |                  |  |
| max_consecutive_c... |              | 5       |          |              |                  |  |
| rollout/             |              |         |          |              |                  |  |
| ep_len_mean          |              | 796     |          |              |                  |  |
| ep_rew_mean          |              | 188     |          |              |                  |  |
| exploration rate     |              | 0.71    |          |              |                  |  |
| time/                |              |         |          |              |                  |  |
| episodes             |              | 5300    |          |              |                  |  |
| fps                  |              | 238     |          |              |                  |  |
| time_elapsed         |              | 1226    |          |              |                  |  |
| total timesteps      |              | 292626  |          |              |                  |  |
| train/               |              |         |          |              |                  |  |
| learning_rate        |              | 0.0001  |          |              |                  |  |
| loss                 |              | 0.155   |          |              |                  |  |
| n_updates            |              | 66906   |          |              |                  |  |
|                      |              |         |          |              |                  |  |
| EP=1,780             | STEP=294,400 | clip=11 | consec=0 | max_consec=5 | avg_clip50=21.16 |  |
| EP=1,790             | STEP=296,370 | clip=17 | consec=0 | max_consec=5 | avg_clip50=20.66 |  |
| EP=1,800             | STEP=298,182 | clip=18 | consec=0 | max_consec=5 | avg_clip50=20.84 |  |
|                      |              |         |          |              |                  |  |
| custom/              |              |         |          |              |                  |  |
| avg_clip50           |              | 21.4    |          |              |                  |  |
| clipped_episode_r... |              | 46.5    |          |              |                  |  |
| consecutive_clipp... |              | 1       |          |              |                  |  |
| max_consecutive_c... |              | 5       |          |              |                  |  |
| rollout/             |              |         |          |              |                  |  |
| ep_len_mean          |              | 806     |          |              |                  |  |
| ep_rew_mean          |              | 191     |          |              |                  |  |
| exploration rate     |              | 0.704   |          |              |                  |  |
| time/                |              |         |          |              |                  |  |
| episodes             |              | 5400    |          |              |                  |  |
| fps                  |              | 238     |          |              |                  |  |
| time_elapsed         |              | 1255    |          |              |                  |  |
| total timesteps      |              | 299144  |          |              |                  |  |
| train/               |              |         |          |              |                  |  |
| learning_rate        |              | 0.0001  |          |              |                  |  |
| loss                 |              | 0.0163  |          |              |                  |  |
| n_updates            |              | 68535   |          |              |                  |  |
|                      |              |         |          |              |                  |  |
| EP=1,810             | STEP=300,385 | clip=27 | consec=1 | max_consec=5 | avg_clip50=20.84 |  |
| EP=1,820             | STEP=302,237 | clip=4  | consec=0 | max_consec=5 | avg_clip50=21.08 |  |
| EP=1,830             | STEP=304,271 | clip=10 | consec=0 | max_consec=5 | avg_clip50=21.90 |  |
|                      |              |         |          |              |                  |  |
| custom/              |              |         |          |              |                  |  |
| avg_clip50           |              | 21.9    |          |              |                  |  |
| clipped_episode_r... |              | 24.5    |          |              |                  |  |
| consecutive_clipp... |              | 1       |          |              |                  |  |
| max_consecutive_c... |              | 5       |          |              |                  |  |
| rollout/             |              |         |          |              |                  |  |
| ep_len_mean          |              | 834     |          |              |                  |  |

|                  |        |
|------------------|--------|
| ep_rew_mean      | 215    |
| exploration rate | 0.697  |
| time/            |        |
| episodes         | 5500   |
| fps              | 238    |
| time_elapsed     | 1283   |
| total timesteps  | 305606 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0612 |
| n_updates        | 70151  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,840 | STEP=306,046 | clip=14 | consec=0 | max_consec=5 | avg_clip50=22.28 |
| EP=1,850 | STEP=308,310 | clip=16 | consec=0 | max_consec=5 | avg_clip50=22.69 |
| EP=1,860 | STEP=310,594 | clip=22 | consec=2 | max_consec=5 | avg_clip50=23.34 |
| EP=1,870 | STEP=312,709 | clip=22 | consec=1 | max_consec=5 | avg_clip50=22.65 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22     |
| clipped_episode_r... | 17     |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 861    |
| ep_rew_mean          | 220    |
| exploration rate     | 0.69   |
| time/                |        |
| episodes             | 5600   |
| fps                  | 237    |
| time_elapsed         | 1315   |
| total timesteps      | 312979 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0872 |
| n_updates            | 71994  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=1,880 | STEP=314,862 | clip=36 | consec=1 | max_consec=5 | avg_clip50=22.65 |
| EP=1,890 | STEP=316,798 | clip=6  | consec=0 | max_consec=5 | avg_clip50=23.11 |
| EP=1,900 | STEP=318,900 | clip=32 | consec=2 | max_consec=5 | avg_clip50=23.70 |

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|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.2   |
| clipped_episode_r... | 24.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 864    |
| ep_rew_mean          | 226    |
| exploration rate     | 0.683  |
| time/                |        |
| episodes             | 5700   |
| fps                  | 237    |
| time_elapsed         | 1344   |
| total timesteps      | 319724 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.052  |
| n_updates            | 73680  |

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|          |              |        |          |              |                  |
|----------|--------------|--------|----------|--------------|------------------|
| EP=1,910 | STEP=320,614 | clip=4 | consec=0 | max_consec=5 | avg_clip50=21.33 |
|----------|--------------|--------|----------|--------------|------------------|



EP=1,920 | STEP=322,423 | clip=23 | consec=1 | max\_consec=5 | avg\_clip50=20.89  
EP=1,930 | STEP=324,331 | clip=32 | consec=3 | max\_consec=5 | avg\_clip50=20.75

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 19.8   |  |
| clipped_episode_r... | 25     |  |
| consecutive_clipp... | 1      |  |
| max_consecutive_c... | 5      |  |
| rollout/             |        |  |
| ep_len_mean          | 849    |  |
| ep_rew_mean          | 215    |  |
| exploration rate     | 0.677  |  |
| time/                |        |  |
| episodes             | 5800   |  |
| fps                  | 237    |  |
| time_elapsed         | 1370   |  |
| total timesteps      | 325895 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0517 |  |
| n_updates            | 75223  |  |

EP=1,940 | STEP=325,978 | clip=8 | consec=0 | max\_consec=5 | avg\_clip50=19.85  
EP=1,950 | STEP=327,902 | clip=34 | consec=3 | max\_consec=5 | avg\_clip50=18.45  
EP=1,960 | STEP=329,774 | clip=20 | consec=1 | max\_consec=5 | avg\_clip50=19.41  
EP=1,970 | STEP=331,687 | clip=10 | consec=0 | max\_consec=5 | avg\_clip50=19.94

|                      |        |  |
|----------------------|--------|--|
| -----                |        |  |
| custom/              |        |  |
| avg_clip50           | 21.1   |  |
| clipped_episode_r... | 42     |  |
| consecutive_clipp... | 2      |  |
| max_consecutive_c... | 5      |  |
| rollout/             |        |  |
| ep_len_mean          | 804    |  |
| ep_rew_mean          | 208    |  |
| exploration rate     | 0.671  |  |
| time/                |        |  |
| episodes             | 5900   |  |
| fps                  | 237    |  |
| time_elapsed         | 1398   |  |
| total timesteps      | 332211 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.169  |  |
| n_updates            | 76802  |  |

EP=1,980 | STEP=333,706 | clip=24 | consec=2 | max\_consec=5 | avg\_clip50=21.20  
EP=1,990 | STEP=335,608 | clip=45 | consec=1 | max\_consec=5 | avg\_clip50=21.85  
EP=2,000 | STEP=337,657 | clip=31 | consec=1 | max\_consec=5 | avg\_clip50=21.47

|                      |       |  |
|----------------------|-------|--|
| -----                |       |  |
| custom/              |       |  |
| avg_clip50           | 21.4  |  |
| clipped_episode_r... | 12    |  |
| consecutive_clipp... | 0     |  |
| max_consecutive_c... | 5     |  |
| rollout/             |       |  |
| ep_len_mean          | 803   |  |
| ep_rew_mean          | 200   |  |
| exploration rate     | 0.665 |  |
| time/                |       |  |

|                 |        |
|-----------------|--------|
| episodes        | 6000   |
| fps             | 237    |
| time_elapsed    | 1428   |
| total timesteps | 338684 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.168  |
| n_updates       | 78420  |

EP=2,010 | STEP=339,591 | clip=25 | consec=2 | max\_consec=5 | avg\_clip50=21.00  
EP=2,020 | STEP=341,261 | clip=4 | consec=0 | max\_consec=5 | avg\_clip50=19.62  
EP=2,030 | STEP=343,396 | clip=27 | consec=2 | max\_consec=5 | avg\_clip50=18.52

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 19.4   |
| clipped_episode_r... | 20.5   |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 818    |
| ep_rew_mean          | 202    |
| exploration rate     | 0.658  |
| time/                |        |
| episodes             | 6100   |
| fps                  | 236    |
| time_elapsed         | 1458   |
| total timesteps      | 345190 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0613 |
| n_updates            | 80047  |

EP=2,040 | STEP=345,454 | clip=32 | consec=3 | max\_consec=5 | avg\_clip50=19.11  
EP=2,050 | STEP=347,466 | clip=22 | consec=2 | max\_consec=5 | avg\_clip50=20.09  
EP=2,060 | STEP=349,714 | clip=32 | consec=1 | max\_consec=5 | avg\_clip50=21.25  
EP=2,070 | STEP=351,739 | clip=24 | consec=3 | max\_consec=5 | avg\_clip50=22.77

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 22.6   |
| clipped_episode_r... | 48     |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 5      |
| rollout/             |        |
| ep_len_mean          | 841    |
| ep_rew_mean          | 204    |
| exploration rate     | 0.651  |
| time/                |        |
| episodes             | 6200   |
| fps                  | 236    |
| time_elapsed         | 1490   |
| total timesteps      | 352654 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.101  |
| n_updates            | 81913  |

EP=2,080 | STEP=353,901 | clip=14 | consec=0 | max\_consec=5 | avg\_clip50=23.23  
EP=2,090 | STEP=356,311 | clip=40 | consec=7 | max\_consec=7 | avg\_clip50=24.91  
EP=2,100 | STEP=358,526 | clip=20 | consec=3 | max\_consec=7 | avg\_clip50=25.78

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 25.3   |
| clipped_episode_r... | 31     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 884    |
| ep_rew_mean          | 235    |
| exploration rate     | 0.644  |
| time/                |        |
| episodes             | 6300   |
| fps                  | 236    |
| time_elapsed         | 1524   |
| total timesteps      | 360101 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.134  |
| n_updates            | 83775  |

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EP=2,110 | STEP=360,587 | clip=10 | consec=0 | max\_consec=7 | avg\_clip50=26.39  
EP=2,120 | STEP=362,294 | clip=4 | consec=0 | max\_consec=7 | avg\_clip50=25.40  
EP=2,130 | STEP=364,262 | clip=26 | consec=1 | max\_consec=7 | avg\_clip50=24.28  
EP=2,140 | STEP=366,075 | clip=2 | consec=0 | max\_consec=7 | avg\_clip50=21.75

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 21.5   |
| clipped_episode_r... | 12.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 864    |
| ep_rew_mean          | 232    |
| exploration rate     | 0.637  |
| time/                |        |
| episodes             | 6400   |
| fps                  | 236    |
| time_elapsed         | 1552   |
| total timesteps      | 366398 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.153  |
| n_updates            | 85349  |

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EP=2,150 | STEP=367,950 | clip=12 | consec=0 | max\_consec=7 | avg\_clip50=20.79  
EP=2,160 | STEP=370,253 | clip=48 | consec=1 | max\_consec=7 | avg\_clip50=22.31  
EP=2,170 | STEP=372,355 | clip=12 | consec=0 | max\_consec=7 | avg\_clip50=23.95

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 24.3 |
| clipped_episode_r... | 38   |
| consecutive_clipp... | 2    |
| max_consecutive_c... | 7    |
| rollout/             |      |
| ep_len_mean          | 882  |
| ep_rew_mean          | 249  |
| exploration rate     | 0.63 |
| time/                |      |
| episodes             | 6500 |
| fps                  | 235  |
| time_elapsed         | 1584 |

|                 |        |
|-----------------|--------|
| total timesteps | 373898 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.243  |
| n_updates       | 87224  |

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|          |              |         |          |              |                  |
|----------|--------------|---------|----------|--------------|------------------|
| EP=2,180 | STEP=374,539 | clip=13 | consec=0 | max_consec=7 | avg_clip50=24.85 |
| EP=2,190 | STEP=376,657 | clip=15 | consec=0 | max_consec=7 | avg_clip50=26.36 |
| EP=2,200 | STEP=378,742 | clip=5  | consec=0 | max_consec=7 | avg_clip50=27.60 |
| EP=2,210 | STEP=380,903 | clip=44 | consec=1 | max_consec=7 | avg_clip50=26.42 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.4   |
| clipped_episode_r... | 44.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 7      |
| rollout/             |        |
| ep_len_mean          | 859    |
| ep_rew_mean          | 244    |
| exploration rate     | 0.623  |
| time/                |        |
| episodes             | 6600   |
| fps                  | 235    |
| time_elapsed         | 1615   |
| total timesteps      | 380949 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.148  |
| n_updates            | 88987  |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=2,220 | STEP=382,902 | clip=9  | consec=0  | max_consec=7  | avg_clip50=26.08 |
| EP=2,230 | STEP=385,474 | clip=36 | consec=10 | max_consec=10 | avg_clip50=29.25 |
| EP=2,240 | STEP=387,932 | clip=19 | consec=0  | max_consec=10 | avg_clip50=30.17 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 29.6   |
| clipped_episode_r... | 18.5   |
| consecutive_clipp... | 0      |
| max_consecutive_c... | 10     |
| rollout/             |        |
| ep_len_mean          | 924    |
| ep_rew_mean          | 284    |
| exploration rate     | 0.615  |
| time/                |        |
| episodes             | 6700   |
| fps                  | 235    |
| time_elapsed         | 1650   |
| total timesteps      | 388732 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.039  |
| n_updates            | 90932  |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=2,250 | STEP=390,267 | clip=39 | consec=6  | max_consec=10 | avg_clip50=30.67 |
| EP=2,260 | STEP=392,724 | clip=22 | consec=16 | max_consec=16 | avg_clip50=30.65 |
| EP=2,270 | STEP=395,158 | clip=24 | consec=6  | max_consec=16 | avg_clip50=31.67 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 30.9 |
| clipped_episode_r... | 42   |

|                      |        |
|----------------------|--------|
| consecutive_clipp... | 1      |
| max_consecutive_c... | 16     |
| rollout/             |        |
| ep_len_mean          | 964    |
| ep_rew_mean          | 298    |
| exploration rate     | 0.607  |
| time/                |        |
| episodes             | 6800   |
| fps                  | 235    |
| time_elapsed         | 1687   |
| total timesteps      | 397438 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0415 |
| n_updates            | 93109  |

EP=2,280 | STEP=397,594 | clip=24 | consec=2 | max\_consec=16 | avg\_clip50=30.62  
EP=2,290 | STEP=400,027 | clip=4 | consec=0 | max\_consec=16 | avg\_clip50=30.77  
EP=2,300 | STEP=402,504 | clip=8 | consec=0 | max\_consec=16 | avg\_clip50=29.98  
EP=2,310 | STEP=404,438 | clip=12 | consec=0 | max\_consec=16 | avg\_clip50=28.16

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 28.4   |
| clipped_episode_r... | 35.5   |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 16     |
| rollout/             |        |
| ep_len_mean          | 978    |
| ep_rew_mean          | 294    |
| exploration rate     | 0.599  |
| time/                |        |
| episodes             | 6900   |
| fps                  | 235    |
| time_elapsed         | 1722   |
| total timesteps      | 405225 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0355 |
| n_updates            | 95056  |

EP=2,320 | STEP=406,987 | clip=52 | consec=4 | max\_consec=16 | avg\_clip50=29.03  
EP=2,330 | STEP=409,412 | clip=30 | consec=5 | max\_consec=16 | avg\_clip50=28.24  
EP=2,340 | STEP=411,100 | clip=8 | consec=0 | max\_consec=16 | avg\_clip50=25.65

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.1   |
| clipped_episode_r... | 21     |
| consecutive_clipp... | 6      |
| max_consecutive_c... | 16     |
| rollout/             |        |
| ep_len_mean          | 979    |
| ep_rew_mean          | 286    |
| exploration rate     | 0.591  |
| time/                |        |
| episodes             | 7000   |
| fps                  | 235    |
| time_elapsed         | 1755   |
| total timesteps      | 413048 |
| train/               |        |
| learning_rate        | 0.0001 |

|           |        |
|-----------|--------|
| loss      | 0.0575 |
| n_updates | 97011  |

|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,350 | STEP=413,767 | clip=28 | consec=8 | max_consec=16 | avg_clip50=27.32 |
| EP=2,360 | STEP=416,277 | clip=28 | consec=3 | max_consec=16 | avg_clip50=28.64 |
| EP=2,370 | STEP=418,399 | clip=14 | consec=0 | max_consec=16 | avg_clip50=27.53 |
| EP=2,380 | STEP=420,552 | clip=15 | consec=0 | max_consec=16 | avg_clip50=27.06 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.6   |
| clipped_episode_r... | 25     |
| consecutive_clipp... | 1      |
| max_consecutive_c... | 16     |
| rollout/             |        |
| ep_len_mean          | 963    |
| ep_rew_mean          | 275    |
| exploration rate     | 0.583  |
| time/                |        |
| episodes             | 7100   |
| fps                  | 235    |
| time_elapsed         | 1790   |
| total timesteps      | 420953 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0521 |
| n_updates            | 98988  |

|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,390 | STEP=422,944 | clip=2  | consec=0 | max_consec=16 | avg_clip50=29.62 |
| EP=2,400 | STEP=425,042 | clip=12 | consec=0 | max_consec=16 | avg_clip50=27.28 |
| EP=2,410 | STEP=426,948 | clip=2  | consec=0 | max_consec=16 | avg_clip50=26.07 |

|                      |        |
|----------------------|--------|
| custom/              |        |
| avg_clip50           | 26.8   |
| clipped_episode_r... | 34     |
| consecutive_clipp... | 2      |
| max_consecutive_c... | 16     |
| rollout/             |        |
| ep_len_mean          | 952    |
| ep_rew_mean          | 280    |
| exploration rate     | 0.576  |
| time/                |        |
| episodes             | 7200   |
| fps                  | 234    |
| time_elapsed         | 1823   |
| total timesteps      | 428357 |
| train/               |        |
| learning_rate        | 0.0001 |
| loss                 | 0.0533 |
| n_updates            | 100839 |

|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,420 | STEP=428,998 | clip=14 | consec=0 | max_consec=16 | avg_clip50=26.54 |
| EP=2,430 | STEP=431,538 | clip=24 | consec=6 | max_consec=16 | avg_clip50=27.03 |
| EP=2,440 | STEP=433,934 | clip=18 | consec=0 | max_consec=16 | avg_clip50=26.93 |
| EP=2,450 | STEP=436,667 | clip=34 | consec=4 | max_consec=16 | avg_clip50=29.13 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 29.1 |
| clipped_episode_r... | 33.5 |
| consecutive_clipp... | 4    |
| max_consecutive_c... | 16   |

|                  |        |  |
|------------------|--------|--|
| rollout/         |        |  |
| ep_len_mean      | 960    |  |
| ep_rew_mean      | 282    |  |
| exploration rate | 0.568  |  |
| time/            |        |  |
| episodes         | 7300   |  |
| fps              | 234    |  |
| time_elapsed     | 1860   |  |
| total timesteps  | 436667 |  |
| train/           |        |  |
| learning_rate    | 0.0001 |  |
| loss             | 0.0441 |  |
| n_updates        | 102916 |  |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,460 | STEP=439,105 | clip=43 | consec=2 | max_consec=16 | avg_clip50=31.41 |
| EP=2,470 | STEP=441,571 | clip=30 | consec=5 | max_consec=16 | avg_clip50=32.76 |
| EP=2,480 | STEP=444,020 | clip=56 | consec=2 | max_consec=16 | avg_clip50=32.92 |

|                      |        |  |
|----------------------|--------|--|
| custom/              |        |  |
| avg_clip50           | 33.3   |  |
| clipped_episode_r... | 43.5   |  |
| consecutive_clipp... | 2      |  |
| max_consecutive_c... | 16     |  |
| rollout/             |        |  |
| ep_len_mean          | 987    |  |
| ep_rew_mean          | 300    |  |
| exploration rate     | 0.559  |  |
| time/                |        |  |
| episodes             | 7400   |  |
| fps                  | 234    |  |
| time_elapsed         | 1898   |  |
| total timesteps      | 445343 |  |
| train/               |        |  |
| learning_rate        | 0.0001 |  |
| loss                 | 0.0325 |  |
| n_updates            | 105085 |  |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,490 | STEP=446,742 | clip=56 | consec=1 | max_consec=16 | avg_clip50=34.72 |
| EP=2,500 | STEP=449,133 | clip=41 | consec=1 | max_consec=16 | avg_clip50=33.09 |
| EP=2,510 | STEP=451,548 | clip=29 | consec=3 | max_consec=16 | avg_clip50=33.22 |

|                      |          |  |
|----------------------|----------|--|
| custom/              |          |  |
| avg_clip50           | 32.8     |  |
| clipped_episode_r... | 24       |  |
| consecutive_clipp... | 1        |  |
| max_consecutive_c... | 16       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.04e+03 |  |
| ep_rew_mean          | 329      |  |
| exploration rate     | 0.551    |  |
| time/                |          |  |
| episodes             | 7500     |  |
| fps                  | 234      |  |
| time_elapsed         | 1935     |  |
| total timesteps      | 453739   |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0385   |  |
| n_updates            | 107184   |  |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,520 | STEP=454,274 | clip=39 | consec=3 | max_consec=16 | avg_clip50=33.18 |
| EP=2,530 | STEP=456,598 | clip=32 | consec=1 | max_consec=16 | avg_clip50=32.35 |
| EP=2,540 | STEP=458,518 | clip=26 | consec=1 | max_consec=16 | avg_clip50=29.04 |
| EP=2,550 | STEP=460,972 | clip=36 | consec=2 | max_consec=16 | avg_clip50=30.68 |

|                      |          |
|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 32       |
| clipped_episode_r... | 62.5     |
| consecutive_clipp... | 4        |
| max_consecutive_c... | 16       |
| rollout/             |          |
| ep_len_mean          | 1.02e+03 |
| ep_rew_mean          | 323      |
| exploration rate     | 0.543    |
| time/                |          |
| episodes             | 7600     |
| fps                  | 234      |
| time_elapsed         | 1972     |
| total timesteps      | 462025   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0853   |
| n_updates            | 109256   |

|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=2,560 | STEP=464,090 | clip=32 | consec=12 | max_consec=16 | avg_clip50=32.99 |
| EP=2,570 | STEP=466,660 | clip=24 | consec=1  | max_consec=16 | avg_clip50=32.34 |
| EP=2,580 | STEP=469,100 | clip=54 | consec=1  | max_consec=16 | avg_clip50=32.17 |

|                      |          |
|----------------------|----------|
| -----                |          |
| custom/              |          |
| avg_clip50           | 34.4     |
| clipped_episode_r... | 30.5     |
| consecutive_clipp... | 4        |
| max_consecutive_c... | 16       |
| rollout/             |          |
| ep_len_mean          | 1.04e+03 |
| ep_rew_mean          | 321      |
| exploration rate     | 0.533    |
| time/                |          |
| episodes             | 7700     |
| fps                  | 234      |
| time_elapsed         | 2013     |
| total timesteps      | 471268   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.056    |
| n_updates            | 111566   |

|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,590 | STEP=471,790 | clip=23 | consec=7 | max_consec=16 | avg_clip50=35.11 |
| EP=2,600 | STEP=474,483 | clip=56 | consec=3 | max_consec=16 | avg_clip50=34.64 |
| EP=2,610 | STEP=477,256 | clip=40 | consec=3 | max_consec=16 | avg_clip50=33.16 |
| EP=2,620 | STEP=479,977 | clip=76 | consec=2 | max_consec=16 | avg_clip50=34.26 |

|                      |          |
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| -----                |          |
| custom/              |          |
| avg_clip50           | 33.4     |
| clipped_episode_r... | 8.5      |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 16       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 332      |



|                  |        |
|------------------|--------|
| exploration rate | 0.525  |
| time/            |        |
| episodes         | 7800   |
| fps              | 233    |
| time_elapsed     | 2052   |
| total timesteps  | 480184 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0622 |
| n_updates        | 113795 |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=2,630 | STEP=482,279 | clip=24 | consec=5  | max_consec=16 | avg_clip50=34.66 |
| EP=2,640 | STEP=484,595 | clip=20 | consec=0  | max_consec=16 | avg_clip50=31.20 |
| EP=2,650 | STEP=487,131 | clip=22 | consec=10 | max_consec=16 | avg_clip50=30.60 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 30       |
| clipped_episode_r... | 55       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 16       |
| rollout/             |          |
| ep_len_mean          | 1.08e+03 |
| ep_rew_mean          | 318      |
| exploration rate     | 0.516    |
| time/                |          |
| episodes             | 7900     |
| fps                  | 233      |
| time_elapsed         | 2088     |
| total timesteps      | 488447   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.271    |
| n_updates            | 115861   |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,660 | STEP=489,741 | clip=80 | consec=2 | max_consec=16 | avg_clip50=30.62 |
| EP=2,670 | STEP=491,871 | clip=10 | consec=0 | max_consec=16 | avg_clip50=28.27 |
| EP=2,680 | STEP=494,694 | clip=74 | consec=2 | max_consec=16 | avg_clip50=31.35 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 33.1     |
| clipped_episode_r... | 23       |
| consecutive_clipp... | 1        |
| max_consecutive_c... | 16       |
| rollout/             |          |
| ep_len_mean          | 1.05e+03 |
| ep_rew_mean          | 322      |
| exploration rate     | 0.508    |
| time/                |          |
| episodes             | 8000     |
| fps                  | 233      |
| time_elapsed         | 2127     |
| total timesteps      | 496792   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.243    |
| n_updates            | 117947   |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,690 | STEP=497,097 | clip=38 | consec=2 | max_consec=16 | avg_clip50=33.44 |
| EP=2,700 | STEP=499,862 | clip=52 | consec=3 | max_consec=16 | avg_clip50=34.66 |
| EP=2,710 | STEP=502,588 | clip=34 | consec=2 | max_consec=16 | avg_clip50=35.93 |

EP=2,720 | STEP=505,190 | clip=13 | consec=0 | max\_consec=16 | avg\_clip50=36.49

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 34.8     |  |
| clipped_episode_r... | 15.5     |  |
| consecutive_clipp... | 0        |  |
| max_consecutive_c... | 16       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.05e+03 |  |
| ep_rew_mean          | 324      |  |
| exploration rate     | 0.499    |  |
| time/                |          |  |
| episodes             | 8100     |  |
| fps                  | 233      |  |
| time_elapsed         | 2167     |  |
| total timesteps      | 505783   |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0473   |  |
| n_updates            | 120195   |  |

EP=2,730 | STEP=507,710 | clip=44 | consec=4 | max\_consec=16 | avg\_clip50=34.99

EP=2,740 | STEP=510,346 | clip=38 | consec=6 | max\_consec=16 | avg\_clip50=35.91

EP=2,750 | STEP=512,405 | clip=9 | consec=0 | max\_consec=16 | avg\_clip50=33.95

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| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 34.4     |  |
| clipped_episode_r... | 47.5     |  |
| consecutive_clipp... | 6        |  |
| max_consecutive_c... | 16       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.07e+03 |  |
| ep_rew_mean          | 351      |  |
| exploration rate     | 0.491    |  |
| time/                |          |  |
| episodes             | 8200     |  |
| fps                  | 233      |  |
| time_elapsed         | 2205     |  |
| total timesteps      | 514283   |  |
| train/               |          |  |
| learning_rate        | 0.0001   |  |
| loss                 | 0.0295   |  |
| n_updates            | 122320   |  |

EP=2,760 | STEP=515,508 | clip=58 | consec=10 | max\_consec=16 | avg\_clip50=34.34

EP=2,770 | STEP=517,862 | clip=41 | consec=6 | max\_consec=16 | avg\_clip50=35.83

EP=2,780 | STEP=520,107 | clip=27 | consec=4 | max\_consec=16 | avg\_clip50=35.45

EP=2,790 | STEP=522,568 | clip=18 | consec=0 | max\_consec=16 | avg\_clip50=34.46

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|----------------------|----------|--|
| -----                |          |  |
| custom/              |          |  |
| avg_clip50           | 34.5     |  |
| clipped_episode_r... | 18       |  |
| consecutive_clipp... | 0        |  |
| max_consecutive_c... | 16       |  |
| rollout/             |          |  |
| ep_len_mean          | 1.07e+03 |  |
| ep_rew_mean          | 352      |  |
| exploration rate     | 0.482    |  |
| time/                |          |  |
| episodes             | 8300     |  |

|                 |        |
|-----------------|--------|
| fps             | 233    |
| time_elapsed    | 2243   |
| total timesteps | 522731 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0509 |
| n_updates       | 124432 |

EP=2,800 | STEP=525,129 | clip=6 | consec=0 | max\_consec=16 | avg\_clip50=36.33  
EP=2,810 | STEP=527,978 | clip=29 | consec=10 | max\_consec=16 | avg\_clip50=34.71  
EP=2,820 | STEP=530,775 | clip=62 | consec=7 | max\_consec=16 | avg\_clip50=35.47

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 35       |
| clipped_episode_r... | 65.5     |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 16       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 360      |
| exploration rate     | 0.473    |
| time/                |          |
| episodes             | 8400     |
| fps                  | 232      |
| time_elapsed         | 2284     |
| total timesteps      | 532001   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0404   |
| n_updates            | 126750   |

EP=2,830 | STEP=533,368 | clip=26 | consec=1 | max\_consec=16 | avg\_clip50=35.15  
EP=2,840 | STEP=535,801 | clip=79 | consec=2 | max\_consec=16 | avg\_clip50=36.40  
EP=2,850 | STEP=538,388 | clip=15 | consec=0 | max\_consec=16 | avg\_clip50=37.70

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.4     |
| clipped_episode_r... | 41       |
| consecutive_clipp... | 9        |
| max_consecutive_c... | 16       |
| rollout/             |          |
| ep_len_mean          | 1.07e+03 |
| ep_rew_mean          | 368      |
| exploration rate     | 0.465    |
| time/                |          |
| episodes             | 8500     |
| fps                  | 232      |
| time_elapsed         | 2324     |
| total timesteps      | 540894   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0232   |
| n_updates            | 128973   |

EP=2,860 | STEP=541,071 | clip=14 | consec=0 | max\_consec=16 | avg\_clip50=38.07  
EP=2,870 | STEP=543,642 | clip=80 | consec=7 | max\_consec=16 | avg\_clip50=37.64  
EP=2,880 | STEP=546,572 | clip=22 | consec=17 | max\_consec=17 | avg\_clip50=39.26  
EP=2,890 | STEP=549,008 | clip=20 | consec=0 | max\_consec=19 | avg\_clip50=38.38

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|---------|--|
| custom/ |  |
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|----------------------|---------|
| avg_clip50           | 39.2    |
| clipped_episode_r... | 8       |
| consecutive_clipp... | 0       |
| max_consecutive_c... | 19      |
| rollout/             |         |
| ep_len_mean          | 1.1e+03 |
| ep_rew_mean          | 377     |
| exploration rate     | 0.456   |
| time/                |         |
| episodes             | 8600    |
| fps                  | 232     |
| time_elapsed         | 2364    |
| total timesteps      | 549928  |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.295   |
| n_updates            | 131231  |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,900 | STEP=551,738 | clip=40 | consec=3 | max_consec=19 | avg_clip50=38.26 |
| EP=2,910 | STEP=553,957 | clip=19 | consec=0 | max_consec=19 | avg_clip50=35.90 |
| EP=2,920 | STEP=556,396 | clip=22 | consec=5 | max_consec=19 | avg_clip50=34.41 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 33.1     |
| clipped_episode_r... | 28.5     |
| consecutive_clipp... | 7        |
| max_consecutive_c... | 19       |
| rollout/             |          |
| ep_len_mean          | 1.06e+03 |
| ep_rew_mean          | 359      |
| exploration rate     | 0.447    |
| time/                |          |
| episodes             | 8700     |
| fps                  | 232      |
| time_elapsed         | 2403     |
| total timesteps      | 558449   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.214    |
| n_updates            | 133362   |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=2,930 | STEP=559,014 | clip=46 | consec=9  | max_consec=19 | avg_clip50=33.53 |
| EP=2,940 | STEP=562,048 | clip=34 | consec=19 | max_consec=19 | avg_clip50=35.72 |
| EP=2,950 | STEP=565,163 | clip=50 | consec=29 | max_consec=29 | avg_clip50=36.13 |
| EP=2,960 | STEP=567,882 | clip=32 | consec=2  | max_consec=33 | avg_clip50=37.89 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 38.2     |
| clipped_episode_r... | 19       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.12e+03 |
| ep_rew_mean          | 376      |
| exploration rate     | 0.437    |
| time/                |          |
| episodes             | 8800     |
| fps                  | 232      |
| time_elapsed         | 2448     |
| total timesteps      | 568591   |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0362 |
| n_updates     | 135897 |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=2,970 | STEP=570,360 | clip=59 | consec=1 | max_consec=33 | avg_clip50=38.73 |
| EP=2,980 | STEP=573,010 | clip=52 | consec=4 | max_consec=33 | avg_clip50=39.80 |
| EP=2,990 | STEP=575,406 | clip=78 | consec=5 | max_consec=33 | avg_clip50=38.35 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37.3     |
| clipped_episode_r... | 20       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 361      |
| exploration rate     | 0.429    |
| time/                |          |
| episodes             | 8900     |
| fps                  | 231      |
| time_elapsed         | 2487     |
| total timesteps      | 576992   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0534   |
| n_updates            | 137997   |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=3,000 | STEP=577,939 | clip=70 | consec=4 | max_consec=33 | avg_clip50=38.95 |
| EP=3,010 | STEP=580,441 | clip=54 | consec=5 | max_consec=33 | avg_clip50=38.67 |
| EP=3,020 | STEP=583,514 | clip=68 | consec=3 | max_consec=33 | avg_clip50=40.03 |
| EP=3,030 | STEP=585,948 | clip=37 | consec=2 | max_consec=33 | avg_clip50=37.17 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37       |
| clipped_episode_r... | 36       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.12e+03 |
| ep_rew_mean          | 383      |
| exploration rate     | 0.42     |
| time/                |          |
| episodes             | 9000     |
| fps                  | 231      |
| time_elapsed         | 2528     |
| total timesteps      | 586221   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0642   |
| n_updates            | 140305   |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=3,040 | STEP=589,063 | clip=20 | consec=0 | max_consec=33 | avg_clip50=40.00 |
| EP=3,050 | STEP=591,826 | clip=49 | consec=4 | max_consec=33 | avg_clip50=39.57 |
| EP=3,060 | STEP=594,239 | clip=17 | consec=0 | max_consec=33 | avg_clip50=40.62 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 40.4 |
| clipped_episode_r... | 54   |
| consecutive_clipp... | 2    |

|                      |          |
|----------------------|----------|
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 397      |
| exploration rate     | 0.41     |
| time/                |          |
| episodes             | 9100     |
| fps                  | 231      |
| time_elapsed         | 2572     |
| total timesteps      | 595925   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0502   |
| n_updates            | 142731   |

EP=3,070 | STEP=596,986 | clip=26 | consec=3 | max\_consec=33 | avg\_clip50=40.21  
EP=3,080 | STEP=599,441 | clip=30 | consec=2 | max\_consec=33 | avg\_clip50=41.02  
EP=3,090 | STEP=602,285 | clip=44 | consec=3 | max\_consec=33 | avg\_clip50=38.47

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 36.8     |
| clipped_episode_r... | 29       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.12e+03 |
| ep_rew_mean          | 384      |
| exploration rate     | 0.401    |
| time/                |          |
| episodes             | 9200     |
| fps                  | 231      |
| time_elapsed         | 2612     |
| total timesteps      | 604598   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0593   |
| n_updates            | 144899   |

EP=3,100 | STEP=604,871 | clip=58 | consec=5 | max\_consec=33 | avg\_clip50=36.78  
EP=3,110 | STEP=607,705 | clip=30 | consec=15 | max\_consec=33 | avg\_clip50=38.27  
EP=3,120 | STEP=610,195 | clip=34 | consec=5 | max\_consec=33 | avg\_clip50=36.92  
EP=3,130 | STEP=612,553 | clip=30 | consec=2 | max\_consec=33 | avg\_clip50=37.44

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 37       |
| clipped_episode_r... | 55.5     |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.09e+03 |
| ep_rew_mean          | 384      |
| exploration rate     | 0.392    |
| time/                |          |
| episodes             | 9300     |
| fps                  | 231      |
| time_elapsed         | 2655     |
| total timesteps      | 613953   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.303    |

|           |        |
|-----------|--------|
| n_updates | 147238 |
|-----------|--------|

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=3,140 | STEP=615,658 | clip=6  | consec=0 | max_consec=33 | avg_clip50=38.58 |
| EP=3,150 | STEP=618,438 | clip=7  | consec=0 | max_consec=33 | avg_clip50=41.41 |
| EP=3,160 | STEP=621,421 | clip=60 | consec=5 | max_consec=33 | avg_clip50=41.80 |
| EP=3,170 | STEP=623,675 | clip=6  | consec=0 | max_consec=33 | avg_clip50=41.82 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 41.8     |
| clipped_episode_r... | 5.5      |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.11e+03 |
| ep_rew_mean          | 394      |
| exploration rate     | 0.383    |
| time/                |          |
| episodes             | 9400     |
| fps                  | 230      |
| time_elapsed         | 2700     |
| total timesteps      | 623675   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.313    |
| n_updates            | 149668   |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=3,180 | STEP=626,097 | clip=20 | consec=0 | max_consec=33 | avg_clip50=42.67 |
| EP=3,190 | STEP=628,876 | clip=24 | consec=1 | max_consec=33 | avg_clip50=42.60 |
| EP=3,200 | STEP=631,947 | clip=58 | consec=8 | max_consec=33 | avg_clip50=42.18 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 41.6     |
| clipped_episode_r... | 75.5     |
| consecutive_clipp... | 2        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.13e+03 |
| ep_rew_mean          | 421      |
| exploration rate     | 0.373    |
| time/                |          |
| episodes             | 9500     |
| fps                  | 230      |
| time_elapsed         | 2745     |
| total timesteps      | 633234   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0227   |
| n_updates            | 152058   |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=3,210 | STEP=634,837 | clip=26 | consec=8 | max_consec=33 | avg_clip50=40.87 |
| EP=3,220 | STEP=638,142 | clip=77 | consec=5 | max_consec=33 | avg_clip50=44.74 |
| EP=3,230 | STEP=641,013 | clip=24 | consec=9 | max_consec=33 | avg_clip50=46.07 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 45.6     |
| clipped_episode_r... | 67.5     |
| consecutive_clipp... | 17       |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.16e+03 |

|                  |        |
|------------------|--------|
| ep_rew_mean      | 441    |
| exploration rate | 0.363  |
| time/            |        |
| episodes         | 9600   |
| fps              | 230    |
| time_elapsed     | 2791   |
| total timesteps  | 643404 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.0706 |
| n_updates        | 154600 |

EP=3,240 | STEP=643,781 | clip=56 | consec=19 | max\_consec=33 | avg\_clip50=46.76  
EP=3,250 | STEP=646,835 | clip=27 | consec=5 | max\_consec=33 | avg\_clip50=47.33  
EP=3,260 | STEP=649,582 | clip=18 | consec=0 | max\_consec=33 | avg\_clip50=47.06  
EP=3,270 | STEP=652,975 | clip=84 | consec=10 | max\_consec=33 | avg\_clip50=48.30

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 48.9     |
| clipped_episode_r... | 48.5     |
| consecutive_clipp... | 14       |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.23e+03 |
| ep_rew_mean          | 471      |
| exploration rate     | 0.352    |
| time/                |          |
| episodes             | 9700     |
| fps                  | 230      |
| time_elapsed         | 2841     |
| total timesteps      | 654379   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0421   |
| n_updates            | 157344   |

EP=3,280 | STEP=655,852 | clip=28 | consec=20 | max\_consec=33 | avg\_clip50=47.43  
EP=3,290 | STEP=658,698 | clip=42 | consec=7 | max\_consec=33 | avg\_clip50=46.33  
EP=3,300 | STEP=662,141 | clip=35 | consec=9 | max\_consec=33 | avg\_clip50=46.49

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 46.6     |
| clipped_episode_r... | 38       |
| consecutive_clipp... | 5        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.25e+03 |
| ep_rew_mean          | 471      |
| exploration rate     | 0.342    |
| time/                |          |
| episodes             | 9800     |
| fps                  | 230      |
| time_elapsed         | 2888     |
| total timesteps      | 664398   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0801   |
| n_updates            | 159849   |

EP=3,310 | STEP=664,791 | clip=14 | consec=0 | max\_consec=33 | avg\_clip50=46.32



|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,320 | STEP=667,783 | clip=78 | consec=10 | max_consec=33 | avg_clip50=45.27 |
| EP=3,330 | STEP=670,756 | clip=18 | consec=0  | max_consec=33 | avg_clip50=45.67 |
| EP=3,340 | STEP=673,471 | clip=14 | consec=0  | max_consec=33 | avg_clip50=44.70 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 45.8     |
| clipped_episode_r... | 35.5     |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.23e+03 |
| ep_rew_mean          | 452      |
| exploration rate     | 0.332    |
| time/                |          |
| episodes             | 9900     |
| fps                  | 229      |
| time_elapsed         | 2934     |
| total timesteps      | 674523   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0518   |
| n_updates            | 162380   |

|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=3,350 | STEP=676,243 | clip=40 | consec=1 | max_consec=33 | avg_clip50=44.46 |
| EP=3,360 | STEP=678,982 | clip=38 | consec=7 | max_consec=33 | avg_clip50=45.19 |
| EP=3,370 | STEP=682,112 | clip=14 | consec=0 | max_consec=33 | avg_clip50=45.05 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 45.6     |
| clipped_episode_r... | 27       |
| consecutive_clipp... | 7        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.21e+03 |
| ep_rew_mean          | 450      |
| exploration rate     | 0.323    |
| time/                |          |
| episodes             | 10000    |
| fps                  | 229      |
| time_elapsed         | 2980     |
| total timesteps      | 684325   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.049    |
| n_updates            | 164831   |

|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,380 | STEP=685,039 | clip=58 | consec=10 | max_consec=33 | avg_clip50=45.59 |
| EP=3,390 | STEP=688,391 | clip=38 | consec=6  | max_consec=33 | avg_clip50=48.28 |
| EP=3,400 | STEP=691,521 | clip=77 | consec=2  | max_consec=33 | avg_clip50=49.73 |
| EP=3,410 | STEP=694,210 | clip=42 | consec=9  | max_consec=33 | avg_clip50=49.96 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 50.2     |
| clipped_episode_r... | 35.5     |
| consecutive_clipp... | 10       |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.22e+03 |
| ep_rew_mean          | 475      |
| exploration rate     | 0.312    |

|                 |        |
|-----------------|--------|
| time/           |        |
| episodes        | 10100  |
| fps             | 229    |
| time_elapsed    | 3028   |
| total timesteps | 694700 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.0454 |
| n_updates       | 167424 |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,420 | STEP=696,852 | clip=31 | consec=3  | max_consec=33 | avg_clip50=46.50 |
| EP=3,430 | STEP=699,710 | clip=64 | consec=13 | max_consec=33 | avg_clip50=46.78 |
| EP=3,440 | STEP=703,068 | clip=78 | consec=23 | max_consec=33 | avg_clip50=47.81 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 45.4     |
| clipped_episode_r... | 20       |
| consecutive_clipp... | 0        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.23e+03 |
| ep_rew_mean          | 472      |
| exploration rate     | 0.302    |
| time/                |          |
| episodes             | 10200    |
| fps                  | 229      |
| time_elapsed         | 3076     |
| total timesteps      | 705004   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.199    |
| n_updates            | 170000   |

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|          |              |          |           |               |                  |
|----------|--------------|----------|-----------|---------------|------------------|
| EP=3,450 | STEP=706,172 | clip=74  | consec=4  | max_consec=33 | avg_clip50=45.81 |
| EP=3,460 | STEP=709,225 | clip=117 | consec=14 | max_consec=33 | avg_clip50=46.46 |
| EP=3,470 | STEP=712,204 | clip=26  | consec=7  | max_consec=33 | avg_clip50=48.47 |
| EP=3,480 | STEP=715,624 | clip=46  | consec=17 | max_consec=33 | avg_clip50=49.71 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 49.7     |
| clipped_episode_r... | 54.5     |
| consecutive_clipp... | 19       |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.27e+03 |
| ep_rew_mean          | 481      |
| exploration rate     | 0.291    |
| time/                |          |
| episodes             | 10300    |
| fps                  | 228      |
| time_elapsed         | 3130     |
| total timesteps      | 716322   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0588   |
| n_updates            | 172830   |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,490 | STEP=718,736 | clip=60 | consec=27 | max_consec=33 | avg_clip50=49.40 |
| EP=3,500 | STEP=721,644 | clip=53 | consec=3  | max_consec=33 | avg_clip50=48.69 |
| EP=3,510 | STEP=724,328 | clip=54 | consec=1  | max_consec=33 | avg_clip50=47.23 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 48.1     |
| clipped_episode_r... | 69       |
| consecutive_clipp... | 8        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.26e+03 |
| ep_rew_mean          | 475      |
| exploration rate     | 0.281    |
| time/                |          |
| episodes             | 10400    |
| fps                  | 228      |
| time_elapsed         | 3178     |
| total timesteps      | 726296   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0783   |
| n_updates            | 175323   |

EP=3,520 | STEP=727,164 | clip=46 | consec=11 | max\_consec=33 | avg\_clip50=47.03  
EP=3,530 | STEP=730,346 | clip=50 | consec=21 | max\_consec=33 | avg\_clip50=47.90  
EP=3,540 | STEP=733,328 | clip=60 | consec=6 | max\_consec=33 | avg\_clip50=46.60  
EP=3,550 | STEP=736,198 | clip=36 | consec=5 | max\_consec=33 | avg\_clip50=46.88

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 47       |
| clipped_episode_r... | 33       |
| consecutive_clipp... | 8        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.25e+03 |
| ep_rew_mean          | 478      |
| exploration rate     | 0.27     |
| time/                |          |
| episodes             | 10500    |
| fps                  | 228      |
| time_elapsed         | 3229     |
| total timesteps      | 737324   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0405   |
| n_updates            | 178080   |

EP=3,560 | STEP=739,223 | clip=38 | consec=6 | max\_consec=33 | avg\_clip50=47.65  
EP=3,570 | STEP=741,719 | clip=95 | consec=1 | max\_consec=33 | avg\_clip50=47.06  
EP=3,580 | STEP=744,713 | clip=64 | consec=5 | max\_consec=33 | avg\_clip50=45.56

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 46.5    |
| clipped_episode_r... | 52      |
| consecutive_clipp... | 2       |
| max_consecutive_c... | 33      |
| rollout/             |         |
| ep_len_mean          | 1.2e+03 |
| ep_rew_mean          | 470     |
| exploration rate     | 0.26    |
| time/                |         |
| episodes             | 10600   |
| fps                  | 228     |

|  |                 |  |        |  |
|--|-----------------|--|--------|--|
|  | time_elapsed    |  | 3276   |  |
|  | total timesteps |  | 747165 |  |
|  | train/          |  |        |  |
|  | learning_rate   |  | 0.0001 |  |
|  | loss            |  | 0.0748 |  |
|  | n_updates       |  | 180541 |  |

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|          |  |              |  |          |  |           |  |               |  |                  |
|----------|--|--------------|--|----------|--|-----------|--|---------------|--|------------------|
| EP=3,590 |  | STEP=747,615 |  | clip=60  |  | consec=4  |  | max_consec=33 |  | avg_clip50=46.41 |
| EP=3,600 |  | STEP=750,527 |  | clip=58  |  | consec=2  |  | max_consec=33 |  | avg_clip50=47.24 |
| EP=3,610 |  | STEP=753,597 |  | clip=100 |  | consec=12 |  | max_consec=33 |  | avg_clip50=48.54 |
| EP=3,620 |  | STEP=756,519 |  | clip=30  |  | consec=9  |  | max_consec=33 |  | avg_clip50=50.38 |

|  |                      |  |          |  |
|--|----------------------|--|----------|--|
|  | custom/              |  |          |  |
|  | avg_clip50           |  | 52.2     |  |
|  | clipped_episode_r... |  | 77       |  |
|  | consecutive_clipp... |  | 14       |  |
|  | max_consecutive_c... |  | 33       |  |
|  | rollout/             |  |          |  |
|  | ep_len_mean          |  | 1.21e+03 |  |
|  | ep_rew_mean          |  | 480      |  |
|  | exploration rate     |  | 0.249    |  |
|  | time/                |  |          |  |
|  | episodes             |  | 10700    |  |
|  | fps                  |  | 227      |  |
|  | time_elapsed         |  | 3327     |  |
|  | total timesteps      |  | 758137   |  |
|  | train/               |  |          |  |
|  | learning_rate        |  | 0.0001   |  |
|  | loss                 |  | 0.0439   |  |
|  | n_updates            |  | 183284   |  |

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|          |  |              |  |         |  |           |  |               |  |                  |
|----------|--|--------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=3,630 |  | STEP=759,553 |  | clip=54 |  | consec=19 |  | max_consec=33 |  | avg_clip50=49.75 |
| EP=3,640 |  | STEP=762,681 |  | clip=48 |  | consec=29 |  | max_consec=33 |  | avg_clip50=49.91 |
| EP=3,650 |  | STEP=765,211 |  | clip=52 |  | consec=4  |  | max_consec=33 |  | avg_clip50=48.45 |
| EP=3,660 |  | STEP=768,179 |  | clip=55 |  | consec=14 |  | max_consec=33 |  | avg_clip50=48.36 |

|  |                      |  |         |  |
|--|----------------------|--|---------|--|
|  | custom/              |  |         |  |
|  | avg_clip50           |  | 48.4    |  |
|  | clipped_episode_r... |  | 55      |  |
|  | consecutive_clipp... |  | 14      |  |
|  | max_consecutive_c... |  | 33      |  |
|  | rollout/             |  |         |  |
|  | ep_len_mean          |  | 1.2e+03 |  |
|  | ep_rew_mean          |  | 484     |  |
|  | exploration rate     |  | 0.239   |  |
|  | time/                |  |         |  |
|  | episodes             |  | 10800   |  |
|  | fps                  |  | 227     |  |
|  | time_elapsed         |  | 3379    |  |
|  | total timesteps      |  | 768483  |  |
|  | train/               |  |         |  |
|  | learning_rate        |  | 0.0001  |  |
|  | loss                 |  | 0.0826  |  |
|  | n_updates            |  | 185870  |  |

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|          |  |              |  |         |  |           |  |               |  |                  |
|----------|--|--------------|--|---------|--|-----------|--|---------------|--|------------------|
| EP=3,670 |  | STEP=771,485 |  | clip=48 |  | consec=24 |  | max_consec=33 |  | avg_clip50=50.68 |
| EP=3,680 |  | STEP=774,368 |  | clip=49 |  | consec=4  |  | max_consec=33 |  | avg_clip50=51.07 |
| EP=3,690 |  | STEP=777,461 |  | clip=10 |  | consec=0  |  | max_consec=33 |  | avg_clip50=50.41 |

|  |         |  |  |  |
|--|---------|--|--|--|
|  | custom/ |  |  |  |
|--|---------|--|--|--|

|                      |          |
|----------------------|----------|
| avg_clip50           | 52.8     |
| clipped_episode_r... | 72       |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.23e+03 |
| ep_rew_mean          | 504      |
| exploration rate     | 0.228    |
| time/                |          |
| episodes             | 10900    |
| fps                  | 226      |
| time_elapsed         | 3436     |
| total timesteps      | 779640   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0429   |
| n_updates            | 188659   |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,700 | STEP=780,363 | clip=54 | consec=6  | max_consec=33 | avg_clip50=52.97 |
| EP=3,710 | STEP=783,309 | clip=12 | consec=0  | max_consec=33 | avg_clip50=52.26 |
| EP=3,720 | STEP=786,198 | clip=54 | consec=10 | max_consec=33 | avg_clip50=50.00 |
| EP=3,730 | STEP=789,235 | clip=58 | consec=7  | max_consec=33 | avg_clip50=51.62 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 50.8     |
| clipped_episode_r... | 40       |
| consecutive_clipp... | 10       |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.22e+03 |
| ep_rew_mean          | 512      |
| exploration rate     | 0.218    |
| time/                |          |
| episodes             | 11000    |
| fps                  | 226      |
| time_elapsed         | 3487     |
| total timesteps      | 790012   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0417   |
| n_updates            | 191252   |

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|          |              |         |          |               |                  |
|----------|--------------|---------|----------|---------------|------------------|
| EP=3,740 | STEP=791,932 | clip=44 | consec=3 | max_consec=33 | avg_clip50=51.65 |
| EP=3,750 | STEP=794,783 | clip=75 | consec=4 | max_consec=33 | avg_clip50=51.92 |
| EP=3,760 | STEP=797,312 | clip=14 | consec=0 | max_consec=33 | avg_clip50=51.20 |

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|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 51.1     |
| clipped_episode_r... | 46.5     |
| consecutive_clipp... | 8        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.21e+03 |
| ep_rew_mean          | 507      |
| exploration rate     | 0.208    |
| time/                |          |
| episodes             | 11100    |
| fps                  | 226      |
| time_elapsed         | 3535     |
| total timesteps      | 800084   |

|               |        |
|---------------|--------|
| train/        |        |
| learning_rate | 0.0001 |
| loss          | 0.0665 |
| n_updates     | 193770 |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,770 | STEP=800,335 | clip=24 | consec=10 | max_consec=33 | avg_clip50=50.44 |
| EP=3,780 | STEP=803,335 | clip=54 | consec=2  | max_consec=33 | avg_clip50=49.26 |
| EP=3,790 | STEP=806,085 | clip=26 | consec=9  | max_consec=33 | avg_clip50=47.88 |
| EP=3,800 | STEP=809,015 | clip=46 | consec=8  | max_consec=33 | avg_clip50=46.85 |

|                      |         |
|----------------------|---------|
| custom/              |         |
| avg_clip50           | 46.9    |
| clipped_episode_r... | 42      |
| consecutive_clipp... | 10      |
| max_consecutive_c... | 33      |
| rollout/             |         |
| ep_len_mean          | 1.2e+03 |
| ep_rew_mean          | 493     |
| exploration rate     | 0.198   |
| time/                |         |
| episodes             | 11200   |
| fps                  | 225     |
| time_elapsed         | 3583    |
| total timesteps      | 809907  |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.261   |
| n_updates            | 196226  |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,810 | STEP=812,101 | clip=42 | consec=18 | max_consec=33 | avg_clip50=48.52 |
| EP=3,820 | STEP=815,118 | clip=88 | consec=5  | max_consec=33 | avg_clip50=49.45 |
| EP=3,830 | STEP=817,616 | clip=46 | consec=2  | max_consec=33 | avg_clip50=46.88 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 46.6     |
| clipped_episode_r... | 74.5     |
| consecutive_clipp... | 3        |
| max_consecutive_c... | 33       |
| rollout/             |          |
| ep_len_mean          | 1.17e+03 |
| ep_rew_mean          | 475      |
| exploration rate     | 0.189    |
| time/                |          |
| episodes             | 11300    |
| fps                  | 225      |
| time_elapsed         | 3630     |
| total timesteps      | 819282   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0416   |
| n_updates            | 198570   |

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|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,840 | STEP=820,210 | clip=77 | consec=7  | max_consec=33 | avg_clip50=47.19 |
| EP=3,850 | STEP=823,571 | clip=46 | consec=17 | max_consec=33 | avg_clip50=48.79 |
| EP=3,860 | STEP=826,225 | clip=39 | consec=6  | max_consec=33 | avg_clip50=46.81 |
| EP=3,870 | STEP=829,210 | clip=58 | consec=16 | max_consec=33 | avg_clip50=47.64 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 48.1 |
| clipped_episode_r... | 52   |

|                      |         |
|----------------------|---------|
| consecutive_clipp... | 17      |
| max_consecutive_c... | 33      |
| rollout/             |         |
| ep_len_mean          | 1.2e+03 |
| ep_rew_mean          | 487     |
| exploration rate     | 0.179   |
| time/                |         |
| episodes             | 11400   |
| fps                  | 225     |
| time_elapsed         | 3678    |
| total timesteps      | 829520  |
| train/               |         |
| learning_rate        | 0.0001  |
| loss                 | 0.0404  |
| n_updates            | 201129  |

-----

|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,880 | STEP=831,949 | clip=58 | consec=26 | max_consec=33 | avg_clip50=49.40 |
| EP=3,890 | STEP=835,054 | clip=53 | consec=36 | max_consec=36 | avg_clip50=51.41 |
| EP=3,900 | STEP=837,719 | clip=52 | consec=2  | max_consec=43 | avg_clip50=48.29 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 49.9     |
| clipped_episode_r... | 71.5     |
| consecutive_clipp... | 8        |
| max_consecutive_c... | 43       |
| rollout/             |          |
| ep_len_mean          | 1.19e+03 |
| ep_rew_mean          | 487      |
| exploration rate     | 0.169    |
| time/                |          |
| episodes             | 11500    |
| fps                  | 225      |
| time_elapsed         | 3728     |
| total timesteps      | 839554   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0221   |
| n_updates            | 203638   |

-----

|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,910 | STEP=840,713 | clip=26 | consec=12 | max_consec=43 | avg_clip50=49.46 |
| EP=3,920 | STEP=843,538 | clip=55 | consec=4  | max_consec=43 | avg_clip50=48.75 |
| EP=3,930 | STEP=846,392 | clip=70 | consec=7  | max_consec=43 | avg_clip50=48.35 |
| EP=3,940 | STEP=849,573 | clip=40 | consec=17 | max_consec=43 | avg_clip50=48.14 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 47.5     |
| clipped_episode_r... | 30       |
| consecutive_clipp... | 18       |
| max_consecutive_c... | 43       |
| rollout/             |          |
| ep_len_mean          | 1.22e+03 |
| ep_rew_mean          | 496      |
| exploration rate     | 0.159    |
| time/                |          |
| episodes             | 11600    |
| fps                  | 224      |
| time_elapsed         | 3777     |
| total timesteps      | 849888   |
| train/               |          |
| learning_rate        | 0.0001   |

|           |        |
|-----------|--------|
| loss      | 0.126  |
| n_updates | 206221 |

-----

|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,950 | STEP=852,611 | clip=54 | consec=27 | max_consec=43 | avg_clip50=49.09 |
| EP=3,960 | STEP=855,330 | clip=22 | consec=2  | max_consec=43 | avg_clip50=47.89 |
| EP=3,970 | STEP=858,365 | clip=72 | consec=8  | max_consec=43 | avg_clip50=49.76 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 50.1     |
| clipped_episode_r... | 37.5     |
| consecutive_clipp... | 12       |
| max_consecutive_c... | 43       |
| rollout/             |          |
| ep_len_mean          | 1.21e+03 |
| ep_rew_mean          | 494      |
| exploration rate     | 0.149    |
| time/                |          |
| episodes             | 11700    |
| fps                  | 224      |
| time_elapsed         | 3826     |
| total timesteps      | 859846   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.224    |
| n_updates            | 208711   |

-----

|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=3,980 | STEP=861,393 | clip=57 | consec=18 | max_consec=43 | avg_clip50=51.79 |
| EP=3,990 | STEP=864,686 | clip=80 | consec=28 | max_consec=43 | avg_clip50=53.39 |
| EP=4,000 | STEP=868,029 | clip=56 | consec=38 | max_consec=43 | avg_clip50=56.50 |
| EP=4,010 | STEP=871,005 | clip=36 | consec=48 | max_consec=48 | avg_clip50=60.19 |

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 60.5     |
| clipped_episode_r... | 55       |
| consecutive_clipp... | 52       |
| max_consecutive_c... | 52       |
| rollout/             |          |
| ep_len_mean          | 1.26e+03 |
| ep_rew_mean          | 539      |
| exploration rate     | 0.137    |
| time/                |          |
| episodes             | 11800    |
| fps                  | 224      |
| time_elapsed         | 3885     |
| total timesteps      | 872107   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0534   |
| n_updates            | 211776   |

-----

|          |              |         |           |               |                  |
|----------|--------------|---------|-----------|---------------|------------------|
| EP=4,020 | STEP=873,736 | clip=44 | consec=2  | max_consec=55 | avg_clip50=56.32 |
| EP=4,030 | STEP=876,546 | clip=30 | consec=8  | max_consec=55 | avg_clip50=54.75 |
| EP=4,040 | STEP=879,652 | clip=58 | consec=18 | max_consec=55 | avg_clip50=54.19 |
| EP=4,050 | STEP=882,283 | clip=51 | consec=6  | max_consec=55 | avg_clip50=49.83 |

|                      |      |
|----------------------|------|
| custom/              |      |
| avg_clip50           | 49.8 |
| clipped_episode_r... | 51   |
| consecutive_clipp... | 6    |
| max_consecutive_c... | 55   |



|                  |          |
|------------------|----------|
| rollout/         |          |
| ep_len_mean      | 1.23e+03 |
| ep_rew_mean      | 532      |
| exploration rate | 0.126    |
| time/            |          |
| episodes         | 11900    |
| fps              | 224      |
| time_elapsed     | 3937     |
| total timesteps  | 882482   |
| train/           |          |
| learning_rate    | 0.0001   |
| loss             | 0.0431   |
| n_updates        | 214370   |

EP=4,060 | STEP=885,763 | clip=58 | consec=16 | max\_consec=55 | avg\_clip50=49.61  
EP=4,070 | STEP=888,794 | clip=18 | consec=0 | max\_consec=55 | avg\_clip50=51.89  
EP=4,080 | STEP=891,812 | clip=22 | consec=10 | max\_consec=55 | avg\_clip50=51.13

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 50.9     |
| clipped_episode_r... | 54.5     |
| consecutive_clipp... | 14       |
| max_consecutive_c... | 55       |
| rollout/             |          |
| ep_len_mean          | 1.26e+03 |
| ep_rew_mean          | 535      |
| exploration rate     | 0.116    |
| time/                |          |
| episodes             | 12000    |
| fps                  | 223      |
| time_elapsed         | 3991     |
| total timesteps      | 893249   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.309    |
| n_updates            | 217062   |

EP=4,090 | STEP=894,918 | clip=61 | consec=20 | max\_consec=55 | avg\_clip50=51.01  
EP=4,100 | STEP=898,263 | clip=40 | consec=1 | max\_consec=55 | avg\_clip50=53.68  
EP=4,110 | STEP=901,892 | clip=58 | consec=11 | max\_consec=55 | avg\_clip50=54.03  
EP=4,120 | STEP=905,597 | clip=76 | consec=21 | max\_consec=55 | avg\_clip50=57.08

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 57.1     |
| clipped_episode_r... | 75.5     |
| consecutive_clipp... | 21       |
| max_consecutive_c... | 55       |
| rollout/             |          |
| ep_len_mean          | 1.32e+03 |
| ep_rew_mean          | 545      |
| exploration rate     | 0.103    |
| time/                |          |
| episodes             | 12100    |
| fps                  | 223      |
| time_elapsed         | 4052     |
| total timesteps      | 905715   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0468   |
| n_updates            | 220178   |

-----  
EP=4,130 | STEP=908,905 | clip=34 | consec=31 | max\_consec=55 | avg\_clip50=60.16  
EP=4,140 | STEP=911,825 | clip=34 | consec=41 | max\_consec=55 | avg\_clip50=59.03  
EP=4,150 | STEP=914,611 | clip=52 | consec=6 | max\_consec=55 | avg\_clip50=56.70  
-----

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 55.7     |
| clipped_episode_r... | 52       |
| consecutive_clipp... | 11       |
| max_consecutive_c... | 55       |
| rollout/             |          |
| ep_len_mean          | 1.32e+03 |
| ep_rew_mean          | 547      |
| exploration rate     | 0.0931   |
| time/                |          |
| episodes             | 12200    |
| fps                  | 223      |
| time_elapsed         | 4103     |
| total timesteps      | 916044   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.11     |
| n_updates            | 222760   |

-----

EP=4,160 | STEP=917,678 | clip=74 | consec=16 | max\_consec=55 | avg\_clip50=55.54  
EP=4,170 | STEP=921,189 | clip=47 | consec=7 | max\_consec=55 | avg\_clip50=52.90  
EP=4,180 | STEP=924,192 | clip=78 | consec=7 | max\_consec=55 | avg\_clip50=51.11  
-----

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 53.4     |
| clipped_episode_r... | 86       |
| consecutive_clipp... | 16       |
| max_consecutive_c... | 55       |
| rollout/             |          |
| ep_len_mean          | 1.37e+03 |
| ep_rew_mean          | 565      |
| exploration rate     | 0.0814   |
| time/                |          |
| episodes             | 12300    |
| fps                  | 222      |
| time_elapsed         | 4162     |
| total timesteps      | 927850   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0356   |
| n_updates            | 225712   |

-----

EP=4,190 | STEP=927,995 | clip=78 | consec=17 | max\_consec=55 | avg\_clip50=54.26  
EP=4,200 | STEP=931,209 | clip=42 | consec=5 | max\_consec=55 | avg\_clip50=56.55  
EP=4,210 | STEP=934,620 | clip=78 | consec=15 | max\_consec=55 | avg\_clip50=57.26  
EP=4,220 | STEP=937,441 | clip=26 | consec=25 | max\_consec=55 | avg\_clip50=55.76  
-----

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 59       |
| clipped_episode_r... | 60.5     |
| consecutive_clipp... | 31       |
| max_consecutive_c... | 55       |
| rollout/             |          |
| ep_len_mean          | 1.33e+03 |
| ep_rew_mean          | 551      |

|                  |        |
|------------------|--------|
| exploration rate | 0.0696 |
| time/            |        |
| episodes         | 12400  |
| fps              | 222    |
| time_elapsed     | 4221   |
| total timesteps  | 939764 |
| train/           |        |
| learning_rate    | 0.0001 |
| loss             | 0.557  |
| n_updates        | 228690 |

EP=4,230 | STEP=940,575 | clip=58 | consec=35 | max\_consec=55 | avg\_clip50=57.30  
EP=4,240 | STEP=943,393 | clip=42 | consec=45 | max\_consec=55 | avg\_clip50=53.42  
EP=4,250 | STEP=946,620 | clip=57 | consec=2 | max\_consec=55 | avg\_clip50=54.44  
EP=4,260 | STEP=949,890 | clip=58 | consec=12 | max\_consec=55 | avg\_clip50=56.47

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 55.9     |
| clipped_episode_r... | 29       |
| consecutive_clipp... | 13       |
| max_consecutive_c... | 55       |
| rollout/             |          |
| ep_len_mean          | 1.32e+03 |
| ep_rew_mean          | 564      |
| exploration rate     | 0.0592   |
| time/                |          |
| episodes             | 12500    |
| fps                  | 222      |
| time_elapsed         | 4273     |
| total timesteps      | 950335   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0546   |
| n_updates            | 231333   |

EP=4,270 | STEP=952,575 | clip=57 | consec=22 | max\_consec=55 | avg\_clip50=57.57  
EP=4,280 | STEP=955,204 | clip=56 | consec=8 | max\_consec=55 | avg\_clip50=55.95  
EP=4,290 | STEP=958,042 | clip=54 | consec=18 | max\_consec=55 | avg\_clip50=57.35

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 55.4     |
| clipped_episode_r... | 60       |
| consecutive_clipp... | 24       |
| max_consecutive_c... | 55       |
| rollout/             |          |
| ep_len_mean          | 1.24e+03 |
| ep_rew_mean          | 558      |
| exploration rate     | 0.0497   |
| time/                |          |
| episodes             | 12600    |
| fps                  | 222      |
| time_elapsed         | 4321     |
| total timesteps      | 959943   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0373   |
| n_updates            | 233735   |

EP=4,300 | STEP=960,984 | clip=52 | consec=28 | max\_consec=55 | avg\_clip50=56.94  
EP=4,310 | STEP=964,053 | clip=49 | consec=38 | max\_consec=55 | avg\_clip50=55.47

EP=4,320 | STEP=967,213 | clip=49 | consec=48 | max\_consec=55 | avg\_clip50=56.34  
EP=4,330 | STEP=969,855 | clip=34 | consec=58 | max\_consec=58 | avg\_clip50=55.68

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 56.9     |
| clipped_episode_r... | 54.5     |
| consecutive_clipp... | 60       |
| max_consecutive_c... | 60       |
| rollout/             |          |
| ep_len_mean          | 1.22e+03 |
| ep_rew_mean          | 558      |
| exploration rate     | 0.039    |
| time/                |          |
| episodes             | 12700    |
| fps                  | 221      |
| time_elapsed         | 4375     |
| total timesteps      | 970664   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0309   |
| n_updates            | 236415   |

EP=4,340 | STEP=973,404 | clip=58 | consec=68 | max\_consec=68 | avg\_clip50=57.28  
EP=4,350 | STEP=976,161 | clip=52 | consec=78 | max\_consec=78 | avg\_clip50=57.32  
EP=4,360 | STEP=979,197 | clip=54 | consec=88 | max\_consec=88 | avg\_clip50=56.60  
EP=4,370 | STEP=982,292 | clip=60 | consec=7 | max\_consec=90 | avg\_clip50=55.41

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 55.7     |
| clipped_episode_r... | 44       |
| consecutive_clipp... | 8        |
| max_consecutive_c... | 90       |
| rollout/             |          |
| ep_len_mean          | 1.23e+03 |
| ep_rew_mean          | 557      |
| exploration rate     | 0.0272   |
| time/                |          |
| episodes             | 12800    |
| fps                  | 221      |
| time_elapsed         | 4435     |
| total timesteps      | 982612   |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.12     |
| n_updates            | 239402   |

EP=4,380 | STEP=985,062 | clip=62 | consec=6 | max\_consec=90 | avg\_clip50=56.46  
EP=4,390 | STEP=988,243 | clip=52 | consec=16 | max\_consec=90 | avg\_clip50=55.21  
EP=4,400 | STEP=991,256 | clip=78 | consec=26 | max\_consec=90 | avg\_clip50=54.12

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 52.8     |
| clipped_episode_r... | 31       |
| consecutive_clipp... | 31       |
| max_consecutive_c... | 90       |
| rollout/             |          |
| ep_len_mean          | 1.25e+03 |
| ep_rew_mean          | 547      |
| exploration rate     | 0.017    |
| time/                |          |

|                 |        |
|-----------------|--------|
| episodes        | 12900  |
| fps             | 221    |
| time_elapsed    | 4488   |
| total timesteps | 992883 |
| train/          |        |
| learning_rate   | 0.0001 |
| loss            | 0.023  |
| n_updates       | 241970 |

---

EP=4,410 | STEP=994,287 | clip=60 | consec=36 | max\_consec=90 | avg\_clip50=53.26  
EP=4,420 | STEP=997,761 | clip=76 | consec=46 | max\_consec=90 | avg\_clip50=54.85  
EP=4,430 | STEP=1,000,682 | clip=80 | consec=56 | max\_consec=90 | avg\_clip50=56.15  
EP=4,440 | STEP=1,003,729 | clip=57 | consec=66 | max\_consec=90 | avg\_clip50=55.60

---

|                      |          |
|----------------------|----------|
| custom/              |          |
| avg_clip50           | 55.5     |
| clipped_episode_r... | 60       |
| consecutive_clipp... | 69       |
| max_consecutive_c... | 90       |
| rollout/             |          |
| ep_len_mean          | 1.26e+03 |
| ep_rew_mean          | 550      |
| exploration rate     | 0.01     |
| time/                |          |
| episodes             | 13000    |
| fps                  | 220      |
| time_elapsed         | 4549     |
| total timesteps      | 1004500  |
| train/               |          |
| learning_rate        | 0.0001   |
| loss                 | 0.0728   |
| n_updates            | 244874   |

---

EP=4,450 | STEP=1,006,546 | clip=49 | consec=76 | max\_consec=90 | avg\_clip50=54.54  
EP=4,460 | STEP=1,009,754 | clip=66 | consec=86 | max\_consec=90 | avg\_clip50=55.91  
EP=4,470 | STEP=1,012,849 | clip=50 | consec=96 | max\_consec=96 | avg\_clip50=56.52

Objetivo conseguido: 100 episodios consecutivos superando el umbral.  
Entrenamiento finalizado.  
max\_consec alcanzado: 100  
Log JSON: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_SpaceInvaders-v0\_sb3\_log.json  
Log CSV: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0/dqn\_SpaceInvaders-v0\_sb3\_log.csv  
Checkpoint PyTorch guardado en: /content/gdrive/My Drive/08\_MIAR/sb3/conlogs/datos/sb3\_1\_spaceinvaders\_v0\_tuned/dqn\_spaceinvaders\_v0\_policy\_latest.pt  
model.save() falló (no es crítico, se usa el .pt): IndexError('tuple index out of range')

## Test uniforme de todas las variantes

Mismo protocolo que el test del modelo base: **120 episodios con política determinista** sobre el entorno de evaluación **sin clipping** de recompensa (se reporta la puntuación real del juego). Los resultados se guardan en `test_results.json` dentro de la carpeta

de cada variante, de modo que esta celda puede re-ejecutarse sin repetir tests ya realizados.

```
In [ ]: # =====
# M8. Test (120 episodios deterministas) de cada variante
# =====

TEST_EPISODES_VARIANTS = 120

def evaluate_variant(run_name, n_episodes=TEST_EPISODES_VARIANTS):
    paths = RUN_PATHS[run_name]
    eval_env = create_eval_env()
    model = build_model(run_name, eval_env, for_eval=True)

    checkpoint = th.load(paths['policy'], map_location=model.device)
    model.policy.load_state_dict(checkpoint['policy_state_dict'])

    rewards = []
    for ep in range(n_episodes):
        obs = eval_env.reset()
        done = False
        total_reward = 0.0
        while not done:
            action, _ = model.predict(obs, deterministic=True)
            obs, reward, done, infos = eval_env.step(action)
            total_reward += float(reward[0])
            done = bool(done[0])
        rewards.append(total_reward)
        if (ep + 1) % 20 == 0:
            print(f' {run_name}: episodio {ep + 1}/{n_episodes} | media parcial

    eval_env.close()
    return rewards

test_results = {}

for run_name in RUN_NAMES:
    paths = RUN_PATHS[run_name]

    if paths['test_json'].exists():
        with open(paths['test_json'], 'r') as f:
            test_results[run_name] = json.load(f)
        print(run_name, '-> resultados de test ya guardados, se reutilizan.')
        continue

    if not paths['policy'].exists():
        print(run_name, '-> sin checkpoint entrenado todavía, se omite.')
        continue

    print('Testeando variante:', RUN_LABELS[run_name])
    rewards = evaluate_variant(run_name)
    test_results[run_name] = {
        'rewards': [float(r) for r in rewards],
        'mean': float(np.mean(rewards)),
        'std': float(np.std(rewards)),
        'max': float(np.max(rewards)),
        'min': float(np.min(rewards)),
    }
```

```

with open(paths['test_json'], 'w') as f:
    json.dump(test_results[run_name], f, indent=2)

print()
for run_name, res in test_results.items():
    print(f"{RUN_LABELS[run_name]:28s} | media={res['mean']:7.1f} | std={res['st

```

Testeando variante: DQN base

Using cuda device

```

base: episodio 20/120 | media parcial: 629.0
base: episodio 40/120 | media parcial: 641.4
base: episodio 60/120 | media parcial: 656.5
base: episodio 80/120 | media parcial: 660.5
base: episodio 100/120 | media parcial: 648.1
base: episodio 120/120 | media parcial: 650.6

```

Testeando variante: Double DQN

Using cuda device

```

double: episodio 20/120 | media parcial: 618.2
double: episodio 40/120 | media parcial: 590.2
double: episodio 60/120 | media parcial: 606.6
double: episodio 80/120 | media parcial: 605.6
double: episodio 100/120 | media parcial: 600.1
double: episodio 120/120 | media parcial: 598.4

```

Testeando variante: Dueling DQN

Using cuda device

```

dueling: episodio 20/120 | media parcial: 576.2
dueling: episodio 40/120 | media parcial: 603.0
dueling: episodio 60/120 | media parcial: 592.1
dueling: episodio 80/120 | media parcial: 612.8
dueling: episodio 100/120 | media parcial: 614.2
dueling: episodio 120/120 | media parcial: 622.0

```

Testeando variante: DQN exploración ajustada

Using cuda device

```

tuned: episodio 20/120 | media parcial: 618.2
tuned: episodio 40/120 | media parcial: 610.4
tuned: episodio 60/120 | media parcial: 616.3
tuned: episodio 80/120 | media parcial: 607.5
tuned: episodio 100/120 | media parcial: 600.2
tuned: episodio 120/120 | media parcial: 591.4

```

|                          |              |            |           |
|--------------------------|--------------|------------|-----------|
| DQN base                 | media= 650.6 | std= 182.0 | max= 1295 |
| Double DQN               | media= 598.4 | std= 147.3 | max= 1280 |
| Dueling DQN              | media= 622.0 | std= 171.0 | max= 1430 |
| DQN exploración ajustada | media= 591.4 | std= 180.7 | max= 1400 |

## Gráficas comparativas de las propuestas

Se generan cuatro gráficas comparando las variantes: **(1)** curvas de aprendizaje (media móvil de 50 episodios), **(2)** progreso hacia el criterio de éxito, **(3)** schedules de exploración y **(4)** recompensa media en test; más una tabla resumen con métricas de eficiencia de muestras (steps necesarios para alcanzar niveles de rendimiento).

```

In [ ]: # =====
# M9. Comparativa: curvas, test y tabla resumen
# =====

if 'test_results' not in globals():
    test_results = {}

```

```

dfs = {}
for run_name in RUN_NAMES:
    csv_path = RUN_PATHS[run_name]['csv']
    if csv_path.exists():
        dfs[run_name] = pd.read_csv(csv_path)
    else:
        print('Sin log de entrenamiento todavía:', run_name)

# --- Gráfica 1: curvas de aprendizaje ---
plt.figure(figsize=(11, 5))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['avg_clip50'], label=RUN_LABELS[run_name])
    plt.axhline(CONSECUTIVE_THRESHOLD, linestyle='--', color='gray', label='Umbral (')
plt.title('Curva de aprendizaje: recompensa clipped media (últimos 50 episodios)')
plt.xlabel('Steps')
plt.ylabel('avg_clip50')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 2: progreso hacia el criterio de éxito ---
plt.figure(figsize=(11, 5))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['max_consecutive_clipped_over_20'], label=RUN_LABELS[run_name])
    plt.axhline(TARGET_CONSECUTIVE_EPISODES, linestyle='--', color='gray', label='Objetivo')
plt.title('Máximo de episodios consecutivos con recompensa clipped > 20')
plt.xlabel('Steps')
plt.ylabel('max_consec')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 3: schedules de exploración ---
plt.figure(figsize=(11, 4))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['exploration_rate'], label=RUN_LABELS[run_name])
plt.title('Evolución de epsilon por variante')
plt.xlabel('Steps')
plt.ylabel('epsilon')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 4: recompensa media en test ---
if len(test_results) > 0:
    names = [RUN_LABELS[k] for k in test_results]
    means = [test_results[k]['mean'] for k in test_results]
    stds = [test_results[k]['std'] for k in test_results]

    plt.figure(figsize=(9, 5))
    plt.bar(names, means, yerr=stds, capsize=6)
    plt.title('Recompensa media en test (120 episodios, política determinista)')
    plt.ylabel('Recompensa media (puntuación real)')
    plt.xticks(rotation=15)
    plt.grid(True, axis='y')
    plt.show()
else:
    print('Aún no hay resultados de test que graficar.')

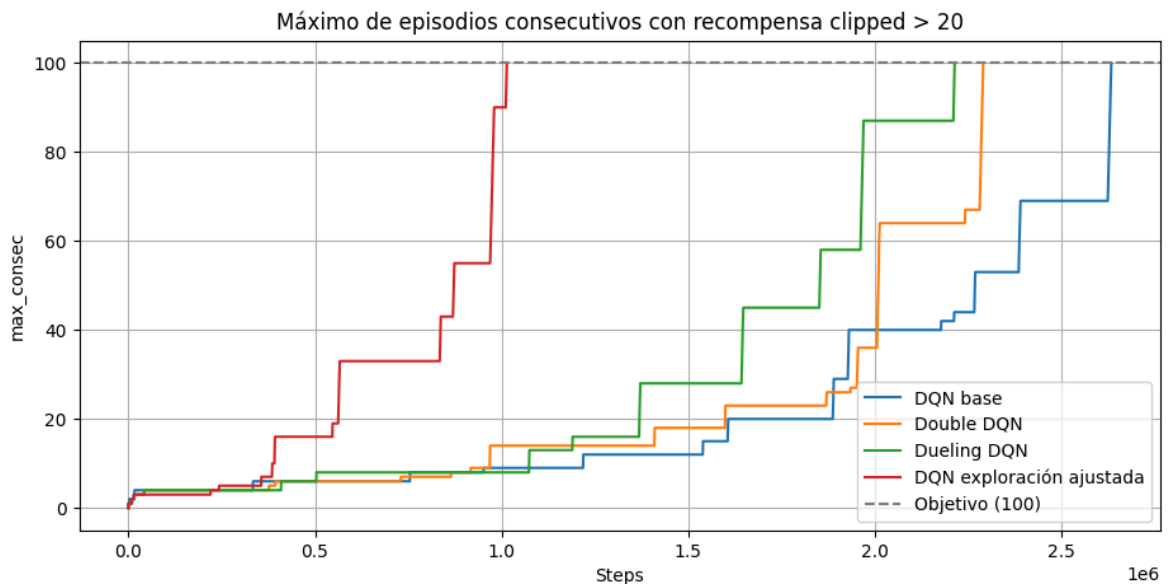
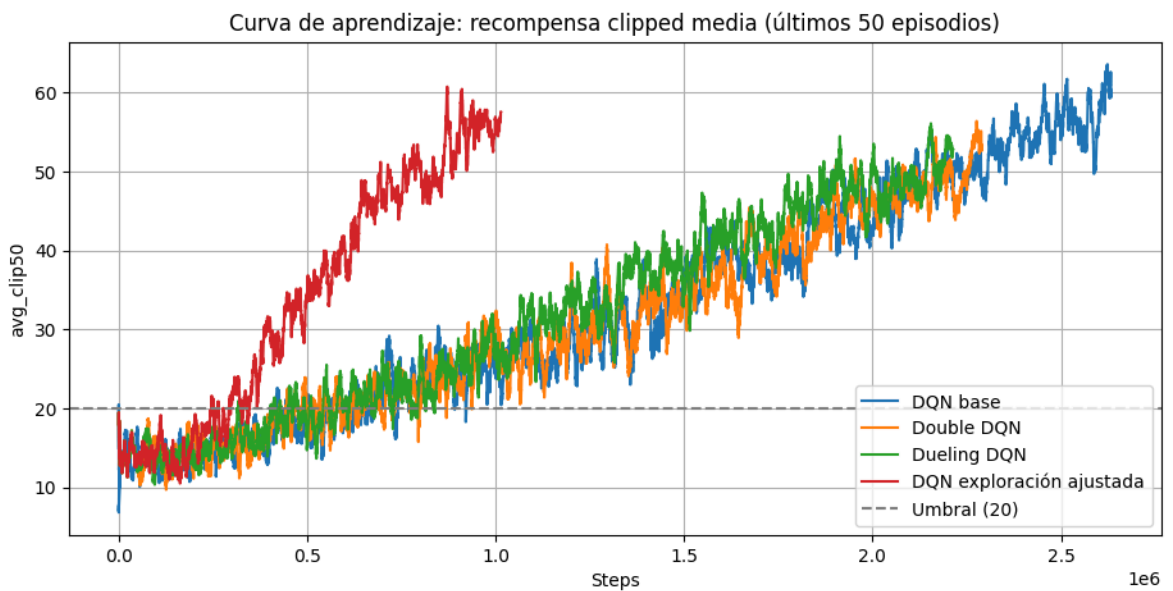
```

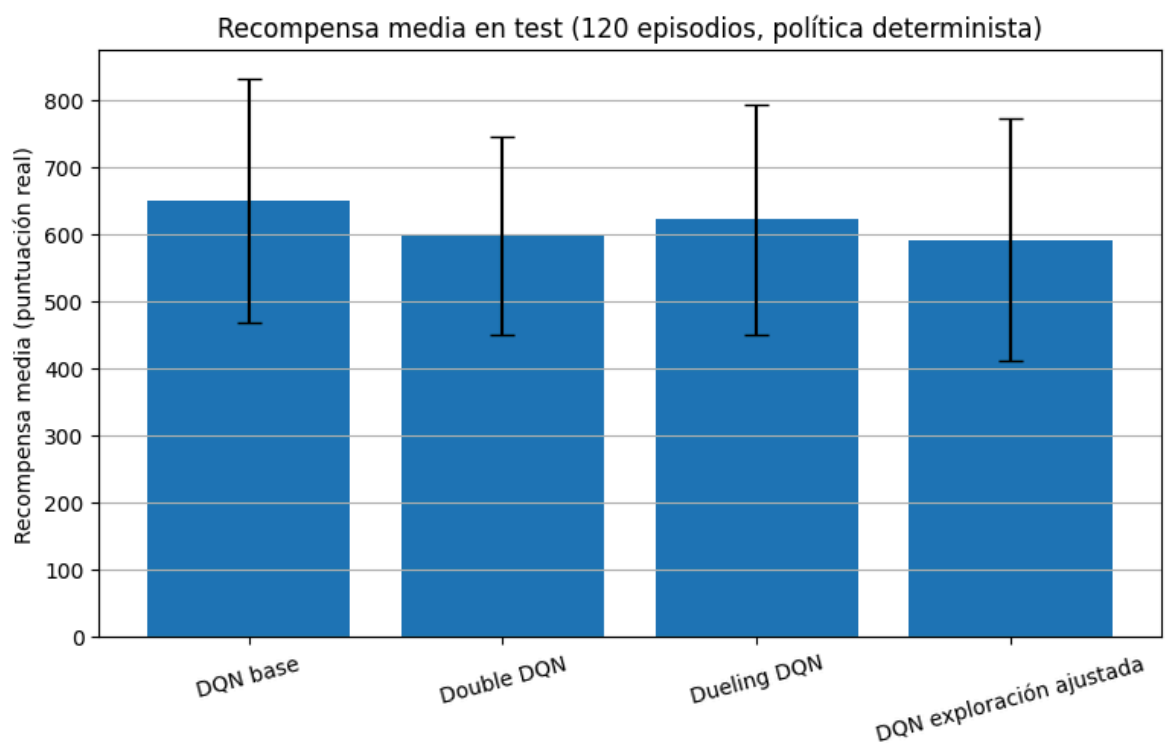
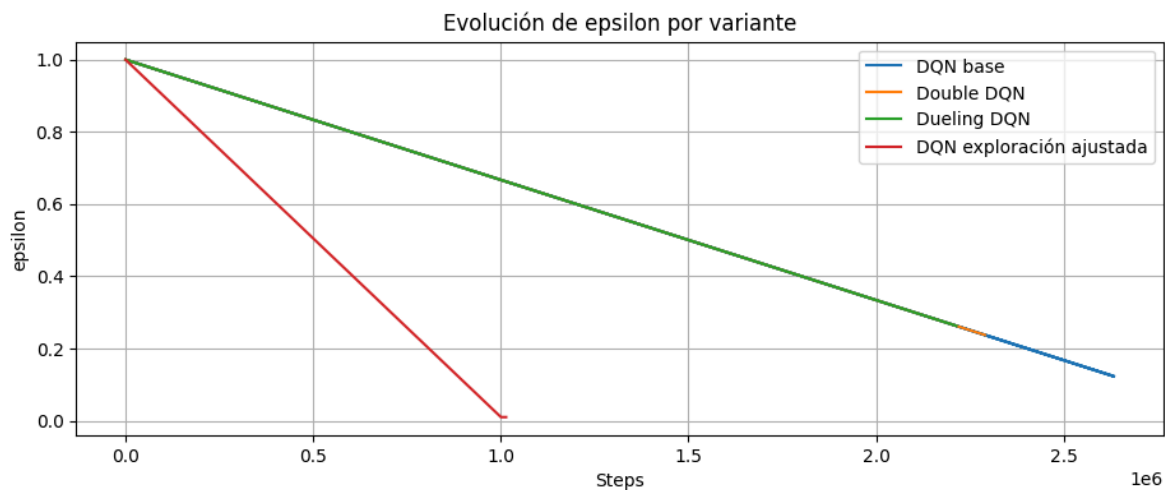


```
# --- Tabla resumen ---
def steps_to_reach(df_run, threshold):
    hits = df_run[df_run['avg_clip50'] >= threshold]
    return int(hits['total_steps'].iloc[0]) if len(hits) else None

rows = []
for run_name, df_run in dfs.items():
    res = test_results.get(run_name, {})
    rows.append({
        'variante': RUN_LABELS[run_name],
        'steps_totales': int(df_run['total_steps'].iloc[-1]),
        'episodios': len(df_run),
        'steps_hasta_avg20': steps_to_reach(df_run, 20),
        'steps_hasta_avg40': steps_to_reach(df_run, 40),
        'mejor_avg_clip50': float(df_run['avg_clip50'].max()),
        'max_consec': int(df_run['max_consecutive_clipped_over_20'].max()),
        'test_media': res.get('mean'),
        'test_std': res.get('std'),
        'test_max': res.get('max'),
    })

summary = pd.DataFrame(rows)
display(summary)
```





|   | variante                 | steps_totales | episodios | steps_hasta_avg20 | steps_hasta_avg40 | mejor_avg_100 |
|---|--------------------------|---------------|-----------|-------------------|-------------------|---------------|
| 0 | DQN base                 | 2633424       | 11961     | 338               | 1557345           | 1770016       |
| 1 | Double DQN               | 2290139       | 10808     | 411061            | 1294769           | 1557345       |
| 2 | Dueling DQN              | 2213601       | 10634     | 314826            | 1370016           | 1557345       |
| 3 | DQN exploración ajustada | 1014048       | 4475      | 245690            | 544860            | 1557345       |



## PARTE 3. Justificación de los parámetros seleccionados y de los resultados

# obtenidos

## 3.1. Entorno y preprocesado

El preprocesado replica el pipeline clásico de DQN para Atari (Mnih et al., 2015), aplicado automáticamente por `make_atari_env`:

| Elemento                                     | Valor                    | Justificación                                                                                                                                                                                   |
|----------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Escala de grises +<br>resize                 | 84×84                    | El color no aporta información decisiva en Space Invaders, y reducir de 210×160×3 a 84×84×1 divide el coste computacional de la red por ~9                                                      |
| Pila de frames                               | 4                        | Una imagen estática no contiene velocidades ni direcciones (proyectiles, desplazamiento de los invasores); el stack de 4 frames restaura aproximadamente la propiedad de Markov del estado      |
| Frame-skip<br>( <code>MaxAndSkipEnv</code> ) | 4                        | Repite cada acción durante 4 frames y toma el máximo píxel a píxel de los 2 últimos (elimina el parpadeo de sprites típico de Atari); acelera la simulación ~4× sin pérdida de control efectivo |
| Clipping de recompensa                       | [-1, 1] en entrenamiento | Homogeneiza la escala de los targets y estabiliza el gradiente (técnica estándar del paper de Nature)                                                                                           |
| <code>FireResetEnv</code>                    | —                        | Ejecuta FIRE automáticamente al inicio del episodio, necesario para que la partida arranque                                                                                                     |

**Aclaración sobre la métrica `clip` de los logs:** es `recompensa_real / 10`, una convención interna del proyecto para seguir el entrenamiento con números comparables a la práctica anterior; **no** es el clipping  $\pm 1$  de Atari (que sí se aplica internamente durante el entrenamiento). En **test** se usa el entorno sin clipping y se reporta la **puntuación real** del juego, que es lo que exige el enunciado.

## 3.2. Red neuronal

Arquitectura convolucional del DQN de Nature (Mnih et al., 2015), implementada como `features_extractor` de SB3 ( `CustomCNN84` ) con `net_arch=[]` para que la única capa posterior al extractor sea la cabeza lineal Q:

| Capa            | Configuración                   | Salida   | Parámetros |
|-----------------|---------------------------------|----------|------------|
| Conv1           | 32 filtros 8×8, stride 4, ReLU  | 20×20×32 | 8.224      |
| Conv2           | 64 filtros 4×4, stride 2, ReLU  | 9×9×64   | 32.832     |
| Conv3           | 64 filtros 3×3, stride 1, ReLU  | 7×7×64   | 36.928     |
| Flatten + Dense | 3136 → 512, ReLU                | 512      | 1.606.144  |
| Cabeza Q        | 512 → 6 (una salida por acción) | 6        | 3.078      |

En total  $\approx 1,69$  M de parámetros. Los strides grandes sustituyen al pooling (reducen resolución preservando información espacial aprendible), y la red produce las 6 Q-values en una sola pasada, lo que permite el `argmax` de la política greedy con una única inferencia.

### 3.3. Hiperparámetros del DQN base

| Parámetro                                               | Valor                                    | Justificación                                                                                                                                     |
|---------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>learning_rate</code>                              | 1e-4                                     | Valor de referencia para Adam en DQN-Atari; con lr mayores la pérdida de Huber oscila y desestabiliza los targets                                 |
| <code>buffer_size</code>                                | 100.000                                  | Compromiso entre diversidad de experiencias y RAM de Colab ( $\sim 2,8$ GB en uint8 con frames $84 \times 84 \times 4$ )                          |
| <code>learning_starts</code>                            | 50.000                                   | Llena el buffer con experiencia (mayoritariamente aleatoria) antes de la primera actualización, evitando sobreajustar a las primeras transiciones |
| <code>batch_size</code>                                 | 32                                       | Valor del paper original: batches pequeños con muchas actualizaciones funcionan bien en DQN                                                       |
| <code>gamma</code>                                      | 0,99                                     | Horizonte efectivo de $\sim 100$ steps, suficiente para conectar causalmente un disparo con su impacto                                            |
| <code>train_freq</code>                                 | 4                                        | Una actualización de gradiente por cada 4 steps de entorno (ratio del paper)                                                                      |
| <code>target_update_interval</code>                     | 10.000                                   | La red target congelada estabiliza los targets del TD-error                                                                                       |
| $\epsilon: 1.0 \rightarrow 0.001$ (lineal, 30 % de 10M) | schedule lineal                          | Exploración intensa al principio, explotación progresiva                                                                                          |
| Criterio de parada                                      | 100 episodios consecutivos con clip > 20 | Exige consistencia: no basta con un episodio bueno aislado                                                                                        |

### 3.4. Resultados del DQN base

- El entrenamiento se detuvo por el **criterio de éxito** tras  $\sim 2,63$ M steps y  $\sim 11.930$  episodios: 100 episodios consecutivos con recompensa clipped > 20.
- Mejor media móvil `avg_clip50` = **63,6** (equivalente a  $\sim 636$  puntos reales);  $\epsilon$  final  $\approx 0,123$ .
- Test (120 episodios, política determinista): media = 662,25 puntos**, máximo = 1.405. El requisito del listado del grupo —**alcanzar más de 20 puntos con reward clipping durante más de 100 episodios consecutivos**— se cumple por partida doble: durante el entrenamiento el agente encadenó 100 episodios consecutivos con recompensa clipped > 20 (es precisamente el criterio de parada implementado),

y en modo test la media equivale a ~66 puntos clipped, más del triple del umbral. Las cuatro configuraciones de la Parte 2 satisfacen igualmente el requisito: todas alcanzaron los 100 episodios consecutivos y obtienen medias de test de 59-65 puntos clipped.

- La curva de aprendizaje muestra tres fases: (i) hasta ~0,5M steps, mejora lenta mientras domina la exploración; (ii) entre 0,5M y 2M, crecimiento sostenido de `avg_clip50` (de ~10 a ~50) a medida que  $\epsilon$  decae; (iii) a partir de ~2M, consistencia suficiente para encadenar los 100 episodios del criterio.
- Un detalle relevante: el agente alcanzó el criterio **explorando todavía al 12 %**. La política greedy subyacente era mejor de lo que reflejaban los episodios de entrenamiento, y por eso el test determinista (662 puntos de media) supera a la media móvil de entrenamiento (~600). Esta observación es precisamente la que motiva la Propuesta 3.

### 3.5. Discusión de las tres propuestas

**Propuesta 1 — Double DQN.** *Qué esperamos:* al eliminar el sesgo de sobreestimación del `max`, los targets son más realistas y el aprendizaje más estable. En la literatura (van Hasselt et al., 2016) Double DQN iguala o supera a DQN en la gran mayoría de juegos de Atari, con ganancias moderadas en Space Invaders. *Resultado obtenido:* Double DQN alcanzó el criterio de éxito en **2,29M steps frente a los 2,63M del base (-13 %)**, y llegó antes a un nivel de juego alto (`avg_clip50`  $\geq 40$  en 1,29M steps frente a 1,56M, -17 %). En el test de 120 episodios obtuvo una media de **598,4 puntos con la desviación típica más baja de las cuatro configuraciones (147,3, un 19 % menos dispersión que el base)**. Es decir: aprendió algo más rápido y juega de forma más *consistente*, exactamente el efecto que predice la corrección del sesgo de sobreestimación (van Hasselt et al., 2016). Su media de test queda por debajo de la del base (650,6), pero el base entrenó un 15 % más de steps y la diferencia (~52 puntos) es del orden de 3 errores estándar con una sola semilla, por lo que no puede considerarse concluyente.

**Propuesta 2 — Dueling DQN.** *Qué esperamos:* mejor eficiencia de muestras, porque cada actualización mejora la estimación de `V(s)` compartida por todas las acciones. Wang et al. (2016) reportan una de las mayores mejoras relativas precisamente en Space Invaders, donde abundan los estados en los que la acción concreta importa poco. *Resultado obtenido:* Dueling fue la más eficiente de las dos mejoras algorítmicas: criterio en **2,21M steps (-16 % vs. base)** y el arranque más rápido de las variantes con schedule estándar (`avg_clip50`  $\geq 20$  en 315k steps, frente a 411k de Double). En test logró la **segunda mejor media (622,0  $\pm$  171,0)** y el **mejor episodio individual de todo el estudio (1.430 puntos)**. El resultado es coherente con Wang et al. (2016), que reportan ganancias de eficiencia especialmente grandes en Space Invaders: al aprender `V(s)` compartido entre acciones, cada actualización aprovecha más la experiencia en los (muchos) estados donde la acción concreta importa poco.

**Propuesta 3 — Exploración ajustada.** *Qué esperamos:* una subida más temprana de la curva de aprendizaje al empezar a explotar antes (el annealing termina en ~1M steps en lugar de seguir al 12 % de aleatoriedad a los 2,6M). Los target updates más frecuentes

(5k) aceleran la propagación de lo aprendido con un leve coste de estabilidad. El riesgo de esta propuesta es converger prematuramente a una política subóptima por falta de exploración tardía. *Resultado obtenido*: el efecto más espectacular del estudio: criterio de éxito en **1,01M steps, 2,6 veces menos que el base**, y nivel alto de juego ( $\text{avg\_clip50} \geq 40$ ) en solo 545k steps frente a 1,56M (2,9×). La curva de aprendizaje (gráfica 1 de M9) se despegaba de las demás desde ~250k steps. Como contrapartida, su media de test es la más baja ( $591,4 \pm 180,7$ ), con dos explicaciones honestas: (i) al parar tan pronto acumuló ~246k actualizaciones de gradiente frente a ~645k del base — su política final está menos refinada; y (ii) el criterio de parada no es del todo comparable entre schedules: esta variante encadenó sus 100 episodios jugando con  $\epsilon = 0,01$  (sin apenas ruido en contra), mientras que base, Double y Dueling lo lograron explorando aún al 12-26 %, un mérito adicional que el criterio no captura.

## 3.6. Comparativa global y conclusiones

**1. Eficiencia de muestras.** Las tres propuestas superan al DQN base: exploración ajustada (1,01M steps hasta el criterio)  $\ll$  Dueling (2,21M)  $\approx$  Double (2,29M)  $<$  base (2,63M). Con el umbral  $\text{avg\_clip50} \geq 40$ , el orden se mantiene (545k / 1,37M / 1,29M / 1,56M). Nota metodológica: el valor `steps_hasta_avg20 = 338` del base es un artefacto de la media móvil de 50 episodios en el arranque (pocos episodios en la ventana), por lo que usamos el umbral 40 como métrica robusta de eficiencia.

**2. Rendimiento final.** El base obtiene la mejor media de test (650,6), seguido de Dueling (622,0), Double (598,4) y tuned (591,4). Pero estas diferencias deben leerse con cautela: el error estándar de cada media es de  $\pm 13$ -17 puntos (std ~147-182 sobre 120 episodios), hay una única semilla, y el diseño experimental iguala la *competencia* alcanzada (todas paran al mismo criterio), no el presupuesto: el base entrenó entre un 15 % y un 160 % más que las demás. La lectura correcta no es "el base es mejor", sino que **las tres mejoras alcanzan un rendimiento final comparable con un coste de entrenamiento sustancialmente menor**. Las cuatro configuraciones superan holgadamente el requisito mínimo del enunciado.

**3. Estabilidad.** Ninguna variante muestra colapsos ni oscilaciones patológicas en sus curvas. Double DQN es la más consistente en test (std 147,3 frente a 171-182 del resto), en línea con la teoría: targets menos sobreestimados producen una política de calidad más uniforme. Dueling combina consistencia razonable con los mejores episodios pico (máx. 1.430).

**4. Coherencia con la literatura.** El orden esperado en eficiencia (Dueling  $\geq$  Double  $\geq$  DQN) se reproduce, incluida la ventaja de Dueling en Space Invaders que reportan Wang et al. (2016). La sorpresa aparente —que ninguna mejora supere al base en media de test— desaparece al considerar el presupuesto: con parada por criterio de competencia, las mejoras convierten su ventaja en *ahorro de cómputo* en lugar de en puntuación final. Un experimento complementario natural sería la comparación a presupuesto fijo (mismos steps para todas), donde la literatura predice que las mejoras superarían también la puntuación del base.

**Conclusión.** Se entrega como mejor modelo el **DQN base (media de test 650,6 con política determinista)**, si bien el resultado central del estudio es que las tres propuestas alcanzan una competencia equivalente con hasta 2,6× menos interacciones con el entorno — en un problema real, donde la experiencia es cara, la variante de exploración ajustada sería la elección preferible.

### 3.7. Limitaciones y trabajo futuro

- **Una sola semilla (42) por variante**, por limitación de cómputo: las diferencias pequeñas entre variantes no son estadísticamente concluyentes. Lo riguroso serían 3–5 semillas por variante y reportar media  $\pm$  desviación entre semillas.
- Las mejoras se han evaluado **por separado** para aislar su efecto; combinarlas (al estilo Rainbow, Hessel et al., 2018), junto con Prioritized Experience Replay y n-step returns, sería la continuación natural.
- La **parada temprana** al alcanzar el criterio favorece comparar eficiencia de muestras, pero no el rendimiento asintótico: con más presupuesto las conclusiones podrían cambiar.
- El test usa una única configuración (120 episodios deterministas con la misma semilla de entorno); una evaluación más robusta usaría semillas de test distintas de la de entrenamiento.

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